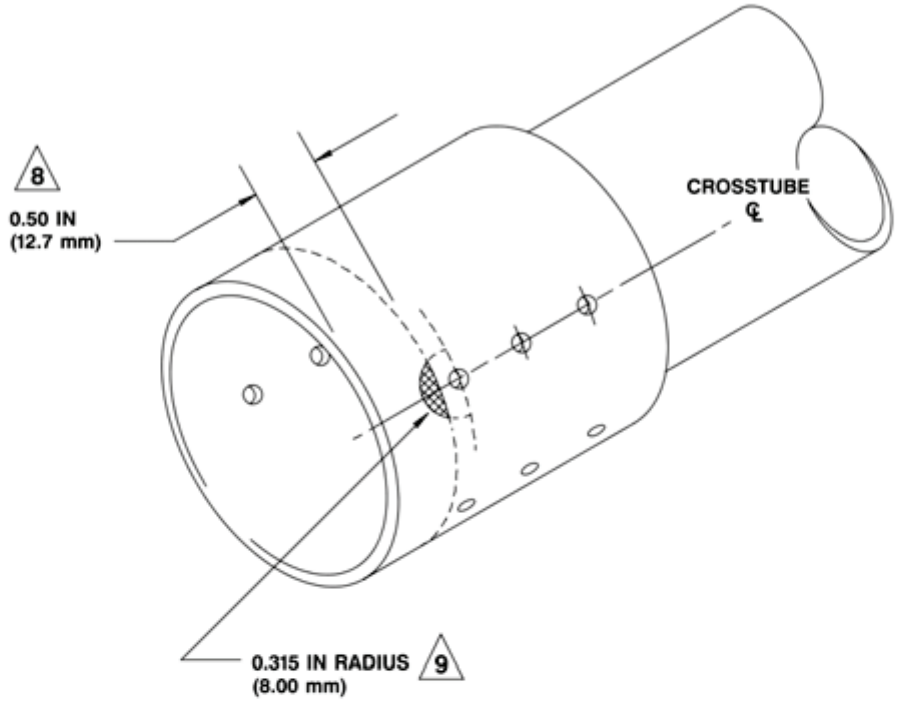


DETAIL C
412-050-030
412-050-034

			(Does not apply to saddles)
DENTS	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No dents allowed. Replace as needed.	Not critical.
HOLES	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No holes allowed. Replace as needed.	Not critical. Plug holes to keep out moisture.
CRACKS	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.
SCRATCHES SCUFFS NICKS	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	Not critical.
	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	
	If damage is greater than established limits, replace as needed.	If damage is greater than established limits, replace as needed.	
CORROSION	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup is limited to 10% of material thickness. Refer to paragraph 32-11.

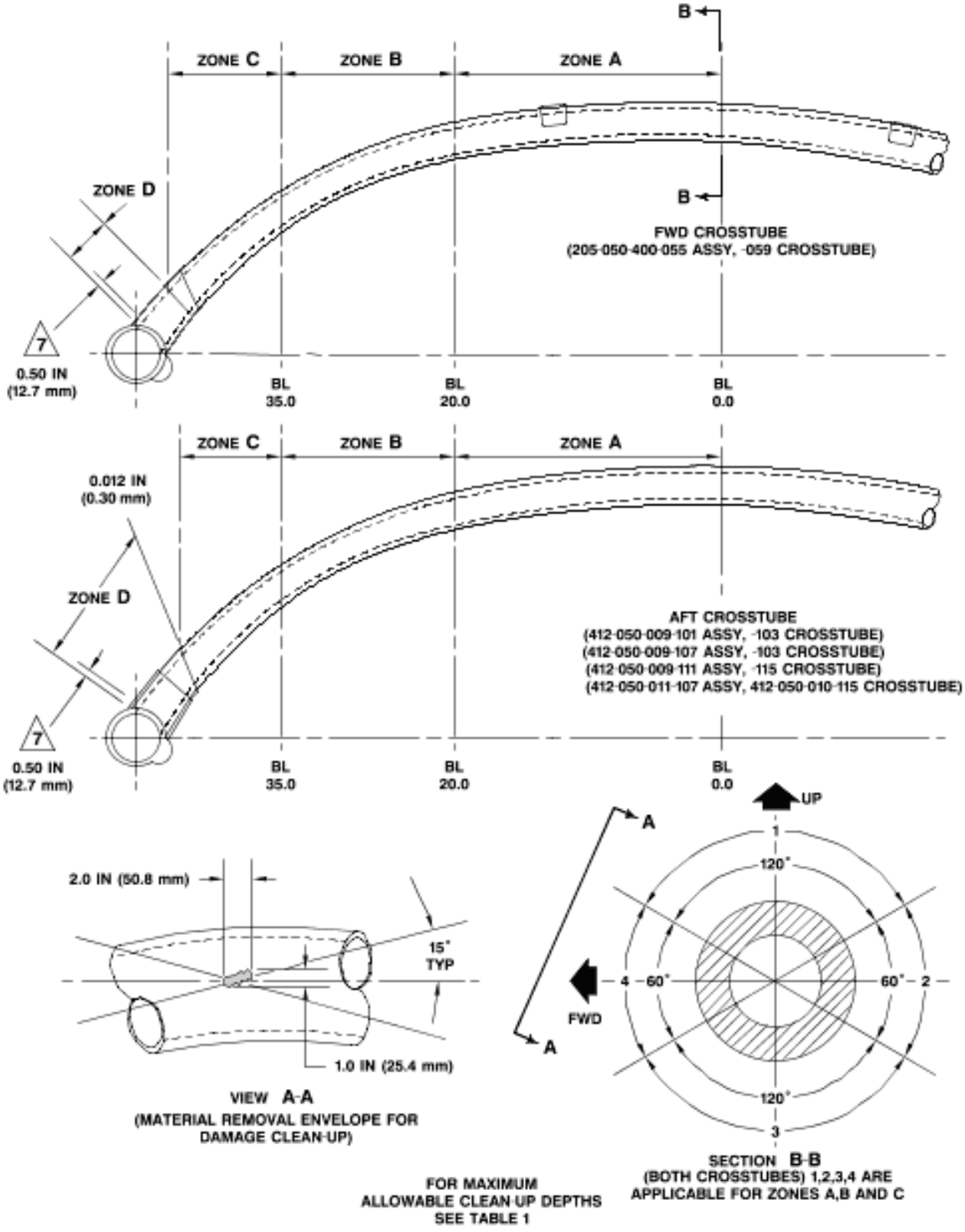
ICN-412-A-320100-B-97499-00006-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 4 of 9)



ICN-412-A-320100-B-97499-00007-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 5 of 9)



ICN-412-A-320100-B-97499-00008-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 6 of 9)

TABLE 1
MAXIMUM ALLOWABLE CLEANUP DEPTH STANDARD AND HIGH SKID GEAR

ZONE	BHT PRODUCTS FWD CROSSTUBE INCHES (mm) △10	BHT PRODUCTS AFT CROSSTUBE INCHES (mm) △11
A1	0.005 (0.13)	0.005 (0.12)
A2	0.015 (0.38)	0.015 (0.38)
A3	0.005 (0.13)	0.005 (0.12)
A4	0.015 (0.38)	0.015 (0.38)
B1	0.015 (0.38)	0.015 (0.38)
B2	0.015 (0.38)	0.015 (0.38)
B3	0.015 (0.38)	0.015 (0.38)
B4	0.015 (0.38)	0.015 (0.38)
C1	0.010 (0.25)	0.012 (0.30)
C2	0.010 (0.25)	0.012 (0.30)
C3	0.010 (0.25)	0.012 (0.30)
C4	0.010 (0.25)	0.012 (0.30)
D △7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 6 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

△7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 1.

△8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note △9 (Figure 32-2, sheet 5).

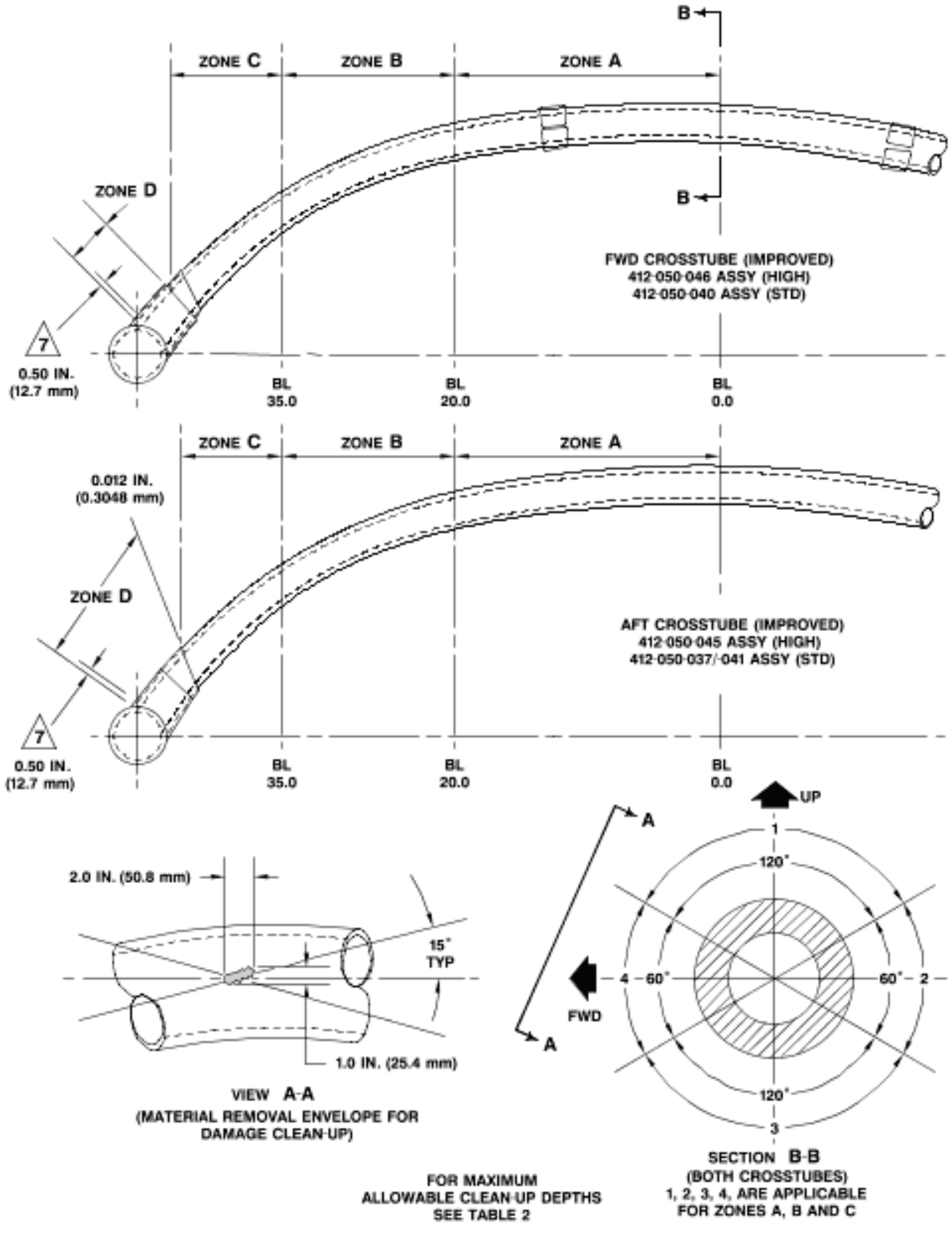
△9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

△10 Forward standard crosstube (BHT 205-050-400-055).
Forward high crosstube (BHT 205-050-403-031).

△11 Aft standard crosstube (BHT 412-050-002-101/103/105/107).
Aft high crosstube (BHT 412-050-011-101/103/105/107).

ICN-412-A-320100-B-97499-00009-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 7 of 9)



ICN-412-A-320100-B-97499-00010-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 8 of 9)

TABLE 2
MAXIMUM ALLOWABLE CLEANUP DEPTH IMPROVED STANDARD AND IMPROVED HIGH SKID GEAR

ZONE	BHT AND AAI PRODUCTS	BHT AND AAI PRODUCTS
	FWD CROSSTUBE INCHES (mm) △10	AFT CROSSTUBE INCHES (mm) △11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
B1 B2 B3 B4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.005 (0.13) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.005 (0.13) 0.012 (0.30)
D △7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 8 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

△7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 2.

△8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note △9 (Figure 32-2, sheet 5).

△9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

△10 Improved forward standard crosstube (BHT 412-050-040-101, AAI 212-320-103 ALL).
Improved forward high crosstube (BHT 412-050-046-101, AAI 212-321-103 ALL).

△11 Improved aft standard crosstube (BHT 412-050-041-109, AAI 412-320-104 ALL).
Improved aft high crosstube (BHT 412-050-045-105, AAI 412-321-106 ALL).

ICN-412-A-320100-B-97499-00011-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 9 of 9)

Landing Gear Assembly - Inspection

Procedure

NOTE

Superficial damage is mechanical damage that does not exceed 0.03 inch (0.7 mm) depth or 1.25 inches (31.7 mm) length. Superficial damage is acceptable where noted.

No cracks in any landing gear parts or attaching parts are acceptable.

1. Inspect landing gear assembly and attaching parts per Figure 1. Inspection procedures for high landing gear are the same as for standard gear, except for crosstube deflection limits.

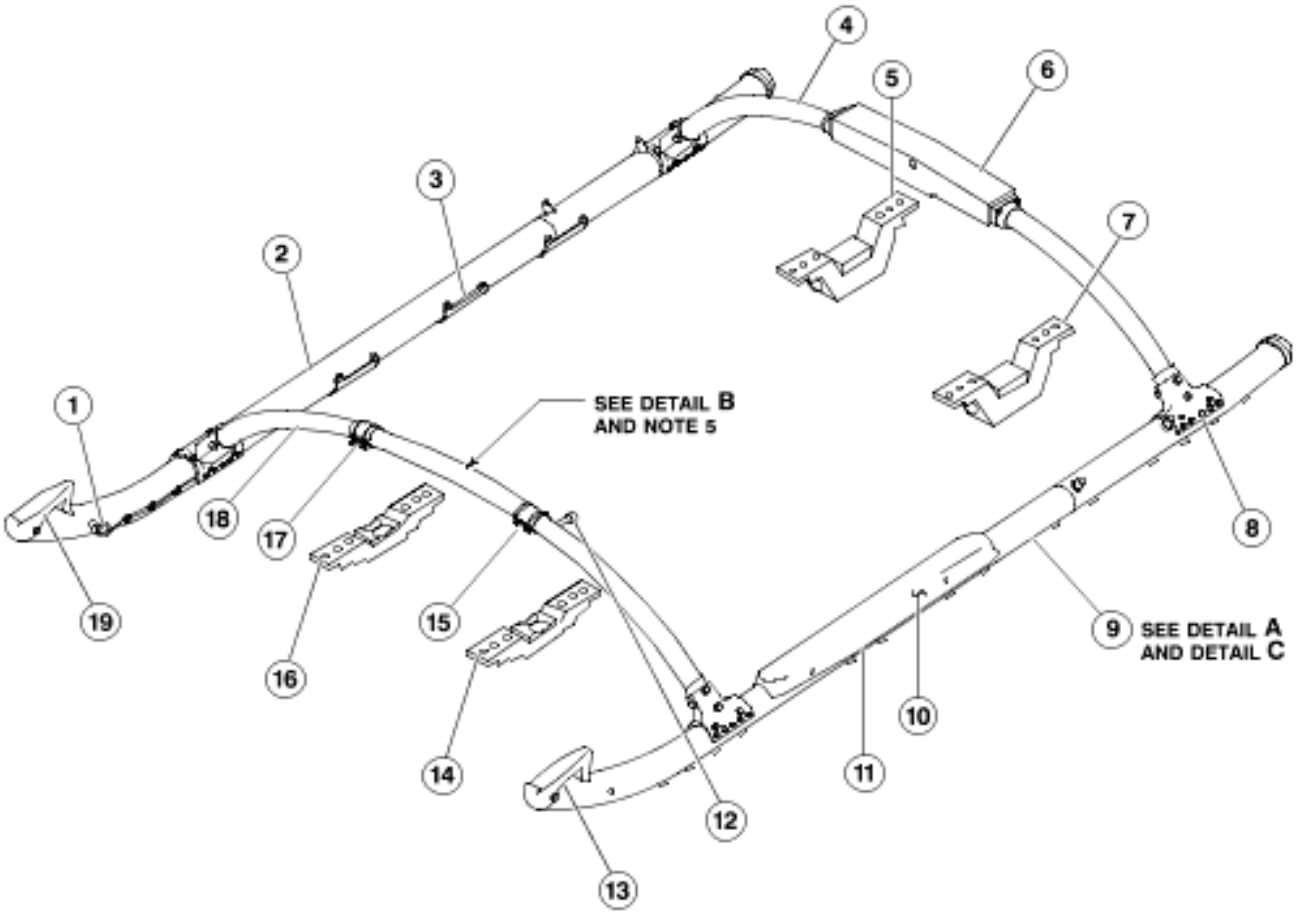
1.1. Inspect landing gear bearing and retaining supports (15 and 17, Figure 1).

1.2. Inspect aft crosstube support beam (6) for secure installation, wear, and deformation.

1.3. Inspect rubber bumpers on landing gear cap fittings (5, 7, 14, 16) for voids or unbonded areas. No one void shall exceed 0.25 square inch (161.3 mm2). The rubber bumper is not repairable.

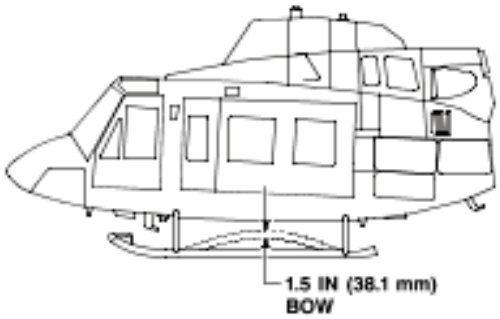
1.4. If required, inspect landing gear assembly spread (Detail B).

1.5. Inspect skid tubes (2 and 9) for security of attachment to crosstubes and for damage (Detail B).



STANDARD SKID LANDING GEAR ASSEMBLY
412-050-001-101, -103, -105, -107, -109 and -111

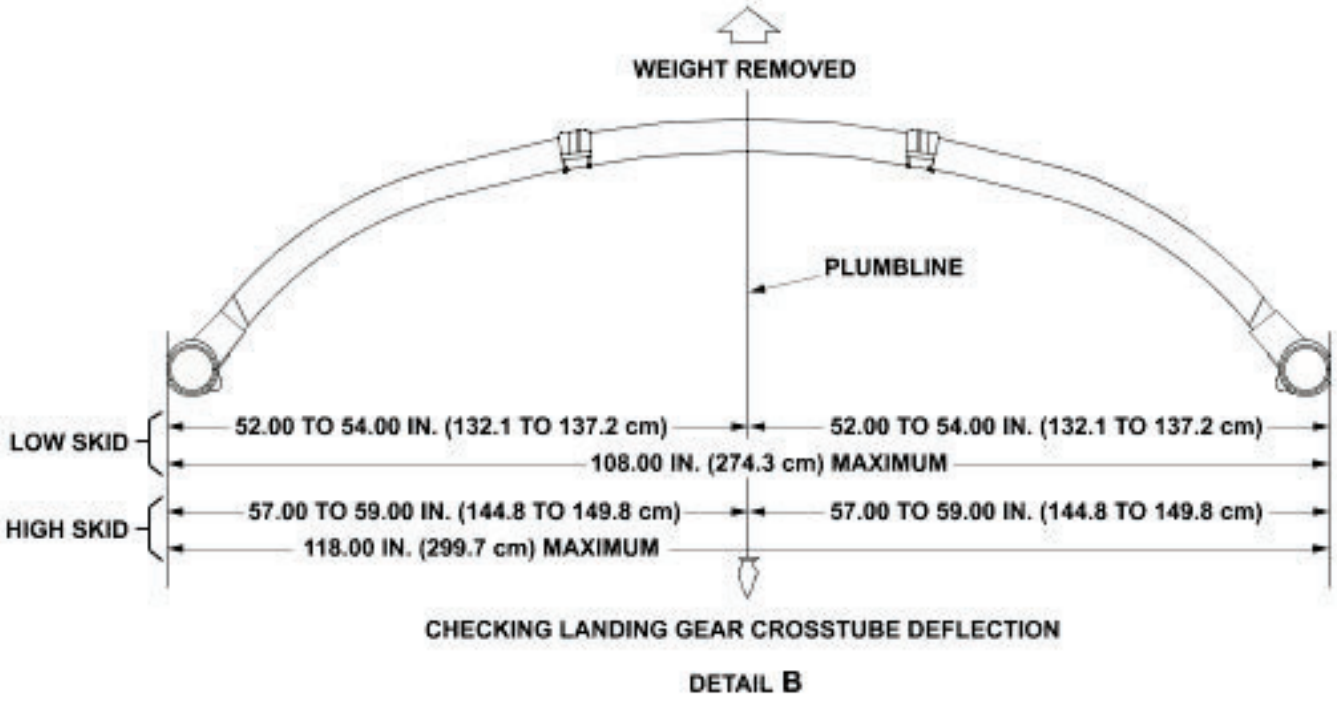
1. Tow ring
2. Skid tube
3. Skid shoe
4. Crosstube
5. Landing gear cap fitting
6. Aft crosstube support beam
7. Landing gear cap fitting
8. Saddle
9. Skid tube
10. Passenger step
11. Skid shoe
12. Passenger step electrical connector
13. Step
14. Landing gear cap fitting
15. Landing gear bearing and retaining support
16. Landing gear cap fitting
17. Landing gear bearing and retaining support
18. Crosstube
19. Step



DETAIL A

ICN-412-A-320100-B-97499-00003-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 1 of 9)

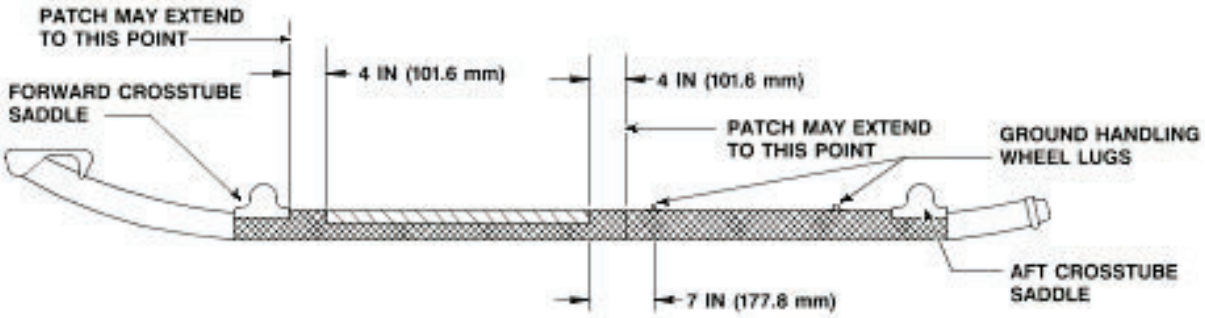


NOTES

1. Deformation (distortion) of either crosstube in excess of limits is cause for replacement of crosstube(s).
2. Crosstube may be deflected on one side only. If limit is exceeded, crosstube is considered scrapped and must be replaced.

ICN-412-A-320100-B-97499-00004-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 2 of 9)



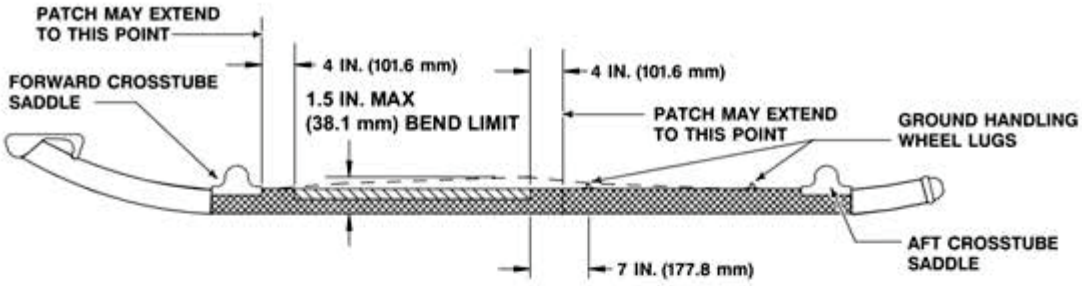
DETAIL C
412-050-023

DAMAGE LOCATION SYMBOLS

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR AREAS ALLOWED		
CRACKS ALLOWED	None	None	None
DENTS AND HOLES	None	4 inches (101.6 mm) Refer to paragraph 32-9.	Not Critical

ICN-412-A-320100-B-97499-00005-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 3 of 9)

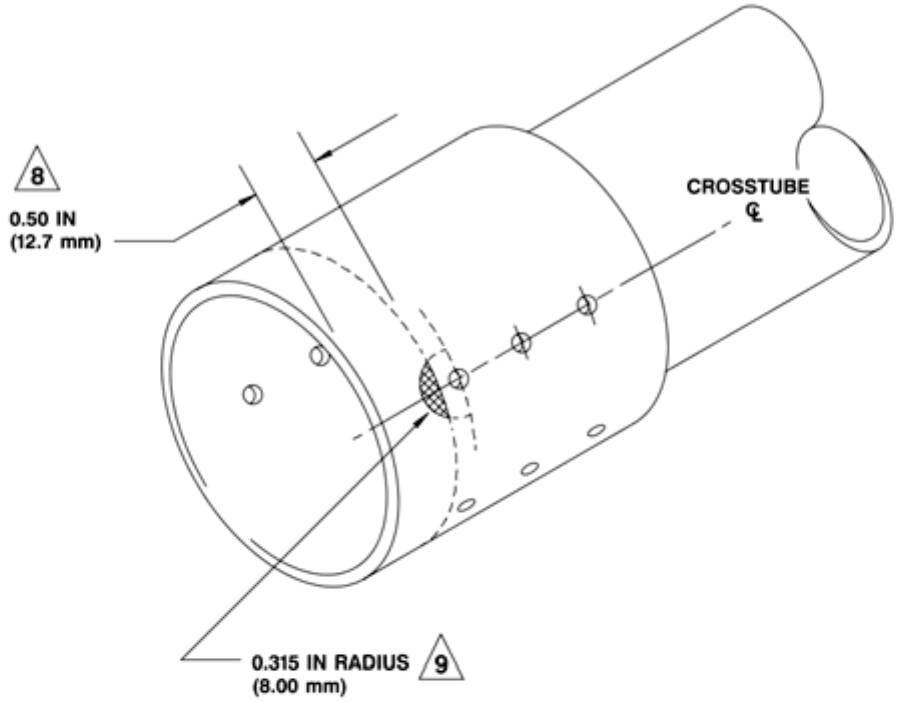


DETAIL C
412-050-030
412-050-034

			(Does not apply to saddles)
DENTS	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No dents allowed. Replace as needed.	Not critical.
HOLES	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No holes allowed. Replace as needed.	Not critical. Plug holes to keep out moisture.
CRACKS	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.
SCRATCHES SCUFFS NICKS	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	Not critical.
	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	
	If damage is greater than established limits, replace as needed.	If damage is greater than established limits, replace as needed.	
CORROSION	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup is limited to 10% of material thickness. Refer to paragraph 32-11.

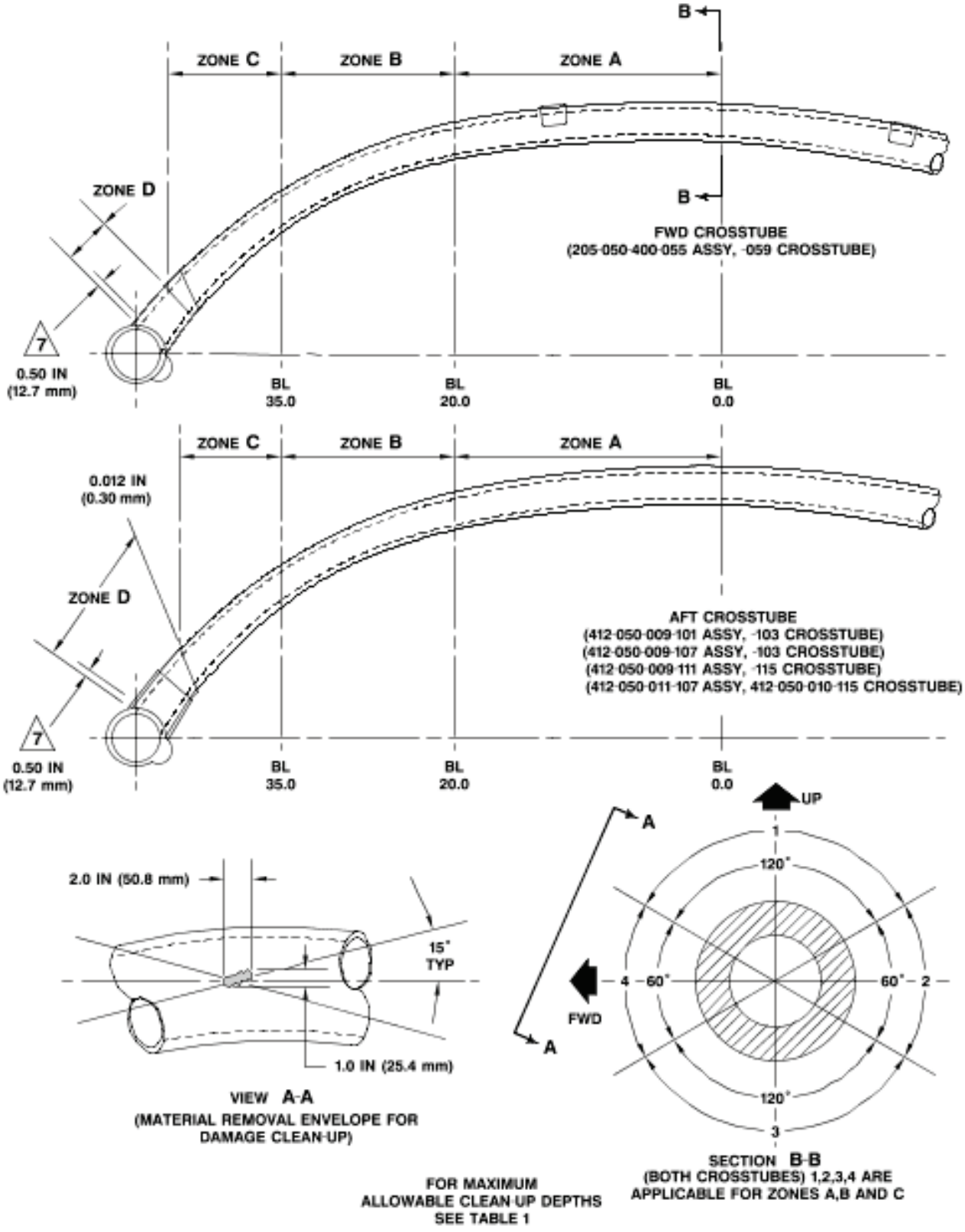
ICN-412-A-320100-B-97499-00006-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 4 of 9)



ICN-412-A-320100-B-97499-00007-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 5 of 9)



ICN-412-A-320100-B-97499-00008-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 6 of 9)

TABLE 1
MAXIMUM ALLOWABLE CLEANUP DEPTH STANDARD AND HIGH SKID GEAR

ZONE	BHT PRODUCTS FWD CROSSTUBE INCHES (mm) △10	BHT PRODUCTS AFT CROSSTUBE INCHES (mm) △11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.12) 0.015 (0.38) 0.005 (0.12) 0.015 (0.38)
B1 B2 B3 B4	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.010 (0.25) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.012 (0.30) 0.012 (0.30)
D △7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 6 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

△7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 1.

△8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note △9 (Figure 32-2, sheet 5).

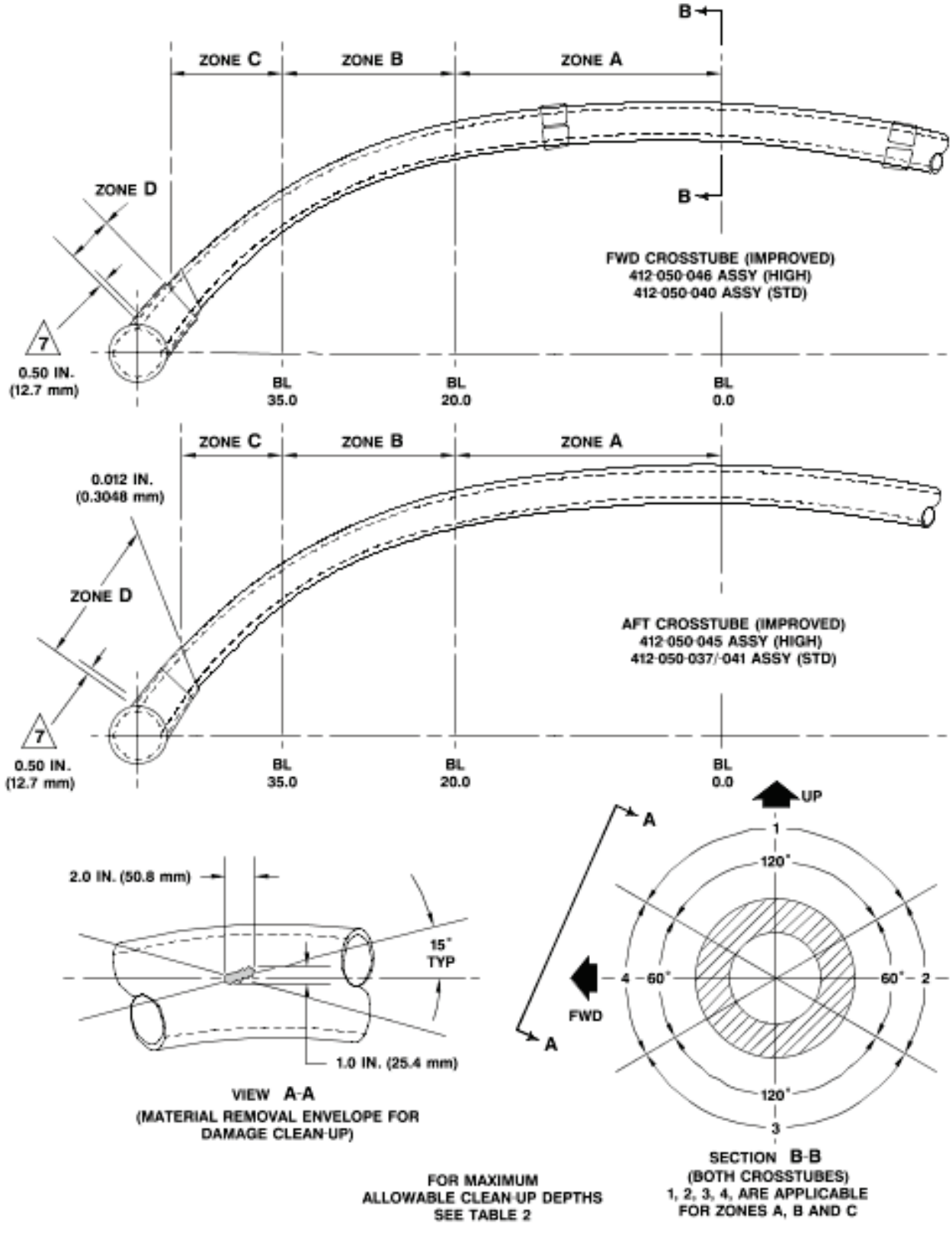
△9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

△10 Forward standard crosstube (BHT 205-050-400-055).
Forward high crosstube (BHT 205-050-403-031).

△11 Aft standard crosstube (BHT 412-050-002-101/103/105/107).
Aft high crosstube (BHT 412-050-011-101/103/105/107).

ICN-412-A-320100-B-97499-00009-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 7 of 9)



ICN-412-A-320100-B-97499-00010-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 8 of 9)

TABLE 2
MAXIMUM ALLOWABLE CLEANUP DEPTH IMPROVED STANDARD AND IMPROVED HIGH SKID GEAR

ZONE	BHT AND AAI PRODUCTS	BHT AND AAI PRODUCTS
	FWD CROSSTUBE INCHES (mm) △10	AFT CROSSTUBE INCHES (mm) △11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
B1 B2 B3 B4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.005 (0.13) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.005 (0.13) 0.012 (0.30)
D △7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 8 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

△7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 2.

△8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note △9 (Figure 32-2, sheet 5).

△9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

△10 Improved forward standard crosstube (BHT 412-050-040-101, AAI 212-320-103 ALL).
Improved forward high crosstube (BHT 412-050-046-101, AAI 212-321-103 ALL).

△11 Improved aft standard crosstube (BHT 412-050-041-109, AAI 412-320-104 ALL).
Improved aft high crosstube (BHT 412-050-045-105, AAI 412-321-106 ALL).

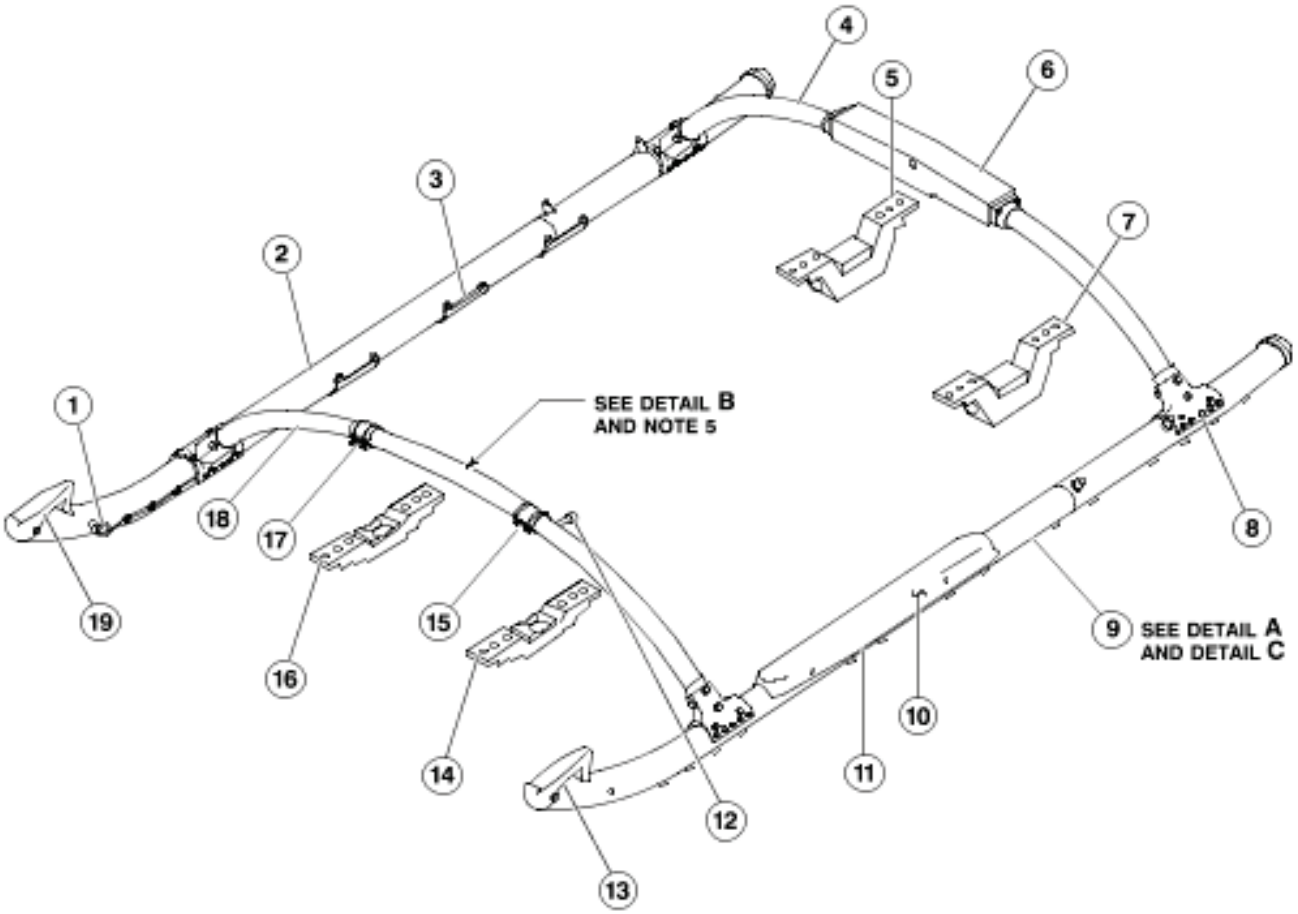
ICN-412-A-320100-B-97499-00011-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 9 of 9)

Landing Gear Assembly - Repair

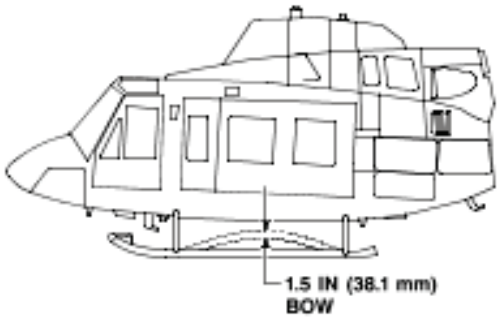
Procedure

1. Repair damaged skid tubes (2 and 9, Figure 1) (412-A-32-01-01-00A-685A-A / 00704).
2. Repair damaged skid shoes (3 and 11) (412-A-32-01-02-00A-685A-A / 00711).
3. Repair damaged passenger step (10) (412-A-32-02-00-00A-685A-A / 00753).
4. Replace components that are damaged in excess of repairable limits.



STANDARD SKID LANDING GEAR ASSEMBLY
412-050-001-101, -103, -105, -107, -109 and -111

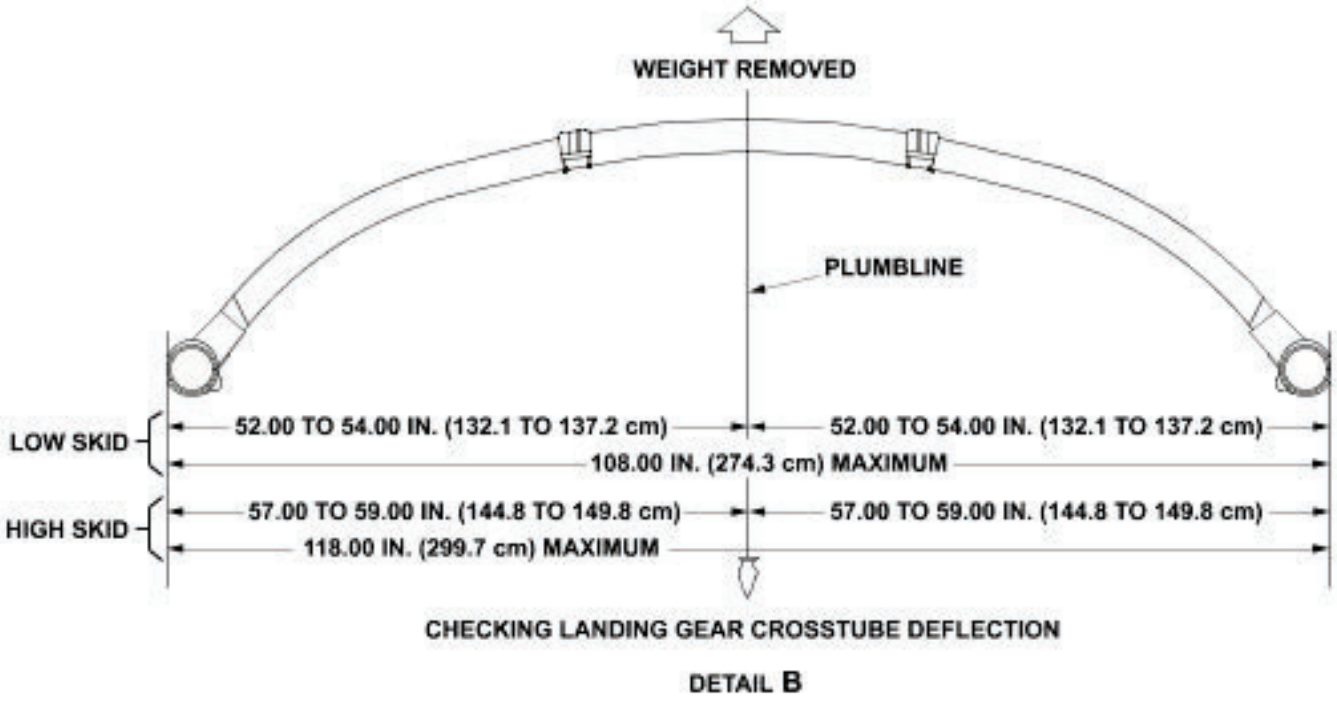
1. Tow ring
2. Skid tube
3. Skid shoe
4. Crosstube
5. Landing gear cap fitting
6. Aft crosstube support beam
7. Landing gear cap fitting
8. Saddle
9. Skid tube
10. Passenger step
11. Skid shoe
12. Passenger step electrical connector
13. Step
14. Landing gear cap fitting
15. Landing gear bearing and retaining support
16. Landing gear cap fitting
17. Landing gear bearing and retaining support
18. Crosstube
19. Step



DETAIL A

ICN-412-A-320100-B-97499-00003-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 1 of 9)

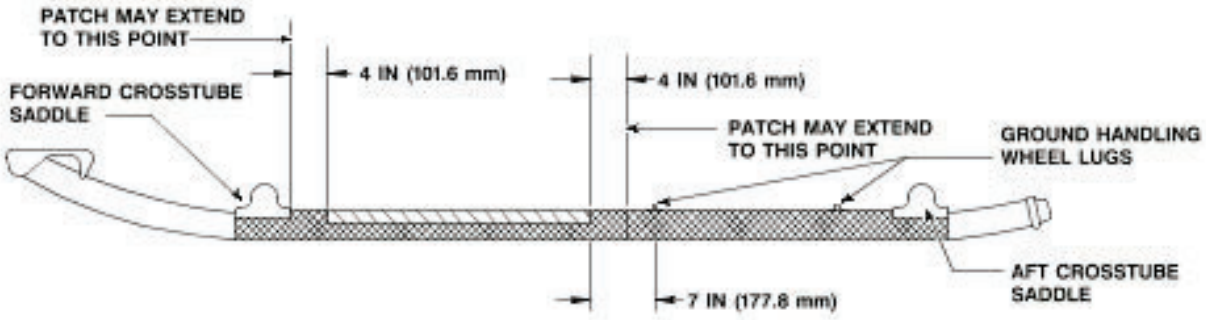


NOTES

1. Deformation (distortion) of either crosstube in excess of limits is cause for replacement of crosstube(s).
2. Crosstube may be deflected on one side only. If limit is exceeded, crosstube is considered scrapped and must be replaced.

ICN-412-A-320100-B-97499-00004-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 2 of 9)



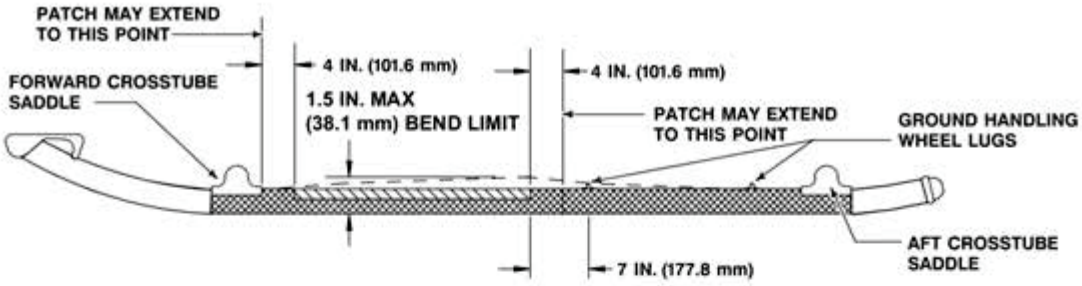
DETAIL C
412-050-023

DAMAGE LOCATION SYMBOLS

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR AREAS ALLOWED		
CRACKS ALLOWED	None	None	None
DENTS AND HOLES	None	4 inches (101.6 mm) Refer to paragraph 32-9.	Not Critical

ICN-412-A-320100-B-97499-00005-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 3 of 9)

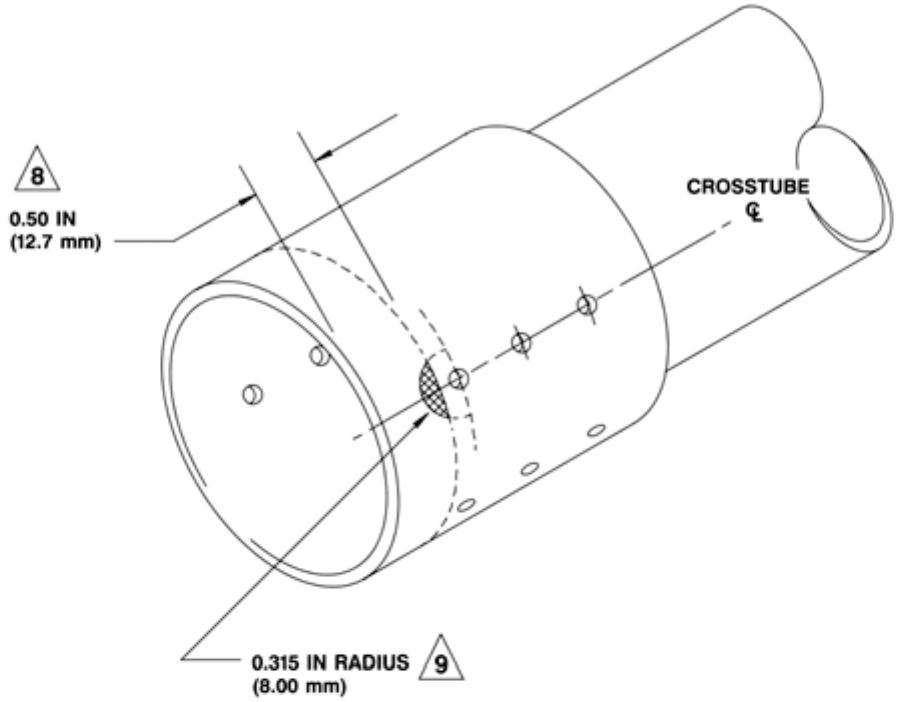


DETAIL C
412-050-030
412-050-034

			(Does not apply to saddles)
DENTS	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No dents allowed. Replace as needed.	Not critical.
HOLES	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No holes allowed. Replace as needed.	Not critical. Plug holes to keep out moisture.
CRACKS	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.
SCRATCHES SCUFFS NICKS	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	Not critical.
	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	
	If damage is greater than established limits, replace as needed.	If damage is greater than established limits, replace as needed.	
CORROSION	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup is limited to 10% of material thickness. Refer to paragraph 32-11.

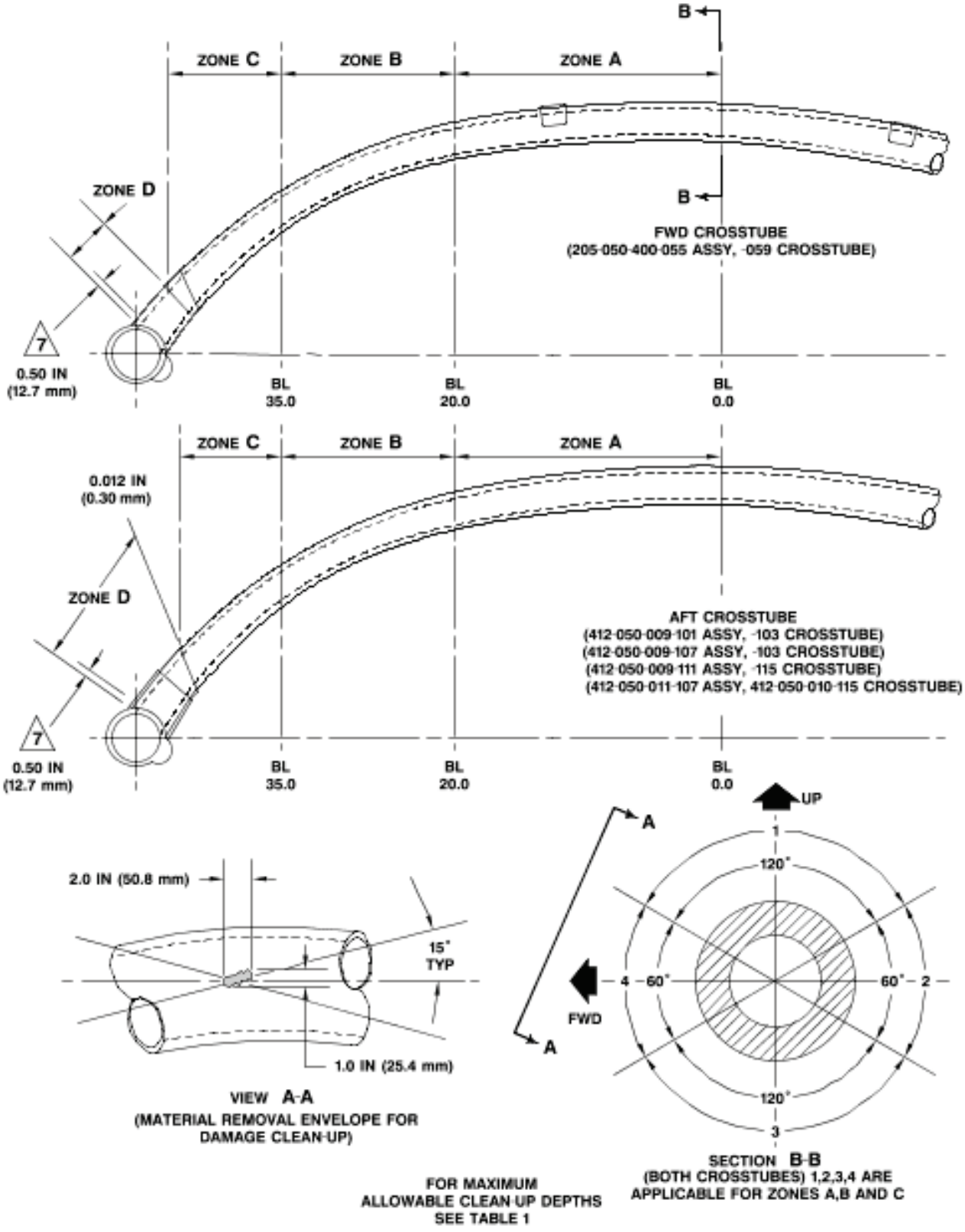
ICN-412-A-320100-B-97499-00006-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 4 of 9)



ICN-412-A-320100-B-97499-00007-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 5 of 9)



ICN-412-A-320100-B-97499-00008-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 6 of 9)

TABLE 1
MAXIMUM ALLOWABLE CLEANUP DEPTH STANDARD AND HIGH SKID GEAR

ZONE	BHT PRODUCTS FWD CROSSTUBE INCHES (mm) △10	BHT PRODUCTS AFT CROSSTUBE INCHES (mm) △11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.12) 0.015 (0.38) 0.005 (0.12) 0.015 (0.38)
B1 B2 B3 B4	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.010 (0.25) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.012 (0.30) 0.012 (0.30)
D △7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 6 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

△7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 1.

△8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note △9 (Figure 32-2, sheet 5).

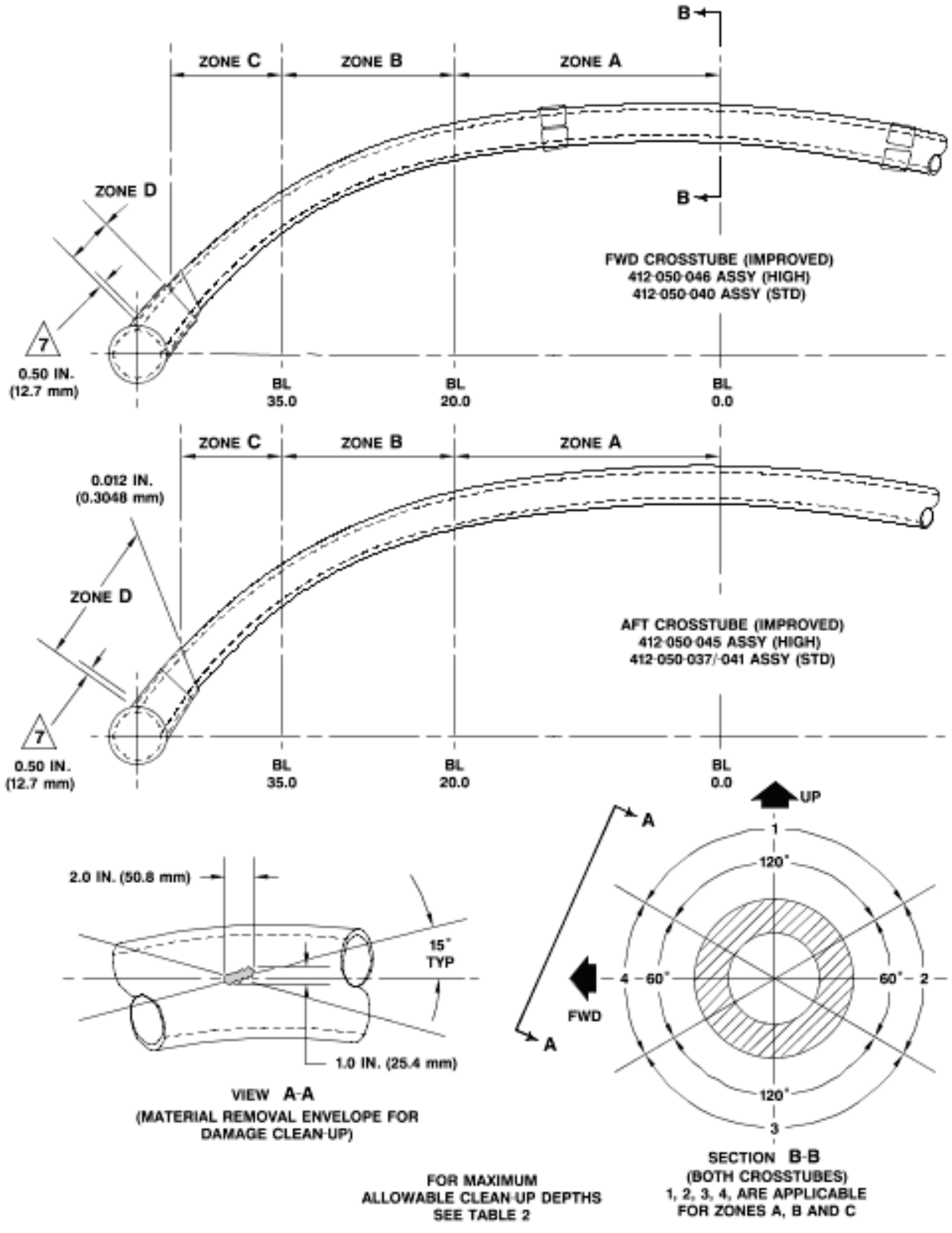
△9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

△10 Forward standard crosstube (BHT 205-050-400-055).
Forward high crosstube (BHT 205-050-403-031).

△11 Aft standard crosstube (BHT 412-050-002-101/103/105/107).
Aft high crosstube (BHT 412-050-011-101/103/105/107).

ICN-412-A-320100-B-97499-00009-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 7 of 9)



ICN-412-A-320100-B-97499-00010-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 8 of 9)

TABLE 2
MAXIMUM ALLOWABLE CLEANUP DEPTH IMPROVED STANDARD AND IMPROVED HIGH SKID GEAR

ZONE	BHT AND AAI PRODUCTS	BHT AND AAI PRODUCTS
	FWD CROSSTUBE INCHES (mm) 10	AFT CROSSTUBE INCHES (mm) 11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
B1 B2 B3 B4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.005 (0.13) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.005 (0.13) 0.012 (0.30)
D 7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 8 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 2.

8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note 9 (Figure 32-2, sheet 5).

9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

10 Improved forward standard crosstube (BHT 412-050-040-101, AAI 212-320-103 ALL).
Improved forward high crosstube (BHT 412-050-046-101, AAI 212-321-103 ALL).

11 Improved aft standard crosstube (BHT 412-050-041-109, AAI 412-320-104 ALL).
Improved aft high crosstube (BHT 412-050-045-105, AAI 412-321-106 ALL).

ICN-412-A-320100-B-97499-00011-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 9 of 9)

Landing Gear Assembly - Installation

Procedure

1. Position landing gear (1, Figure 1) under helicopter.

2. Slowly lower the helicopter on each crosstube (4 and 18, Figure 2) to prevent damage.

3. Make sure that forward crosstube support structure (23, Figure 1) properly engages bearing and retaining support (30). Also ensure bearing and retaining support (29) properly engages opposite support (not illustrated).

4. Make sure that aft crosstube support beam (7) properly engages support in fuselage.

5. Position forward cap fitting (32) on forward crosstube support structure (23).

6. Install bolt (38) and aluminum washer (37), bolt (39) and aluminum washer (40), and bolt (41) and aluminum washer (42).

7. Install bolt (36) and aluminum washer (35), bolt (34) and aluminum washers (33 and 24), and nut (25).

8. Install bolt (43), aluminum washer (44 and 27), and nut (26).

9. Install forward cap fitting (45) in same manner outlined in preceding steps.

10. Position aft cap fitting (21) on landing gear support structure (4).

11. Install bolt (10) and aluminum washer (9).

12. Install bolt (15) and aluminum washer (16).

13. Install bolt (12) and aluminum washer (11).

14. Install bolt (17) and aluminum washer (18).

15. Install bolt (14), aluminum washer (13 and 6), and nut (5).

16. Install bolt (19), aluminum washer (20 and 2), and nut (3).

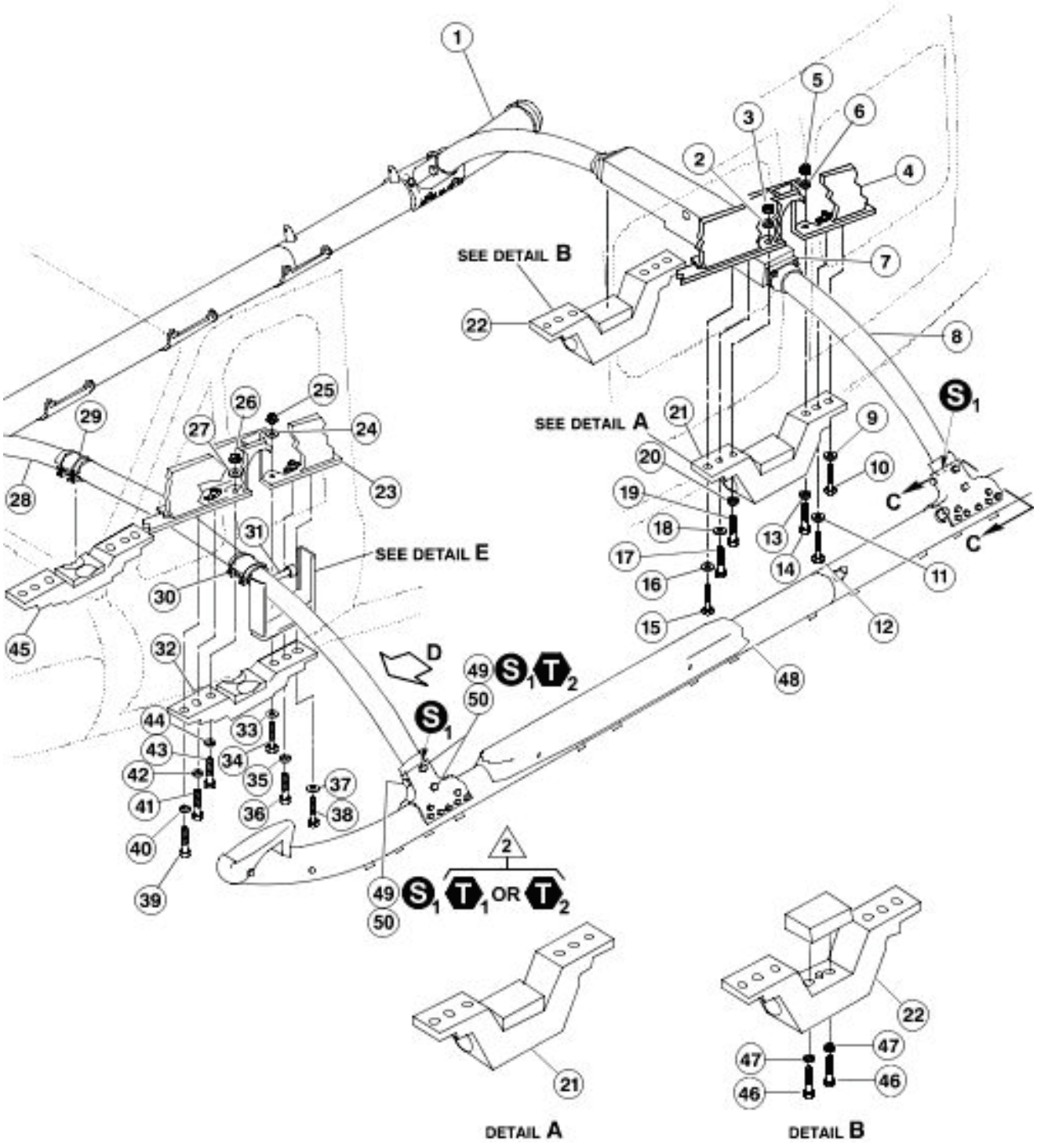
17. Install aft cap fitting (22) in same manner outlined in preceding steps.

18. If applicable, install the weight-on-gear switch 3420S1 (60) as follows:

18.1. Cut and discard the lockwire or cable tie attaching the support assembly (57) and weight-on-gear switch 3420S1 (60) to the bracket assembly (56).

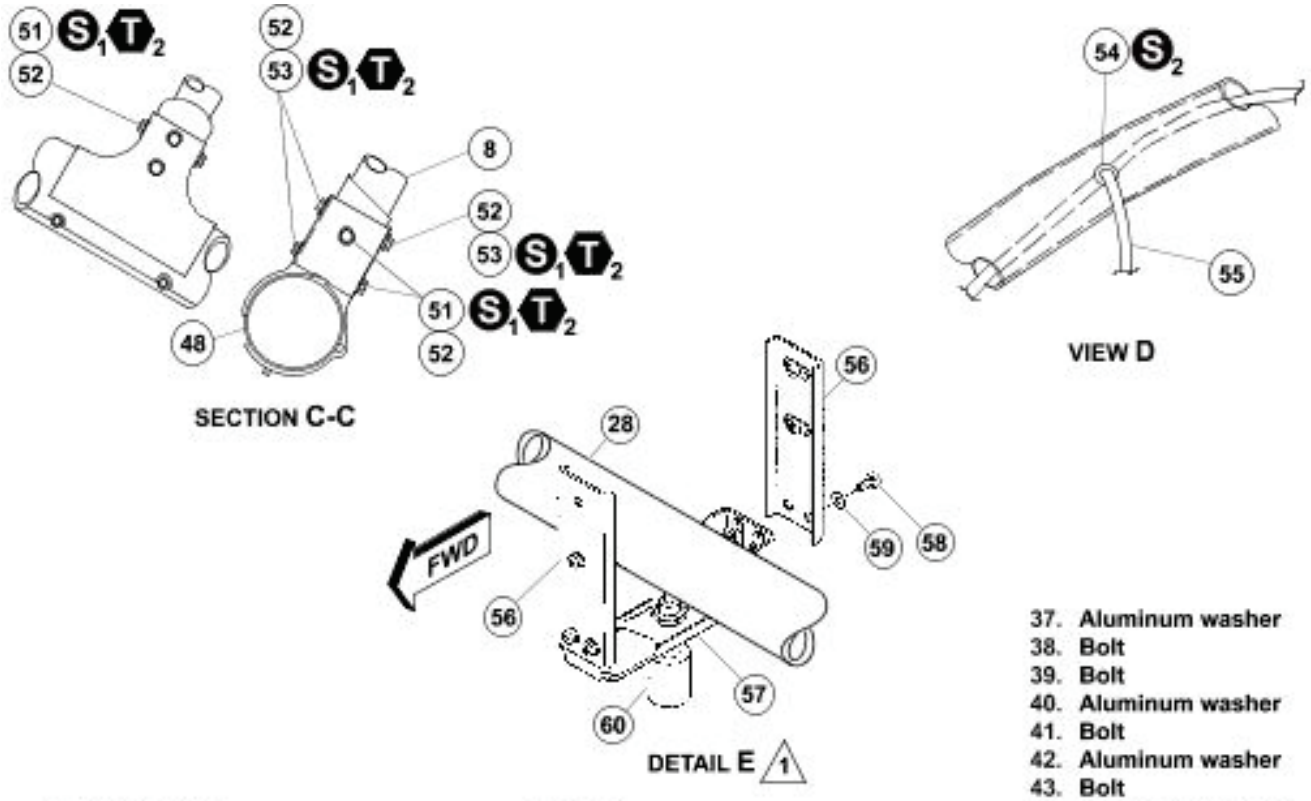
18.2. Attach the support assembly (57) to the two bracket assemblies with the four screws (58) and washers (59). Tighten the screws.

18.3. Do an adjustment of the weight-on-gear switch 3420S1 (60) (412-A-96-07-00-03A-271A-A / 02156).



ICN-412-A-320100-B-97499-00001-A-001-01

Figure 1. Landing Gear (Standard) (Sheet 1 of 2)



1. Landing gear
2. Aluminum washer
3. Nut
4. Landing gear support structure
5. Nut
6. Aluminum washer
7. Aft crosstube support beam
8. Aft crosstube
9. Aluminum washer
10. Bolt
11. Aluminum washer
12. Bolt
13. Aluminum washer
14. Bolt
15. Bolt
16. Aluminum washer
17. Bolt
18. Aluminum washer
19. Bolt
20. Aluminum washer
21. Aft cap fitting
22. Aft cap fitting
23. Forward crosstube support structure
24. Aluminum washer
25. Nut
26. Nut
27. Aluminum washer
28. Forward crosstube
29. Bearing and retaining support
30. Bearing and retaining support
31. Power step electrical connector
32. Forward cap fitting
33. Aluminum washer
34. Bolt
35. Aluminum washer
36. Bolt
37. Aluminum washer
38. Bolt
39. Bolt
40. Aluminum washer
41. Bolt
42. Aluminum washer
43. Bolt
44. Aluminum washer
45. Forward cap fitting
46. Bolt
47. Washer
48. Skid tube
49. Bolt
50. Washer
51. Bolt
52. Washer
53. Bolt
54. Grommet
55. Cable assembly
56. Bracket assembly
57. Support assembly
58. Screw
59. Washer
60. Weight-on-gear switch (3420S1)

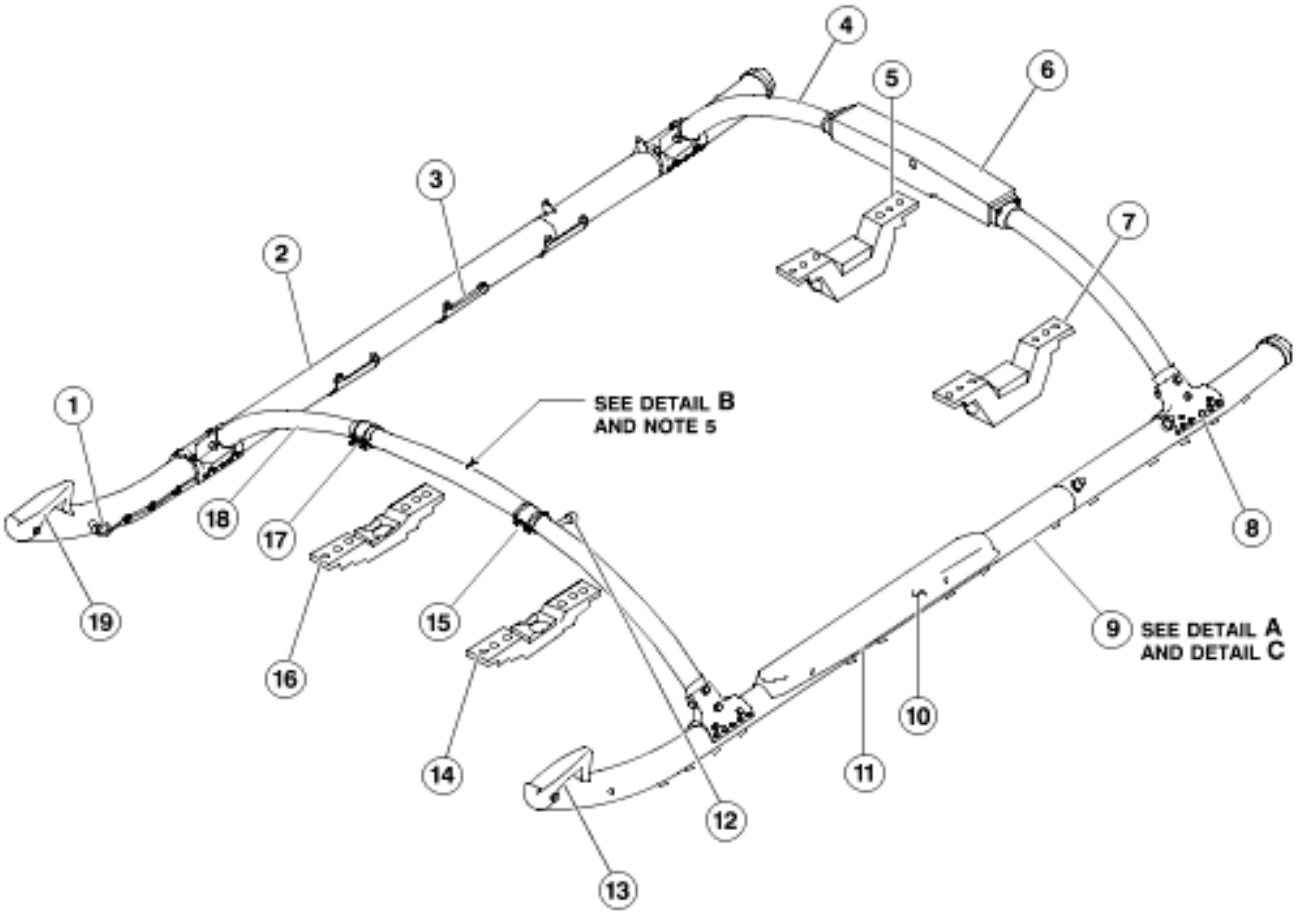
S_1 SEALANT (C-392) S_2 SEALANT (C-308) T_1 50 TO 70 IN-LBS (5.65 TO 7.91 Nm) T_2 160 TO 190 IN-LBS (18.08 TO 21.47 Nm)

NOTES

1. Weight-on-gear switch 3420S1 installed on helicopters S/N 36462 and subsequent. It is also installed on helicopters S/N 36248 through 36461 with the weather radar kit (412-705-034) or weight-on-gear retrofit kit (BHT-412-SI-84) installed.
2. For 1/4-28 bolts, torque per T_1 and for 3/8-24 bolts, torque per T_2.

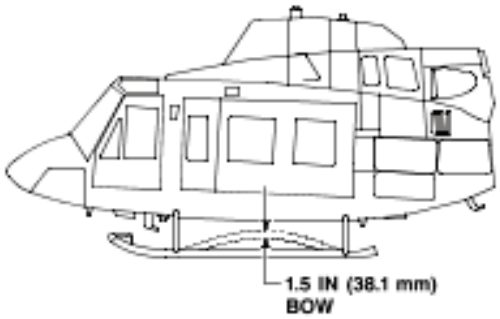
ICN-412-A-320100-B-97499-00002-A-001-01

Figure 1. Landing Gear (Standard) (Sheet 2 of 2)



STANDARD SKID LANDING GEAR ASSEMBLY
412-050-001-101, -103, -105, -107, -109 and -111

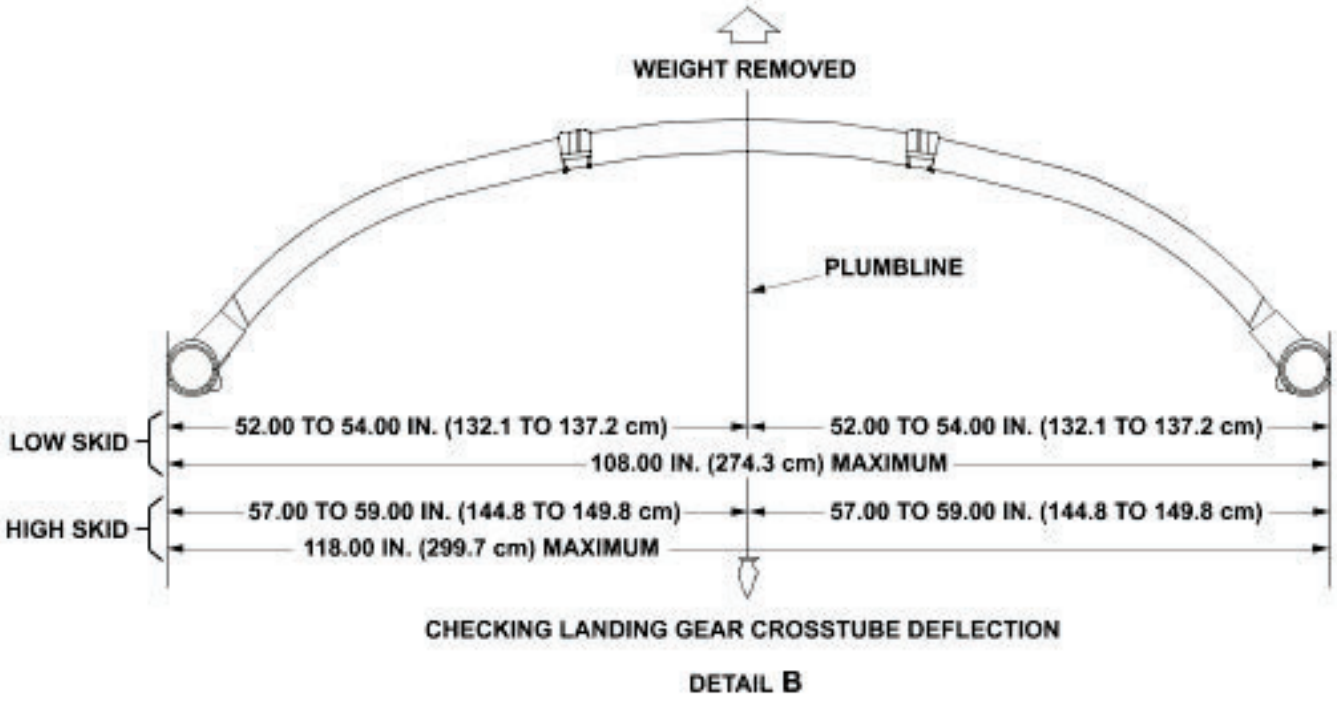
1. Tow ring
2. Skid tube
3. Skid shoe
4. Crosstube
5. Landing gear cap fitting
6. Aft crosstube support beam
7. Landing gear cap fitting
8. Saddle
9. Skid tube
10. Passenger step
11. Skid shoe
12. Passenger step electrical connector
13. Step
14. Landing gear cap fitting
15. Landing gear bearing and retaining support
16. Landing gear cap fitting
17. Landing gear bearing and retaining support
18. Crosstube
19. Step



DETAIL A

ICN-412-A-320100-B-97499-00003-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 1 of 9)

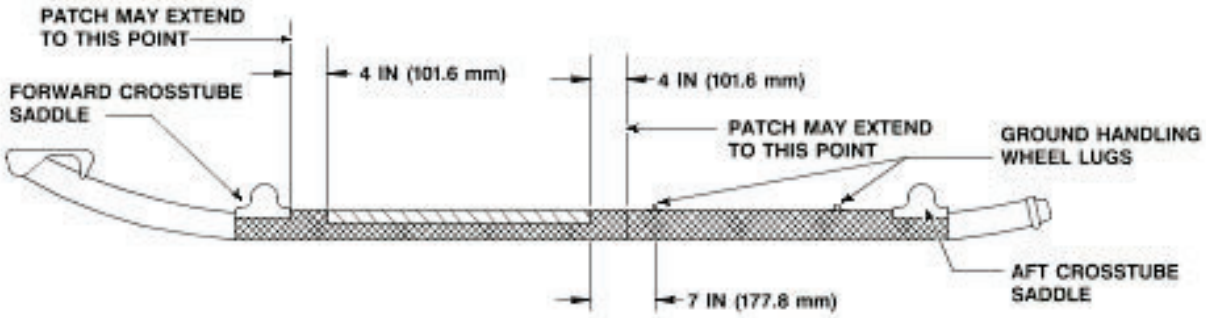


NOTES

1. Deformation (distortion) of either crosstube in excess of limits is cause for replacement of crosstube(s).
2. Crosstube may be deflected on one side only. If limit is exceeded, crosstube is considered scrapped and must be replaced.

ICN-412-A-320100-B-97499-00004-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 2 of 9)



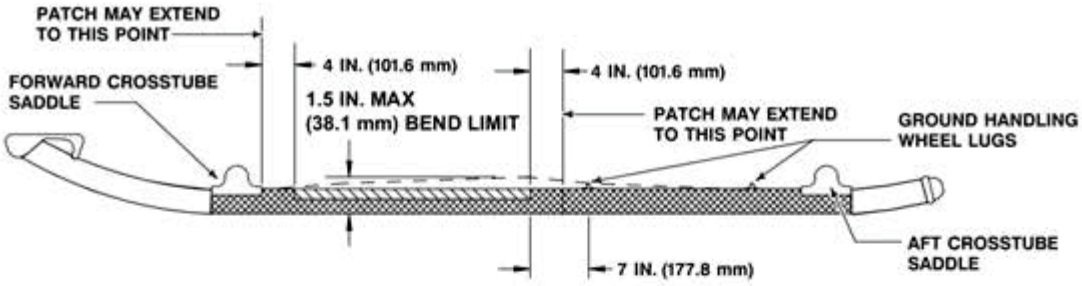
DETAIL C
412-050-023

DAMAGE LOCATION SYMBOLS

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR AREAS ALLOWED		
CRACKS ALLOWED	None	None	None
DENTS AND HOLES	None	4 inches (101.6 mm) Refer to paragraph 32-9.	Not Critical

ICN-412-A-320100-B-97499-00005-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 3 of 9)

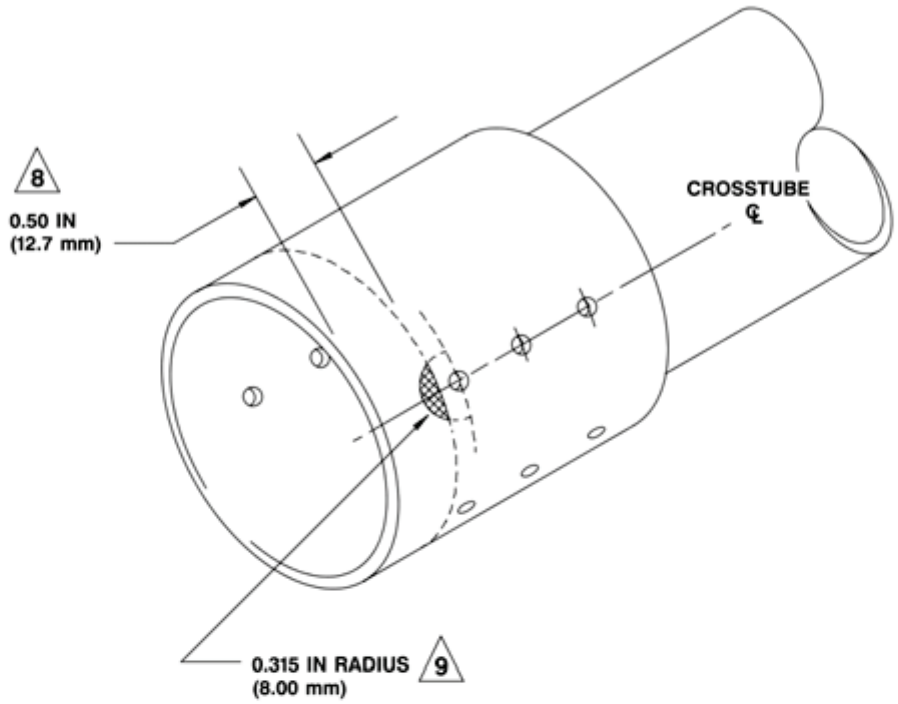


DETAIL C
412-050-030
412-050-034

			(Does not apply to saddles)
DENTS	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No dents allowed. Replace as needed.	Not critical.
HOLES	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No holes allowed. Replace as needed.	Not critical. Plug holes to keep out moisture.
CRACKS	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.
SCRATCHES SCUFFS NICKS	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	Not critical.
	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	
	If damage is greater than established limits, replace as needed.	If damage is greater than established limits, replace as needed.	
CORROSION	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup is limited to 10% of material thickness. Refer to paragraph 32-11.

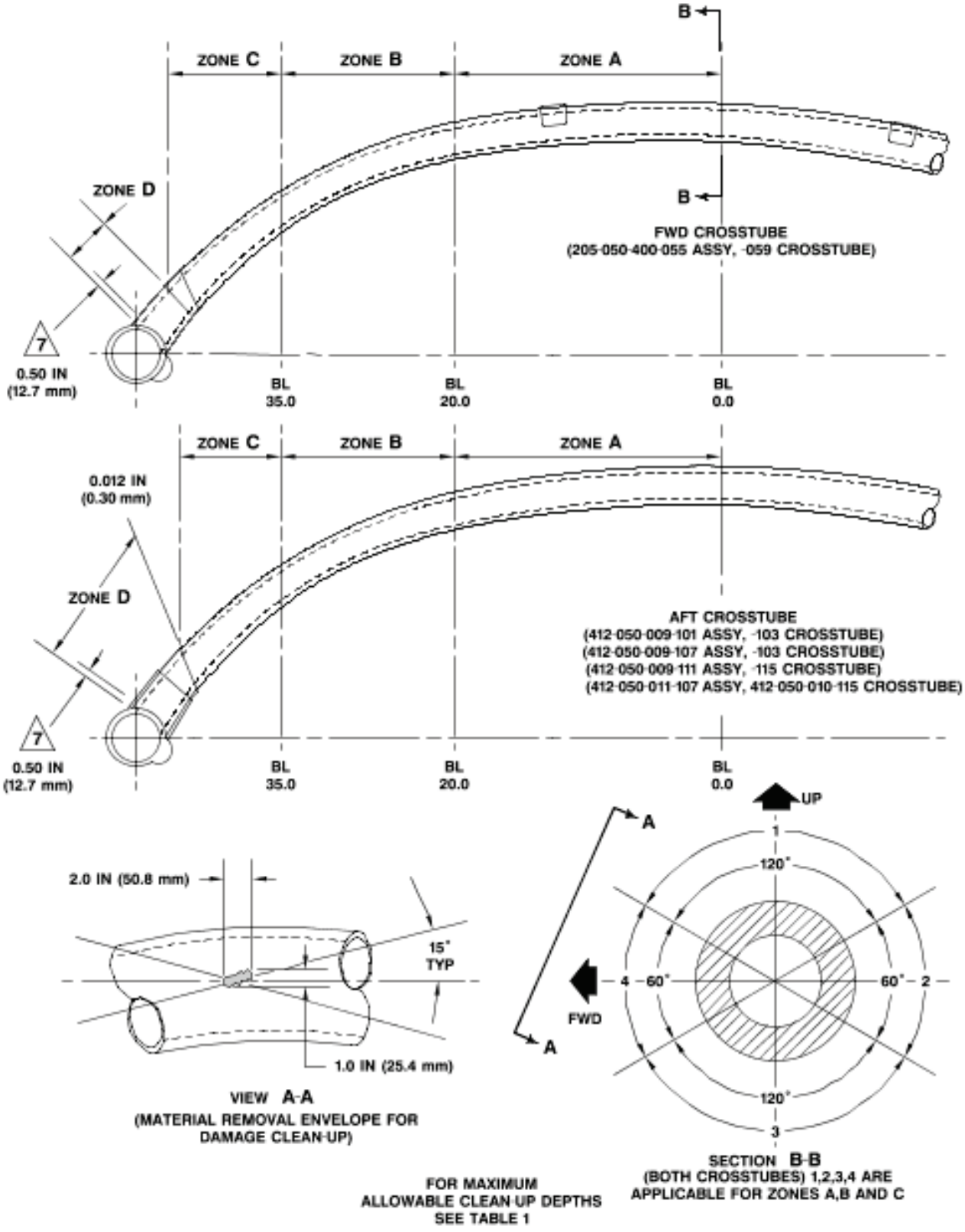
ICN-412-A-320100-B-97499-00006-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 4 of 9)



ICN-412-A-320100-B-97499-00007-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 5 of 9)



ICN-412-A-320100-B-97499-00008-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 6 of 9)

TABLE 1
MAXIMUM ALLOWABLE CLEANUP DEPTH STANDARD AND HIGH SKID GEAR

ZONE	BHT PRODUCTS FWD CROSSTUBE INCHES (mm) △10	BHT PRODUCTS AFT CROSSTUBE INCHES (mm) △11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.12) 0.015 (0.38) 0.005 (0.12) 0.015 (0.38)
B1 B2 B3 B4	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.010 (0.25) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.012 (0.30) 0.012 (0.30)
D △7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 6 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

△7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 1.

△8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note △9 (Figure 32-2, sheet 5).

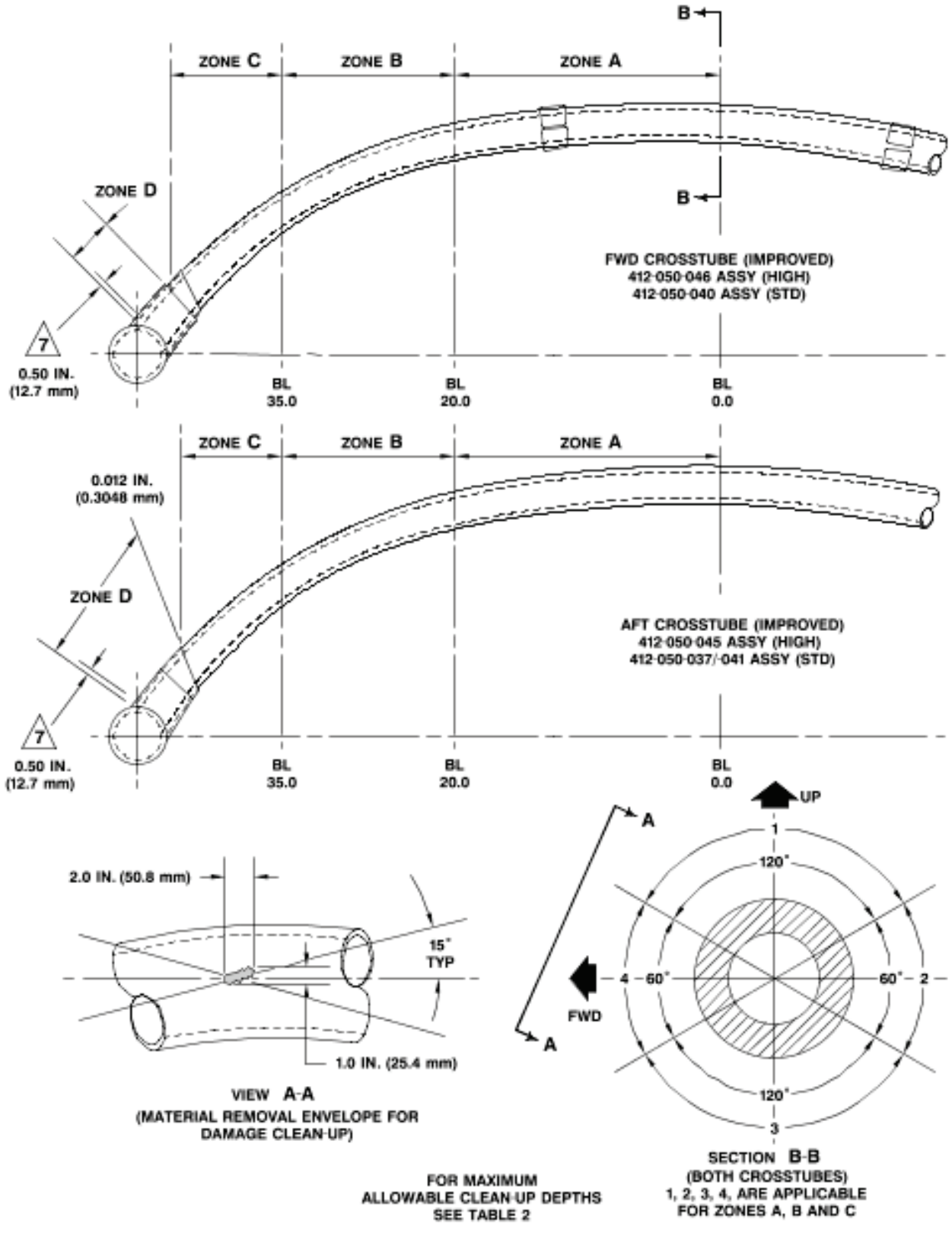
△9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

△10 Forward standard crosstube (BHT 205-050-400-055).
Forward high crosstube (BHT 205-050-403-031).

△11 Aft standard crosstube (BHT 412-050-002-101/103/105/107).
Aft high crosstube (BHT 412-050-011-101/103/105/107).

ICN-412-A-320100-B-97499-00009-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 7 of 9)



ICN-412-A-320100-B-97499-00010-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 8 of 9)

TABLE 2
MAXIMUM ALLOWABLE CLEANUP DEPTH IMPROVED STANDARD AND IMPROVED HIGH SKID GEAR

ZONE	BHT AND AAI PRODUCTS	BHT AND AAI PRODUCTS
	FWD CROSSTUBE INCHES (mm) 10	AFT CROSSTUBE INCHES (mm) 11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
B1 B2 B3 B4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.005 (0.13) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.005 (0.13) 0.012 (0.30)
D 7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 8 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 2.

8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note 9 (Figure 32-2, sheet 5).

9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

10 Improved forward standard crosstube (BHT 412-050-040-101, AAI 212-320-103 ALL).
Improved forward high crosstube (BHT 412-050-046-101, AAI 212-321-103 ALL).

11 Improved aft standard crosstube (BHT 412-050-041-109, AAI 412-320-104 ALL).
Improved aft high crosstube (BHT 412-050-045-105, AAI 412-321-106 ALL).

ICN-412-A-320100-B-97499-00011-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 9 of 9)

Skid Tubes - Description

NOTE

Maintenance procedures for skid tubes installed on high landing gear and standard gear are the same.

Maintenance procedures for left and right skid tubes are the same.

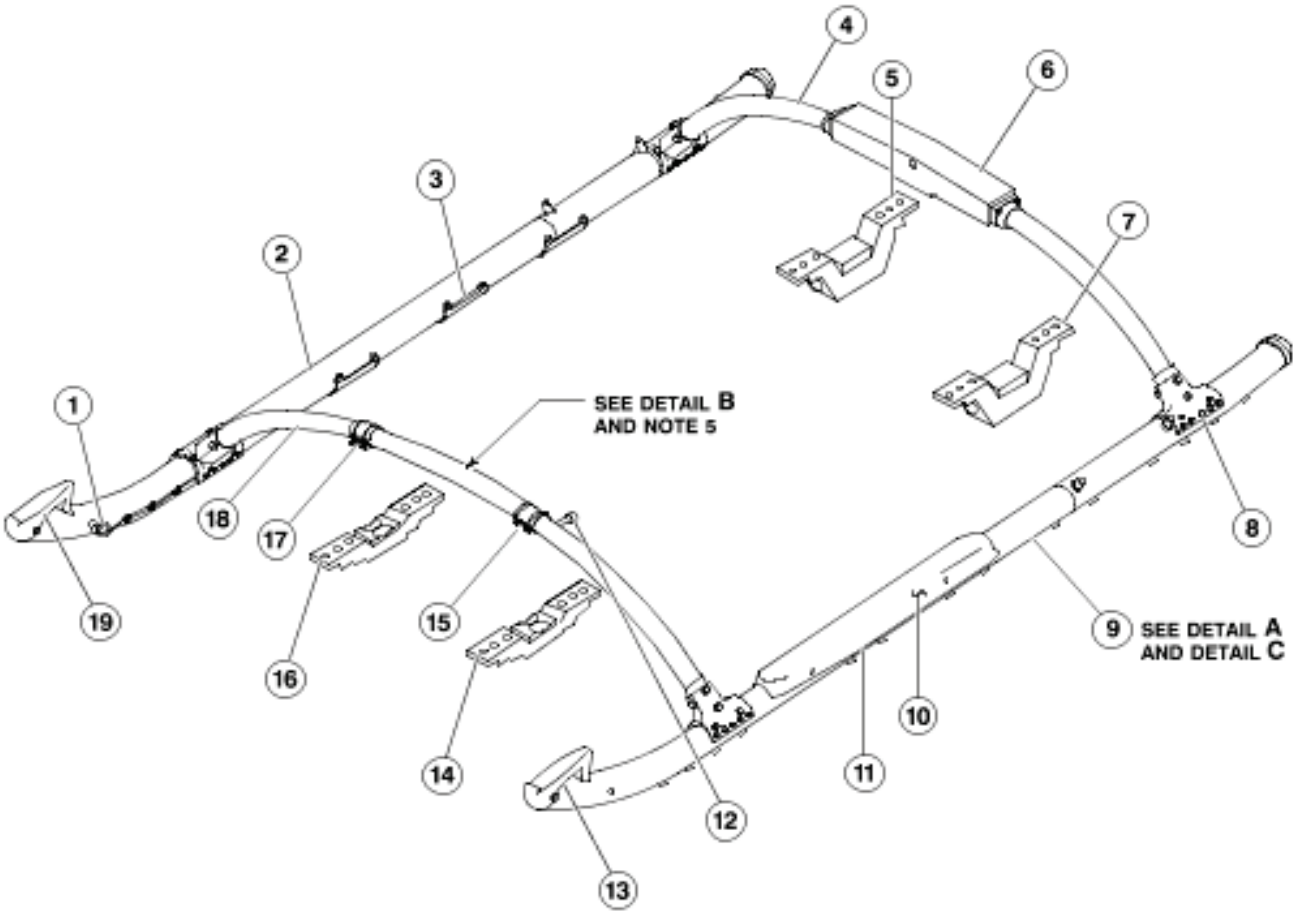
The landing gear skid tubes are mounted on ends of two arched crosstubes, which are secured to the fuselage structure. Each skid tube is fitted with forward end step, tow ring, two saddles with sockets for crosstubes, rear end cap, two eyebolt fittings for mounting ground handling wheels, and fore and aft skid shoes to absorb wear that would otherwise be inflicted to bottom of skid tubes during takeoff and landing.

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Skid Tubes - Removal

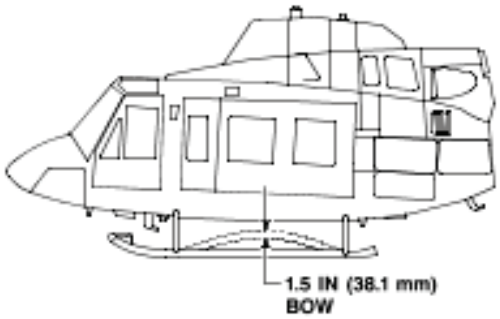
Procedure

1. Remove landing gear (412-A-32-01-00-00A-520A-A / 00697).
2. If electrically powered passenger step is installed, disconnect passenger step electrical connector (12, Figure 1) from crosstube (18).
3. Remove bolt (9, Figure 2).
4. Remove bolt (5).
5. Cut sealant at intersection of skid tube forward and aft saddles (2 and 6) and forward and aft crosstubes (3 and 7) with a sharp plastic scraper.
6. Remove skid tube (1) from forward and aft crosstubes (3 and 7). Use a rubber mallet to tap skid tubes and forward and aft saddles (2 and 6) off crosstubes, working both ends equally.



STANDARD SKID LANDING GEAR ASSEMBLY
412-050-001-101, -103, -105, -107, -109 and -111

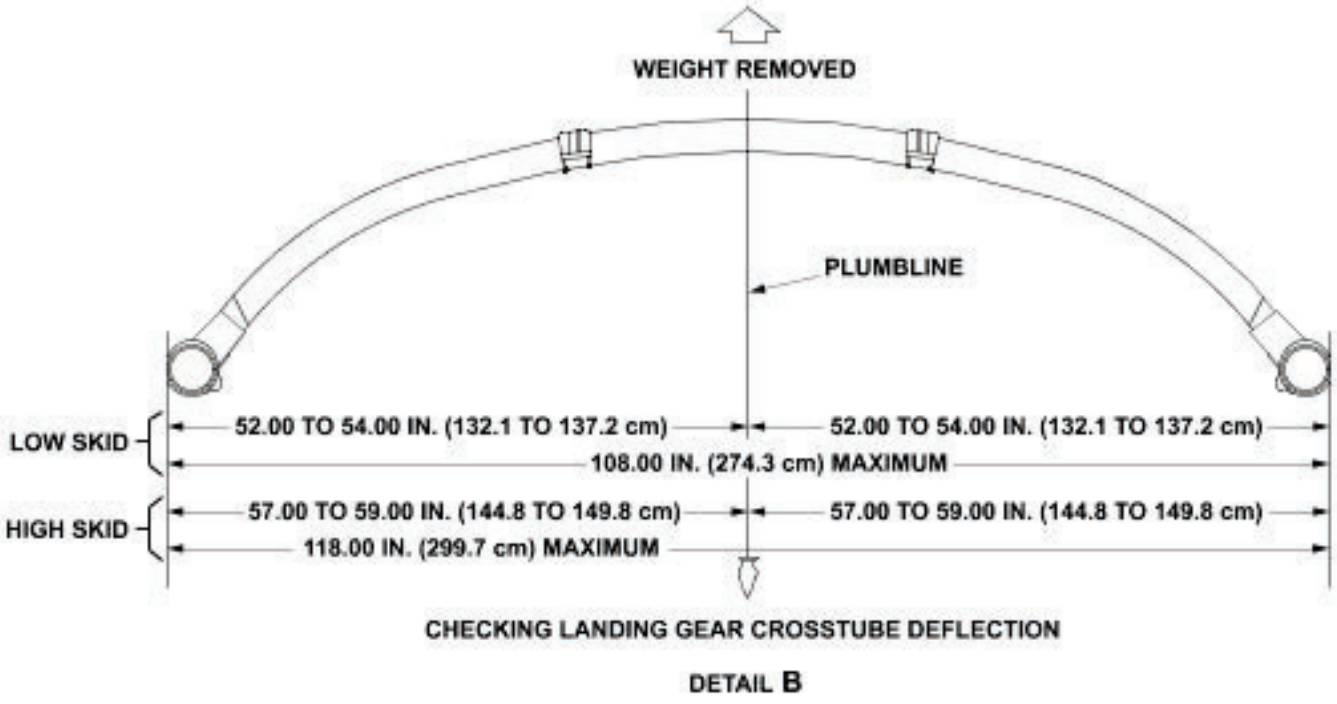
1. Tow ring
2. Skid tube
3. Skid shoe
4. Crosstube
5. Landing gear cap fitting
6. Aft crosstube support beam
7. Landing gear cap fitting
8. Saddle
9. Skid tube
10. Passenger step
11. Skid shoe
12. Passenger step electrical connector
13. Step
14. Landing gear cap fitting
15. Landing gear bearing and retaining support
16. Landing gear cap fitting
17. Landing gear bearing and retaining support
18. Crosstube
19. Step



DETAIL A

ICN-412-A-320100-B-97499-00003-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 1 of 9)

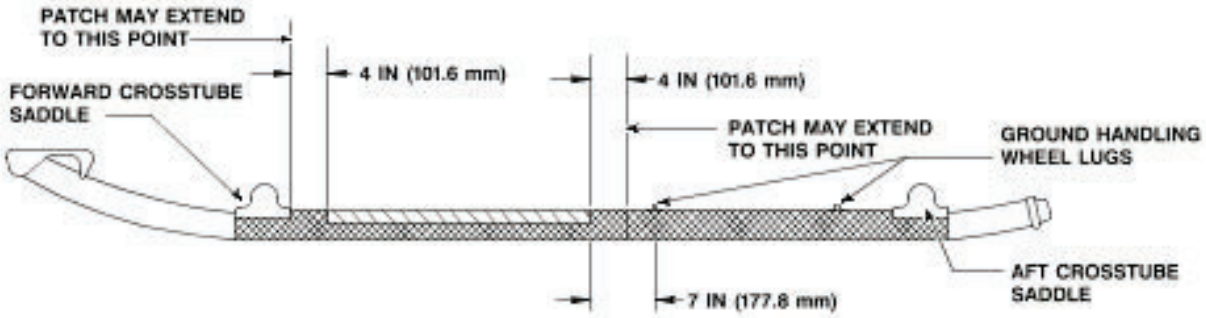


NOTES

1. Deformation (distortion) of either crosstube in excess of limits is cause for replacement of crosstube(s).
2. Crosstube may be deflected on one side only. If limit is exceeded, crosstube is considered scrapped and must be replaced.

ICN-412-A-320100-B-97499-00004-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 2 of 9)



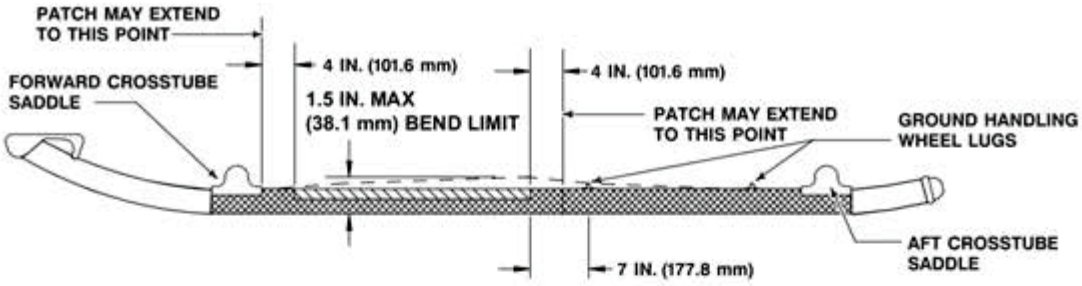
DETAIL C
412-050-023

DAMAGE LOCATION SYMBOLS

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR AREAS ALLOWED		
CRACKS ALLOWED	None	None	None
DENTS AND HOLES	None	4 inches (101.6 mm) Refer to paragraph 32-9.	Not Critical

ICN-412-A-320100-B-97499-00005-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 3 of 9)

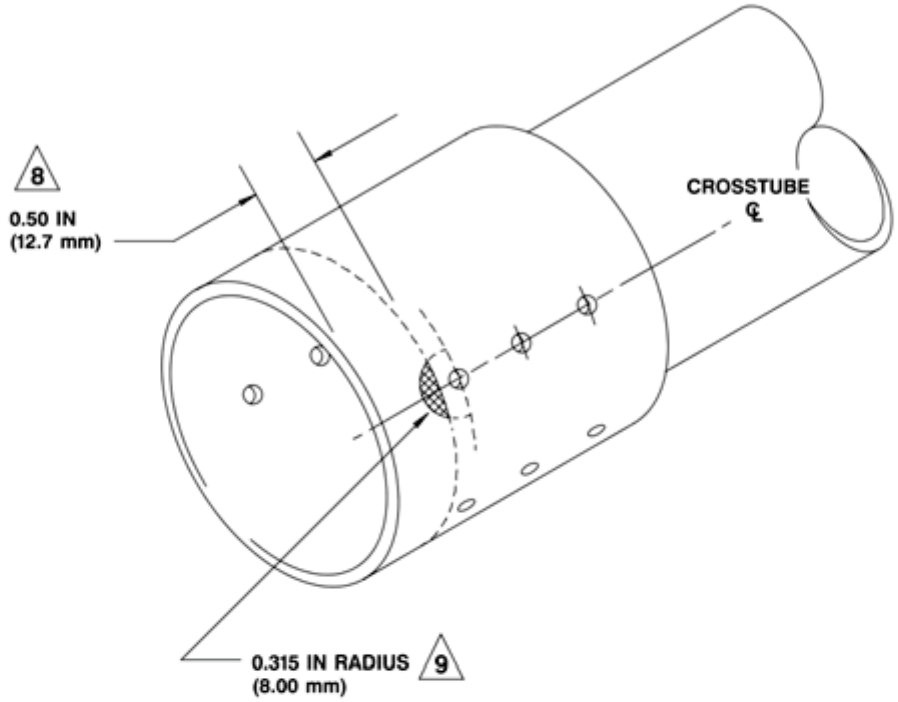


DETAIL C
412-050-030
412-050-034

			(Does not apply to saddles)
DENTS	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No dents allowed. Replace as needed.	Not critical.
HOLES	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No holes allowed. Replace as needed.	Not critical. Plug holes to keep out moisture.
CRACKS	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.
SCRATCHES SCUFFS NICKS	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	Not critical.
	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	
	If damage is greater than established limits, replace as needed.	If damage is greater than established limits, replace as needed.	
CORROSION	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup is limited to 10% of material thickness. Refer to paragraph 32-11.

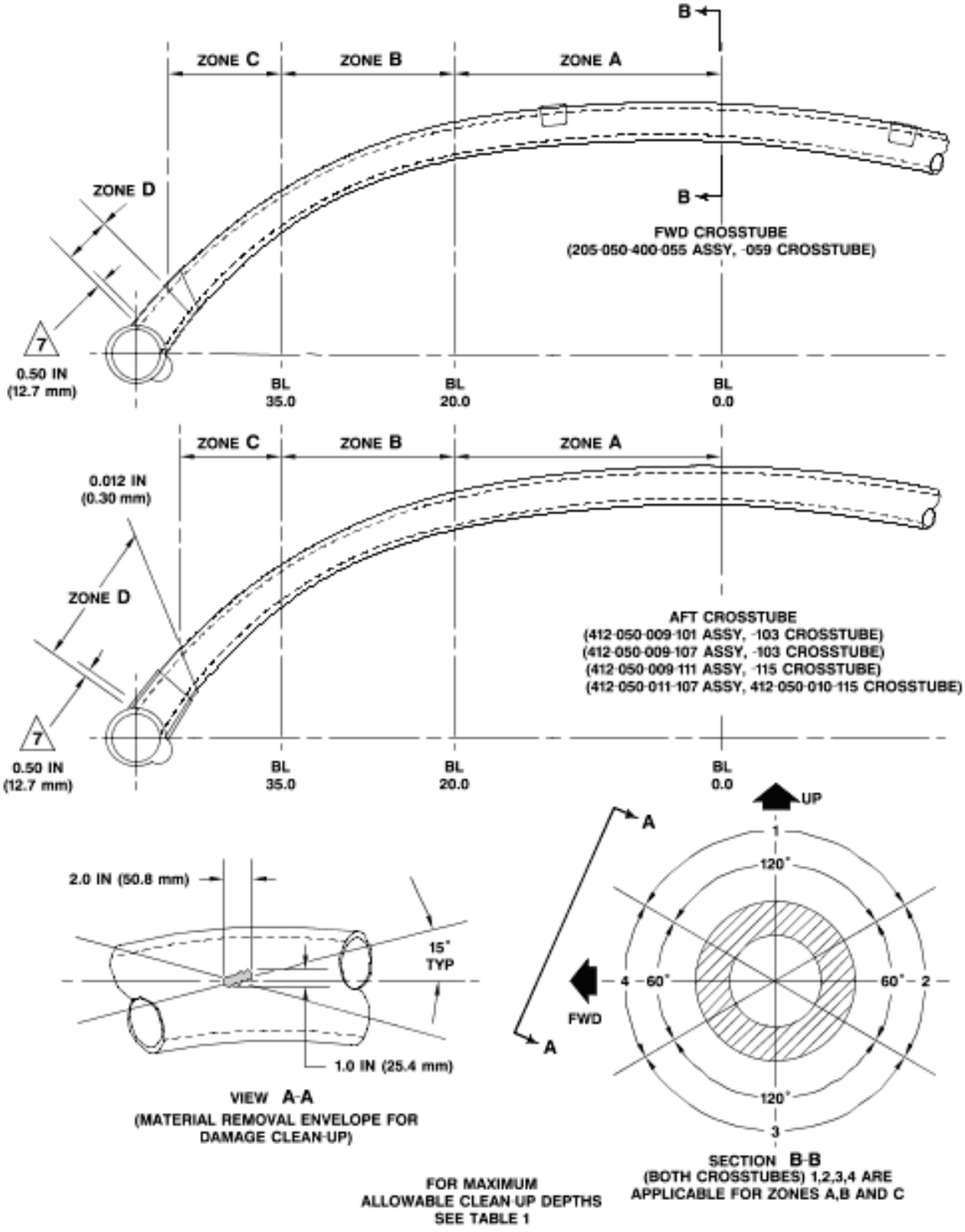
ICN-412-A-320100-B-97499-00006-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 4 of 9)



ICN-412-A-320100-B-97499-00007-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 5 of 9)



ICN-412-A-320100-B-97499-00008-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 6 of 9)

TABLE 1
MAXIMUM ALLOWABLE CLEANUP DEPTH STANDARD AND HIGH SKID GEAR

ZONE	BHT PRODUCTS FWD CROSSTUBE INCHES (mm) △10	BHT PRODUCTS AFT CROSSTUBE INCHES (mm) △11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.12) 0.015 (0.38) 0.005 (0.12) 0.015 (0.38)
B1 B2 B3 B4	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.010 (0.25) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.012 (0.30) 0.012 (0.30)
D △7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 6 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

△7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 1.

△8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note △9 (Figure 32-2, sheet 5).

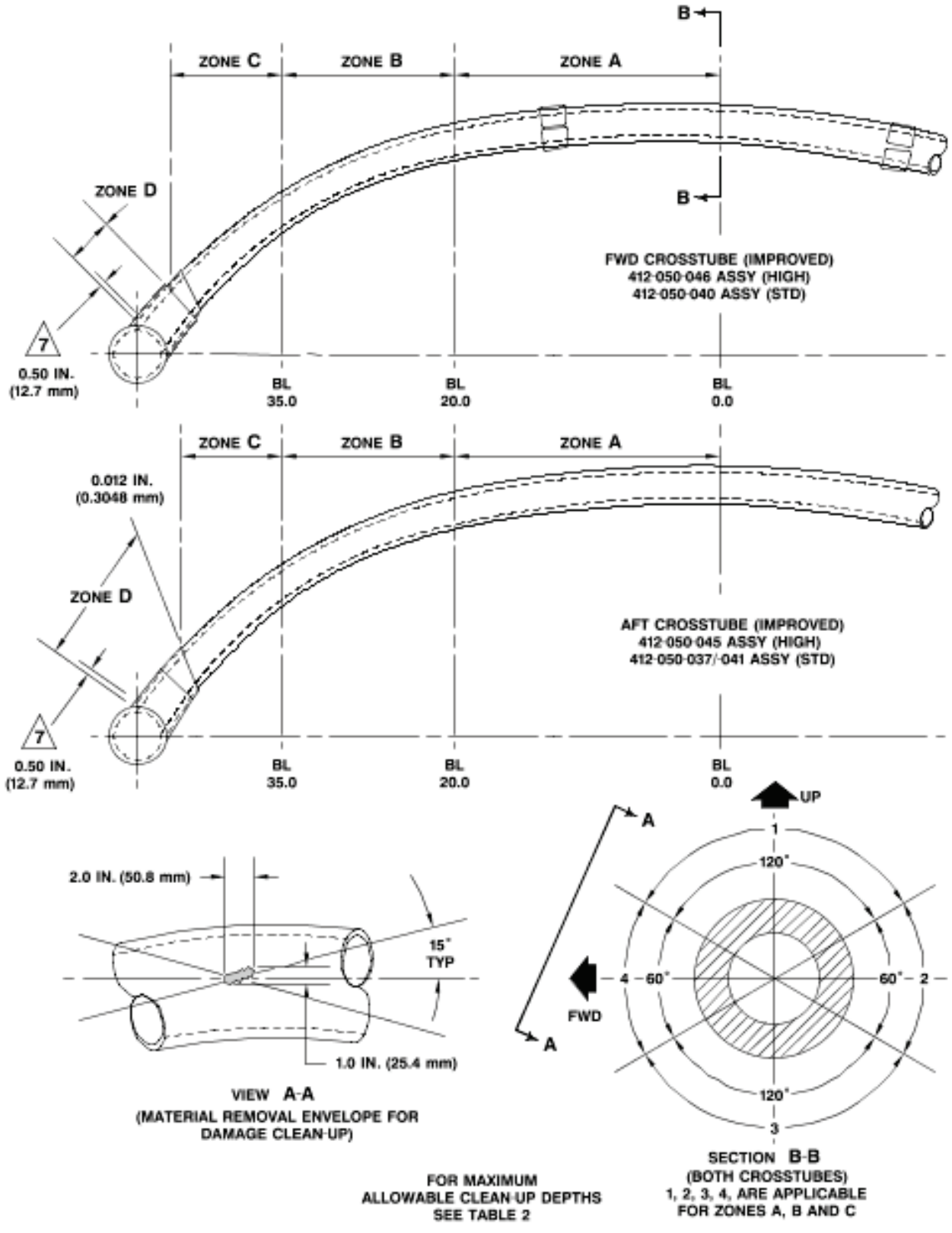
△9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

△10 Forward standard crosstube (BHT 205-050-400-055).
Forward high crosstube (BHT 205-050-403-031).

△11 Aft standard crosstube (BHT 412-050-002-101/103/105/107).
Aft high crosstube (BHT 412-050-011-101/103/105/107).

ICN-412-A-320100-B-97499-00009-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 7 of 9)



ICN-412-A-320100-B-97499-00010-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 8 of 9)

TABLE 2
MAXIMUM ALLOWABLE CLEANUP DEPTH IMPROVED STANDARD AND IMPROVED HIGH SKID GEAR

ZONE	BHT AND AAI PRODUCTS	BHT AND AAI PRODUCTS
	FWD CROSSTUBE INCHES (mm) 10	AFT CROSSTUBE INCHES (mm) 11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
B1 B2 B3 B4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.005 (0.13) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.005 (0.13) 0.012 (0.30)
D 7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 8 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 2.

8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note 9 (Figure 32-2, sheet 5).

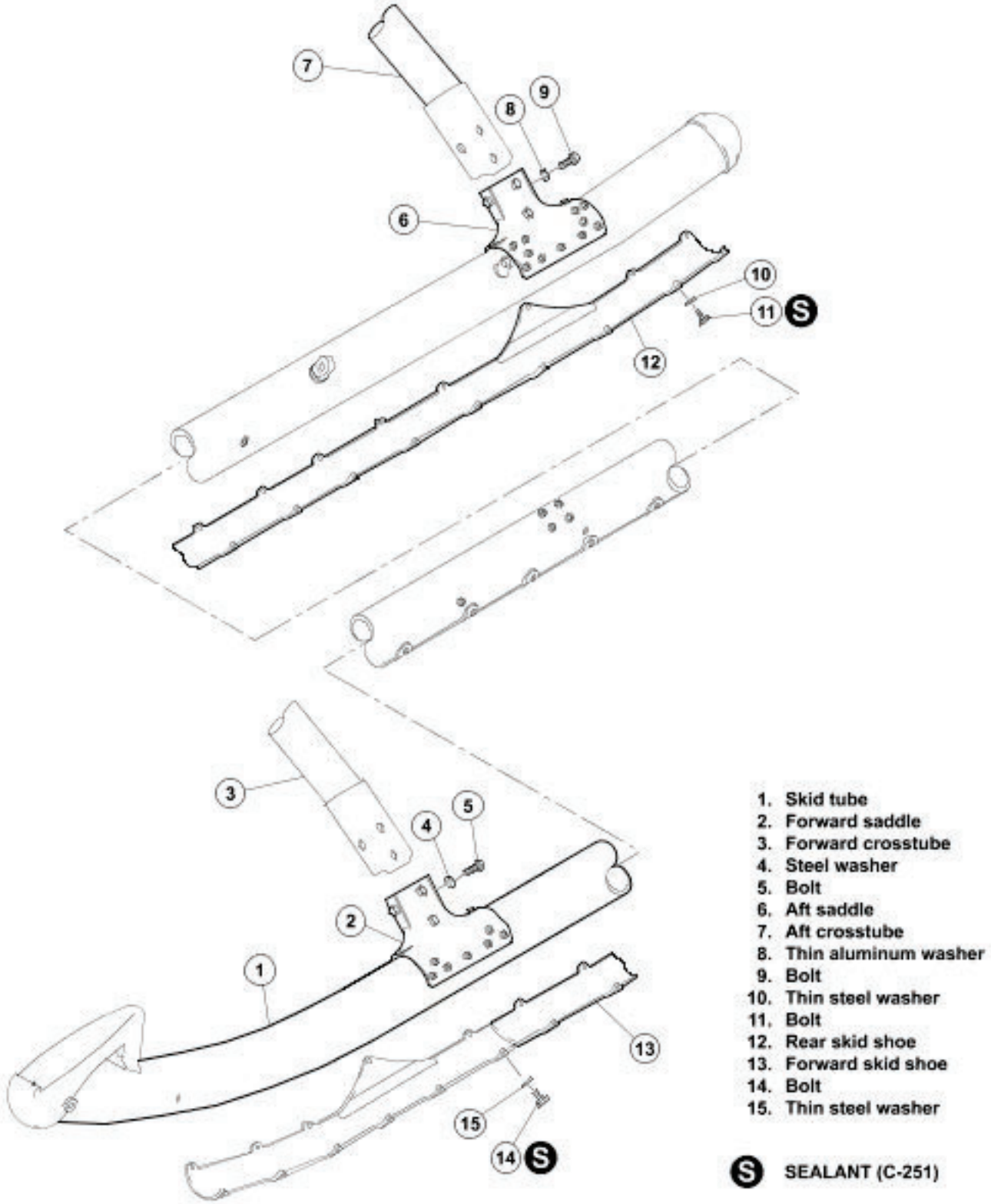
9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

10 Improved forward standard crosstube (BHT 412-050-040-101, AAI 212-320-103 ALL).
Improved forward high crosstube (BHT 412-050-046-101, AAI 212-321-103 ALL).

11 Improved aft standard crosstube (BHT 412-050-041-109, AAI 412-320-104 ALL).
Improved aft high crosstube (BHT 412-050-045-105, AAI 412-321-106 ALL).

ICN-412-A-320100-B-97499-00011-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 9 of 9)



ICN-412-A-320101-B-97499-00001-A-001-01

Figure 2. Skid Tube and Attaching Parts (Sheet 1 of 1)

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Skid Tubes - Inspection

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Nonslip compound	C-223	AR	

Procedure

NOTE

Superficial mechanical damage to skid tubes (2 and 9, Figure 1) is acceptable without repair. Sharp corners and gouges should be repaired within allowable limits (Detail C). Distortion (deflection) in excess of limits shown (Detail A) is cause for replacement. For BHT 412-050-023 skid tubes, dents and/or holes in skid tubes up to 4 inches (101.6 mm) in diameter in the area indicated in Figure 2 are acceptable, if repaired as described in 412-A-32-01-01-00A-685A-A / 00704. For BHT 412-050-030/-034 and AAI skid tubes, dents and/or holes in skid tubes up to 3 inches (76.2 mm) in diameter in the area indicated in Figure 3 are acceptable, if repaired as described in 412-A-32-01-01-00A-685A-A / 00704. No repairs (patches) are allowed under skid shoes.

1. Inspect skid tubes (2 and 9, Figure 1), including skid shoes (3 and 11) and saddles (8) for dents, holes, cracks, scratches, scuffs, and corrosion (Detail C).

NOTE

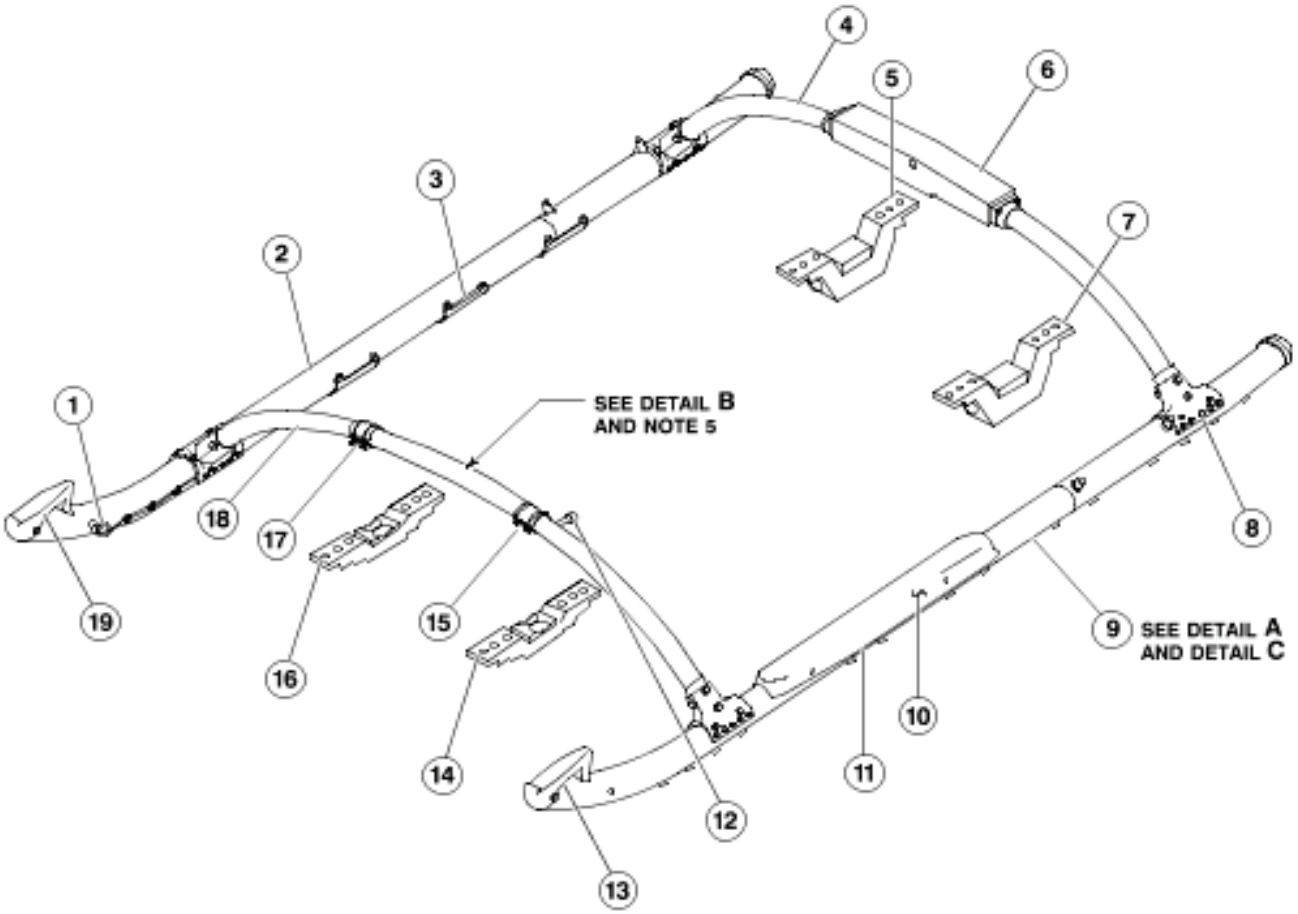
Superficial mechanical damage on tow rings (1) is acceptable without repair. Distorted and/or loose fitting tow rings are cause to replace tow rings and, if attaching holes in skid tubes (2 and 9) are elongated, replace skid tube(s).

2. Inspect tow rings (1) for condition, damage, and security of attachment.

3. Inspect skid tubes (2 and 9) for deformation and out-of-round condition.

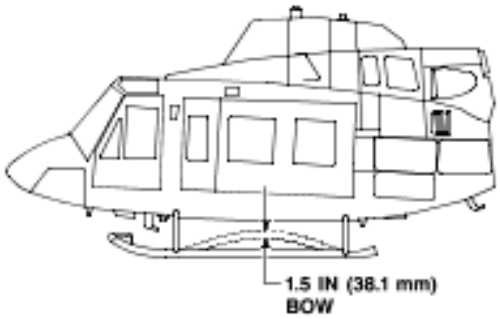
4. Inspect for missing paint, primer, and nonslip compound (C-223). Restore as required.

5. Apply electrical power to passenger step (10) and raise away from skid tube (9). Inspect actuator, support levers, step, and attaching hardware for condition.



STANDARD SKID LANDING GEAR ASSEMBLY
412-050-001-101, -103, -105, -107, -109 and -111

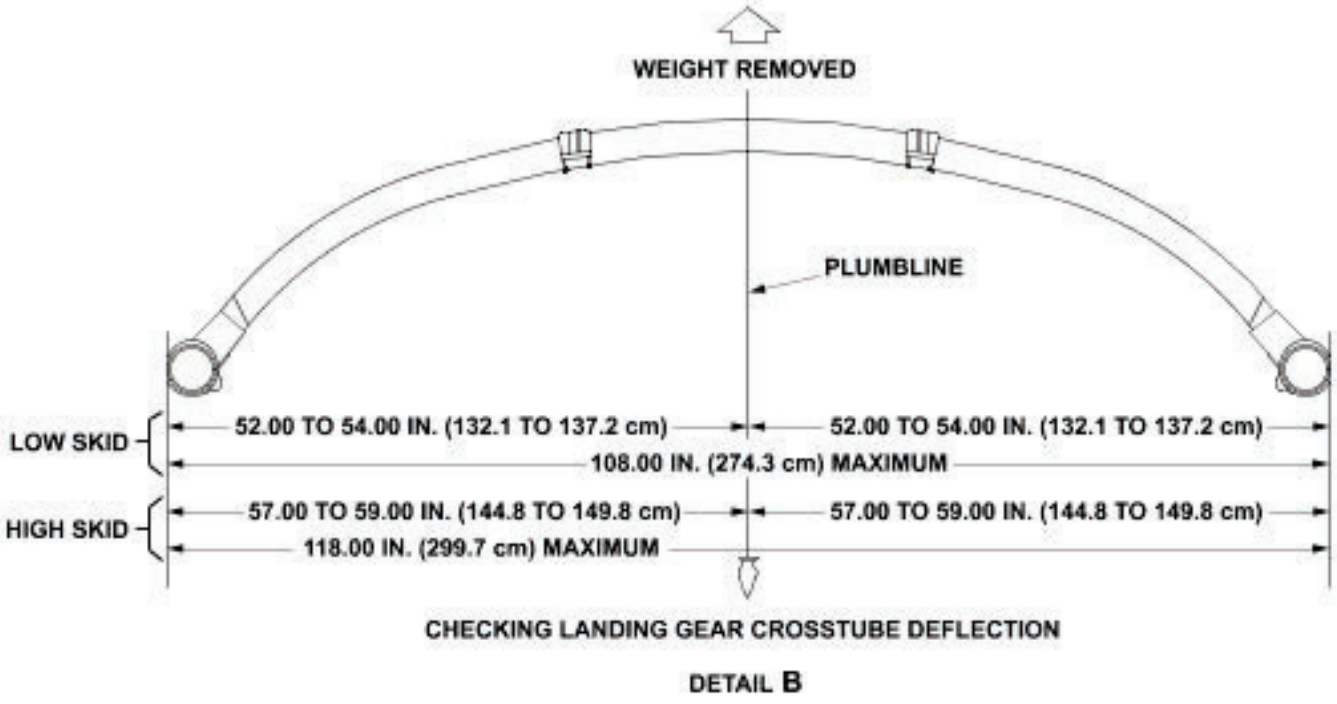
1. Tow ring
2. Skid tube
3. Skid shoe
4. Crosstube
5. Landing gear cap fitting
6. Aft crosstube support beam
7. Landing gear cap fitting
8. Saddle
9. Skid tube
10. Passenger step
11. Skid shoe
12. Passenger step electrical connector
13. Step
14. Landing gear cap fitting
15. Landing gear bearing and retaining support
16. Landing gear cap fitting
17. Landing gear bearing and retaining support
18. Crosstube
19. Step



DETAIL A

ICN-412-A-320100-B-97499-00003-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 1 of 9)

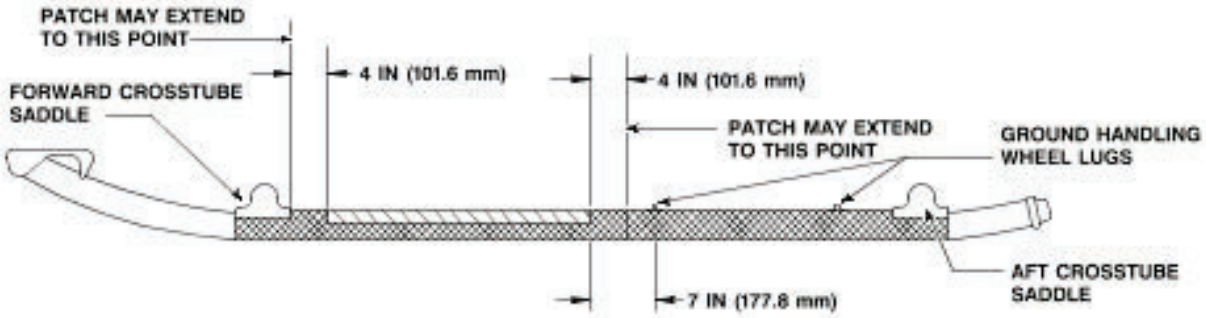


NOTES

1. Deformation (distortion) of either crosstube in excess of limits is cause for replacement of crosstube(s).
2. Crosstube may be deflected on one side only. If limit is exceeded, crosstube is considered scrapped and must be replaced.

ICN-412-A-320100-B-97499-00004-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 2 of 9)



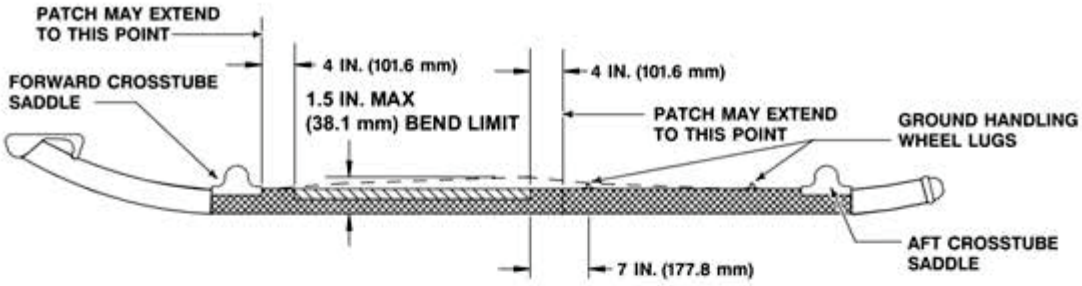
DETAIL C
412-050-023

DAMAGE LOCATION SYMBOLS

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR AREAS ALLOWED		
CRACKS ALLOWED	None	None	None
DENTS AND HOLES	None	4 inches (101.6 mm) Refer to paragraph 32-9.	Not Critical

ICN-412-A-320100-B-97499-00005-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 3 of 9)

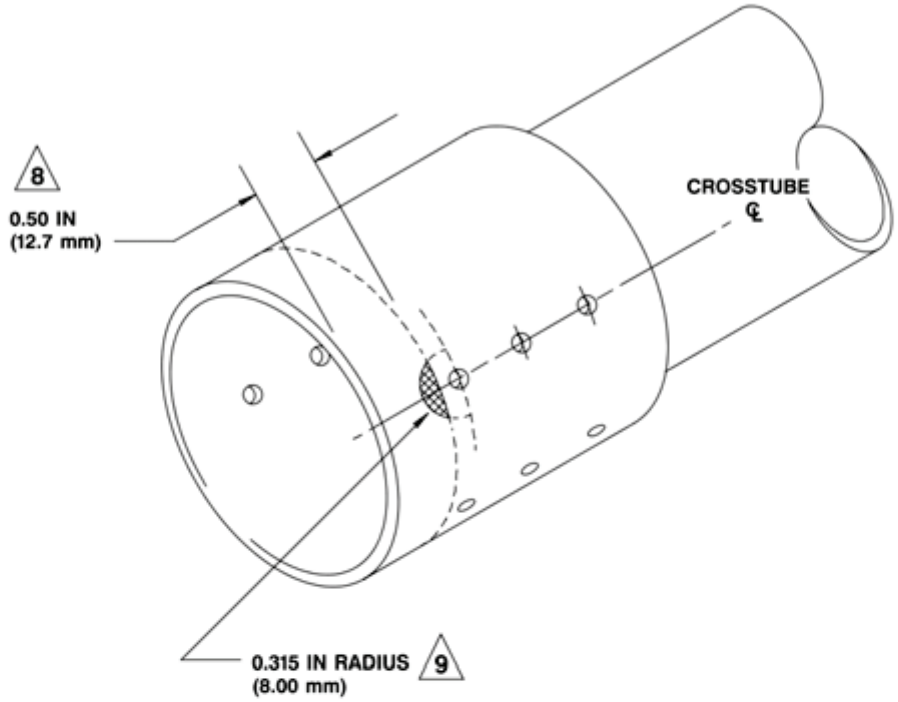


DETAIL C
412-050-030
412-050-034

			(Does not apply to saddles)
DENTS	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No dents allowed. Replace as needed.	Not critical.
HOLES	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No holes allowed. Replace as needed.	Not critical. Plug holes to keep out moisture.
CRACKS	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.
SCRATCHES SCUFFS NICKS	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	Not critical.
	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	
	If damage is greater than established limits, replace as needed.	If damage is greater than established limits, replace as needed.	
CORROSION	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup is limited to 10% of material thickness. Refer to paragraph 32-11.

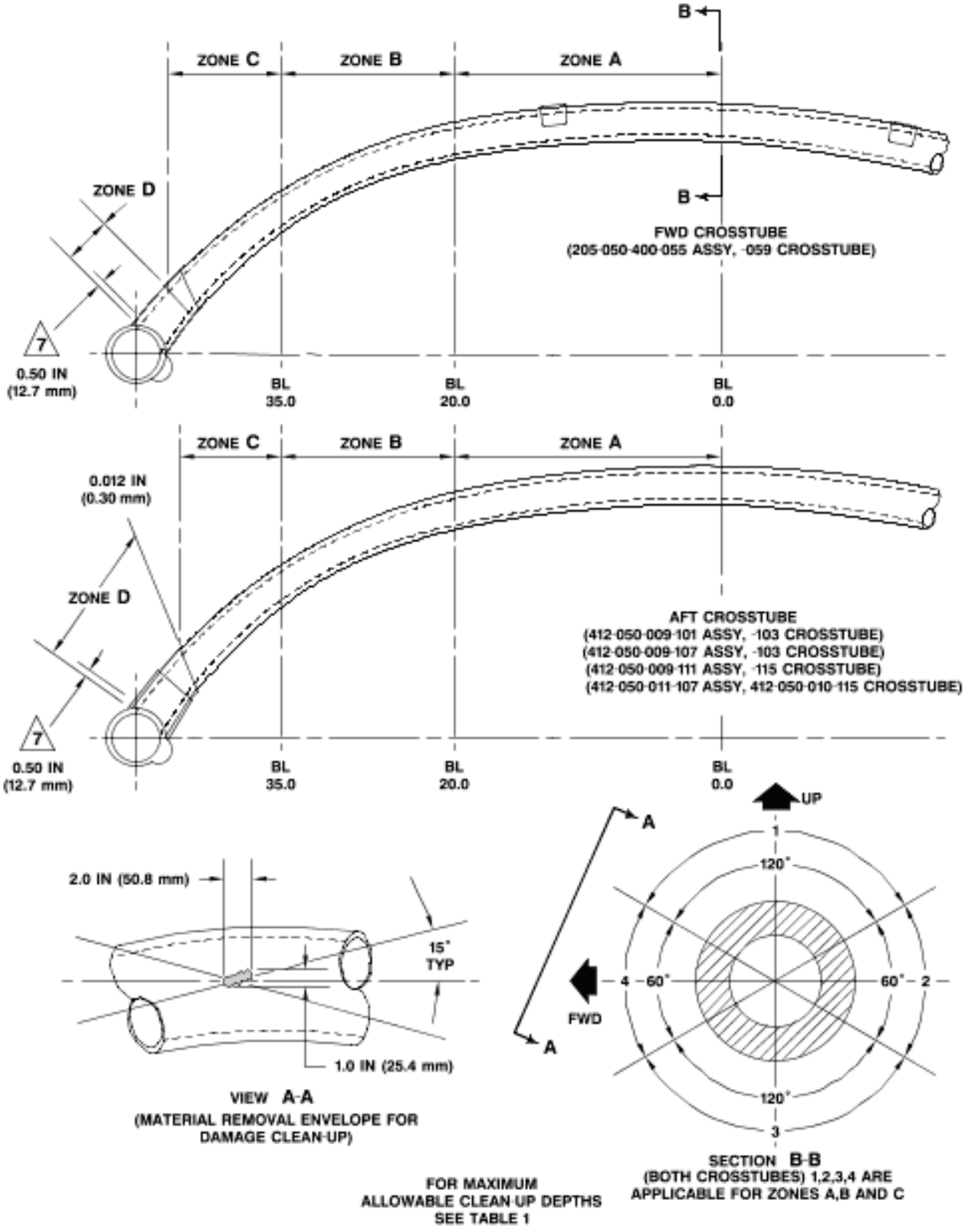
ICN-412-A-320100-B-97499-00006-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 4 of 9)



ICN-412-A-320100-B-97499-00007-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 5 of 9)



ICN-412-A-320100-B-97499-00008-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 6 of 9)

TABLE 1
MAXIMUM ALLOWABLE CLEANUP DEPTH STANDARD AND HIGH SKID GEAR

ZONE	BHT PRODUCTS FWD CROSSTUBE INCHES (mm) △10	BHT PRODUCTS AFT CROSSTUBE INCHES (mm) △11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.12) 0.015 (0.38) 0.005 (0.12) 0.015 (0.38)
B1 B2 B3 B4	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.010 (0.25) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.012 (0.30) 0.012 (0.30)
D △7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 6 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

△7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 1.

△8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note △9 (Figure 32-2, sheet 5).

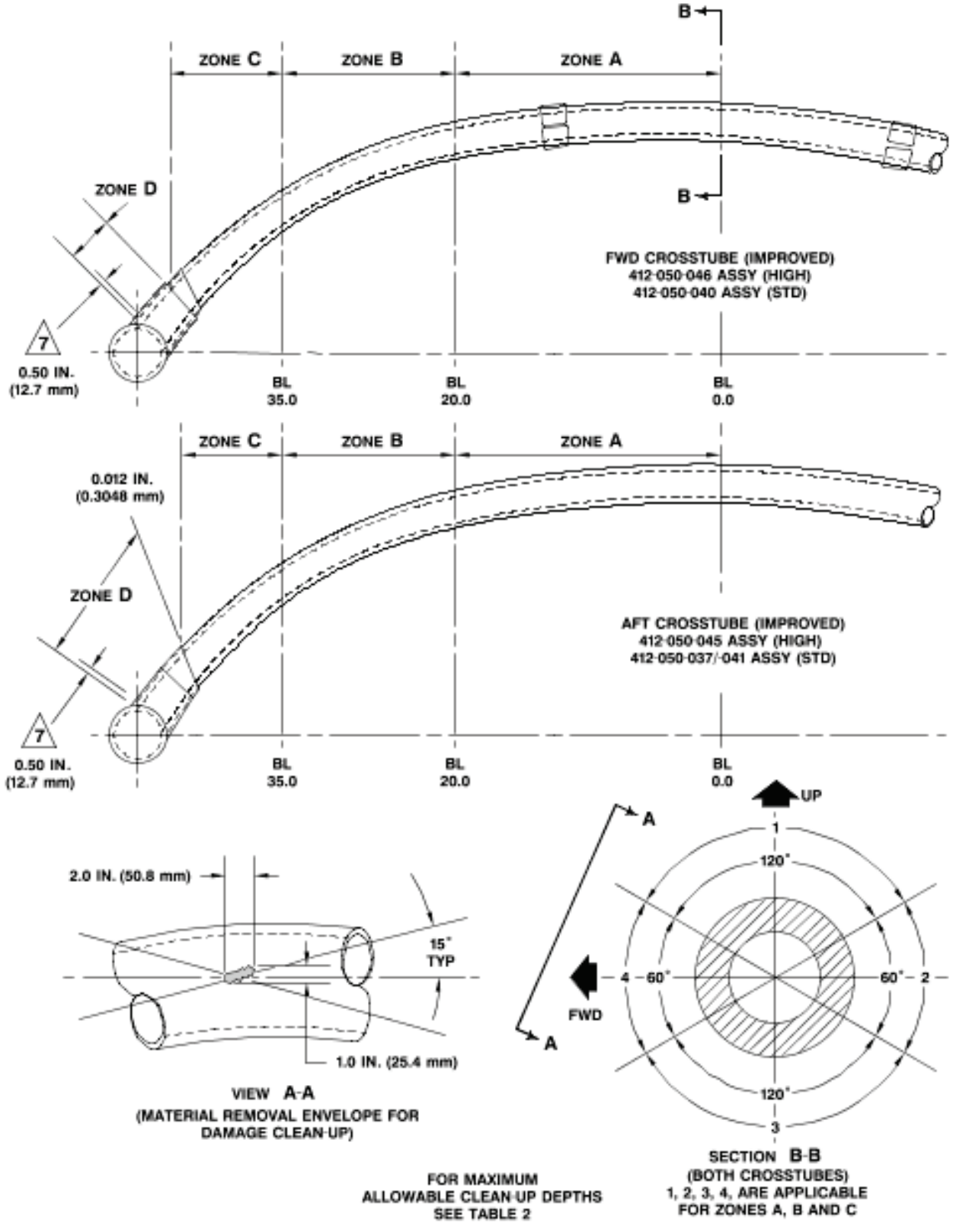
△9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

△10 Forward standard crosstube (BHT 205-050-400-055).
Forward high crosstube (BHT 205-050-403-031).

△11 Aft standard crosstube (BHT 412-050-002-101/103/105/107).
Aft high crosstube (BHT 412-050-011-101/103/105/107).

ICN-412-A-320100-B-97499-00009-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 7 of 9)



ICN-412-A-320100-B-97499-00010-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 8 of 9)

TABLE 2
MAXIMUM ALLOWABLE CLEANUP DEPTH IMPROVED STANDARD AND IMPROVED HIGH SKID GEAR

ZONE	BHT AND AAI PRODUCTS	BHT AND AAI PRODUCTS
	FWD CROSSTUBE INCHES (mm) 10	AFT CROSSTUBE INCHES (mm) 11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
B1 B2 B3 B4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.005 (0.13) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.005 (0.13) 0.012 (0.30)
D 7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 8 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 2.

8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note 9 (Figure 32-2, sheet 5).

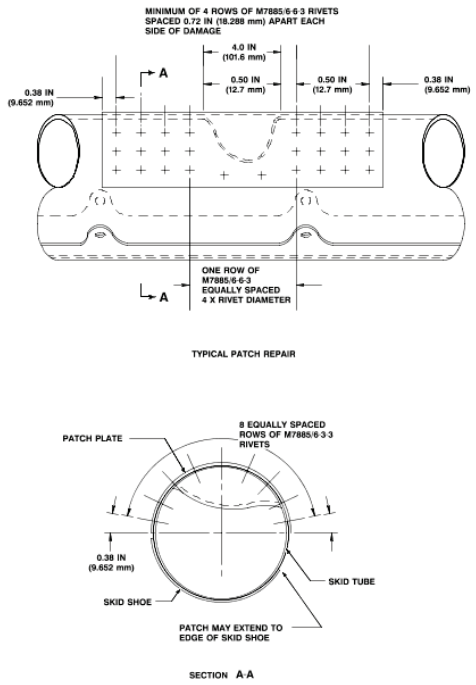
9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

10 Improved forward standard crosstube (BHT 412-050-040-101, AAI 212-320-103 ALL).
Improved forward high crosstube (BHT 412-050-046-101, AAI 212-321-103 ALL).

11 Improved aft standard crosstube (BHT 412-050-041-109, AAI 412-320-104 ALL).
Improved aft high crosstube (BHT 412-050-045-105, AAI 412-321-106 ALL).

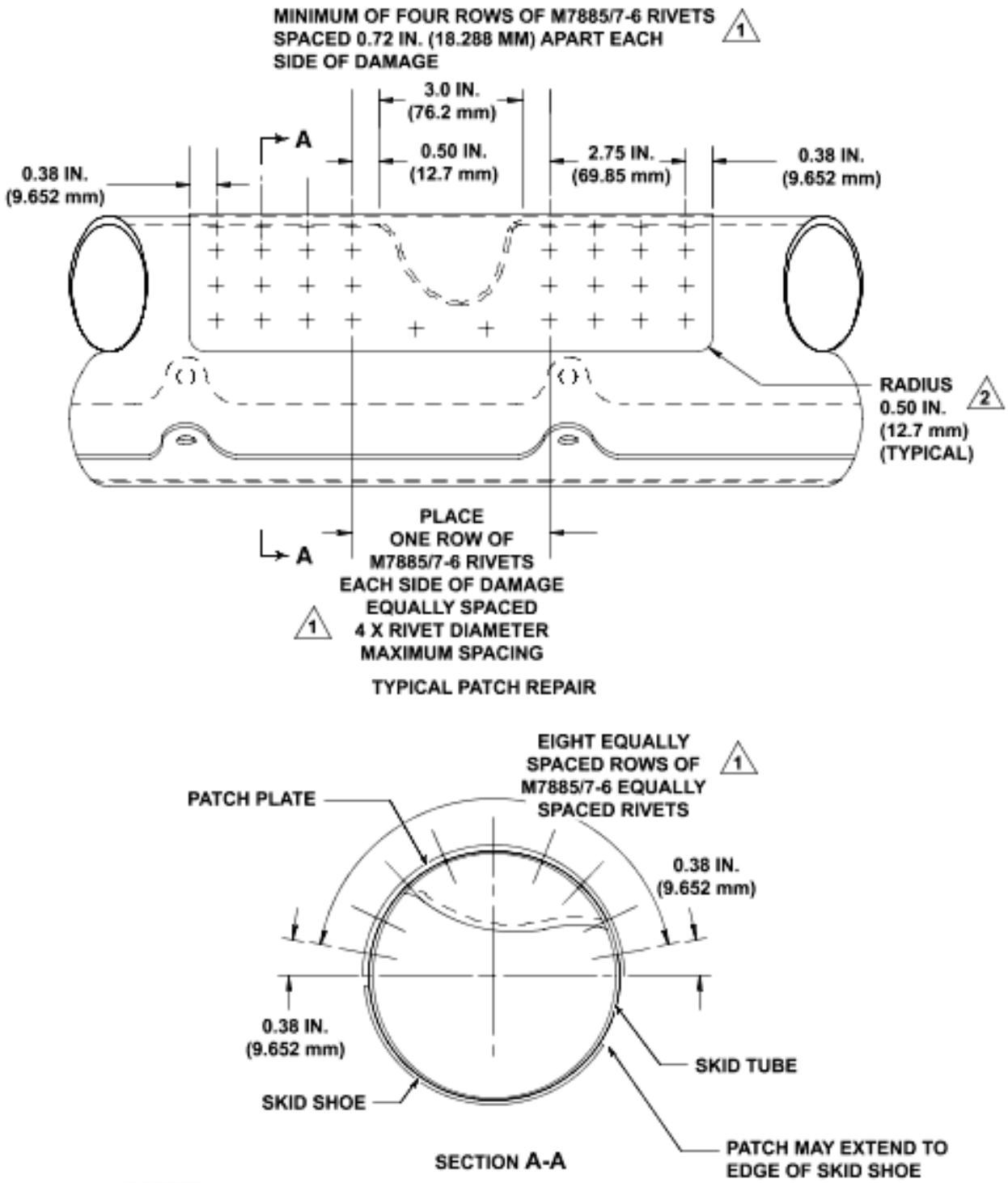
ICN-412-A-320100-B-97499-00011-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 9 of 9)



ICN-412-A-320101-B-97499-00002-A-001-01

Figure 2. Skid Tube (412-050-023) Patch Installation (Sheet 1 of 1)



NOTES

1. Use appropriate grip length of rivets to accommodate doubler and skid tube thickness.
2. Break all sharp edges on doubler.

ICN-412-A-320101-B-97499-00003-A-001-01

Figure 3. Skid Tube (412-050-030/-034) Patch Installation (Sheet 1 of 1)

Skid Tubes - Patch Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Chemical film material	C-100	AR	
Epoxy polyamide primer	C-204	AR	
Nonslip compound	C-223	AR	
Sealant	C-308	AR	

Procedure

NOTE

Repair to skid tube is limited to patching top of skid tube in areas indicated in Figure 1 and Figure 2 or Figure 3, and repair of skid shoes (Figure 4) (412-A-32-01-02-00A-685A-A / 00711).

1. If dents are present in authorized repair area shown in Figure 1, polish out raised areas that would interfere with application of patch.

2. If holes are present in authorized repair area shown in Figure 1, smooth rough edges of holes to remove sharp edges and raised areas that would interfere with application of patch.

3. For skid tube 412-050-023, fabricate a patch to cover dent or hole from 2024 aluminum alloy tube, WW-T-700/2,T3 or 2024 aluminum sheet, per QQ-A-250/4, T73 with 0.095 inch (2.41 mm) wall thickness and 4.0 inches (101.6 mm) inside diameter. See Figure 2 for required patch size.

4. For skid tubes 412-050-030/-034 and AAI skid tubes, fabricate a patch to cover dent or hole from 2024 aluminum alloy tube, WW-T-700/7,T3 or 2024 aluminum sheet, per QQ-A-250/4, T73 with 0.125 inch (3.17 mm) wall thickness and 3.0 inches (76.2 mm) inside diameter. See Figure 3 for required patch size.

5. Secure patch to skid tube and drill holes for M7885/6-6-3 rivets as shown in Figure 2 or Figure 3, as applicable. Remove patch and deburr holes.

6. Clean patch and mating area of skid tube. Apply (brush-on) chemical film material (C-100) to bare aluminum surfaces of skid tube and patch.

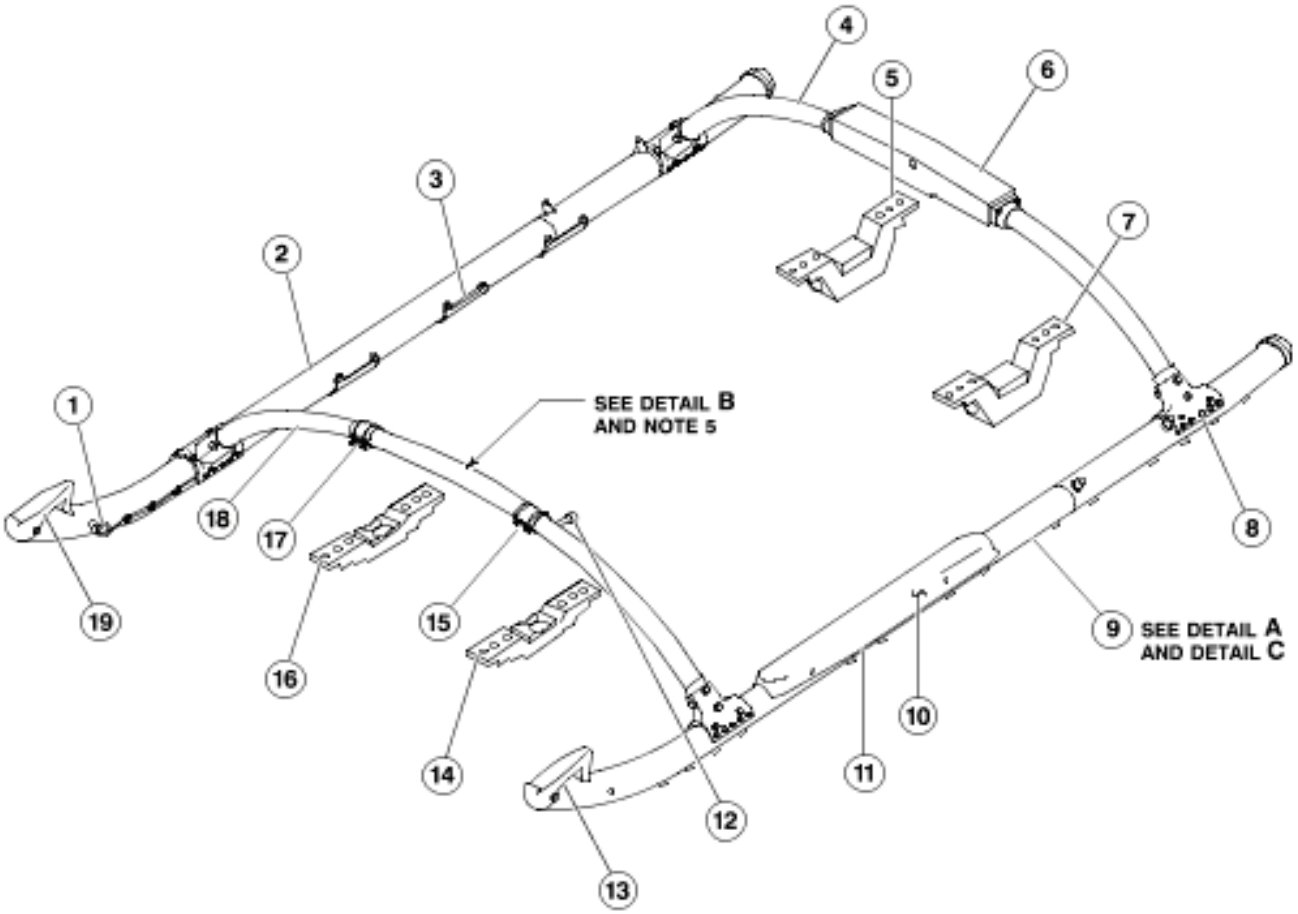
7. Apply coat of epoxy polyamide primer (C-204) to both surfaces of patch and to mating surface of skid tube.

8. Coat mating surfaces of patch and skid tube with sealant (C-308).

9. Position patch on skid tube and install rivets (M7885/6-6-3).

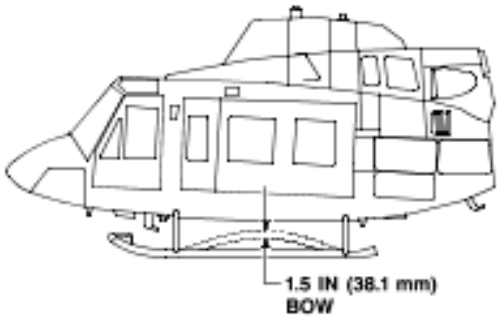
10. Apply coat of epoxy polyamide primer (C-204) to repair area and paint to match existing finish on skid tube.

11. Apply two coats of nonslip compound (C-223) to appropriate areas.



STANDARD SKID LANDING GEAR ASSEMBLY
412-050-001-101, -103, -105, -107, -109 and -111

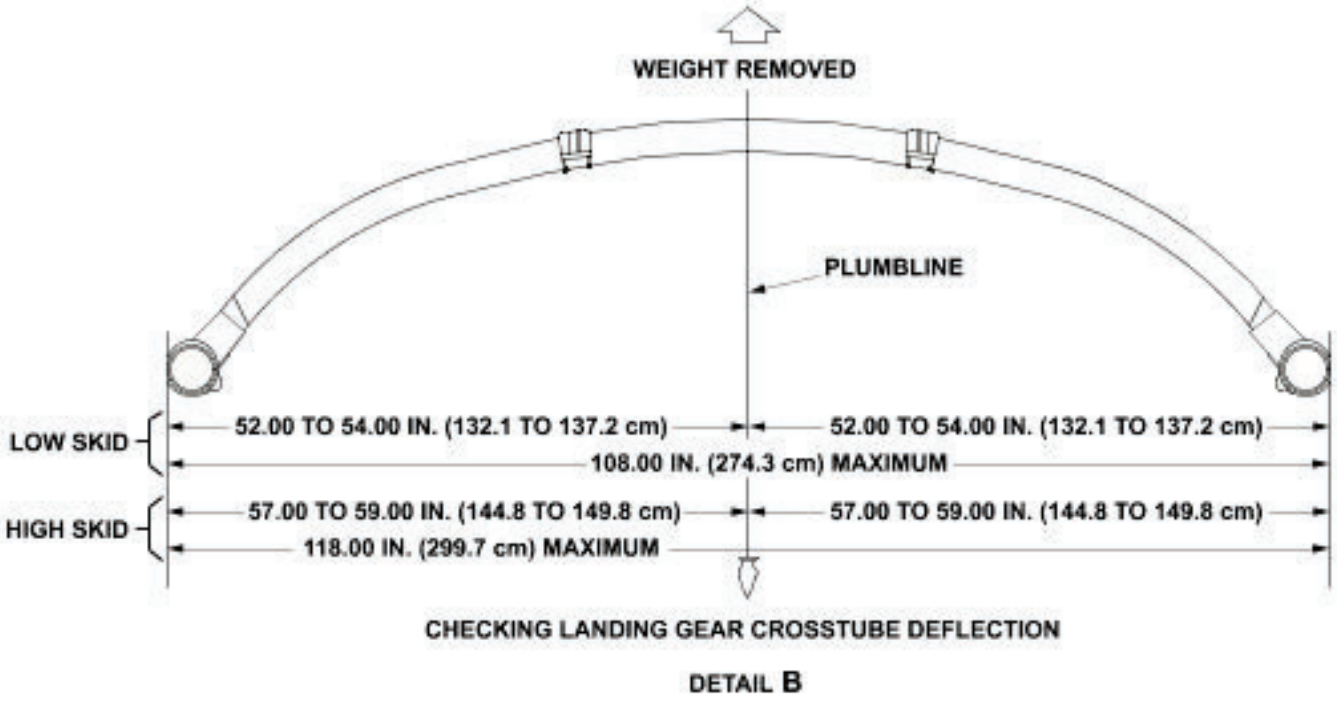
1. Tow ring
2. Skid tube
3. Skid shoe
4. Crosstube
5. Landing gear cap fitting
6. Aft crosstube support beam
7. Landing gear cap fitting
8. Saddle
9. Skid tube
10. Passenger step
11. Skid shoe
12. Passenger step electrical connector
13. Step
14. Landing gear cap fitting
15. Landing gear bearing and retaining support
16. Landing gear cap fitting
17. Landing gear bearing and retaining support
18. Crosstube
19. Step



DETAIL A

ICN-412-A-320100-B-97499-00003-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 1 of 9)

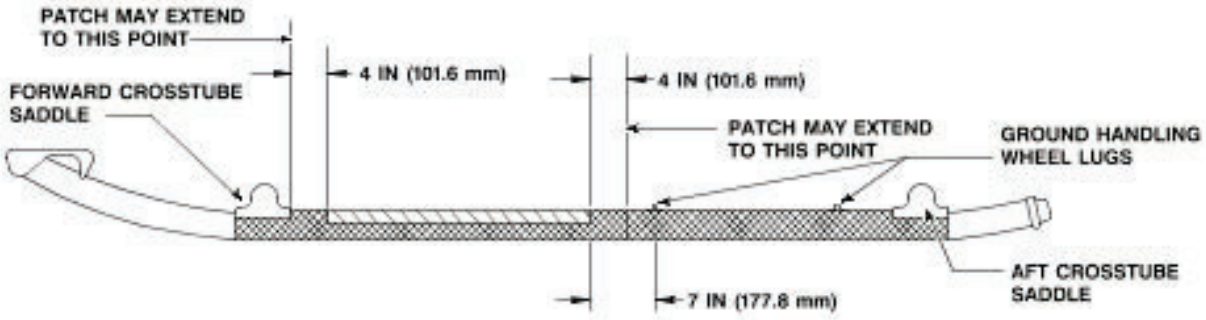


NOTES

1. Deformation (distortion) of either crosstube in excess of limits is cause for replacement of crosstube(s).
2. Crosstube may be deflected on one side only. If limit is exceeded, crosstube is considered scrapped and must be replaced.

ICN-412-A-320100-B-97499-00004-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 2 of 9)



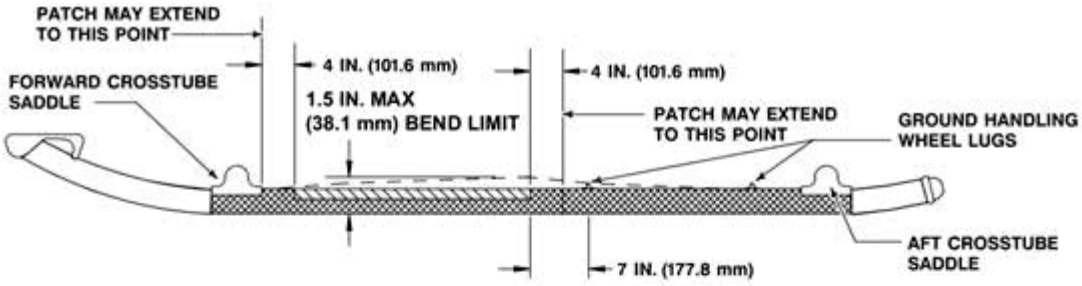
DETAIL C
412-050-023

DAMAGE LOCATION SYMBOLS

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR AREAS ALLOWED		
CRACKS ALLOWED	None	None	None
DENTS AND HOLES	None	4 inches (101.6 mm) Refer to paragraph 32-9.	Not Critical

ICN-412-A-320100-B-97499-00005-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 3 of 9)

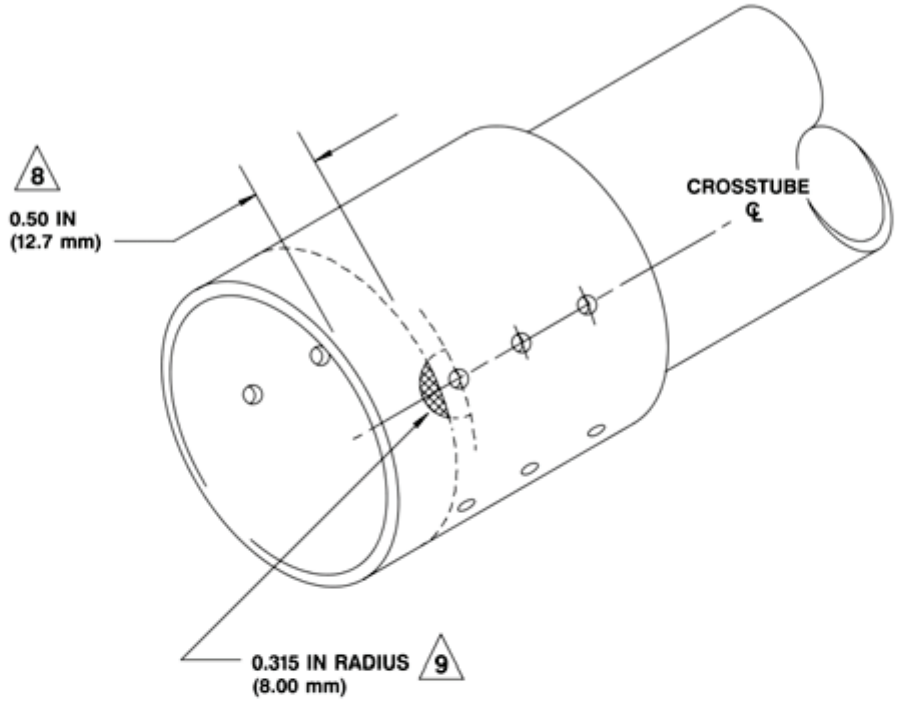


DETAIL C
412-050-030
412-050-034

			(Does not apply to saddles)
DENTS	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No dents allowed. Replace as needed.	Not critical.
HOLES	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No holes allowed. Replace as needed.	Not critical. Plug holes to keep out moisture.
CRACKS	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.
SCRATCHES SCUFFS NICKS	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	Not critical.
	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	
	If damage is greater than established limits, replace as needed.	If damage is greater than established limits, replace as needed.	
CORROSION	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup is limited to 10% of material thickness. Refer to paragraph 32-11.

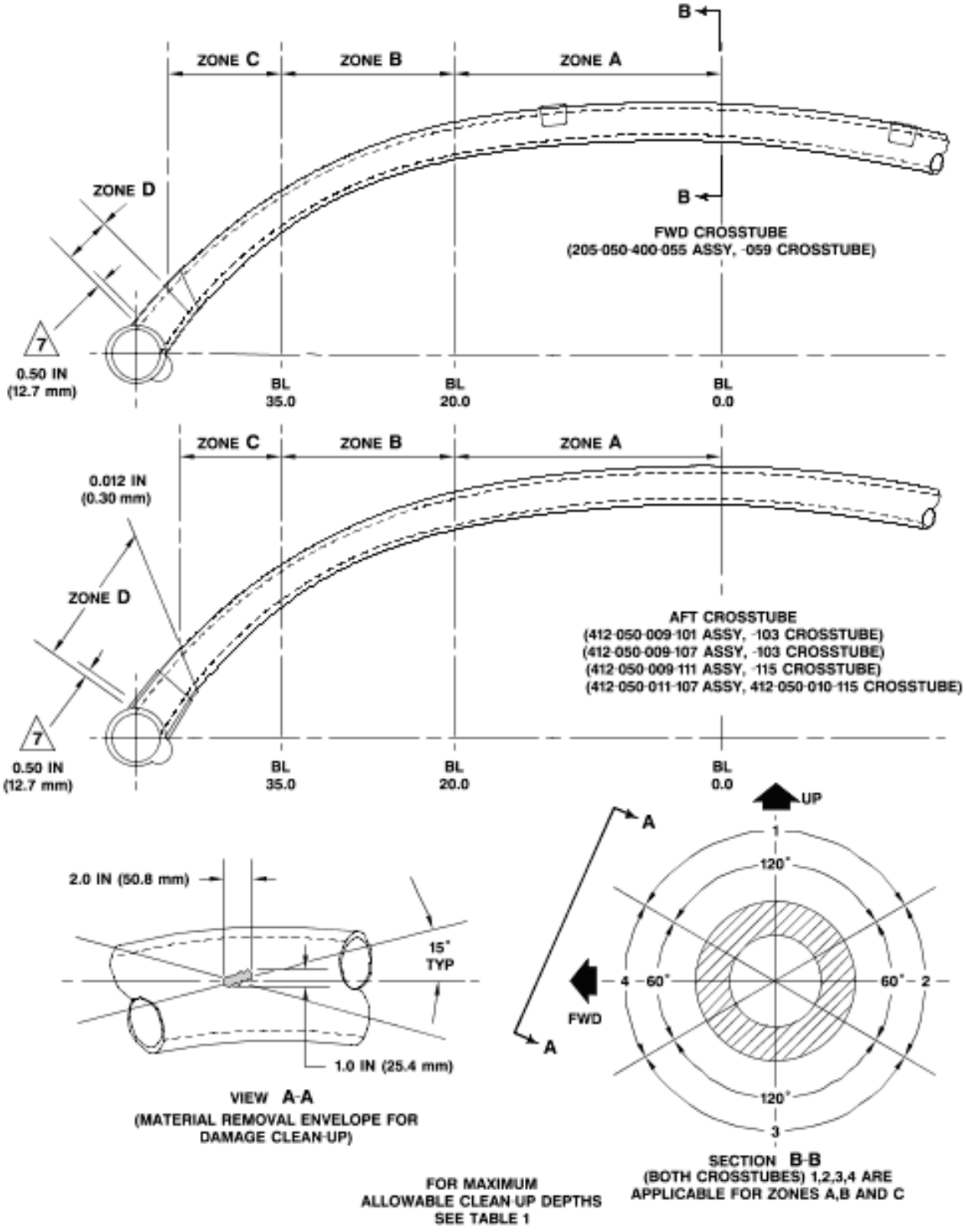
ICN-412-A-320100-B-97499-00006-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 4 of 9)



ICN-412-A-320100-B-97499-00007-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 5 of 9)



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Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 6 of 9)

TABLE 1
MAXIMUM ALLOWABLE CLEANUP DEPTH STANDARD AND HIGH SKID GEAR

ZONE	BHT PRODUCTS FWD CROSSTUBE INCHES (mm) △10	BHT PRODUCTS AFT CROSSTUBE INCHES (mm) △11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.12) 0.015 (0.38) 0.005 (0.12) 0.015 (0.38)
B1 B2 B3 B4	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.010 (0.25) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.012 (0.30) 0.012 (0.30)
D △7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 6 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

△7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 1.

△8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note △9 (Figure 32-2, sheet 5).

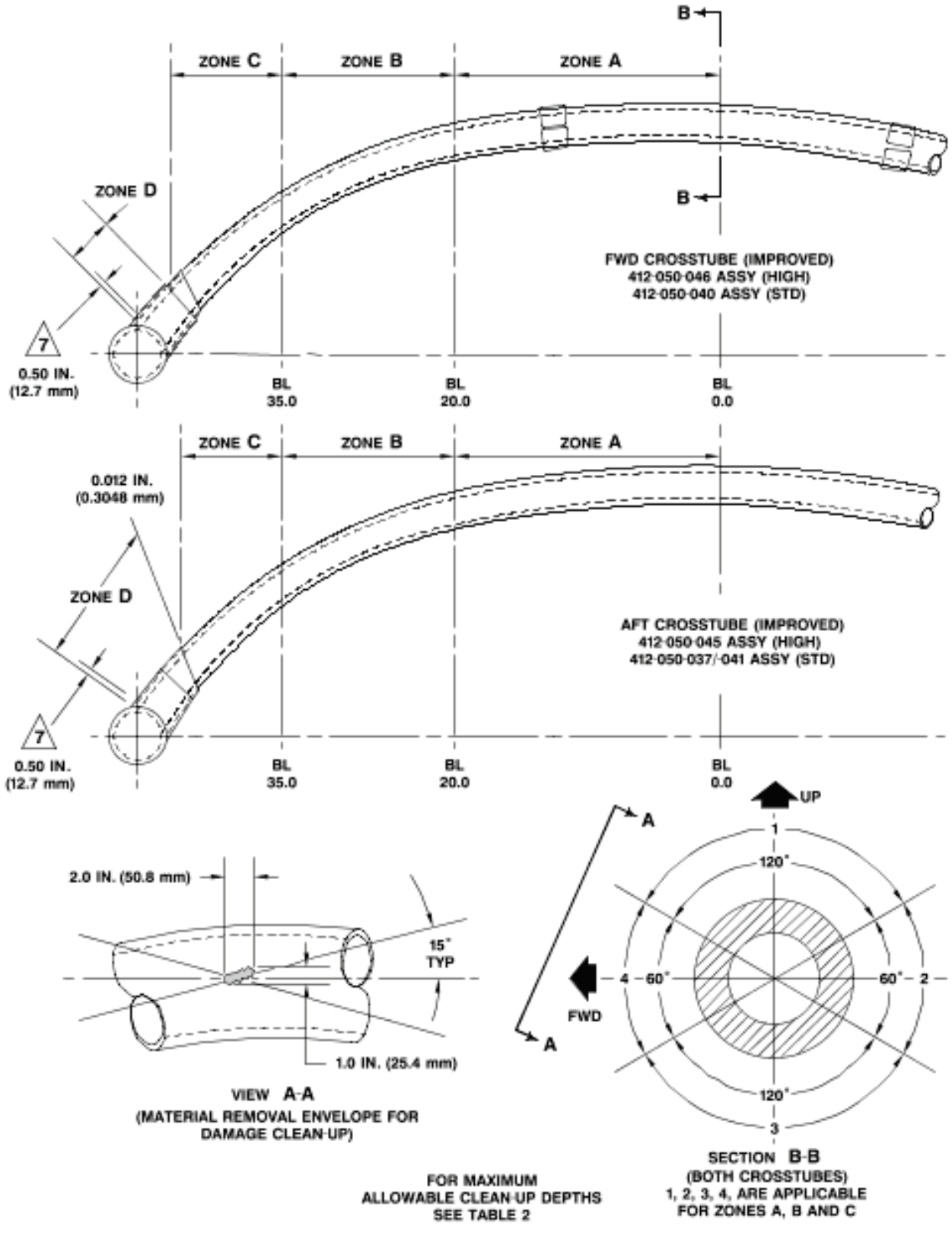
△9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

△10 Forward standard crosstube (BHT 205-050-400-055).
Forward high crosstube (BHT 205-050-403-031).

△11 Aft standard crosstube (BHT 412-050-002-101/103/105/107).
Aft high crosstube (BHT 412-050-011-101/103/105/107).

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Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 7 of 9)



ICN-412-A-320100-B-97499-00010-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 8 of 9)

TABLE 2
MAXIMUM ALLOWABLE CLEANUP DEPTH IMPROVED STANDARD AND IMPROVED HIGH SKID GEAR

ZONE	BHT AND AAI PRODUCTS	BHT AND AAI PRODUCTS
	FWD CROSSTUBE INCHES (mm) 10	AFT CROSSTUBE INCHES (mm) 11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
B1 B2 B3 B4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.005 (0.13) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.005 (0.13) 0.012 (0.30)
D 7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 8 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 2.

8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note 9 (Figure 32-2, sheet 5).

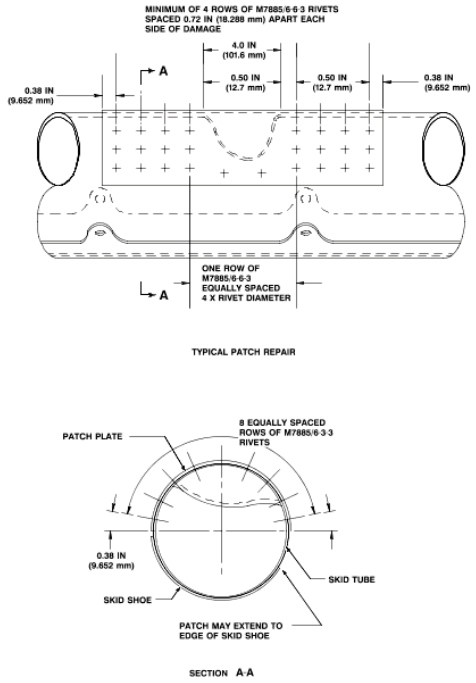
9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

10 Improved forward standard crosstube (BHT 412-050-040-101, AAI 212-320-103 ALL).
Improved forward high crosstube (BHT 412-050-046-101, AAI 212-321-103 ALL).

11 Improved aft standard crosstube (BHT 412-050-041-109, AAI 412-320-104 ALL).
Improved aft high crosstube (BHT 412-050-045-105, AAI 412-321-106 ALL).

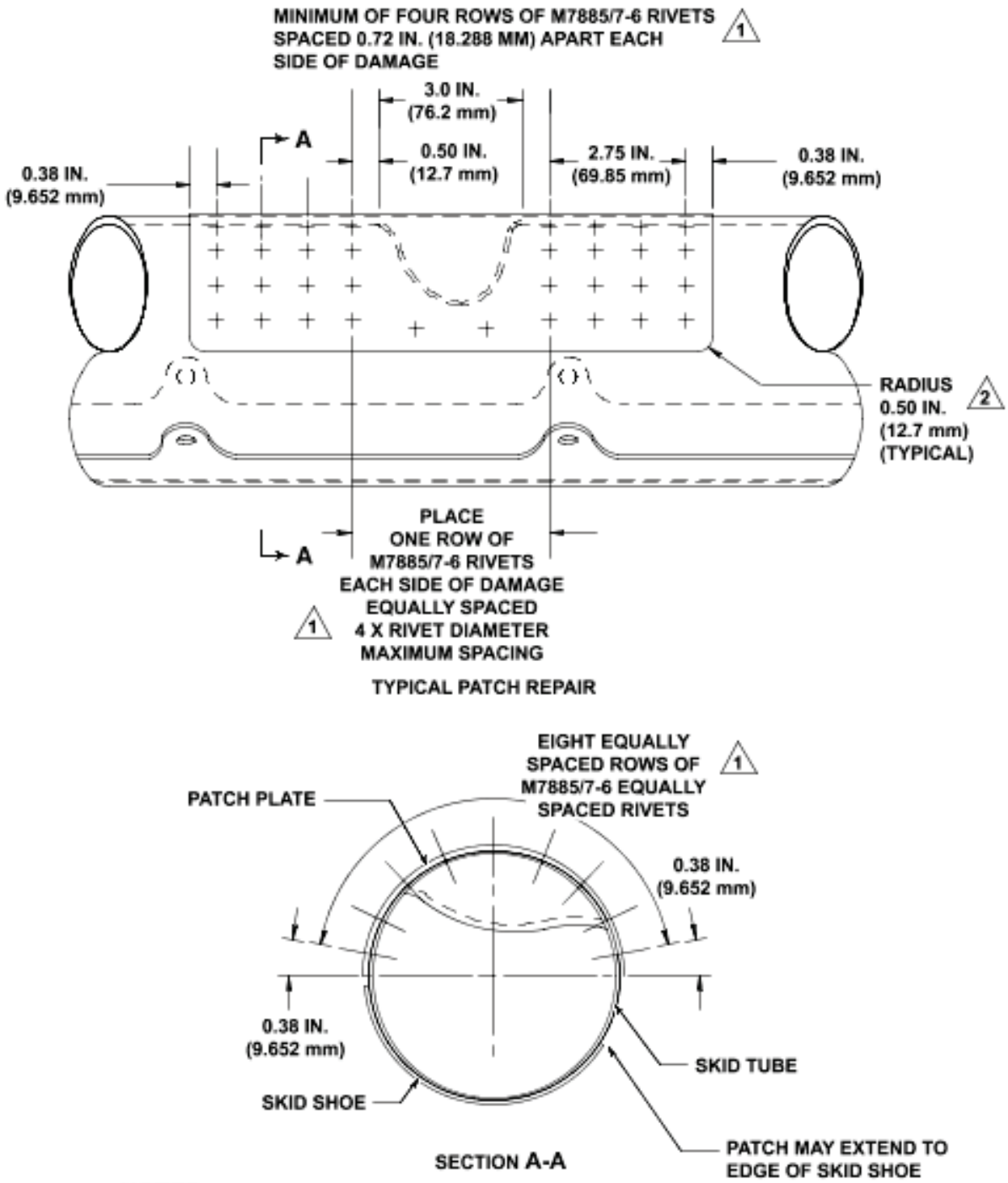
ICN-412-A-320100-B-97499-00011-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 9 of 9)



ICN-412-A-320101-B-97499-00002-A-001-01

Figure 2. Skid Tube (412-050-023) Patch Installation (Sheet 1 of 1)



NOTES

1. Use appropriate grip length of rivets to accommodate doubler and skid tube thickness.
2. Break all sharp edges on doubler.

ICN-412-A-320101-B-97499-00003-A-001-01

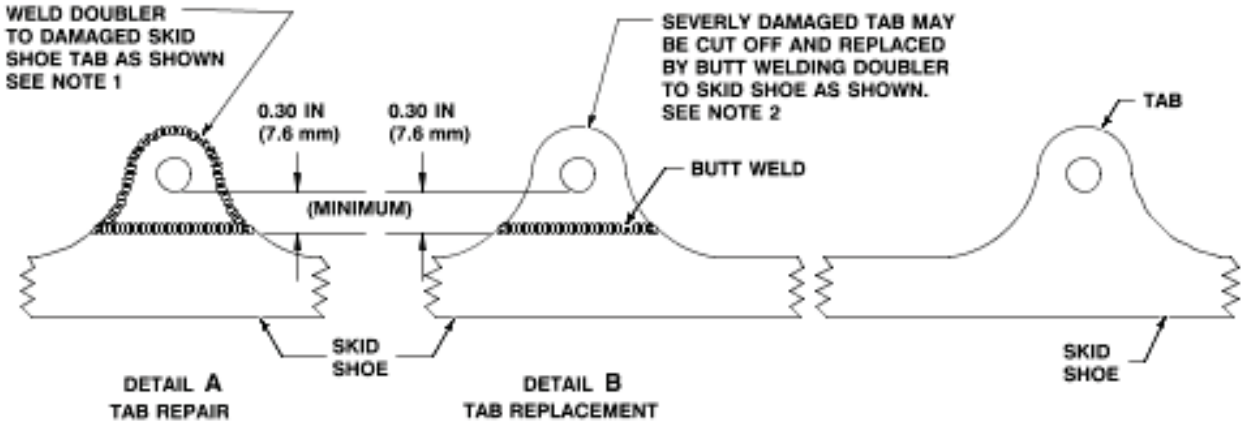
Figure 3. Skid Tube (412-050-030/-034) Patch Installation (Sheet 1 of 1)

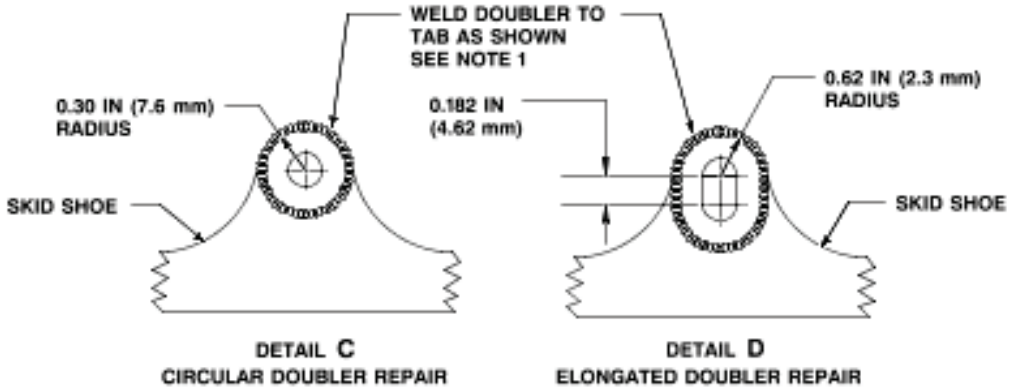
NOTE 1

Fabricate doubler of normalized 4130 steel, MIL-S-18729, 0.063 inch (1.60 mm) thickness and of same configuration as damaged tab.

NOTE 2

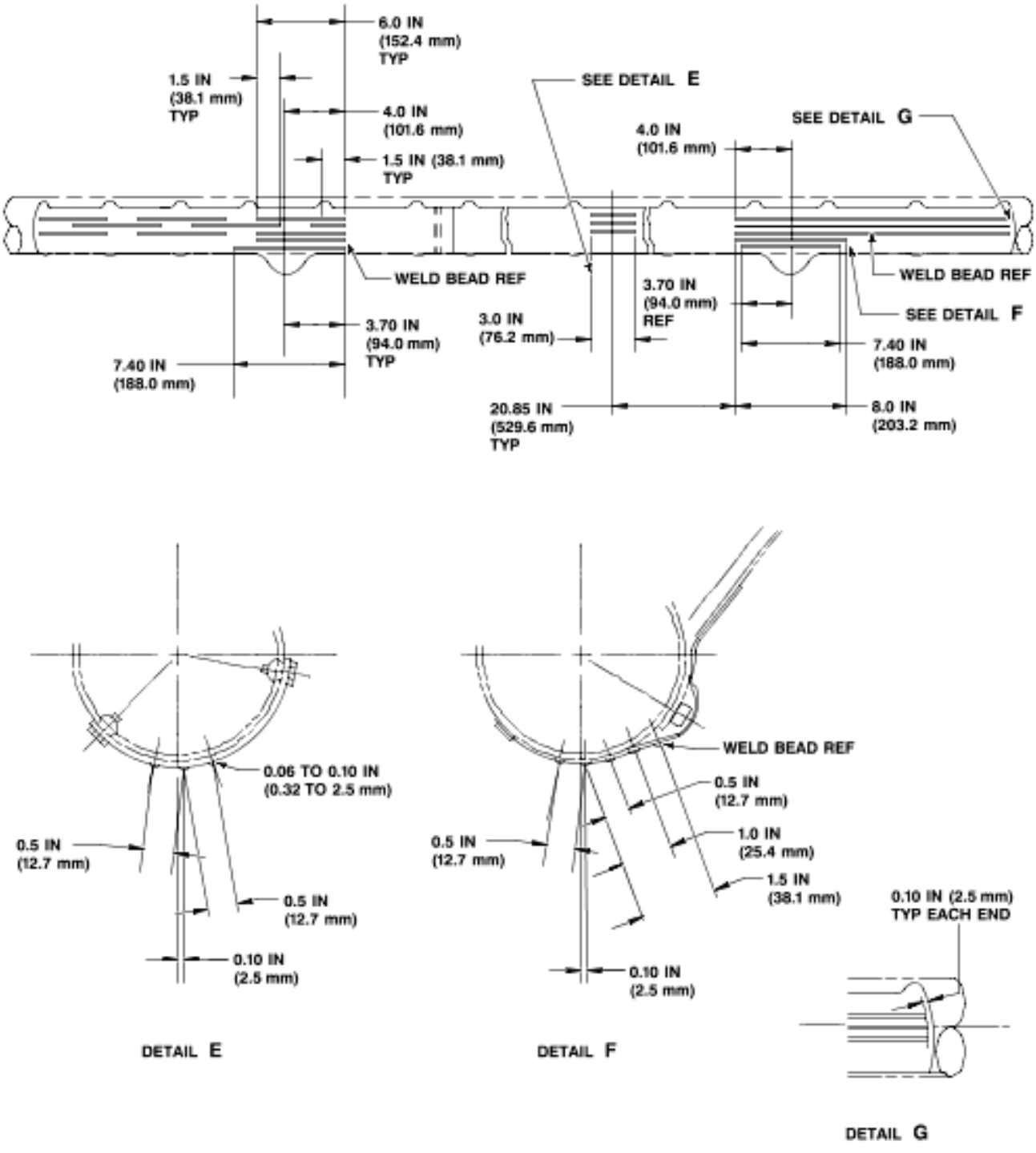
Fabricate doubler of normalized 4130 steel, MIL-S-18720, 0.063 inch (1.60 mm) thickness and of same configuration as removed tab. Grind weld smooth on side adjacent to skid tube.





ICN-412-A-320101-B-97499-00004-A-001-01

Figure 4. Skid Shoe Repair Methods (Sheet 1 of 2)



ICN-412-A-320101-B-97499-00005-A-001-01

Figure 4. Skid Shoe Repair Methods (Sheet 2 of 2)

Skid Tubes - Corrosion Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Chemical film material	C-100	AR	
Epoxy polyamide primer	C-204	AR	
Nonslip compound	C-223	AR	

Procedure

1. Repair slight damage or corrosion as follows:

1.1. Remove scratches, scuffs, nicks, or corrosion by sanding or polishing.

1.2. Apply (brush on) chemical film material (C-100) to bare aluminum of skid tubes and patch.

1.3. Apply one coat of epoxy polyamide primer (C-204) or equivalent, to the affected skid tube area.

1.4. Paint the reworked area to match the rest of the tube and apply two coats of nonslip compound (C-223) to appropriate areas.

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Skid Tubes - Walkway Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Nonslip compound	C-223	AR	
Abrasive cloth or paper	C-406	AR	
Clean cloth	C-516	AR	

Procedure

1. Clean the area to be repaired by lightly sanding with abrasive cloth or paper (C-406) and wipe with a clean cloth (C-516).
2. Mask the area that will not receive nonslip compound (C-223).
3. Apply nonslip compound (C-223) to the required area. Allow a minimum of 30 minutes drying time between coats. The first coat shall be a light coat and the second coat shall be a heavy cross coat.
4. Remove masking paper when process is finished.

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Skid Tubes - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-001	AR	
Sealant	C-308	AR	

Procedure

1. Lubricate ends of crosstubes (3 and 7, Figure 1) with grease (C-001).

2. If electrically powered passenger step (10, Figure 2) is installed, position passenger step electrical connector (12) in crosstube (18).

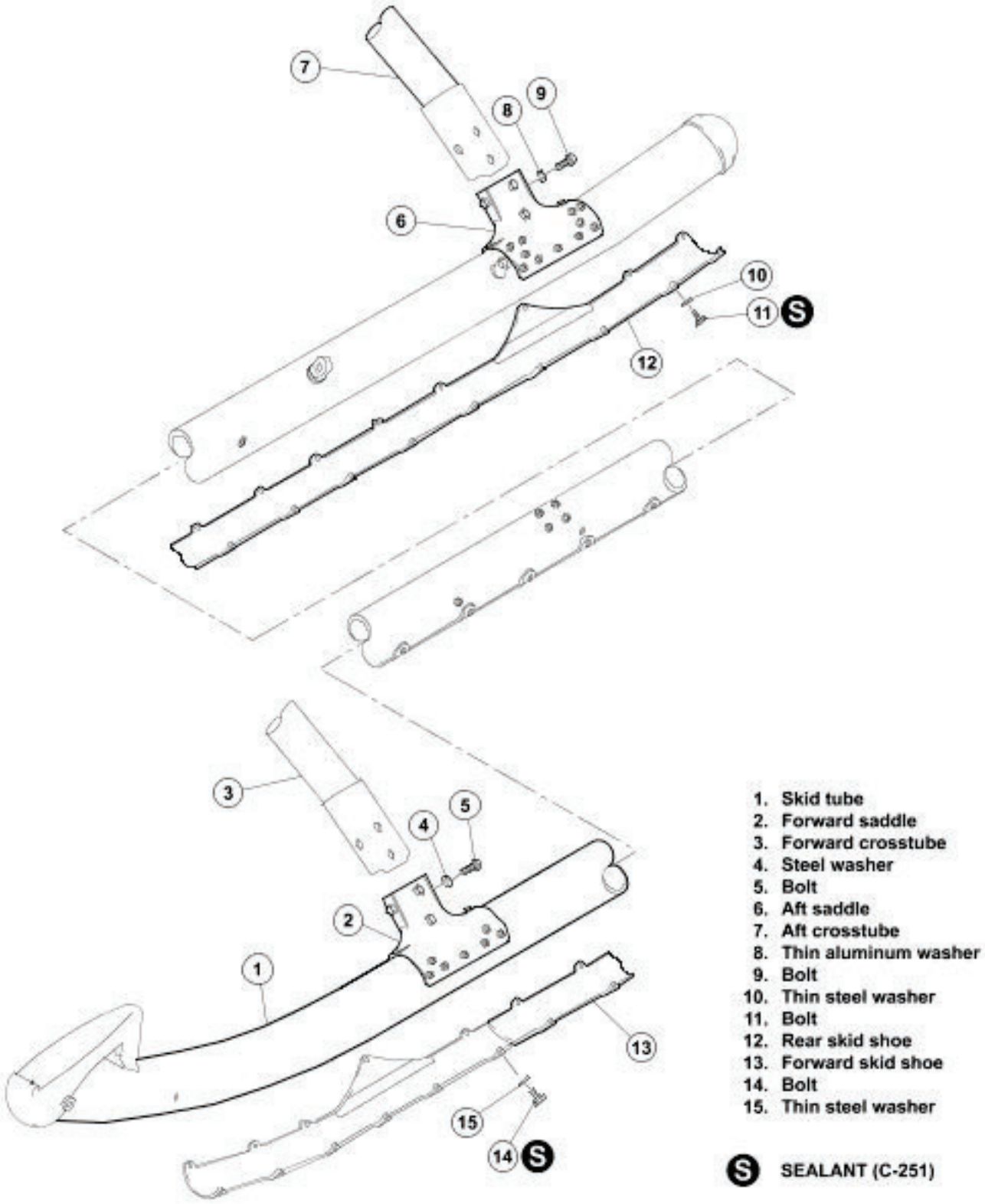
3. Position skid tube (1, Figure 1) on forward and aft crosstubes (3 and 7). Use rubber mallet if necessary. Push skid tube on both crosstubes as evenly as possible and align holes for bolts (5 and 9).

4. Install bolt (5) and steel washer (4).

5. Install bolt (9) and thin aluminum washer (8).

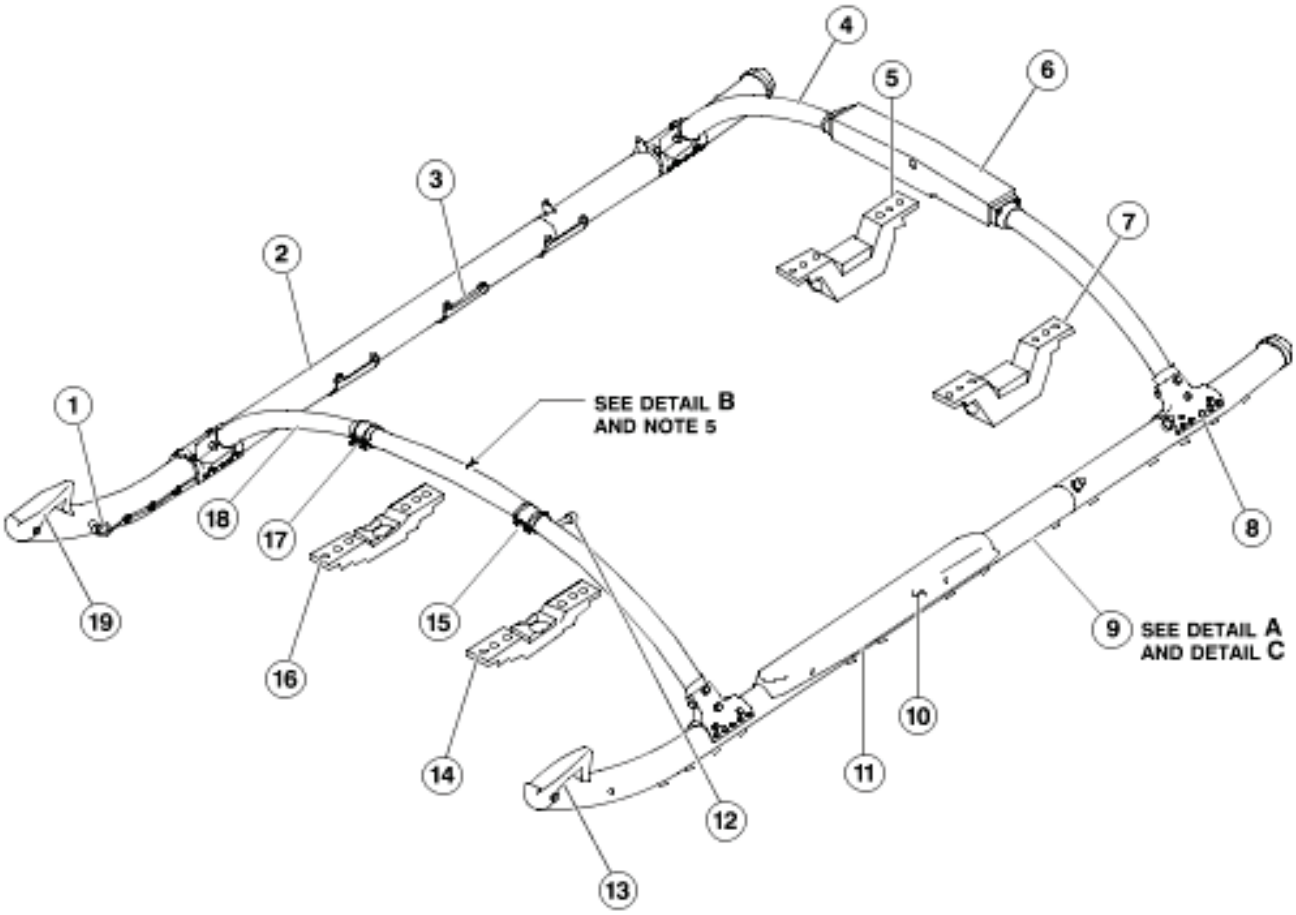
6. Apply sealant (C-308) at intersection of crosstubes (3 and 7) and forward and aft saddles (2 and 6).

7. Install landing gear (412-A-32-01-00-00A-720A-A / 00700).



ICN-412-A-320101-B-97499-00001-A-001-01

Figure 1. Skid Tube and Attaching Parts (Sheet 1 of 1)



STANDARD SKID LANDING GEAR ASSEMBLY
412-050-001-101, -103, -105, -107, -109 and -111

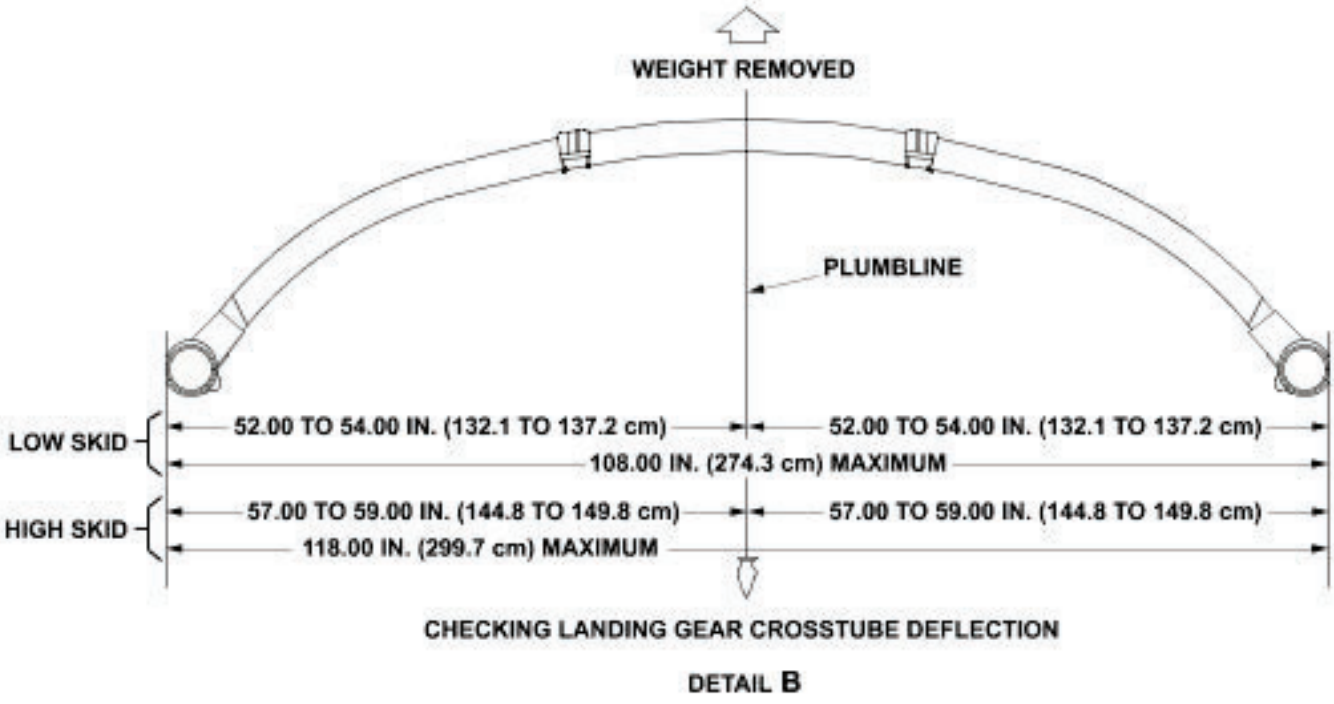
1. Tow ring
2. Skid tube
3. Skid shoe
4. Crosstube
5. Landing gear cap fitting
6. Aft crosstube support beam
7. Landing gear cap fitting
8. Saddle
9. Skid tube
10. Passenger step
11. Skid shoe
12. Passenger step electrical connector
13. Step
14. Landing gear cap fitting
15. Landing gear bearing and retaining support
16. Landing gear cap fitting
17. Landing gear bearing and retaining support
18. Crosstube
19. Step



DETAIL A

ICN-412-A-320100-B-97499-00003-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 1 of 9)

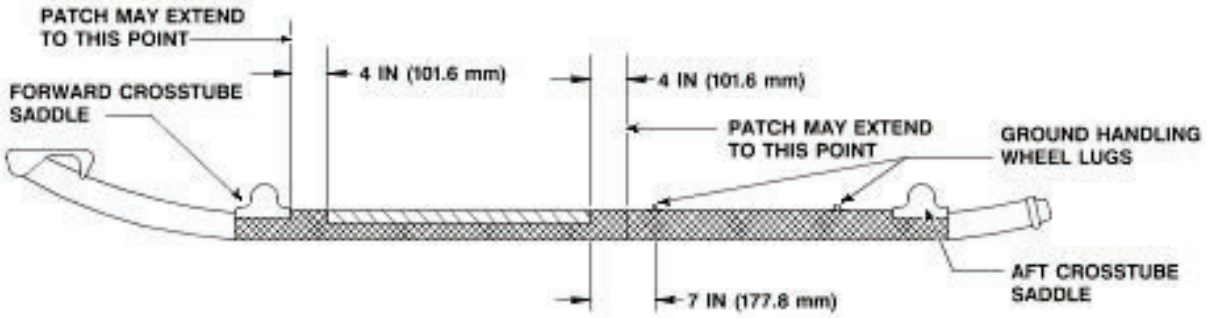


NOTES

1. Deformation (distortion) of either crosstube in excess of limits is cause for replacement of crosstube(s).
2. Crosstube may be deflected on one side only. If limit is exceeded, crosstube is considered scrapped and must be replaced.

ICN-412-A-320100-B-97499-00004-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 2 of 9)



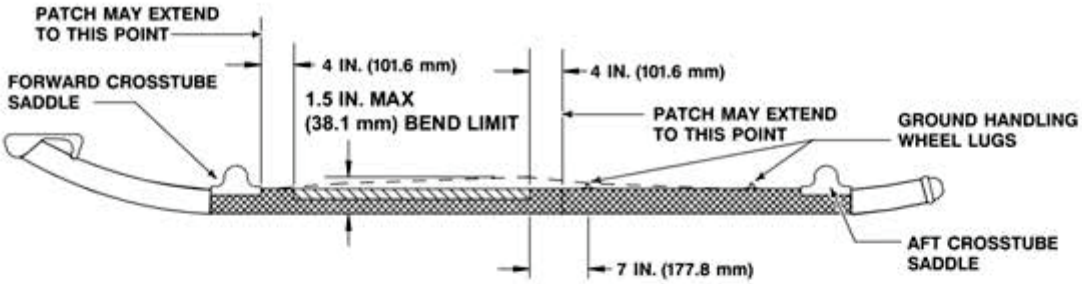
DETAIL C
412-050-023

DAMAGE LOCATION SYMBOLS

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR AREAS ALLOWED		
CRACKS ALLOWED	None	None	None
DENTS AND HOLES	None	4 inches (101.6 mm) Refer to paragraph 32-9.	Not Critical

ICN-412-A-320100-B-97499-00005-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 3 of 9)

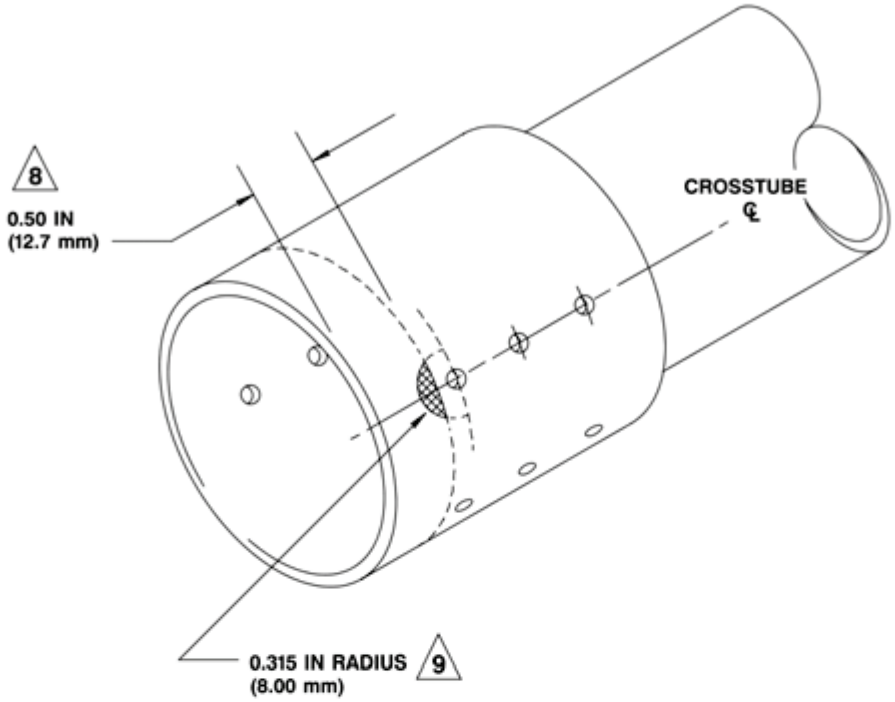


DETAIL C
412-050-030
412-050-034

			(Does not apply to saddles)
DENTS	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No dents allowed. Replace as needed.	Not critical.
HOLES	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No holes allowed. Replace as needed.	Not critical. Plug holes to keep out moisture.
CRACKS	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.
SCRATCHES SCUFFS NICKS	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	Not critical.
	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	
	If damage is greater than established limits, replace as needed.	If damage is greater than established limits, replace as needed.	
CORROSION	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup is limited to 10% of material thickness. Refer to paragraph 32-11.

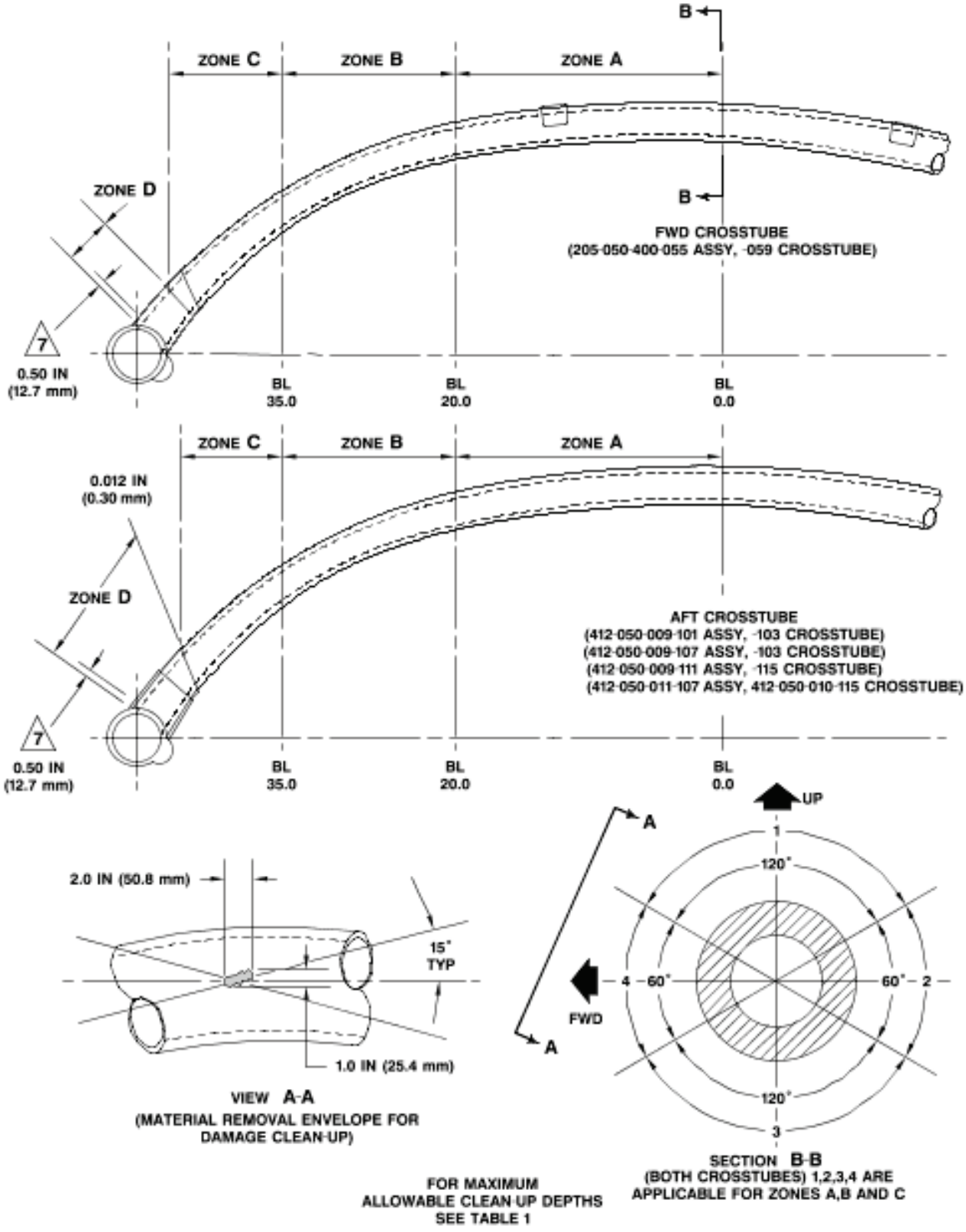
ICN-412-A-320100-B-97499-00006-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 4 of 9)



ICN-412-A-320100-B-97499-00007-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 5 of 9)



ICN-412-A-320100-B-97499-00008-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 6 of 9)

TABLE 1
MAXIMUM ALLOWABLE CLEANUP DEPTH STANDARD AND HIGH SKID GEAR

ZONE	BHT PRODUCTS FWD CROSSTUBE INCHES (mm) △10	BHT PRODUCTS AFT CROSSTUBE INCHES (mm) △11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.12) 0.015 (0.38) 0.005 (0.12) 0.015 (0.38)
B1 B2 B3 B4	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.010 (0.25) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.012 (0.30) 0.012 (0.30)
D △7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 6 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

△7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 1.

△8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note △9 (Figure 32-2, sheet 5).

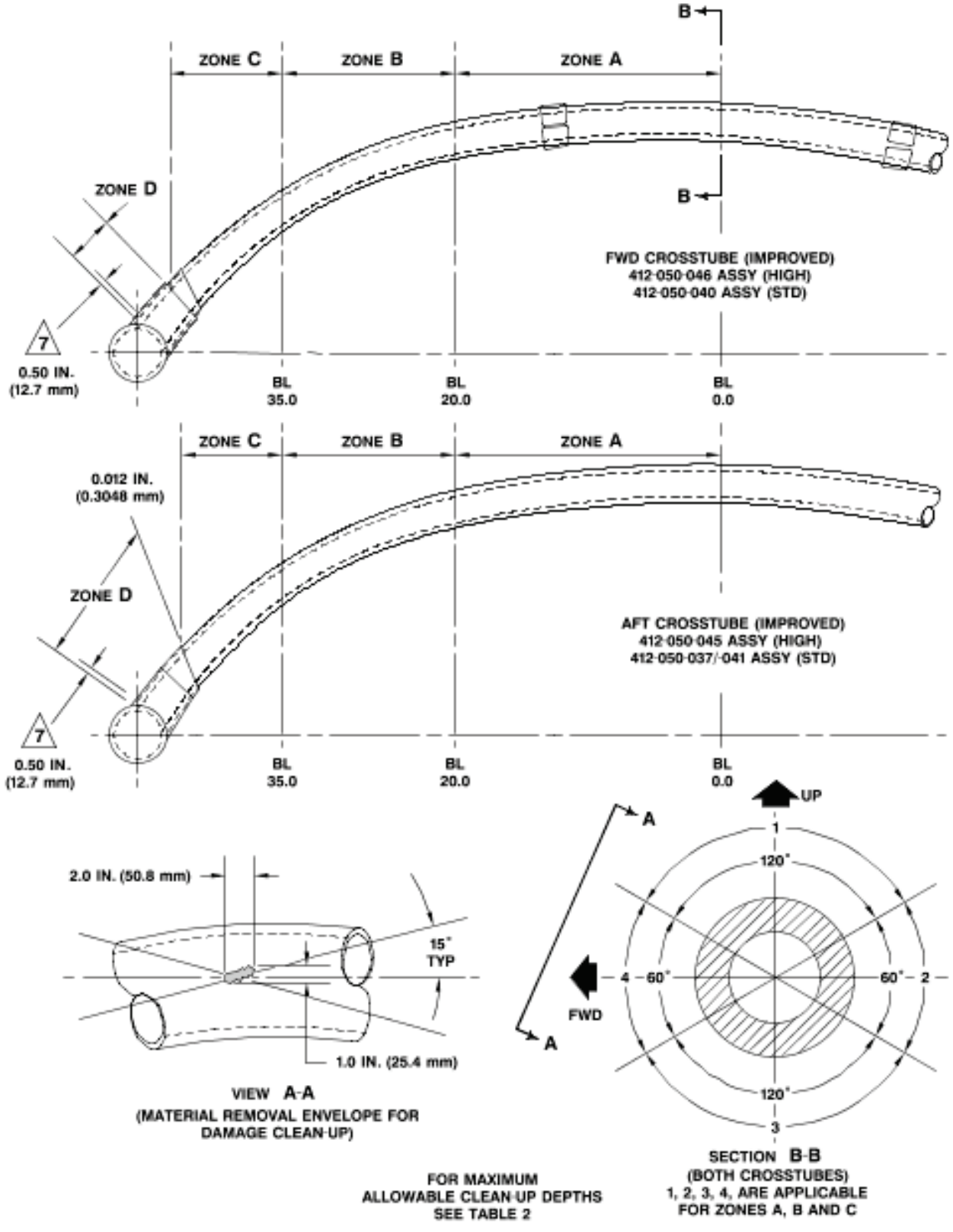
△9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

△10 Forward standard crosstube (BHT 205-050-400-055).
Forward high crosstube (BHT 205-050-403-031).

△11 Aft standard crosstube (BHT 412-050-002-101/103/105/107).
Aft high crosstube (BHT 412-050-011-101/103/105/107).

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Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 7 of 9)



ICN-412-A-320100-B-97499-00010-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 8 of 9)

TABLE 2
MAXIMUM ALLOWABLE CLEANUP DEPTH IMPROVED STANDARD AND IMPROVED HIGH SKID GEAR

ZONE	BHT AND AAI PRODUCTS	BHT AND AAI PRODUCTS
	FWD CROSSTUBE INCHES (mm) △10	AFT CROSSTUBE INCHES (mm) △11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
B1 B2 B3 B4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.005 (0.13) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.005 (0.13) 0.012 (0.30)
D △7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 8 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

△7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 2.

△8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note △9 (Figure 32-2, sheet 5).

△9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

△10 Improved forward standard crosstube (BHT 412-050-040-101, AAI 212-320-103 ALL).
Improved forward high crosstube (BHT 412-050-046-101, AAI 212-321-103 ALL).

△11 Improved aft standard crosstube (BHT 412-050-041-109, AAI 412-320-104 ALL).
Improved aft high crosstube (BHT 412-050-045-105, AAI 412-321-106 ALL).

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Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 9 of 9)

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Skid Shoes - Description

NOTE

Distortion damage to skid shoes is acceptable if repaired by reforming, welding on doublers, and/or welding on new tabs as described in 412-A-32-01-02-00A-685A-A / 00711.

NOTE

Maintenance procedures for skid shoes for both high gear and standard gear are the same.

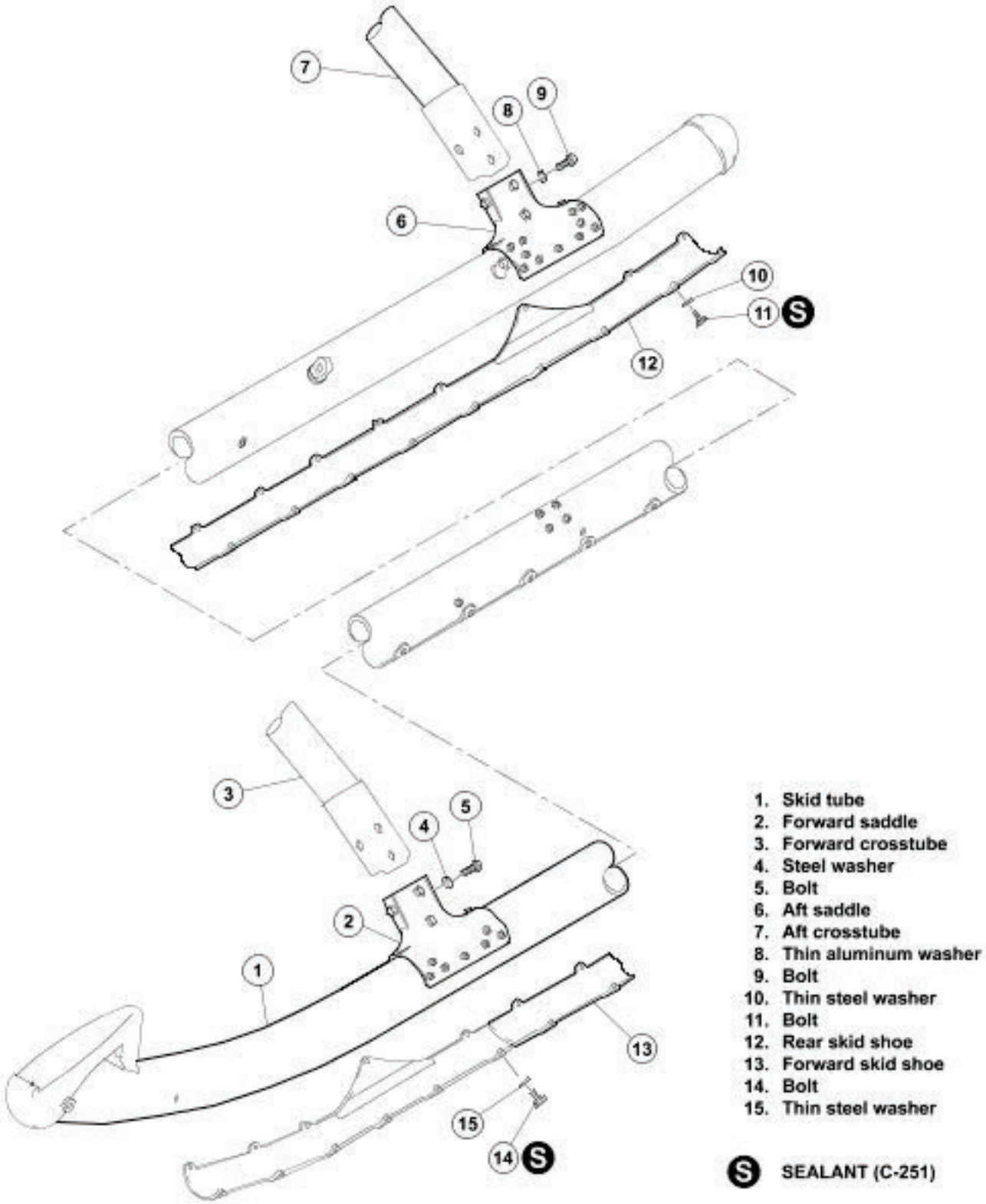
Maintenance procedures for left and right skid shoes are the same.

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Skid Shoes - Removal

Procedure

1. Support helicopter with hoist, stands, or ground handling gear.
2. Remove bolts (14, Figure 1). Remove forward skid shoe (13).
3. Remove bolts (11) and remove rear skid shoe (12).



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Figure 1. Skid Tube and Attaching Parts (Sheet 1 of 1)

Skid Shoes - Inspection

Procedure

1. Inspect for deformation, cracks, broken tang, and elongated holes.
2. Inspect for missing primer and cadmium plating. Restore as required.
3. Inspect for excessive corrosion.
4. Inspect beads for condition and excessive wear.

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Skid Shoes - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Cadmium plating solution	C-108	AR	
Epoxy polyamide primer	C-204	AR	
Polyurethane coating	C-245	AR	
Ammonium nitrate	C-350	AR	

Procedure



PRIOR TO ANY WELDING REPAIR, REMOVE THE SKID SHOE FROM SKID TUBE TO AVOID DAMAGE TO THE ALUMINUM SKID TUBE FROM EXCESSIVE HEAT.

1. Heat skid shoe and reshape to remove dents that are not closer than 2 inches (50.8 mm) from borium weld beads.

2. Repair skid shoe tabs with holes that are elongated in excess of 0.030 inch (0.76 mm) as follows:

2.1. Fabricate a circular or elongated doubler, as shown in Figure 1, Detail C and Detail D. Use 4130 steel 0.050 inch (1.27 mm) thick that has been normalized per MIL-S-18729.

NOTE

Cadmium plated stripped parts are very susceptible to corrosion and shall immediately be repaired and primed.

2.2. Remove cadmium plating from repair area on skid shoes prior to welding.

2.2.1. Mix a solution of 16 ounces (473 ml) of ammonium nitrate (C-350) per gallon of water.

2.2.2. Maintain solution at a temperature of 120°F (49°C). Immerse section or complete skid shoes into solution until all cadmium plating has been removed. Rinse skid shoes in clean water and allow to air dry.

2.3. Weld doubler to skid shoe as shown in Detail C and Detail D.

3. Repair skid shoe tabs that are damaged in excess of limits noted in Step 2. as follows:

3.1. Heat and reshape tab to original configuration if possible. After tab is reshaped, fabricate a doubler from 4130 steel 0.063 inch (1.60 mm) thick that has been normalized per MIL-S-18729 and weld in place, as shown in Detail E.

3.2. If tab cannot be reshaped to original configuration, cut tab off parallel to skid shoe. Fabricate a new tab and weld in place, as shown in Detail F.

4. Check bottom of skid shoes for evidence of weld bead. Do not use skid shoes that have the weld beads worn off (Detail F).

NOTE

Transverse cracks across weld beads are permissible and should not be cause for repair or replacement of skid shoes.

5. Reform bead as follows:

5.1. Remove cadmium plating from skid shoes per Step 2.2..

5.2. Weld beads 0.06 to 0.10 inch (1.5 to 2.5 mm) high along the skid shoe using acetylene and a 0.125 inch (3.18 mm) hard facing weld tube (borium) (material available from Stoody Manufacturing Co., Whittier, CA 90608 or DT600 HM, 0.125 inch (3.175 mm) hard facing weld tube from Reed Tool Co., Houston, TX 77011).

5.3. Hot reform the skid shoe, as required, to fit the contour of the skid tube.

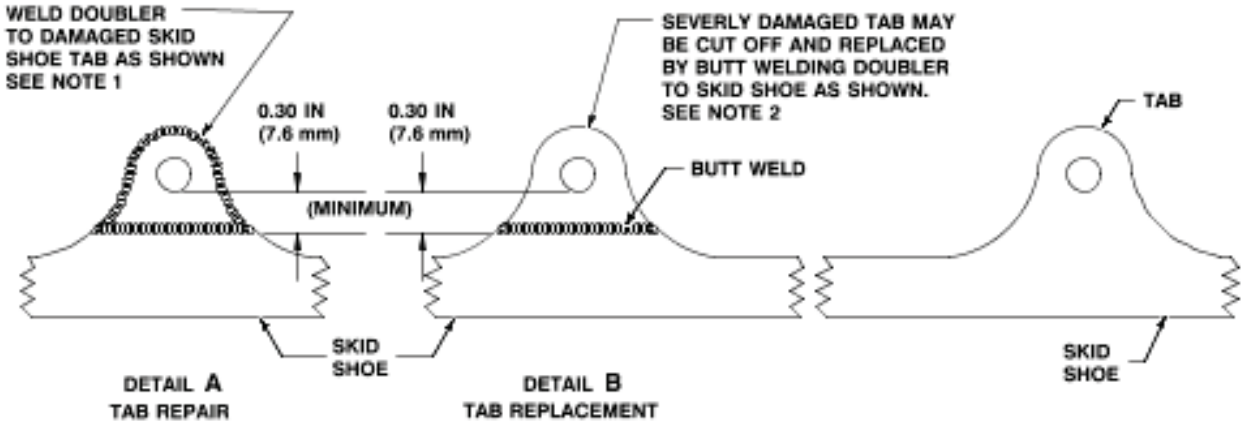
6. After repair and clean-up of the weld deposits, apply cadmium plating solution (C-108) per MIL-STD-865 or two coats of epoxy polyamide primer (C-204). When dry, apply two or three coats of polyurethane coating (C-245) to match the original finish.

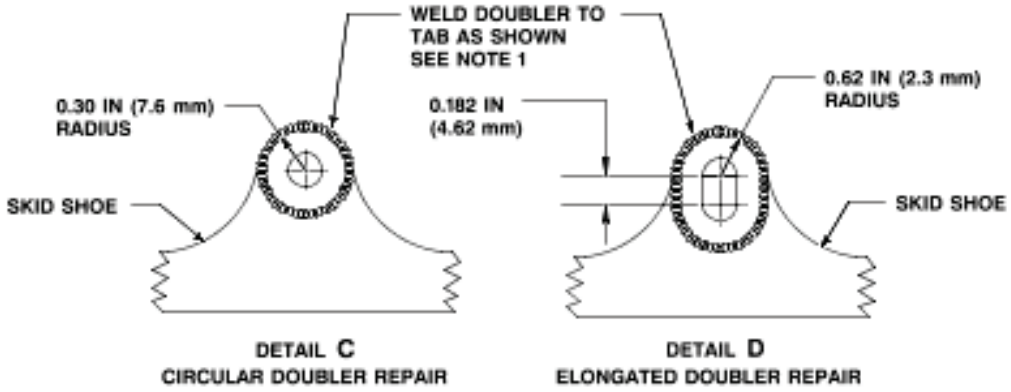
NOTE 1

Fabricate doubler of normalized 4130 steel, MIL-S-18729, 0.063 inch (1.60 mm) thickness and of same configuration as damaged tab.

NOTE 2

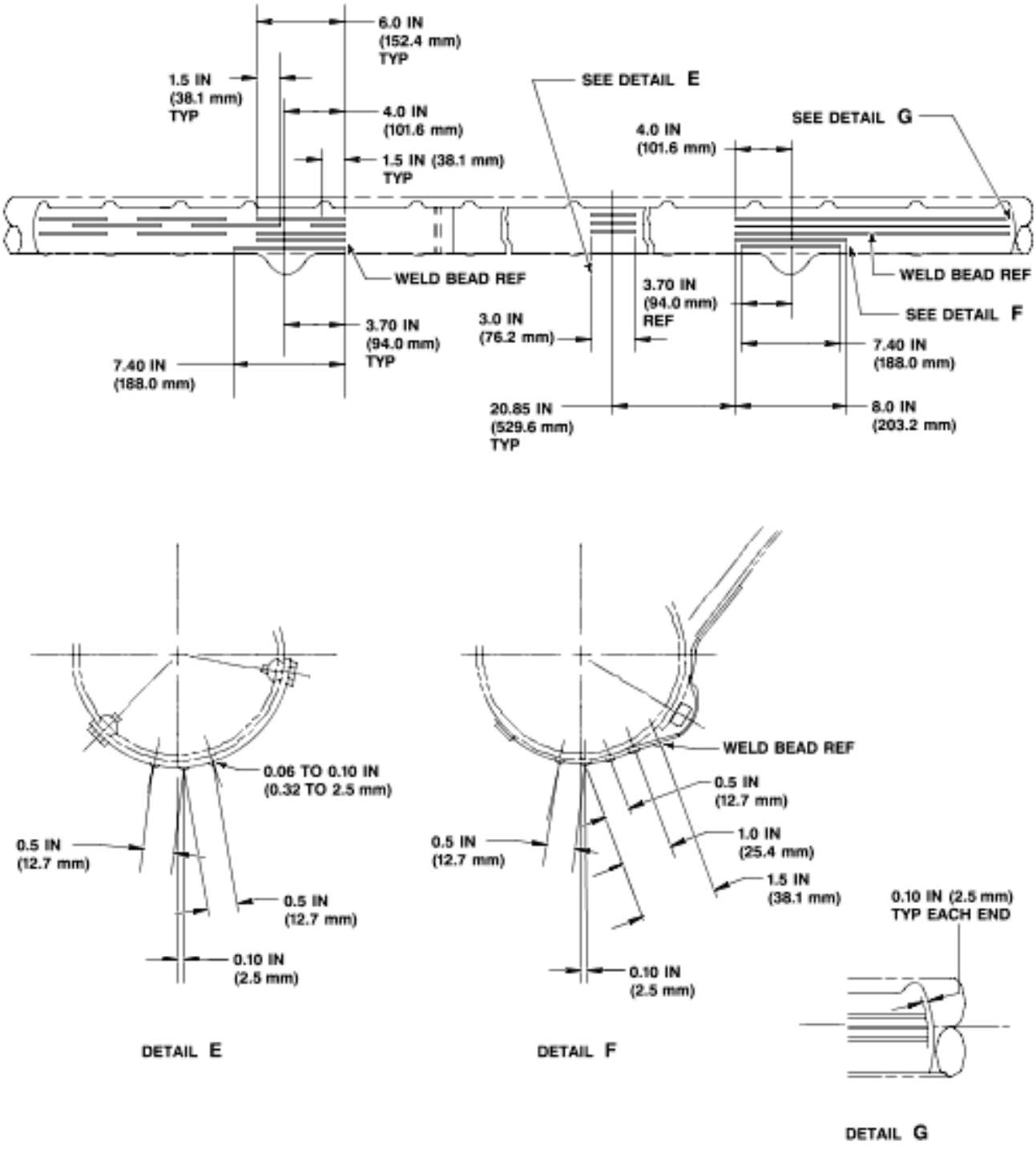
Fabricate doubler of normalized 4130 steel, MIL-S-18720, 0.063 inch (1.60 mm) thickness and of same configuration as removed tab. Grind weld smooth on side adjacent to skid tube.





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Figure 1. Skid Shoe Repair Methods (Sheet 1 of 2)



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Figure 1. Skid Shoe Repair Methods (Sheet 2 of 2)

Skid Shoes - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-251	AR	
Barrier tape	C-456	AR	

Procedure

1. Apply barrier tape (C-456) to the entire faying surface of the rear and forward skid shoes (12 and 13, Figure 1). Overlap the tape 0.25 inch (6.35 mm). Trim the tape around the periphery of the skid shoe.

2. Support the helicopter with the hoist or with the ground handling gear.

3. Position the forward skid shoe (13) on the skid tube (1).

4. Install the bolts (14) and the thin steel washers (15).

5. Apply sealant (C-251) around the bolt (14) heads.

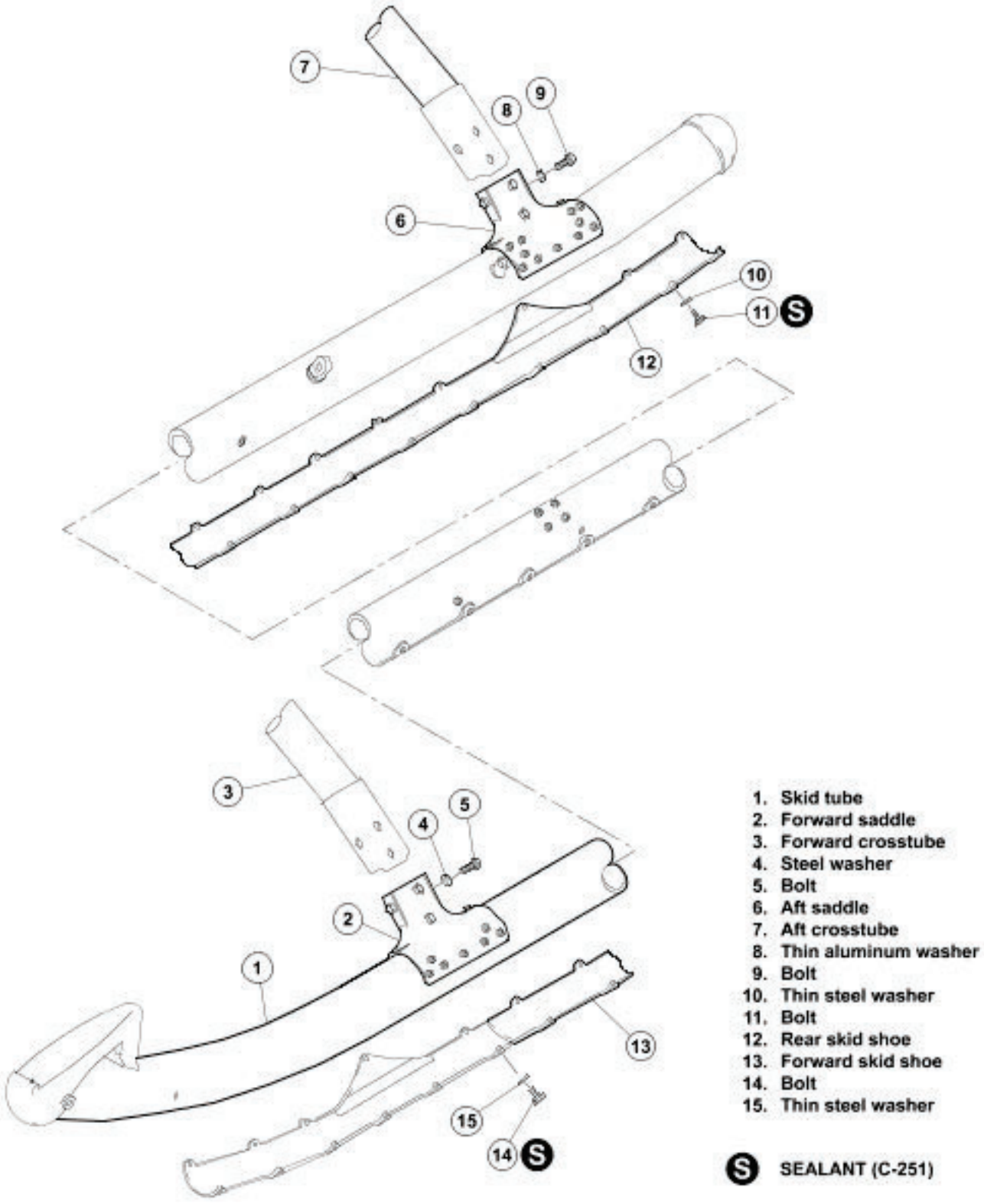
6. Position the rear skid shoe (12) on the skid tube (1).

7. Install the bolts (11) and the thin steel washers (10).

8. Apply sealant (C-251) around the bolt (11) heads.

9. Apply sealant (C-251) to the faying surface of the skid tube (1) and rear and forward skid shoes (12 and 13).

10. Lower the helicopter.



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Figure 1. Skid Tube and Attaching Parts (Sheet 1 of 1)

Fixed Step - Description

NOTE

Maintenance procedures for the fixed step on the high gear and standard gear are the same.

Maintenance procedures for the left and right steps are the same.

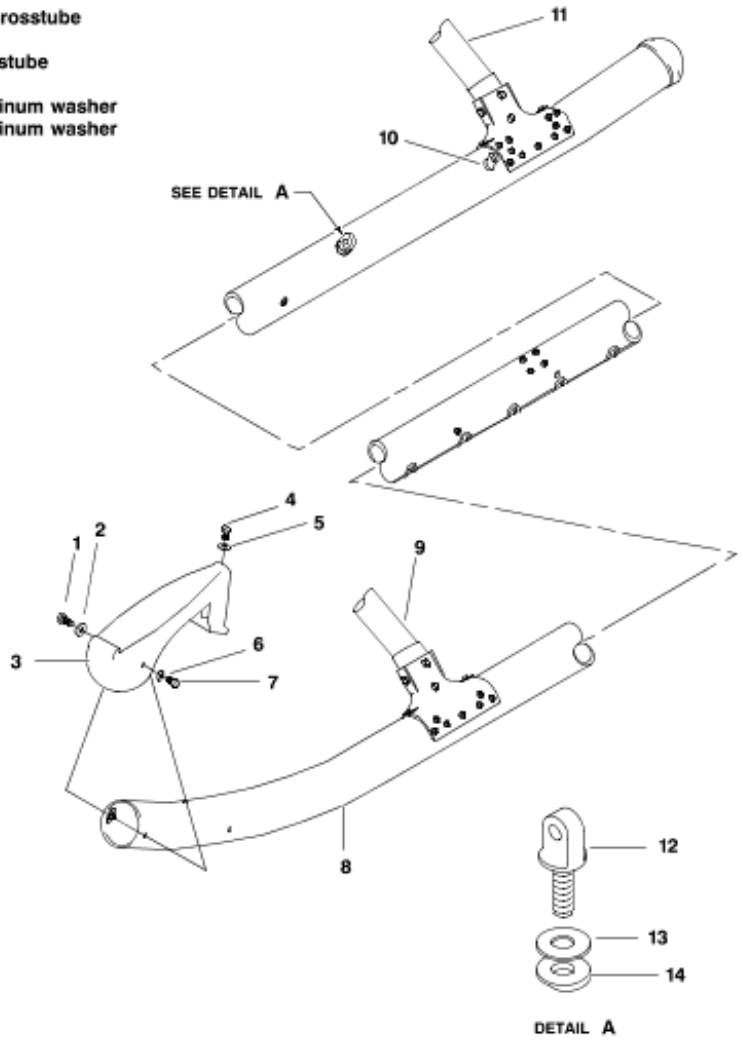
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Fixed Step - Removal

Procedure

1. Remove the bolts (1, 4, and 7, Figure 1) and the thin aluminum washers (2, 5, and 6).
2. Cut the sealant at the intersection of the step (3) and skid tube (8) with a sharp plastic scraper.
3. Remove the step (3) from the skid tube (8).

1. Bolt
2. Thin aluminum washer
3. Step
4. Bolt
5. Thin aluminum washer
6. Thin aluminum washer
7. Bolt
8. Skid tube
9. Forward crosstube
10. Eyebolt
11. Rear crosstube
12. Eyebolt
13. Thin aluminum washer
14. Thin aluminum washer



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Figure 1. Skid Tube Fixed Step and Eyebolt (Sheet 1 of 1)

Fixed Step - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	

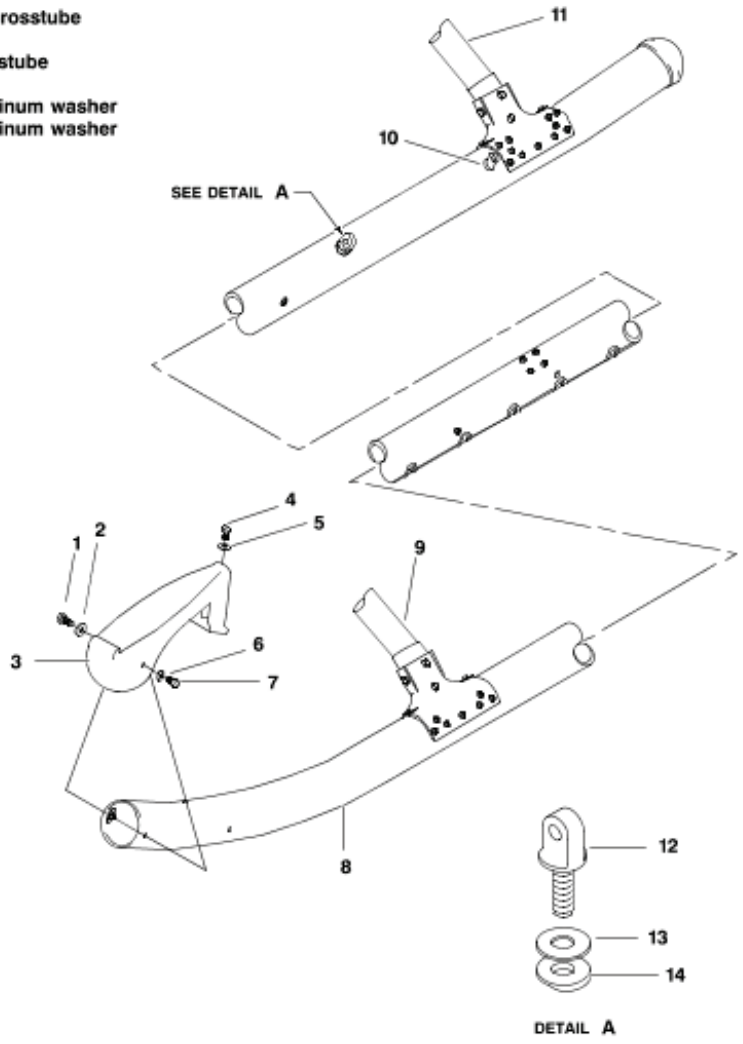
Procedure

1. Position the step (3, Figure 1) on the skid tube (8) for installation.

1.1. Install the bolts (1, 4, and 7), and the thin aluminum washers (2, 5, and 6).

1.2. Apply a bead of sealant (C-308) to the intersection of the step (3) and the skid tube (8).

1. Bolt
2. Thin aluminum washer
3. Step
4. Bolt
5. Thin aluminum washer
6. Thin aluminum washer
7. Bolt
8. Skid tube
9. Forward crosstube
10. Eyebolt
11. Rear crosstube
12. Eyebolt
13. Thin aluminum washer
14. Thin aluminum washer



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Figure 1. Skid Tube Fixed Step and Eyebolt (Sheet 1 of 1)

Eyebolt (Ground Handling Wheels Attach Points) - Description

NOTE

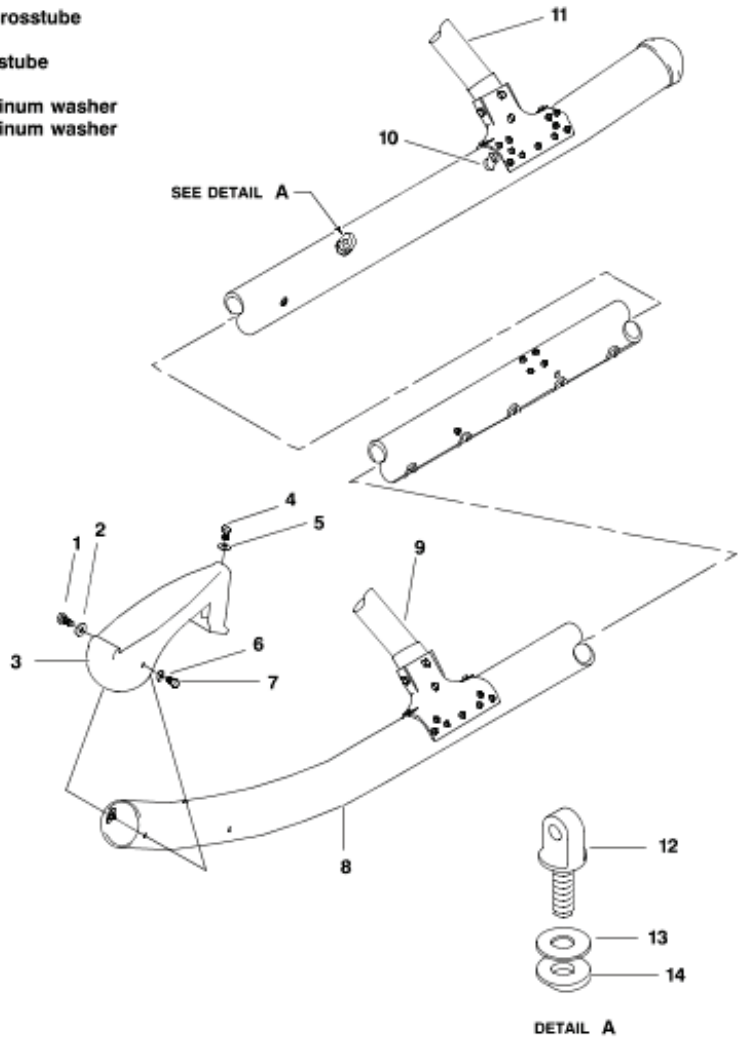
Maintenance procedures for eyebolt are the same for high gear and standard gear.

Maintenance procedures for eyebolt (12, Figure 1) are given. Procedures for eyebolt (10) are the same.

NOTE

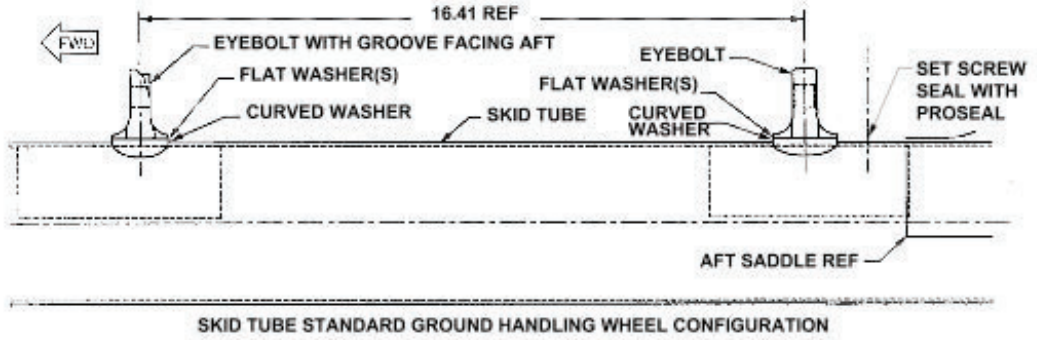
The Aeronautical Accessories Inc (AAI) replacement skid tubes come equipped to accept standard ground handling wheels. When using ground handling wheels 204-050-200-005/-015, the skid gear ground handling eyebolts need to be relocated as shown in Figure 2.

1. Bolt
2. Thin aluminum washer
3. Step
4. Bolt
5. Thin aluminum washer
6. Thin aluminum washer
7. Bolt
8. Skid tube
9. Forward crosstube
10. Eyebolt
11. Rear crosstube
12. Eyebolt
13. Thin aluminum washer
14. Thin aluminum washer

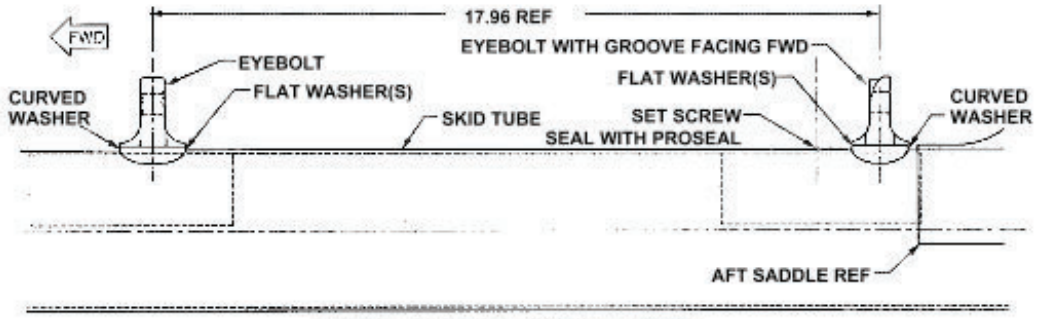


ICN-412-A-320103-B-97499-00001-A-001-01

Figure 1. Skid Tube Fixed Step and Eyebolt (Sheet 1 of 1)



SKID TUBE STANDARD GROUND HANDLING WHEEL CONFIGURATION



ALTERNATE GROUND HANDLING WHEELS
P/N 204-050-200-005/-115

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Figure 2. Skid Tube Ground Handling Configurations (Sheet 1 of 1)

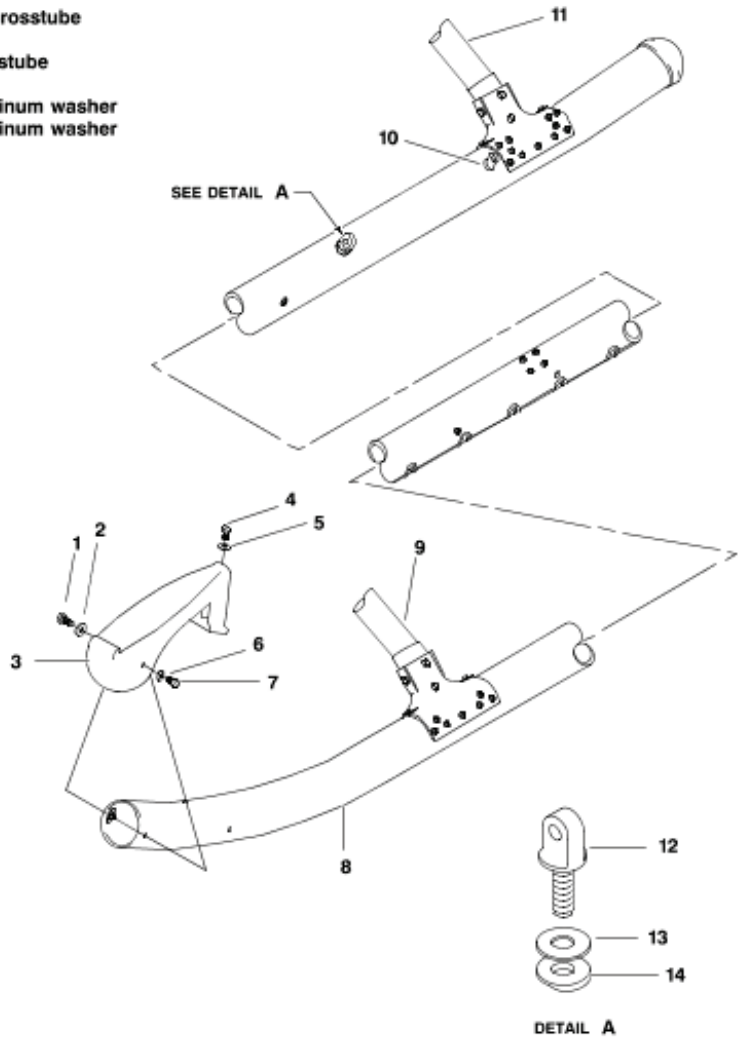
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Eyebolt (Ground Handling Wheels Attach Points) - Removal

Procedure

1. Turn eyebolt (12, Figure 1) counterclockwise to disengage threads.
2. Remove eyebolt (12) and thin aluminum washers (13 and 14).

1. Bolt
2. Thin aluminum washer
3. Step
4. Bolt
5. Thin aluminum washer
6. Thin aluminum washer
7. Bolt
8. Skid tube
9. Forward crosstube
10. Eyebolt
11. Rear crosstube
12. Eyebolt
13. Thin aluminum washer
14. Thin aluminum washer



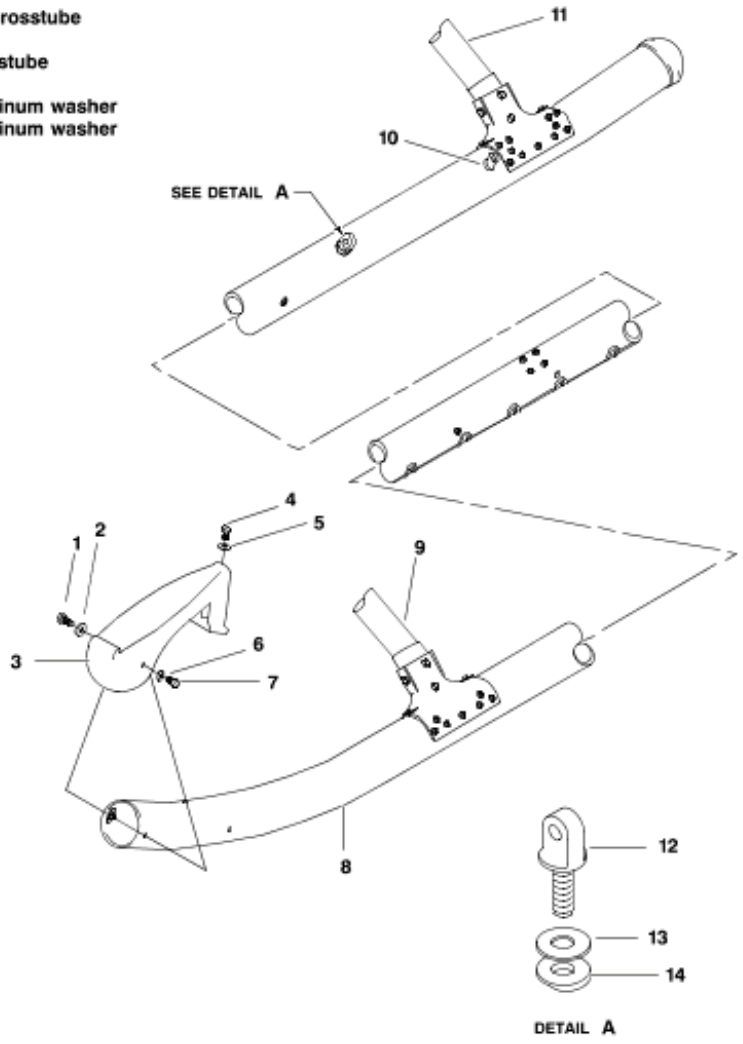
ICN-412-A-320103-B-97499-00001-A-001-01

Figure 1. Skid Tube Fixed Step and Eyebolt (Sheet 1 of 1)

Eyebolt (Ground Handling Wheels Attach Points) - Inspection and Repair Procedure

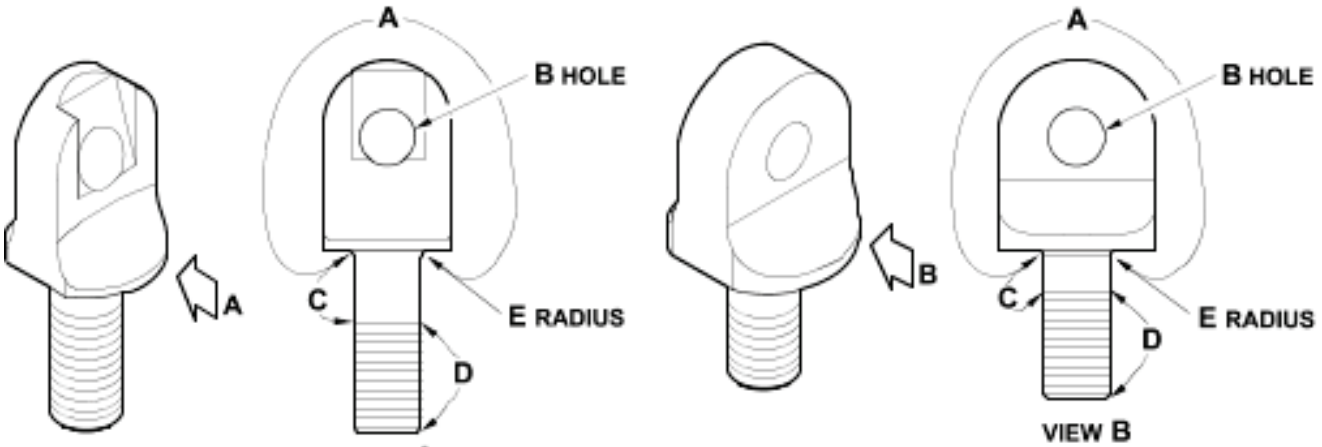
1. Inspect eyebolts (10 and 12, Figure 1) for mechanical and corrosion damage (Figure 2).

1. Bolt
2. Thin aluminum washer
3. Step
4. Bolt
5. Thin aluminum washer
6. Thin aluminum washer
7. Bolt
8. Skid tube
9. Forward crosstube
10. Eyebolt
11. Rear crosstube
12. Eyebolt
13. Thin aluminum washer
14. Thin aluminum washer



ICN-412-A-320103-B-97499-00001-A-001-01

Figure 1. Skid Tube Fixed Step and Eyebolt (Sheet 1 of 1)



NO.	REF LTR	CHARACTERISTIC	INSPECTION METHOD	LIMIT
1.	A	Mechanical damage	Visual/ measure	0.010 inch (0.25 mm) maximum depth. 1 2
2.	B	Mechanical damage	Visual/ measure	0.005 inch (0.13 mm) maximum depth and 90° maximum length. 2 3
3.	C	Mechanical damage	Visual/ measure	0.005 inch (0.13 mm) maximum depth and 120° maximum length. 2 3
4.	C	Bending	Visual	None permitted.
5.	D	Thread damage	Visual	None permitted. 4
6.	D	Bending	Visual	None permitted.
7.	E	Mechanical damage	Visual	None permitted.

NOTES

1. Maximum of four repairs per area.
2. To repair the damage, blend smooth and touchup repair the cadmium plating (BHT-ALL-SPM, Chapter 3).
3. Maximum of one repair per area.
4. If required, touchup repair the cadmium plating (BHT-ALL-SPM, Chapter 3).
5. All dimensions, wear and damage limits provided are limits required after repair except where indicated.

ICN-412-A-320104-B-97499-00002-A-001-01

Figure 2. Eyebolt - Damage Limits (Sheet 1 of 1)

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Eyebolt (Ground Handling Wheels Attach Points) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	
Sealant	C-251	AR	

Procedure

1. Position eyebolt (12, Figure 1) and thin aluminum washers (13 and 14) on skid tube (8).

2. Install eyebolt (12) by turning clockwise to engage threads in mounting hole of skid tube (8). Align axis of hole in eyebolt parallel with longitudinal axis of skid tube.

3. Torque the eyebolt (12) to 40 to 58 foot-pounds (54.2 to 78.6 Nm) and check that axis of hole in eyebolt is parallel to longitudinal axis of skid tube (8).

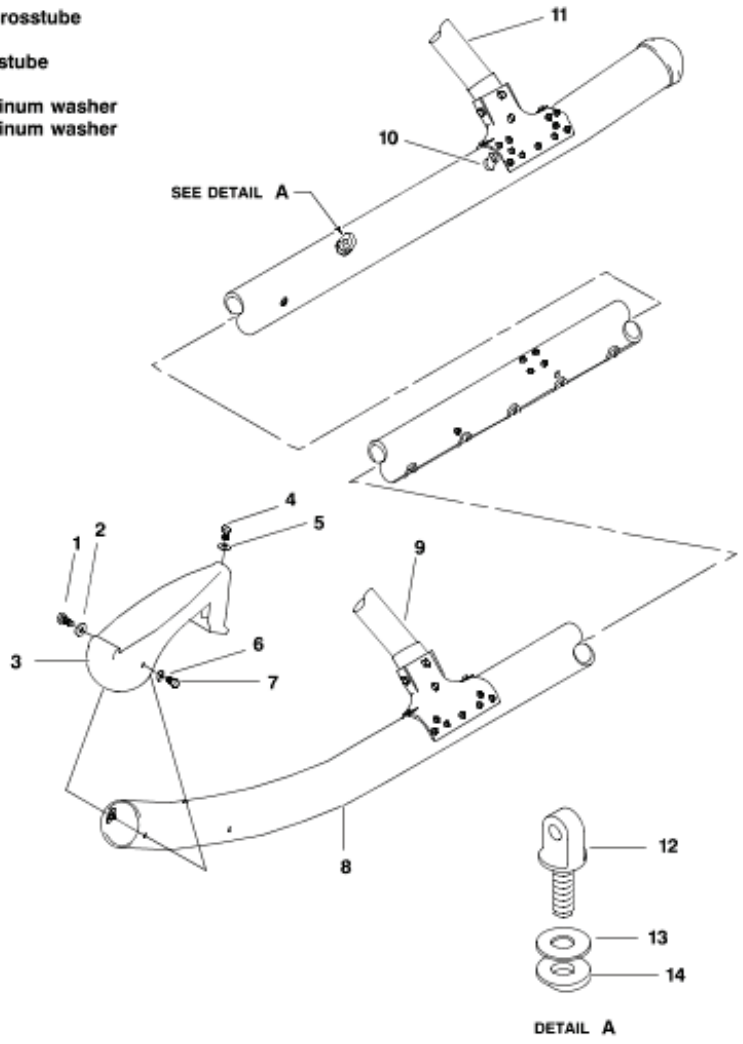
NOTE

A maximum of four washers (AN960JD1016L) may be used under the eyebolt for alignment.

4. If not aligned, undo torque and with the use of washers, retorque to 40 to 58 foot-pounds (54.2 to 78.6 Nm), and confirm alignment described in previous step.

5. If required, reinstall setscrew in alternate eyebolt hole and add sealant.

1. Bolt
2. Thin aluminum washer
3. Step
4. Bolt
5. Thin aluminum washer
6. Thin aluminum washer
7. Bolt
8. Skid tube
9. Forward crosstube
10. Eyebolt
11. Rear crosstube
12. Eyebolt
13. Thin aluminum washer
14. Thin aluminum washer



ICN-412-A-320103-B-97499-00001-A-001-01

Figure 1. Skid Tube Fixed Step and Eyebolt (Sheet 1 of 1)

Landing Gear Crosstube Assemblies - Removal

Procedure

1. Remove the landing gear assembly from under the helicopter (412-A-32-01-00-00A-520A-A / 00697).

2. Remove the hardware securing the extremities of the crosstube assembly from the front or aft saddles on the left or right skid tubes as required.



DO NOT DAMAGE THE SURFACE OF THE CROSSTUBES.

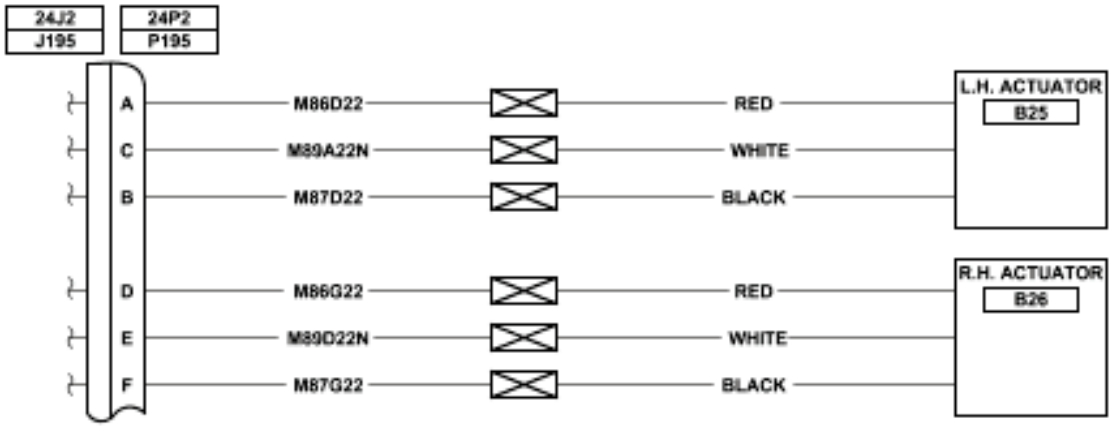
3. Using a plastic scraper, remove the sealant from the upper edges of the left and right skid tube saddles.



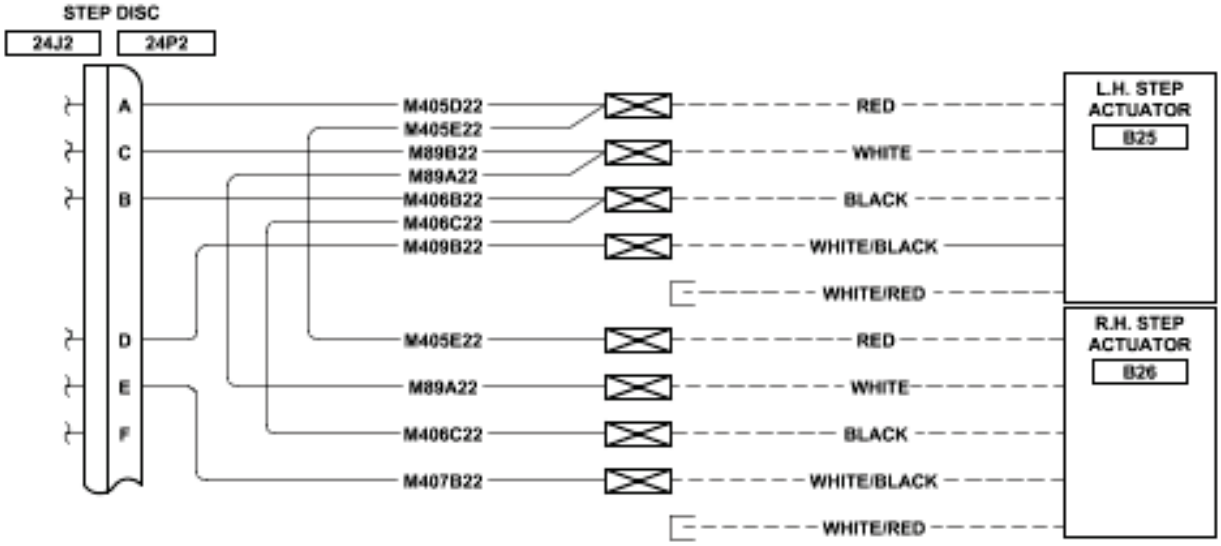
BE CAREFUL NOT TO DAMAGE THE ELECTRIC WIRES WHEN THE FORWARD CROSSTUBE IS MOVED AWAY FROM THE FRONT SADDLE.

4. If applicable, tag and disconnect the electric wires from each step disconnect connector 24P2/P195 or from the LH and RH step actuators B25/B26 (Figure 1).

5. Apply downward force on skid tube near each saddle. Move each crosstube slightly in a fore-and-aft direction to release the ends of the crosstubes from their mating saddles.



HELICOPTERS S/N 36001 THROUGH 36124
HELICOPTERS S/N 33001 AND SUBSEQUENT



HELICOPTERS S/N 36125 AND SUBSEQUENT

ICN-412-A-320105-B-97499-00001-A-001-01

Figure 1. Wiring Diagram Passenger Step (Sheet 1 of 1)

Landing Gear Crosstube Assemblies - Inspection

Procedure

NOTE

All crosstube damage is unacceptable and shall be repaired. Refer to 412-A-32-01-05-00A-685A-A / 00722 for repair criteria.

1. Inspect crosstube for condition and damage such as dents, cracks, nicks, scratches, and corrosion.

2. Inspect each extremity of crosstube for fretting damage and corrosion.

3. Inspect each support or clamp assembly at BL 14.0 for damage and condition.

4. Inspect the sealant, clamps, and rivets for looseness and the bondline for indication of voids.

NOTE

If support or rivets appear to be loose, remove support (fasteners for riveted supports are installed in cold-worked holes that are not damage tolerant) and inspect surface of crosstube for condition.

5. Inspect aft crosstube support beam for proper alignment on crosstube.

NOTE

Uneven fretting or wear on clamp assemblies at BL 14.0 may indicate that the beam is not positioned correctly and will need realignment.

6. Remove aft crosstube support beam and inspect upper and lower supports for looseness.

7. Inspect upper support bondline for voids.

8. Inspect crosstube for missing paint and primer. Restore as required (BHT-ALL-SPM, Standard Practices Manual).

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Landing Gear Crosstube Assemblies - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Chemical film material	C-100	AR	
Epoxy polyamide primer	C-204	AR	
Isopropyl alcohol	C-385	AR	
Abrasive pad	C-407	AR	

Procedure

1. Repair all damage (scratches, nicks, dents) and corrosion as follows (Table 1):



ABRASIVE BLENDING-OUT STROKES MUST BE PARALLEL TO THE CROSSTUBE, NOT CIRCUMFERENTIAL.

2. Polish the affected area in a longitudinal direction using abrasive pad (C-407). Make sure the limits presented in Table 1 and Figure 1 are not exceeded.

3. Clean repaired area with isopropyl alcohol (C-385), apply chemical film material (C-100) on bare aluminum areas per MIL-DTD-81706 or equivalent, and reprime with epoxy polyamide primer (C-204) or equivalent.

4. Paint repaired area to match existing finish (BHT-ALL-SPM, Standard Practices Manual).

Table 1. Crosstubes - Damages and Repair Limits

COMPONENT	MAXIMUM ALLOWABLE DAMAGE AND REPAIR LIMITS (Damage and corrosion may be repaired per 412-A-32-01-05-00A-685A-A / 00722)	
	SCRATCHES, NICKS, DENTS AND CORROSION	CRACKS/HOLES
Crosstube	See Figure 1, sheets 5 through 9	None
Forward Support	See Figure 2	None
Aft Center Support	See Figure 2	None
Clamp Assembly	To a depth of 0.005 inch (0.127 mm)	None

NOTES:

Superficial damage on crosstubes is unacceptable. All damage on crosstubes shall be repaired. Observe limitations shown in Figure 1, Section B-B. Generous radii and/or fairing shall be used in repair.

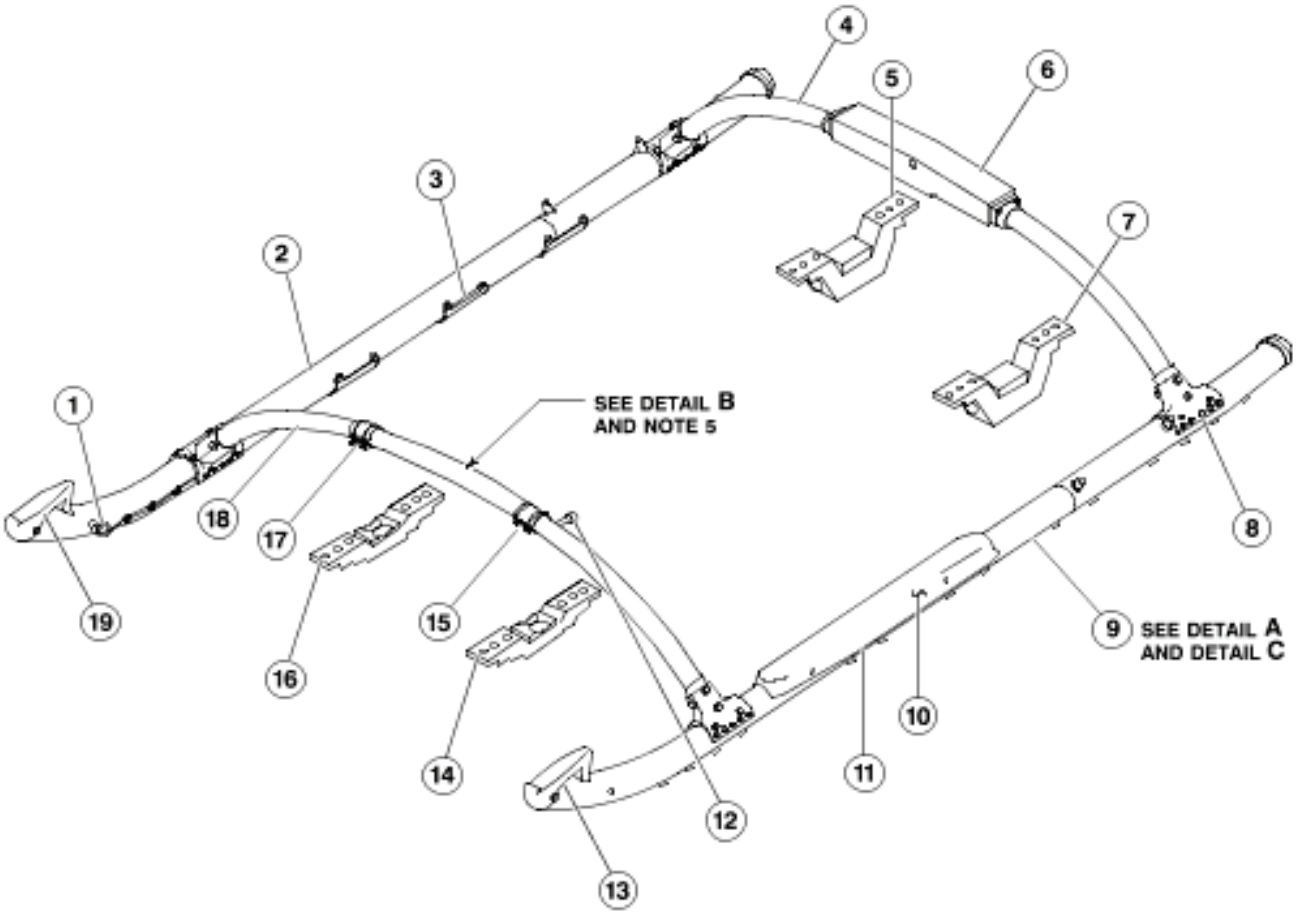
Damage limits apply to the surface areas of the crosstube as indicated by Figure 1, Section B-B. Circumferential grooves or grind marks are not acceptable.

Scratches, nicks, dents or corrosion shall be polished out in the longitudinal direction within limits shown in Figure 1.

All corrosion shall be cleaned up to twice the visible corrosion depth and shall not exceed the damage limits shown in Figure 1.

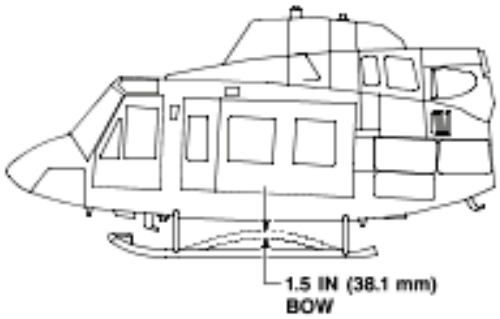
The orientation of scratches and corrosion damage is limited by the dimensions shown in Figure 1, View A-A.

The minimal longitudinal distance between repairs is 3.0 inches (76.2 mm).



STANDARD SKID LANDING GEAR ASSEMBLY
412-050-001-101, -103, -105, -107, -109 and -111

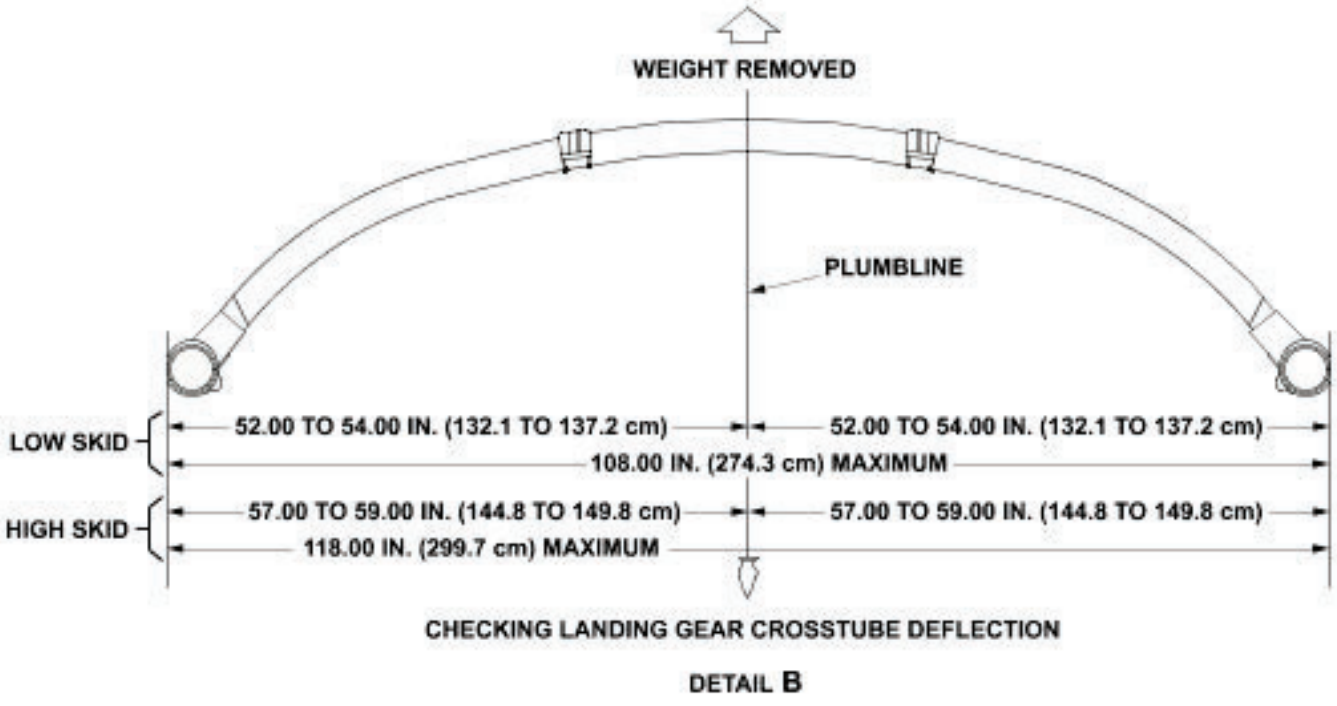
1. Tow ring
2. Skid tube
3. Skid shoe
4. Crosstube
5. Landing gear cap fitting
6. Aft crosstube support beam
7. Landing gear cap fitting
8. Saddle
9. Skid tube
10. Passenger step
11. Skid shoe
12. Passenger step electrical connector
13. Step
14. Landing gear cap fitting
15. Landing gear bearing and retaining support
16. Landing gear cap fitting
17. Landing gear bearing and retaining support
18. Crosstube
19. Step



DETAIL A

ICN-412-A-320100-B-97499-00003-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 1 of 9)

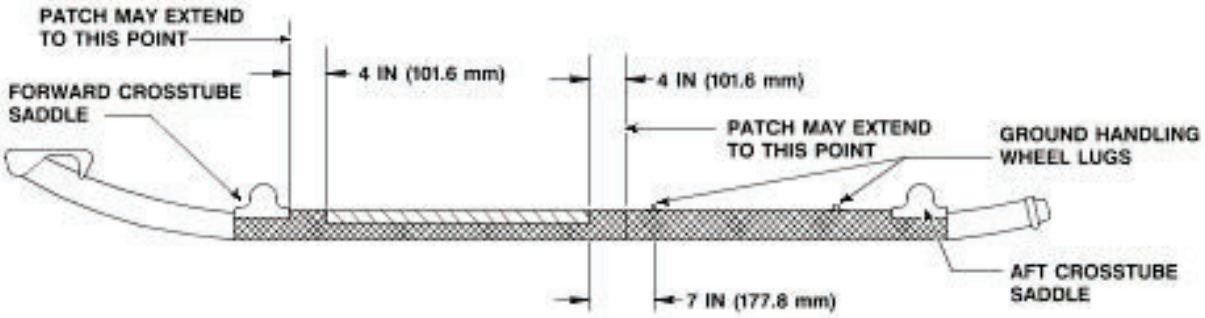


NOTES

1. Deformation (distortion) of either crosstube in excess of limits is cause for replacement of crosstube(s).
2. Crosstube may be deflected on one side only. If limit is exceeded, crosstube is considered scrapped and must be replaced.

ICN-412-A-320100-B-97499-00004-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 2 of 9)



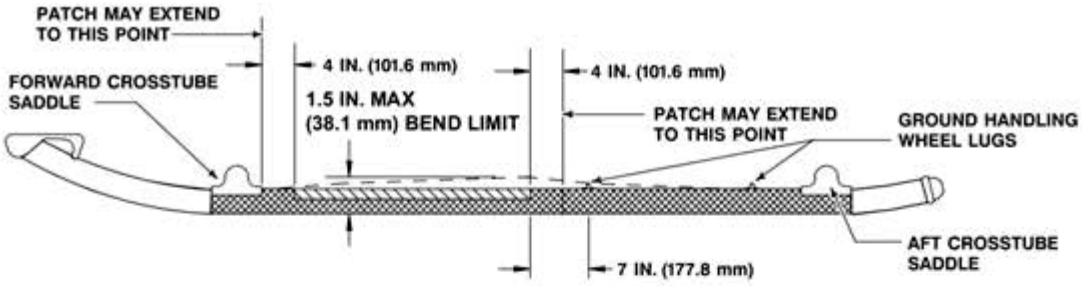
DETAIL C
412-050-023

DAMAGE LOCATION SYMBOLS

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR AREAS ALLOWED		
CRACKS ALLOWED	None	None	None
DENTS AND HOLES	None	4 inches (101.6 mm) Refer to paragraph 32-9.	Not Critical

ICN-412-A-320100-B-97499-00005-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 3 of 9)

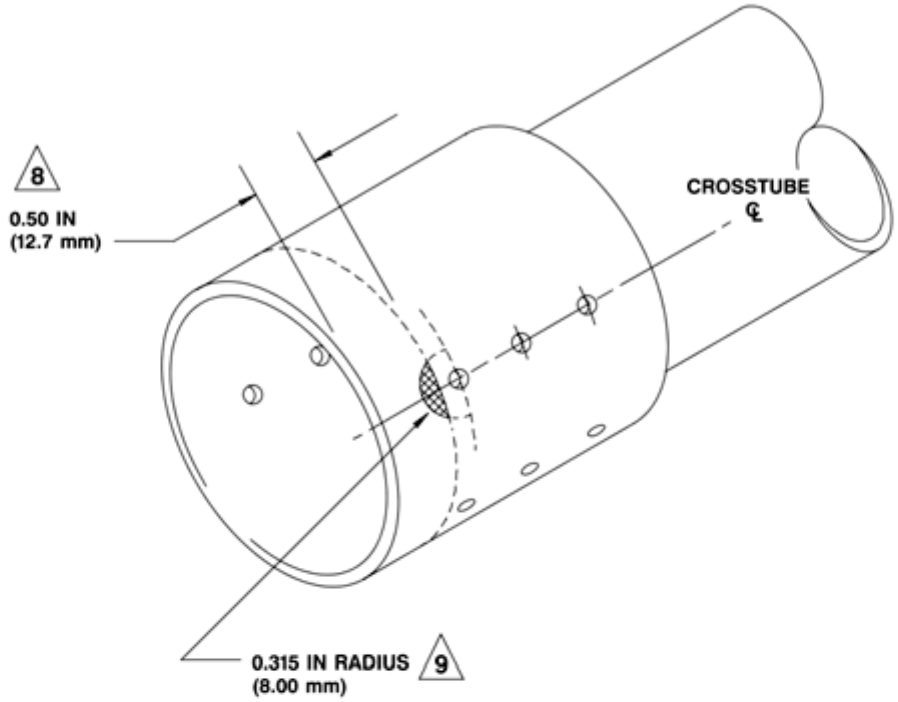


DETAIL C
412-050-030
412-050-034

			(Does not apply to saddles)
DENTS	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No dents allowed. Replace as needed.	Not critical.
HOLES	Repair damage up to 3.0 inches (76.2 mm) in diameter.	No holes allowed. Replace as needed.	Not critical. Plug holes to keep out moisture.
CRACKS	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.	No cracks allowed. Replace as needed.
SCRATCHES SCUFFS NICKS	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	If damage is less than 0.03 inch (0.7 mm) deep and/or 1.25 inches (31.8 mm) length, no repair is needed.	Not critical.
	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	If damage is between 0.03 to 0.05 inch (0.8 to 1.2 mm) deep and/or 1.25 to 1.65 inches (31.8 to 41.9 mm) length, repair.	
	If damage is greater than established limits, replace as needed.	If damage is greater than established limits, replace as needed.	
CORROSION	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup limits are 0.05 inch (1.3 mm) depth and 1.65 inches (41.9 mm) length. Refer to paragraph 32-11.	No corrosion allowed. Corrosion cleanup is limited to 10% of material thickness. Refer to paragraph 32-11.

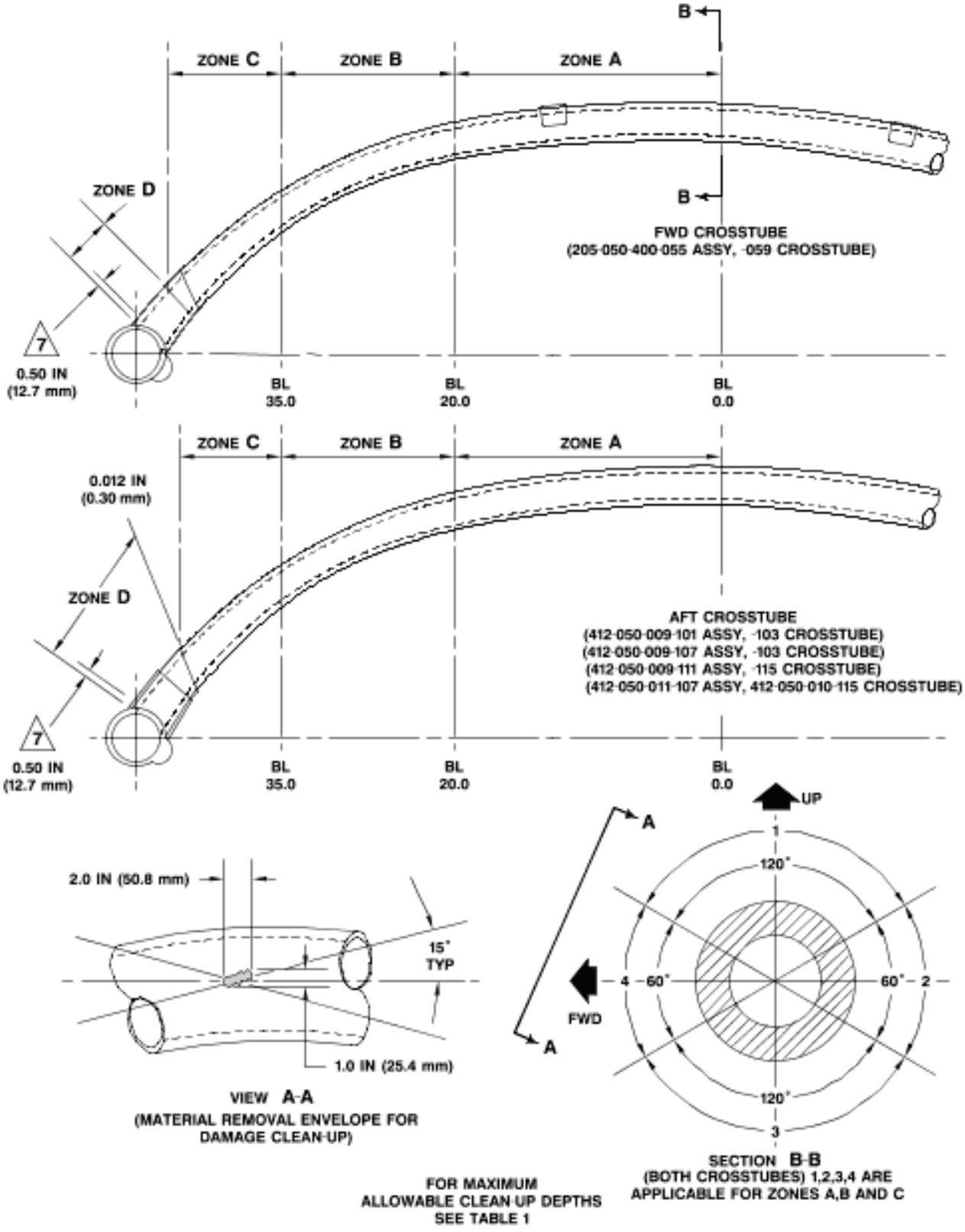
ICN-412-A-320100-B-97499-00006-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 4 of 9)



ICN-412-A-320100-B-97499-00007-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 5 of 9)



ICN-412-A-320100-B-97499-00008-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 6 of 9)

TABLE 1
MAXIMUM ALLOWABLE CLEANUP DEPTH STANDARD AND HIGH SKID GEAR

ZONE	BHT PRODUCTS FWD CROSSTUBE INCHES (mm) △10	BHT PRODUCTS AFT CROSSTUBE INCHES (mm) △11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.12) 0.015 (0.38) 0.005 (0.12) 0.015 (0.38)
B1 B2 B3 B4	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)	0.015 (0.38) 0.015 (0.38) 0.015 (0.38) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.010 (0.25) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.012 (0.30) 0.012 (0.30)
D △7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 6 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

△7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 1.

△8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note △9 (Figure 32-2, sheet 5).

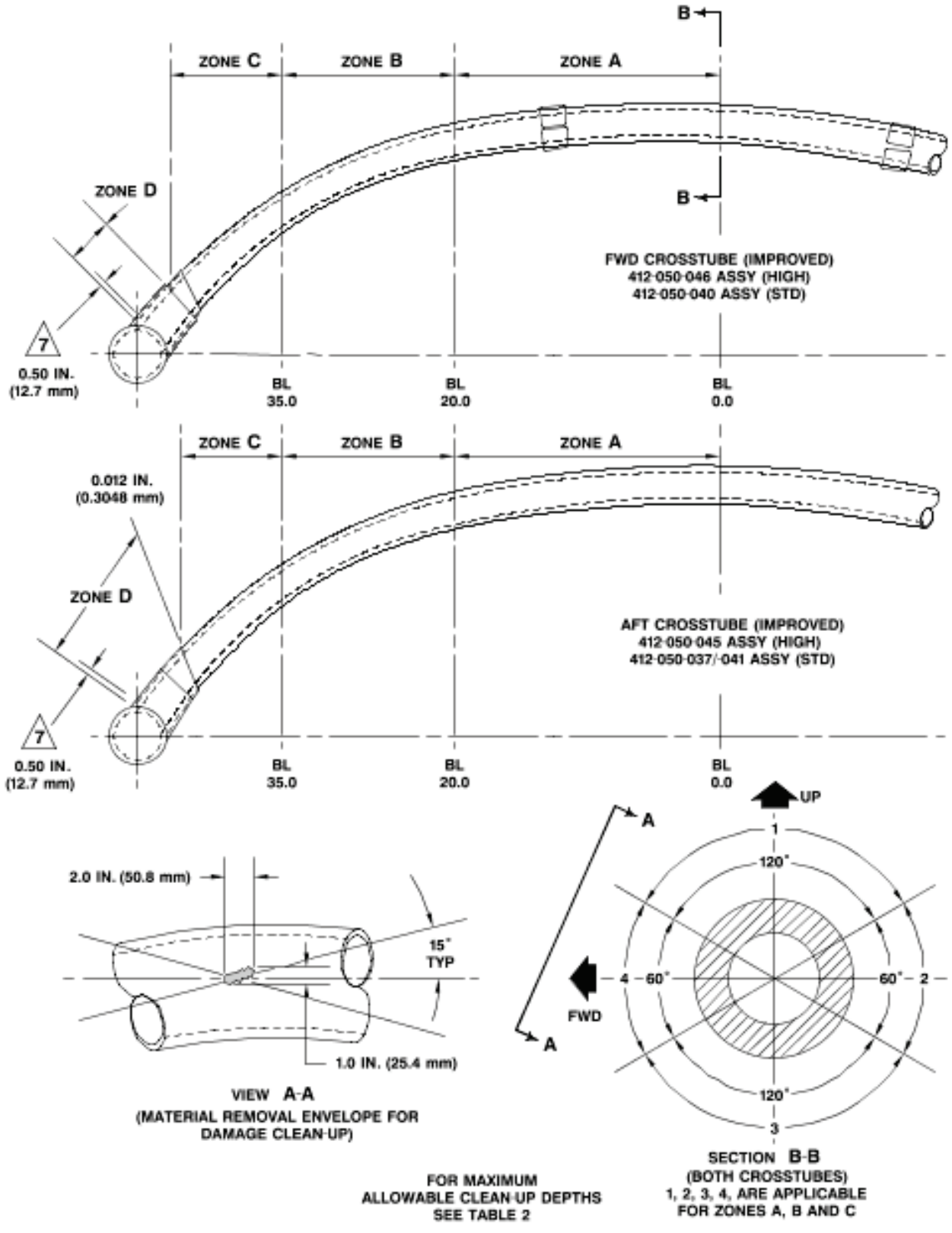
△9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

△10 Forward standard crosstube (BHT 205-050-400-055).
Forward high crosstube (BHT 205-050-403-031).

△11 Aft standard crosstube (BHT 412-050-002-101/103/105/107).
Aft high crosstube (BHT 412-050-011-101/103/105/107).

ICN-412-A-320100-B-97499-00009-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 7 of 9)



ICN-412-A-320100-B-97499-00010-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 8 of 9)

TABLE 2
MAXIMUM ALLOWABLE CLEANUP DEPTH IMPROVED STANDARD AND IMPROVED HIGH SKID GEAR

ZONE	BHT AND AAI PRODUCTS	BHT AND AAI PRODUCTS
	FWD CROSSTUBE INCHES (mm) ⚠10	AFT CROSSTUBE INCHES (mm) ⚠11
A1 A2 A3 A4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
B1 B2 B3 B4	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)	0.005 (0.13) 0.015 (0.38) 0.005 (0.13) 0.015 (0.38)
C1 C2 C3 C4	0.010 (0.25) 0.010 (0.25) 0.005 (0.13) 0.010 (0.25)	0.012 (0.30) 0.012 (0.30) 0.005 (0.13) 0.012 (0.30)
D ⚠7	0.030 (0.76)	0.038 (0.96)

NOTES

1. See Figure 32-2, sheet 8 for zone definition.
2. All dimensions are in inches (except as noted).
3. Repairs limited to one repair per "quadrant", two at any cross section.
4. Log location and depth of all repairs for each crosstube.
5. Repair limits are lifetime accumulation at any location.
6. Repair in Zone D not to impinge on more than one fastener hole.

⚠7 Allowable depth of repair is 0.050 inch (1.27 mm) for bottom 0.50 inch (12.7 mm) of area as indicated except within 0.315 inch (8.00 mm) radius of center of any fastener hole. Otherwise limits for Zone D are as shown in Table 2.

⚠8 0.050 inch (12.7 mm) maximum damage allowed on first 0.050 inch (12.7 mm) of crosstube except as indicated by note ⚠9 (Figure 32-2, sheet 5).

⚠9 Maximum allowable damage in shaded area is determined in Zone D inspection limits (Figure 32-2, sheet 5).

⚠10 Improved forward standard crosstube (BHT 412-050-040-101, AAI 212-320-103 ALL).
Improved forward high crosstube (BHT 412-050-046-101, AAI 212-321-103 ALL).

⚠11 Improved aft standard crosstube (BHT 412-050-041-109, AAI 412-320-104 ALL).
Improved aft high crosstube (BHT 412-050-045-105, AAI 412-321-106 ALL).

ICN-412-A-320100-B-97499-00011-A-001-01

Figure 1. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 9 of 9)

FORWARD LANDING GEAR SUPPORTS



205-050-132



205-050-133

DAMAGE LOCATION SYMBOLS



TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR DEPTH
MECHANICAL	0.020 IN. (0.50 mm)
CORROSION DAMAGE (AFTER CLEANUP)	0.020 IN. (0.50 mm)
MAXIMUM AREA PER FULL DEPTH REPAIR	1.0 SQ. IN. (645.1 Sq. mm)
NUMBER OF REPAIRS	2
EDGE CHAMFER	0.050 IN. (1.27 mm)
BORE DAMAGE	0.020 IN. (0.50 mm) for 1/4 circumference

ICN-412-A-320105-B-97499-00002-A-001-01

Figure 2. Landing Gear Support Damage Limits (Sheet 1 of 3)

IMPROVED FORWARD LANDING GEAR SUPPORT



412-050-038

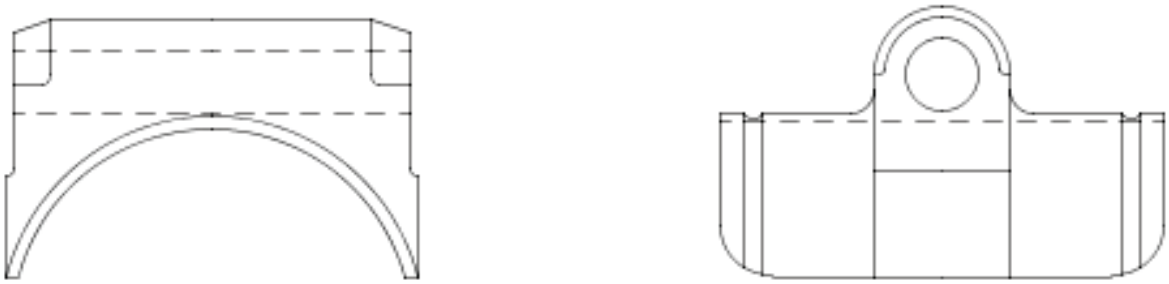
DAMAGE LOCATION SYMBOLS

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR DEPTH
MECHANICAL	0.020 IN. (0.50 mm)
CORROSION DAMAGE (AFTER CLEANUP)	0.020 IN. (0.50 mm)
MAXIMUM AREA PER FULL DEPTH REPAIR	1.0 SQ. IN. (645.1 Sq. mm)
NUMBER OF REPAIRS	2
EDGE CHAMFER	0.050 IN. (1.27 mm)
BORE DAMAGE	0.020 IN. (0.50 mm) for 1/4 circumference

ICN-412-A-320105-B-97499-00003-A-001-01

Figure 2. Landing Gear Support Damage Limits (Sheet 2 of 3)

AFT LANDING GEAR SUPPORTS



412-050-006



412-050-007

DAMAGE LOCATION SYMBOLS



TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR DEPTH
MECHANICAL	0.020 IN. (0.50 mm)
CORROSION DAMAGE (AFTER CLEANUP)	0.020 IN. (0.50 mm)
MAXIMUM AREA PER FULL DEPTH REPAIR	1.0 SQ. IN. (645.1 Sq. mm)
NUMBER OF REPAIRS	2
EDGE CHAMFER	0.050 IN. (1.27 mm)
BORE DAMAGE	0.020 IN. (0.50 mm) for 1/4 circumference

ICN-412-A-320105-B-97499-00004-A-001-01

Figure 2. Landing Gear Support Damage Limits (Sheet 3 of 3)

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Landing Gear Crosstube Assemblies - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	
Sealant	C-251	AR	

Procedure

1. Install aft crosstube (8, Figure 1) to left skid tube (48) as follows:

NOTE

Install all fasteners and grommets with sealant (C-251).

1.1. Insert aft crosstube into aft saddle of skid tube (48), and secure with three bolts (53), three bolts (51), and six washers (52).

1.2. Torque bolts .

2. If applicable, install the powered step cable assembly (55) as follows:

2.1. Route the wires for the powered step through the crosstube (Figure 2).

2.2. Install the grommet (54, Figure 1) on the wires and install the grommet in the crosstube or skid tubes, as applicable.

2.3. Remove the tags and connect the wires to the step disconnect connector 24P2/P195 or the LH and RH step actuators B25/B26, as shown in Figure 2.



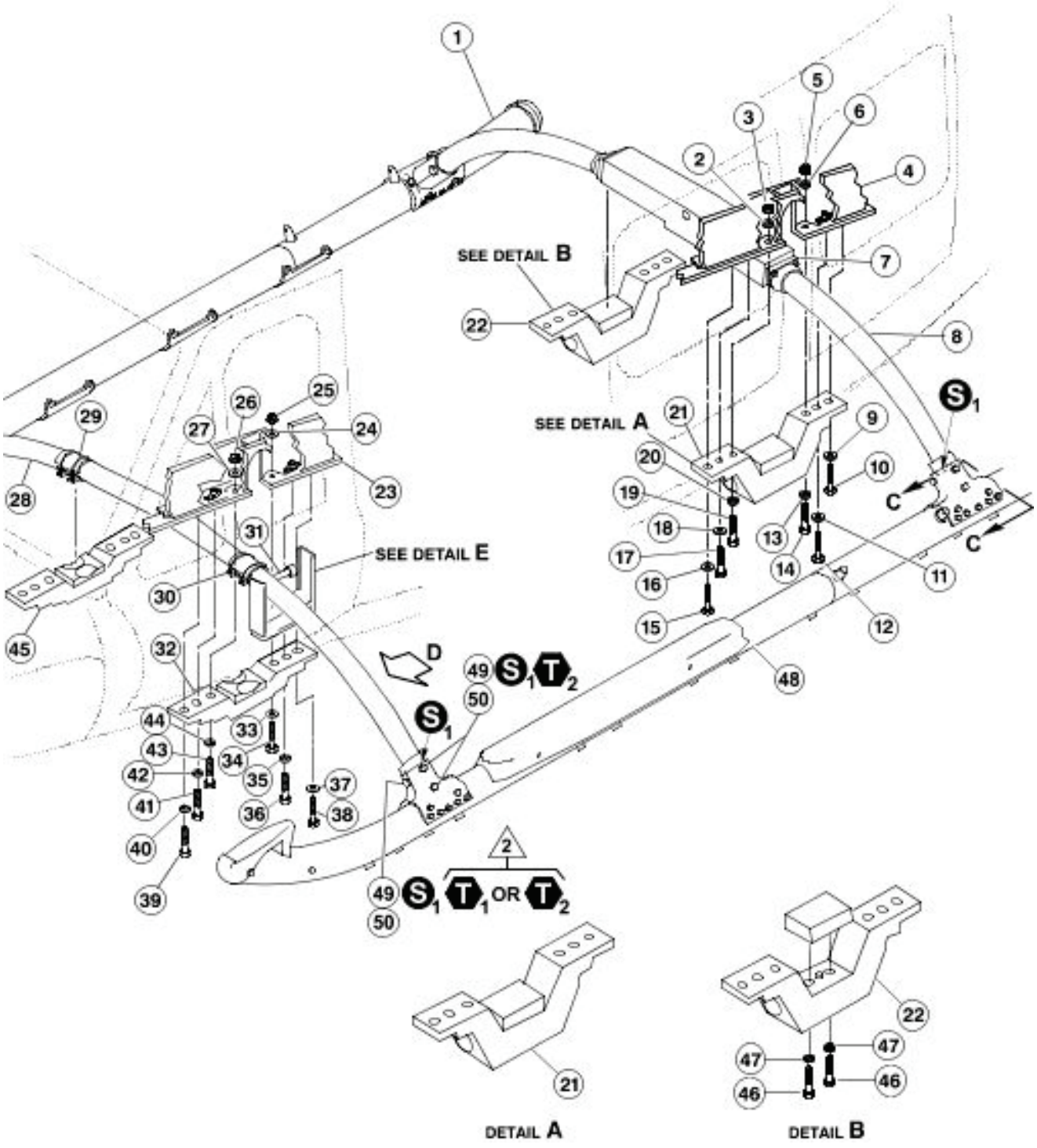
USE CARE TO AVOID CHAFING, CUTTING, OR BREAKING CABLE ASSEMBLY WHEN PERFORMING NEXT STEP.

3. Insert forward crosstube (28, Figure 1) into forward saddle of left skid tube (48). Secure crosstube and saddle with six bolts (49) and six washers (50). Torque the bolts , as shown in Figure 1.

4. Apply a bead of sealant (C-251) around joint of saddles and crosstubes.

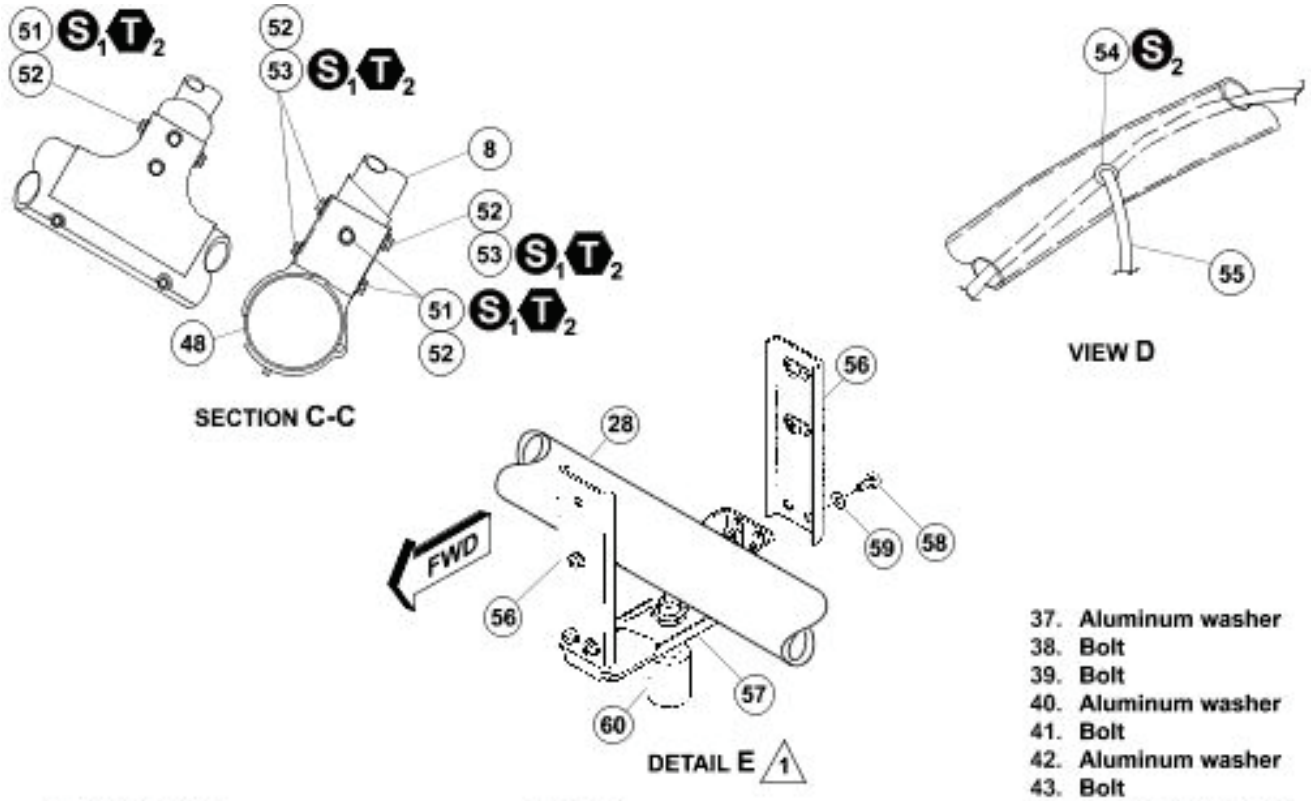
5. Install crosstubes to right skid tube same as left skid tube.

6. If applicable, apply sealant (C-308) to the grommet (54) and cable assembly (55).



ICN-412-A-320100-B-97499-00001-A-001-01

Figure 1. Landing Gear (Standard) (Sheet 1 of 2)



1. Landing gear
2. Aluminum washer
3. Nut
4. Landing gear support structure
5. Nut
6. Aluminum washer
7. Aft crosstube support beam
8. Aft crosstube
9. Aluminum washer
10. Bolt
11. Aluminum washer
12. Bolt
13. Aluminum washer
14. Bolt
15. Bolt
16. Aluminum washer
17. Bolt
18. Aluminum washer
19. Bolt
20. Aluminum washer
21. Aft cap fitting
22. Aft cap fitting
23. Forward crosstube support structure
24. Aluminum washer
25. Nut
26. Nut
27. Aluminum washer
28. Forward crosstube
29. Bearing and retaining support
30. Bearing and retaining support
31. Power step electrical connector
32. Forward cap fitting
33. Aluminum washer
34. Bolt
35. Aluminum washer
36. Bolt
37. Aluminum washer
38. Bolt
39. Bolt
40. Aluminum washer
41. Bolt
42. Aluminum washer
43. Bolt
44. Aluminum washer
45. Forward cap fitting
46. Bolt
47. Washer
48. Skid tube
49. Bolt
50. Washer
51. Bolt
52. Washer
53. Bolt
54. Grommet
55. Cable assembly
56. Bracket assembly
57. Support assembly
58. Screw
59. Washer
60. Weight-on-gear switch (3420S1)

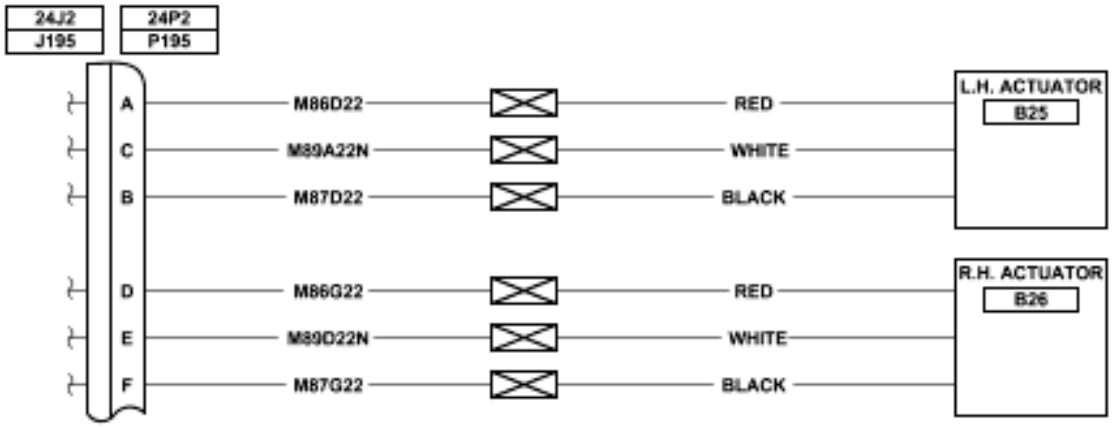
S_1 SEALANT (C-392) S_2 SEALANT (C-308) T_1 50 TO 70 IN-LBS (5.65 TO 7.91 Nm) T_2 160 TO 190 IN-LBS (18.08 TO 21.47 Nm)

NOTES

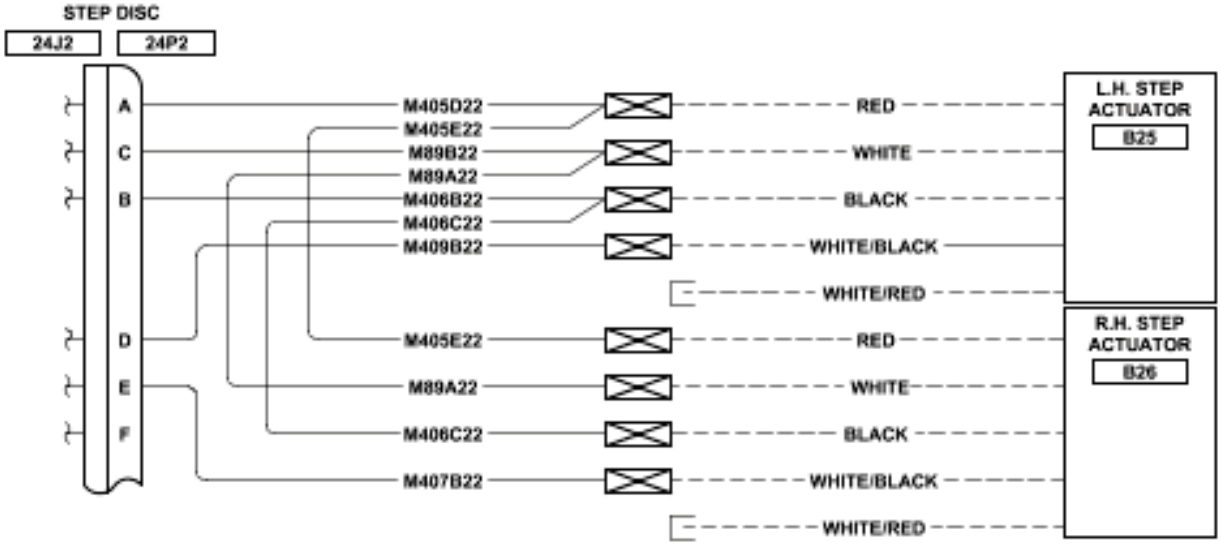
1. Weight-on-gear switch 3420S1 installed on helicopters S/N 36462 and subsequent. It is also installed on helicopters S/N 36248 through 36461 with the weather radar kit (412-705-034) or weight-on-gear retrofit kit (BHT-412-SI-84) installed.
2. For 1/4-28 bolts, torque per T_1 and for 3/8-24 bolts, torque per T_2.

ICN-412-A-320100-B-97499-00002-A-001-01

Figure 1. Landing Gear (Standard) (Sheet 2 of 2)



HELICOPTERS S/N 36001 THROUGH 36124
HELICOPTERS S/N 33001 AND SUBSEQUENT



HELICOPTERS S/N 36125 AND SUBSEQUENT

ICN-412-A-320105-B-97499-00001-A-001-01

Figure 2. Wiring Diagram Passenger Step (Sheet 1 of 1)

Aft Crosstube Support Beam - Description

NOTE

Maintenance procedures for aft crosstube support beam for high gear and standard gear are the same.

The aft crosstube support beam is constructed of forged aluminum alloy incorporating an upper and lower support for attachment of support beam to crosstube. This arrangement provides a pivot point between fuselage and aft crosstube at centerline of helicopter to dampen vibrations caused by ground resonance.

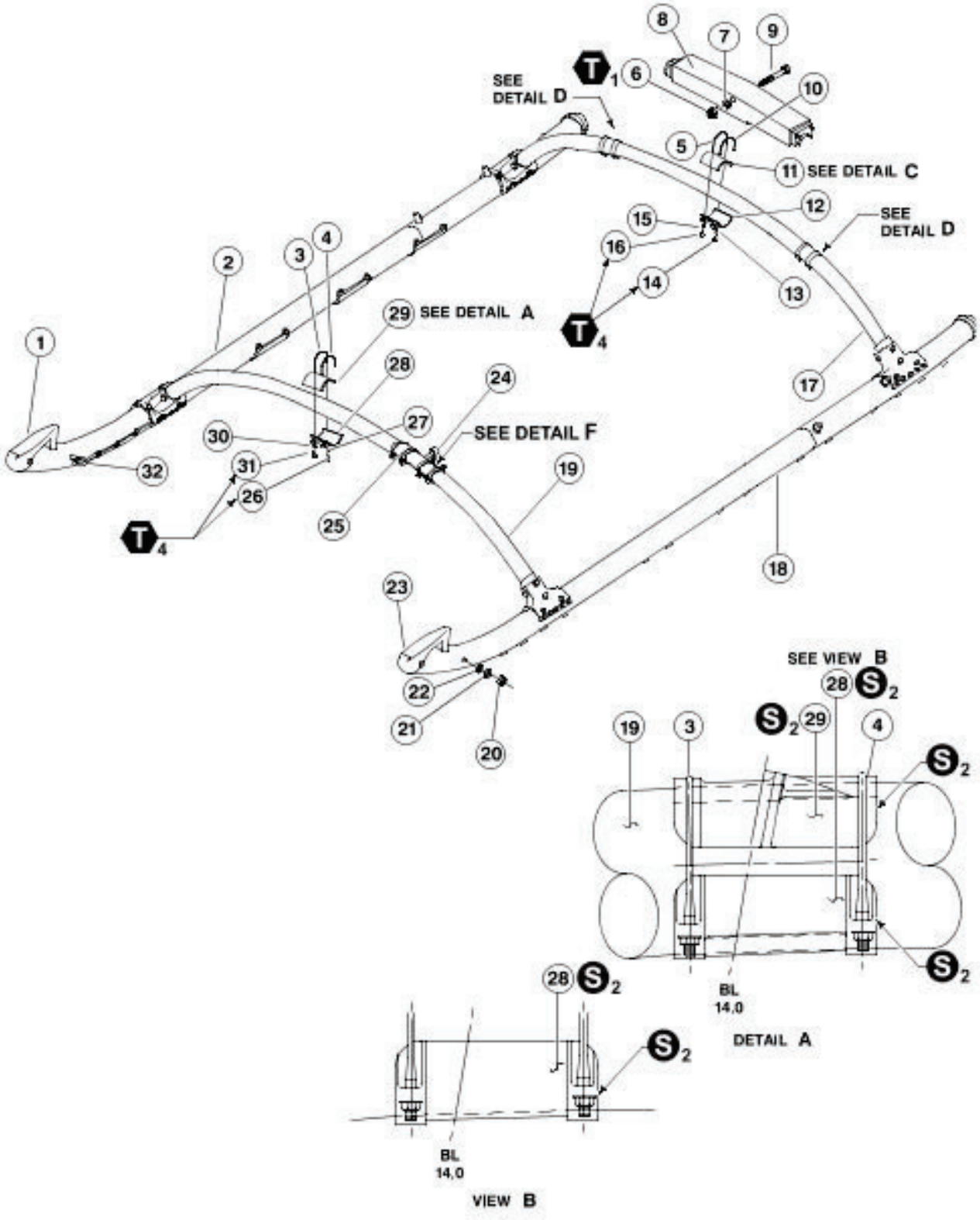
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Aft Crosstube Support Beam - Removal

Procedure

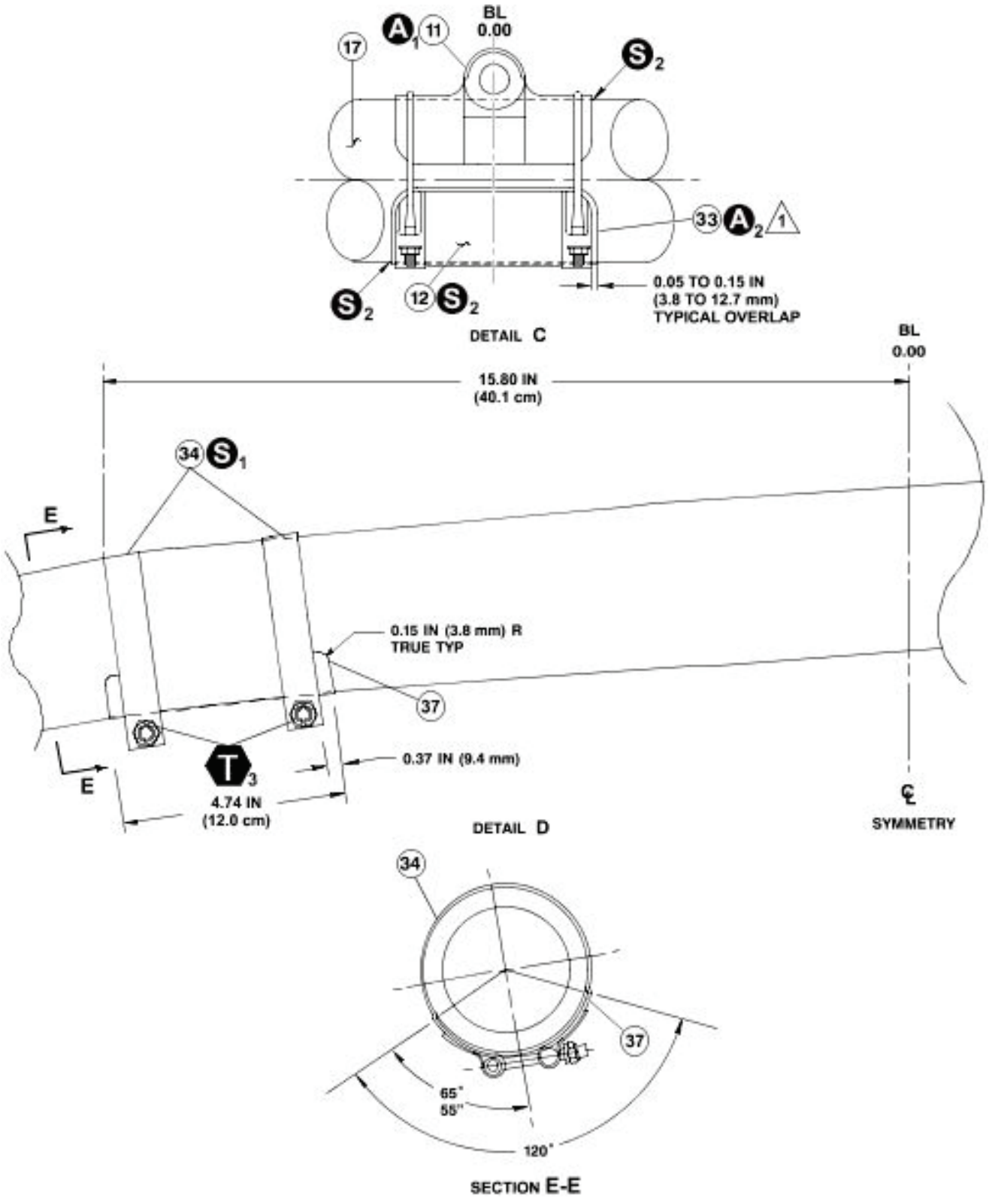
1. Remove landing gear (412-A-32-01-00-00A-520A-A / 00697).

2. Remove nut (6, Figure 1) thin steel washer (7), and bolt (9). Remove aft crosstube support beam (8).



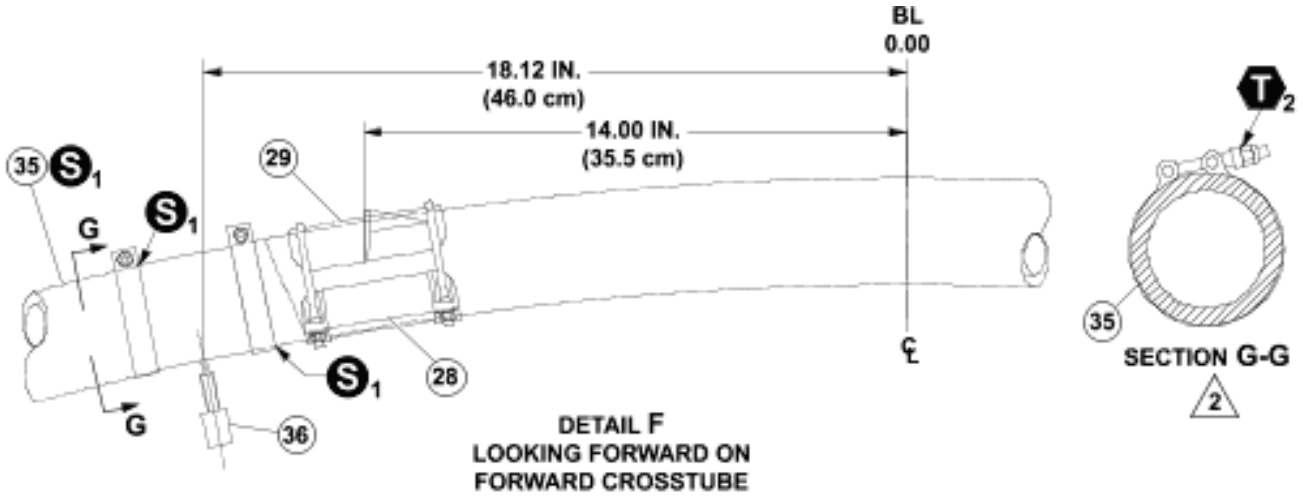
ICN-412-A-320106-B-97499-00004-A-001-01

Figure 1. Landing Gear Assembly (Sheet 1 of 3)



ICN-412-A-320106-B-97499-00005-A-001-01

Figure 1. Landing Gear Assembly (Sheet 2 of 3)



1. Strap
2. Right skid tube
3. U-bolt
4. U-bolt
5. U-bolt
6. Nut
7. Thin steel washer
8. Aft crosstube support beam
9. Bolt
10. U-bolt
11. Aft crosstube upper support
12. Aft crosstube lower support
13. Steel washer
14. Nut
15. Steel washer
16. Nut
17. Aft crosstube
18. Left skid tube
19. Forward crosstube
20. Nut
21. Aluminum washer
22. Aluminum Washer
23. Step
24. Power step electrical connector
25. Left forward crosstube support
26. Nut
27. Steel washer
28. Forward crosstube lower support
29. Forward crosstube upper support
30. Steel washer
31. Nut
32. Tow fitting
33. Abrasion strip
34. Clamp assembly
35. Clamp assembly
36. Weight-on-gear switch
37. Abrasion strip

A_1 ADHESIVE (C-317)

A_2 ADHESIVE (C-571)

S_1 SEALANT (C-251)

S_2 SEALANT (C-308)

T_1 10 TO 30 IN-LBS (1.13 TO 3.39 Nm)

T_2 40 IN-LBS (4.57 Nm)

T_3 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T_4 80 TO 100 IN-LBS (9.04 TO 11.30 Nm)

NOTES

1. Abrasion strip (33) must be removed for installation of AAI aft crosstube lower support half (12)

2. Weight-on-gear switch 3420S1 installed on helicopters S/N 36462 and subsequent. It is also installed on helicopters S/N 36248 through 36461 with the weather radar kit (412-705-034) or weight-on-gear retrofit kit (BHT-412-SI-84) installed.

ICN-412-A-320106-B-97499-00006-A-001-01

Figure 1. Landing Gear Assembly (Sheet 3 of 3)

Aft Crosstube Support Beam - Inspection and Repair

Procedure

NOTE

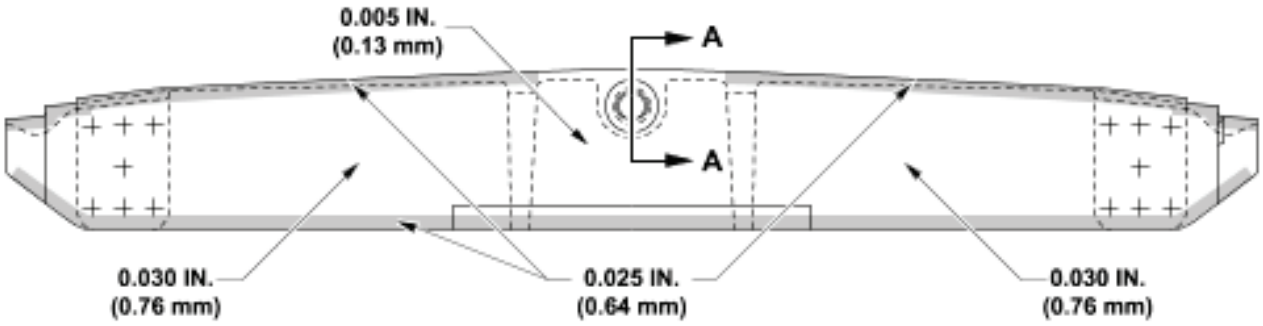
Superficial mechanical damage to aft crosstube support beam is acceptable without repair. Repair is limited to replacement of two bushings.

1. There are six configurations for the support beam and they are listed in the table that follows. The damage and repair limits are shown in Figure 1 for each configuration.

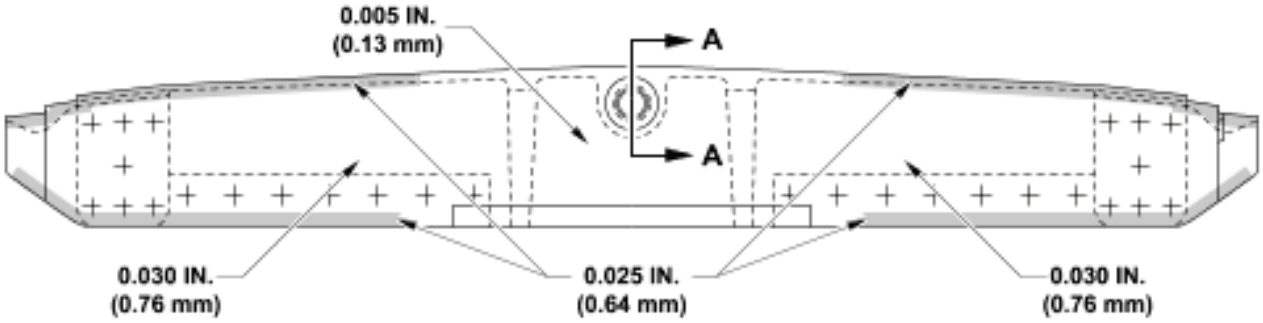
BEAM ASSEMBLY	NEXT ASSEMBLY
412-050-003-101	412-050-002-101 412-050-002-103 412-050-011-101 412-050-011-103
412-050-003-105	412-050-002-105 412-050-011-105
412-050-003-109	412-050-011-107 412-050-041-105 412-050-045-105
412-050-003-111	412-050-041-105 412-050-045-105
412-050-003-113	412-050-002-107 412-050-041-105 412-050-045-105
604-030-001/-201	412-320-XXX

1.1. Inspect support beam for damage and repair limits per Figure 1. If support beam has damage exceeding limits shown in Figure 1, discard and replace with a new one. No cracks are allowed in any circumstance.

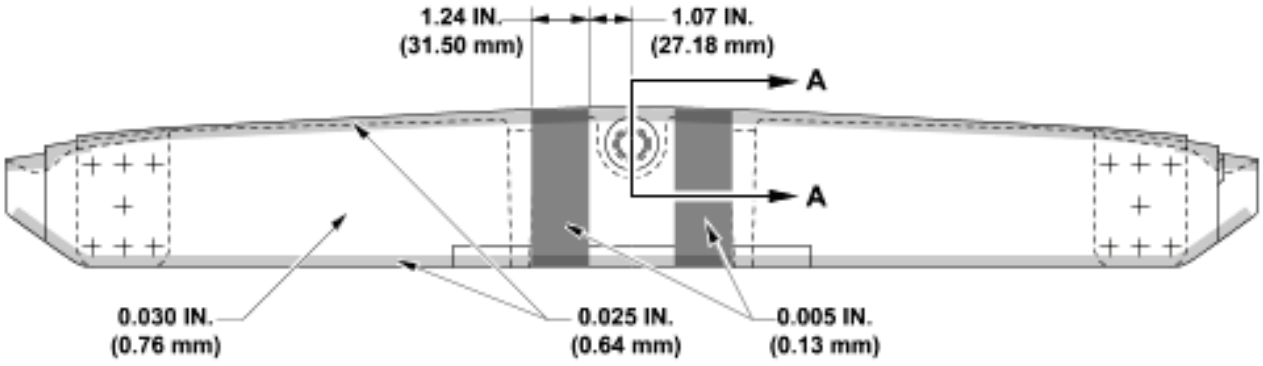
1.2. In case of a lightning strike, inspect support beam and blend affected area(s) to remove damage without exceeding the limits shown in Figure 1.



412-050-003-101



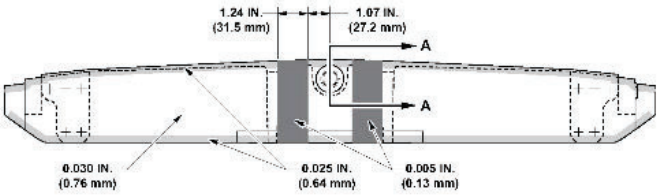
412-050-003-105



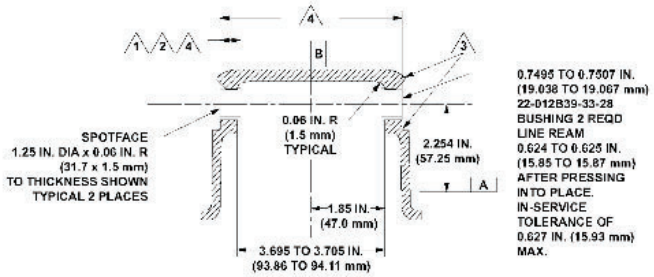
412-050-003-109

ICN-412-A-320106-B-97499-00001-A-001-01

Figure 1. Support Beam - Damage and Repair Limits (Sheet 1 of 3)



412-050-003-111/-113 AND 604-030-001/-201



SECTION A-A
BOLT LOCATION

The bushing hole can be reworked as follows:

	OVERSIZE	OUTER BUSHING DIAMETER	BUSHING P/N
1ST O/S	0.010 INCH (0.25 mm)	0.7595 TO 0.7605 INCH (19.292 TO 19.316 mm)	22X012B39-33-28
2ND O/S	0.020 INCH (0.50 mm)	0.7695 TO 0.7705 INCH (19.546 TO 19.570 mm)	22Y012B39-33-28
3RD O/S	0.031 INCH (0.79 mm)	0.7807 TO 0.7817 INCH (19.830 TO 19.855 mm)	22H012B39-33-28

BUSHING HOLES OVERSIZE

ICN-412-A-320106-B-97499-00002-A-001-01

Figure 1. Support Beam - Damage and Repair Limits (Sheet 2 of 3)

NOTES

BOLT LOCATION

1. Install bushings with wet epoxy polyamide primer per BHT-ALL-SPM.

2. Ensure each bushing does not protrude from the fitting face after rework. Otherwise, install a shorter bushing so that the bushing ends flush with fitting or 0.016 inch (0.41 mm) inside.

3. Ensure 0.06 inch (1.5 mm) radius is maintained, no sharp edges.

4. Spot face permissible on each outer face of fitting to remove corrosion. For support beam 412-050-003-101/-105, 0.005 inch (0.13 mm) maximum material removal per spotface and for support beam 412-050-003-109/-111/-113 and AAI 604-030-001/-201, 0.030 inch (0.76 mm) maximum material removal is allowed. Omit organic finish in this area.

5. Add NAS1149D10xxK, or AN960-816L, or AN960C816L washer with the minimum thickness required to ensure the nut does not bottom out on the bolt's thread at installation after rework. During the installation, after rework, torque nut without clamping the lugs. Rotate the nut counterclockwise to align the cotter pin with the first hole. Protect the nut, bolt head, and washer with anticorrosion compound.

CORROSION PROTECTION AFTER ANY REWORK

6. Blended areas shall have a surface finish of 63 RMS or better.

7. No power tools allowed.

8. All reworked areas shall be protected with touch-up alodine and touch-up polyamide epoxy primer, and then touch-up painted.

LIMITATIONS FOR CORROSION, NICKS, SCRATCHES AND DENTS

9. One repair per cross section.

10. Two inches (50.8 mm) minimum between two damages in the longitudinal direction.

11. Material removed is total for inside and outside at same location.

12. Blend in the part's longitudinal axis.

ICN-412-A-320106-B-97499-00003-A-001-01

Figure 1. Support Beam - Damage and Repair Limits (Sheet 3 of 3)

Aft Crosstube Support Beam - Installation

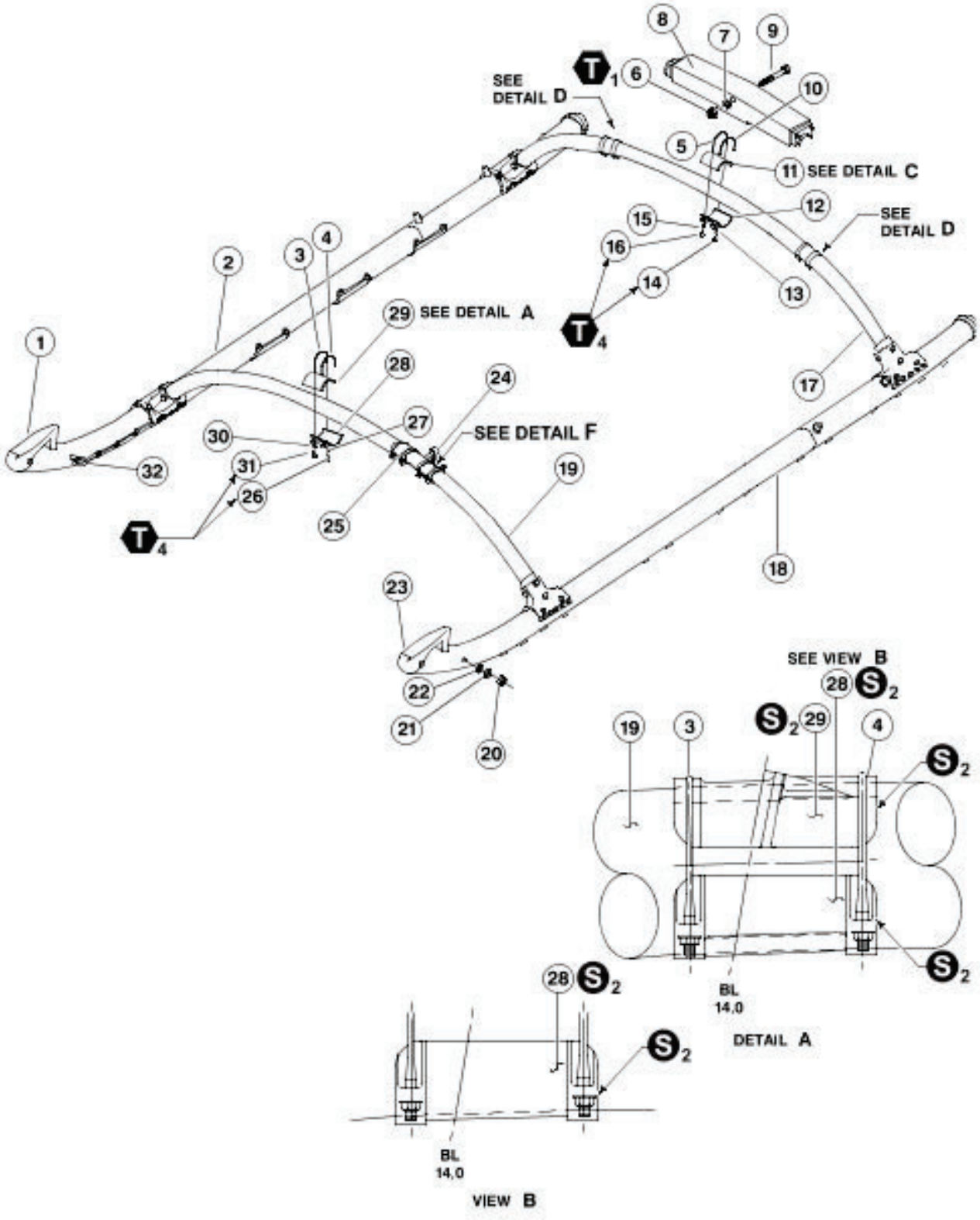
Procedure

1. Position aft crosstube support beam (8, Figure 1) on aft crosstube (17).
2. Align hole in aft crosstube support beam (8) with hole in aft upper crosstube attachment support (11).



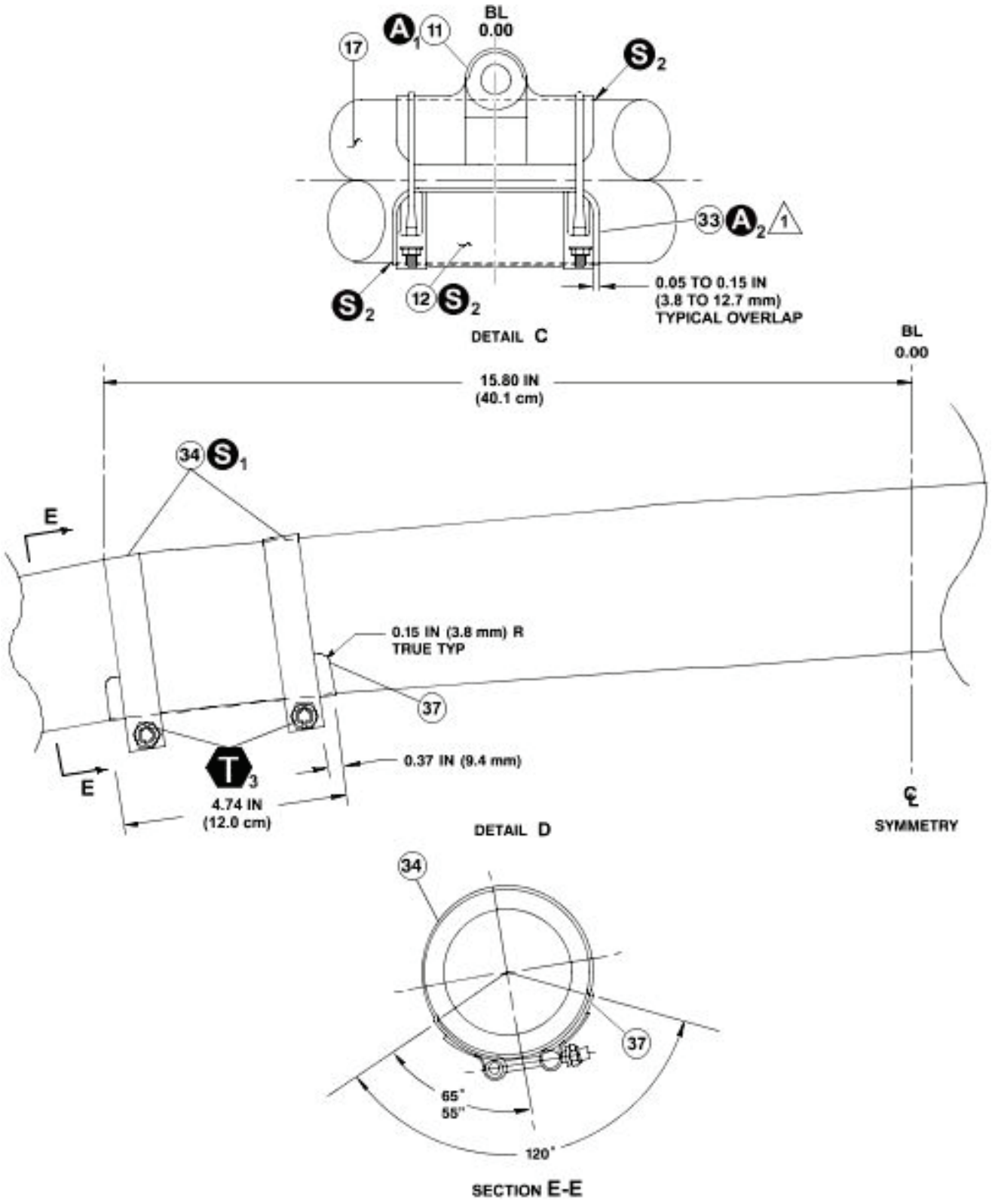
MAKE SURE THE WASHER (7) STAYS IN PLACE ON THE SHANK OF THE BOLT (9) DURING INSTALLATION OF THE NUT (6). OTHERWISE, THE WASHER COULD DROP INTO THE RADIUS BETWEEN THE SHANK AND THREADED PORTION OF THE BOLT AND CAUSE DAMAGE TO THE PARTS.

3. Install bolt (9), thin steel washer (7), and nut (6). Torque the nut and install the cotter pin.
4. Install landing gear (412-A-32-01-00-00A-720A-A / 00700).



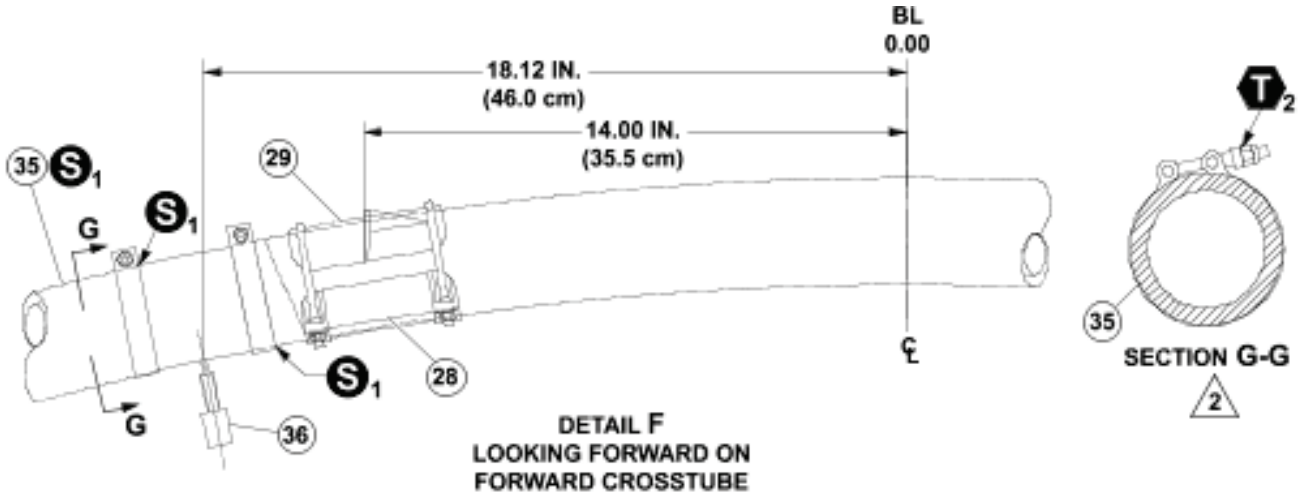
ICN-412-A-320106-B-97499-00004-A-001-01

Figure 1. Landing Gear Assembly (Sheet 1 of 3)



ICN-412-A-320106-B-97499-00005-A-001-01

Figure 1. Landing Gear Assembly (Sheet 2 of 3)



1. Strap
2. Right skid tube
3. U-bolt
4. U-bolt
5. U-bolt
6. Nut
7. Thin steel washer
8. Aft crosstube support beam
9. Bolt
10. U-bolt
11. Aft crosstube upper support
12. Aft crosstube lower support
13. Steel washer
14. Nut
15. Steel washer
16. Nut
17. Aft crosstube
18. Left skid tube
19. Forward crosstube
20. Nut
21. Aluminum washer
22. Aluminum Washer
23. Step
24. Power step electrical connector
25. Left forward crosstube support
26. Nut
27. Steel washer
28. Forward crosstube lower support
29. Forward crosstube upper support
30. Steel washer
31. Nut
32. Tow fitting
33. Abrasion strip
34. Clamp assembly
35. Clamp assembly
36. Weight-on-gear switch
37. Abrasion strip

A_1 ADHESIVE (C-317)

A_2 ADHESIVE (C-571)

S_1 SEALANT (C-251)

S_2 SEALANT (C-308)

T_1 10 TO 30 IN-LBS (1.13 TO 3.39 Nm)

T_2 40 IN-LBS (4.57 Nm)

T_3 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T_4 80 TO 100 IN-LBS (9.04 TO 11.30 Nm)

NOTES

1. Abrasion strip (33) must be removed for installation of AAI aft crosstube lower support half (12)

2. Weight-on-gear switch 3420S1 installed on helicopters S/N 36462 and subsequent. It is also installed on helicopters S/N 36248 through 36461 with the weather radar kit (412-705-034) or weight-on-gear retrofit kit (BHT-412-SI-84) installed.

ICN-412-A-320106-B-97499-00006-A-001-01

Figure 1. Landing Gear Assembly (Sheet 3 of 3)

Crosstube Supports - Description

NOTE

The Aeronautical Accessories Inc (AAI) aft crosstube assemblies (412-320-104/ 412-321-104) is similar to the Bell Helicopter Textron (BHT) crosstube assembly 412-050-045-ALL, however, different material is used on faying surfaces of abrasion strips (clamps) (BL 14.0), and lower support (BL 0.0). Refer to 412-A-32-01-09-00A-520A-A / 00740, 412-A-32-01-09-00A-280A-A / 00741, 412-A-32-01-09-00A-720A-A / 00742, 412-A-32-01-09-01A-520A-A / 00743, 412-A-32-01-09-01A-280A-A / 00744 and 412-A-32-01-09-01A-720A-A / 00745 for replacement of supports on AAI crosstubes.

The BHT aft crosstube assemblies 412-050-045-ALL and the AAI aft crosstube assemblies 412-320-104/412-321-104 have a limited life cycle.

Maintenance procedures for forward crosstube attachment supports of high gear and standard gear are the same.

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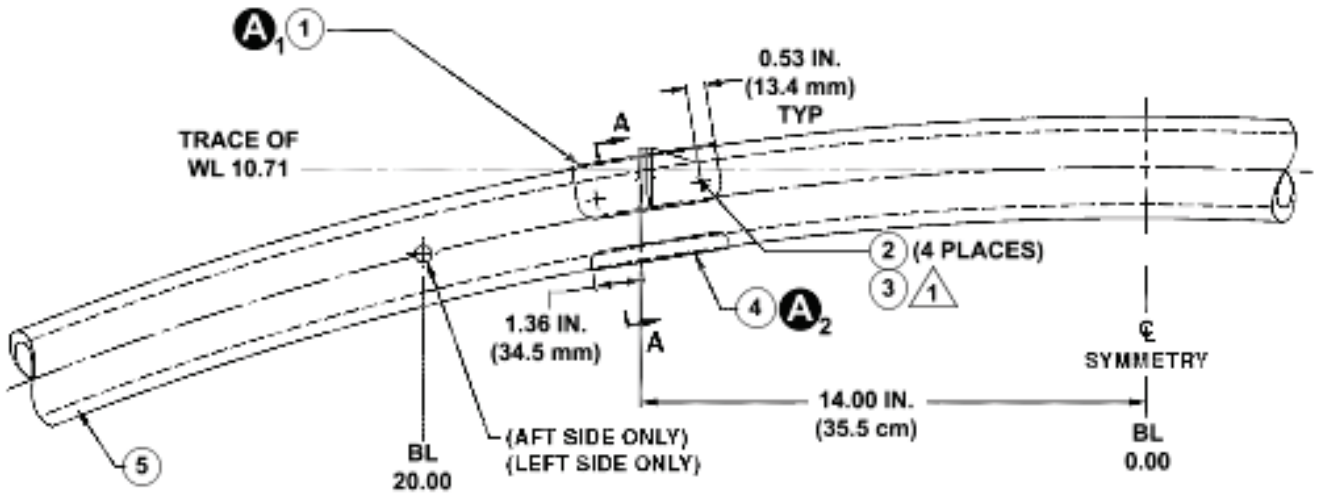
BHT Crosstube Forward Supports - Description

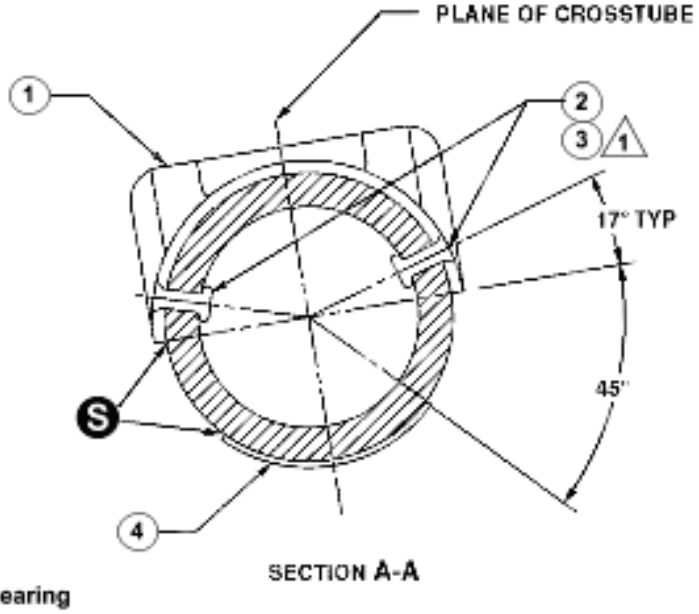
There are two BHT designs for the forward crosstube supports. The standard forward crosstube 205-050-400-055 and high forward crosstube 205-050-403-031 are fitted with an upper and lower support that are bonded to the crosstube.

The standard forward crosstube 412-050-040-101 and high forward crosstube 412-050-046-101 are fitted with a single upper support that is riveted to the crosstube. An abrasion strip is bonded to the lower surface.

The holes in the crosstube that accept the rivets are cold worked (Figure 1). An abrasion strip 206-050-301 is bonded to the underside of the crosstube at BL 14.0. Replacement of the forward crosstube support rivets must only be accomplished using the same size rivets.

Damage to the cold-worked holes up to 0.005 inch (0.12 mm) deep may be repaired by reaming the hole to 0.205 to 0.209 inch (5.21 to 5.31 mm) and installing a CR4652CW-6 fastener of the correct length. Contact Product Support Engineering for damage greater than 0.005 inch (0.12 mm) deep. Further damage assessment by Product Support Engineering is required before rework is attempted.





SECTION A-A

1. Support bearing
2. Rivet
3. Rivet
4. Abrasion strip
5. Crosstube

A1 ADHESIVE (C-252) A2 ADHESIVE (C-571) S SEALANT (C-392)

NOTE

1 Cold-worked hole can be reamed to 0.205 to 0.209 inch (5.21 to 5.31 mm) diameter to clean maximum damage of 0.005 inch (0.12 mm) depth, before installation of oversize rivet.

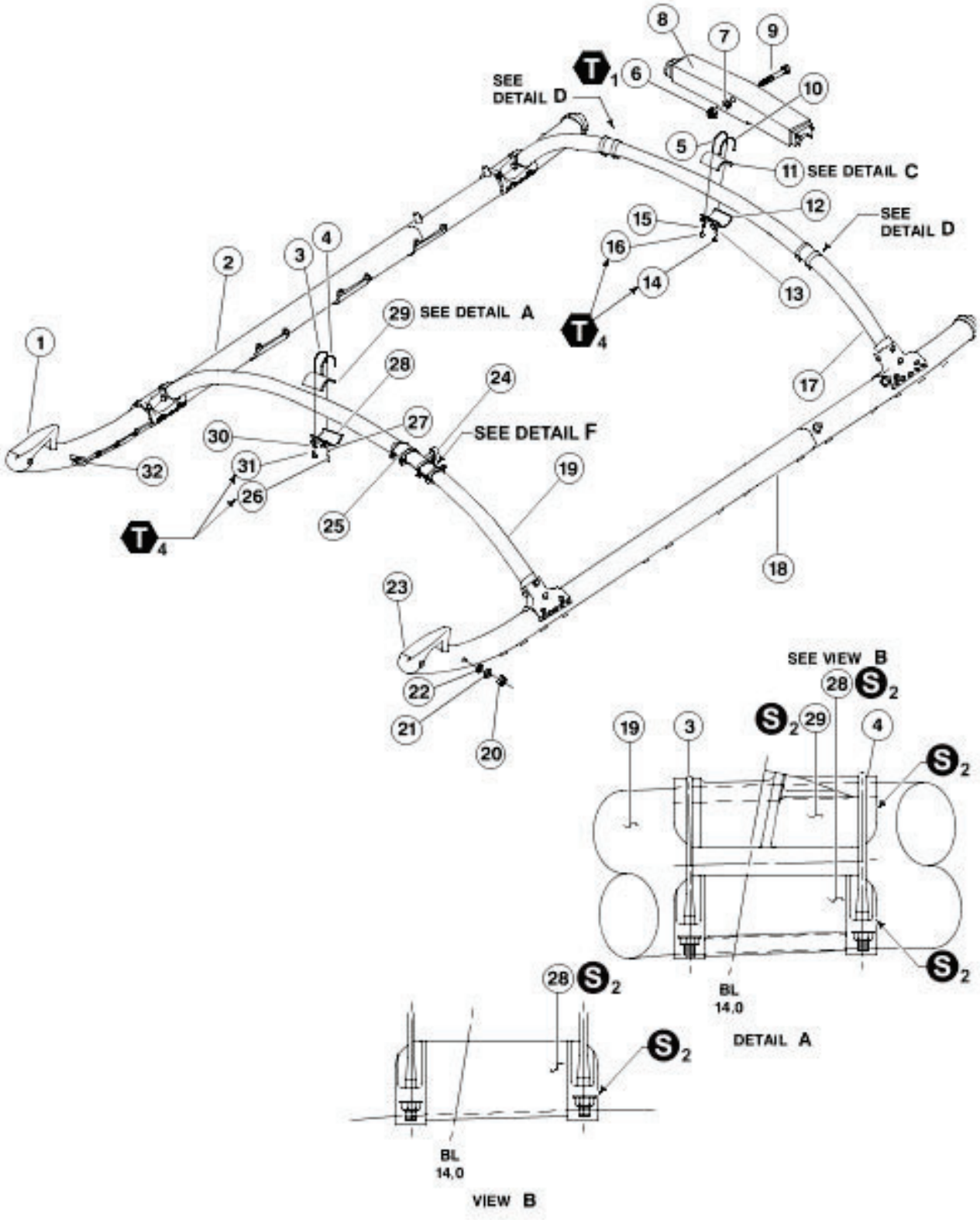
ICN-412-A-320107-B-97499-00001-A-001-01

Figure 1. Cold-work Hole Damage - Forward Crosstube (412-050-040-101/412-050-046-101) (Sheet 1 of 1)

Two-piece Support with U-Clamps (Crosstubes 205-050-400-055 and 205-050-403-031) - Removal

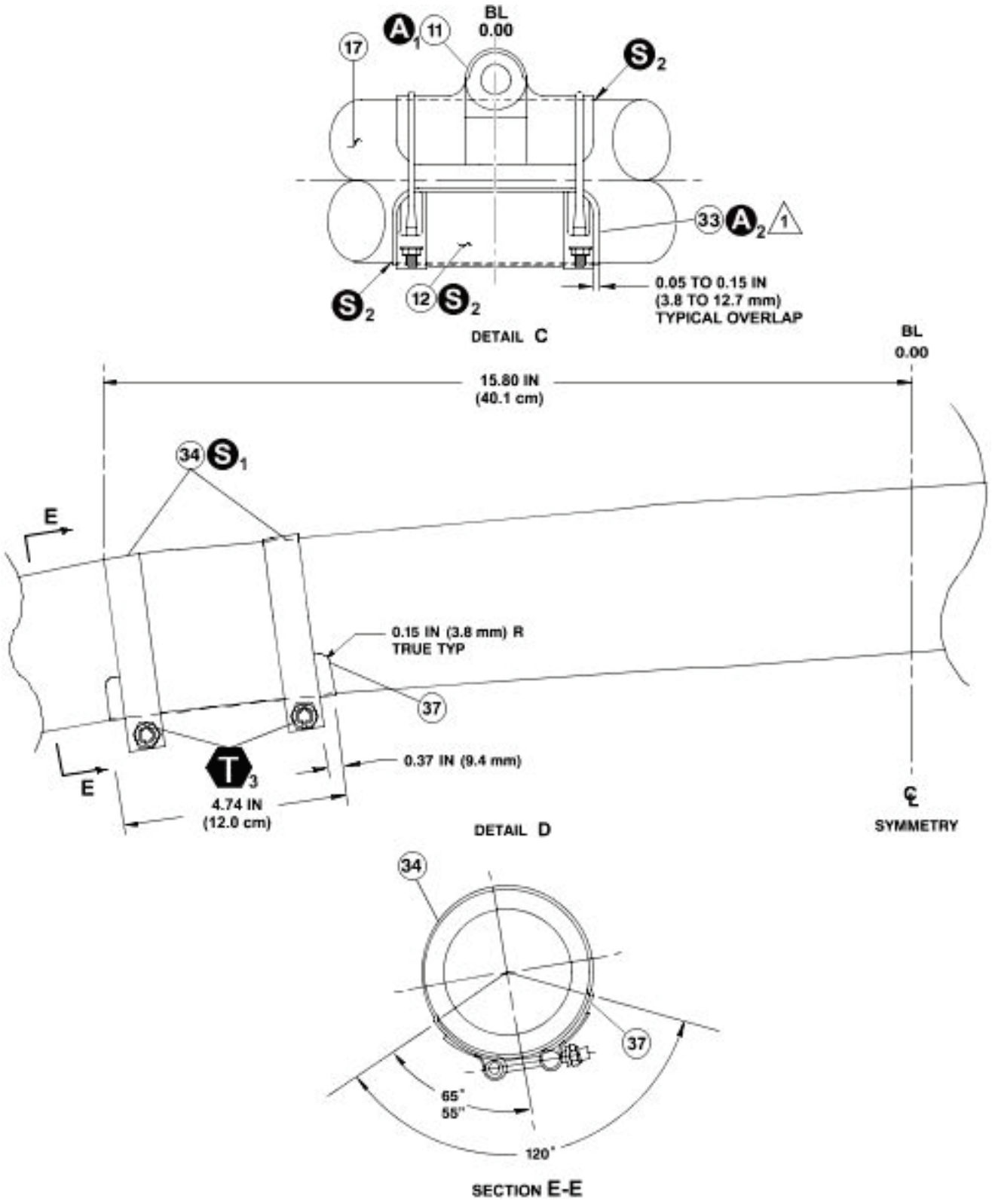
Procedure

1. Remove landing gear (412-A-32-01-00-00A-520A-A / 00697).
2. Remove nuts (26 and 31, Figure 1), steel washers (27 and 30), and U-bolts (3 and 4).
3. Remove forward crosstube supports (28 and 29).



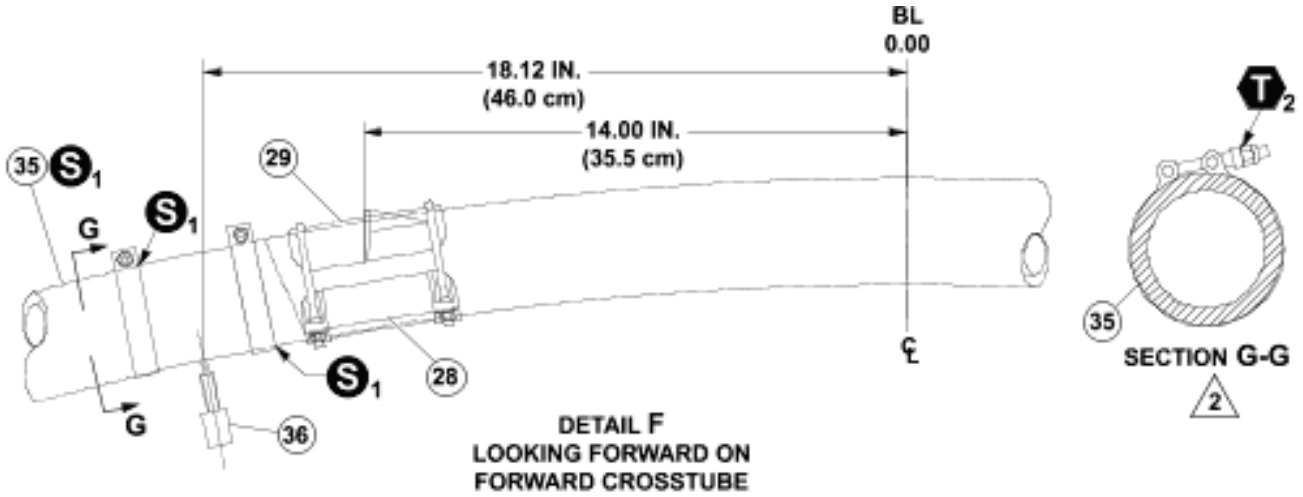
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Figure 1. Landing Gear Assembly (Sheet 1 of 3)



ICN-412-A-320106-B-97499-00005-A-001-01

Figure 1. Landing Gear Assembly (Sheet 2 of 3)



1. Strap
2. Right skid tube
3. U-bolt
4. U-bolt
5. U-bolt
6. Nut
7. Thin steel washer
8. Aft crosstube support beam
9. Bolt
10. U-bolt
11. Aft crosstube upper support
12. Aft crosstube lower support
13. Steel washer
14. Nut
15. Steel washer
16. Nut
17. Aft crosstube
18. Left skid tube
19. Forward crosstube
20. Nut
21. Aluminum washer
22. Aluminum Washer
23. Step
24. Power step electrical connector
25. Left forward crosstube support
26. Nut
27. Steel washer
28. Forward crosstube lower support
29. Forward crosstube upper support
30. Steel washer
31. Nut
32. Tow fitting
33. Abrasion strip
34. Clamp assembly
35. Clamp assembly
36. Weight-on-gear switch
37. Abrasion strip

A_1 ADHESIVE (C-317)

A_2 ADHESIVE (C-571)

S_1 SEALANT (C-251)

S_2 SEALANT (C-308)

T_1 10 TO 30 IN-LBS (1.13 TO 3.39 Nm)

T_2 40 IN-LBS (4.57 Nm)

T_3 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T_4 80 TO 100 IN-LBS (9.04 TO 11.30 Nm)

NOTES

1. Abrasion strip (33) must be removed for installation of AAI aft crosstube lower support half (12)

2. Weight-on-gear switch 3420S1 installed on helicopters S/N 36462 and subsequent. It is also installed on helicopters S/N 36248 through 36461 with the weather radar kit (412-705-034) or weight-on-gear retrofit kit (BHT-412-SI-84) installed.

ICN-412-A-320106-B-97499-00006-A-001-01

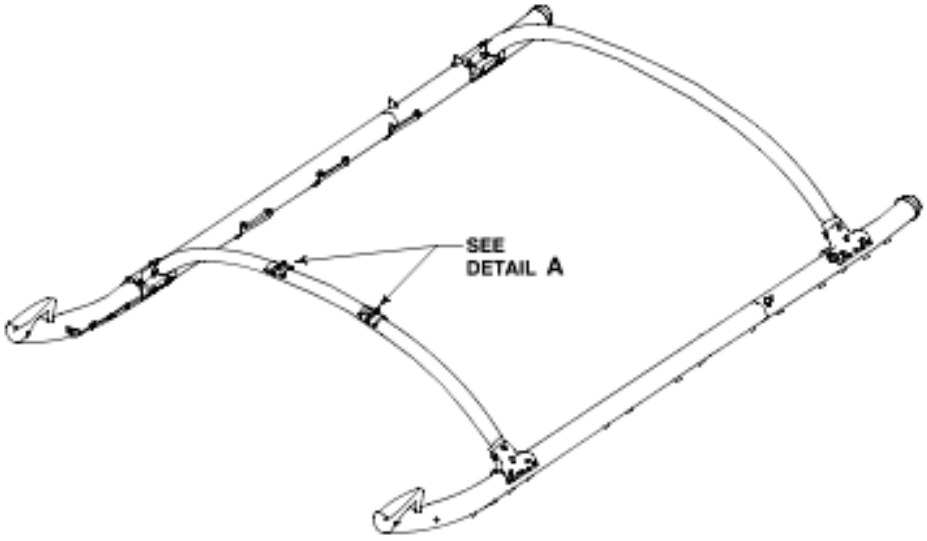
Figure 1. Landing Gear Assembly (Sheet 3 of 3)

One-piece Riveted Support (Crosstubes 412-050-040-101 and 412-050-046-101) - Removal Procedure

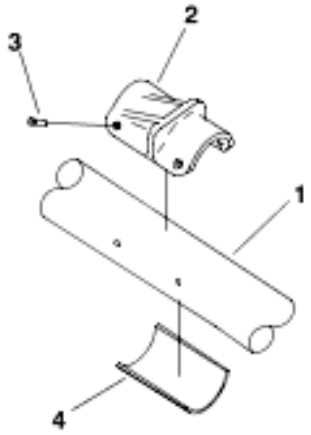


ONE-PIECE CROSSTUBE/SUPPORT FASTENER HOLES ARE COLD WORKED. REPLACEMENT OF THE SUPPORTS MUST ONLY BE ACCOMPLISHED USING THE SAME SIZE FASTENERS. IF DAMAGE HAS OCCURRED TO THE COLD-WORKED HOLE(S), DAMAGE UP TO 0.005 INCH (0.12 MM) DEEP MAY BE REPAIRED BY REAMING THE HOLE TO 0.205 TO 0.209 INCH (5.21 TO 5.30 MM) AND INSTALLING A CR4652CW-6 FASTENER OF THE CORRECT LENGTH. CONTACT PRODUCT SUPPORT ENGINEERING FOR DAMAGE GREATER THAN 0.005 INCH (0.12 MM) DEEP.

1. Remove landing gear (412-A-32-01-00-00A-520A-A / 00697).
2. Using caution, remove rivets (3, Figure 1) (BHT-ALL-SPM, Standard Practices Manual).
3. Remove support (2).



1. Forward crosstube
2. Support
3. Rivet
4. Abrasion strip



DETAIL A

ICN-412-A-320107-B-97499-00002-A-001-01

Figure 1. One-piece Riveted Supports (Sheet 1 of 1)

BHT Crosstube Forward Supports - Inspection and Repair

Procedure

NOTE

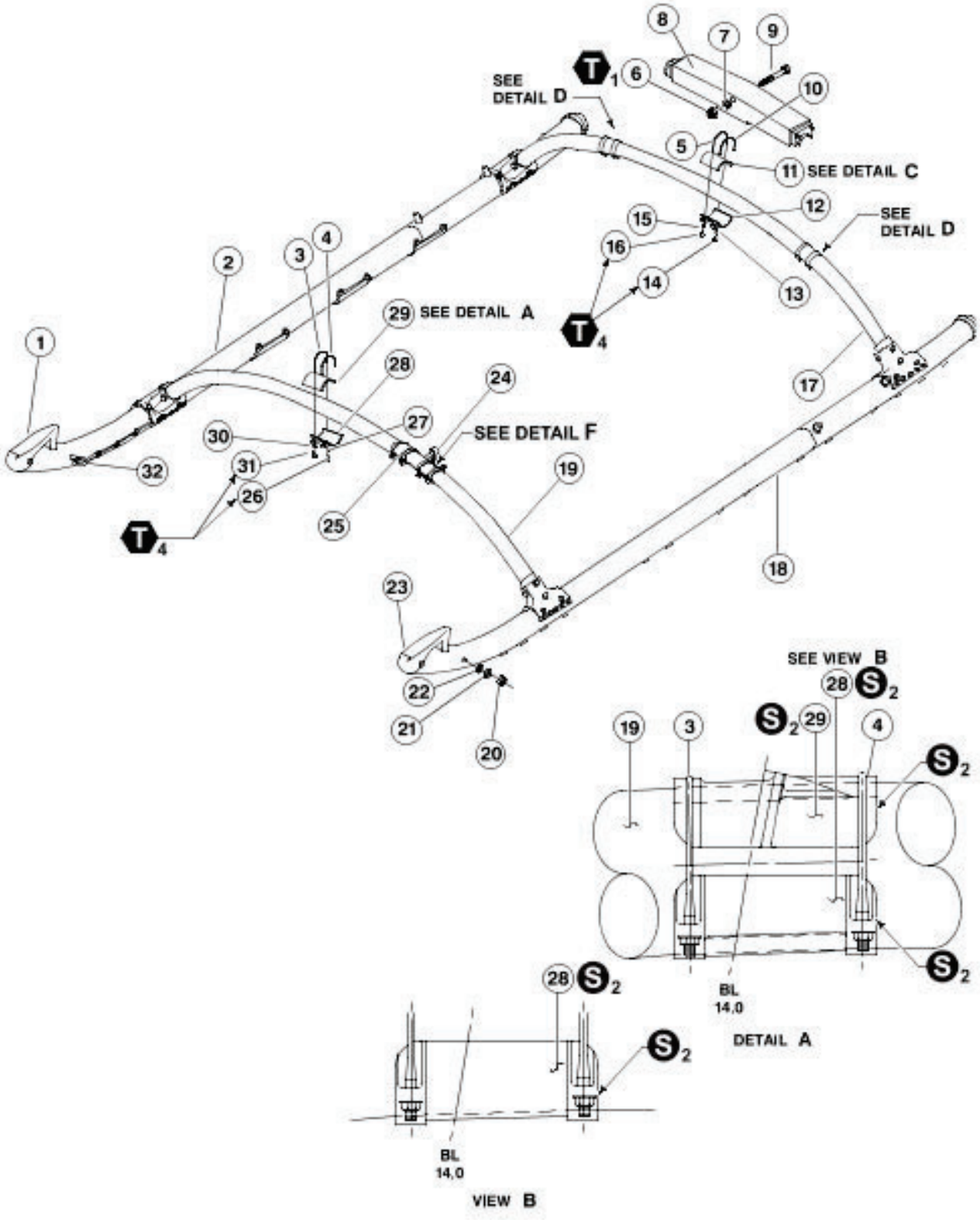
Superficial mechanical damage to bearing and retaining supports is acceptable without repair. Sharp corners and gouges should be repaired within allowable limits. Distortion and/or wear that can be detected visually is not acceptable.

1. Inspect forward crosstube supports (28 and 29, Figure 1) for mechanical and corrosion damage (Figure 2).

2. Inspect support 412-050-038 for mechanical and corrosion damage (Figure 2).

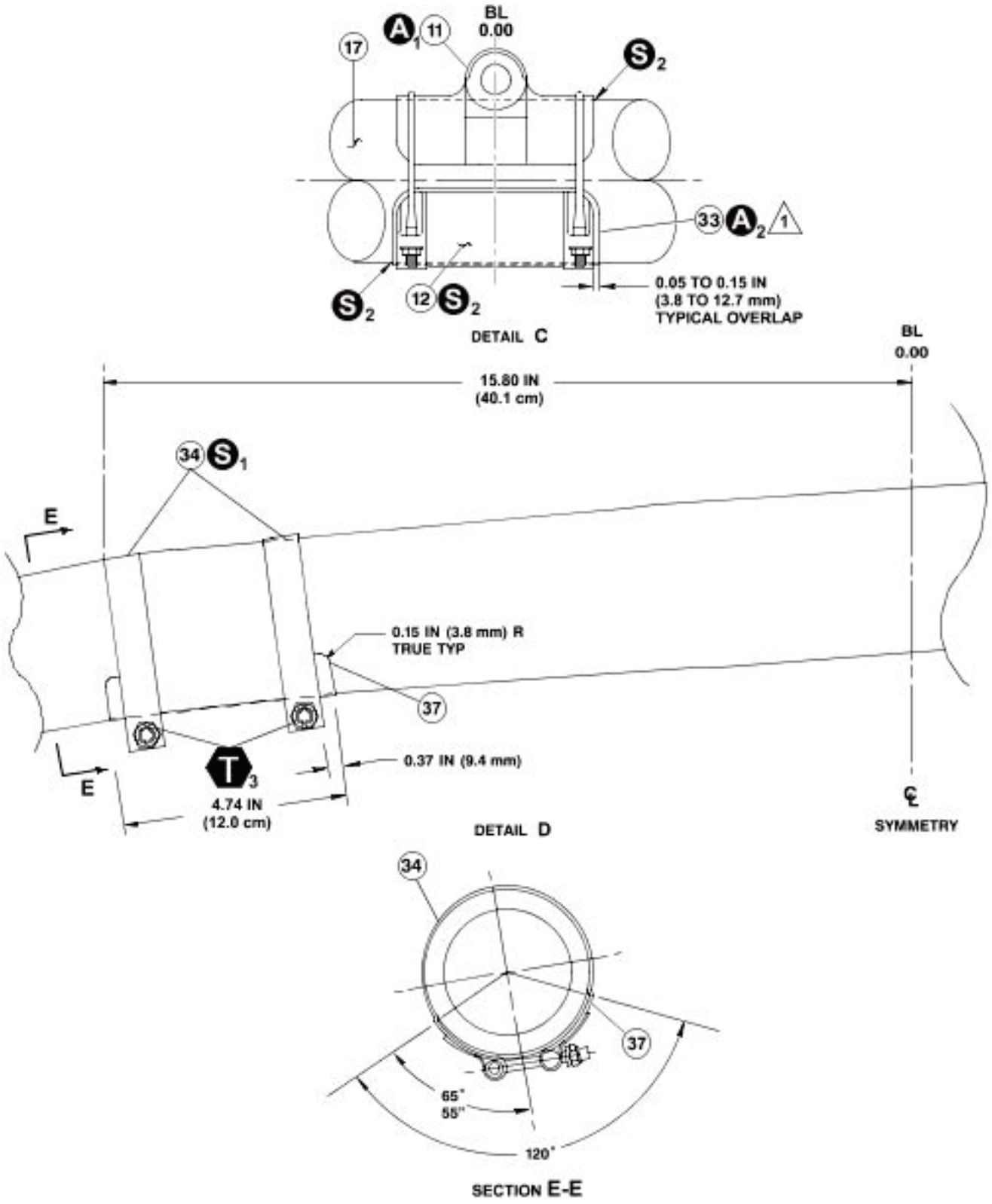
3. Inspect inner adhesive film of support 412-050-038 for condition and security. No voids permitted.

4. Scratches, dents, and corrosion may be polished out to maximum depths, as shown in Figure 2. No cracks are acceptable.



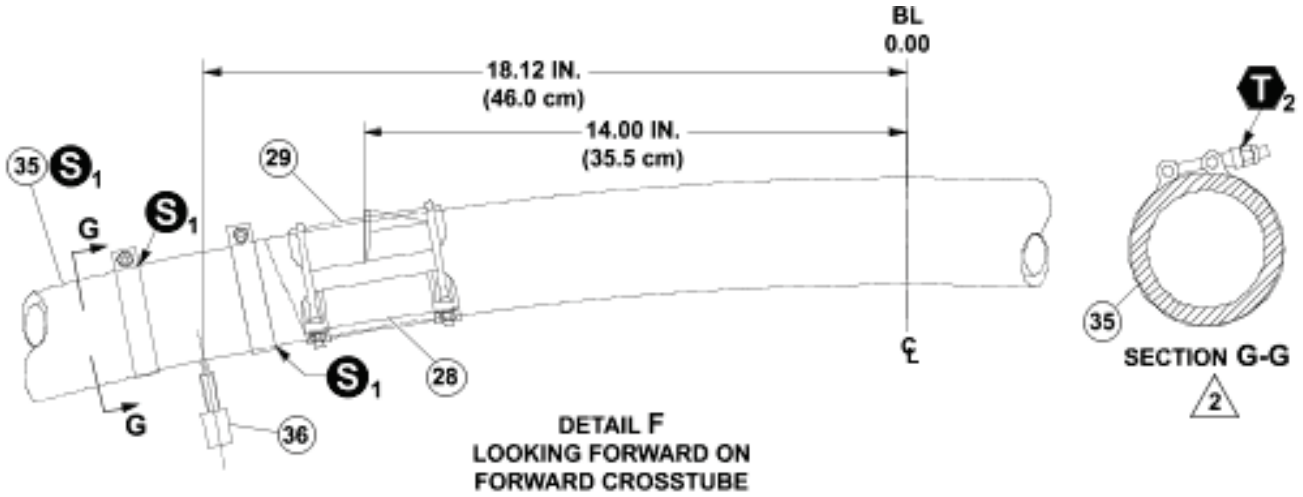
ICN-412-A-320106-B-97499-00004-A-001-01

Figure 1. Landing Gear Assembly (Sheet 1 of 3)



ICN-412-A-320106-B-97499-00005-A-001-01

Figure 1. Landing Gear Assembly (Sheet 2 of 3)



1. Strap
2. Right skid tube
3. U-bolt
4. U-bolt
5. U-bolt
6. Nut
7. Thin steel washer
8. Aft crosstube support beam
9. Bolt
10. U-bolt
11. Aft crosstube upper support
12. Aft crosstube lower support
13. Steel washer
14. Nut
15. Steel washer
16. Nut
17. Aft crosstube
18. Left skid tube
19. Forward crosstube
20. Nut
21. Aluminum washer
22. Aluminum Washer
23. Step
24. Power step electrical connector
25. Left forward crosstube support
26. Nut
27. Steel washer
28. Forward crosstube lower support
29. Forward crosstube upper support
30. Steel washer
31. Nut
32. Tow fitting
33. Abrasion strip
34. Clamp assembly
35. Clamp assembly
36. Weight-on-gear switch
37. Abrasion strip

A_1 ADHESIVE (C-317)

A_2 ADHESIVE (C-571)

S_1 SEALANT (C-251)

S_2 SEALANT (C-308)

T_1 10 TO 30 IN-LBS (1.13 TO 3.39 Nm)

T_2 40 IN-LBS (4.57 Nm)

T_3 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T_4 80 TO 100 IN-LBS (9.04 TO 11.30 Nm)

NOTES

1. Abrasion strip (33) must be removed for installation of AAI aft crosstube lower support half (12)

2. Weight-on-gear switch 3420S1 installed on helicopters S/N 36462 and subsequent. It is also installed on helicopters S/N 36248 through 36461 with the weather radar kit (412-705-034) or weight-on-gear retrofit kit (BHT-412-SI-84) installed.

ICN-412-A-320106-B-97499-00006-A-001-01

Figure 1. Landing Gear Assembly (Sheet 3 of 3)

FORWARD LANDING GEAR SUPPORTS



205-050-132



205-050-133

DAMAGE LOCATION SYMBOLS



TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR DEPTH
MECHANICAL	0.020 IN. (0.50 mm)
CORROSION DAMAGE (AFTER CLEANUP)	0.020 IN. (0.50 mm)
MAXIMUM AREA PER FULL DEPTH REPAIR	1.0 SQ. IN. (645.1 Sq. mm)
NUMBER OF REPAIRS	2
EDGE CHAMFER	0.050 IN. (1.27 mm)
BORE DAMAGE	0.020 IN. (0.50 mm) for 1/4 circumference

ICN-412-A-320105-B-97499-00002-A-001-01

Figure 2. Landing Gear Support Damage Limits (Sheet 1 of 3)

IMPROVED FORWARD LANDING GEAR SUPPORT



412-050-038

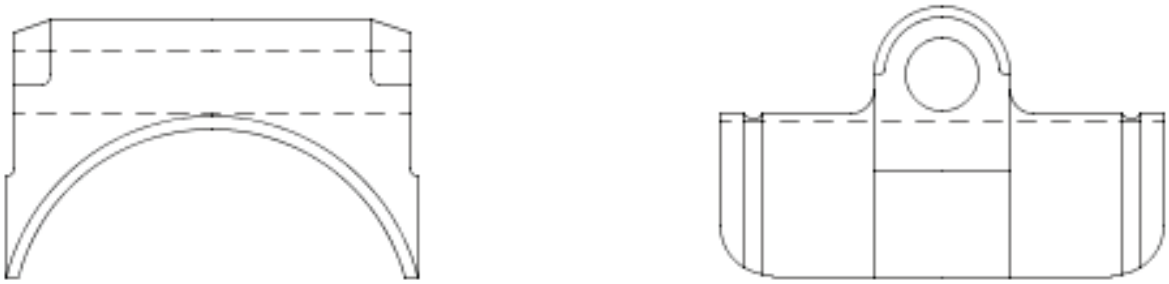
DAMAGE LOCATION SYMBOLS

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR DEPTH
MECHANICAL	0.020 IN. (0.50 mm)
CORROSION DAMAGE (AFTER CLEANUP)	0.020 IN. (0.50 mm)
MAXIMUM AREA PER FULL DEPTH REPAIR	1.0 SQ. IN. (645.1 Sq. mm)
NUMBER OF REPAIRS	2
EDGE CHAMFER	0.050 IN. (1.27 mm)
BORE DAMAGE	0.020 IN. (0.50 mm) for 1/4 circumference

ICN-412-A-320105-B-97499-00003-A-001-01

Figure 2. Landing Gear Support Damage Limits (Sheet 2 of 3)

AFT LANDING GEAR SUPPORTS



412-050-006



412-050-007

DAMAGE LOCATION SYMBOLS



TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR DEPTH
MECHANICAL	0.020 IN. (0.50 mm)
CORROSION DAMAGE (AFTER CLEANUP)	0.020 IN. (0.50 mm)
MAXIMUM AREA PER FULL DEPTH REPAIR	1.0 SQ. IN. (645.1 Sq. mm)
NUMBER OF REPAIRS	2
EDGE CHAMFER	0.050 IN. (1.27 mm)
BORE DAMAGE	0.020 IN. (0.50 mm) for 1/4 circumference

ICN-412-A-320105-B-97499-00004-A-001-01

Figure 2. Landing Gear Support Damage Limits (Sheet 3 of 3)

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Two-piece Support with U-Clamps (Crosstubes 205-050-400-055 and 205-050-403-031) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	
MEK	C-309	AR	
Abrasive cloth or paper	C-423	AR	
Clean cloth	C-516	AR	

Procedure

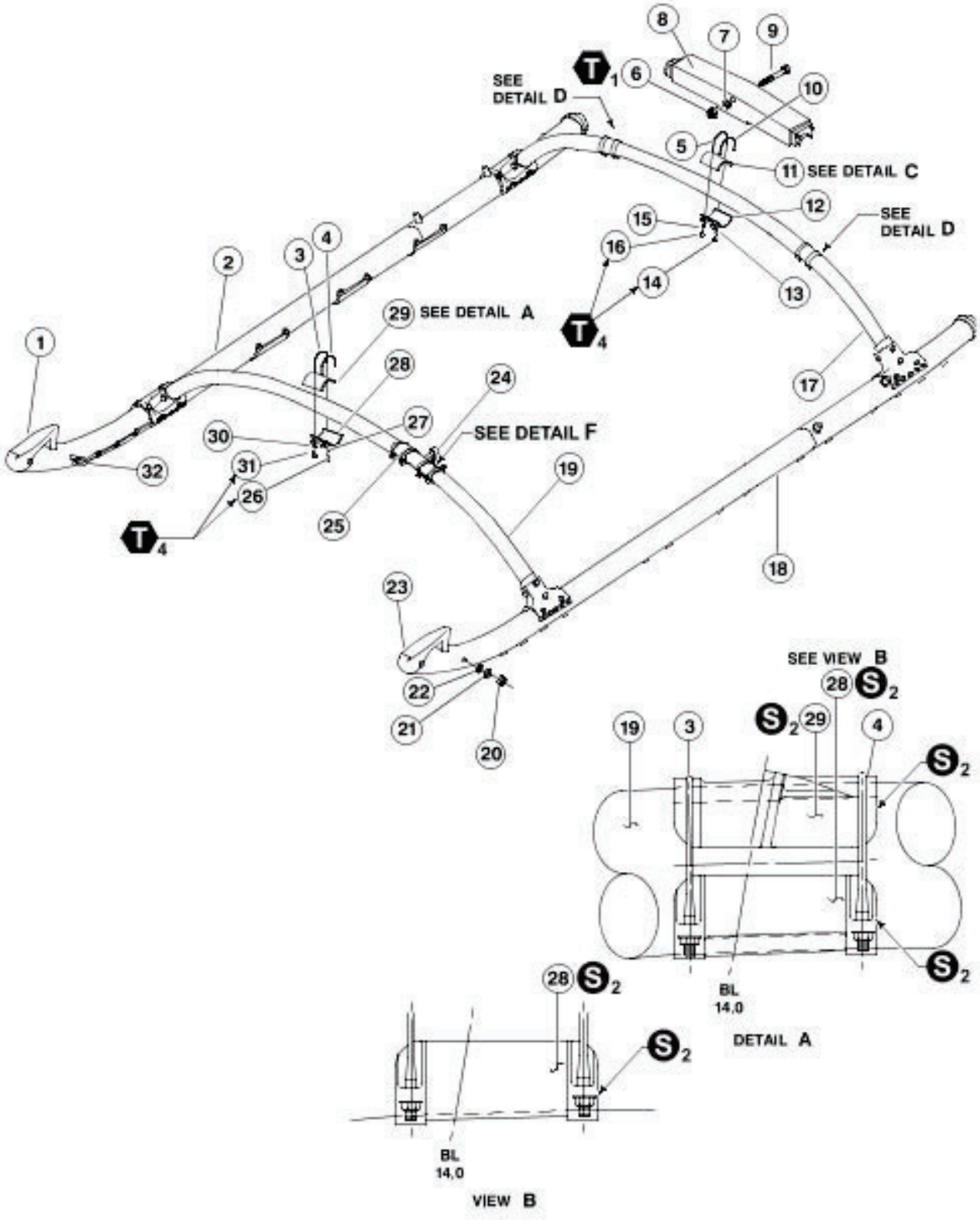


WHEN CLEANING FAYING SURFACES, DO NOT USE ANY TOOL OR SUBSTANCE THAT COULD SCRATCH TUBE OR SUPPORTS.



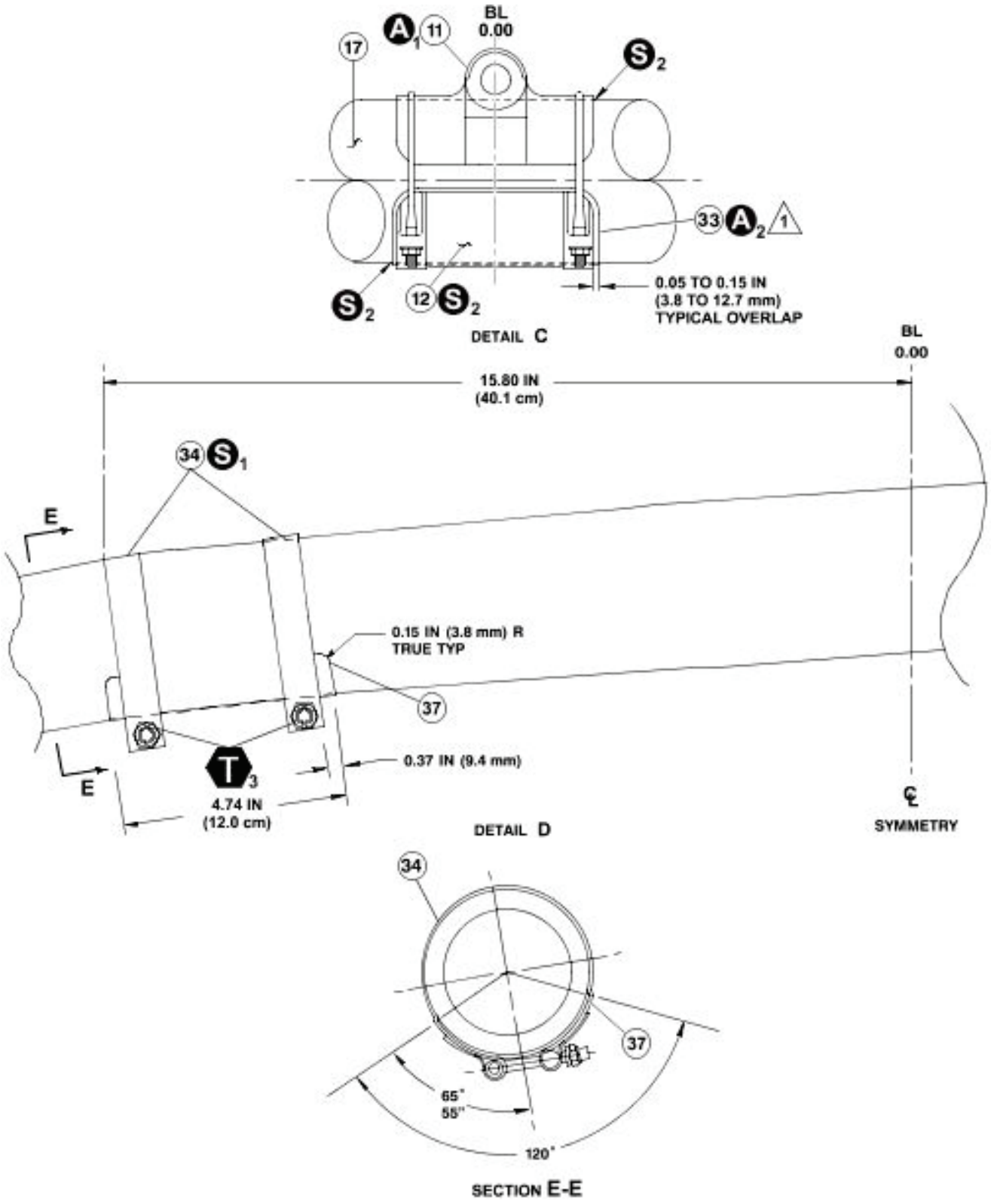
ABRADE CROSSTUBE SURFACE IN LONGITUDINAL DIRECTION. MAKE SURE YOU DO NOT ABRADE THE CROSSTUBE METAL SURFACE.

1. Clean mating surfaces by lightly abrading the forward crosstube (19, Figure 1) and forward crosstube supports (28 and 29) with 400 to 600 grit abrasive cloth or paper (C-423). Remove sanding residue with a clean cloth (C-516) dampened with MEK (C-309).
2. Coat mating surfaces of forward crosstube (19) and forward crosstube supports (28 and 29) and bond using sealant (C-308).
3. Position forward crosstube supports (28 and 29) and U-bolts (3 and 4) on forward crosstube (19) at location shown in Detail A.
4. Install two steel washers (27) and nuts (26). Install steel washers (30) and nuts (31).
5. Torque nuts (26 and 31) within 30 minutes after application of adhesive ◍.
6. Apply a bead of sealant (C-308) around periphery of forward crosstube supports (28 and 29).
7. Check torque on nuts (26 and 31) after adhesive is cured (24 hours) and apply torque as required ◍.
8. Install landing gear (412-A-32-01-00-00A-720A-A / 00700).



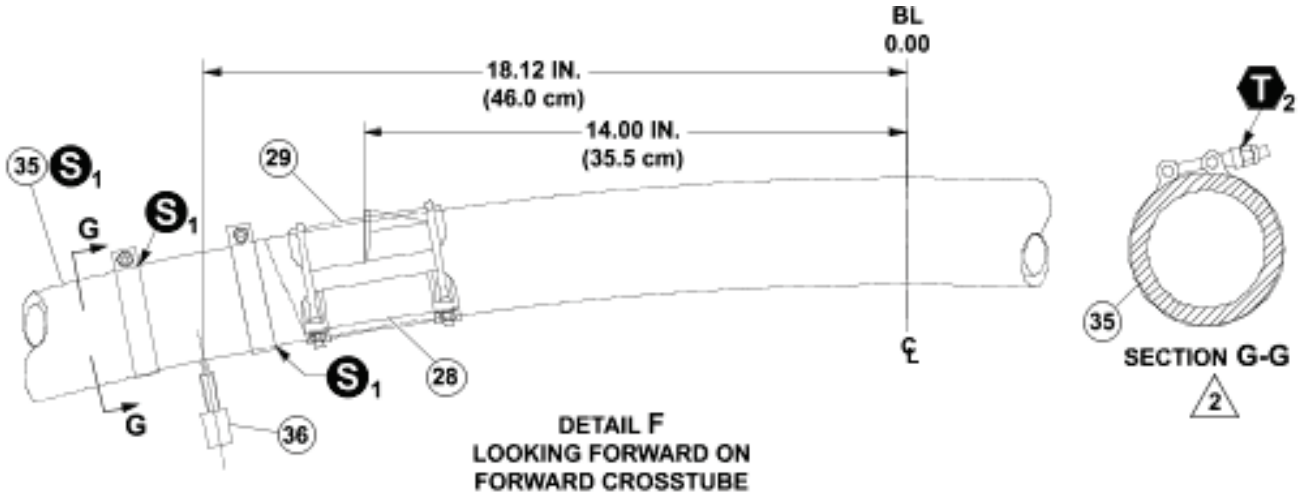
ICN-412-A-320106-B-97499-00004-A-001-01

Figure 1. Landing Gear Assembly (Sheet 1 of 3)



ICN-412-A-320106-B-97499-00005-A-001-01

Figure 1. Landing Gear Assembly (Sheet 2 of 3)



1. Strap
2. Right skid tube
3. U-bolt
4. U-bolt
5. U-bolt
6. Nut
7. Thin steel washer
8. Aft crosstube support beam
9. Bolt
10. U-bolt
11. Aft crosstube upper support
12. Aft crosstube lower support
13. Steel washer
14. Nut
15. Steel washer
16. Nut
17. Aft crosstube
18. Left skid tube
19. Forward crosstube
20. Nut
21. Aluminum washer
22. Aluminum Washer
23. Step
24. Power step electrical connector
25. Left forward crosstube support
26. Nut
27. Steel washer
28. Forward crosstube lower support
29. Forward crosstube upper support
30. Steel washer
31. Nut
32. Tow fitting
33. Abrasion strip
34. Clamp assembly
35. Clamp assembly
36. Weight-on-gear switch
37. Abrasion strip

A1 ADHESIVE (C-317)

A2 ADHESIVE (C-571)

S1 SEALANT (C-251)

S2 SEALANT (C-308)

T1 10 TO 30 IN-LBS (1.13 TO 3.39 Nm)

T2 40 IN-LBS (4.57 Nm)

T3 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T4 80 TO 100 IN-LBS (9.04 TO 11.30 Nm)

NOTES

1. Abrasion strip (33) must be removed for installation of AAI aft crosstube lower support half (12)

2. Weight-on-gear switch 3420S1 installed on helicopters S/N 36462 and subsequent. It is also installed on helicopters S/N 36248 through 36461 with the weather radar kit (412-705-034) or weight-on-gear retrofit kit (BHT-412-SI-84) installed.

ICN-412-A-320106-B-97499-00006-A-001-01

Figure 1. Landing Gear Assembly (Sheet 3 of 3)

One-piece Riveted Support (Crosstubes 412-050-040-101 and 412-050-046-101) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Epoxy polyamide primer	C-204	AR	
Sealant	C-251	AR	
Adhesive film	C-252	AR	
Adhesive primer	C-253	AR	
Drycleaning solvent	C-304	AR	
Adhesive	C-317	AR	
Denatured alcohol	C-326	AR	
Abrasive pad	C-407	AR	
Abrasive cloth or paper	C-423	AR	
Dry blasting abrasive	C-474	AR	
Release fabric	C-478	AR	
Clean cloth	C-516	AR	
Adhesive	C-571	AR	

Procedure

1. Replace support (2, Figure 1) and abrasion strip (4) on forward crosstube (1) (Left Buttock Line (LBL) 14.0 and Right Buttock Line (RBL) 14.0) as follows:

1.1. Prepare faying surface of support (2) as follows:

1.1.1. Remove oils and grease with drycleaning solvent (C-304).

1.1.2. Blast the faying surface with 80 grit dry blasting abrasive (C-474) at 50 to 80 PSI (344.7 to 551.5 kPa).

1.1.3. Flush the area with drycleaning solvent (C-304) to remove the blasting residue.

1.1.4. Apply adhesive primer (C-253) to cleaned area of the support (2). Cure at 65 to 80°F (18.3 to 26.6°C) for a minimum of 2 hours.

1.1.5. Apply adhesive film (C-252) to the adhesive primed surface. Place release fabric (C-478) to the inside faying surface of the support (2).

NOTE

Provide a film or other release barrier to the caul sheet to prevent bonding to the adhesive film.

1.1.6. Place a caul sheet over the adhesive surface to provide a uniform and smooth surface when cured.

1.1.7. Cure at a maximum bondline temperature of 290°F (143.3°C) with a maximum temperature rate of 1°F (0.5°C) per minute. Cure time shall be 60 minutes minimum at a bondline pressure of 20 to 28 inches (67.8 to 94.8 kPa) of mercury (vacuum) or 20 to 125 PSI (137.9 to 861.8 kPa).

1.2. Prepare forward crosstube upper surface (LBL 14.0 and RBL 14.0) as follows:

1.2.1. Remove primer from forward crosstube (1) in the area 2.5 inches (63.5 mm) either side of BL 14.0, 5.0 inches (127.0 mm) wide.

1.2.2. Prepare faying surface area of forward crosstube (1) 0.50 inch (12.7 mm) beyond the edge of support (2) by dry blasting abrasive (C-474) using 220 grit aluminum oxide at 30 to 40 PSI (206.8 to 275.8 kPa).



AVOID CONTACT BETWEEN THE PAINTED SURFACE AND THE ALCOHOL.

1.2.3. Wipe with clean cloth dampened with denatured alcohol (C-326). Wipe dry with a clean cloth (C-516).



DO NOT CURE ADHESIVE PRIMER AT ELEVATED TEMPERATURES.

1.2.4. Apply adhesive primer (C-253) to cleaned area of the forward crosstube (1). Cure at room temperature for a minimum of 2 hours.

1.3. Install support as follows:

1.3.1. Remove release fabric (C-478) from inner surface of support (2).

1.3.2. Lightly sand adhesive with 180 to 200 abrasive cloth or paper (C-423). Wipe with clean cloth dampened with denatured alcohol (C-326). Wipe dry with a clean cloth (C-516).



NEW CROSSTUBE/SUPPORT FASTENER HOLES ARE COLD WORKED. REPLACEMENT OF THE SUPPORTS MUST ONLY BE ACCOMPLISHED USING THE SAME SIZE FASTENERS. IF DAMAGE HAS OCCURRED TO THE COLD-WORKED HOLE(S), A DAMAGE ASSESSMENT BY PRODUCT SUPPORT ENGINEERING IS REQUIRED BEFORE REWORK/REPAIR IS ATTEMPTED.

1.3.3. Bond support (2) in place on forward crosstube (1) using adhesive (C-317) and wet install rivets (3) with sealant (C-251). Remove excess adhesive and cure at room temperature.

1.3.4. Apply epoxy polyamide primer (C-204) to the entire unprimed area of the forward crosstube (1), the edges of the adhesive bondline, and the edges of the support (2). Apply fillet seal using sealant (C-251) all around support and adhesive edges.

1.4. Install abrasion strip (4) as follows:



WHEN PREPARING THE LOWER SURFACE OF CROSSTUBE, MAKE SURE THAT THE epoxy polyamide primer (C-204) IS NOT REMOVED.



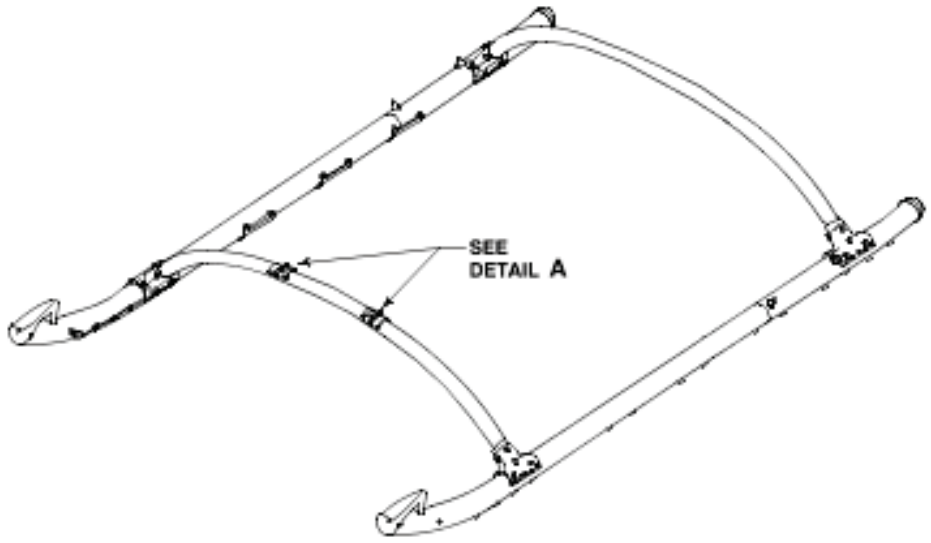
DO NOT USE AN ALTERNATE ADHESIVE FOR THIS APPLICATION.

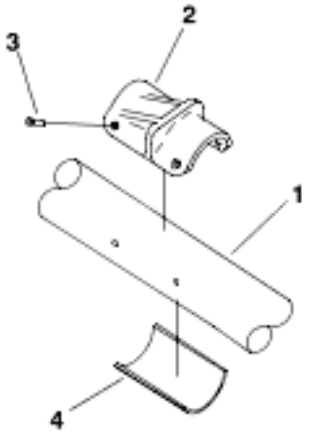
1.4.1. Lightly abrade the lower surface of the forward crosstube (1) common to the abrasion strip (4) using abrasive pad (C-407). Wipe with clean cloth dampened with denatured alcohol (C-326). Wipe dry with a clean cloth (C-516).

1.4.2. Bond the abrasion strip (4) in place using adhesive (C-571). Cure at room temperature for a minimum of 24 hours.

1.4.3. Seal edges of abrasion strip (4) using sealant (C-251).

2. Install landing gear (412-A-32-01-00-00A-720A-A / 00700).





DETAIL A

1. Forward crosstube
2. Support
3. Rivet
4. Abrasion strip

ICN-412-A-320107-B-97499-00002-A-001-01

Figure 1. One-piece Riveted Supports (Sheet 1 of 1)

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BHT Crosstube Aft Supports - Maintenance

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Adhesive film	C-252	AR	
Adhesive primer	C-253	AR	
Adhesive	C-317	AR	
Sealant	C-251	AR	

Procedure

1. There are two methods for bonding the band style clamp at BL 14.0. The first method installs clamp (330C400-338S) to the crosstube using sealant MIL-S-81733. The second method uses the same clamp with a composite material bonded to the inside surface for enhanced corrosion protection. The clamp assembly is reidentified as 412-050-045-115/-117. This method also incorporates an abrasion strip that is bonded to the lower surface of the crosstube prior to the bonding of the clamp assembly to the crosstube using MIL-S-81733.

2. In the center position BL 0.0, the upper support has a adhesive primer (C-253) and a ply of adhesive film (C-252) bonded to the inside faying surface. This upper support is bonded to the crosstube surface prepared with adhesive primer (C-253). The support is finally bonded to the prepared surface using adhesive (C-317).

3. The lower support is bonded to an abrasion strip (Larson L101) (raw material size 0.020 x 5.0 x 6.0 inches) that is bonded to the crosstube. This support is bonded into position on the lower surface of the crosstube using sealant (C-251).

4. The AAI lower support P/N 604-026-003 comes with black nylon abrasion material bonded to the support. The Larson L101 abrasion strip is not used with the support.

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BHT Aft Crosstube Supports - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Denatured alcohol	C-326	AR	
Clean cloth	C-516	AR	

Procedure

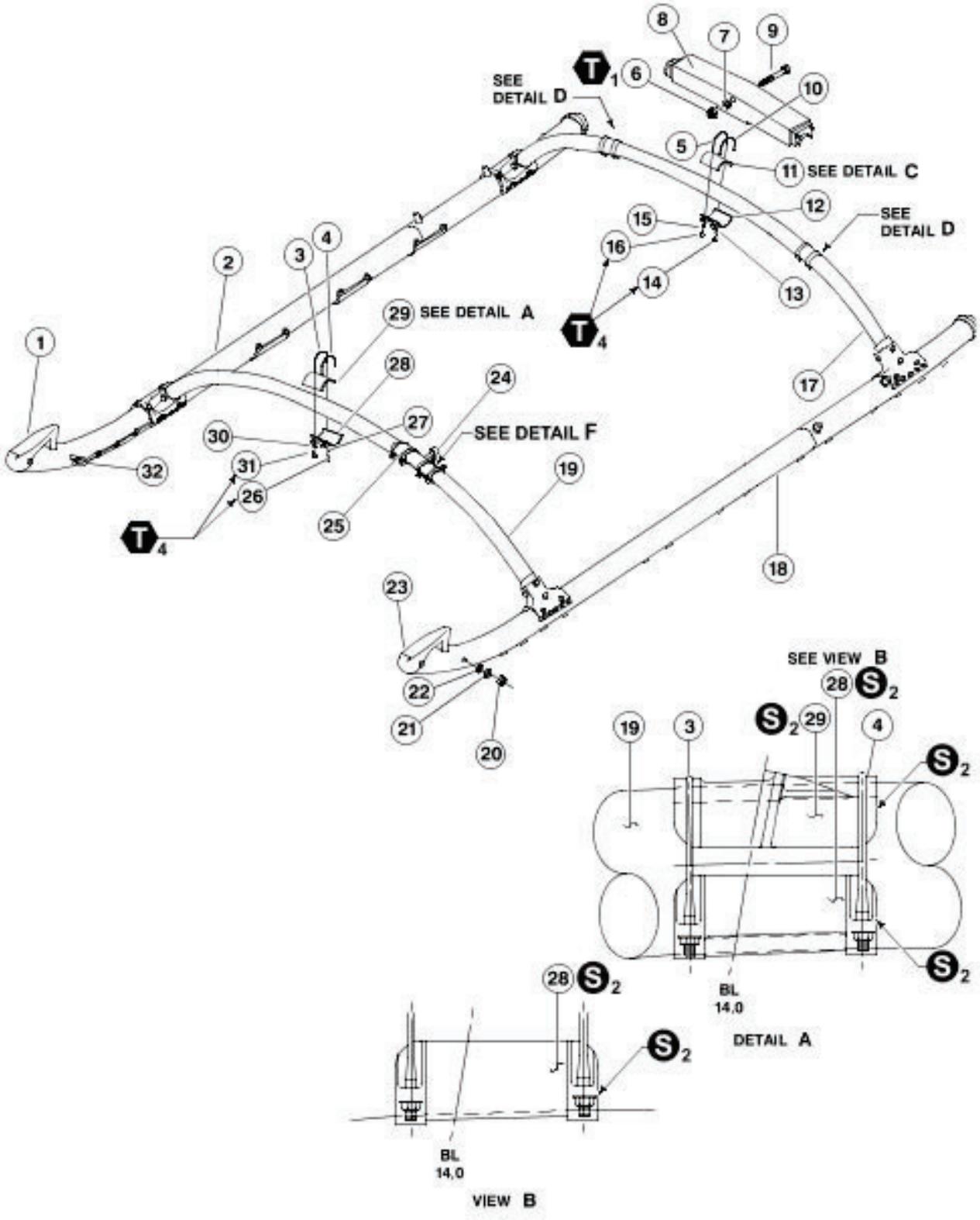
NOTE

Maintenance procedures for aft crosstube supports on high gear and standard gear are the same.

NOTE

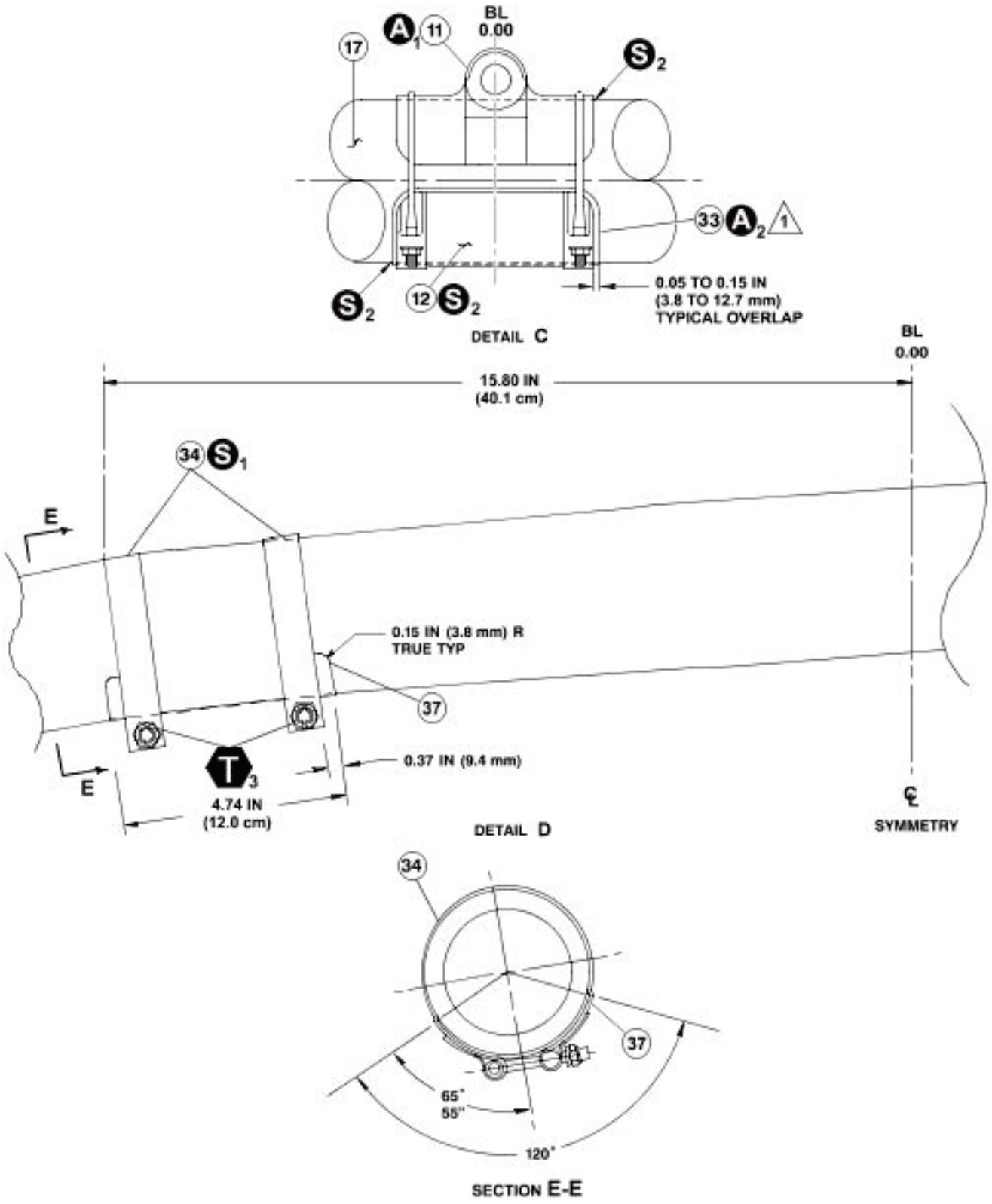
The upper center support on the aft crosstubes is bonded with structural adhesive and is not designed for removal. The following removal procedure is supplied in case bondline separation is detected during routine inspection.

1. Remove aft crosstube support beam (8, Figure 1) (412-A-32-01-06-00A-520A-A / 00725).
2. Remove nuts (14 and 16), steel washers (13 and 15), and U-bolts (5 and 10).
3. Remove the aft crosstube supports (11 and 12) from the aft crosstube (17). Use of a heat gun to soften the adhesive may facilitate removal. Do not exceed 180°F (82°C).
4. Remove the nuts from clamp assembly (34).
5. Carefully remove the clamp assembly (34) from the aft crosstube (17) and abrasion strip (37). Use of a heat gun to soften the adhesive may facilitate removal. Do not exceed 180°F (82°C).
6. Using a plastic scraper, remove the old adhesive from aft crosstube (17) and clamp assembly (34). Clean with a clean cloth (C-516) and denatured alcohol (C-326).



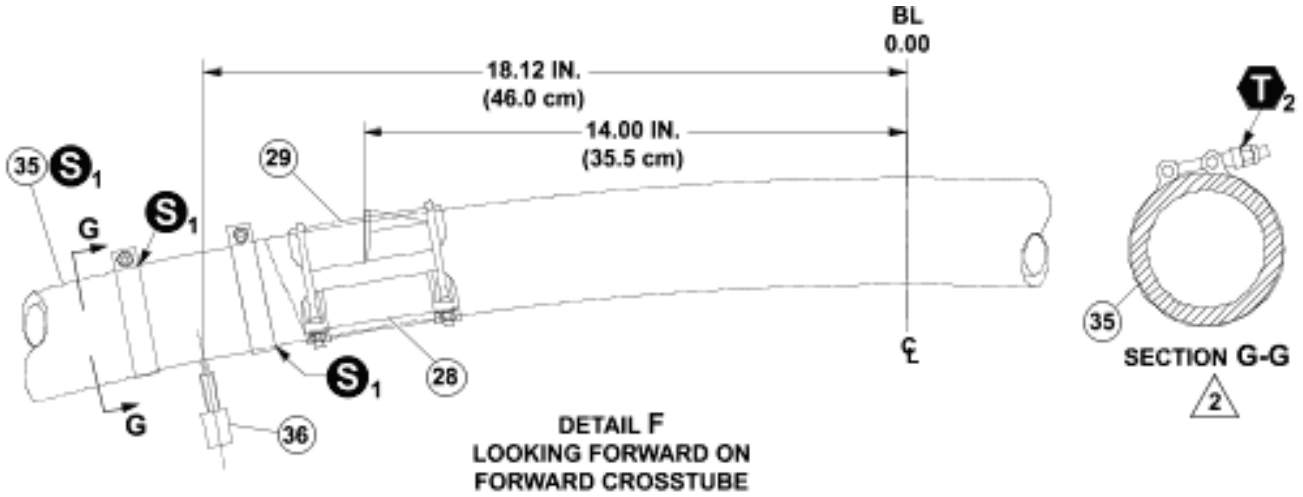
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Figure 1. Landing Gear Assembly (Sheet 1 of 3)



ICN-412-A-320106-B-97499-00005-A-001-01

Figure 1. Landing Gear Assembly (Sheet 2 of 3)



1. Strap
2. Right skid tube
3. U-bolt
4. U-bolt
5. U-bolt
6. Nut
7. Thin steel washer
8. Aft crosstube support beam
9. Bolt
10. U-bolt
11. Aft crosstube upper support
12. Aft crosstube lower support
13. Steel washer
14. Nut
15. Steel washer
16. Nut
17. Aft crosstube
18. Left skid tube
19. Forward crosstube
20. Nut
21. Aluminum washer
22. Aluminum Washer
23. Step
24. Power step electrical connector
25. Left forward crosstube support
26. Nut
27. Steel washer
28. Forward crosstube lower support
29. Forward crosstube upper support
30. Steel washer
31. Nut
32. Tow fitting
33. Abrasion strip
34. Clamp assembly
35. Clamp assembly
36. Weight-on-gear switch
37. Abrasion strip

A_1 ADHESIVE (C-317)

A_2 ADHESIVE (C-571)

S_1 SEALANT (C-251)

S_2 SEALANT (C-308)

T_1 10 TO 30 IN-LBS (1.13 TO 3.39 Nm)

T_2 40 IN-LBS (4.57 Nm)

T_3 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T_4 80 TO 100 IN-LBS (9.04 TO 11.30 Nm)

NOTES

1. Abrasion strip (33) must be removed for installation of AAI aft crosstube lower support half (12)

2. Weight-on-gear switch 3420S1 installed on helicopters S/N 36462 and subsequent. It is also installed on helicopters S/N 36248 through 36461 with the weather radar kit (412-705-034) or weight-on-gear retrofit kit (BHT-412-SI-84) installed.

ICN-412-A-320106-B-97499-00006-A-001-01

Figure 1. Landing Gear Assembly (Sheet 3 of 3)

Abrasion Strip (Larson L101) - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Plastic (Phenolic) Wedge	Commercial	1	
Protective Gloves	Commercial	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Cooling agent	C-137	AR	

Procedure



DO NOT USE ANY TYPE OF SANDPAPER OR METAL TOOLS TO REMOVE THE ABRASION STRIP. THIS CAN DAMAGE THE SURFACE OF THE CROSSTUBE.

DO NOT USE HEAT TO REMOVE THE ABRASION STRIP. HEATING MORE THAN 250°F (121°C) WILL ADVERSELY AFFECT THE CONDITION OF THE CROSSTUBE.

1. For the aft crosstube, remove the lower support (BL 0.00) (412-A-32-01-08-01A-520A-A / 00736).
2. Using the plastic (phenolic) wedge, remove any remaining sealant along the edges of the Larson L101 abrasion strip (2, Figure 1).



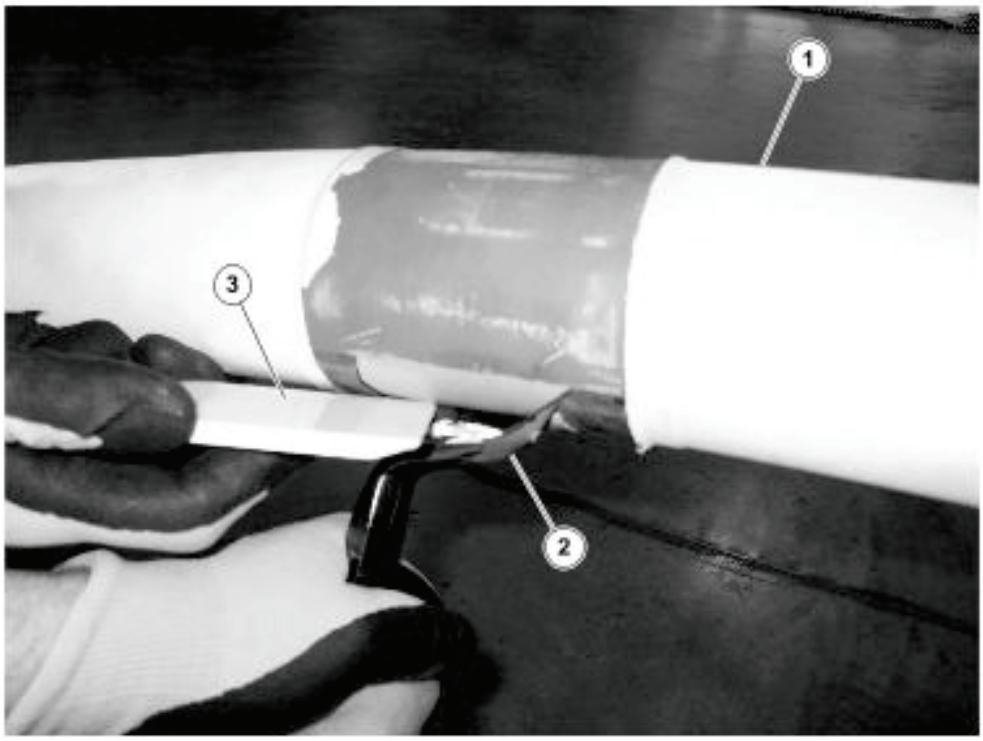
REFER TO THE MANUFACTURERS INSTRUCTIONS FOR THE PROPER USE OF cooling agent (C-137). THE TEMPERATURES OF THE COOLING AGENT CAN REACH -55°F (-48°C). WEAR protective gloves.

3. Apply cooling agent (C-137) to one edge of the Larson L101 abrasion strip (2). Use a brief application (or until frost appears).
4. Initiate peeling of the Larson L101 abrasion strip (2) with the plastic (phenolic) wedge (3).
5. Once peeling has been initiated, grip the abrasion strip (2) with pliers.

NOTE

Keep the bondline of the adhesive frozen. This will make removal easier.

6. Apply cooling agent (C-137) to the bondline of the adhesive. Use a brief application (or until frost appears).
7. Use the plastic (phenolic) wedge (3) and continue to gently pull the abrasion strip (2) with pliers.
8. Repeat Step 5. through Step 7. until the abrasion strip (2) is completely removed.



1. Aft crosstube
2. Larson L101 abrasion strip
3. Plastic wedge

ICN-412-A-320108-B-97499-00001-A-001-01

Figure 1. Abrasion Strip (Typical) - Removal (Sheet 1 of 1)

BHT Aft Crosstube Supports - Inspection and Repair

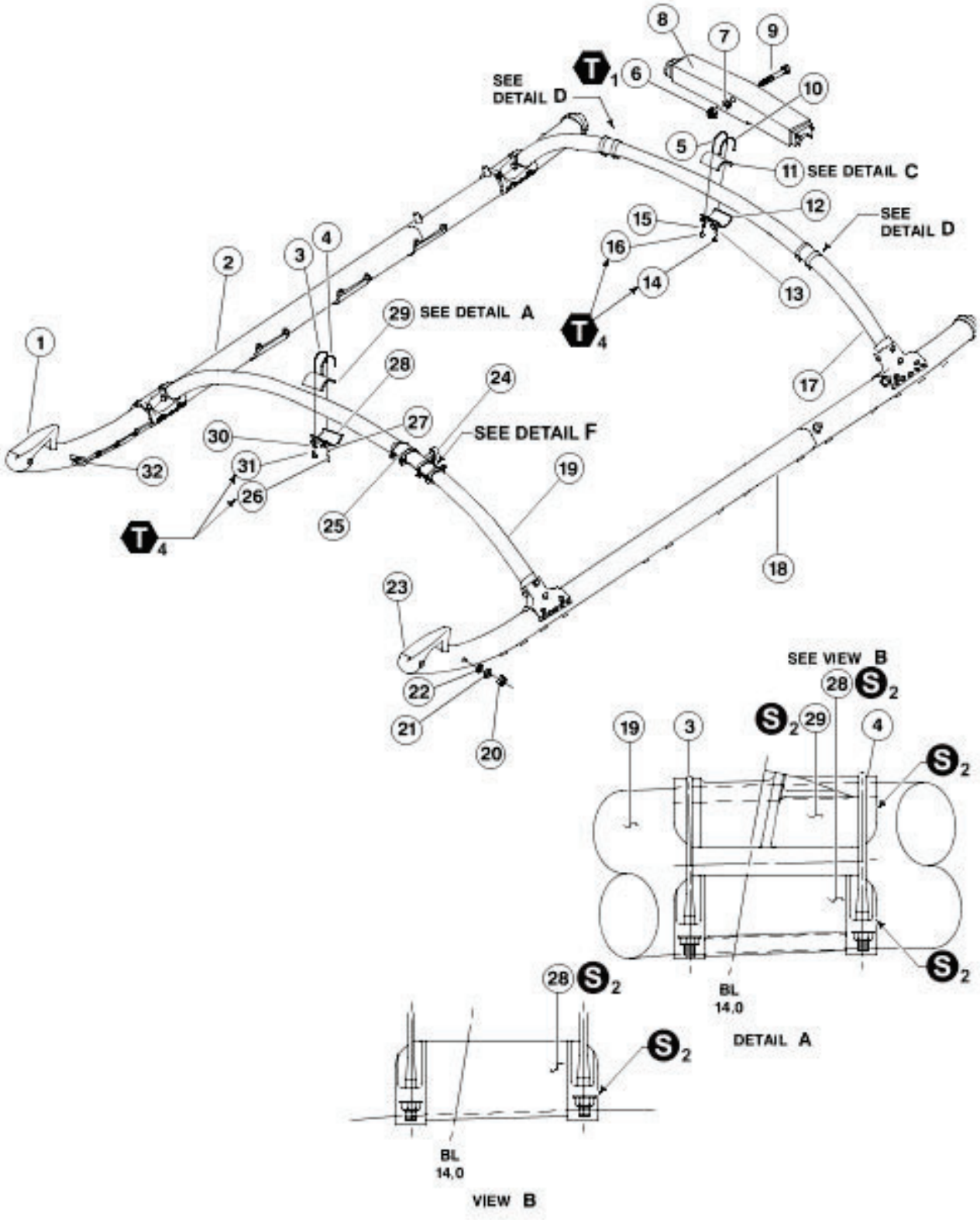
Procedure

1. Inspect aft crosstube supports (11 and 12, Figure 1) for mechanical and corrosion damage. Inspect abrasion strip (33), if installed, under aft lower crosstube support (12) for looseness or deteriorated adhesive. To repair adhesive, refer to BHT-ALL-SPM, Standard Practices Manual.

NOTE

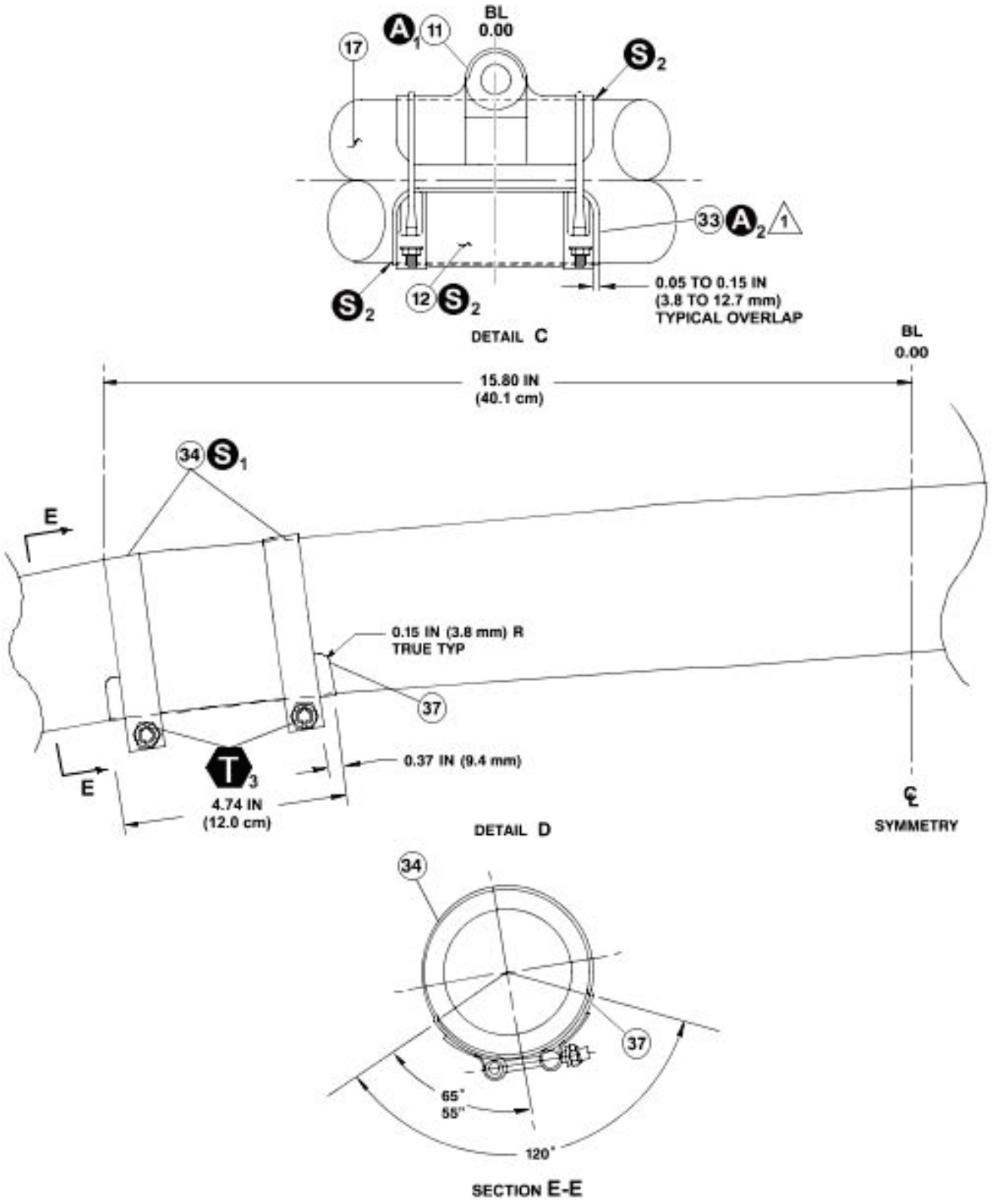
Scratches, dents, and corrosion may be polished out to maximum depths, as shown in Figure 2. No cracks are acceptable.

2. Inspect LBL 14.0 and RBL 14.0 clamp assemblies (34, Figure 1) for security, mechanical and corrosion damage. Inspect abrasion strips (33) underneath the clamp assemblies for looseness or deteriorated adhesive. To repair adhesive, refer to BHT-ALL-SPM, Standard Practices Manual.



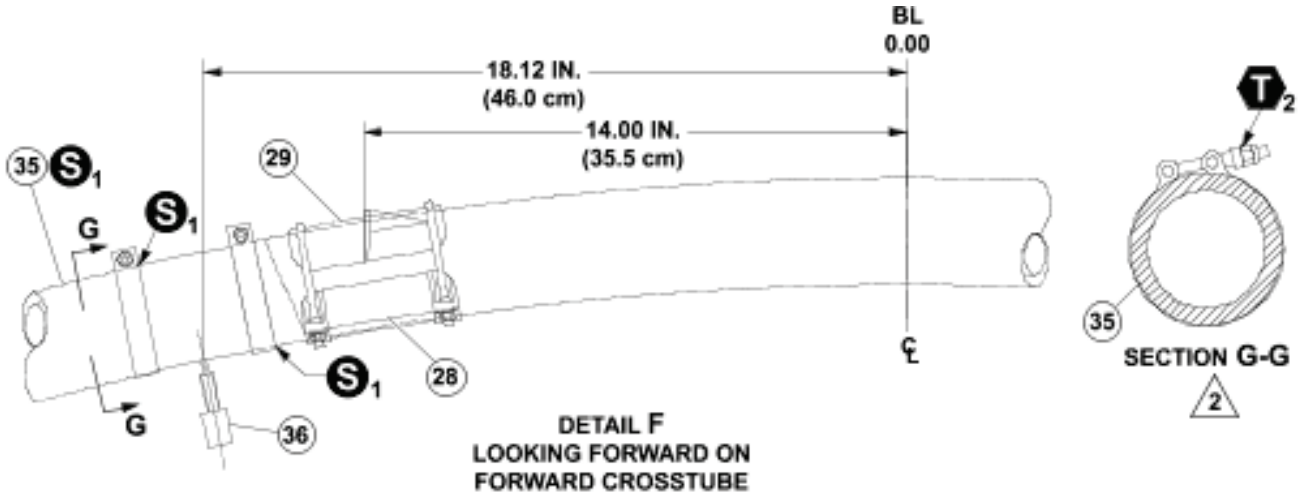
ICN-412-A-320106-B-97499-00004-A-001-01

Figure 1. Landing Gear Assembly (Sheet 1 of 3)



ICN-412-A-320106-B-97499-00005-A-001-01

Figure 1. Landing Gear Assembly (Sheet 2 of 3)



1. Strap
2. Right skid tube
3. U-bolt
4. U-bolt
5. U-bolt
6. Nut
7. Thin steel washer
8. Aft crosstube support beam
9. Bolt
10. U-bolt
11. Aft crosstube upper support
12. Aft crosstube lower support
13. Steel washer
14. Nut
15. Steel washer
16. Nut
17. Aft crosstube
18. Left skid tube
19. Forward crosstube
20. Nut
21. Aluminum washer
22. Aluminum Washer
23. Step
24. Power step electrical connector
25. Left forward crosstube support
26. Nut
27. Steel washer
28. Forward crosstube lower support
29. Forward crosstube upper support
30. Steel washer
31. Nut
32. Tow fitting
33. Abrasion strip
34. Clamp assembly
35. Clamp assembly
36. Weight-on-gear switch
37. Abrasion strip

A_1 ADHESIVE (C-317)

A_2 ADHESIVE (C-571)

S_1 SEALANT (C-251)

S_2 SEALANT (C-308)

T_1 10 TO 30 IN-LBS (1.13 TO 3.39 Nm)

T_2 40 IN-LBS (4.57 Nm)

T_3 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T_4 80 TO 100 IN-LBS (9.04 TO 11.30 Nm)

NOTES

1. Abrasion strip (33) must be removed for installation of AAI aft crosstube lower support half (12)

2. Weight-on-gear switch 3420S1 installed on helicopters S/N 36462 and subsequent. It is also installed on helicopters S/N 36248 through 36461 with the weather radar kit (412-705-034) or weight-on-gear retrofit kit (BHT-412-SI-84) installed.

ICN-412-A-320106-B-97499-00006-A-001-01

Figure 1. Landing Gear Assembly (Sheet 3 of 3)

FORWARD LANDING GEAR SUPPORTS



205-050-132



205-050-133

DAMAGE LOCATION SYMBOLS



TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR DEPTH
MECHANICAL	0.020 IN. (0.50 mm)
CORROSION DAMAGE (AFTER CLEANUP)	0.020 IN. (0.50 mm)
MAXIMUM AREA PER FULL DEPTH REPAIR	1.0 SQ. IN. (645.1 Sq. mm)
NUMBER OF REPAIRS	2
EDGE CHAMFER	0.050 IN. (1.27 mm)
BORE DAMAGE	0.020 IN. (0.50 mm) for 1/4 circumference

ICN-412-A-320105-B-97499-00002-A-001-01

Figure 2. Landing Gear Support Damage Limits (Sheet 1 of 3)

IMPROVED FORWARD LANDING GEAR SUPPORT



412-050-038

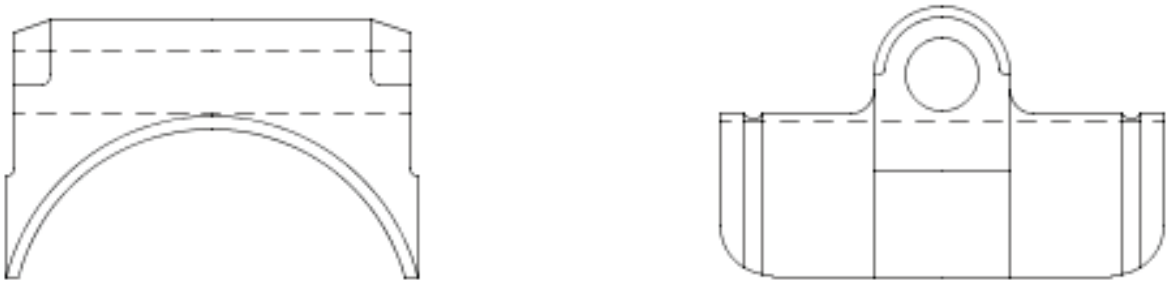
DAMAGE LOCATION SYMBOLS

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR DEPTH
MECHANICAL	0.020 IN. (0.50 mm)
CORROSION DAMAGE (AFTER CLEANUP)	0.020 IN. (0.50 mm)
MAXIMUM AREA PER FULL DEPTH REPAIR	1.0 SQ. IN. (645.1 Sq. mm)
NUMBER OF REPAIRS	2
EDGE CHAMFER	0.050 IN. (1.27 mm)
BORE DAMAGE	0.020 IN. (0.50 mm) for 1/4 circumference

ICN-412-A-320105-B-97499-00003-A-001-01

Figure 2. Landing Gear Support Damage Limits (Sheet 2 of 3)

AFT LANDING GEAR SUPPORTS



412-050-006



412-050-007

DAMAGE LOCATION SYMBOLS



TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR DEPTH
MECHANICAL	0.020 IN. (0.50 mm)
CORROSION DAMAGE (AFTER CLEANUP)	0.020 IN. (0.50 mm)
MAXIMUM AREA PER FULL DEPTH REPAIR	1.0 SQ. IN. (645.1 Sq. mm)
NUMBER OF REPAIRS	2
EDGE CHAMFER	0.050 IN. (1.27 mm)
BORE DAMAGE	0.020 IN. (0.50 mm) for 1/4 circumference

ICN-412-A-320105-B-97499-00004-A-001-01

Figure 2. Landing Gear Support Damage Limits (Sheet 3 of 3)

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BHT Aft Crosstube Supports - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Primer	C-063	AR	
Epoxy polyamide primer	C-204	AR	
Sealant	C-251	AR	
Adhesive film	C-252	AR	
Adhesive primer	C-253	AR	
Sealant	C-308	AR	
Acetone	C-316	AR	
Adhesive	C-317	AR	
Denatured alcohol	C-326	AR	
Adhesive film	C-364	AR	
Sealant	C-251	AR	
Abrasive pad	C-407	AR	
Abrasive cloth or paper	C-423	AR	
Dry blasting abrasive	C-474	AR	
Release fabric	C-478	AR	
Abrasive	C-496	AR	
Clean cloth	C-516	AR	
Adhesive	C-571	AR	

Procedure



A NEW CROSSTUBE HAS AN ADHESIVE COMPOSITE BOND AROUND CENTER OF TUBE WHERE SUPPORTS ARE INSTALLED. PROTECT THIS AREA WHILE HANDLING TUBE UNTIL TUBE IS INSTALLED ON HELICOPTER.



WHEN CLEANING MATING SURFACES, DO NOT USE ANY TOOL OR SUBSTANCE THAT COULD SCRATCH TUBE OR SUPPORTS.

1. If you are installing a new BHT support, go to Step 3.

2. If you are installing the same support, proceed as follows:

2.1. Remove all existing adhesive and primer from the support mating surface by abrading with 400 to 600 grit abrasive cloth or paper (C-423).

2.2. Wipe faying surface of upper support with clean cloth dampened with denatured alcohol (C-326) and wipe dry.

NOTE

Primed part can be kept in a controlled environment for up to 48 hours before the next step is accomplished. Controlled environment is defined as between 65 and 80°F (18.3 and 26.6°C) with relative humidity not exceeding 60% and should be monitored during that period of time.

2.3. Apply adhesive primer (C-253) to cleaned surface. Cure at a temperature between 65 and 80°F (18.3 and 26.6°C) for 2 hours.

2.4. Apply adhesive film (C-252) to the support primed surface. Place release fabric (C-478) over adhesive coated surface.

NOTE

Use a release film or other release barrier between the adhesive film and caul sheet, to prevent bonding to the adhesive film.

2.5. Use a caul sheet over the adhesive surface to provide a uniform and smooth surface when finished.

2.6. Cure adhesive film (C-252) for 30 minutes minimum at a temperature 65 to 80°F (18.3 to 26.6°C) plus 60 minutes between 275 and 290°F (135.0 and 143.3°C). Ramp-up and cool down rates should be between 1 and 10°F (0.6 and 5.5°C) maximum per minute. Preferred pressure to caul sheet to be 20 PSI (138 kPa) or 20 to 28 inches (68 to 94 kPa) of mercury (vacuum).

3. Prepare the aft crosstube upper surface (BL 0.0) as follows:

3.1. Remove primer from the crosstube in the area 2.5 inches (63.5 mm) on either side of BL 0.0 5.0 inches (127.0 mm) wide.

3.2. Prepare faying surface area of crosstube 0.50 inch (12.7 mm) beyond the edge of support assembly with dry blasting abrasive (C-474) using 220-grit aluminum oxide at 30 to 40 PSI (206.9 to 275.8 kPa).

3.3. Wipe with clean cloth (C-516) dampened with denatured alcohol (C-326).

NOTE

Primed part can be kept in a controlled environment for up to 48 hours before next step is accomplished. Controlled environment is defined as between 65 and 80°F (18.3 and 26.6°C) with relative humidity not exceeding 60% and should be monitored during that period of time.

3.4. Apply adhesive primer (C-253) to cleaned surface. Cure at a temperature between 65 and 80°F (18.3 and 26.6°C) for 2 hours. Do not cure at elevated temperatures.

4. Install aft crosstube upper support (11, Figure 1) as follows:

4.1. Remove release fabric (C-478) from inner surface of support. Lightly sand adhesive with 180 to 200 grit abrasive cloth or paper (C-423). Wipe with clean cloth (C-516) dampened with denatured alcohol (C-326).

4.2. Bond support in place on crosstube using adhesive (C-317).

4.3. Apply bond pressure using the aft crosstube lower support (12) with U-bolts (5 and 10), steel washers (13 and 15), and nuts (14 and 16). Torque the nuts to 15 to 20 inch-pounds (1.69 to 2.26 Nm). Retorque as required to maintain noted torque as adhesive is squeezed out. Remove excess adhesive and cure at room temperature.

4.4. Within 30 minutes of application of the adhesive, torque nuts (14 and 16) .

4.5. Apply a bead of sealant (C-308) around periphery of supports (11 and 12).

4.6. Check torque on U-bolts (5 and 10) after adhesive has cured (24 hours). Retorque as required.

4.7. Install aft crosstube support beam (412-A-32-01-06-00A-720A-A / 00727).

4.8. Apply epoxy polyamide primer (C-204) to the entire unprimed area of the crosstube, the edges of the adhesive bond, and the edges of the upper support assembly. Apply fillet seal using sealant (C-251) all around support assembly and adhesive edges.

5. Install aft crosstube lower support (12) as follows:



WHILE ACCOMPLISHING THE FOLLOWING STEP, DO NOT REMOVE PRIMER FROM MATING SURFACES.

NOTE

If installing AAI lower support 604-026-003 (with black nylon), do not install the Larson L101 abrasion strip (33). If already installed, remove the Larson L101 abrasion strip (412-A-32-01-08-02A-520A-A / 00737) and go to Step 5.4..

5.1. Size the Larson L101 abrasion strip (33) (raw material size 0.020 x 5.0 x 6.0 inches (0.51 x 127 x 152.4 mm)) to match the size of the lower support allowing 0.05 to 0.15 inch (1.3 to 3.8 mm).

5.2. Lightly abrade the surface of the crosstube using abrasive pad (C-407). Do not remove epoxy primer from crosstube, wipe with clean cloth dampened with denatured alcohol (C-326), then wipe dry with clean cloth.

5.3. Position abrasion strip (33) to be centered on the support (12). Bond the abrasion strip (33) in place on crosstube using adhesive (C-571). Cure at room temperature for a minimum of 24 hours.

5.4. Install the lower support (12) with sealant (C-308). Apply a uniform coat of sealant to each surface. Clamp with U-bolts attaching lower support (12) to crosstube (17). Torque U-bolts 80 to 100 inch-pounds (9.04 to 11.30 Nm) within 30 minutes of application of sealant. Fair-in the sealant around the periphery of the support. Ensure that all edges of the abrasion strip (33) have sealant applied to them. Check torque on the U-bolts after the sealant has cured. Retorque as required. Apply epoxy polyamide primer (C-204) to all surfaces.

6. Prepare the bare clamp assemblies (LBL 14.0 and RBL 14.0) for adhesive film as follows:

6.1. Degrease the clamp assembly by wiping with a clean cloth dampened with acetone (C-316).

6.2. Prepare the clamp assembly surface by wet blasting using abrasive (C-496). Use a slurry of 30 to 50% abrasive (C-496) in water. Blast uniformly at 30 to 80 PSI (206.84 to 551.58 kPa), adjusting the blasting pressure to provide adequate cleaning without warping.

NOTE

Parts prepared with the wet abrasive blast method must be bonded or adhesive primed within 96 hours after processing.

6.3. Spray or flush rinse to remove blasting residue. Perform a final rinse using a clean water spray or by flushing at room temperature to 160°F (71.11°C).

NOTE

Do not allow parts to dry before rinsing.

6.4. Apply primer (C-063), Class 5 to the cleaned inside surface of the clamp assembly. Pre-cure for 60 minutes at room temperature.

6.5. Apply one ply of adhesive film (C-364) to the inner surface, taking care to prevent air entrapment under the film.

6.6. Remove the protective backing from the adhesive film (C-364).

NOTE

Use a caul sheet over the adhesive surface to provide a uniform and smooth surface for curing.

6.7. Cure the adhesive at 340 to 355°F (171.1 to 179.1°C) with a minimum temperature rise rate of 3°F (1.6°C) per minute. Cure for 60 to 90 minutes at a maximum preferred bondline pressure of 50 PSI (101.80 inches of mercury).

7. Prepare the aft crosstube lower surfaces (LBL 14.0 and RBL 14.0) as follows:

7.1. Size the abrasion strip (37) (raw material size 0.02 x 3.38 x 4.74 inches (0.51 x 85.85 x 120.40 mm)) as shown in Figure 1, Detail D (length), and Section A-A (width) to fit between the clamp assembly (34) and the crosstube. Round the corners of the abrasion strip as shown in Detail D to prevent lifting and peeling.

7.2. Lightly abrade the primer on the lower surface of the crosstube common to the abrasion strip (BL 16.17 to BL 11.43) using abrasive cloth or paper (C-423). Abrade the crosstube in a longitudinal direction, not circumferential. Do not remove the primer from the crosstube. Wipe with denatured alcohol (C-326) and dry with a clean cloth (C-516).

7.3. Position the abrasion strip (37) on the lower surface of the crosstube between BL 16.17 and BL 11.43, so it will be centered under the clamp assembly (34) as shown in Figure 1, Section A-A.

7.4. Bond the abrasion strip to the crosstube as follows:

7.4.1. Mix adhesive (C-571) (100 parts A to 30 parts B) and apply a thin coat of the adhesive to both the crosstube and the abrasion strip. Allow adhesive to become tacky (wait 15 to 20 minutes) before assembly.

7.4.2. Place the abrasion strip (37) on the crosstube and rub in place.

NOTE

Use a release film or other release barrier between the abrasion strip and caul sheet to prevent bonding to the adhesive film.

7.4.3. Use a caul sheet formed with mating contour to apply a uniform pressure on abrasion strip.

7.4.4. Remove adhesive squeeze-out with acetone (C-316).

7.4.5. Allow to cure for 10 minutes minimum before next operation. Full cure requires 24 hours at 65°F (18.33°C).

7.4.6. Seal edges using sealant (C-251).

8. Assemble the clamp assembly to the crosstube as follows:

8.1. Prepare the faying surfaces of the aft crosstube (17, Figure 1) (upper surface), the abrasion strip (37) on the lower surface, and the clamp assembly (34) by lightly abrading/cleaning with 3M Scotchbrite, Grade, fine. Do not remove epoxy primer from the crosstube. Wipe the area clean with denatured alcohol (C-326), and dry with a clean cloth (C-516).

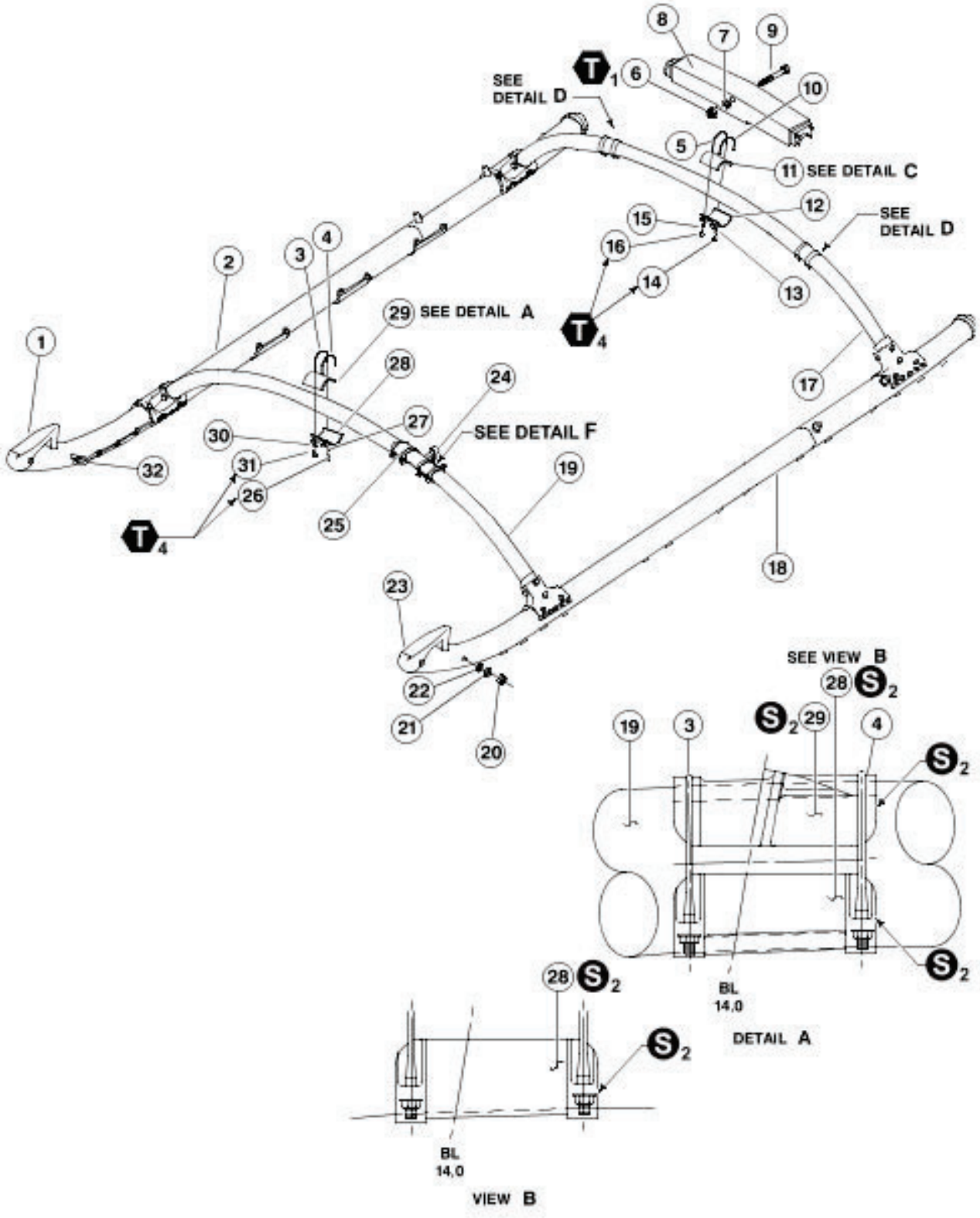
8.2. Apply a uniform coat of sealant (C-251) to the faying surfaces of the aft crosstube (17), clamp assembly (34), and abrasion strip (37).

8.3. Install the clamp assembly (34) into position on the crosstube (17). When properly placed, the top outside edge of the clamp assembly will sit at BL 15.80 as shown in Figure 1, Detail D. Torque the clamp fasteners to 50 to 70 inch-pounds (5.65 to 7.91 Nm).

8.4. Fair in the sealant squeeze-out around the periphery of the clamp assembly. Assure that the edges of the abrasion strip (37) have sealant applied to them. Apply additional sealant as required to ensure a uniform seal.

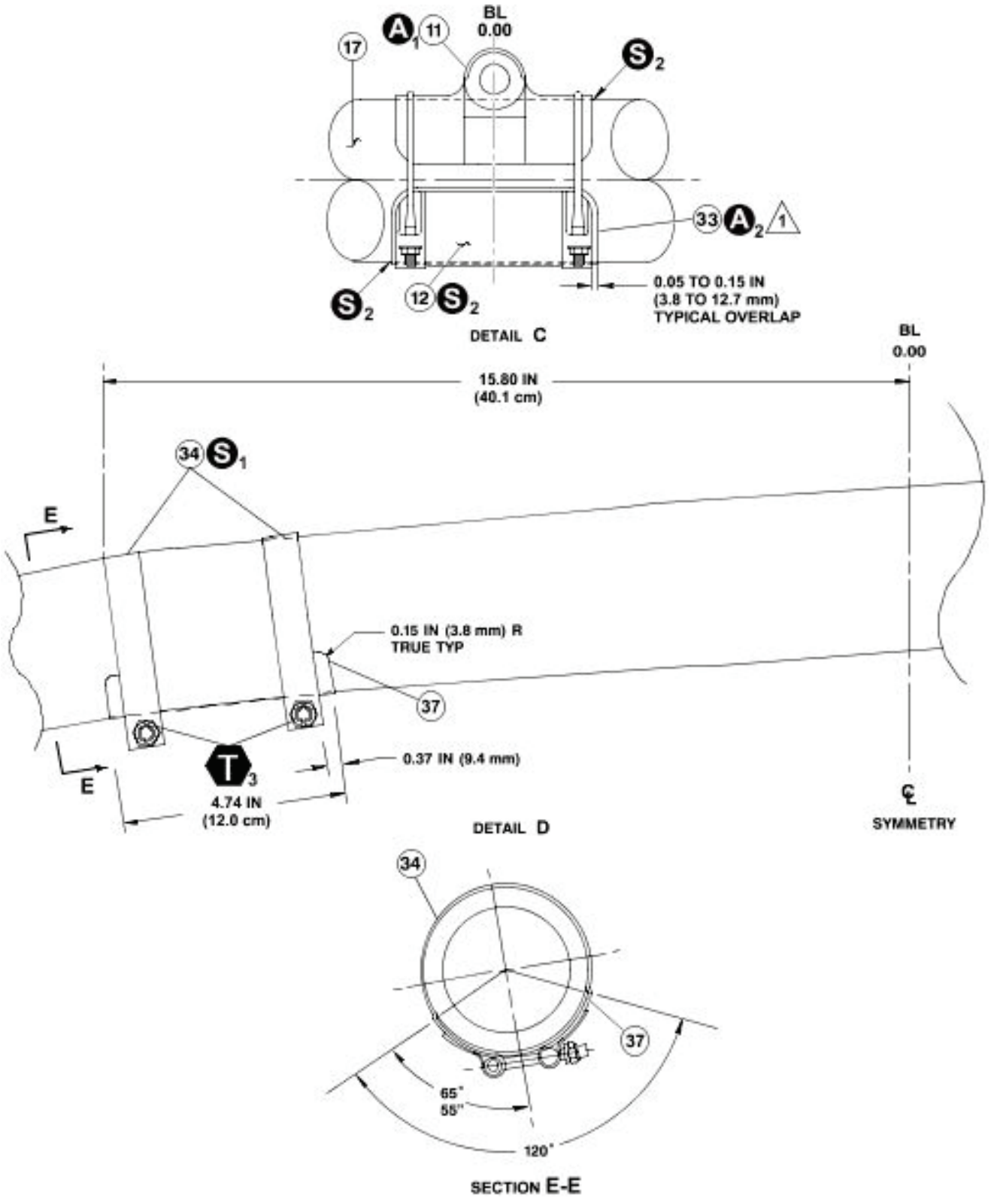
8.5. Check the torque on the clamp fasteners after the sealant is cured, and retorque as required.

8.6. Apply epoxy polyamide primer (C-204) to all surfaces.



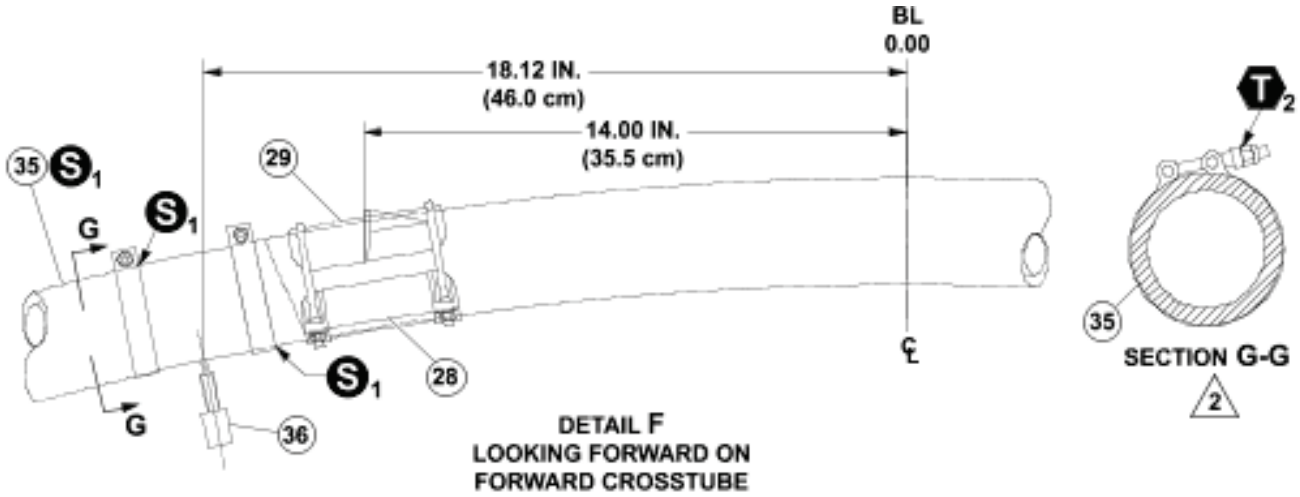
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Figure 1. Landing Gear Assembly (Sheet 1 of 3)



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Figure 1. Landing Gear Assembly (Sheet 2 of 3)



1. Strap
2. Right skid tube
3. U-bolt
4. U-bolt
5. U-bolt
6. Nut
7. Thin steel washer
8. Aft crosstube support beam
9. Bolt
10. U-bolt
11. Aft crosstube upper support
12. Aft crosstube lower support
13. Steel washer
14. Nut
15. Steel washer
16. Nut
17. Aft crosstube
18. Left skid tube
19. Forward crosstube
20. Nut
21. Aluminum washer
22. Aluminum Washer
23. Step
24. Power step electrical connector
25. Left forward crosstube support
26. Nut
27. Steel washer
28. Forward crosstube lower support
29. Forward crosstube upper support
30. Steel washer
31. Nut
32. Tow fitting
33. Abrasion strip
34. Clamp assembly
35. Clamp assembly
36. Weight-on-gear switch
37. Abrasion strip

A_1 ADHESIVE (C-317)

A_2 ADHESIVE (C-571)

S_1 SEALANT (C-251)

S_2 SEALANT (C-308)

T_1 10 TO 30 IN-LBS (1.13 TO 3.39 Nm)

T_2 40 IN-LBS (4.57 Nm)

T_3 50 TO 70 IN-LBS (5.65 TO 7.91 Nm)

T_4 80 TO 100 IN-LBS (9.04 TO 11.30 Nm)

NOTES

1. Abrasion strip (33) must be removed for installation of AAI aft crosstube lower support half (12)

2. Weight-on-gear switch 3420S1 installed on helicopters S/N 36462 and subsequent. It is also installed on helicopters S/N 36248 through 36461 with the weather radar kit (412-705-034) or weight-on-gear retrofit kit (BHT-412-SI-84) installed.

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Figure 1. Landing Gear Assembly (Sheet 3 of 3)

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AAI Crosstube Aft (BL 0.0) and Forward (BL 14.0) Supports - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Denatured alcohol	C-326	AR	
Clean cloth	C-516	AR	

Procedure

NOTE

The upper center support on the aft crosstubes is bonded with structural adhesive and is not designed for removal. The following removal procedure is supplied in case bondline is detected during routine inspection.

1. Remove the support U-bolts from the crosstube assembly.

2. Remove the supports from the crosstube assembly. Use of a heat gun to soften the adhesive may facilitate removal.

3. Using a plastic scraper, remove the old adhesive from crosstube and supports. Clean with a clean cloth (C-516) and denatured alcohol (C-326).

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AAI Crosstube Aft (BL 0.0) and Forward (BL 14.0) Supports - Inspection and Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Adhesive	C-322	AR	

Procedure

1. Inspect the inner surface coating material for deterioration (pealing, cracking, or wear).

NOTE

Supports 604-050-001/-003 have a black nylon coating and supports 205-050-132-001/-003 and 205-050-133-001/-003 have an EC2216 adhesive (C-322) coating.

2. Replace supports if nylon coating has separated or has worn through.

NOTE

For EC2216 coated supports, two coats of adhesive (C-322) may be applied for restoration of the protective barrier.

3. Allow 24 hours for the adhesive to cure.

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AAI Crosstube Aft (BL 0.0) and Forward (BL 14.0) Supports - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Denatured alcohol	C-326	AR	
Sealant	C-251	AR	
Abrasive cloth or paper	C-423	AR	
Clean cloth	C-516	AR	

Procedure

NOTE

Do not reinstall any AAI support if uncoated. AAI coated supports must be used for reinstallation. Replace coated supports if damaged or worn. (This note is not applicable to alternate BHT uncoated supports.)

1. Scuff the faying surface of the crosstube and the supports with 180 grit abrasive cloth or paper (C-423). Do not remove paint from the crosstube or nylon/EC2216 coating from the supports.

2. Clean the sanding residue from the crosstube and supports using denatured alcohol (C-326) and a clean cloth (C-516).

3. Apply one layer approximately 0.06 inch (1.6 mm) thick of sealant (C-251) to inside surface of the supports prior to assembly.

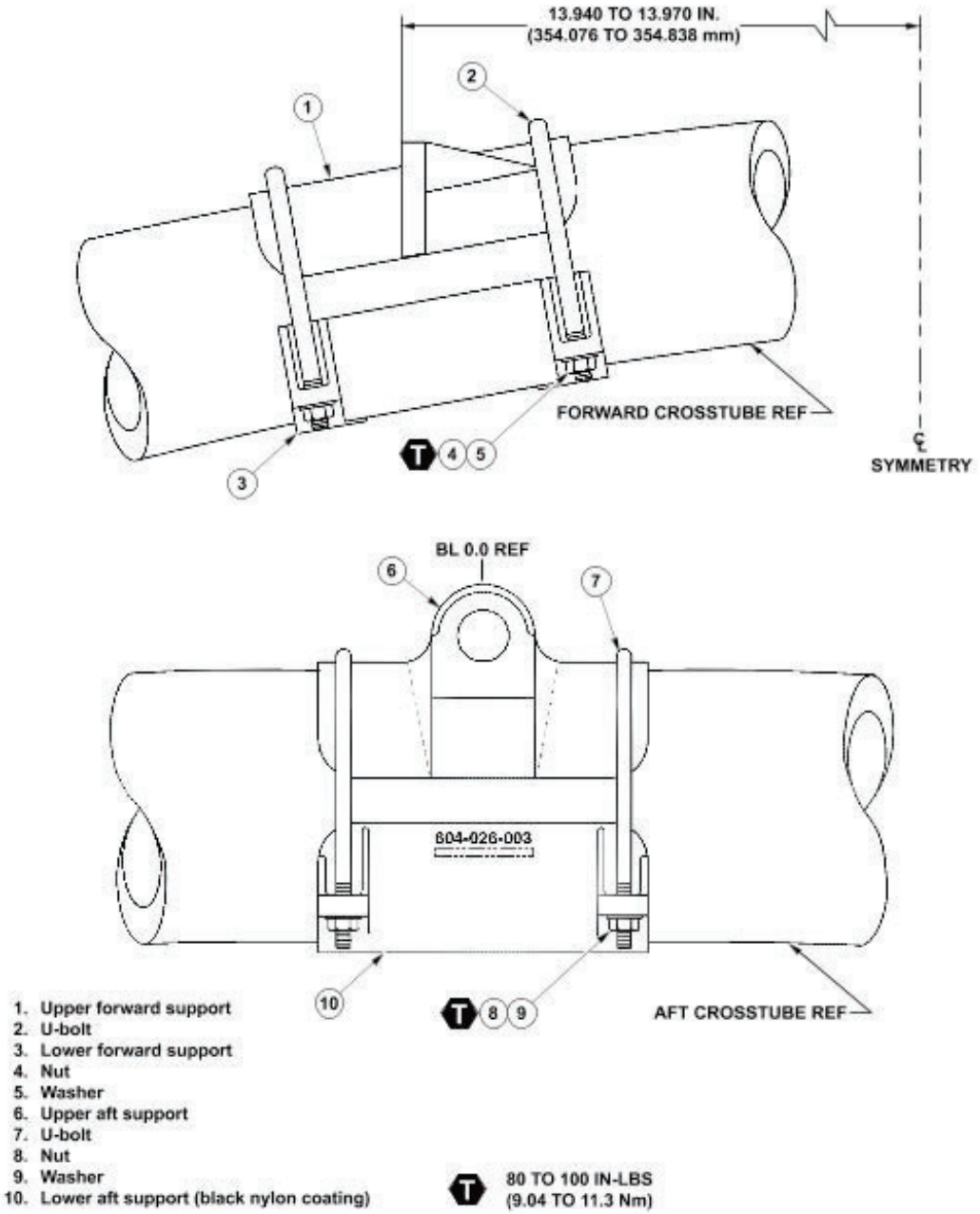
4. Reassemble the upper and lower supports and the U-bolts on the crosstube (Figure 1).

5. Remove excess sealant (C-251) and wipe clean after assembly.



FLIGHT/LANDING OPERATIONS PRIOR TO FULL CURE (72 HOURS AT ROOM TEMPERATURE) COULD RESULT IN CROSSTUBE SUPPORT MOVEMENT.

6. Torque the U-bolt nuts to between 80 and 100 inch-pounds (9.0 to 11.0 Nm) within 30 minutes after application of sealant or adhesive. Check torque on the U-bolts after the cure (72 hours at room temperature). Retorque as required.



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Figure 1. AAI Aft/Forward Crosstube Assemblies (Sheet 1 of 1)

AAI Crosstube Aft Clamp Assembly (BL 14.0) - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Cooling agent	C-137	AR	
Denatured alcohol	C-326	AR	
Clean cloth	C-516	AR	

Procedure

1. Remove the nuts from clamp assembly (1, Figure 1).

2. Remove the clamp assembly (1) from the crosstube assembly. Use of a heat gun to soften the adhesive may facilitate removal.

3. Using a plastic scraper, remove the old adhesive from crosstube and clamp assembly (1). Clean with a clean cloth (C-516) and denatured alcohol (C-326).



REFER TO THE MANUFACTURERS INSTRUCTIONS FOR THE PROPER USE OF cooling agent (C-137). THE TEMPERATURES OF THE COOLING AGENT CAN REACH -55°F (-48°C) . WEAR PROTECTIVE GLOVES.



DO NOT USE ANY TYPE OF SANDPAPER OR METAL TOOLS TO REMOVE THE ABRASION STRIP. THIS CAN DAMAGE THE SURFACE OF THE CROSSTUBE.

DO NOT USE HEAT TO REMOVE THE ABRASION STRIP. HEATING MORE THAN 250°F (121°C) WILL ADVERSELY AFFECT THE CONDITION OF THE CROSSTUBE.

4. If necessary, remove the abrasion strip (2) as follows:

4.1. Apply cooling agent (C-137) to one edge of the abrasion strip (2). Use a brief application (or until frost appears).

4.2. Initiate peeling of the abrasion strip (2) with the plastic wedge.

4.3. Once peeling has been initiated, grip the abrasion stirp (2) with pliers.

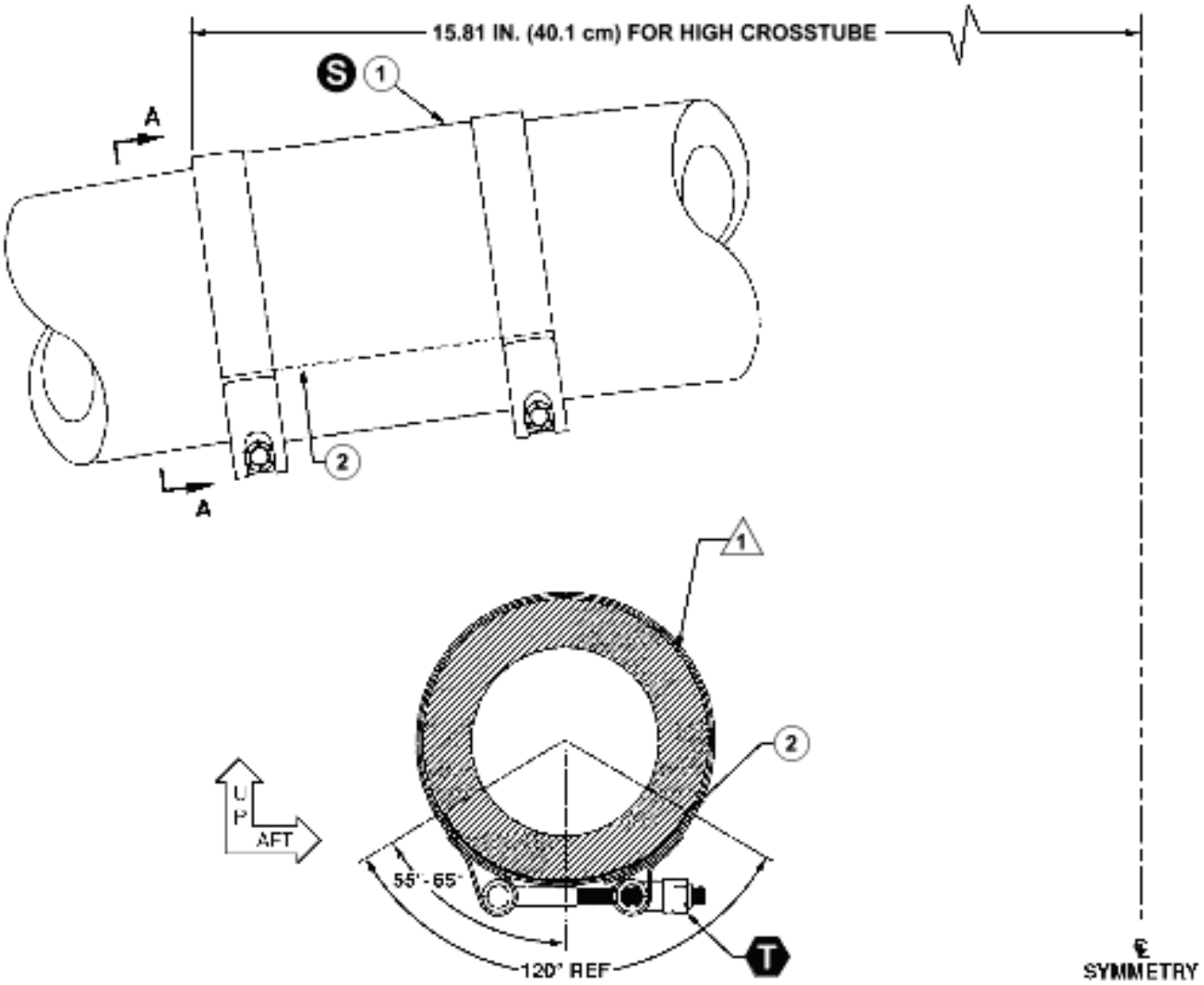
NOTE

Keep the bondline of the adhesive frozen. This will make removal easier.

4.4. Apply cooling agent (C-137) to the bondline of the adhesive. use a brief application (or until frost appears).

4.5. Use the plastic wedge and continue to gently pull the abrasion strip (2) with pliers.

4.6. Repeat Step 4.1. through Step 4.5. ntil the abrasion strip (2) is completely removed.



SECTION A-A
CLAMP ASSEMBLY

1. Clamp assembly
2. Abrasion strip

S SEALANT (C-308)

T 30 TO 40 IN-LBS (3.4 TO 4.5 Nm)

NOTE

1 Apply layer approximately 0.06 inch (1.5 mm) thick of sealant to inside surface of clamp prior to assembly with crosstube. Remove excess and wipe clean sealant (C-308) to have no voids around edge.

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Figure 1. Clamp Assembly - Aft Crosstube (Sheet 1 of 1)

AAI Crosstube Aft Clamp Assembly (BL 14.0) - Inspection and Repair

Procedure

1. Inspect the nylon coating on the faying surface of the clamp for condition.
2. Replace if coating has separated or has worn through.

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AAI Crosstube Aft Clamp Assembly (BL 14.0) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	
Acetone	C-316	AR	
Denatured alcohol	C-326	AR	
Sealant	C-251	AR	
Abrasive cloth or paper	C-423	AR	
Clean cloth	C-516	AR	
Adhesive	C-571	AR	

Procedure

NOTE

Do not reinstall any clamp if uncoated. Coated clamps must be used for reinstallation. Replace if damaged or worn.

1. f required, install a new abrasion strip (2, Figure 1) on the aft crosstube lower surfaces as follows:

1.1. Size the abrasion strip (2) (raw material size 0.02 x 3.38 x 4.74 inches (0.51 x 85.85 x 120.40 mm)) as shown in Figure 1, to fit between the clamp assembly (1) and the crosstube.

1.2. Lightly abrade the primer on the lower surface of the crosstube common to the abrasion strip (BL 16.17 to BL 11.43) using abrasive cloth or paper (C-423). Abrade the crosstube in a longitudinal direction not circumferential. Do not remove the primer from the crosstube. Wipe with denatured alcohol (C-326) and dry with a clean cloth (C-516).

1.3. Position the abrasion strip (2) on the lower surface of the crosstube between BL 16.17 and BL 11.43, so it will be centered under the clamp assembly (1), as shown in Figure 1.

1.4. Bond the abrasion stirp (2) to the crosstube as follows:

1.4.1. Mix adhesive (C-571) (100 parts A to 30 parts B) and apply a thin coat of the adhesive to both the crosstube and the abrasion strip. Allow adhesive to become tacky (wait 15 to 20 minutes) before assembly.

1.4.2. Place the abrasion strip (2) on the crosstube and rub in place.

NOTE

Use a release film or other release barrier between the abrasion strip and caul sheet to prevent bonding to the adhesive film.

1.4.3. Use a caul sheet formed with mating contour to apply a uniform pressure on abrasion strip.

1.4.4. Remove adhesive squeeze-out with acetone (C-316).

1.4.5. Allow to cure for 10 minutes minimum before next operation. Full cure required 24 hours at 65°F (18.33°C).

1.4.6. Seal edges using sealant (C-251).

2. Scuff the faying surface of the crosstube abrasion strip (2) and the clamp assembly (1) with 180 grit abrasive cloth or paper (C-423). Do not remove paint from the crosstube or nylon coating from the clamps.

3. Clean the sanding residue from the crosstube and clamps using denatured alcohol (C-326) and a clean cloth (C-516).

4. Apply a layer approximately 0.06 inch (1.5 mm) thick of sealant (C-308) to inside surface of the clamp.

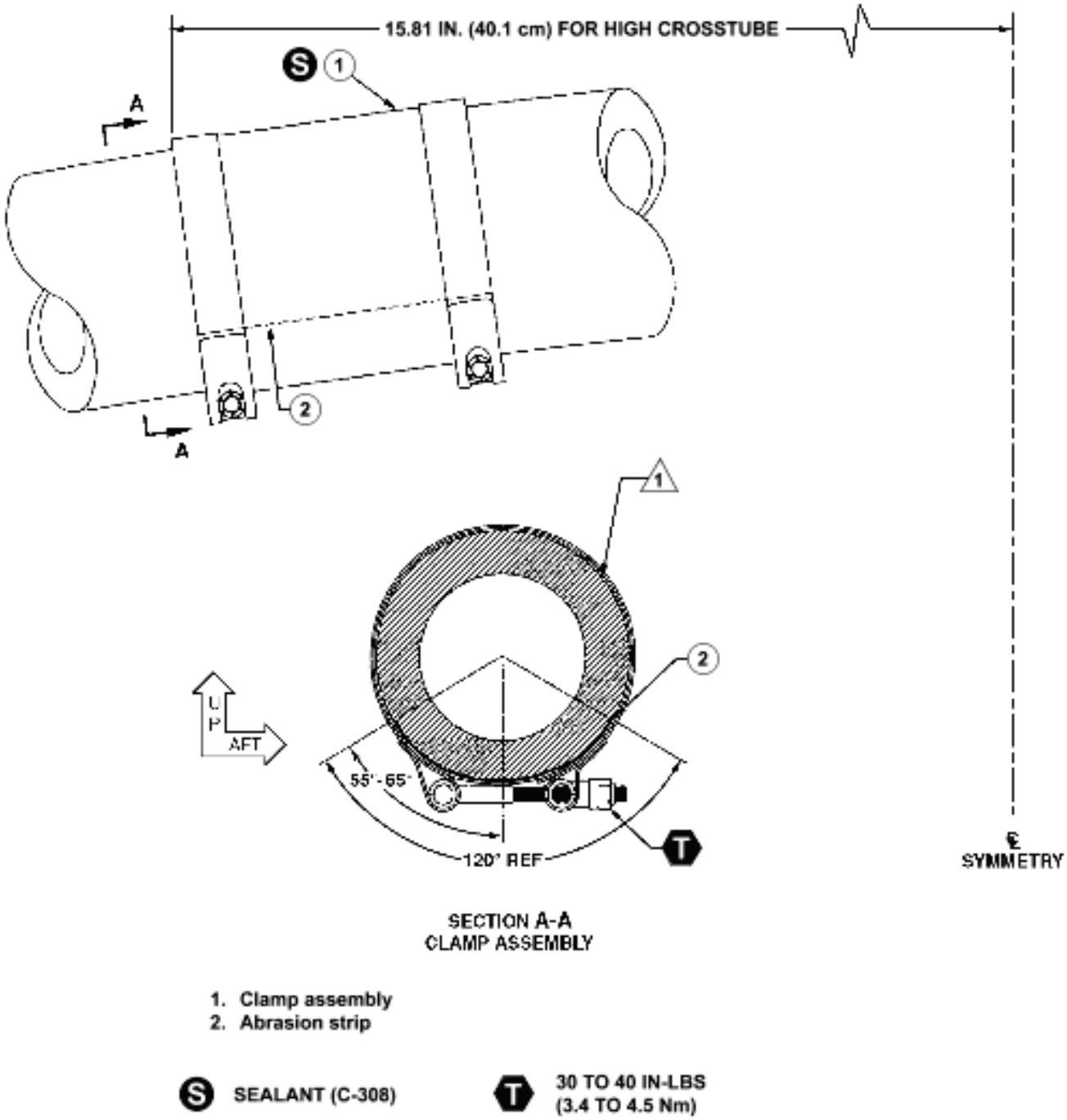
5. Reassemble the clamp assembly (1) on the crosstube (Figure 1).

6. Remove excess sealant and wipe clean.

NOTE

Use of a wire or string may be required to keep the clamp assemblies from sliding down the crosstube when torque is being applied.

7. Torque the nuts 30 to 40 inch-pounds (3.4 to 4.5 Nm) within 30 minutes after application of sealant. Check torque after the sealant has cured (72 hours at room temperature). Retorque as required.



SECTION A-A
CLAMP ASSEMBLY

1. Clamp assembly
2. Abrasion strip

S SEALANT (C-308)

T 30 TO 40 IN-LBS (3.4 TO 4.5 Nm)

NOTE

1 Apply layer approximately 0.06 inch (1.5 mm) thick of sealant to inside surface of clamp prior to assembly with crosstube. Remove excess and wipe clean sealant (C-308) to have no voids around edge.

ICN-412-A-320109-B-97499-00002-A-001-01

Figure 1. Clamp Assembly - Aft Crosstube (Sheet 1 of 1)

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Tail Skid - Description

A tubular steel tail skid is attached on the lower aft section of the tailboom. The purpose of the tail skid is to warn the pilot of a tail-low attitude when landing.

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Tail Skid - Removal

Procedure

NOTE

Do not remove packing (8) if it is suitable for further service.

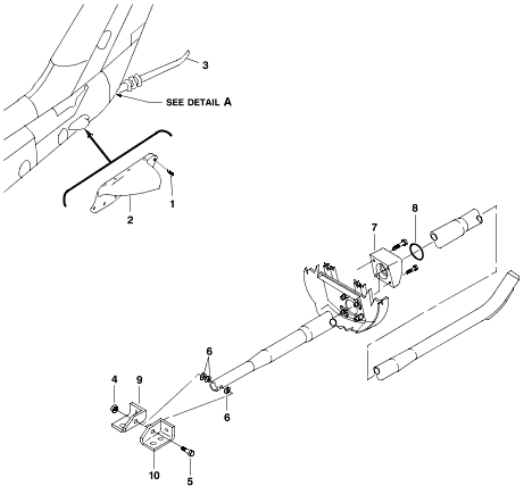
Do not remove support block (7) if only tail skid (3) is to be replaced.

1. Remove screws (1, Figure 1) and remove navigational light cover (2) on both sides of the tailboom to gain access to tail skid (3).

2. Remove any installed ballast (412-A-08-40-20-00A-520A-A / 00098).

3. Remove nut (4), bolt (5), and washers (6).

4. Pull tail skid (3) aft through support block (7).



1. Screw
2. Navigational light cover
3. Tail skid
4. Nut
5. Bolt
6. Washer
7. Support block
8. Packing
9. Bracket
10. Bracket

ICN-412-A-322000-B-97499-00001-A-001-01

Figure 1. Tail Skid (Sheet 1 of 1)

Tail Skid - Inspection and Repair

Procedure

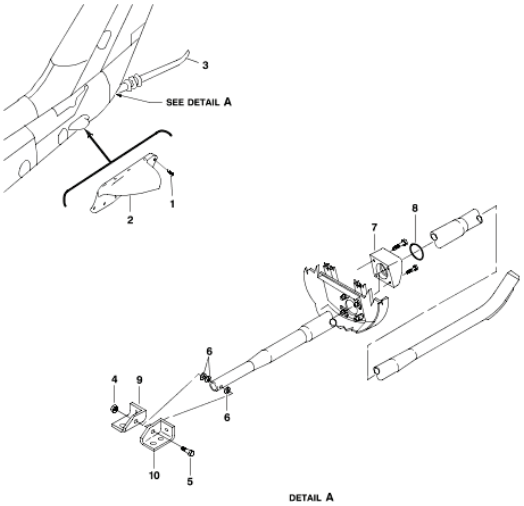
1. Inspect tail skid (3, Figure 1) for dents. Smooth dents up to 0.050 inch (1.27 mm) deep and 0.375 inch (9.50 mm) in diameter anywhere along tail skid length are acceptable without repair.

2. Inspect tail skid (3) for scratches (Figure 2). Longitudinal scratches up to 0.005 inch (0.12 mm) deep and 0.50 inch (12.7 mm) long are acceptable without repair. No circumferential scratches are allowed without repair. Scratches may be polished to a depth of 0.005 inch (0.12 mm).

3. Inspect tail skid (3, 412-A-32-01-02-00A-720A-A / 00712) for cracks. No cracks are acceptable.

4. Inspect ballast (if installed) and attaching components for security.

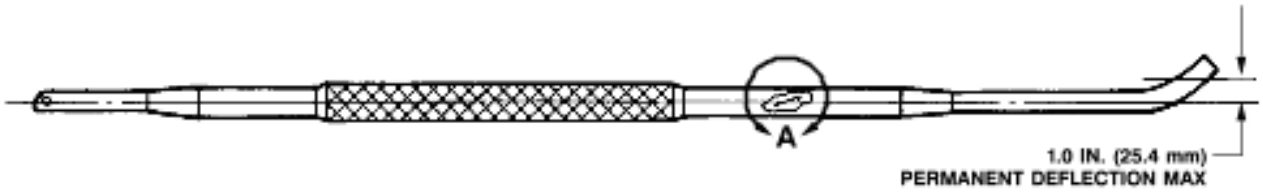
5. Inspect support block (7) and brackets (9 and 10) for cracks. No cracks are acceptable.



1. Screw
2. Navigational light cover
3. Tail skid
4. Nut
5. Bolt
6. Washer
7. Support block
8. Packing
9. Bracket
10. Bracket

ICN-412-A-322000-B-97499-00001-A-001-01

Figure 1. Tail Skid (Sheet 1 of 1)

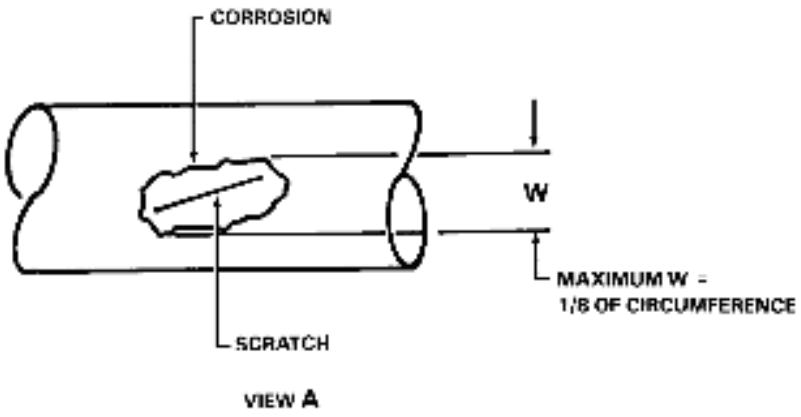


DAMAGE AREA REPAIR SYMBOLS

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR AREAS	
MECHANICAL	0.010 In. (0.25 mm)	0.020 In. (0.50 mm)
CORROSION		
BEFORE CLEANUP	0.005 In. (0.12 mm)	0.010 In. (0.25 mm)
AFTER CLEANUP	0.010 In. (0.25 mm)	0.020 In. (0.50 mm)
MAXIMUM AREA PER FULL DEPTH REPAIR	0.25 Sq. In. ($1.6\ cm^2$)	0.50 Sq. In. ($3.2\ cm^2$)
MAXIMUM LENGTH	1.50 In. (38.1 mm)	1.50 In. (38.1 mm)
NUMBER OF REPAIR AREAS	3	5
DENTS	None	Note 1

ICN-412-A-322000-B-97499-00002-A-001-01

Figure 2. Tail Skid Damage Limits (Sheet 1 of 2)



NOTES

1. Smooth dents up to 0.125 inch (3.17 mm) deep and up to 0.60 inch (15.2 mm) in diameter are acceptable without repair.
2. Scratches and corrosion shall be polished out to the depth of damage limits.
3. Radial damage across the tube, (W - view A), is limited to 1/8 of circumference.

ICN-412-A-322000-B-97499-00003-A-001-01

Figure 2. Tail Skid Damage Limits (Sheet 2 of 2)

Tail Skid - Installation

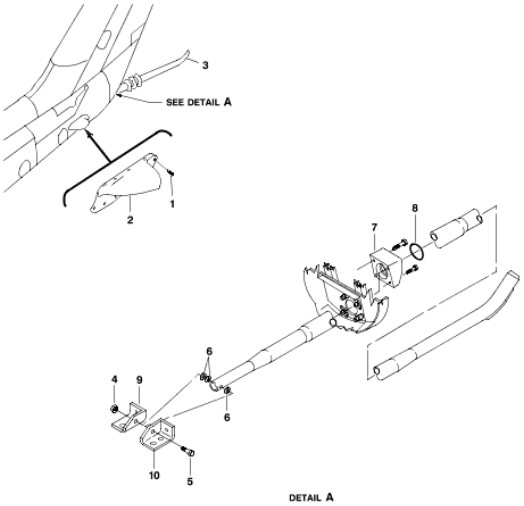
Procedure

1. If not previously accomplished, lubricate packing (8, Figure 1) with petroleum jelly or soapy water and install packing in support block (7).

2. Insert tail skid (3) through packing (8) and position in brackets (9 and 10), which are installed on bulkhead inside tailboom. Align tail skid with center of vertical fin and determine number of washers (6) required on each side of tail skid. Partially withdraw tail skid and secure washers on brackets with masking tape. Position tail skid in brackets and install bolt (5) and nut (4).

3. Install ballast on tail skid (3) (412-A-08-40-20-00A-720A-A / 00099).

4. Install navigational light cover (2) on both sides of tailboom and secure with screws (1).



1. Screw
2. Navigational light cover
3. Tail skid
4. Nut
5. Bolt
6. Washer
7. Support block
8. Packing
9. Bracket
10. Bracket

ICN-412-A-322000-B-97499-00001-A-001-01

Figure 1. Tail Skid (Sheet 1 of 1)

Passenger Step (Electrically Power Operated) - Description



DO NOT ACTUATE STEP WHEN A LOAD IS APPLIED.

The passenger step elevates for passengers to enter and exit, and retracts for flight. The installation consists of two arms, step, actuator, and electrical system. The passenger step is mounted to skid tube on each side of helicopter. The passenger step is actuated by a STEP switch (RAISE and STOW), located on upper left side of pedestal. The actuator retains passenger step in raised and lowered positions.

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Passenger Step - Removal

Procedure

1. Remove bolt (21, Figure 1) attaching step assembly (1) to forward arm (14) by removing nut (33), washer (32), and bushings (17 and 18).

2. Remove bolt (3) attaching step assembly (1) to aft arm (30) by removing nut (29), washer (28), and bushings (26 and 27). Remove step assembly.

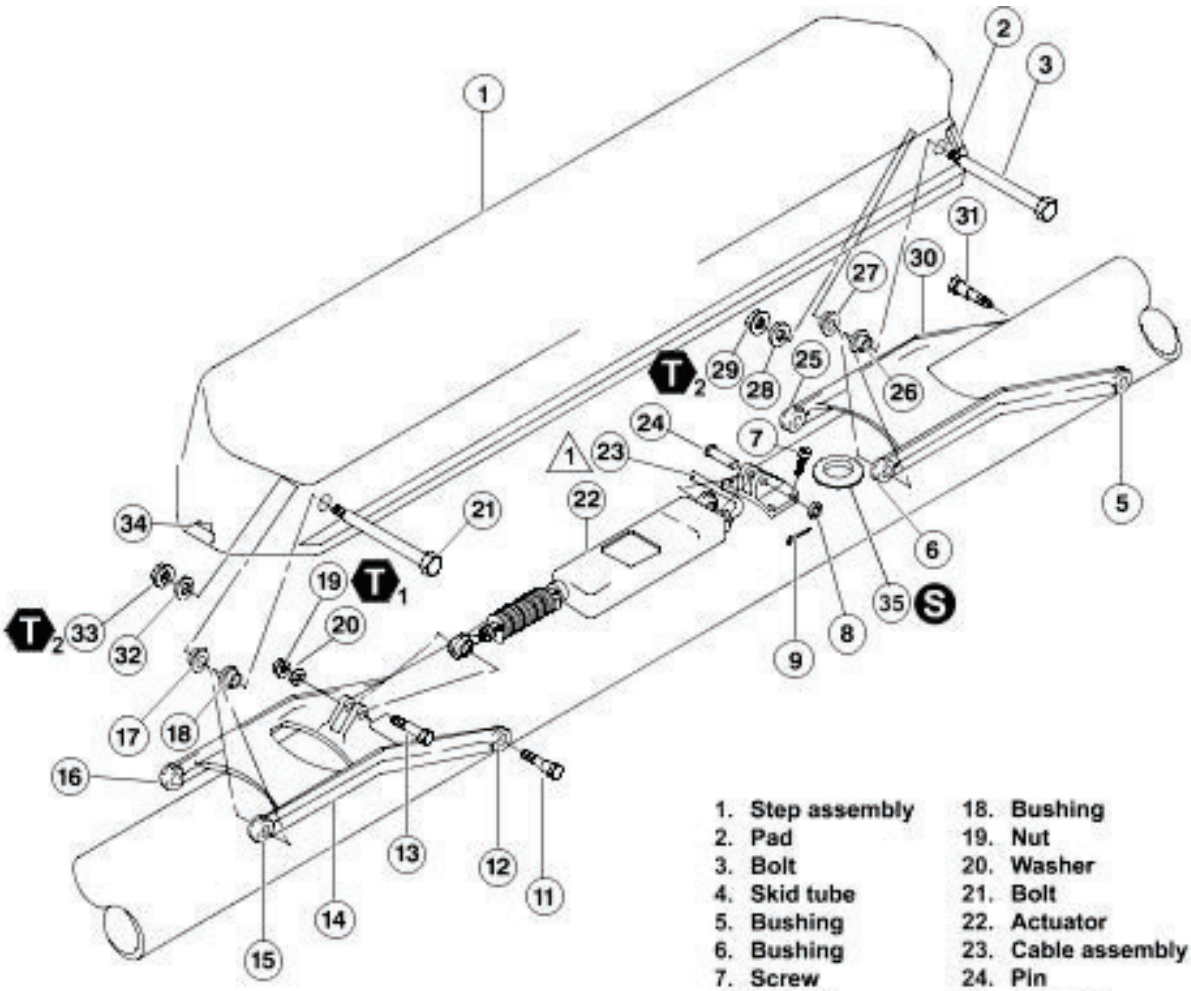
3. Remove the sealant from the grommet (35) and pull the cable assembly (23) from the crosstube (8) until the splices are visible. Cut the wire at the splices.

4. Remove bolt (13), nut (19), and washer (20) attaching actuator to forward arm (14).

5. Remove cotter pin (9), washer (8), and pin (24) attaching actuator (22) to actuator bracket (10). Remove actuator.

6. Remove bolts (11 and 31) attaching forward and aft arms (14 and 30) to skid tube (4). Remove arms.

7. Remove screws (7) attaching actuator bracket (10) to skid tube (4). Remove actuator bracket.



1. Step assembly
2. Pad
3. Bolt
4. Skid tube
5. Bushing
6. Bushing
7. Screw
8. Washer
9. Cotter pin
10. Actuator bracket
11. Bolt
12. Bushing
13. Bolt
14. Forward arm
15. Bushing
16. Bushing
17. Bushing
18. Bushing
19. Nut
20. Washer
21. Bolt
22. Actuator
23. Cable assembly
24. Pin
25. Bushing
26. Bushing
27. Bushing
28. Washer
29. Nut
30. Aft arm
31. Bolt
32. Washer
33. Nut
34. Pad forward
35. Grommet

S SEALANT (C-308)

T1 20 TO 25 IN-LBS (2.3 TO 2.8 Nm)

T2 25 TO 30 IN-LBS (2.8 TO 3.3 Nm)

NOTE

1 Remove sealant from grommet (35) and pull cable assembly (23). Cut wires as close as possible to splices.

ICN-412-A-320200-B-97499-00001-A-001-01

Figure 1. Passenger Step (Electrically Power Operated) (Sheet 1 of 1)

Passenger Step - Inspection

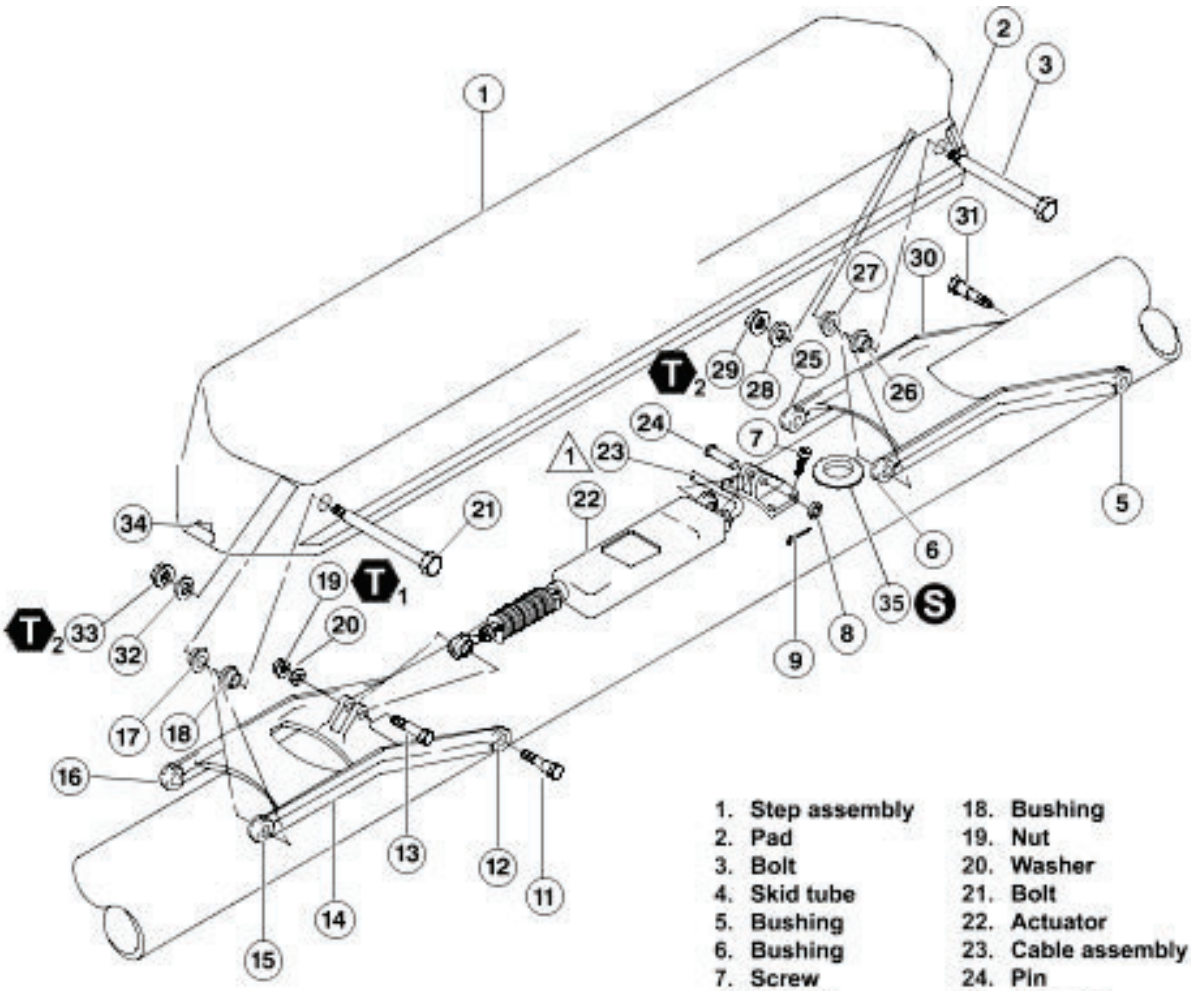
Procedure

1. Check actuator (22, Figure 1) for proper operation (BHT-412-FM).
2. Inspect actuator boot for tears, cracks, or deterioration.

NOTE

Superficial mechanical damage is acceptable without repair. Cracked parts are not acceptable and shall be replaced. Superficial corrosion damage shall be removed. Parts exhibiting corrosion damage deeper than superficial shall be replaced.

3. Inspect step assembly (1), arms (14 and 30), and actuator bracket (10) for mechanical and corrosion damage.
4. Inspect bushings (5, 6, 12, 15, 16, 17, 18, 25, 26, and 27) for secure installation and wear. Loose or worn bushing are not acceptable and shall be replaced.
5. Inspect actuator rod and bearings for axial and radial play.



1. Step assembly
2. Pad
3. Bolt
4. Skid tube
5. Bushing
6. Bushing
7. Screw
8. Washer
9. Cotter pin
10. Actuator bracket
11. Bolt
12. Bushing
13. Bolt
14. Forward arm
15. Bushing
16. Bushing
17. Bushing
18. Bushing
19. Nut
20. Washer
21. Bolt
22. Actuator
23. Cable assembly
24. Pin
25. Bushing
26. Bushing
27. Bushing
28. Washer
29. Nut
30. Aft arm
31. Bolt
32. Washer
33. Nut
34. Pad forward
35. Grommet

S SEALANT (C-308)

T1 20 TO 25 IN-LBS (2.3 TO 2.8 Nm)

T2 25 TO 30 IN-LBS (2.8 TO 3.3 Nm)

NOTE

1 Remove sealant from grommet (35) and pull cable assembly (23). Cut wires as close as possible to splices.

ICN-412-A-320200-B-97499-00001-A-001-01

Figure 1. Passenger Step (Electrically Power Operated) (Sheet 1 of 1)

Passenger Step - Repair

Procedure

1. Passenger step and attaching parts that are damaged beyond limits in 412-A-32-02-00-00A-280A-A / 00752 shall be replaced.

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Passenger Step - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	
Sealant	C-320	AR	

Procedure

1. Position actuator bracket (10, Figure 1) on skid tube (4) and install attaching screws (7) with sealant (C-320) applied.

2. Position forward and aft arms (14 and 30) on skid tube (4) and install attaching bolts (11 and 31) with sealant (C-320).

3. Preset actuator (22) to full electrical retract dimension of 11.25 inches (28.5 mm) between attachment bolt hole centers. Adjust actuator rod end as necessary to obtain required dimension.

4. Position actuator (22) in actuator bracket (10) and forward arm (14). Note marking THIS SIDE INBOARD. Use lower hole of spring loaded rear anchor for mounting.

5. Install pin (24), washer (8), and cotter pin (9) to attach actuator (22) to actuator bracket (10).

6. Install bolt (13), washer (20), and nut (19) to attach actuator (22) to forward arm (14). Torque the nut .

7. Position step assembly (1) on forward and aft arms (14 and 30) and install attaching washers (28 and 32), nuts (29 and 33), and bolts (3 and 21). Torque the nuts .

NOTE

Do not cap or connect the test wires (white/ red and white/black) until the step rigging (Step 12. to Step 15.) is completed.

8. Install the grommet (35) on the actuator wires.

9. Prepare the wires and connect with splices as shown in Figure 2 (BHT-ELEC-SPM, Chapter 4, Electrical Standard Practices Manual).

10. Fully extend actuator (22) through microswitch stop. Clearance between forward arm (14) and skid tube (4) should be 0.06 to 0.12 inch (1.5 to 3.0 mm).

11. Check clearance at extreme ends (fore and aft) of step assembly (1) with assembly in fully retracted position. Aft end of step assembly should contact skid tube (4). Forward end of step should have a gap of approximately 0.25 inch (6.4 mm).

12. Check clearance fore and aft between skid tube (4) and step assembly (1). Adjust actuator (22) to maintain 0.09 inch (2.3 mm) clearance.

13. With step raised, select STOW. While step is in transit, 28 VDC shall be present on the actuator white/ black test wire and shall switch to ground reference when step has reached its fully stowed position.

NOTE

Voltage between white/black or white/red test wires will be present as long as actuator is not fully extended or fully retracted.

14. If 28 VDC remains present on test wires after extend or retract procedure is complete, actuator will require adjustment of rod end bearing.

15. Remove bolt (13), washer (20), and nut (19) securing actuator (22) to forward arm (14) and adjust as required, until 28 VDC switches to ground reference during extend or retract procedure.

16. Once rigging is completed with satisfactory results, cap and stow test wires.

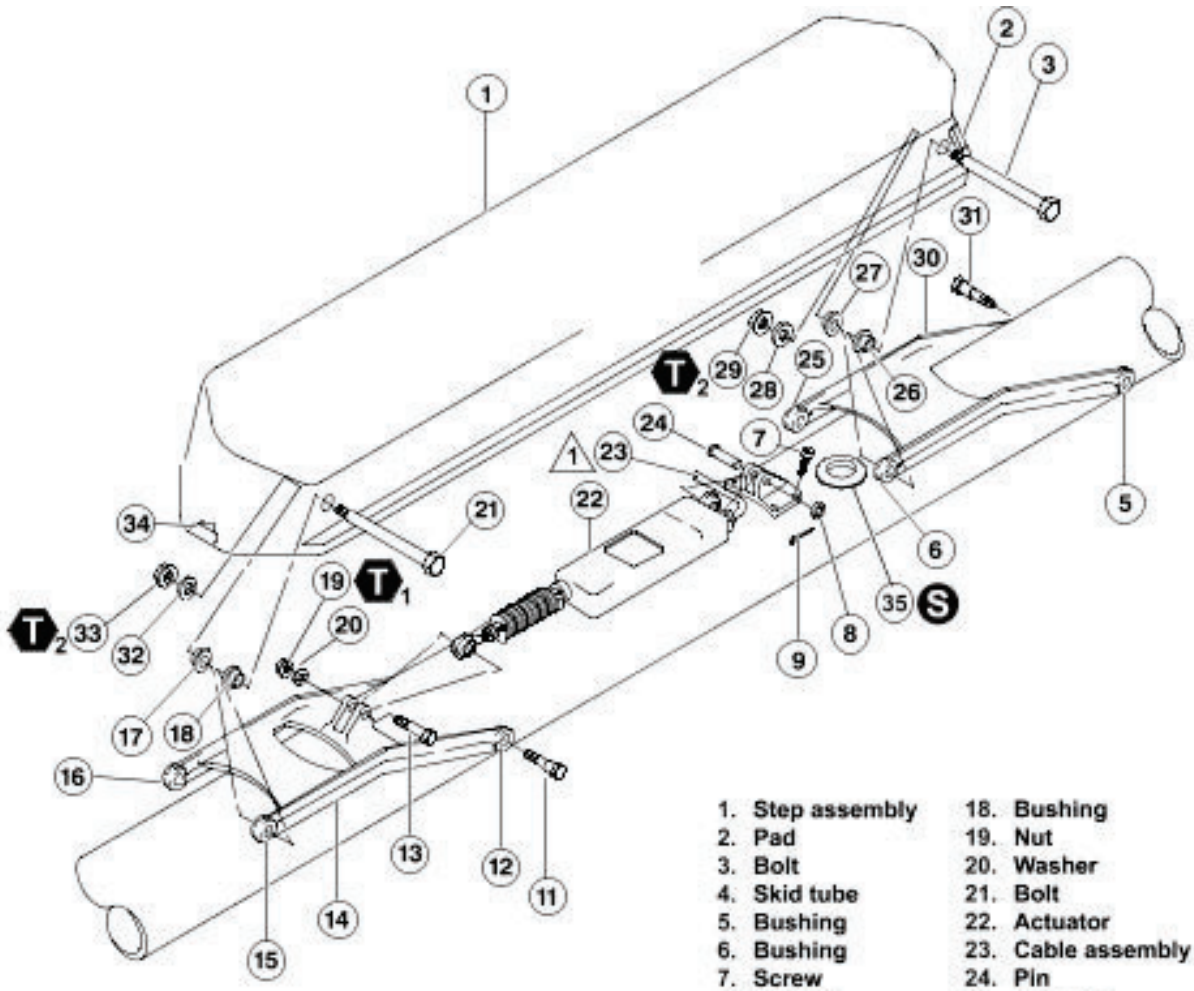
17. Carefully push the wires and splices inside the skid tube (4) . Install the grommet (35) and seal with sealant (C-308).

18. Apply vinyl tape pads (2 and 34).



DO NOT OPERATE STEP MORE THAN TWO COMPLETE CYCLES IN A 5 MINUTE PERIOD.

19. Perform operational check of passenger step (BHT-412-FM).



1. Step assembly
2. Pad
3. Bolt
4. Skid tube
5. Bushing
6. Bushing
7. Screw
8. Washer
9. Cotter pin
10. Actuator bracket
11. Bolt
12. Bushing
13. Bolt
14. Forward arm
15. Bushing
16. Bushing
17. Bushing
18. Bushing
19. Nut
20. Washer
21. Bolt
22. Actuator
23. Cable assembly
24. Pin
25. Bushing
26. Bushing
27. Bushing
28. Washer
29. Nut
30. Aft arm
31. Bolt
32. Washer
33. Nut
34. Pad forward
35. Grommet

S SEALANT (C-308)

T1 20 TO 25 IN-LBS (2.3 TO 2.8 Nm)

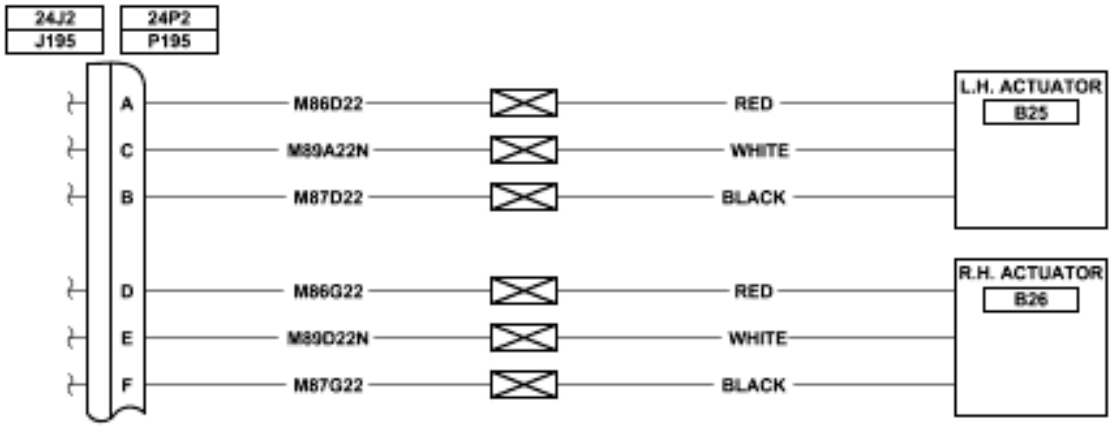
T2 25 TO 30 IN-LBS (2.8 TO 3.3 Nm)

NOTE

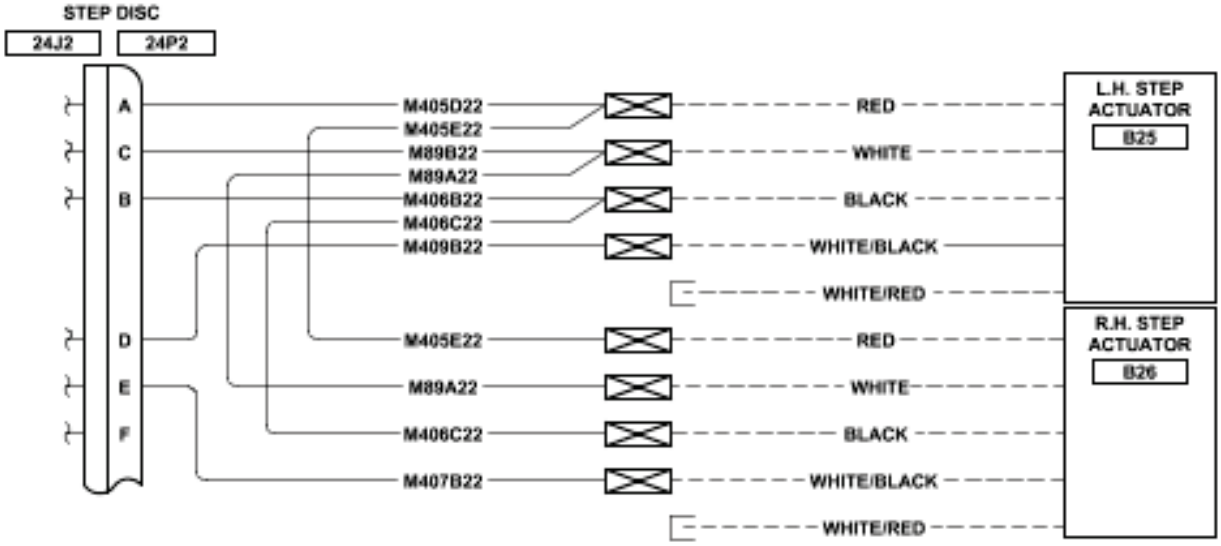
1. Remove sealant from grommet (35) and pull cable assembly (23). Cut wires as close as possible to splices.

ICN-412-A-320200-B-97499-00001-A-001-01

Figure 1. Passenger Step (Electrically Power Operated) (Sheet 1 of 1)



HELICOPTERS S/N 36001 THROUGH 36124
HELICOPTERS S/N 33001 AND SUBSEQUENT



HELICOPTERS S/N 36125 AND SUBSEQUENT

ICN-412-A-320105-B-97499-00001-A-001-01

Figure 2. Wiring Diagram Passenger Step (Sheet 1 of 1)

Ground Handling Wheels - Description

NOTE

Overhaul instructions for ground handling wheels 214-706-104-003 contained in this chapter are limited to inspection and replacement of faulty parts.

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Ground Handling Wheels - Inspection

Procedure

1. Inspect wheel and tire assembly (1 and 5, Figure 1) for tire damage and deterioration. Check both tires for inflation to 75 PSI (517 kPa). Inspect wheels for corrosion, cracks, and secure installation on axles (4 and 24).

2. Inspect pump assembly (2) for adequate supply of hydraulic fluid, corrosion, obvious mechanical damage, and secure installation on cradle assembly (11).

3. Inspect cradle assembly (11) and all detail parts shown in View B for obvious damage, and secure installation.

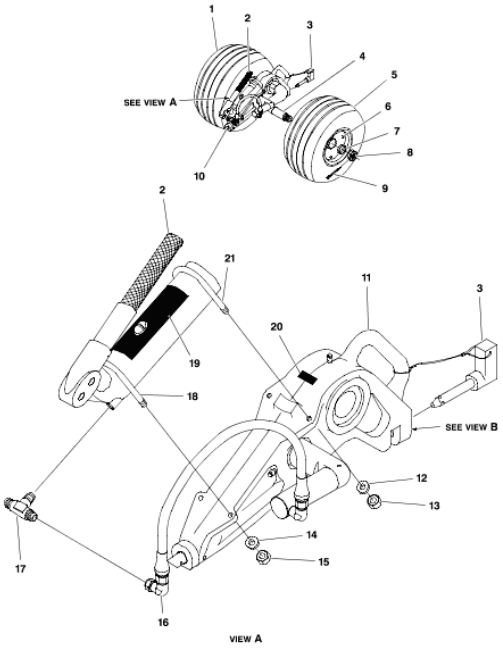
4. Inspect bushings (47, 48, 49, and 51) for obvious wear and for secure installation.

5. Inspect decals (19 and 20) for legibility and secure installation.

6. Inspect pin (10) and spring pin (50) to assure they are not sheared and that pin is adequately secured.

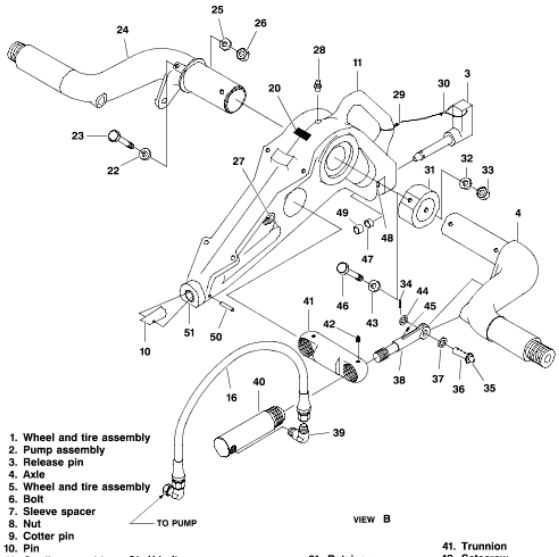
7. Inspect lubrication fittings (27, 28, and 35) and also fitting (not illustrated) that is opposite lubrication fitting (35) for secure installation and for damage.

8. Perform operational check of ground handling wheels. Install ground handling wheel assembly on helicopter landing gear skid and raise helicopter. Inspect for proper engagement and operation of release pin (3) and pin (10). Inspect for hydraulic leaks. If wheel assembly will not raise helicopter, and the pump has supply of hydraulic fluid, the cause is probably internal hydraulic leak in pump assembly (2).



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

ICN-412-A-320300-B-97499-00002-A-001-01

Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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Ground Handling Wheels - Lubrication

Preliminary requirements

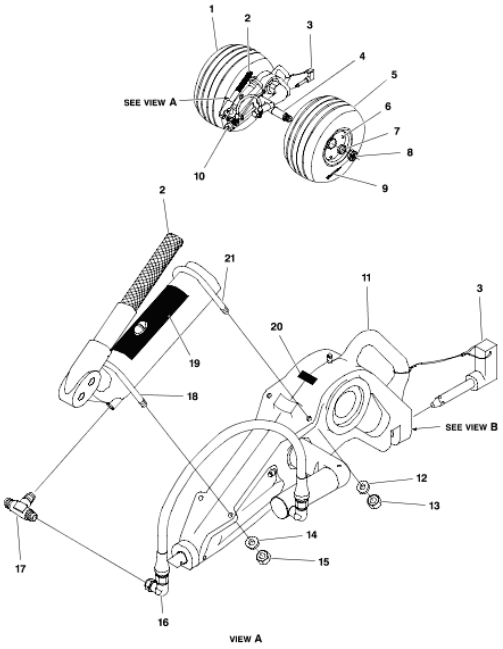
Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-001	AR	

Procedure

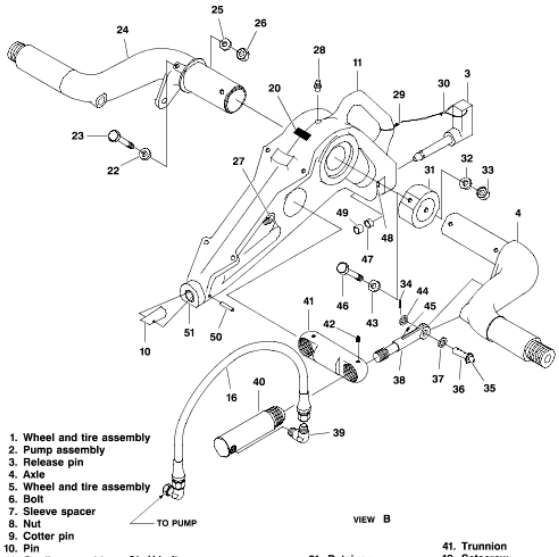
1. Use hand grease gun and lubricate with grease (C-001).

2. Lubricate fittings (27, 28, and 35, Figure 1). Also lubricate the fitting that is opposite fitting (35) (not illustrated).



ICN-412-A-320300-B-97499-00001-A-001-01

Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

ICN-412-A-320300-B-97499-00002-A-001-01

Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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Wheel and Tire Assembly - Removal

Procedure

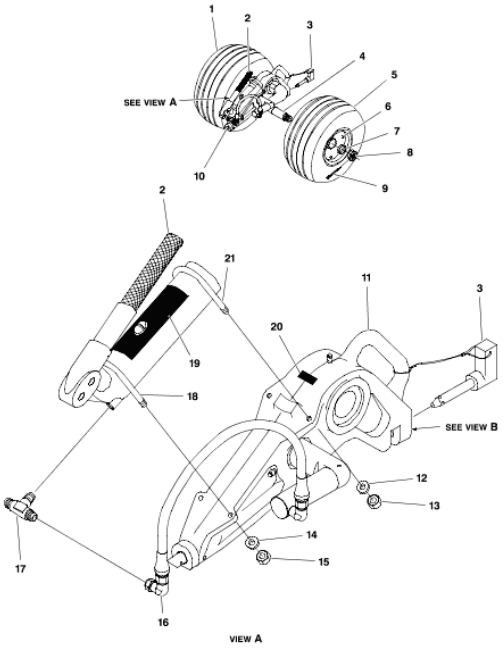
WARNING

DEFLATE TIRE, IF INSTALLED ON GROUND HANDLING ASSEMBLY, PRIOR TO REMOVING NUT (8, Figure 1). IF BOLTS (6) ARE NOT PROPERLY INSTALLED, AN INFLATED TIRE MAY CAUSE WHEEL HALVES TO SEPARATE WHEN NUT IS REMOVED.

NOTE

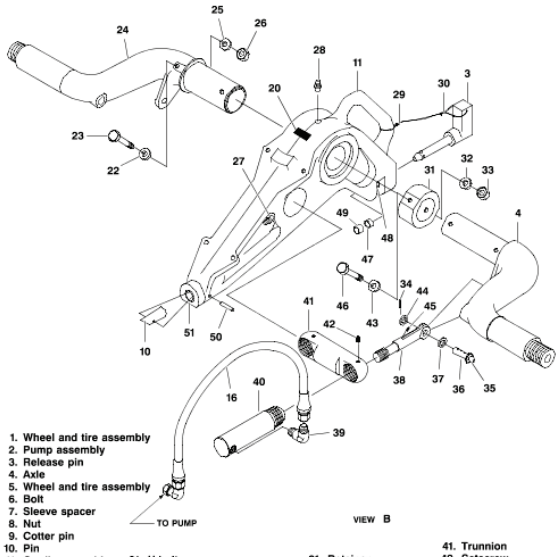
Removal procedure for wheel and tire assembly (5, Figure 1) is given. Removal procedure for wheel and tire assembly (1) is the same.

1. Deflate tire, if installed.
2. Remove cotter pin (9).
3. Remove nut (8).
4. Remove sleeve spacer (7) from axle (4).
5. Remove wheel and tire assembly (5).



ICN-412-A-320300-B-97499-00001-A-001-01

Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

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Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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Wheel and Tire Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-001	AR	

Procedure

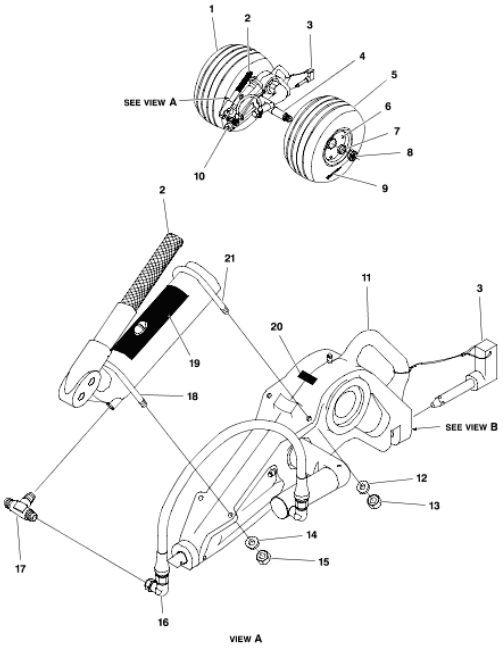
NOTE

Installation procedure for wheel and tire assembly (5, Figure 1) is given. Installation procedure for wheel and tire assembly (1) is the same.

1. Apply a bead of grease (C-001) around wheel bearing and apply a film of grease to axle (4).

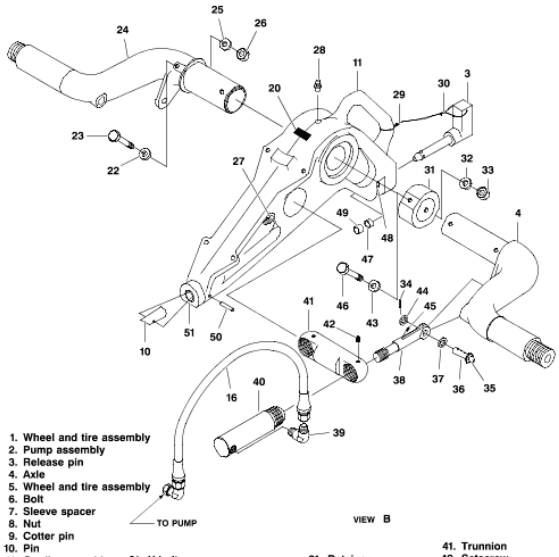
2. Position wheel and tire assembly on axle (4). Install sleeve spacer (7), nut (8), and cotter pin (9).

3. Inflate tires to 75 PSI (517 kPa).



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

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Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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Tubeless Tire - Removal

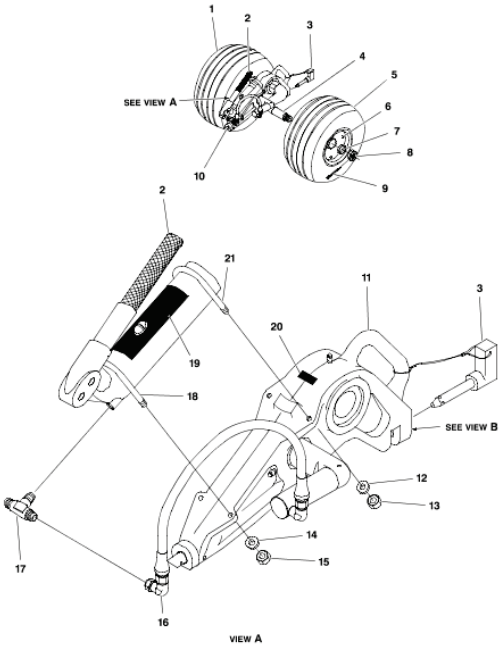
Procedure

1. Remove wheel and tire assembly (412-A-32-03-00-01A-520A-A / 00758).
2. Remove valve core.



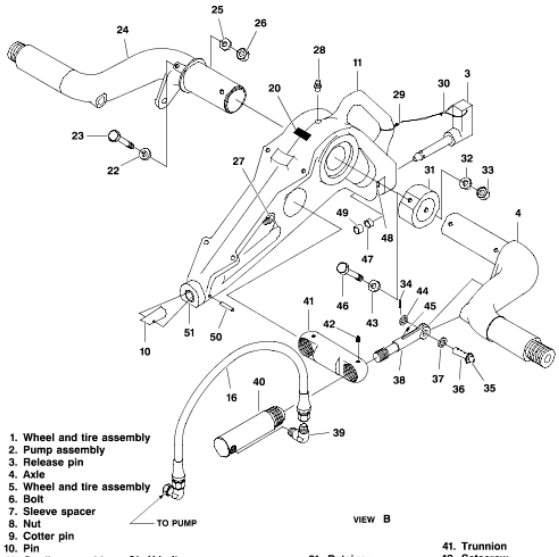
MAKE SURE THAT TIRE IS DEFLATED PRIOR TO REMOVING BOLTS IN FOLLOWING STEPS.

3. Remove bolts (6, Figure 1).
4. Separate wheel halves and remove tire.
5. Remove and discard nitrile wheel seal.
6. Remove valve stem.



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

ICN-412-A-320300-B-97499-00002-A-001-01

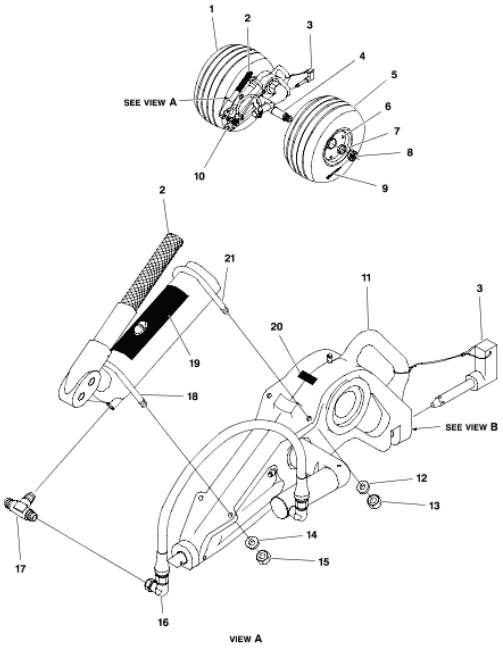
Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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Tubeless Tire - Installation

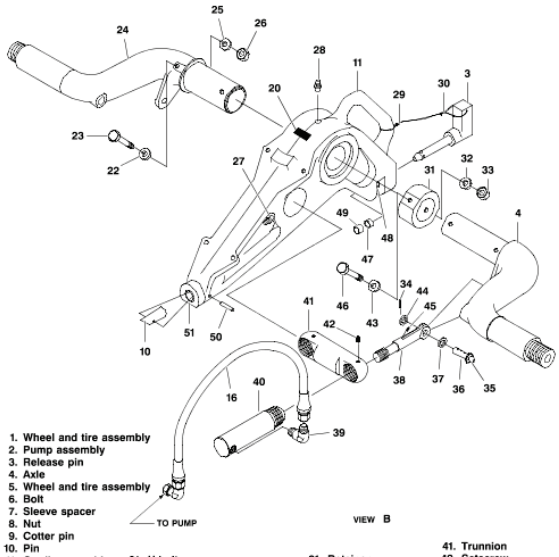
Procedure

1. Install new nitrile wheel seal on inboard machined surface of wheel half.
2. Install new valve stem.
3. Place tire on wheel half. Place opposite wheel half on tire and align holes for bolts (6, Figure 1).
4. Install bolts (6) with steel washer and nut. Torque nuts 125 to 150 inch-pounds (14.13 to 16.94 Nm).
5. Inflate tire to 75 PSI (517 kPa).
6. Install wheel and tire assembly (412-A-32-03-00-01A-720A-A / 00759).



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

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Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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Applicable to: ALL
ECCN EAR99

Tube Tire - Removal

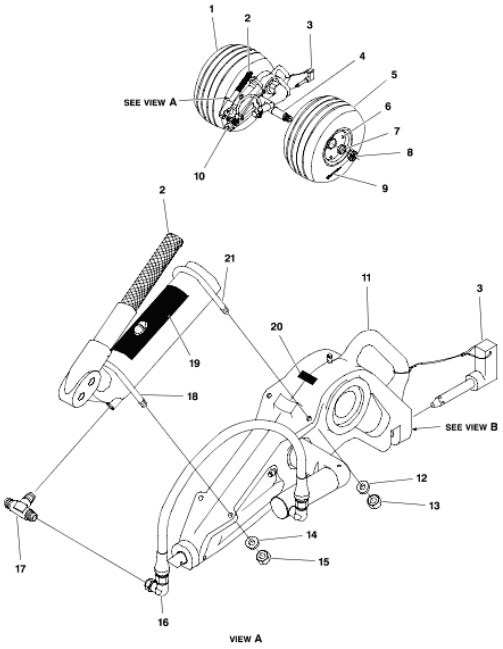
Procedure

1. Remove wheel and tire assembly (412-A-32-03-00-01A-520A-A / 00758).
2. Remove valve core.



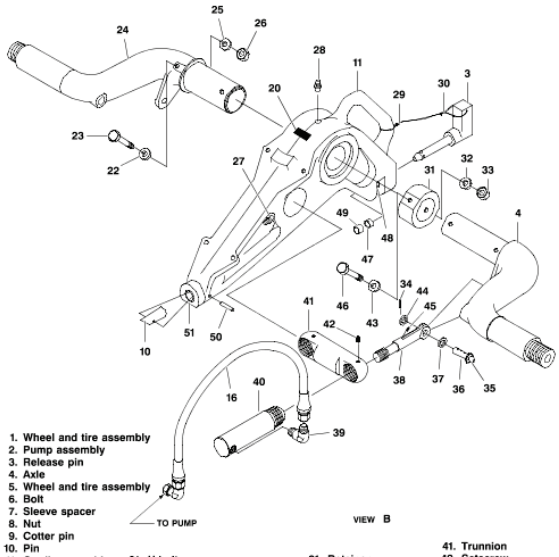
MAKE SURE THAT TIRE IS DEFLATED PRIOR TO REMOVING BOLTS IN FOLLOWING STEPS.

3. Remove bolts (6, Figure 1).
4. Separate wheel halves and remove tire and tube.
5. If installed, remove and discard nitrile wheel seal.



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

ICN-412-A-320300-B-97499-00002-A-001-01

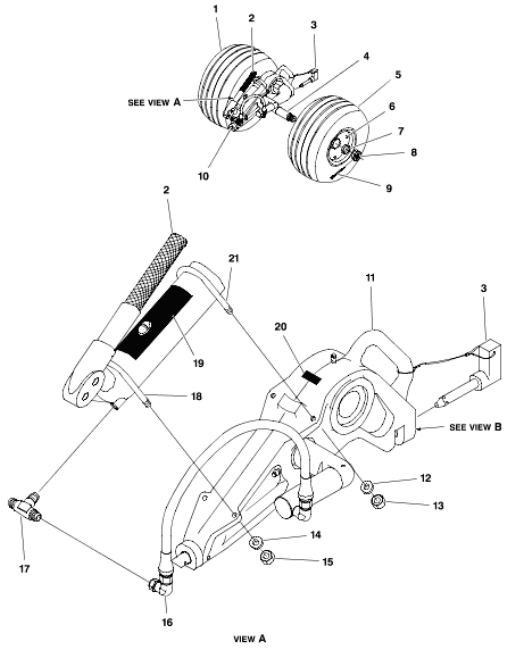
Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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Tube Tire - Installation

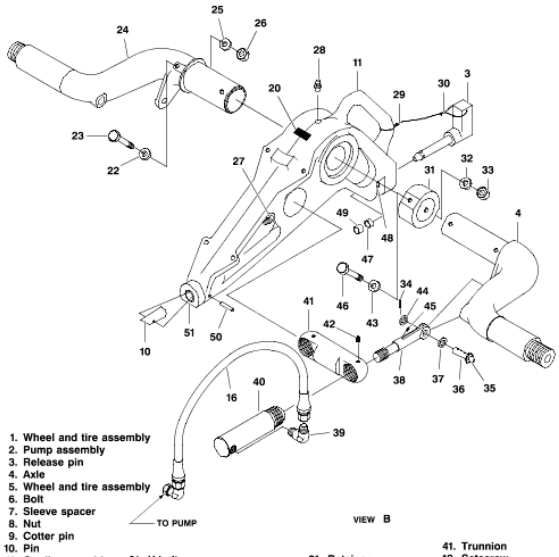
Procedure

1. Insert tube into tire.
2. Position tire and tube on wheel half. Position valve stem through hole in wheel.
3. Place opposite wheel half on tire and align holes for bolts (6, Figure 1).
4. Install bolts (6) with steel washer and nut. Torque nuts 125 to 150 inch-pounds (14.13 to 16.94 Nm).
5. Inflate tire to 75 PSI (517 kPa).
6. Install wheel and tire assembly (412-A-32-03-00-01A-720A-A / 00759).



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

ICN-412-A-320300-B-97499-00002-A-001-01

Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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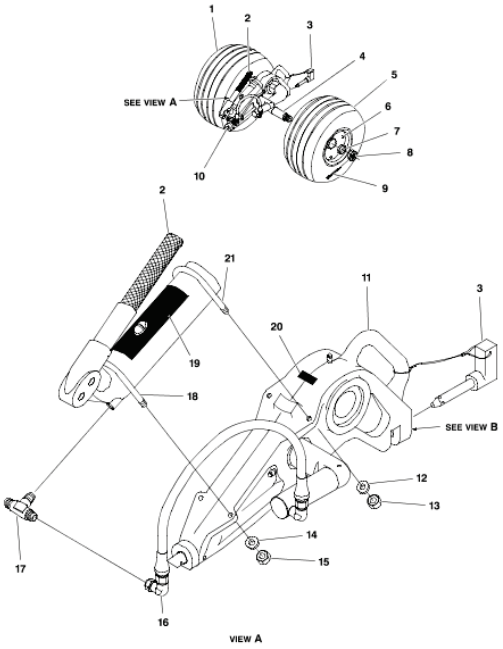
Pump Assembly - Description

For repair of pump assembly (2, Figure 1) (BU 953 or BU 953B) and to obtain service kit (JS0953), refer to:

Porter - Ferguson, Inc.

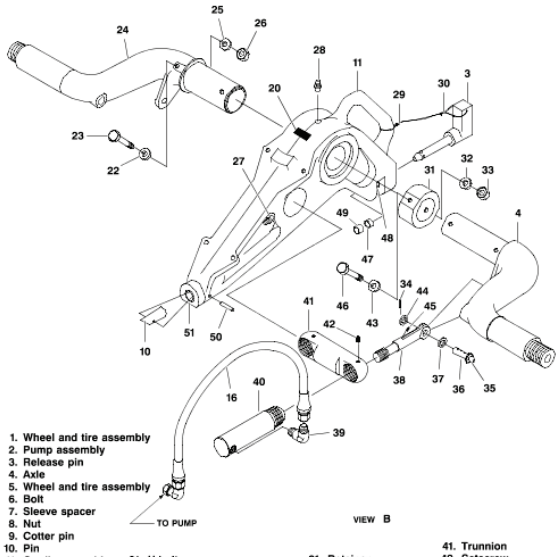
Beverly, Massachusetts, 01915

FAX (508) 921-5560



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

ICN-412-A-320300-B-97499-00002-A-001-01

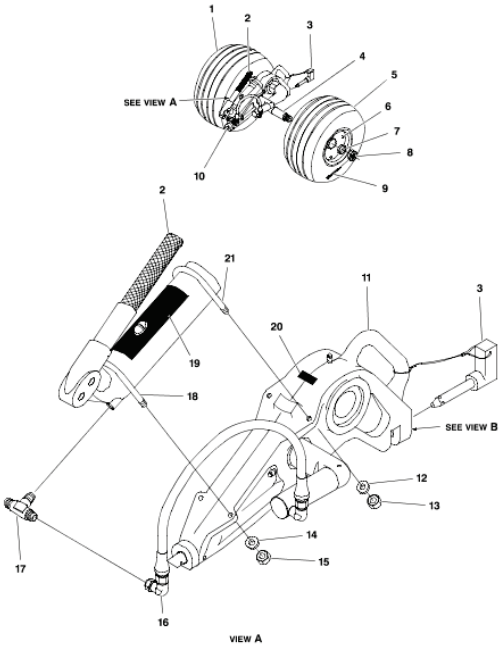
Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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Pump Assembly - Removal

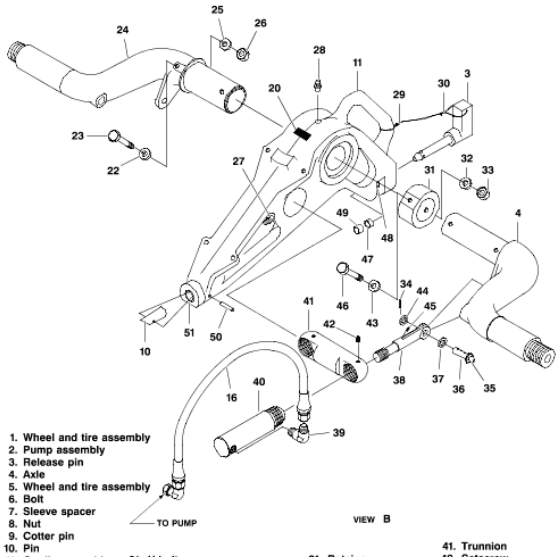
Procedure

1. Remove hose assembly (16, Figure 1) from tee (17). Remove hose assembly (not illustrated) from opposite side of tee.
2. Remove two nuts (13), two thin steel washers (12), and U-bolt (21).
3. Remove two nuts (15), two thin steel washers (14), and U-bolt (18).
4. Remove pump assembly (2).



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

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Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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Pump Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-002).
Hydraulic fluid	C-072	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-072).
Tape	C-596	AR	

Procedure

1. Apply tape (C-596) to threads of tee (17, Figure 1) and install tee in pump assembly (2).

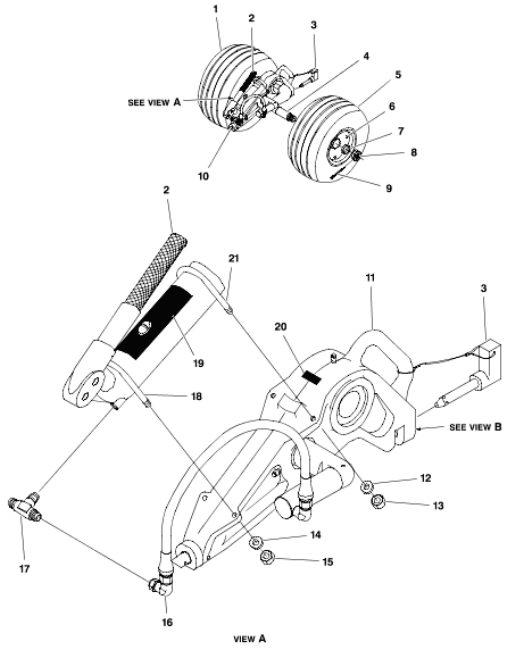
2. Position pump assembly (2) on cradle assembly (11). Install U-bolt (21), two thin steel washers (12), and nuts (13). Install U-bolt (18), two thin steel washers (14), and two nuts (15).

3. Install hose assembly (16) on tee (17). Install hose assembly (not illustrated) on opposite side of tee.

4. Fill pump with hydraulic fluid (C-002) or hydraulic fluid (C-072).

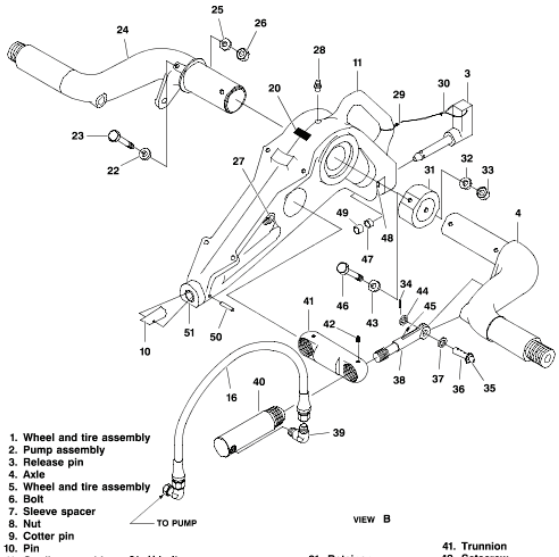
5. Bleed air from hydraulic fluid in pump assembly (2) lines.

6. Perform operational check of pump assembly (2). Install ground handling wheel assembly on helicopter and raise helicopter. Check for hydraulic leaks.



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

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Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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Decals - Removal and Installation

Preliminary requirements

Consumables, materials and expendables

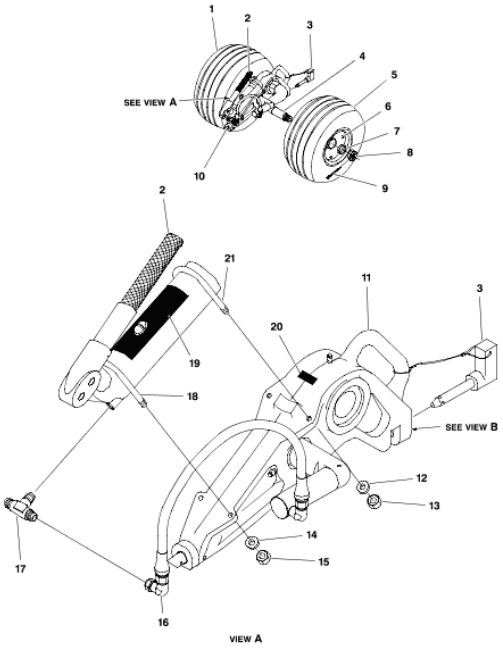
Nomenclature	Identification No.	Qty	Remarks
Abrasive cloth or paper	C-423	AR	

Procedure

1. Remove old decals (19 and 20, Figure 1).

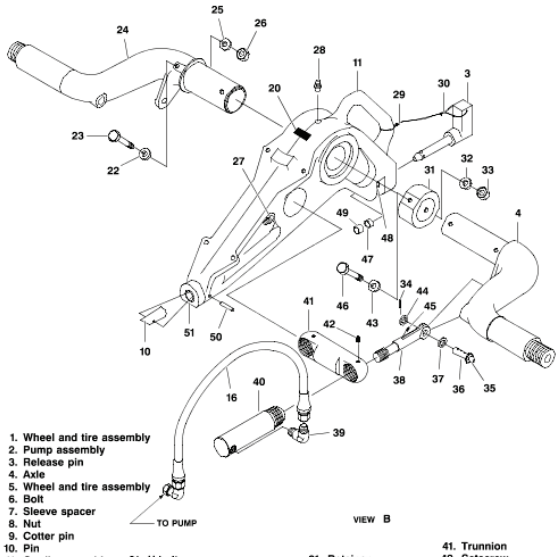
2. Clean area where decals will be installed with fine grit abrasive cloth or paper (C-423).

3. Install new decals (20) on pump cradle assembly (11).



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

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Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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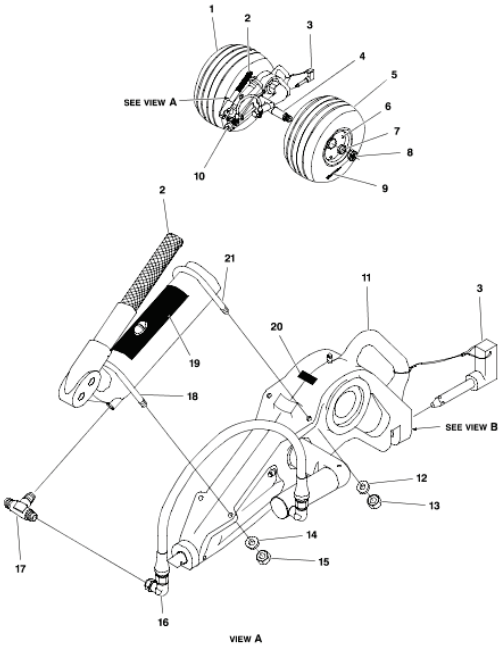
Ram Assembly and Trunnion - Removal

Procedure

NOTE

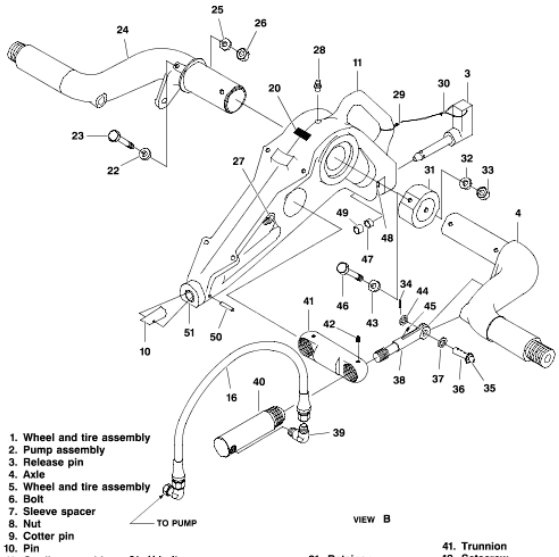
Removal procedure for ram assembly (40, Figure 1) is given. Removal procedure for opposite ram assembly (not illustrated) is the same.

1. Remove hose assembly (16) from elbow (39).
2. Remove elbow (39).
3. Remove lubrication pin (36), lubrication fitting (35), thin steel washer (44) from clevis (38) by removing cotter pin (34) and thin steel washer (37).
4. Remove setscrew (45).
5. Remove clevis (38) from ram assembly (40) by unscrewing.
6. Remove setscrew (42).
7. Remove ram assembly (40) by unscrewing.
8. Remove opposite ram assembly (not illustrated) in the same manner.
9. Remove trunnion (41) from cradle assembly (11) by sliding trunnion from cradle.



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

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Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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Ram Assembly and Trunnion - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-002).
Hydraulic fluid	C-072	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-072).
Tape	C-596	AR	

Procedure

NOTE

Installation procedures for ram assembly (40, Figure 1) is given. Installation procedures for opposite ram assembly (not illustrated) is the same.

1. If not previously accomplished, position trunnion (41) in cradle assembly (11).

2. Thread ram assembly (40) into trunnion (41) until it bottoms, then back out until port for elbow (39) is outboard. Install setscrew (42).

3. Thread clevis (38) into ram assembly (40) until clevis maintains a dimension of 1.48 inches (37.6 mm). Install setscrew (45).

4. Position clevis (38) on either side of hole in flange of axle (4). Install lubrication pin (36) with lubrication fitting (35) (facing out) through clevis (38) and flange of axle (4) with thin steel washer (37 and 44) on either side of clevis.

5. Install cotter pin (34).

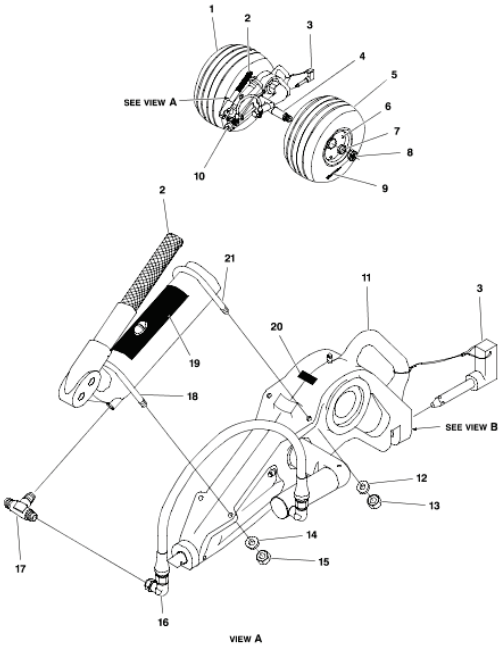
6. Coat threads of elbow (39) with tape (C-596) and install elbow in ram assembly (40).

7. Install hose assembly (16) onto elbow (39).

8. Fill pump assembly (2) with hydraulic fluid (C-002) or hydraulic fluid (C-072).

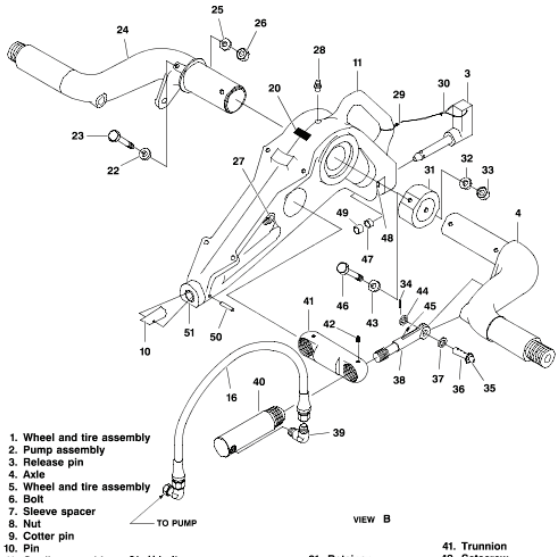
9. Bleed hydraulic pump assembly (2).

10. Perform operational check of ram. Install ground handling wheel assembly on helicopter and raise helicopter. Check for hydraulic fluid leaks.



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

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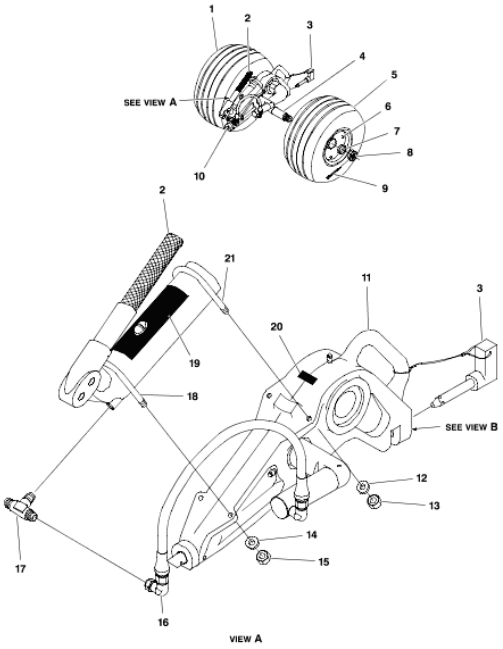
Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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Axles - Removal

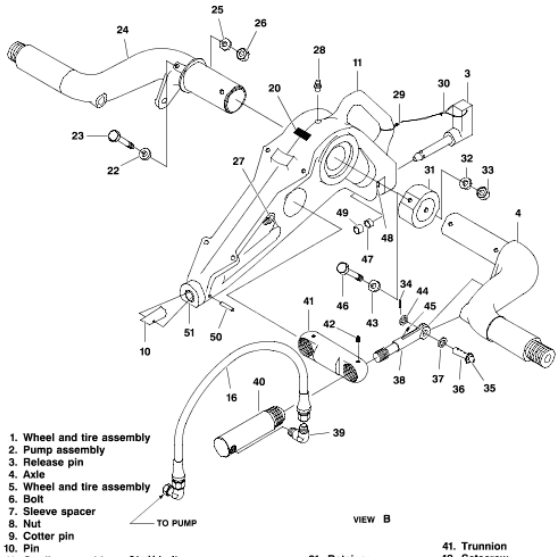
Procedure

1. Remove wheel and tire assemblies (1 and 5, Figure 1) (412-A-32-03-00-01A-520A-A / 00758).
2. Remove lubrication pin (36) and opposite lubrication pin (not illustrated) (412-A-32-03-00-06A-520A-A / 00768).
3. Remove nut (26), bolt (23), and thin steel washers (22 and 25).
4. Remove nut (33), bolt (46), and thin steel washers (32 and 43).
5. Remove axles (4 and 24).



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

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Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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Axles - Installation

Procedure

NOTE

If axles (4 and 24, Figure 1) have not been drilled for bolts (23 and 46), accomplish Step 1. through Step 8.. If axles have been drilled, accomplished Step 1. then proceed to Step 9..

1. Operate pump assembly (2) to fully extend ram assembly (40) and opposite ram (not illustrated).

2. Position axle (24) in cradle assembly (11) from helicopters right side. Position axle (4) in cradle assembly from helicopters left side with retainer (31) positioned on axle with larger dimension from bolt hole to edge of retainer spacer facing inboard.

3. Loosen setscrew (45) and adjust clevis (38) to hold axle (4) at 1.48 inches (37.5 mm) dimension, as shown in Figure 2, View B.

4. Adjust setscrew (45, Figure 1) to contact axle (4) and allow lubrication pin (36) to be installed. Install lubrication pin, thin steel washers (37 and 44), and cotter pin (34).

5. Adjust clevis at base of axle (24) in same manner outlined in Step 3. and Step 4..

6. Position axles (4 and 24) to 5.214 inches (132.43 mm) dimensions, as shown in Figure 2, View A.

7. Line drill axles (4 and 24, Figure 1) for bolts (23 and 46). Make holes 0.316 to 0.322 inches (8.03 to 8.17 mm) in diameter.

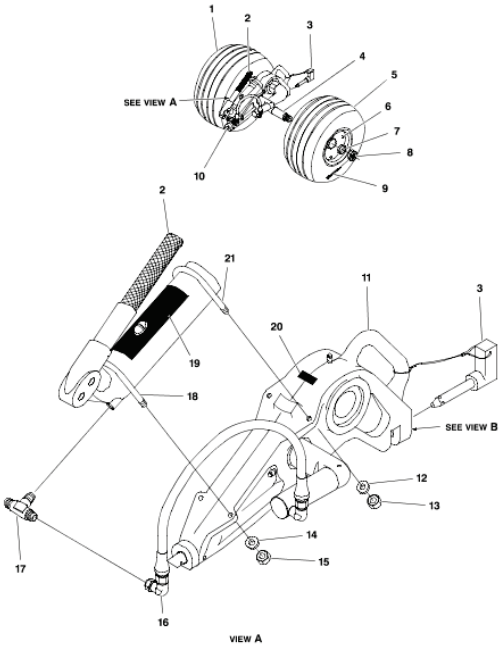
8. Remove axles (4 and 24) and remove chips from drilling operation, then position axles on cradle assembly (11) in original position.

9. Align holes in retainer (31) and axles (4 and 24). Install bolt (46), thin steel washers (32 and 43), and nut (33). Install bolt (23), thin steel washers (22 and 25), and nut (26).

10. Install wheel and tire assemblies (1 and 5) (412-A-32-03-00-01A-720A-A / 00759).

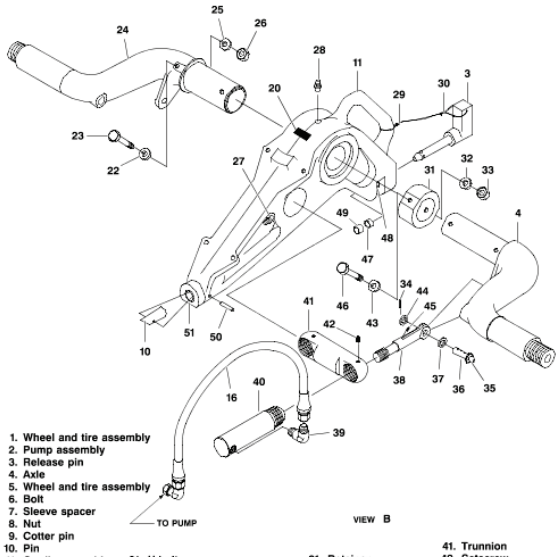
11. Lubricate ground handling wheels (412-A-32-03-00-00A-240A-A / 00757).

12. Perform operational check of ground handling wheel assembly. Install ground handling wheel assembly on helicopter and raise helicopter. Check for proper operation of ground handling wheel assembly.



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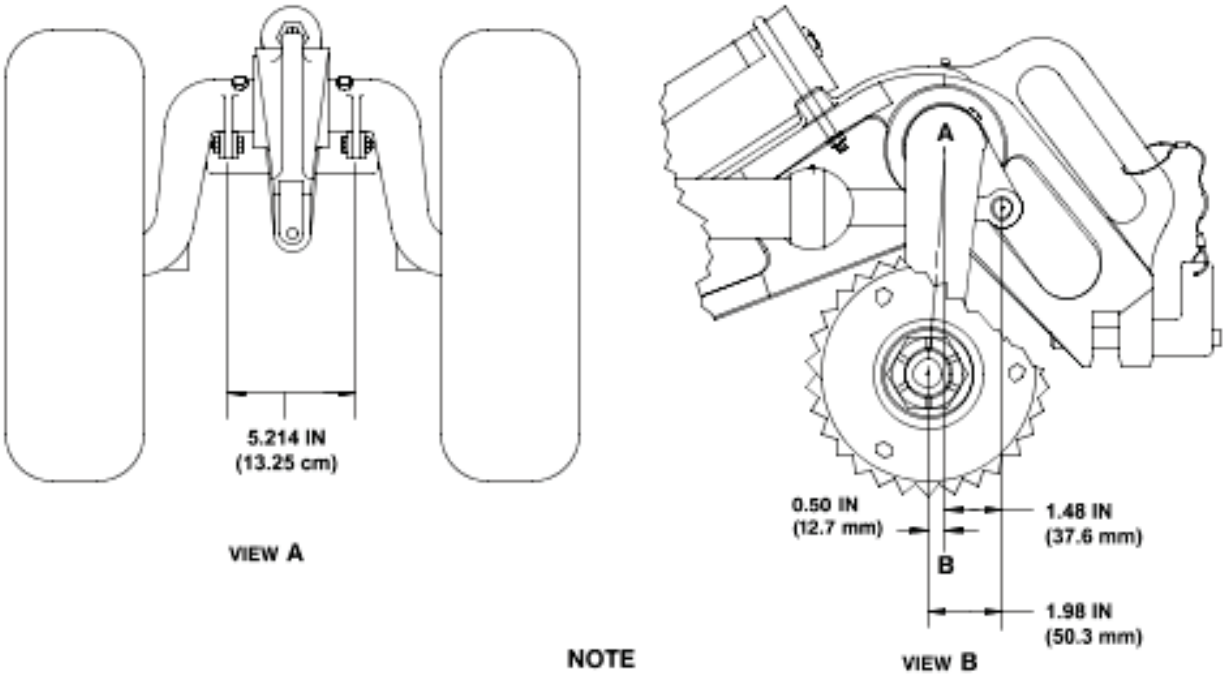
Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

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Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)



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Figure 2. Ground Handling Wheels Axle Installation (Sheet 1 of 1)

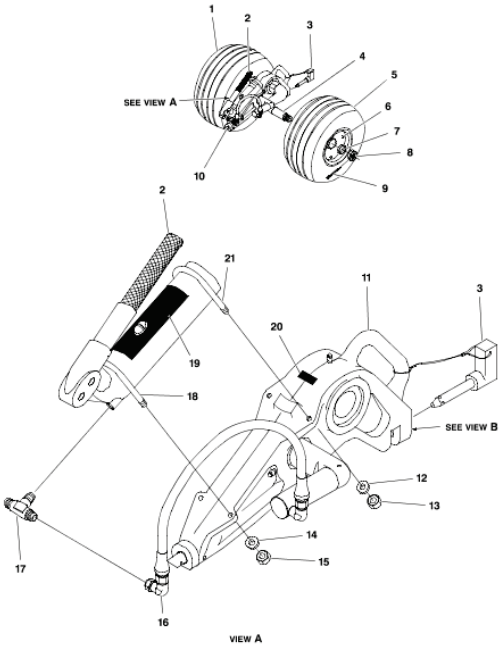
Bushings - Removal

Procedure

1. Remove wheels, pump assembly (2, Figure 1), axles (4 and 24), and other components as necessary to gain access to bushings (47, 48, 49, and 51) (412-A-32-03-00-01A-520A-A / 00758, 412-A-32-03-00-04A-520A-A / 00765, and 412-A-32-03-00-07A-520A-A / 00770). If bushing (51) must be replaced, remove spring pin (50) and pin (10).

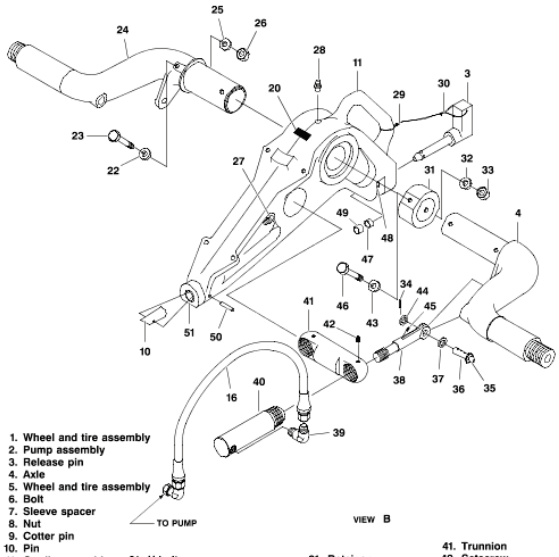
2. Heat cradle assembly (11) with heat lamp in area where bushings are to be removed.

3. Press or pull bushings (47, 48, 49, or 51) from cradle.



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

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Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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Bushings - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Zinc chromate primer	C-201	AR	

Procedure

1. Clean bore in cradle assembly (11, Figure 1) where bushings (47, 48, 49, or 51) will be installed.

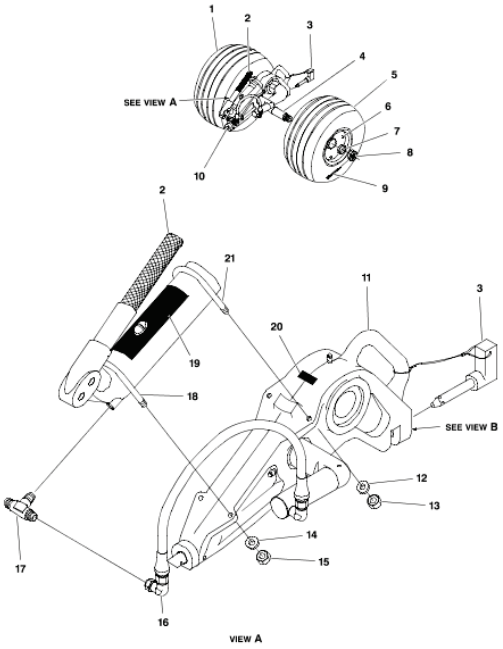
2. Heat cradle assembly (11) with a heat lamp in area where bushings are to be installed.

3. Coat bushings with zinc chromate primer (C-201) and press into cradle assembly (11) while primer is wet. Install bushing (47) first and ensure a recess of 0.060 inch (1.52 mm) gap from surface of cradle. Outboard faces of bushings (49 and 51) must be flush with surface of cradle. Bushing (48) must be flush with cradle surface at both ends. If bushings protrude past cradle surface, face off bushing.

4. Line ream bushings (47, 48, and 49) to 0.5000 to 0.5100 inch (12.7 to 12.954 mm). Check release pin (3) from slip fit in bushings.

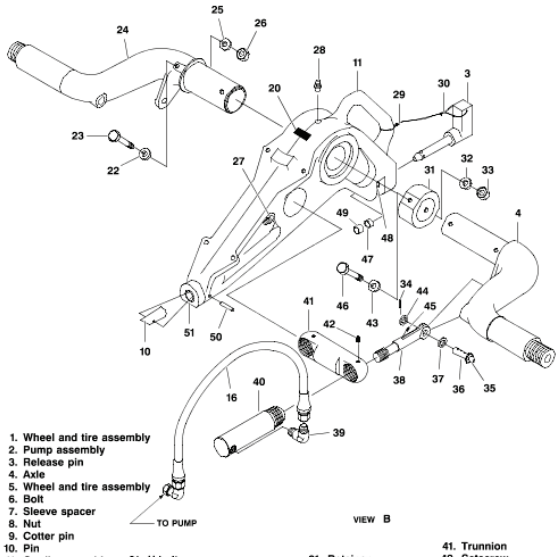
5. Ream bushing (51) 0.4995 to 0.5005 inch (12.688 to 12.712 mm). Position pin (10) in bushing with bevel facing down. Install spring pin (50).

6. Install wheels, axles (4 and 24), pump assembly (2), and other components that were removed to expose bushings (412-A-32-03-00-01A-720A-A / 00759, 412-A-32-03-00-04A-720A-A / 00766, and 412-A-32-03-00-07A-720A-A / 00771).



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Figure 1. Ground Handling Wheel Assembly (Sheet 1 of 2)



1. Wheel and tire assembly
2. Pump assembly
3. Release pin
4. Axle
5. Wheel and tire assembly
6. Bolt
7. Sleeve spacer
8. Nut
9. Cotter pin
10. Pin
11. Cradle assembly
12. Thin steel washer
13. Nut
14. Thin steel washer
15. Nut
16. Hose assembly
17. Tee
18. U-bolt
19. Decal
20. Decal
21. U-bolt
22. Thin steel washer
23. Bolt
24. Axle
25. Thin steel washer
26. Nut
27. Lubrication fitting
28. Lubrication fitting
29. Wire rope swaging sleeve
30. Wire rope
31. Retainer
32. Thin steel washer
33. Nut
34. Cotter pin
35. Lubrication fitting
36. Lubrication pin
37. Thin steel washer
38. Clevis
39. Elbow
40. Ram assembly
41. Trunnion
42. Setscrew
43. Thin steel washer
44. Thin steel washer
45. Setscrew
46. Bolt
47. Bushing
48. Bushing
49. Bushing
50. Spring pin
51. Bushing

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Figure 1. Ground Handling Wheel Assembly (Sheet 2 of 2)

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CHAPTER 52
DOORS AND WINDOWS

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Doors and Windows - General Description	DMC-412-A-52-00-00-00A-028A-A / 00775	2023-05-31	ALL
Access Doors, Panels, Cowlings, Fairings, and Covers - Installation Requirements - Description	DMC-412-A-52-00-00-00A-042A-A / 00776	2023-05-31	ALL
Access Doors, Panels, Cowlings, Fairings, and Covers - Seal Installation	DMC-412-A-52-00-00-00A-720A-A / 00777	2023-05-31	ALL
Crew Doors - Description and Maintenance	DMC-412-A-52-10-00-00A-042B-A / 00778	2023-05-31	ALL
Crew Doors - Removal	DMC-412-A-52-10-00-00A-520B-A / 00778.1	2023-08-28	ALL
Crew Doors - Inspection	DMC-412-A-52-10-00-00A-280A-A / 00780	2023-08-28	ALL
Crew Doors - Repair	DMC-412-A-52-10-00-00A-685A-A / 00781	2023-08-28	ALL
Crew Doors - Installation	DMC-412-A-52-10-00-00A-720B-A / 00781.1	2023-08-28	ALL
Crew Door Jettison Mechanism - Removal	DMC-412-A-52-10-00-01A-520B-A / 00781.2	2023-08-28	ALL
Crew Door Jettison Mechanism - Installation	DMC-412-A-52-10-00-01A-720B-A / 00781.3	2023-08-28	ALL
Crew Doors - Operational Check	DMC-412-A-52-10-00-00A-320A-A / 00785	2023-08-28	ALL
Crew Door Latches - Removal	DMC-412-A-52-10-00-02A-520A-A / 00786	2023-08-28	ALL
Crew Door Latches - Installation	DMC-412-A-52-10-00-02A-720A-A / 00787	2023-08-28	ALL
Crew Door Hinges - Removal	DMC-412-A-52-10-00-03A-520A-A / 00788	2023-08-28	ALL
Crew Door Hinges - Inspection	DMC-412-A-52-10-00-03A-280A-A / 00789	2023-05-31	ALL
Crew Door Hinges - Installation	DMC-412-A-52-10-00-03A-720A-A / 00790	2023-08-28	ALL
Crew Door Hinges - Repair	DMC-412-A-52-10-00-03A-685A-A / 00791	2023-05-31	ALL
Composite Crew Doors - Description and Maintenance	DMC-412-A-52-11-00-00A-042B-A / 00792	2023-05-31	ALL
Crew Doors - Removal	DMC-412-A-52-11-00-01A-520A-A / 00793	2023-08-28	ALL
Crew Doors - Inspection	DMC-412-A-52-11-00-01A-280A-A / 00794	2023-05-31	ALL
Crew Doors - Repair	DMC-412-A-52-11-00-01A-685A-A / 00795	2023-05-31	ALL
Crew Doors - Installation	DMC-412-A-52-11-00-01A-720A-A / 00796	2023-08-28	ALL
Crew Door Jettison Mechanism - Description	DMC-412-A-52-11-00-02A-042A-A / 00797	2023-08-28	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Crew Door Hinges - Removal	DMC-412-A-52-11-00-02A-520A-A / 00798	2023-05-31	ALL
Crew Door Hinges - Inspection	DMC-412-A-52-11-00-02A-280A-A / 00799	2023-05-31	ALL
Crew Door Hinges - Repair	DMC-412-A-52-11-00-02A-685A-A / 00800	2023-05-31	ALL
Crew Door Hinges - Installation	DMC-412-A-52-11-00-02A-720A-A / 00801	2023-05-31	ALL
Crew Door Cover - Removal	DMC-412-A-52-11-00-03A-520A-A / 00802	2023-05-31	ALL
Crew Door Cover - Installation	DMC-412-A-52-11-00-03A-720A-A / 00803	2023-05-31	ALL
Crew Door Handle - Removal	DMC-412-A-52-11-00-04A-520A-A / 00804	2023-05-31	ALL
Crew Door Handle - Installation	DMC-412-A-52-11-00-04A-720A-A / 00805	2023-05-31	ALL
Bell Crank Assembly - Removal	DMC-412-A-52-11-00-05A-520A-A / 00806	2023-05-31	ALL
Bell Crank Assembly - Installation	DMC-412-A-52-11-00-05A-720A-A / 00807	2023-05-31	ALL
Plunger Assembly - Removal	DMC-412-A-52-11-00-06A-520A-A / 00808	2023-05-31	ALL
Plunger Assembly - Installation	DMC-412-A-52-11-00-06A-720A-A / 00809	2023-05-31	ALL
Passenger/Cargo Area Access Doors - General Description	DMC-412-A-52-12-00-00A-042A-A / 00810	2023-05-31	ALL
Passenger/Cargo Door - Description	DMC-412-A-52-12-00-01A-042A-A / 00811	2023-05-31	ALL
Passenger/Cargo Door - Removal	DMC-412-A-52-12-00-01A-520A-A / 00812	2023-05-31	ALL
Passenger/Cargo Door - Inspection	DMC-412-A-52-12-00-01A-280A-A / 00813	2023-05-31	ALL
Passenger/Cargo Door - Repair	DMC-412-A-52-12-00-01A-685A-A / 00814	2023-05-31	ALL
Passenger/Cargo Door - Installation	DMC-412-A-52-12-00-01A-720A-A / 00815	2023-05-31	ALL
Passenger/Cargo Door - Operational Check and Adjustment	DMC-412-A-52-12-00-01A-320A-A / 00816	2023-05-31	ALL
Passenger/Cargo Door Latches - Removal	DMC-412-A-52-12-00-02A-520A-A / 00817	2023-05-31	ALL
Passenger/Cargo Door Latches - Installation	DMC-412-A-52-12-00-02A-720A-A / 00818	2023-05-31	ALL
Cargo Hinged Panels - Description	DMC-412-A-52-30-00-00A-042A-A / 00819	2023-05-31	ALL
Cargo Hinged Panels - Removal	DMC-412-A-52-30-00-00A-520A-A / 00820	2023-05-31	ALL
Cargo Hinged Panels - Inspection	DMC-412-A-52-30-00-00A-280A-A / 00821	2023-05-31	ALL
Cargo Hinged Panels - Repair	DMC-412-A-52-30-00-00A-685A-A / 00822	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Cargo Hinged Panels - Installation	DMC-412-A-52-30-00-00A-720A-A / 00823	2023-05-31	ALL
Cargo Hinged Panel Latching Mechanism - Removal	DMC-412-A-52-30-00-01A-520A-A / 00824	2023-05-31	ALL
Cargo Hinged Panel Latching Mechanism - Installation	DMC-412-A-52-30-00-01A-720A-A / 00825	2023-05-31	ALL
Baggage Compartment Door - Description	DMC-412-A-52-30-01-00A-042A-A / 00826	2023-05-31	ALL
Baggage Compartment Door - Removal	DMC-412-A-52-30-01-00A-520A-A / 00827	2023-05-31	ALL
Baggage Compartment Door - Inspection	DMC-412-A-52-30-01-00A-280A-A / 00828	2023-05-31	ALL
Baggage Compartment Door - Repair	DMC-412-A-52-30-01-00A-685A-A / 00829	2023-05-31	ALL
Baggage Compartment Door - Installation	DMC-412-A-52-30-01-00A-720A-A / 00830	2023-05-31	ALL
Nose Access Doors - Description	DMC-412-A-52-40-00-01A-042A-A / 00831	2023-05-31	ALL
Nose Access Doors - Removal	DMC-412-A-52-40-00-01A-520A-A / 00832	2023-05-31	ALL
Nose Access Doors - Inspection	DMC-412-A-52-40-00-01A-280A-A / 00833	2023-05-31	ALL
Nose Access Doors - Repair	DMC-412-A-52-40-00-01A-685A-A / 00834	2023-05-31	ALL
Nose Access Doors - Installation	DMC-412-A-52-40-00-01A-720A-A / 00835	2023-05-31	ALL
Miscellaneous Access Doors and Panels - Description	DMC-412-A-52-40-00-02A-042A-A / 00836	2023-05-31	ALL
Miscellaneous Access Doors and Panels - Removal	DMC-412-A-52-40-00-02A-520A-A / 00837	2023-05-31	ALL
Miscellaneous Access Doors and Panels - Inspection	DMC-412-A-52-40-00-02A-280A-A / 00838	2023-05-31	ALL
Miscellaneous Access Doors and Panels - Repair	DMC-412-A-52-40-00-02A-685A-A / 00839	2023-05-31	ALL
Miscellaneous Access Doors and Panels - Installation	DMC-412-A-52-40-00-02A-720A-A / 00840	2023-05-31	ALL
Tail Rotor Driveshaft Access Covers - Description	DMC-412-A-52-40-00-03A-042A-A / 00841	2023-05-31	ALL
Tail Rotor Driveshaft Access Covers - Removal	DMC-412-A-52-40-00-03A-520A-A / 00842	2023-05-31	ALL
Tail Rotor Driveshaft Access Covers - Inspection	DMC-412-A-52-40-00-03A-280A-A / 00843	2023-05-31	ALL
Tail Rotor Driveshaft Access Covers - Repair	DMC-412-A-52-40-00-03A-685A-A / 00844	2023-05-31	ALL
Tail Rotor Driveshaft Access Covers - Installation	DMC-412-A-52-40-00-03A-720A-A / 00845	2023-05-31	ALL
Cabin Roof Windows - Description	DMC-412-A-52-01-00-01A-042A-A / 00846	2023-05-31	ALL
Cabin Roof Windows - Removal	DMC-412-A-52-01-00-01A-520A-A / 00847	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Cabin Roof Windows - Inspection	DMC-412-A-52-01-00-01A-280A-A / 00848	2023-05-31	ALL
Cabin Roof Windows - Repair	DMC-412-A-52-01-00-01A-685A-A / 00849	2023-05-31	ALL
Cabin Roof Windows - Installation	DMC-412-A-52-01-00-01A-720A-A / 00850	2023-05-31	ALL
Upper Crew Door Windows - Description	DMC-412-A-52-01-00-02A-042A-A / 00851	2023-05-31	ALL
Upper Crew Door Windows - Removal	DMC-412-A-52-01-00-02A-520A-A / 00852	2023-05-31	ALL
Upper Crew Door Windows - Inspection	DMC-412-A-52-01-00-02A-280A-A / 00853	2023-05-31	ALL
Upper Crew Door Windows - Repair	DMC-412-A-52-01-00-02A-685A-A / 00854	2023-05-31	ALL
Upper Crew Door Windows - Installation	DMC-412-A-52-01-00-02A-720A-A / 00855	2023-05-31	ALL
Forward Crew Door Windows - Description	DMC-412-A-52-01-00-03A-042A-A / 00856	2023-05-31	ALL
Forward Crew Door Windows - Removal	DMC-412-A-52-01-00-03A-520A-A / 00857	2023-05-31	ALL
Forward Crew Door Windows - Inspection	DMC-412-A-52-01-00-03A-280A-A / 00858	2023-05-31	ALL
Forward Crew Door Windows - Repair	DMC-412-A-52-01-00-03A-685A-A / 00859	2023-05-31	ALL
Forward Crew Door Windows - Installation	DMC-412-A-52-01-00-03A-720A-A / 00860	2023-05-31	ALL
Lower Crew Door Windows - Description	DMC-412-A-52-01-00-04A-042A-A / 00861	2023-05-31	ALL
Lower Crew Door Windows - Removal	DMC-412-A-52-01-00-04A-520A-A / 00862	2023-05-31	ALL
Lower Crew Door Windows - Inspection	DMC-412-A-52-01-00-04A-280A-A / 00863	2023-05-31	ALL
Lower Crew Door Windows - Repair	DMC-412-A-52-01-00-04A-685A-A / 00864	2023-05-31	ALL
Lower Crew Door Windows - Installation	DMC-412-A-52-01-00-04A-720A-A / 00865	2023-05-31	ALL
Cargo Hinged Panel Door Windows - Description	DMC-412-A-52-01-00-05A-042A-A / 00866	2023-05-31	ALL
Cargo Hinged Panel Door Windows - Removal	DMC-412-A-52-01-00-05A-520A-A / 00867	2023-05-31	ALL
Cargo Hinged Panel Door Windows - Inspection	DMC-412-A-52-01-00-05A-280A-A / 00868	2023-05-31	ALL
Cargo Hinged Panel Door Windows - Repair	DMC-412-A-52-01-00-05A-685A-A / 00869	2023-05-31	ALL
Cargo Hinged Panel Door Windows - Installation	DMC-412-A-52-01-00-05A-720A-A / 00870	2023-05-31	ALL
Passenger Door Windows - Description	DMC-412-A-52-01-00-06A-042A-A / 00871	2023-05-31	ALL
Passenger Door Windows - Removal	DMC-412-A-52-01-00-06A-520A-A / 00872	2023-05-31	ALL

Document title	Data module code / Seq number		Issue date	Applicable to
Passenger Door Windows - Inspection	DMC-412-A-52-01-00-06A-280A-A / 00873		2023-05-31	ALL
Passenger Door Windows - Installation	DMC-412-A-52-01-00-06A-720A-A / 00874		2023-05-31	ALL
Lower Nose Section Windows - Description	DMC-412-A-52-01-00-07A-042A-A / 00875		2023-05-31	ALL
Lower Nose Section Windows - Removal	DMC-412-A-52-01-00-07A-520A-A / 00876		2023-05-31	ALL
Lower Nose Section Windows - Inspection	DMC-412-A-52-01-00-07A-280A-A / 00877		2023-05-31	ALL
Lower Nose Section Windows - Repair	DMC-412-A-52-01-00-07A-685A-A / 00878		2023-05-31	ALL
Lower Nose Section Windows - Installation	DMC-412-A-52-01-00-07A-720A-A / 00879		2023-05-31	ALL
Composite Crew Door Windows - Description	DMC-412-A-52-01-00-10A-042A-A / 00880		2023-05-31	ALL
Composite Crew Door Windows - Removal	DMC-412-A-52-01-00-10A-520A-A / 00881		2023-05-31	ALL
Composite Crew Door Windows - Inspection	DMC-412-A-52-01-00-10A-280A-A / 00882		2023-05-31	ALL
Composite Crew Door Windows - Repair	DMC-412-A-52-01-00-10A-685A-A / 00883		2023-05-31	ALL
Composite Crew Door Windows - Installation	DMC-412-A-52-01-00-10A-720A-A / 00884		2023-05-31	ALL
Windshields - Description	DMC-412-A-52-01-00-08A-042A-A / 00885		2023-05-31	ALL
Windshields - Removal	DMC-412-A-52-01-00-08A-520A-A / 00886		2023-05-31	ALL
Windshields - Inspection	DMC-412-A-52-01-00-08A-280A-A / 00887		2023-05-31	ALL
Windshields - Cleaning	DMC-412-A-52-01-00-08A-251A-A / 00888		2023-05-31	ALL
Windshields - Repair	DMC-412-A-52-01-00-08A-685A-A / 00889		2023-05-31	ALL
Windshields - Installation	DMC-412-A-52-01-00-08A-720A-A / 00890		2023-05-31	ALL
Heated Windshield Kit (412-706-010/-138) - Description	DMC-412-A-52-01-00-09A-042A-A / 00891		2023-05-31	ALL
Heated Windshield Kit (412-706-010/-138) - Removal	DMC-412-A-52-01-00-09A-520A-A / 00892	C	2025-11-18	ALL
Heated Windshield Kit (412-706-010/-138) - Cleaning	DMC-412-A-52-01-00-09A-251A-A / 00893		2023-05-31	ALL
Heated Windshield Kit (412-706-010/-138) - Inspection	DMC-412-A-52-01-00-09A-280A-A / 00894	C	2025-11-18	ALL
Heated Windshield Kit (412-706-010/-138) - Repair	DMC-412-A-52-01-00-09A-685A-A / 00895		2023-05-31	ALL
Heated Windshield Kit (412-706-010/-138) - Installation	DMC-412-A-52-01-00-09A-720A-A / 00896	C	2025-11-18	ALL

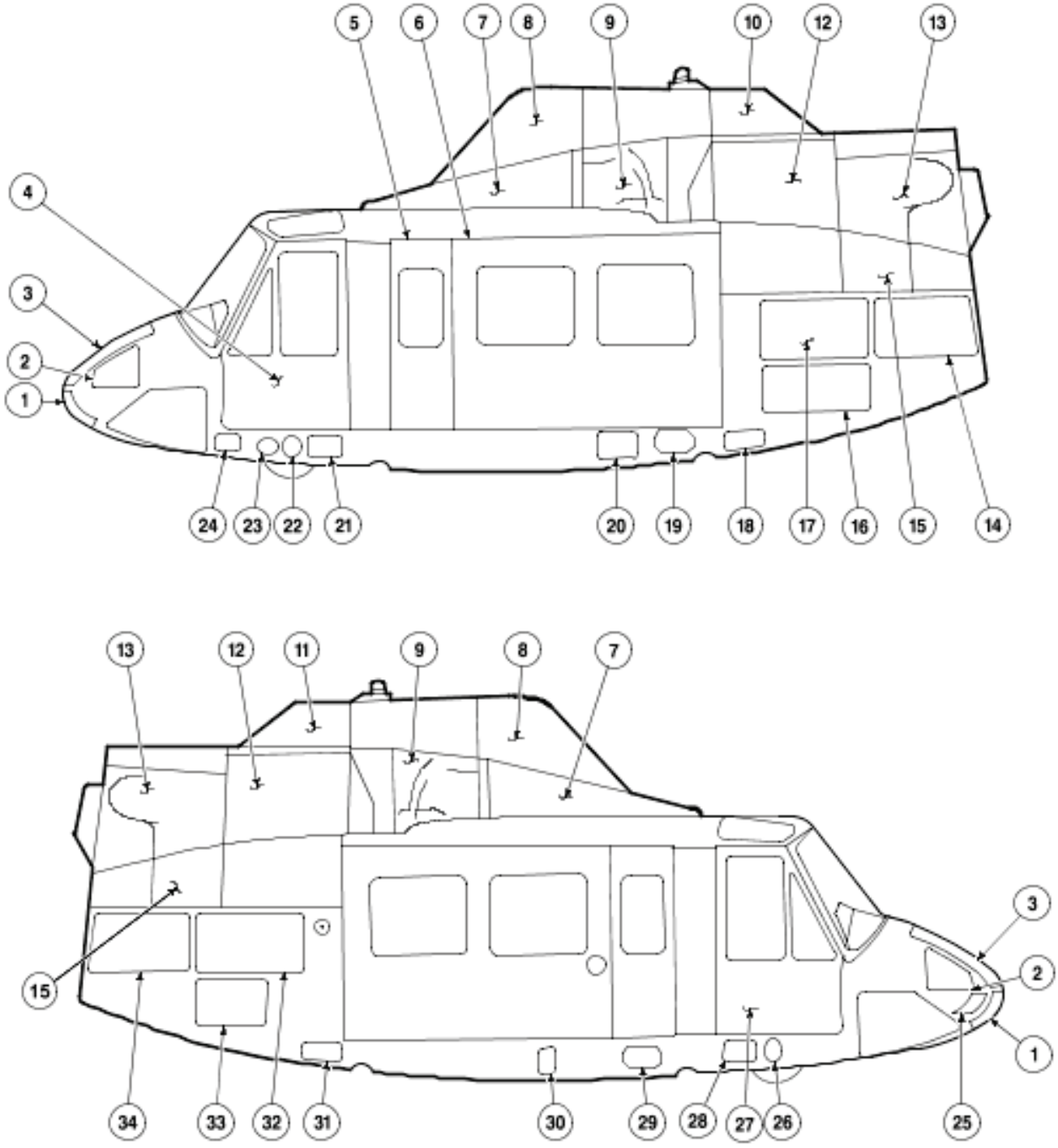
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Doors and Windows - General Description

This chapter provides maintenance data for crew doors, passenger/cargo doors, access doors and panels, tail rotor driveshaft covers, and windows.

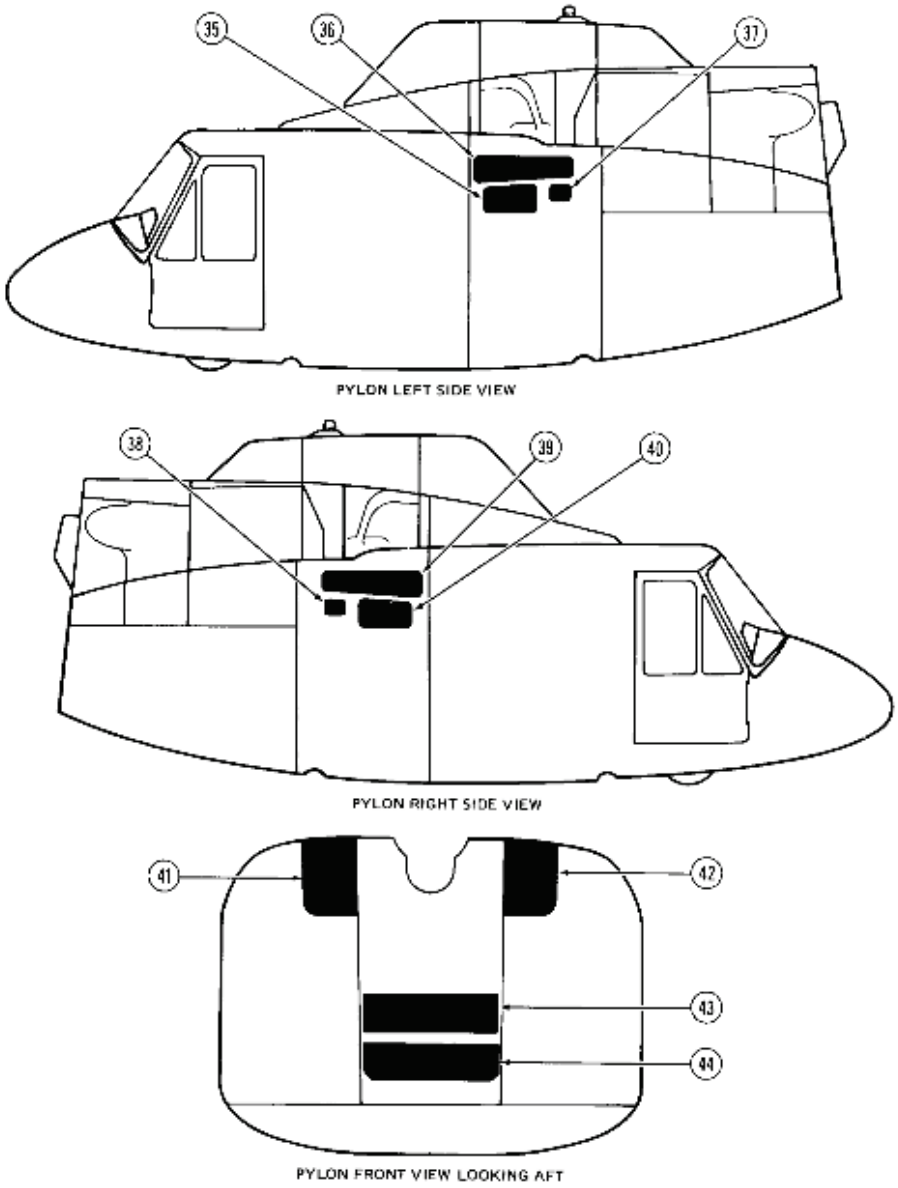
The doors covered in this chapter consist of doors, panels, cowlings, and fairings illustrated in Figure 1. The legend in Figure 1 includes two columns. The left column identifies individual doors, panels, etc. The right column lists major components which are accessible when door, panel, etc. is removed.

The windows (Figure 2) covered in this chapter consist of cockpit and passenger area transparent panels (windshields and windows).



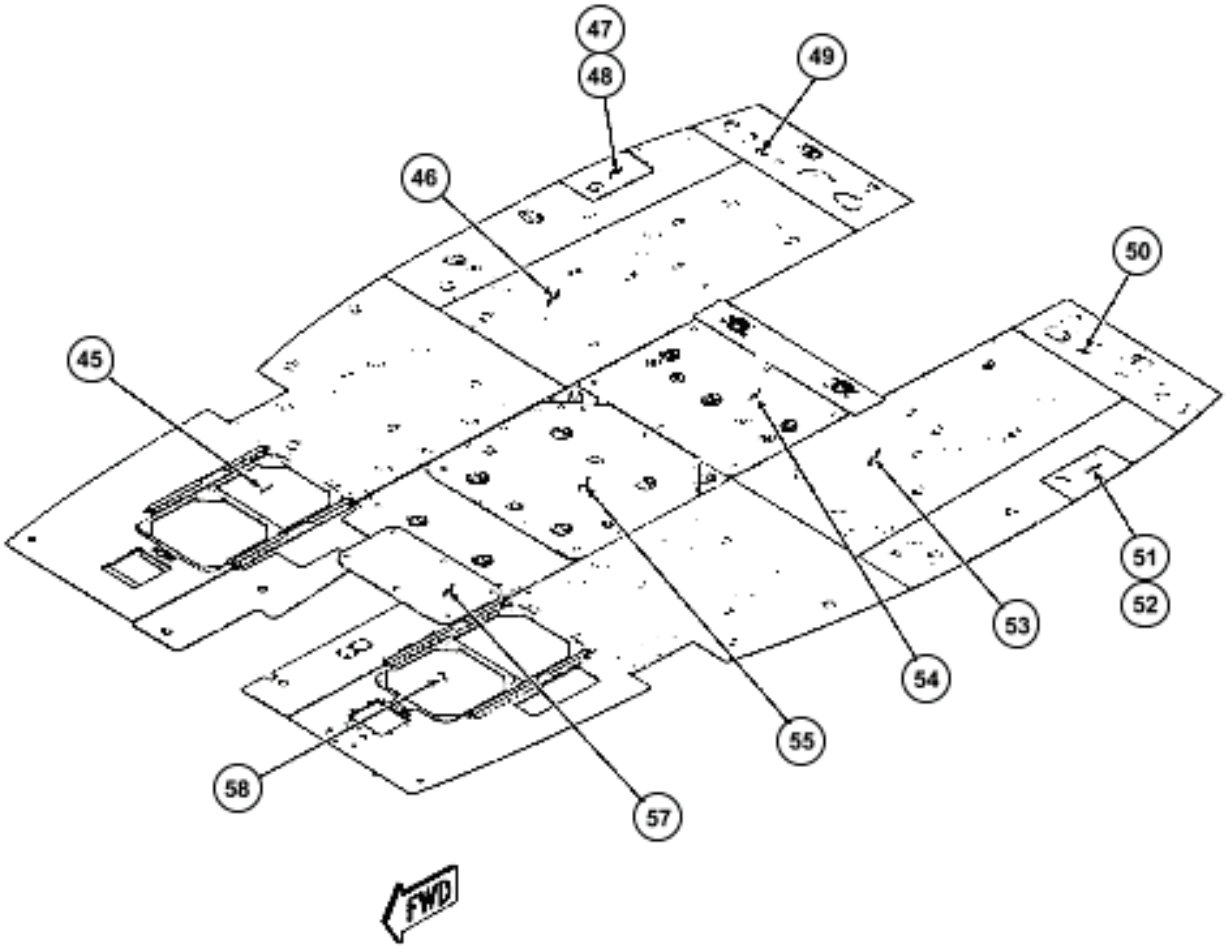
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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 1 of 6)



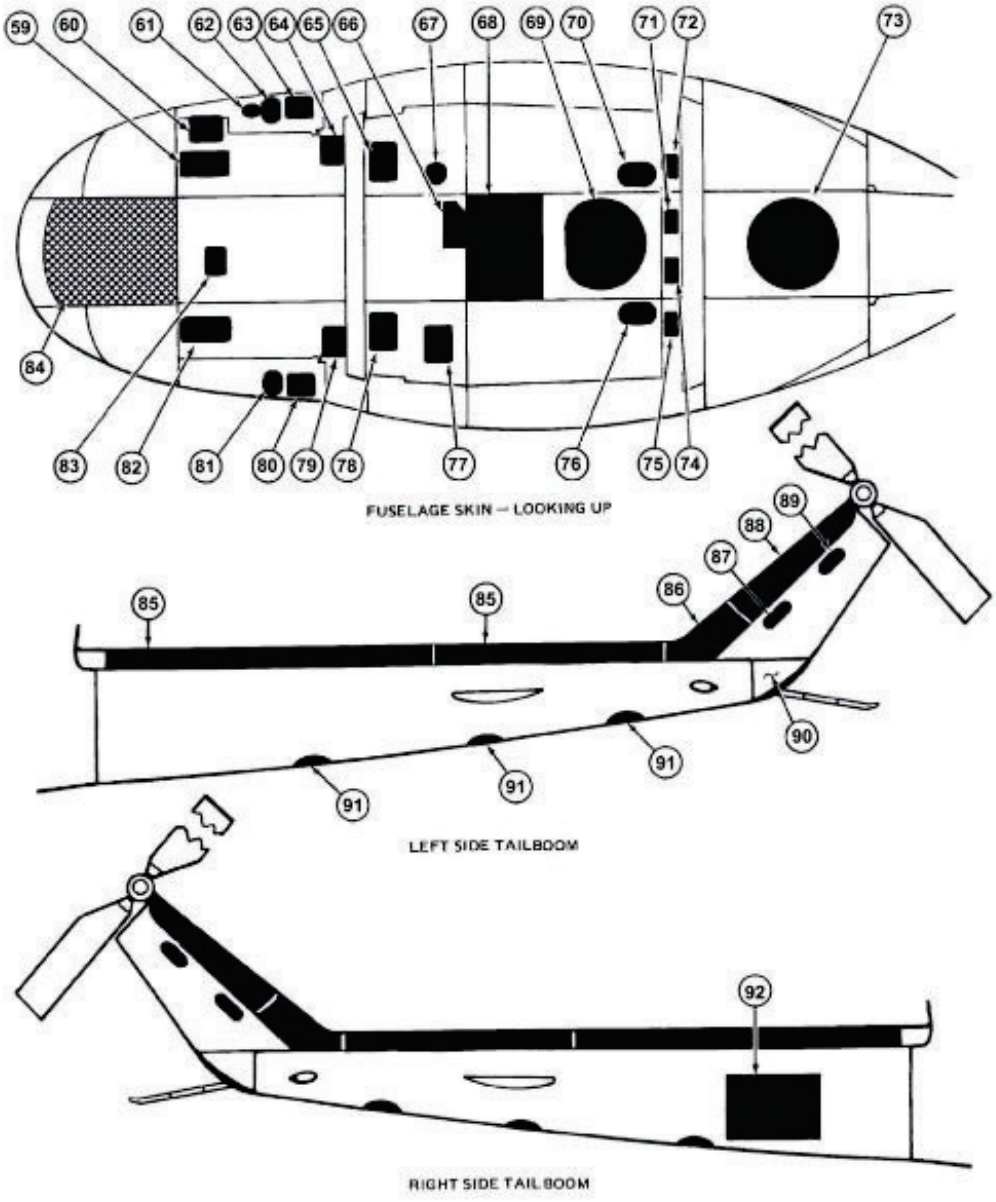
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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 2 of 6)



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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 3 of 6)



ICN-412-A-520000-B-97499-00004-A-001-01

Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 4 of 6)

INDEX NO.	NAME	ACCESS TO
1.	Lower nose door	Battery
2.	Cover	Automatic direction finder
3.	Upper nose door	Avionics equipment
4.	Crew door (copilot)	Copilot station
5.	Cargo hinged panel	Cabin
6.	Passenger cargo door	Cabin
7.	Transmission fairing	Hydraulic reservoirs, rotating controls
8.	Forward pylon fairing assembly	Transmission, swashplate, hydraulic reservoirs, hyd system 2 pump
9.	Fairing, engine inlet	Engine inlet, fuel controls, starter generator
10.	Aft pylon fairing assembly	Transmission, rotating controls
11.	Door	LH/RH engine
12.	Door	LH/RH engine power section
13.	Engine oil cooler support cowling assembly	Engine reduction gearbox
14.	Door	Radar altimeter
15.	Lower engine cowl assembly	LH/RH engine power section, fire extinguisher container
16.	Door	Electrical compartment
17.	Door	Navigation equipment
18.	Door	Avionics and electrical wiring
19.	Door	Hydraulic system 1 test connectors
20.	Door	Hydraulic system 2 test connectors
21.	Door	Cyclic flight controls
22.	Door	Flight controls
23.	Door	Flight controls
24.	Door	Flight controls
25.	Door	Auxiliary power unit receptacle
26.	Door	
27.	Crew door (pilot)	Pilot station
28.	Door	
29.	Door	
30.	Door	
31.	Door	Heater duct
32.	Door	
33.	Door	Heater variable control mixing valve
34.	Door	Tail rotor servo actuator, heater duct noise suppressor
35.	Door	Transmission, flight controls, lift link
36.	Door, pylon bulkhead	Transmission mounts
37.	Door	Transmission oil lines
38.	Door	Transmission oil line
39.	Door, pylon bulkhead	Transmission mounts
40.	Door	Hydraulic system 1 pump
41.	Cover	Engine accessory gearbox and drain valve, electric chip detector
42.	Cover	Engine accessory gearbox and drain valve, electric chip detector
43.	Cover	Flight controls hydraulic actuators
44.	Cover	Hydraulic integrated modules and filters

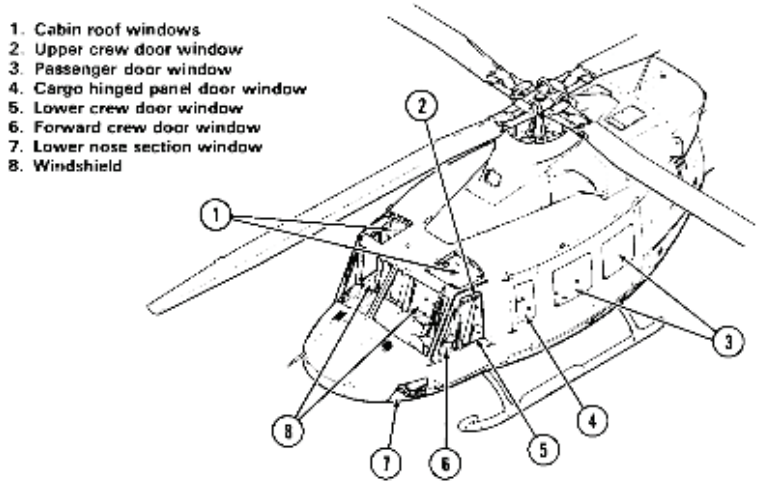
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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 5 of 6)

INDEX NO.	NAME	ACCESS TO
45.	Panel	Fore and aft force gradient, magnetic brake, flight controls
46.	Panel	Fuel lines and fuel cell
47.	Panel	
48.	Door	
49.	Panel	Fuel lines
50.	Panel	Fuel lines
51.	Panel	
52.	Door	
53.	Panel	Fuel cell
54.	Panel	Flight controls
55.	Panel	AFCS actuators, fwd fuel crossover valve
56.	Panel	Collective jackshaft
57.	Door	Flight controls
58.	Panel	Force gradient, flight controls
59.	Panel	Flight controls
60.	Door	Flight controls
61.	Door	Flight controls
62.	Door	Flight controls
63.	Door	Flight controls
64.	Door	Flight controls
65.	Door	Fuel lines, electrical wiring
66.	Door	Forward fuel crossover valve
67.	Door	Fuel line
68.	Door	VHF antenna
69.	Opening, cargo hook	Cargo hook, hydraulic lines, transmission sump
70.	Door	Fuel boost pump, fuel cell drain
71.	Door	Fuel lines, aft crossover and main
72.	Door	Fuel lines, aft crossover and main
73.	Door	Droop compensator control, Hydraulic lines, aft center fuel cell
74.	Door	Fuel lines, main and aft crossover
75.	Door	Fuel lines, main and aft crossover
76.	Door	Fuel boost pump, fuel cell drain
77.	Door	Fuel lines, heater controls, duct, mixing valve
78.	Door	AFCS, flight controls
79.	Door	Flight controls, collective
80.	Door	Flight controls, cyclic
81.	Door	Flight controls
82.	Door	Flight controls, searchlight
83.	Door	Flight controls, jackshaft
84.	Lower nose access panel	Electrical equipment
85.	Cover	Driveshaft, tail rotor
86.	Cover	Intermediate gearbox
87.	Door	Flight controls
88.	Cover	Driveshaft, tail rotor
89.	Door	Flight controls
90.	Fairing	Tail skid
91.	Door	Flight controls
92.	Door	Baggage compartment

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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 6 of 6)



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Figure 2. Windows and Windshields (Sheet 1 of 1)

Access Doors, Panels, Cowlings, Fairings, and Covers - Installation Requirements - Description

Refer to Table 1 to identify doors, panels, cowlings, fairings, and covers which shall be installed for ground run, jacking, towing, and flight.

Table 1. Installation Requirements for Access Doors, Panels, Cowlings, Fairings, and Covers

Figure 1 INDEX NO.	NAME	GROUND RUN	REQUIRED TO BE IN PLACE FOR:		
			JACKING	TOWING	FLIGHT
1	Lower nose door	No	No	No	Yes
2	Cover	No	No	No	Yes
3	Upper nose door	No	No	No	Yes
4	Crew door, copilot	No	No	No	(1)
5	Cargo hinged panel	No	No	No	(1)
6	Passenger cargo door	No	No	No	(1)
7	Transmission fairings	No	No	No	Yes
8	Forward pylon fairing assembly	No	No	No	Yes
9	Fairing, engine inlet	No	No	No	Yes
10	Aft pylon fairing assembly	No	No	No	Yes
11	Door	No	No	No	Yes
12	Door	(2)	No	No	Yes
13	Engine oil cooler support cowling assembly	No	No	No	Yes
14	Door	No	No	No	Yes
15	Lower engine cowl assembly	No	No	No	Yes
16	Door	No	No	No	Yes
17	Door	No	No	No	Yes
18	Door	No	No	No	Yes
19	Door	No	No	No	Yes
20	Door	No	No	No	Yes
21	Door	No	No	No	Yes
22	Door	No	No	No	Yes
23	Door	No	No	No	Yes
24	Door	No	No	No	Yes
25	Door	No	No	No	Yes
26	Door	No	No	No	Yes
27	Crew door (pilot)	No	No	No	(1)
28	Door	No	No	No	Yes
29	Door	No	No	No	Yes
30	Door	No	No	No	Yes
31	Door	No	No	No	Yes
32	Door	No	No	No	Yes
33	Door	No	No	No	Yes
34	Door	No	No	No	Yes
35	Door	(3)	No	No	Yes
36	Door, pylon bulkhead	(3)	No	No	Yes
37	Door	(3)	No	No	Yes
38	Door	(3)	No	No	Yes

Table 1. Installation Requirements for Access Doors, Panels, Cowlings, Fairings, and Covers (continued)

Figure 1 INDEX NO.	NAME	GROUND RUN	REQUIRED TO BE IN PLACE FOR: JACKING	TOWING	FLIGHT
39	Door, pylon bulkhead	(3)	No	No	Yes
40	Door	(3)	No	No	Yes
41	Cover	(3)	No	No	Yes
42	Cover	(3)	No	No	Yes
43	Cover	(3)	No	No	Yes
44	Cover	(3)	No	No	Yes
45	Panel	Yes	No	No	Yes
46	Panel	Yes	No	Yes	Yes
47	Panel	Yes	No	No	Yes
48	Door	No	No	No	Yes
49	Panel	Yes	Yes	Yes	Yes
50	Panel	Yes	Yes	Yes	Yes
51	Panel	Yes	No	No	Yes
52	Door	No	No	No	Yes
53	Panel	Yes	No	Yes	Yes
54	Panel	Yes	Yes	Yes	Yes
55	Panel	Yes	Yes	Yes	Yes
56	Panel	Yes	Yes	No	Yes
57	Panel	Yes	Yes	Yes	Yes
58	Panel	Yes	No	No	Yes
59	Panel	No	No	No	Yes
60	Door	No	No	No	Yes
61	Door	No	No	No	Yes
62	Door	No	No	No	Yes
63	Door	No	No	No	Yes
64	Door	Yes	No	No	Yes
65	Door	Yes	No	No	Yes
66	Door	Yes	No	No	Yes
67	Door	No	No	No	Yes
68	Door	Yes	No	No	Yes
69	Opening, cargo hook	No	No	No	No
70	Door	Yes	No	No	Yes
71	Door	Yes	No	No	Yes
72	Door	Yes	No	No	Yes
73	Door	Yes	No	No	Yes
74	Door	Yes	No	No	Yes
75	Door	Yes	No	No	Yes
76	Door	Yes	No	No	Yes
77	Door	Yes	No	No	Yes
78	Door	Yes	No	No	Yes
79	Door	Yes	No	No	Yes
80	Door	No	No	No	Yes

Table 1. Installation Requirements for Access Doors, Panels, Cowlings, Fairings, and Covers (continued)

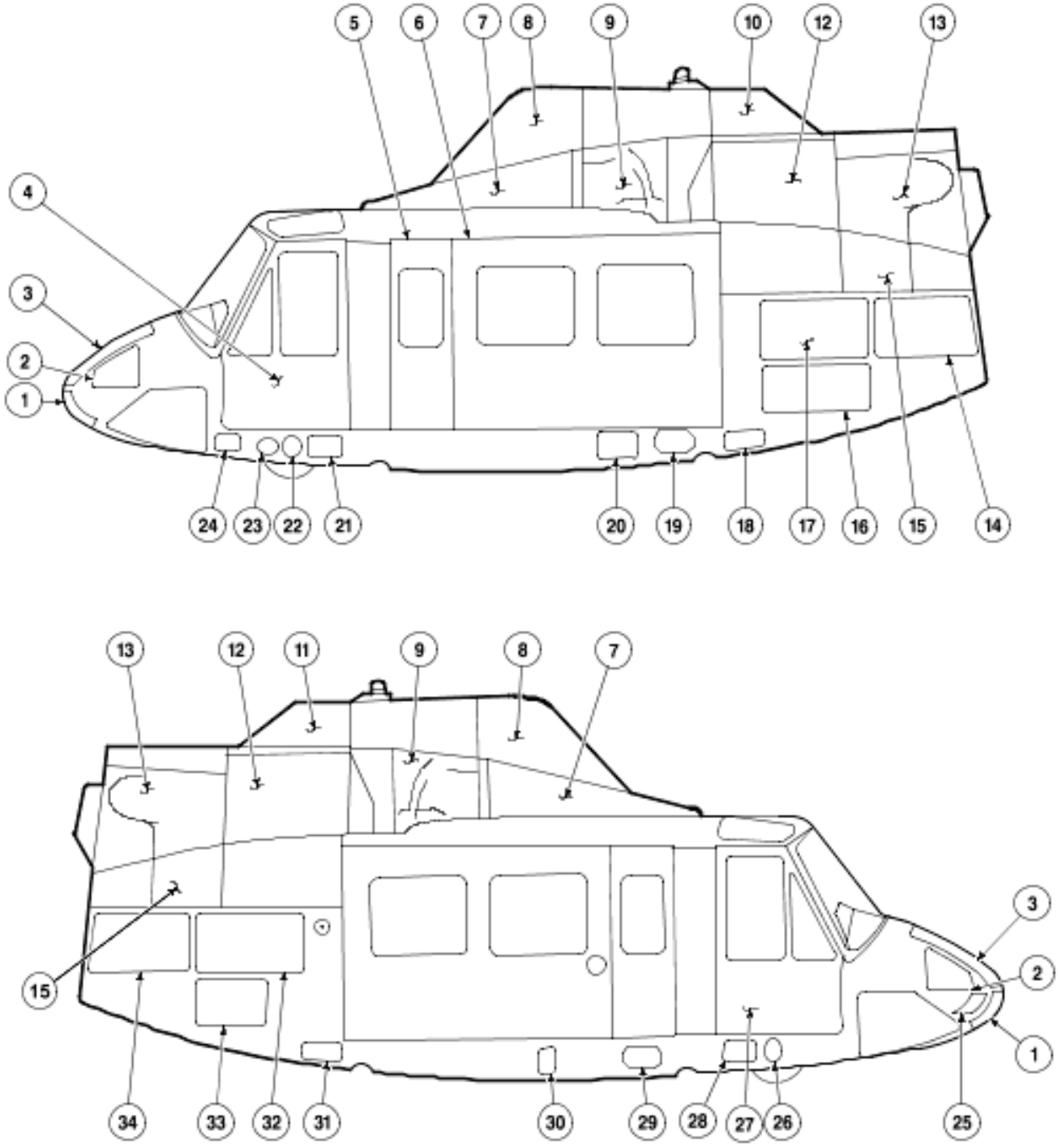
Figure 1 INDEX NO.	NAME	GROUND RUN	REQUIRED TO BE IN PLACE FOR:		
			JACKING	TOWING	FLIGHT
81	Door	No	No	No	Yes
82	Door	No	No	No	Yes
83	Door	No	No	No	Yes
84	Lower nose access panel	No	No	No	Yes
85	Cover	Yes	No	No	Yes
86	Cover	Yes	No	No	Yes
87	Door	Yes	No	No	Yes
88	Cover	Yes	No	No	Yes
89	Door	Yes	No	No	Yes
90	Fairing	No	No	No	Yes
91	Door	Yes	No	No	Yes
92	Door	Yes	No	No	Yes

NOTES:

1 Refer to the applicable BHT-412-FM.

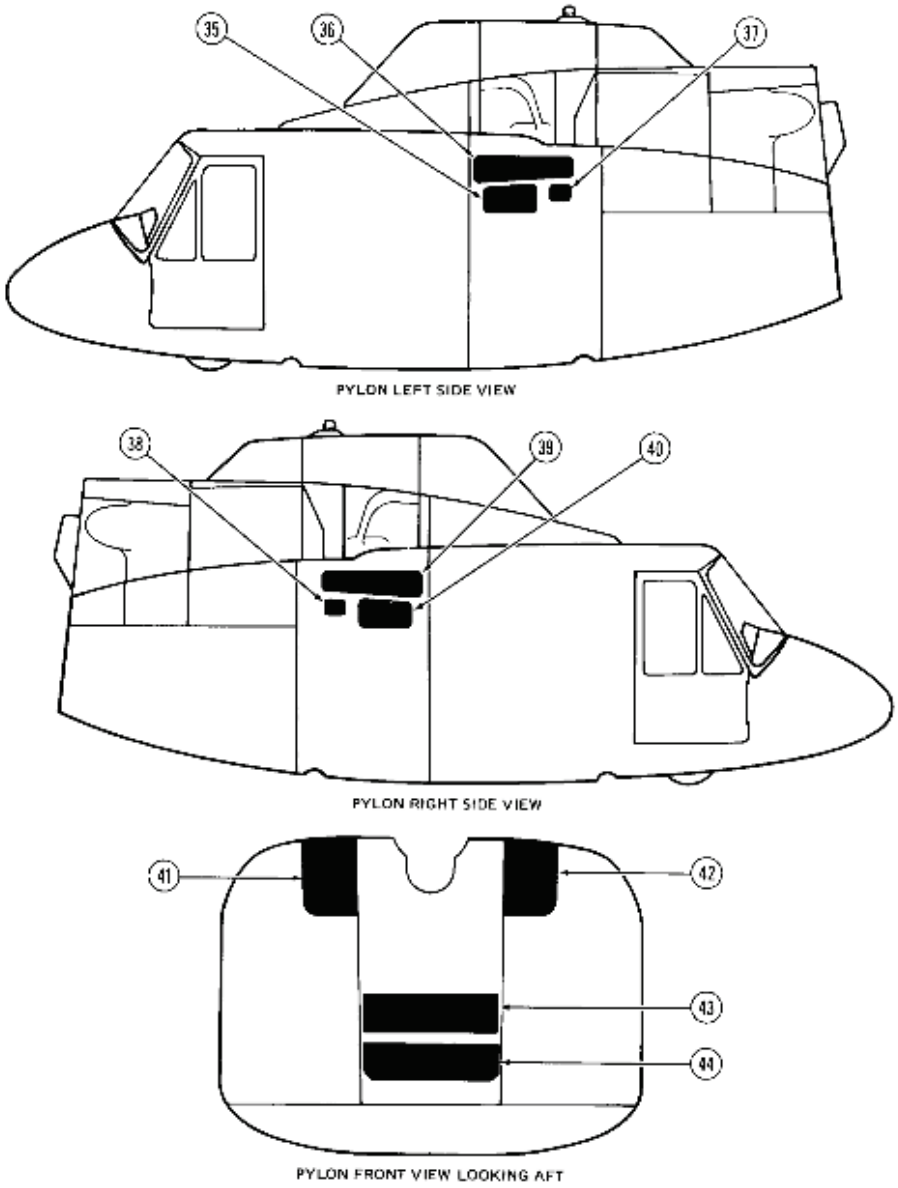
2 Door (12) can be removed during ground run to facilitate maintenance for a maximum of 30 minutes.

3 Can be removed during ground run to facilitate maintenance if collective control is at flat pitch. Secure all loose wiring.



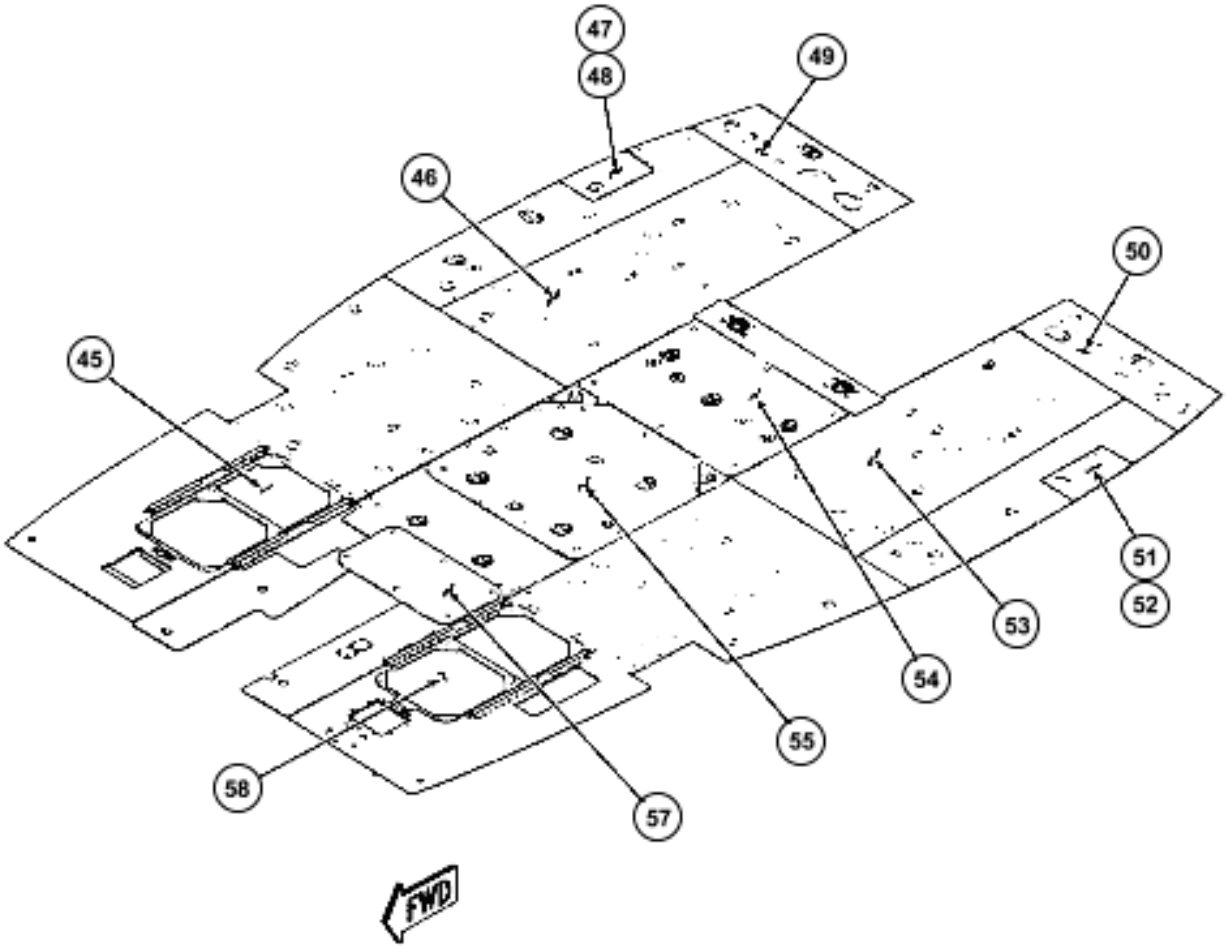
ICN-412-A-520000-B-97499-00001-A-001-01

Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 1 of 6)



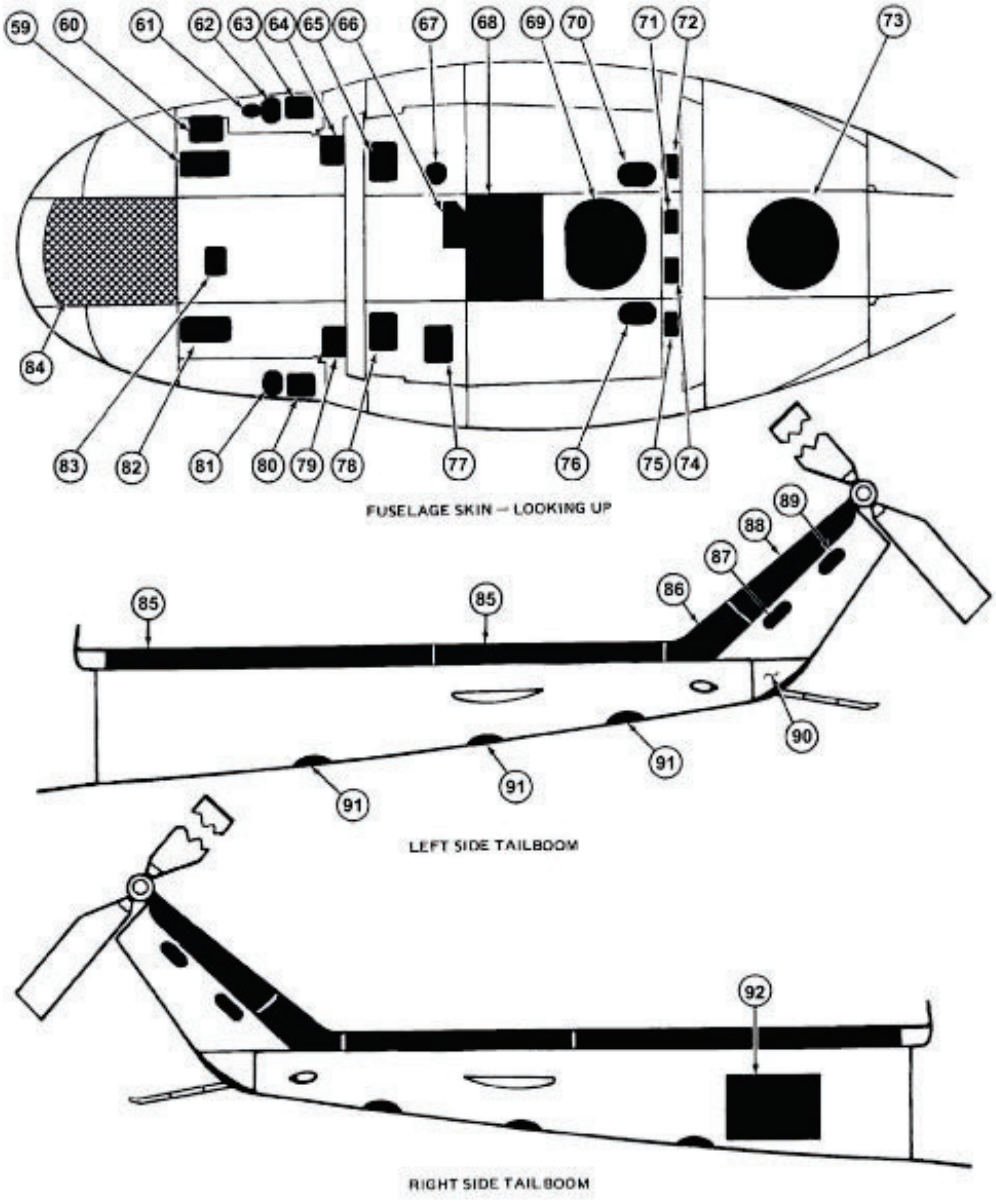
ICN-412-A-520000-B-97499-00002-A-001-01

Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 2 of 6)



ICN-412-A-520000-B-97499-00003-A-001-01

Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 3 of 6)



ICN-412-A-520000-B-97499-00004-A-001-01

Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 4 of 6)

INDEX NO.	NAME	ACCESS TO
1.	Lower nose door	Battery
2.	Cover	Automatic direction finder
3.	Upper nose door	Avionics equipment
4.	Crew door (copilot)	Copilot station
5.	Cargo hinged panel	Cabin
6.	Passenger cargo door	Cabin
7.	Transmission fairing	Hydraulic reservoirs, rotating controls
8.	Forward pylon fairing assembly	Transmission, swashplate, hydraulic reservoirs, hyd system 2 pump
9.	Fairing, engine inlet	Engine inlet, fuel controls, starter generator
10.	Aft pylon fairing assembly	Transmission, rotating controls
11.	Door	LH/RH engine
12.	Door	LH/RH engine power section
13.	Engine oil cooler support cowling assembly	Engine reduction gearbox
14.	Door	Radar altimeter
15.	Lower engine cowl assembly	LH/RH engine power section, fire extinguisher container
16.	Door	Electrical compartment
17.	Door	Navigation equipment
18.	Door	Avionics and electrical wiring
19.	Door	Hydraulic system 1 test connectors
20.	Door	Hydraulic system 2 test connectors
21.	Door	Cyclic flight controls
22.	Door	Flight controls
23.	Door	Flight controls
24.	Door	Flight controls
25.	Door	Auxiliary power unit receptacle
26.	Door	
27.	Crew door (pilot)	Pilot station
28.	Door	
29.	Door	
30.	Door	
31.	Door	Heater duct
32.	Door	
33.	Door	Heater variable control mixing valve
34.	Door	Tail rotor servo actuator, heater duct noise suppressor
35.	Door	Transmission, flight controls, lift link
36.	Door, pylon bulkhead	Transmission mounts
37.	Door	Transmission oil lines
38.	Door	Transmission oil line
39.	Door, pylon bulkhead	Transmission mounts
40.	Door	Hydraulic system 1 pump
41.	Cover	Engine accessory gearbox and drain valve, electric chip detector
42.	Cover	Engine accessory gearbox and drain valve, electric chip detector
43.	Cover	Flight controls hydraulic actuators
44.	Cover	Hydraulic integrated modules and filters

ICN-412-A-520000-B-97499-00005-A-001-01

Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 5 of 6)

INDEX NO.	NAME	ACCESS TO
45.	Panel	Fore and aft force gradient, magnetic brake, flight controls
46.	Panel	Fuel lines and fuel cell
47.	Panel	
48.	Door	
49.	Panel	Fuel lines
50.	Panel	Fuel lines
51.	Panel	
52.	Door	
53.	Panel	Fuel cell
54.	Panel	Flight controls
55.	Panel	AFCS actuators, fwd fuel crossover valve
56.	Panel	Collective jackshaft
57.	Door	Flight controls
58.	Panel	Force gradient, flight controls
59.	Panel	Flight controls
60.	Door	Flight controls
61.	Door	Flight controls
62.	Door	Flight controls
63.	Door	Flight controls
64.	Door	Flight controls
65.	Door	Fuel lines, electrical wiring
66.	Door	Forward fuel crossover valve
67.	Door	Fuel line
68.	Door	VHF antenna
69.	Opening, cargo hook	Cargo hook, hydraulic lines, transmission sump
70.	Door	Fuel boost pump, fuel cell drain
71.	Door	Fuel lines, aft crossover and main
72.	Door	Fuel lines, aft crossover and main
73.	Door	Droop compensator control, Hydraulic lines, aft center fuel cell
74.	Door	Fuel lines, main and aft crossover
75.	Door	Fuel lines, main and aft crossover
76.	Door	Fuel boost pump, fuel cell drain
77.	Door	Fuel lines, heater controls, duct, mixing valve
78.	Door	AFCS, flight controls
79.	Door	Flight controls, collective
80.	Door	Flight controls, cyclic
81.	Door	Flight controls
82.	Door	Flight controls, searchlight
83.	Door	Flight controls, jackshaft
84.	Lower nose access panel	Electrical equipment
85.	Cover	Driveshaft, tail rotor
86.	Cover	Intermediate gearbox
87.	Door	Flight controls
88.	Cover	Driveshaft, tail rotor
89.	Door	Flight controls
90.	Fairing	Tail skid
91.	Door	Flight controls
92.	Door	Baggage compartment

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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 6 of 6)

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Access Doors, Panels, Cowlings, Fairings, and Covers - Seal Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Adhesive	C-300	AR	
MEK	C-309	AR	
Adhesive	C-311	AR	
Abrasive cloth or paper	C-423	AR	

Procedure

NOTE

Seals may be either of rubber or silicone composition. Bonding procedure is different for each type. Seals subjected to fuel or oil contamination (used in pylon transmission and engine compartment) should be of polysulfide or neoprene (rubber) type construction.

1. Determine type of seal to be installed as follows: Cut a small specimen from damaged seal and subject it to flame. Silicone seals are more rapidly affected by flame and will leave a gray ash residue. Rubber type seals are much more fire resistant and will leave a black ash residue.

2. Remove all paint and primer from metal surface where seal is to be applied.

3. Clean new seal and metal where it is to be applied with MEK (C-309) or equivalent and dry with clean cloth.

NOTE

It is necessary to thoroughly clean surfaces before sanding to preclude working any foreign matter into pores of material.

4. Sand mating surfaces of seal and metal with 180 grit abrasive cloth or paper (C-423).

5. Clean sanded surfaces with MEK (C-309) or equivalent and dry with a clean cloth.

6. Bond rubber type (black ash) seals as follows:

6.1. Apply an even coat of adhesive (C-311) to mating surfaces of seal and metal.

6.2. Allow adhesive (C-311) to air dry 10 to 15 minutes at 75°F (24°C) or above. Adhesive should be aggressively tacky and adhere (but not transfer) to finger when touched.

6.3. Repeat Step 6.1. and Step 6.2.. Second coat is mandatory to obtain good adhesion.

6.4. Start at one end and roll on seal. Press seal down to expel all air from bond area to ensure proper contact and adhesion.

6.5. Air cure for a minimum of 4 hours at 75°F (24°C) or above.

7. Bond silicone composition (gray ash) seals as follows:



DO NOT CAP adhesive (C-300) AFTER MIXTURE. MIXTURE RELEASES HYDROGEN GAS WHICH CAN CAUSE HIGH PRESSURES. POT LIFE OF THIS ADHESIVE IS 6 HOURS.

7.1. Apply one even coat of adhesive (C-300) to bond surfaces of seal and metal.

7.2. Air dry at 75°F (24°C) or above for at least 1 hour, but not to exceed 8 hours.

7.3. Start at one end and roll on seal. Press seal down to expel all air from bond area to ensure proper contact and adhesion.

7.4. Air cure for 7 days at 75°F (24°C) or for 18 hours at 110 to 115°F (43 to 46°C).

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Crew Doors - Description and Maintenance

The two crew doors are hinged on the forward side and are equipped with a latch assembly, which may be operated from either side of the door, to secure the door in the closed position. Each door incorporates three transparent acrylic plastic windows: forward, upper, and lower (adjustable) windows. In an emergency, the doors may be jettisoned by pulling the EMERGENCY RELEASE handle mounted inside the cabin, forward of each door.

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Crew Doors - Removal

Procedure



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT (412-A-96-00-01-02A-042A-A / 01874).

NOTE

The procedure that follow is for the crew door on the left side. The procedure is the same for the crew door on the right side, except where noted.

1. Open the nose doors and/or panels, as applicable (412-A-52-40-00-01A-520A-A / 00832).

2. Disconnect the battery and/or the external DC power from the helicopter.

3. Pull down on the outer handle (20, Figure 1) and open the crew door.

NOTE

The step that follows is only applicable to the left crew door.

Applicable: SN: 37002-37999 or 38001-38999 or 39101-39999:

4. If the helicopter is equipped with the Vertical Reference Operations (VRO) kit (412-706-135), disconnect the VRO connector (3121P1) (45) from the VRO receptacle (3121J1) (46).

5. Pull on the ejection handle (48, Figure 2) to retract the lower and upper hinge pins (33 and 40).

6. Remove the crew door.

NOTE

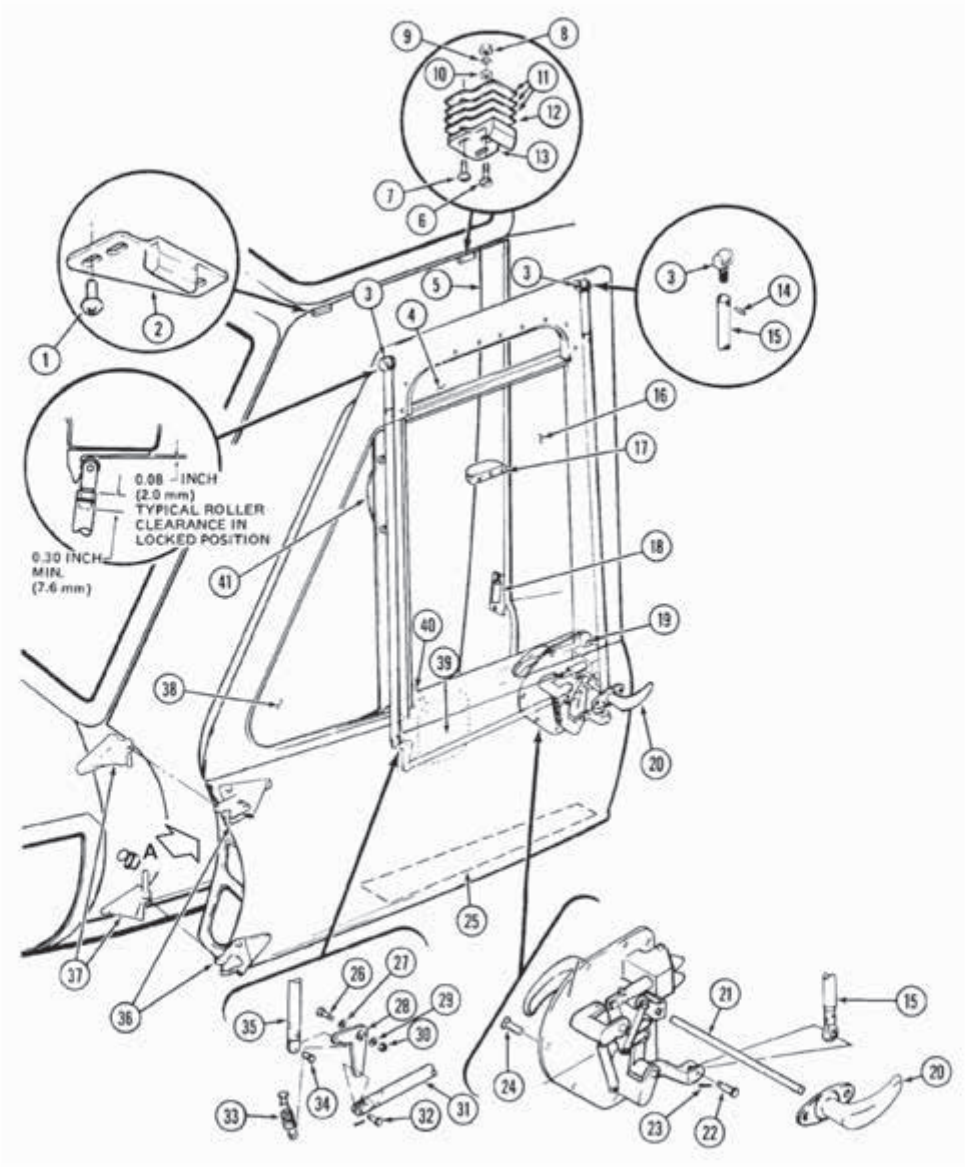
The step that follows is only applicable to the left crew door.

Applicable: SN: 37002-37999 or 38001-38999 or 39101-39999:

7. If the helicopter is equipped with the VRO kit (412-706-135), do as follows:

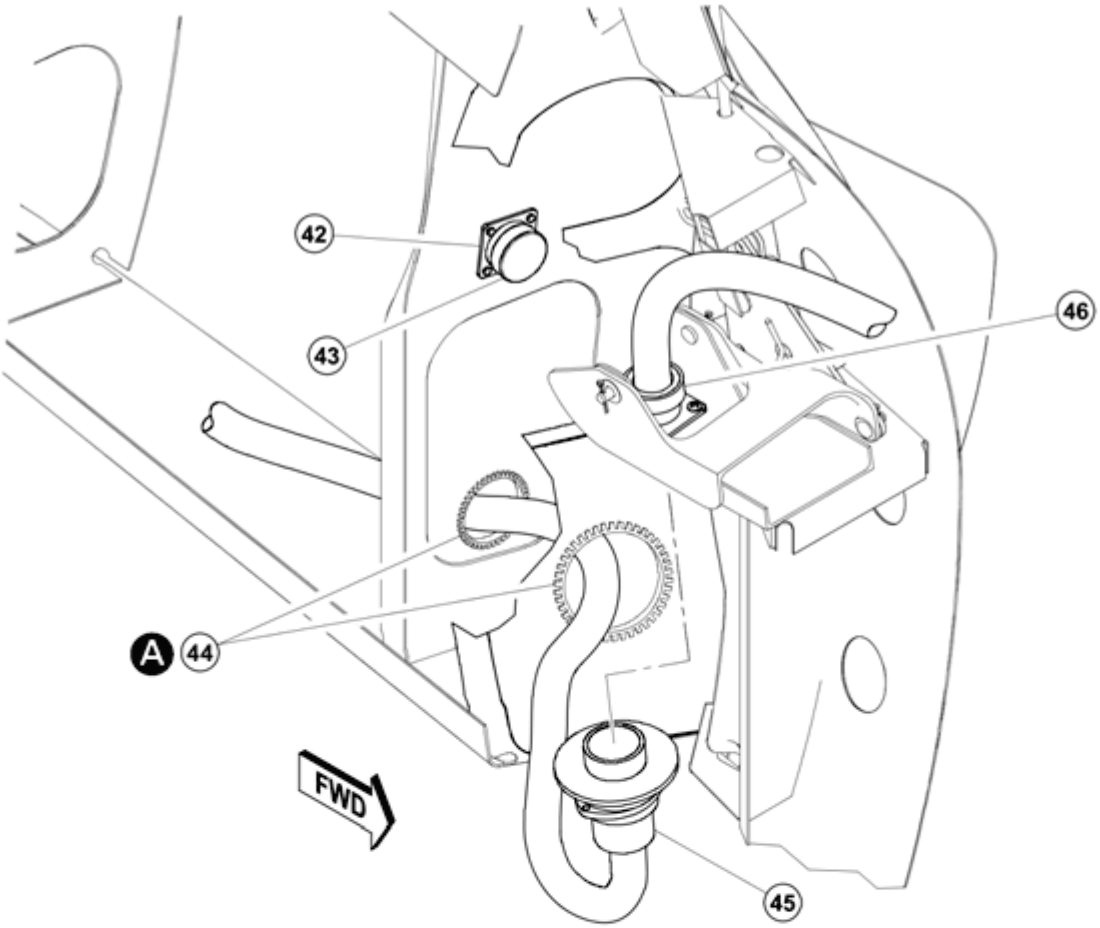
7.1. Remove the protective cap (43, Figure 1) from the dummy receptacle (42) and put it on the VRO receptacle (3121J1) (46).

7.2. Connect the VRO connector (3121P1) (45) to the dummy receptacle (42).



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Figure 1. Crew Door - Removal/Installation (Sheet 1 of 2)



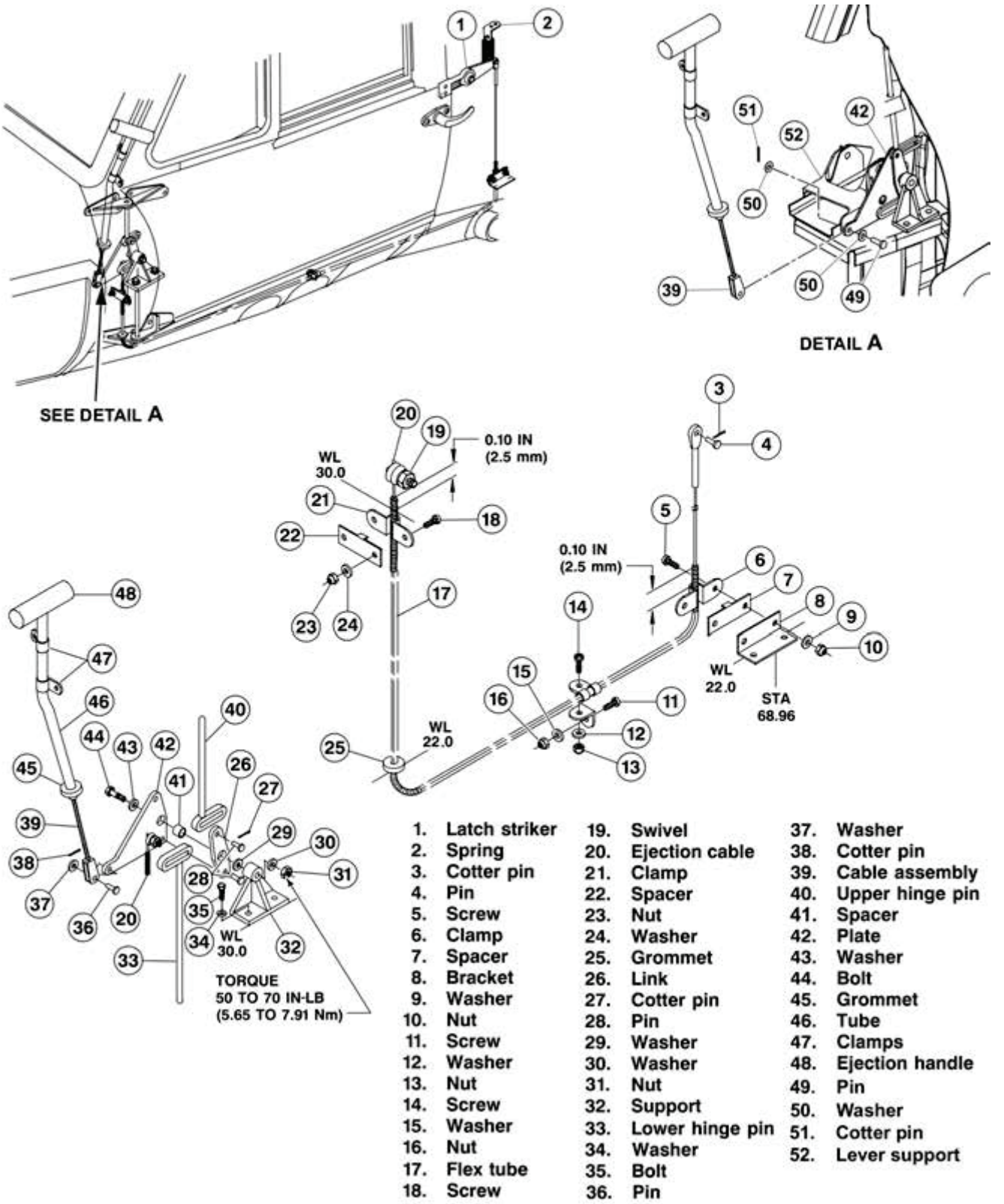
VIEW A
S/N 37002 through 37999, 38001 through 38999 and 39101 through 39999 installed with Vertical Reference Operations (VRO) kit (412-706-135)

1. Screw
2. Upper forward striker
3. Latch rod roller
4. Upper window
5. Door post
6. Screw
7. Screw
8. Nut
9. Washer
10. Washer
11. Backing plates
12. Backing plate
13. Upper aft striker
14. Spring pin
15. Latch rod
16. Sliding window
17. Lift handle
18. Latch striker
19. Latch assembly
20. Outer handle
21. Latch shaft
22. Pin
23. Cotter pin
24. Screw
25. Lower access door
26. Shouldered bolt
27. Washer
28. Bellcrank
29. Washer
30. Nut
31. Rod
32. Pin
33. Spring
34. Pin
35. Latch rod
36. Hinge halves
37. Hinge halves
38. Forward window
39. Inner access door
40. Adjustable stop
41. Closing handle
42. Dummy receptacle
43. Protective cap
44. Grommet
45. VRO connector (3121P1)
46. VRO receptacle (3121J1)

A ADHESIVE (C-301)

ICN-412-A-521000-B-97499-02854-A-001-01

Figure 1. Crew Door - Removal/Installation (Sheet 2 of 2)



1. Latch striker
2. Spring
3. Cotter pin
4. Pin
5. Screw
6. Clamp
7. Spacer
8. Bracket
9. Washer
10. Nut
11. Screw
12. Washer
13. Nut
14. Screw
15. Washer
16. Nut
17. Flex tube
18. Screw
19. Swivel
20. Ejection cable
21. Clamp
22. Spacer
23. Nut
24. Washer
25. Grommet
26. Link
27. Cotter pin
28. Pin
29. Washer
30. Washer
31. Nut
32. Support
33. Lower hinge pin
34. Washer
35. Bolt
36. Pin
37. Washer
38. Cotter pin
39. Cable assembly
40. Upper hinge pin
41. Spacer
42. Plate
43. Washer
44. Bolt
45. Grommet
46. Tube
47. Clamps
48. Ejection handle
49. Pin
50. Washer
51. Cotter pin
52. Lever support

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Figure 2. Crew Door Jettison Mechanism - Removal/Installation (Sheet 1 of 1)

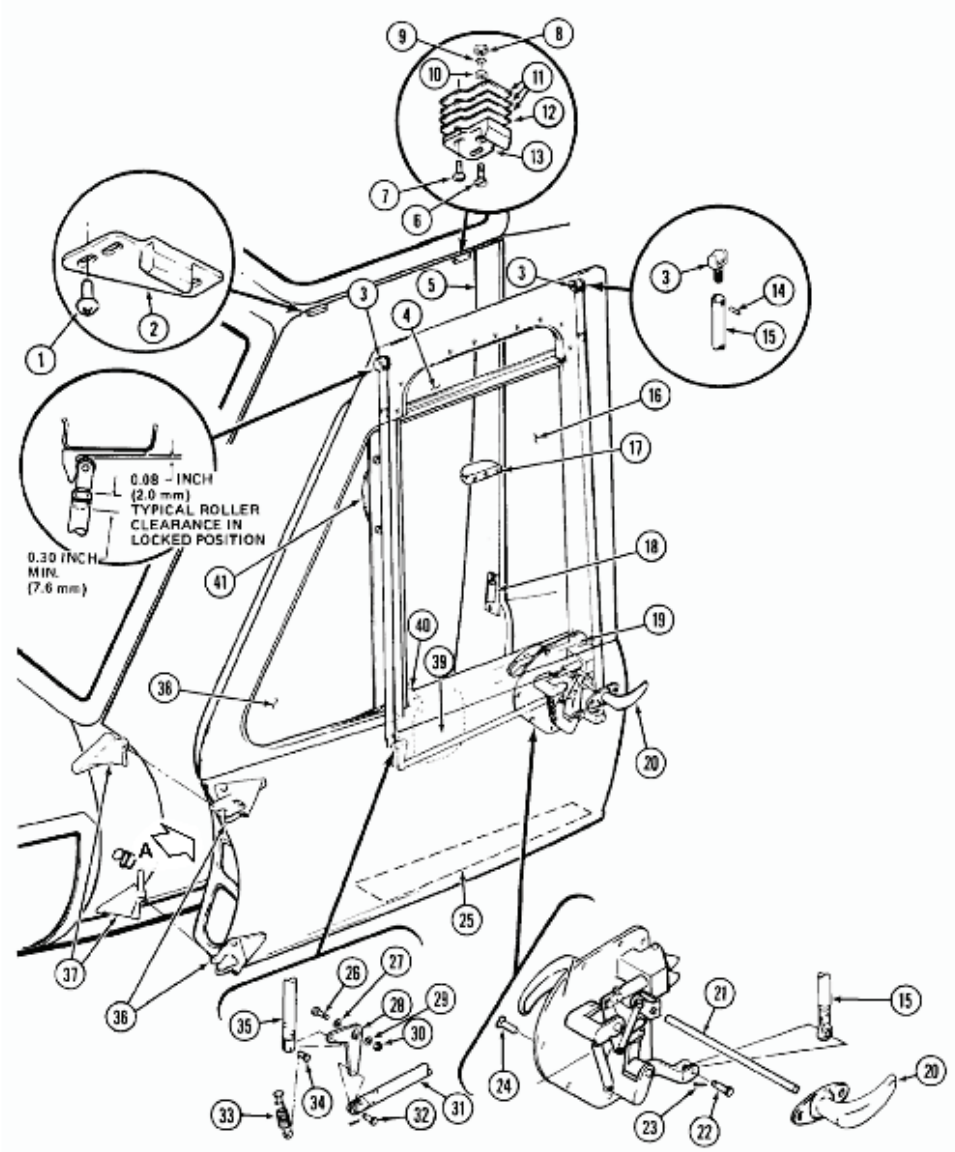
Crew Doors - Inspection

Procedure

NOTE

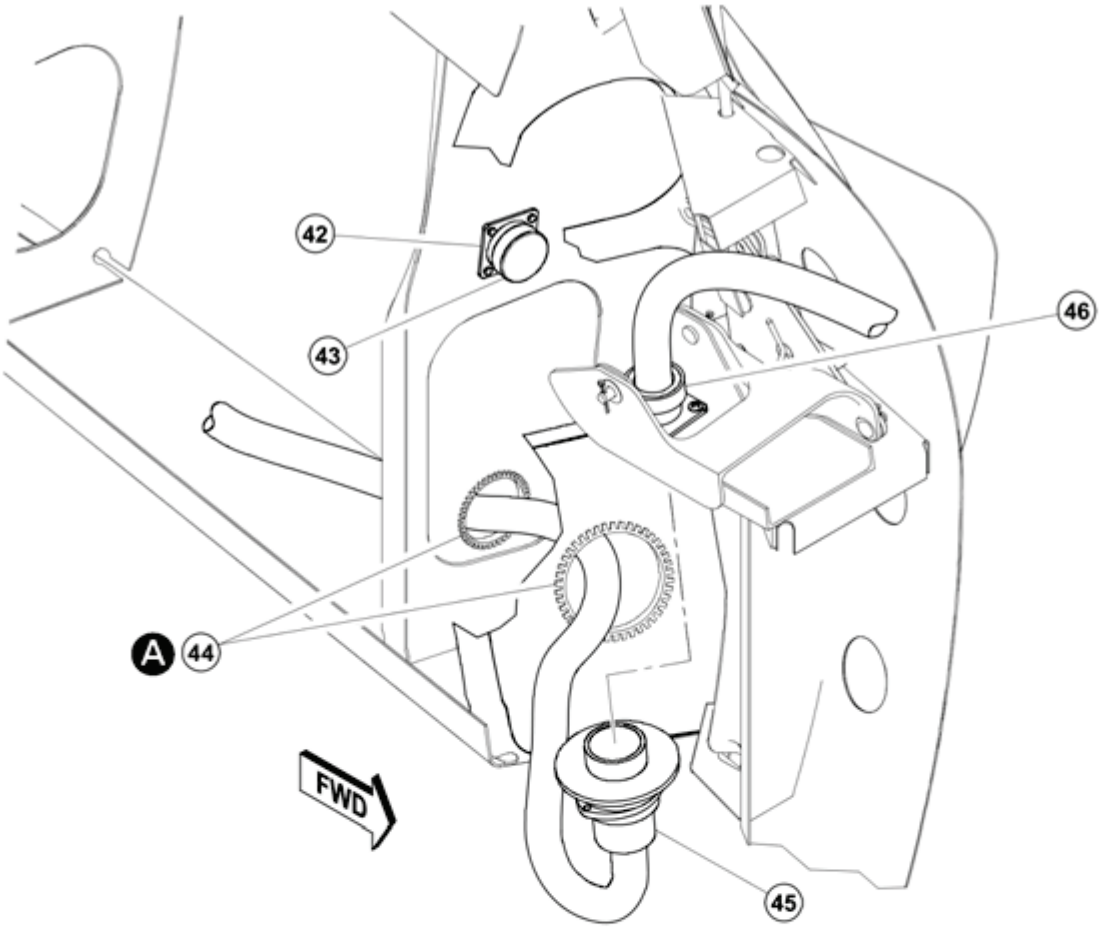
The procedure that follows is for the crew door on the left side. The procedure is the same for the crew door on the right side.

1. *Do an operational check for the crew doors (412-A-52-10-00-00A-320A-A / 00785).*
2. *Examine the visible parts of the door latch mechanism (Figure 1) for correct installation, corrosion, and damage.*
3. *Examine the visible parts of the crew door jettison mechanism (Figure 2) for correct installation, corrosion, and damage.*
4. *Examine the crew door structure and hinges for cracks and/or distortion.*
5. *Examine the crew door windows (412-A-52-01-00-01A-280A-A / 00848, 412-A-52-01-00-02A-280A-A / 00853, and 412-A-52-01-00-03A-280A-A / 00858).*
6. *Examine the crew door seals for correct bonding and for deterioration.*
7. *Examine the grommet (44, Figure 1) for unbonded condition. Replace as necessary (412-A-52-10-00-00A-685A-A / 00781).*



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Figure 1. Crew Door - Removal/Installation (Sheet 1 of 2)



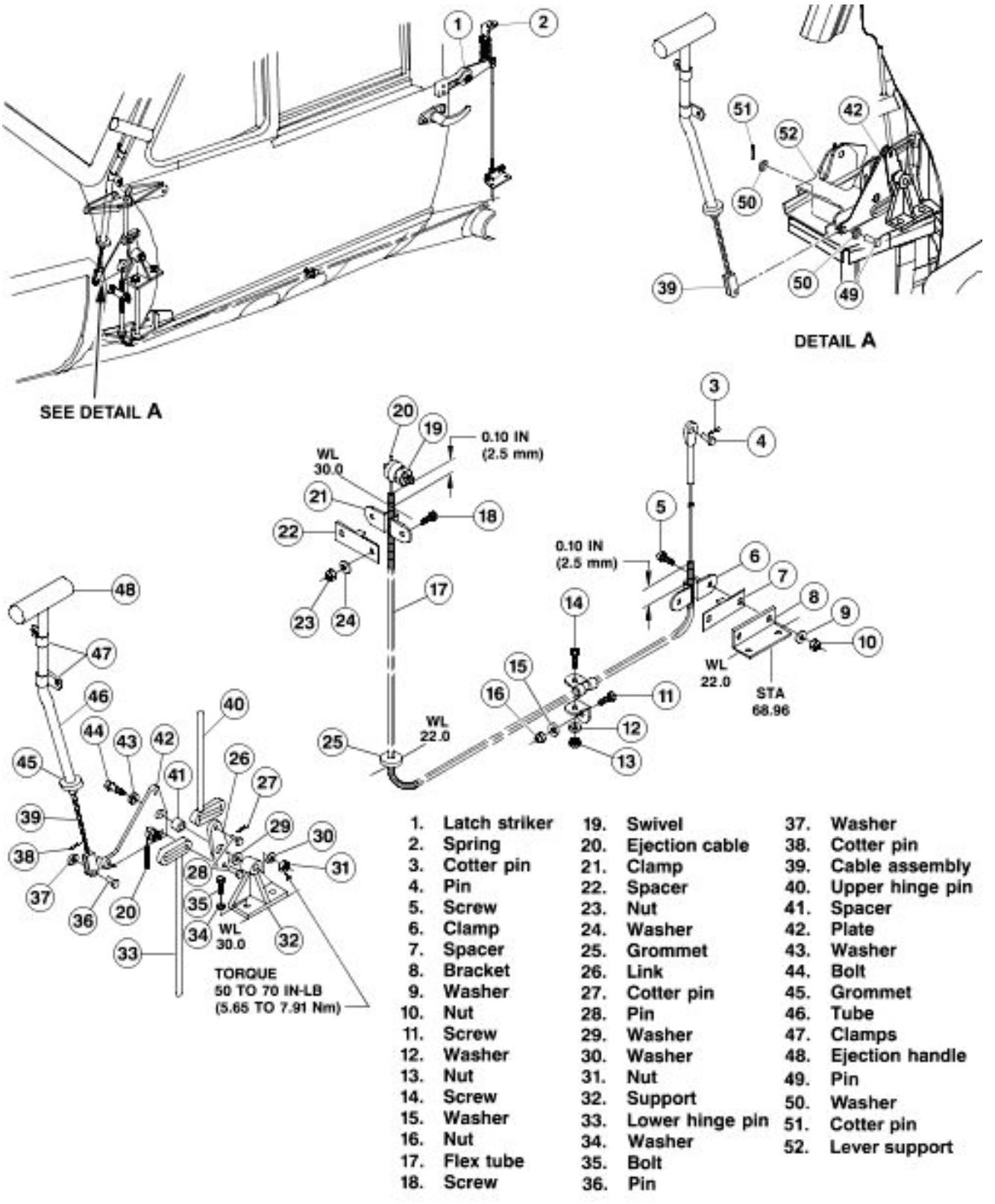
VIEW A
S/N 37002 through 37999, 38001 through 38999 and 39101 through 39999 installed with Vertical Reference Operations (VRO) kit (412-706-135)

1. Screw
2. Upper forward striker
3. Latch rod roller
4. Upper window
5. Door post
6. Screw
7. Screw
8. Nut
9. Washer
10. Washer
11. Backing plates
12. Backing plate
13. Upper aft striker
14. Spring pin
15. Latch rod
16. Sliding window
17. Lift handle
18. Latch striker
19. Latch assembly
20. Outer handle
21. Latch shaft
22. Pin
23. Cotter pin
24. Screw
25. Lower access door
26. Shouldered bolt
27. Washer
28. Bellcrank
29. Washer
30. Nut
31. Rod
32. Pin
33. Spring
34. Pin
35. Latch rod
36. Hinge halves
37. Hinge halves
38. Forward window
39. Inner access door
40. Adjustable stop
41. Closing handle
42. Dummy receptacle
43. Protective cap
44. Grommet
45. VRO connector (3121P1)
46. VRO receptacle (3121J1)

A ADHESIVE (C-301)

ICN-412-A-521000-B-97499-02854-A-001-01

Figure 1. Crew Door - Removal/Installation (Sheet 2 of 2)



1.	Latch striker	19.	Swivel	37.	Washer
2.	Spring	20.	Ejection cable	38.	Cotter pin
3.	Cotter pin	21.	Clamp	39.	Cable assembly
4.	Pin	22.	Spacer	40.	Upper hinge pin
5.	Screw	23.	Nut	41.	Spacer
6.	Clamp	24.	Washer	42.	Plate
7.	Spacer	25.	Grommet	43.	Washer
8.	Bracket	26.	Link	44.	Bolt
9.	Washer	27.	Cotter pin	45.	Grommet
10.	Nut	28.	Pin	46.	Tube
11.	Screw	29.	Washer	47.	Clamps
12.	Washer	30.	Washer	48.	Ejection handle
13.	Nut	31.	Nut	49.	Pin
14.	Screw	32.	Support	50.	Washer
15.	Washer	33.	Lower hinge pin	51.	Cotter pin
16.	Nut	34.	Washer	52.	Lever support
17.	Flex tube	35.	Bolt		
18.	Screw	36.	Pin		

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Figure 2. Crew Door Jettison Mechanism - Removal/Installation (Sheet 1 of 1)

Crew Doors - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Adhesive	C-301	AR	

Procedure

NOTE

Repair of the crew door latch and jettison mechanism is not permitted to replace the unserviceable parts.

1. Replace unserviceable parts in the crew door latch mechanism (412-A-52-10-00-02A-520A-A / 00786 and 412-A-52-10-00-02A-720A-A / 00787).

2. *Replace unserviceable parts in the crew door jettison mechanism (*412-A-52-10-00-01A-520B-A / 00781.2 *and* 412-A-52-10-00-01A-720B-A / 00781.3*).*

3. *Repair the crew door for structural damage. Refer to the BHT-MED-SRM-1 and to the FAA Publication AC 43.13-1B Aircraft Inspection and Repair Manual for structural maintenance.*

4. Repair damaged crew door windows (412-A-52-01-00-01A-685A-A / 00849, 412-A-52-01-00-02A-685A-A / 00854, and 412-A-52-01-00-03A-685A-A / 00859).

5. Install new crew door seals to replace damaged or missing seals (412-A-52-00-00-00A-720A-A / 00777).

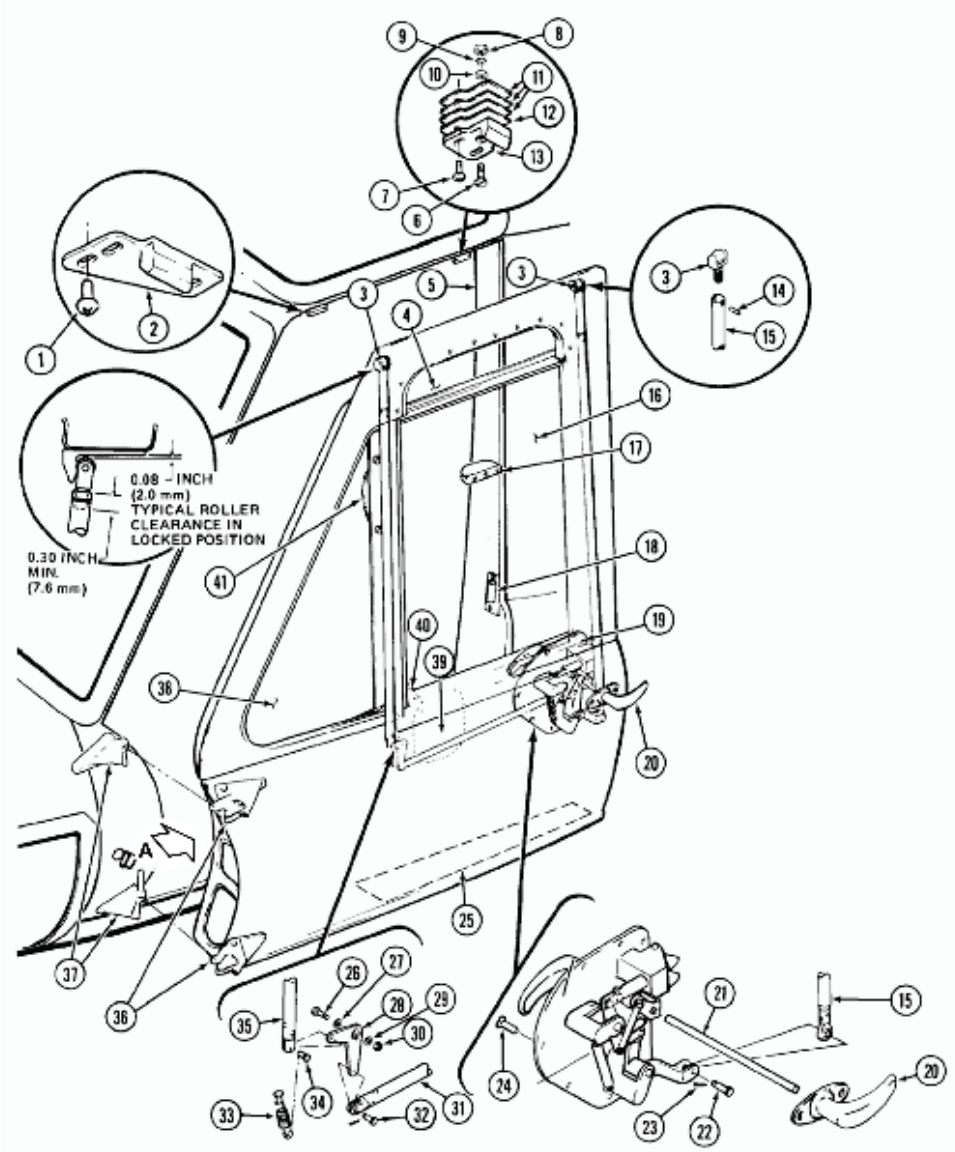
6. Replace the grommet (44, Figure 1) as follows:

6.1. Remove the grommet (44) from the left crew door.

6.2. Remove any trace of adhesive.

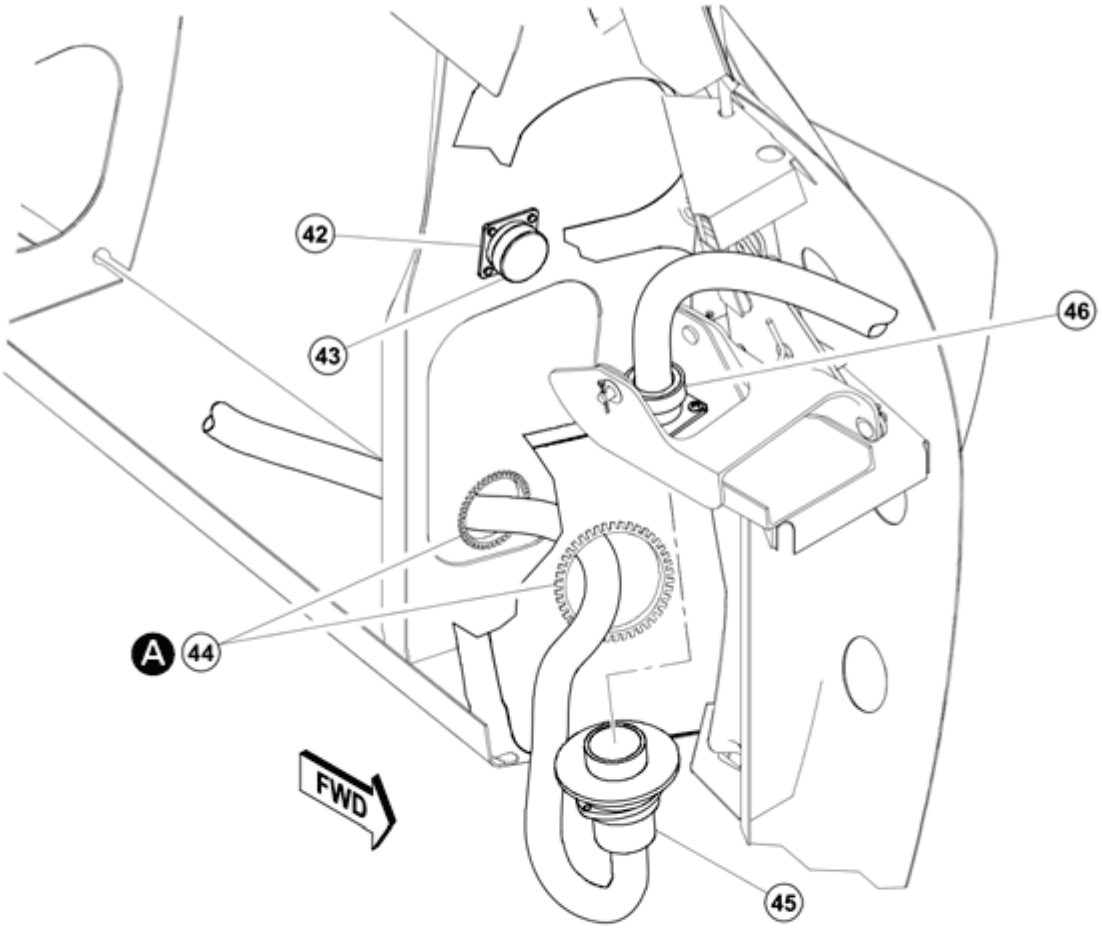
6.3. Install a new grommet (44) on the left crew door with adhesive (C-301).

6.4. *Remove unwanted adhesive.*



ICN-412-A-521000-B-97499-00001-A-001-01

Figure 1. Crew Door - Removal/Installation (Sheet 1 of 2)



VIEW A
S/N 37002 through 37999, 38001 through 38999 and 39101 through 39999 installed with Vertical Reference Operations (VRO) kit (412-706-135)

1. Screw
2. Upper forward striker
3. Latch rod roller
4. Upper window
5. Door post
6. Screw
7. Screw
8. Nut
9. Washer
10. Washer
11. Backing plates
12. Backing plate
13. Upper aft striker
14. Spring pin
15. Latch rod
16. Sliding window
17. Lift handle
18. Latch striker
19. Latch assembly
20. Outer handle
21. Latch shaft
22. Pin
23. Cotter pin
24. Screw
25. Lower access door
26. Shouldered bolt
27. Washer
28. Bellcrank
29. Washer
30. Nut
31. Rod
32. Pin
33. Spring
34. Pin
35. Latch rod
36. Hinge halves
37. Hinge halves
38. Forward window
39. Inner access door
40. Adjustable stop
41. Closing handle
42. Dummy receptacle
43. Protective cap
44. Grommet
45. VRO connector (3121P1)
46. VRO receptacle (3121J1)

A ADHESIVE (C-301)

ICN-412-A-521000-B-97499-02854-A-001-01

Figure 1. Crew Door - Removal/Installation (Sheet 2 of 2)

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Crew Doors - Installation

Procedure

WARNING

OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT (412-A-96-00-01-02A-042A-A / 01874).

NOTE

The procedure that follow is for the crew door on the left side. The procedure is the same for the crew door on the right side, except where noted.

NOTE

The step that follows is only applicable to the left crew door.

Applicable: SN: 37002-37999 or 38001-38999 or 39101-39999:

1. If the helicopter is equipped with the Vertical Reference Operations (VRO) kit (412-706-135), do as follows:

1.1. Disconnect the VRO connector (3121P1) (45, Figure 1) from the dummy receptacle (42).

1.2. Remove the protective cap (43) from the VRO receptacle (3121J1) and put it on the dummy receptacle (42).

2. Put the crew door in position on the helicopter.

3. Pull the ejection handle (48, Figure 2) to retract the lower and upper hinge pins (33 and 40).

4. Align the hinge halves (36 and 37, Figure 1) and the ejection handle (48, Figure 2).

5. Make sure that the lower and upper hinge pins (33 and 40, Figure 1) engage the hinge halves properly.

6. Close the crew door slowly and observe the action of the latch assembly (19, Figure 1).

7. Slowly move the outer handle (20) to lock in position. Observe the engagement of the two latch rod rollers (3) with the upper forward and aft strikers (2 and 13) and for a clearance of 0.08 inch (2.03 mm) above each roller when fully extended.

8. Adjust the latch rod rollers (3) to obtain a clearance of 0.08 inch (2.03 mm), if required. Maintain a minimum thread engagement of 0.30 inch (7.62 mm), as illustrated.

9. Adjust the position of the strikers (2, 13, and 18), if required.

10. When the adjustment is satisfactory, install the spring pin (14) to secure each latch rod roller (3).

NOTE

The step that follows is only applicable to the left crew door.

Applicable: SN: 37002-37999 or 38001-38999 or 39101-39999:

11. If the helicopter is equipped with the VRO kit (412-706-135), connect the VRO connector (3121P1) (45) to the VRO receptacle (3121J1) (46).

12. Connect the battery and/or the external DC power to the helicopter.

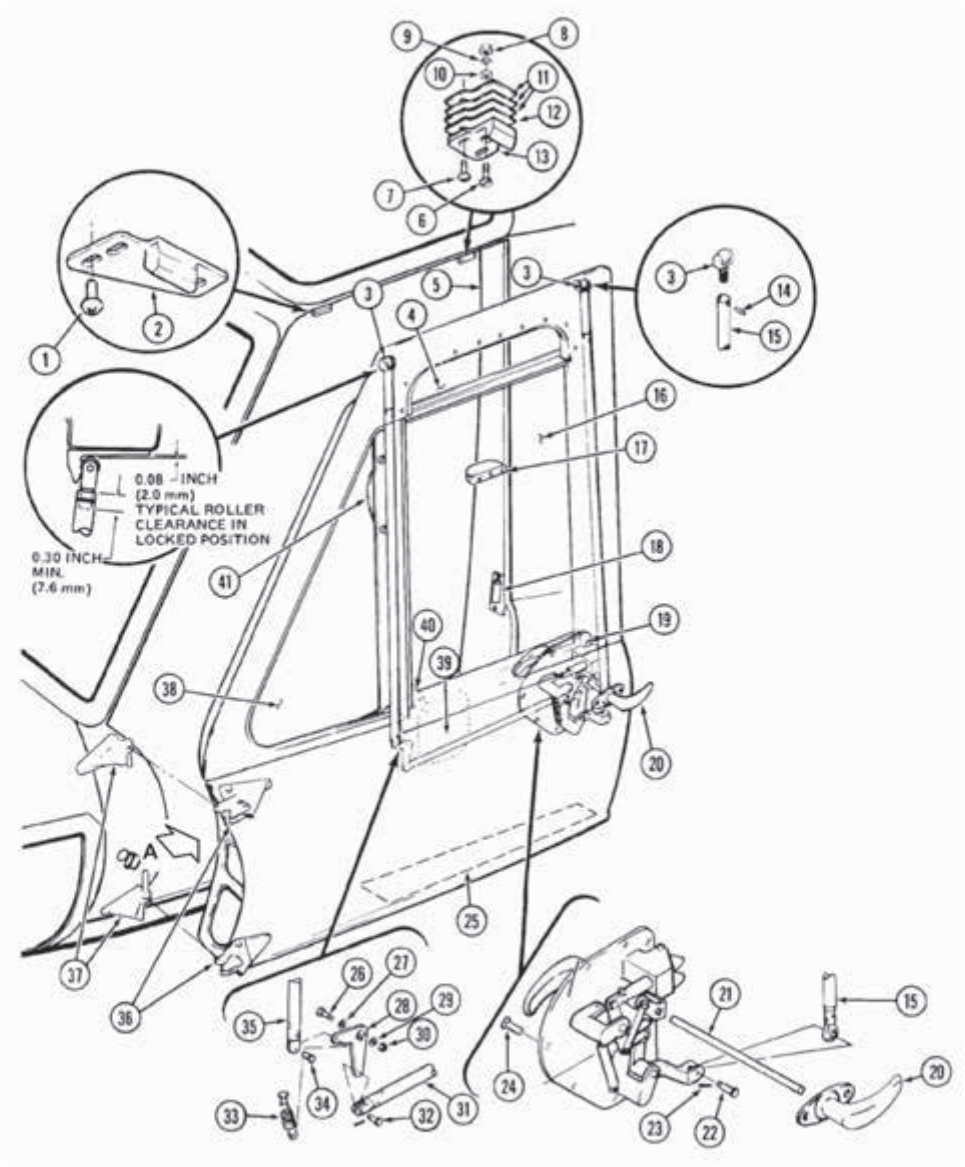
13. Close the nose doors and/or panels, as applicable (412-A-52-40-00-01A-720A-A / 00835).

Applicable: SN: 37002-37999:

14. Do an operational check of the VRO system BHT-412-MMS-EPI, Chapter 95, BHT-412-MMS-EPI, Chapter 95 and BHT-412-MMS-EPI, Chapter 95.

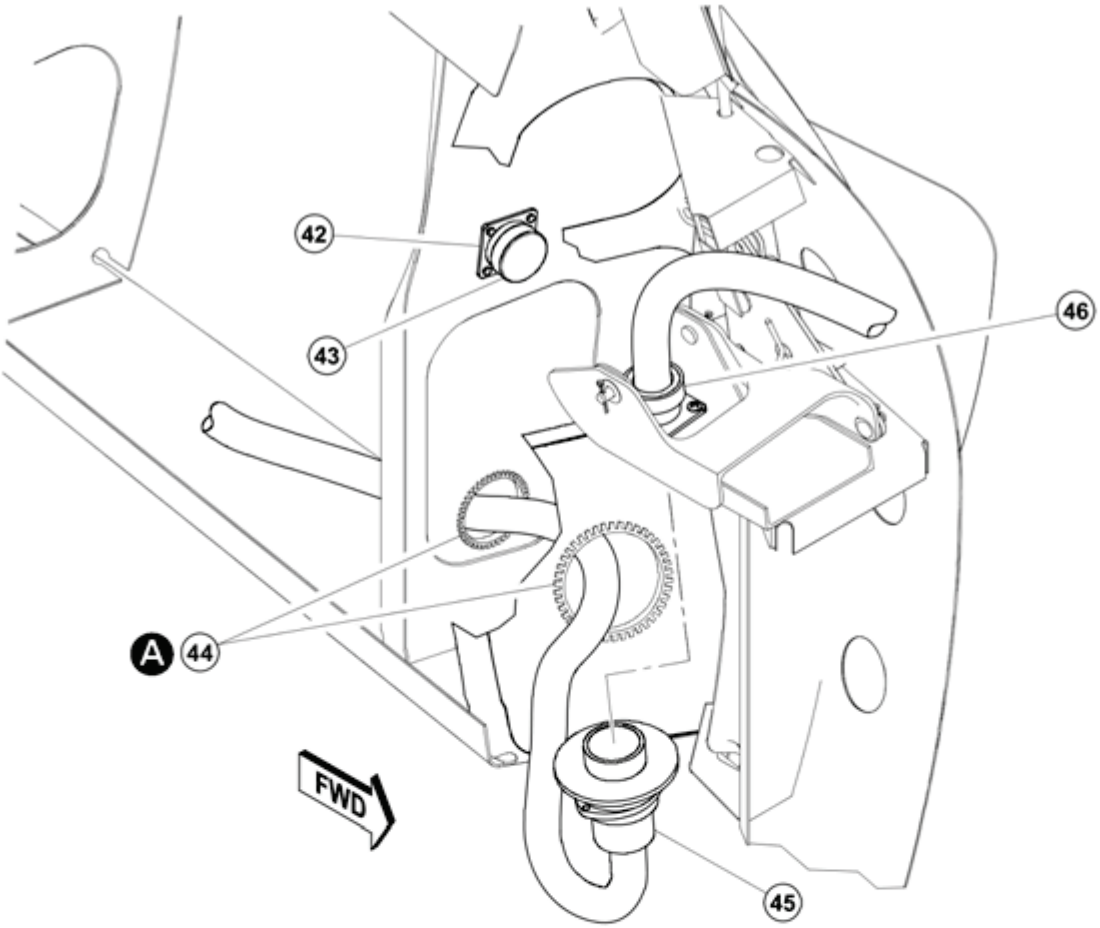
Applicable: SN: 38001-38999 or 39101-39999:

15. Do an operational check of the VRO system BHT-412-MMS-EPX, Chapter 95.



ICN-412-A-521000-B-97499-02853-A-001-01

Figure 1. Crew Door - Removal/Installation (Sheet 1 of 2)



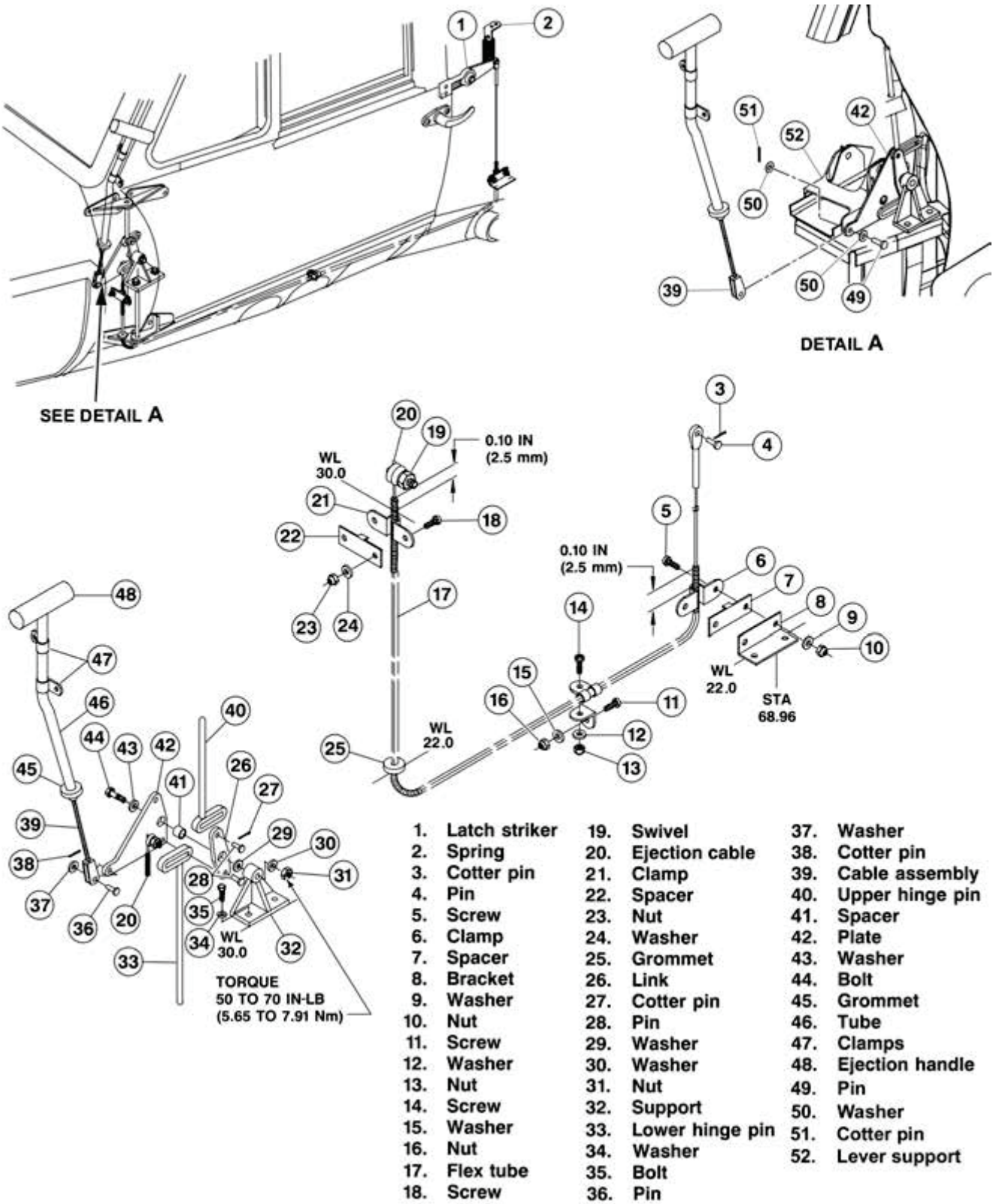
VIEW A
S/N 37002 through 37999, 38001 through 38999 and 39101 through 39999 installed with Vertical Reference Operations (VRO) kit (412-706-135)

1. Screw
2. Upper forward striker
3. Latch rod roller
4. Upper window
5. Door post
6. Screw
7. Screw
8. Nut
9. Washer
10. Washer
11. Backing plates
12. Backing plate
13. Upper aft striker
14. Spring pin
15. Latch rod
16. Sliding window
17. Lift handle
18. Latch striker
19. Latch assembly
20. Outer handle
21. Latch shaft
22. Pin
23. Cotter pin
24. Screw
25. Lower access door
26. Shouldered bolt
27. Washer
28. Bellcrank
29. Washer
30. Nut
31. Rod
32. Pin
33. Spring
34. Pin
35. Latch rod
36. Hinge halves
37. Hinge halves
38. Forward window
39. Inner access door
40. Adjustable stop
41. Closing handle
42. Dummy receptacle
43. Protective cap
44. Grommet
45. VRO connector (3121P1)
46. VRO receptacle (3121J1)

A ADHESIVE (C-301)

ICN-412-A-521000-B-97499-02854-A-001-01

Figure 1. Crew Door - Removal/Installation (Sheet 2 of 2)



1. Latch striker
2. Spring
3. Cotter pin
4. Pin
5. Screw
6. Clamp
7. Spacer
8. Bracket
9. Washer
10. Nut
11. Screw
12. Washer
13. Nut
14. Screw
15. Washer
16. Nut
17. Flex tube
18. Screw
19. Swivel
20. Ejection cable
21. Clamp
22. Spacer
23. Nut
24. Washer
25. Grommet
26. Link
27. Cotter pin
28. Pin
29. Washer
30. Washer
31. Nut
32. Support
33. Lower hinge pin
34. Washer
35. Bolt
36. Pin
37. Washer
38. Cotter pin
39. Cable assembly
40. Upper hinge pin
41. Spacer
42. Plate
43. Washer
44. Bolt
45. Grommet
46. Tube
47. Clamps
48. Ejection handle
49. Pin
50. Washer
51. Cotter pin
52. Lever support

ICN-412-A-521000-B-97499-02855-A-001-01

Figure 2. Crew Door Jettison Mechanism - Removal/Installation (Sheet 1 of 1)

Crew Door Jettison Mechanism - Removal

Procedure

NOTE

The procedure that follows is for the crew door on the left side. The procedure is the same for the crew door on the right side, except where noted.

1. Remove the crew door (412-A-52-10-00-00A-520B-A / 00778.1).

2. Remove the pin (36, Figure 1) and detach the cable assembly (39) from the plate (42). When required, detach the two clamps (47) that secure the tube (46) to the support angle and pull the handle assembly up through the grommet (45).

NOTE

The step that follows is only applicable to the left crew door.

Applicable: SN: 37002-37999 or 38001-38999 or 39101-39999:

3. If the helicopter is equipped with the Vertical Reference Operations (VRO) kit (412-706-135), do as follows:

3.1. Remove the cotter pin (51). Discard the cotter pin.

3.2. Remove the washer (50) and the pin (49) and disconnect the cable assembly (39) from the plate (42) and the lever support (52).

3.3. If required, remove the two clamps (47) that secure the tube (46) to the support angle and pull the handle assembly up through the grommet (45).

4. Remove the swivel (19) from the plate (42). Pull the swivel from the end of the ejection cable (20). Keep the nut and the washer with the swivel.

5. Remove the two bolts (35), washers (34), and support (32). Pull the lower and upper hinge pins (33 and 40) free of the hinge bushings. Remove the assembled jettison mechanism from the inboard side of the nose structure.

6. Disassemble the jettison mechanism as follows:

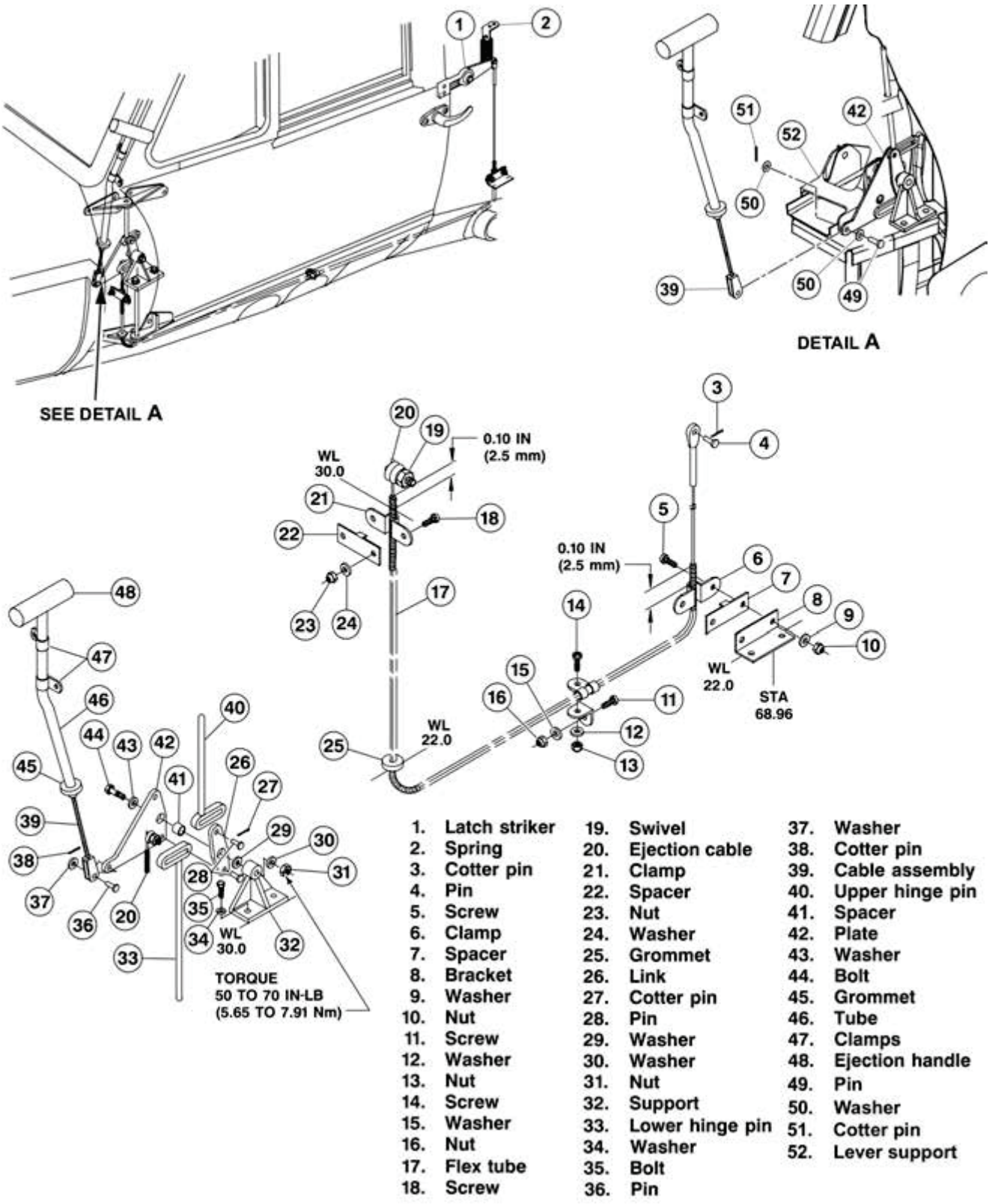
6.1. Remove the cotter pins (27) to release the pins (28). Remove the pins (28) to detach the lower and upper hinge pins (33 and 40) from the plate (42) and the link (26).

6.2. Remove the bolt (44), nut (31), washers (29, 30, and 43), and spacer (41) to separate the plate (42) and the link (26) from the support (32).

7. Remove the ejection cable (20) as follows:

7.1. To disconnect the aft end fork of the ejection cable (20) from the latch striker (1), remove the cotter pin (3) and the pin (4).

7.2. Pull the cable aft out of the flex tube (17) and leave the tube in place.



1. Latch striker
2. Spring
3. Cotter pin
4. Pin
5. Screw
6. Clamp
7. Spacer
8. Bracket
9. Washer
10. Nut
11. Screw
12. Washer
13. Nut
14. Screw
15. Washer
16. Nut
17. Flex tube
18. Screw
19. Swivel
20. Ejection cable
21. Clamp
22. Spacer
23. Nut
24. Washer
25. Grommet
26. Link
27. Cotter pin
28. Pin
29. Washer
30. Washer
31. Nut
32. Support
33. Lower hinge pin
34. Washer
35. Bolt
36. Pin
37. Washer
38. Cotter pin
39. Cable assembly
40. Upper hinge pin
41. Spacer
42. Plate
43. Washer
44. Bolt
45. Grommet
46. Tube
47. Clamps
48. Ejection handle
49. Pin
50. Washer
51. Cotter pin
52. Lever support

ICN-412-A-521000-B-97499-02855-A-001-01

Figure 1. Crew Door Jettison Mechanism - Removal/Installation (Sheet 1 of 1)

Crew Door Jettison Mechanism - Installation

Procedure

NOTE

The procedure that follows is for the crew door on the left side. The procedure is the same for the crew door on the right side, except where noted.

1. Check the ends of the flex tube (17, Figure 1) for proper installation as follows:

1.1. The aft end of the flex tube (17) should be positioned vertically at station 68.96 in the door post, secured between the clamp (6) and the spacer (7), and attached to the bracket (8) on the structure with the two screws (5), washers (9), and nuts (10). The end of the tube should extend 0.10 inch (2.54 mm) above the clamp.

1.2. The forward end of the flex tube (17) should pass up through the grommet (25) in the nose structure forward of the door opening and be secured between the clamp (21) and the spacer (22), attached to the structure with the two screws (18), washers (24), and nuts (23). The end of the tube should extend 0.10 inch (2.54 mm) above the horizontal angle at WL 30.

2. Thread the plain end of the ejection cable (20) through the flex tube (17) from the aft end.

NOTE

The forward end of ejection cable (20) will be attached during installation of jettison mechanism (Step 4.).

3. Attach the cable fork terminal on the aft end of the latch striker (1) with the pin (4) and secure with the cotter pin (3). Verify that the spring (2) is attached between the striker and the bracket in the door post.

4. Install the jettison mechanism as follows:

4.1. Insert the spacer (41) through the pivot holes of the plate (42) and the link (26).

4.2. Place the standard steel washer (43) on the bolt (44). Insert the bolt from the plate side through the spacer (41).

4.3. Place the thin aluminum washer (29) on the bolt next to the spacer (41). Insert the bolt (44) through the support (32).

4.4. Install the nut (31) and the aluminum washer (30) on the outboard end of the bolt. Torque the nut 50 to 70 inch-pounds (5.65 to 7.91 Nm).

4.5. Position the slotted ends of the hinge pins (33 and 40) between the plate (42) and the link (26). Align the holes and install the pins (28). Secure the pins with the cotter pins (27).

4.6. Position the assembly in the nose structure, with the hinge pins inserted in the hinge bushings and the support (32), resting on the horizontal structural member at WL 30.

4.7. Align the support (32) to the holes with the plate nuts. Install the two bolts (35) with the aluminum washers (34). Torque the bolts 20 to 25 inch-pounds (2.26 to 2.82 Nm).

4.8. Position the ejection handle (48) and insert the cable tube (46) through the grommet (45) at WL 35.56.

4.9. Align the tube clamps (47) to the mounting holes in the angle and attach with the two screws, washers, and nuts.

4.10. Connect the fork terminal of the cable assembly (39) to the forward end of the plate (42) with the pin (36), washer (37), and cotter pin (38).

NOTE

The step that follows is only applicable to the left crew door.

Applicable: SN: 37002-37999 or 38001-38999 or 39101-39999:

4.11. If the helicopter is equipped with the Vertical Reference Operations (VRO) kit (412-706-135), do as follows:

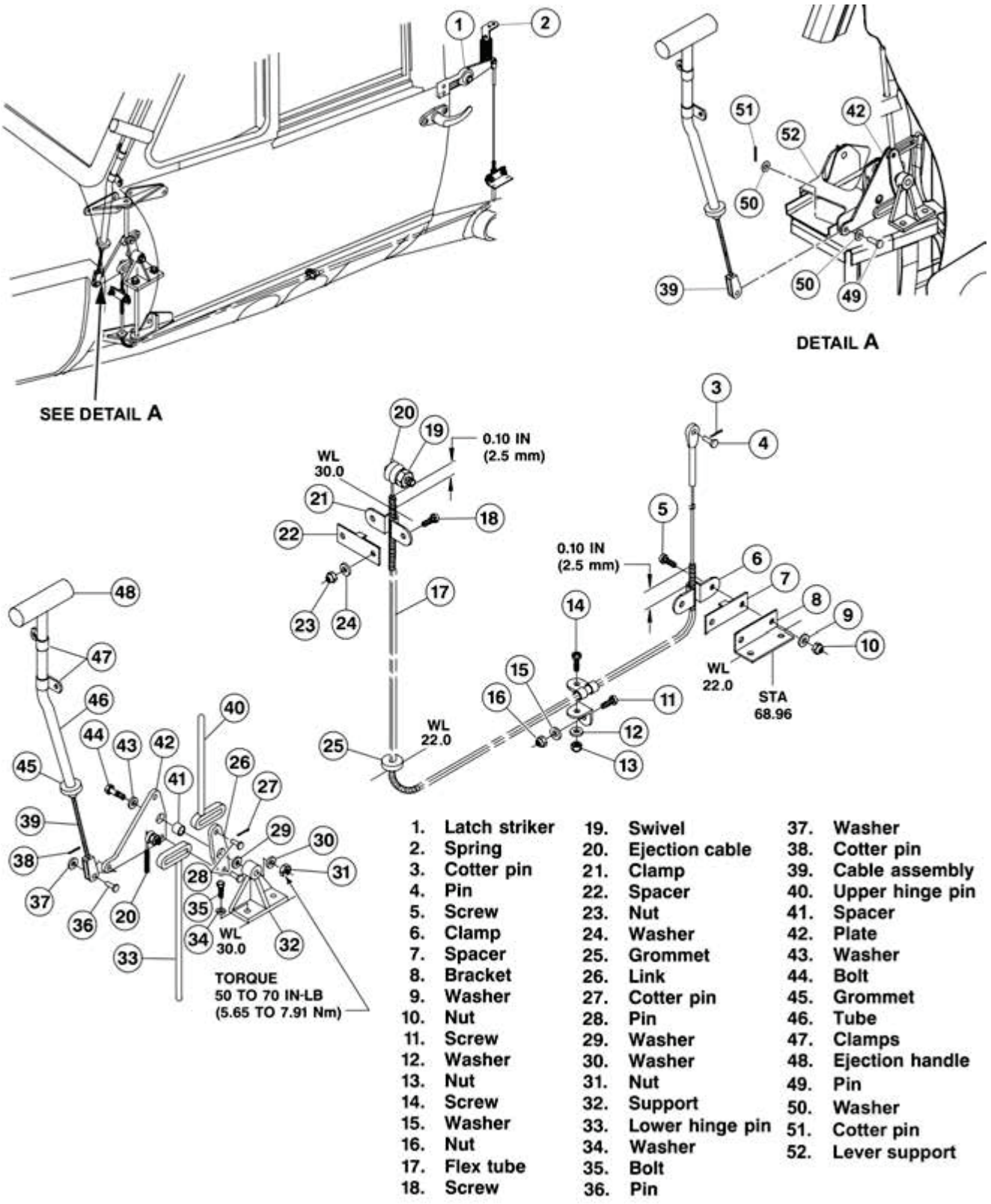
4.11.1. If previously removed, install the handle assembly through the grommet (45). Install the tube (46) with the two clamps (47).

4.11.2. Connect the cable assembly (39) to the plate (42) and the lever support (52) with the washer (50) and the pin (49). Safety the pin with a new cotter pin (51).

5. Connect the ejection cable (20) with the swivel (19) to the hole with the bushing in the middle of the plate (42).

6. Adjust the cable length to remove any slack with the hinge pins extended and the latch striker (1) held down in latching position. Tighten the nut on the swivel (19) to hold the end of the cable securely.

7. Do an operational check (412-A-52-10-00-00A-320A-A / 00785).



1. Latch striker
2. Spring
3. Cotter pin
4. Pin
5. Screw
6. Clamp
7. Spacer
8. Bracket
9. Washer
10. Nut
11. Screw
12. Washer
13. Nut
14. Screw
15. Washer
16. Nut
17. Flex tube
18. Screw
19. Swivel
20. Ejection cable
21. Clamp
22. Spacer
23. Nut
24. Washer
25. Grommet
26. Link
27. Cotter pin
28. Pin
29. Washer
30. Washer
31. Nut
32. Support
33. Lower hinge pin
34. Washer
35. Bolt
36. Pin
37. Washer
38. Cotter pin
39. Cable assembly
40. Upper hinge pin
41. Spacer
42. Plate
43. Washer
44. Bolt
45. Grommet
46. Tube
47. Clamps
48. Ejection handle
49. Pin
50. Washer
51. Cotter pin
52. Lever support

ICN-412-A-521000-B-97499-02855-A-001-01

Figure 1. Crew Door Jettison Mechanism - Removal/Installation (Sheet 1 of 1)

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Crew Doors - Operational Check

Procedure

NOTE

The procedure that follows is for the crew door on the left side. The procedure is the same for the crew door on the right side.

1. *Put the door in the closed and locked position.*

2. *Make sure that the door is correctly latched. The two vertical latch rods (15 and 35,* Figure 1*) must be extended to engage the strikers (2 and 13) at the top of the door opening with a clearance of* 0.08 inches (2.0 mm) *above the top of each latch rod roller (3), as illustrated. The latch assembly (19) must fully engage the latch striker (18).*

3. *Turn the outer handle (20) clockwise to the open position. The latch assembly (19) must disengage from the latch striker (18) and the latch rod rollers (3) must disengage from the strikers (2 and 13) to let the door open.*

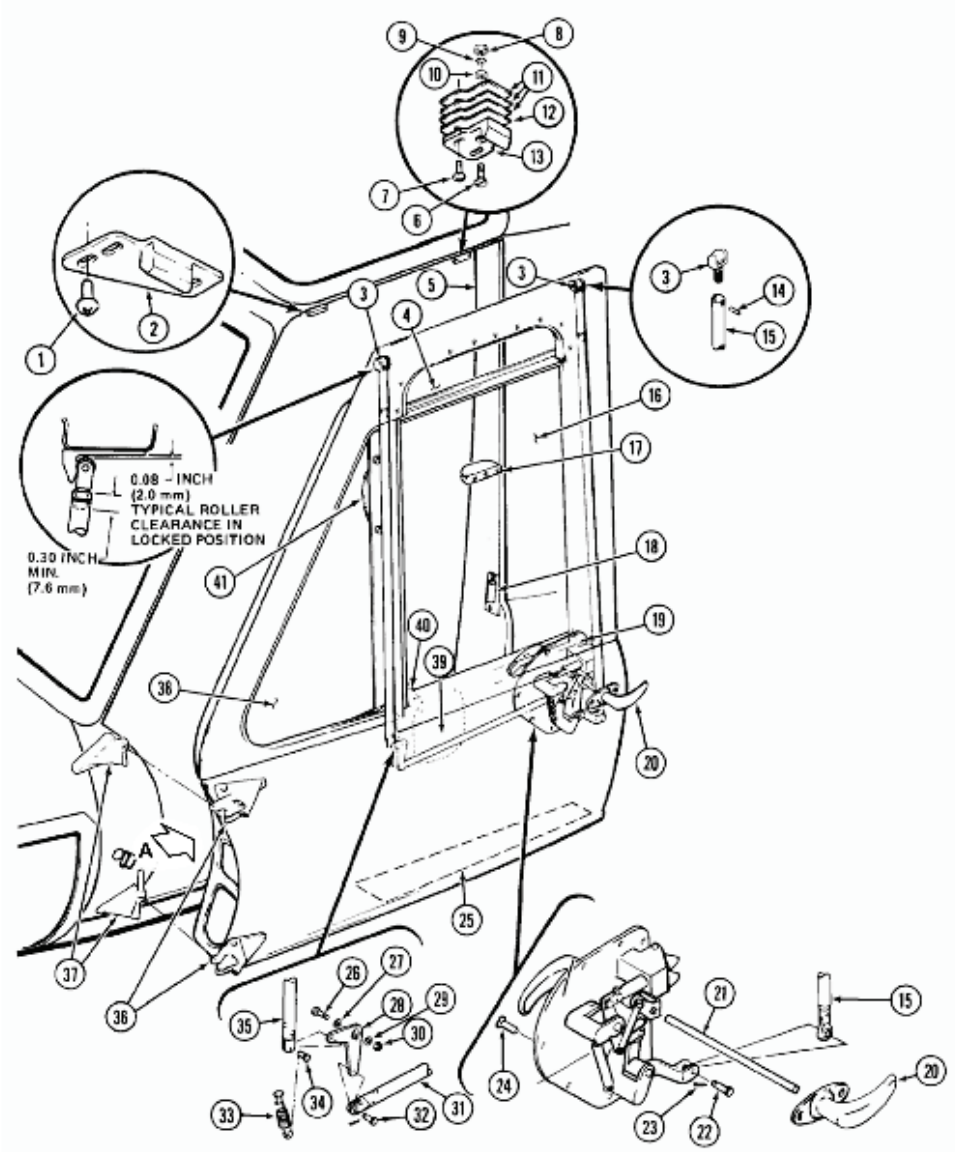
4. *Close the door and turn the outer handle (20) to the horizontal position. The outer handle (20) must operate smoothly and the door must be held tightly against the fuselage.*

5. *Examine the jettison mechanism as follows:*

5.1. Close and lock the door with the outer handle (20).

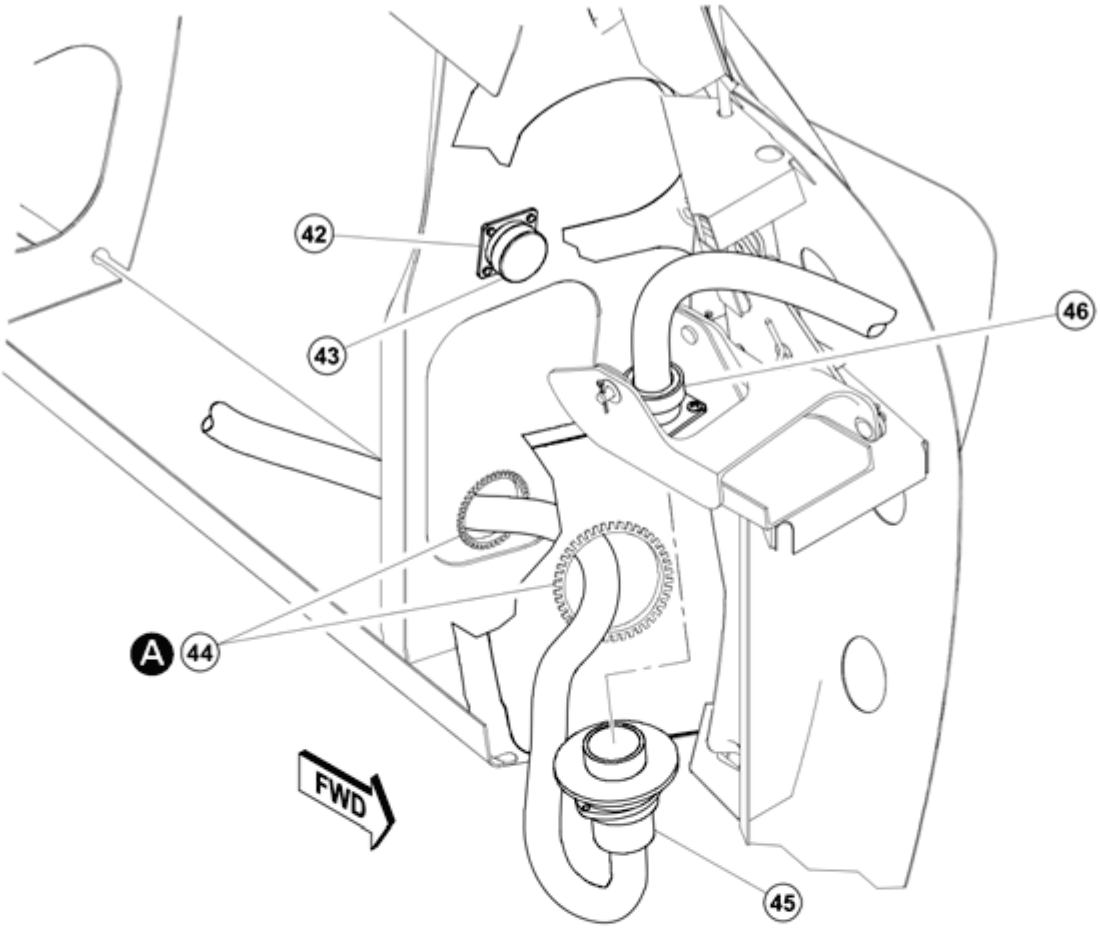
5.2. *Put a person near the helicopter to hold the door. Pull the ejection handle (48,* Figure 2*). The hinge pins (33 and 40) must retract smoothly and release the hinges. The latch striker (1) must pivot up to release the rear door latch. If necessary, push outward on the door to disconnect the door from the fuselage.*

5.3. *Install the crew door (*412-A-52-10-00-00A-720B-A / 00781.1*).*



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Figure 1. Crew Door - Removal/Installation (Sheet 1 of 2)



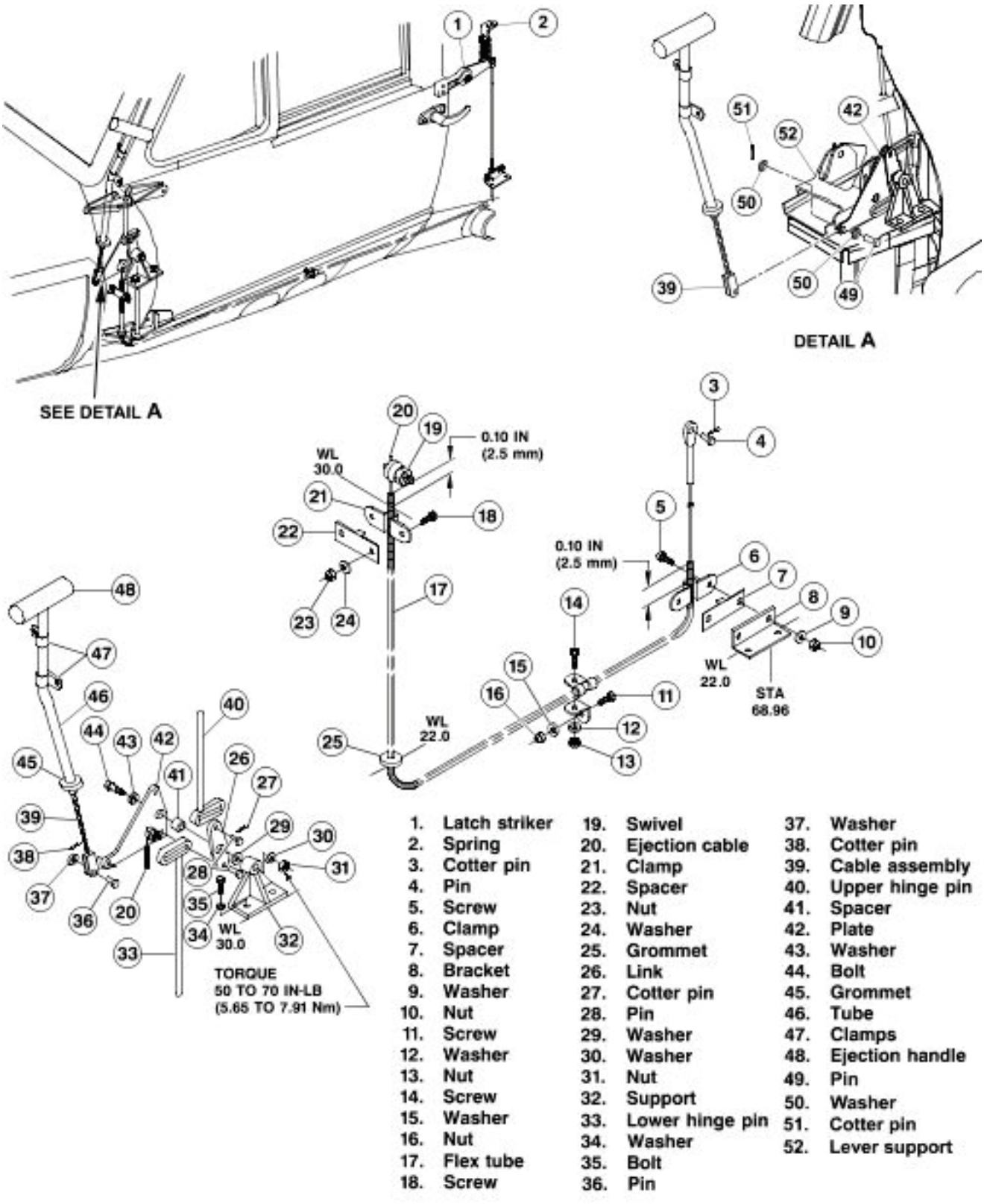
VIEW A
S/N 37002 through 37999, 38001 through 38999 and 39101 through 39999 installed with Vertical Reference Operations (VRO) kit (412-706-135)

1. Screw
2. Upper forward striker
3. Latch rod roller
4. Upper window
5. Door post
6. Screw
7. Screw
8. Nut
9. Washer
10. Washer
11. Backing plates
12. Backing plate
13. Upper aft striker
14. Spring pin
15. Latch rod
16. Sliding window
17. Lift handle
18. Latch striker
19. Latch assembly
20. Outer handle
21. Latch shaft
22. Pin
23. Cotter pin
24. Screw
25. Lower access door
26. Shouldered bolt
27. Washer
28. Bellcrank
29. Washer
30. Nut
31. Rod
32. Pin
33. Spring
34. Pin
35. Latch rod
36. Hinge halves
37. Hinge halves
38. Forward window
39. Inner access door
40. Adjustable stop
41. Closing handle
42. Dummy receptacle
43. Protective cap
44. Grommet
45. VRO connector (3121P1)
46. VRO receptacle (3121J1)

A ADHESIVE (C-301)

ICN-412-A-521000-B-97499-02854-A-001-01

Figure 1. Crew Door - Removal/Installation (Sheet 2 of 2)



1.	Latch striker	19.	Swivel	37.	Washer
2.	Spring	20.	Ejection cable	38.	Cotter pin
3.	Cotter pin	21.	Clamp	39.	Cable assembly
4.	Pin	22.	Spacer	40.	Upper hinge pin
5.	Screw	23.	Nut	41.	Spacer
6.	Clamp	24.	Washer	42.	Plate
7.	Spacer	25.	Grommet	43.	Washer
8.	Bracket	26.	Link	44.	Bolt
9.	Washer	27.	Cotter pin	45.	Grommet
10.	Nut	28.	Pin	46.	Tube
11.	Screw	29.	Washer	47.	Clamps
12.	Washer	30.	Washer	48.	Ejection handle
13.	Nut	31.	Nut	49.	Pin
14.	Screw	32.	Support	50.	Washer
15.	Washer	33.	Lower hinge pin	51.	Cotter pin
16.	Nut	34.	Washer	52.	Lever support
17.	Flex tube	35.	Bolt		
18.	Screw	36.	Pin		

ICN-412-A-521000-B-97499-00003-A-001-01

Figure 2. Crew Door Jettison Mechanism - Removal/Installation (Sheet 1 of 1)

Crew Door Latches - Removal

Procedure

NOTE

Remove the latch parts only to the extent necessary for replacement or other maintenance.

1. *Remove and discard the cotter pin (23, Figure 1). Remove the pin (22) and disconnect the latch rod (15) from the latch arm. Remove the spring pin (14) and disconnect the latch rod (15) from the latch rod roller (3).*

2. Remove the inner access door (39) and the plug button from the inner face of the door.

3. *Remove the pin (32), and disconnect the rod (31) from the bellcrank (28).*

4. *Remove the screws (24). Hold the latch shaft (21) and move the latch assembly (19) off the shaft.*

5. *Remove the screws that attach the outer handle (20) to the door and remove the outer handle (20) and the latch shaft (21) .*

6. *Remove the spring (33) and the pin (34). Disconnect the latch rod (35) from the bellcrank (28).*

7. Remove the shouldered bolt (26), washers (27 and 29), and nut (30). Remove the bellcrank (28).

8. *Remove the spring pin (14) and disconnect the latch rod (35) from the roller (3).*

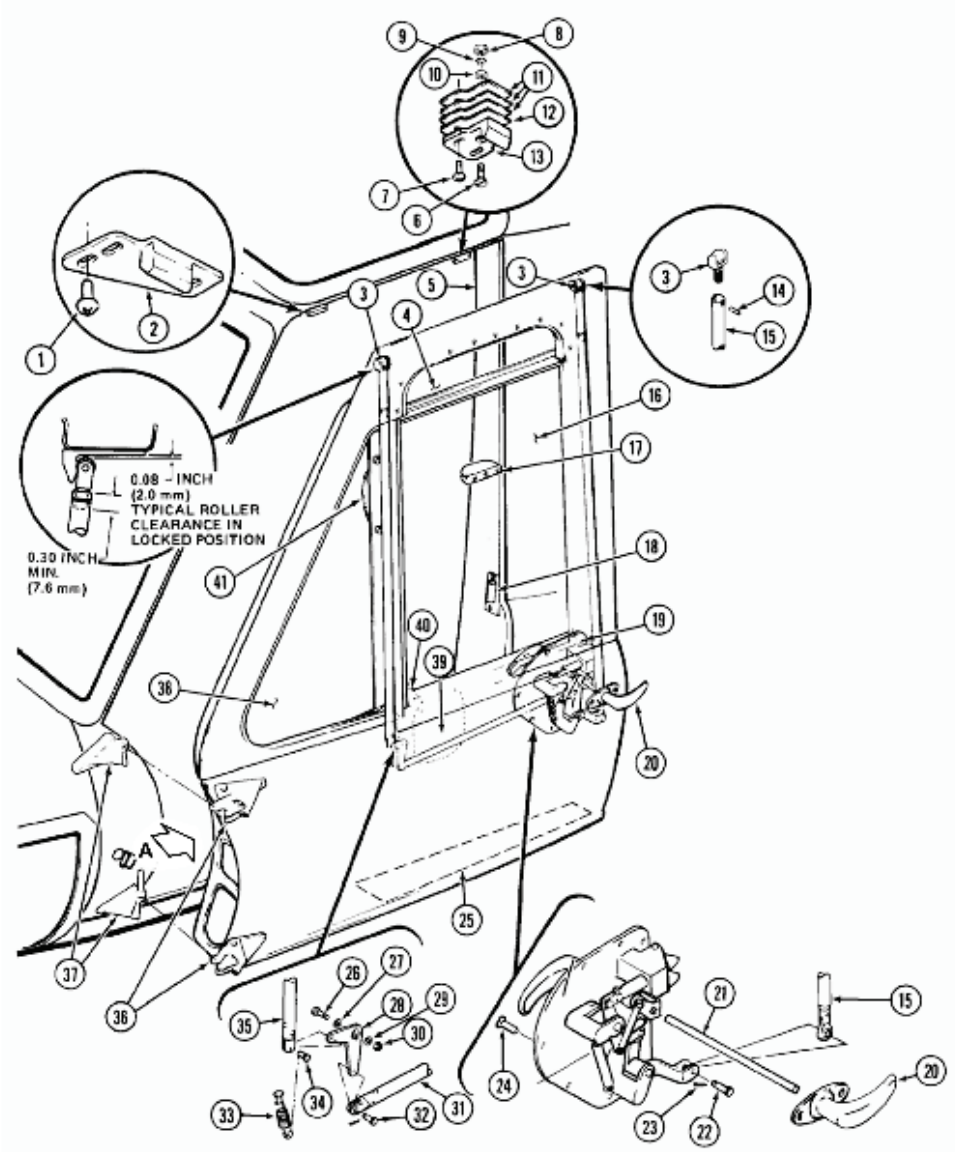
9. Remove the screws (1) and the upper forward striker (2).

10. *Remove the screws (6 and 7), upper aft striker (13), and backing plates (11 and 12). Index the backing plates (11 and 12) for reinstallation in the same position.*

11. *Remove and discard the cotter pin (3, Figure 2). Remove the pin (4) and disconnect the ejection cable (20) from the latch striker (1).*

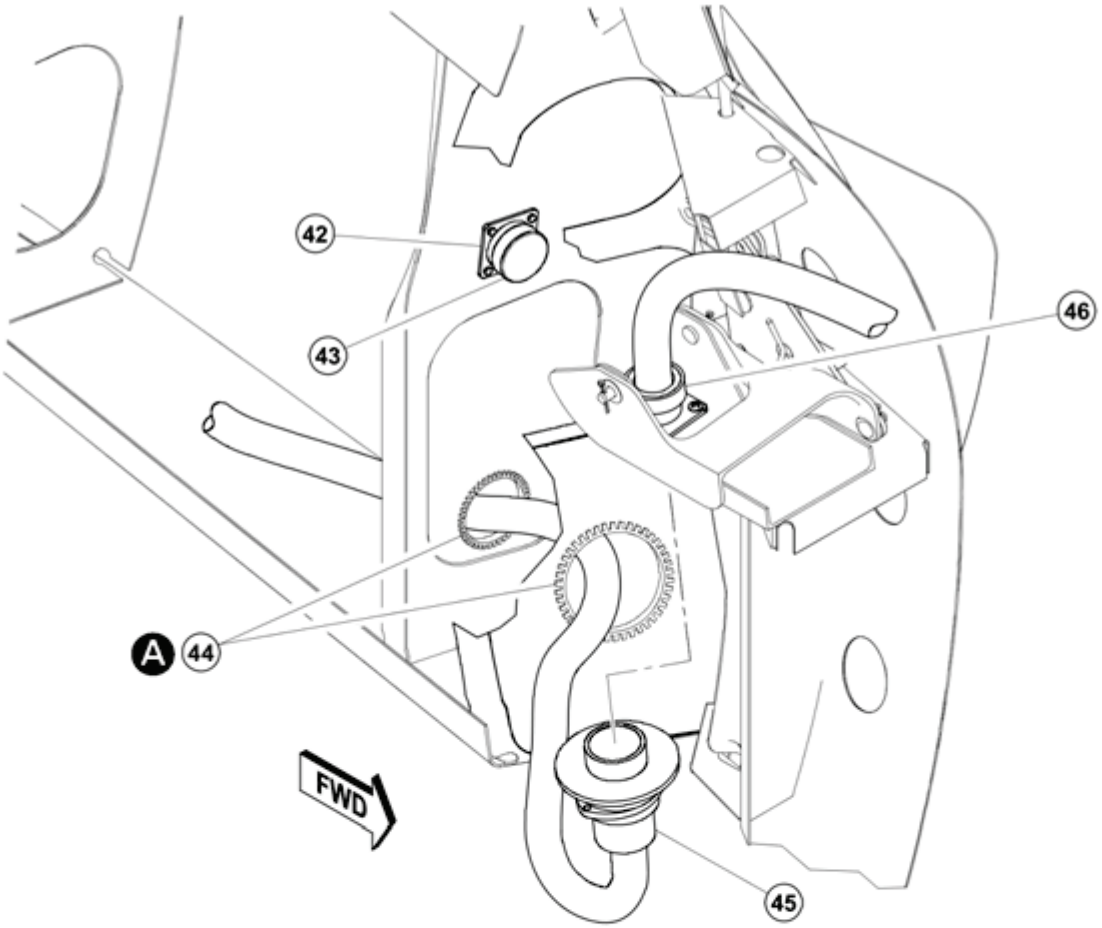
12. Remove spring (2) from the latch striker (1).

13. *Remove the bolt and the latch striker (1).*



ICN-412-A-521000-B-97499-00001-A-001-01

Figure 1. Crew Door - Removal/Installation (Sheet 1 of 2)



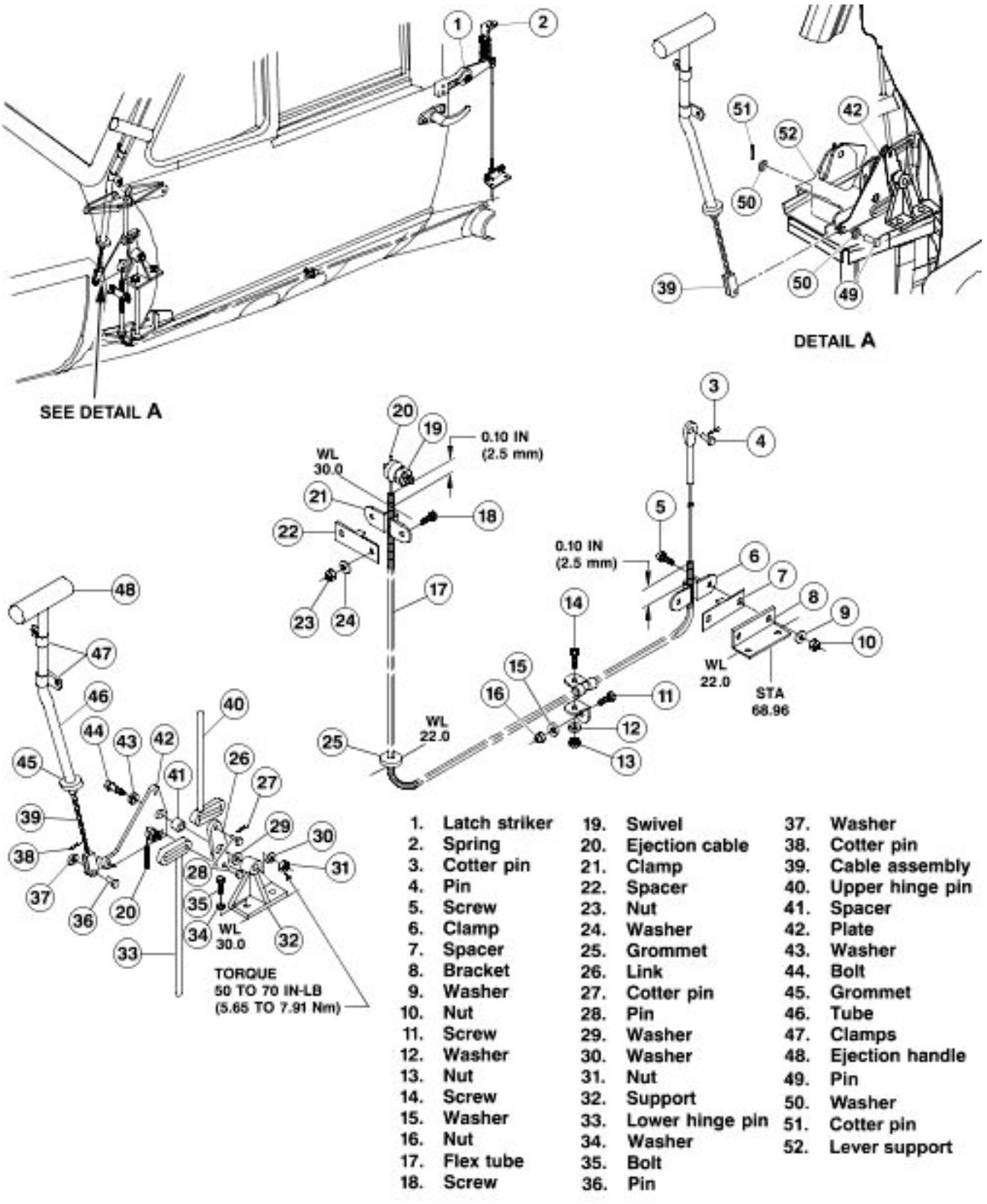
VIEW A
S/N 37002 through 37999, 38001 through 38999 and 39101 through 39999 installed with Vertical Reference Operations (VRO) kit (412-706-135)

1. Screw
2. Upper forward striker
3. Latch rod roller
4. Upper window
5. Door post
6. Screw
7. Screw
8. Nut
9. Washer
10. Washer
11. Backing plates
12. Backing plate
13. Upper aft striker
14. Spring pin
15. Latch rod
16. Sliding window
17. Lift handle
18. Latch striker
19. Latch assembly
20. Outer handle
21. Latch shaft
22. Pin
23. Cotter pin
24. Screw
25. Lower access door
26. Shouldered bolt
27. Washer
28. Bellcrank
29. Washer
30. Nut
31. Rod
32. Pin
33. Spring
34. Pin
35. Latch rod
36. Hinge halves
37. Hinge halves
38. Forward window
39. Inner access door
40. Adjustable stop
41. Closing handle
42. Dummy receptacle
43. Protective cap
44. Grommet
45. VRO connector (3121P1)
46. VRO receptacle (3121J1)

A ADHESIVE (C-301)

ICN-412-A-521000-B-97499-02854-A-001-01

Figure 1. Crew Door - Removal/Installation (Sheet 2 of 2)



1.	Latch striker	19.	Swivel	37.	Washer
2.	Spring	20.	Ejection cable	38.	Cotter pin
3.	Cotter pin	21.	Clamp	39.	Cable assembly
4.	Pin	22.	Spacer	40.	Upper hinge pin
5.	Screw	23.	Nut	41.	Spacer
6.	Clamp	24.	Washer	42.	Plate
7.	Spacer	25.	Grommet	43.	Washer
8.	Bracket	26.	Link	44.	Bolt
9.	Washer	27.	Cotter pin	45.	Grommet
10.	Nut	28.	Pin	46.	Tube
11.	Screw	29.	Washer	47.	Clamps
12.	Washer	30.	Washer	48.	Ejection handle
13.	Nut	31.	Nut	49.	Pin
14.	Screw	32.	Support	50.	Washer
15.	Washer	33.	Lower hinge pin	51.	Cotter pin
16.	Nut	34.	Washer	52.	Lever support
17.	Flex tube	35.	Bolt		
18.	Screw	36.	Pin		

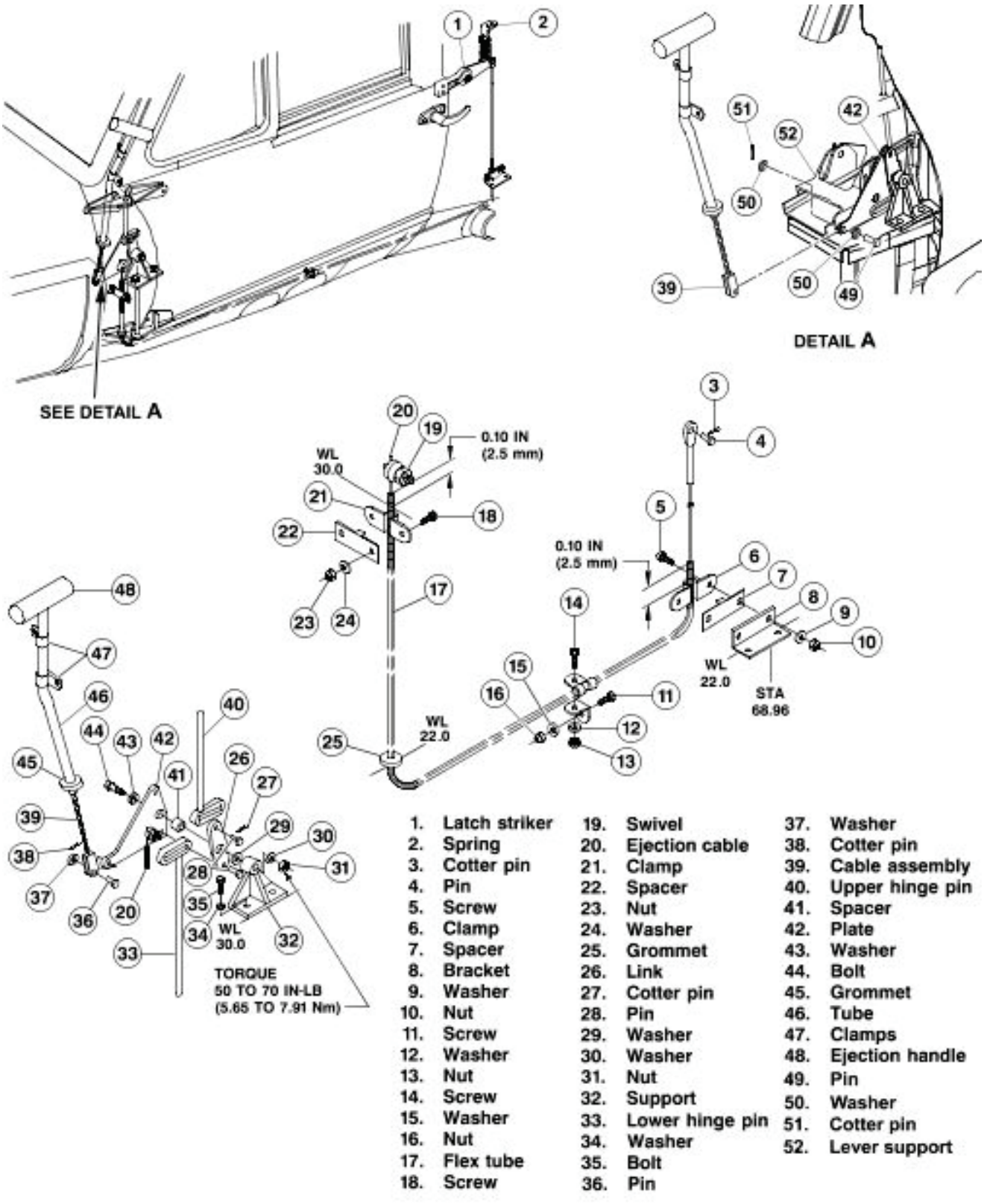
ICN-412-A-521000-B-97499-00003-A-001-01

Figure 2. Crew Door Jettison Mechanism - Removal/Installation (Sheet 1 of 1)

Crew Door Latches - Installation

Procedure

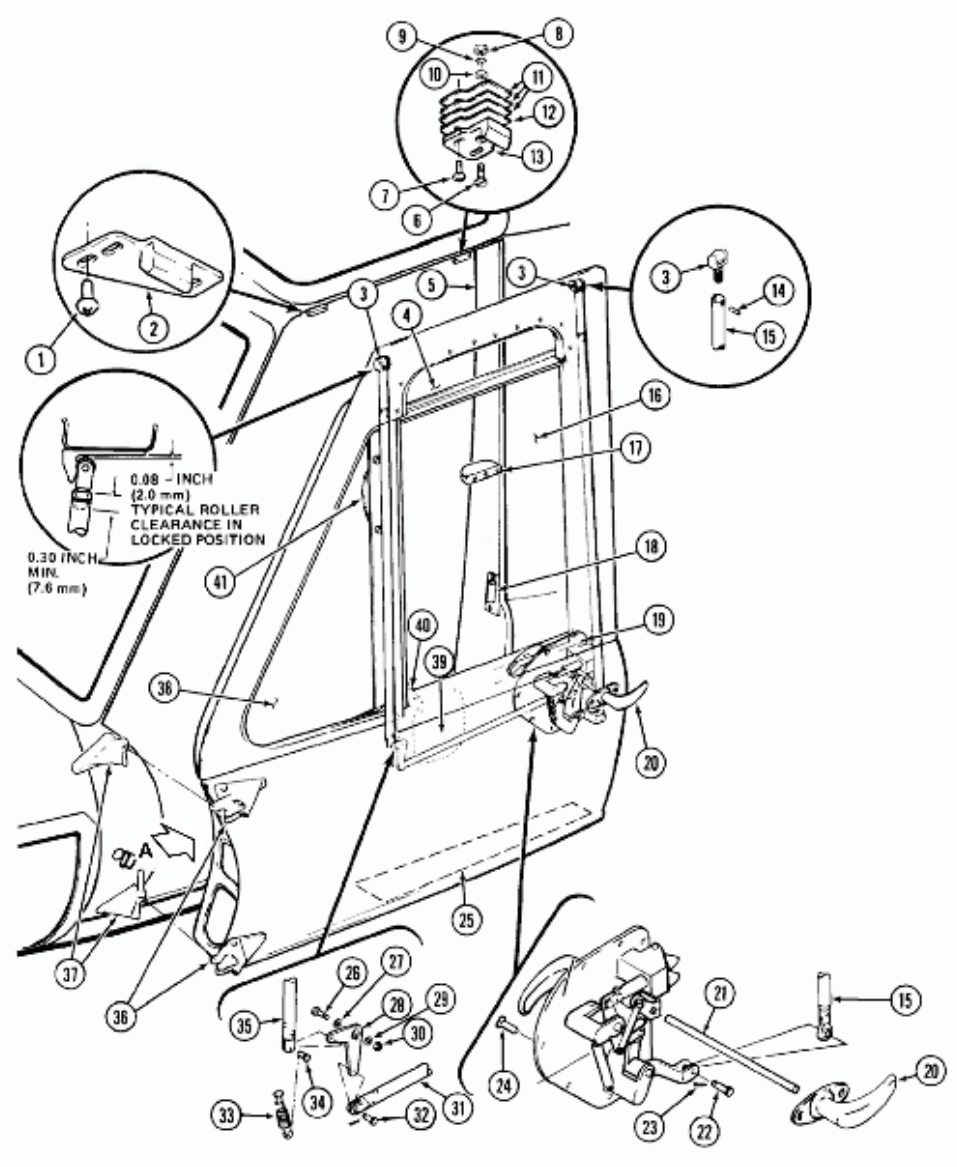
1. *Put the latch striker (1, Figure 1) on the door post and install the bolt.*
2. Install the spring (2).
3. *Put the ejection cable (20) on the latch striker (1). Install the pin (4). Safety the pin (4) with the new cotter pin (3).*
4. *Put the backing plates (11 and 12, Figure 2) on the door frame in the same position identified during the removal. Put the upper aft striker (13) on the backing plates (11 and 12) and install the screws (6 and 7), washers (9 and 10), and nut (8).*
5. *Put the upper forward striker (2) on the door frame and install the screws (1).*
6. *Put the latch assembly (19) and the inboard handle in the door and install nine screws (24).*
7. *Put the latch shaft (21) and the outer handle (20) on the door. Install the outer handle (20) to the door with the two screws.*
8. *Put the rod (31) in the door and engage the rod (31) end into the latch assembly (19).*
9. *Put the bellcrank (28) in the door and install the shouldered bolt (26), washers (27 and 29), and nut (30).*
10. *Put the rod (31) clevis on the bellcrank (28) and install the pin (32). Safety the pin (32) with the new cotter pin.*
11. *Put the latch rod (15) in the door and engage the latch rod (15) into the latch rod roller (3). Do not install the spring pin (14) at this time.*
12. *Put the latch rod (15) on the latch assembly (19) and install the pin (22). Safety the pin (22) with the new cotter pin (23).*
13. *Put the latch rod (35) in the door and engage the latch rod (35) into the latch rod roller (3) as given in Step 11. Do not install the spring pin at this time.*
14. *Put the latch rod (35) on the bellcrank (28) and install the pin (34). Safety the pin (34) with the new cotter pin.*
15. *Install the spring (33) on the latch rod (35) with the pin (34).*
16. *Close the crew door and make sure of the position of the strikers (2 and 13) and the latch rod rollers (3). Adjust the latch rods (15 and 35) in latch rod rollers (3) to get* 0.08 inch-pounds (0.0 Nm) *clearance as shown in Figure 2. Also adjust the strikers (2 and 13) to hold the crew door tightly in the closed position. Make sure that the threads of the latch rods (15 and 35) are engaged* 0.30 inch-pounds (0.0 Nm) *in the latch rod rollers (3) as shown in Figure 2. Put the spring pins (14) in the latch rods (15 and 35) to install the latch rod rollers (3).*
17. *Install the inner access door (39) and plug button on the crew door.*
18. *Do an operational check of the crew door (412-A-52-10-00-00A-320A-A / 00785).*



1.	Latch striker	19.	Swivel	37.	Washer
2.	Spring	20.	Ejection cable	38.	Cotter pin
3.	Cotter pin	21.	Clamp	39.	Cable assembly
4.	Pin	22.	Spacer	40.	Upper hinge pin
5.	Screw	23.	Nut	41.	Spacer
6.	Clamp	24.	Washer	42.	Plate
7.	Spacer	25.	Grommet	43.	Washer
8.	Bracket	26.	Link	44.	Bolt
9.	Washer	27.	Cotter pin	45.	Grommet
10.	Nut	28.	Pin	46.	Tube
11.	Screw	29.	Washer	47.	Clamps
12.	Washer	30.	Washer	48.	Ejection handle
13.	Nut	31.	Nut	49.	Pin
14.	Screw	32.	Support	50.	Washer
15.	Washer	33.	Lower hinge pin	51.	Cotter pin
16.	Nut	34.	Washer	52.	Lever support
17.	Flex tube	35.	Bolt		
18.	Screw	36.	Pin		

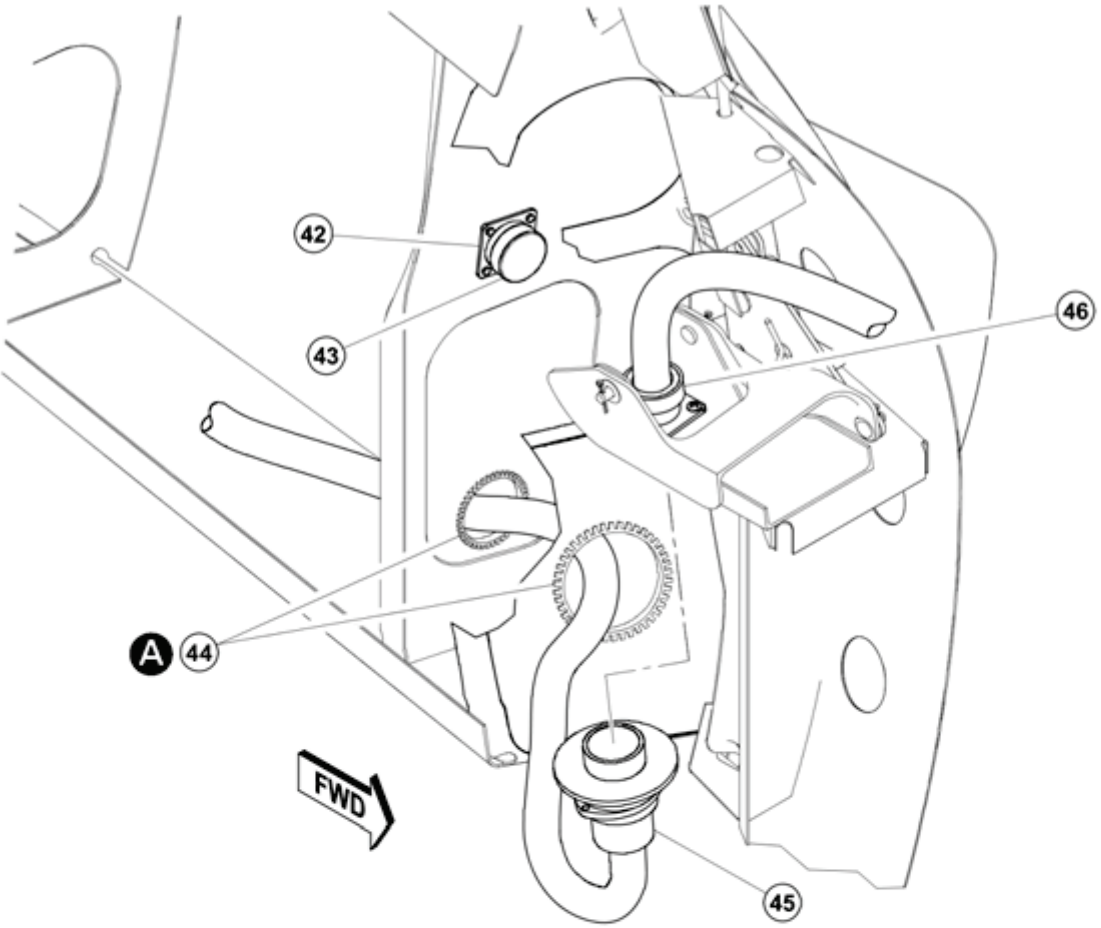
ICN-412-A-521000-B-97499-00003-A-001-01

Figure 1. Crew Door Jettison Mechanism - Removal/Installation (Sheet 1 of 1)



ICN-412-A-521000-B-97499-00001-A-001-01

Figure 2. Crew Door - Removal/Installation (Sheet 1 of 2)



VIEW A
S/N 37002 through 37999, 38001 through 38999 and 39101 through 39999 installed with Vertical Reference Operations (VRO) kit (412-706-135)

1. Screw
2. Upper forward striker
3. Latch rod roller
4. Upper window
5. Door post
6. Screw
7. Screw
8. Nut
9. Washer
10. Washer
11. Backing plates
12. Backing plate
13. Upper aft striker
14. Spring pin
15. Latch rod
16. Sliding window
17. Lift handle
18. Latch striker
19. Latch assembly
20. Outer handle
21. Latch shaft
22. Pin
23. Cotter pin
24. Screw
25. Lower access door
26. Shouldered bolt
27. Washer
28. Bellcrank
29. Washer
30. Nut
31. Rod
32. Pin
33. Spring
34. Pin
35. Latch rod
36. Hinge halves
37. Hinge halves
38. Forward window
39. Inner access door
40. Adjustable stop
41. Closing handle
42. Dummy receptacle
43. Protective cap
44. Grommet
45. VRO connector (3121P1)
46. VRO receptacle (3121J1)

A ADHESIVE (C-301)

ICN-412-A-521000-B-97499-02854-A-001-01

Figure 2. Crew Door - Removal/Installation (Sheet 2 of 2)

Crew Door Hinges - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	

Procedure

NOTE

This procedure is applicable to the upper and lower hinge half assemblies. The procedure that follows is for the upper hinge half assemblies (1 and 2, Figure 1) and the differences for the lower hinge half assemblies (3 and 4) are identified.

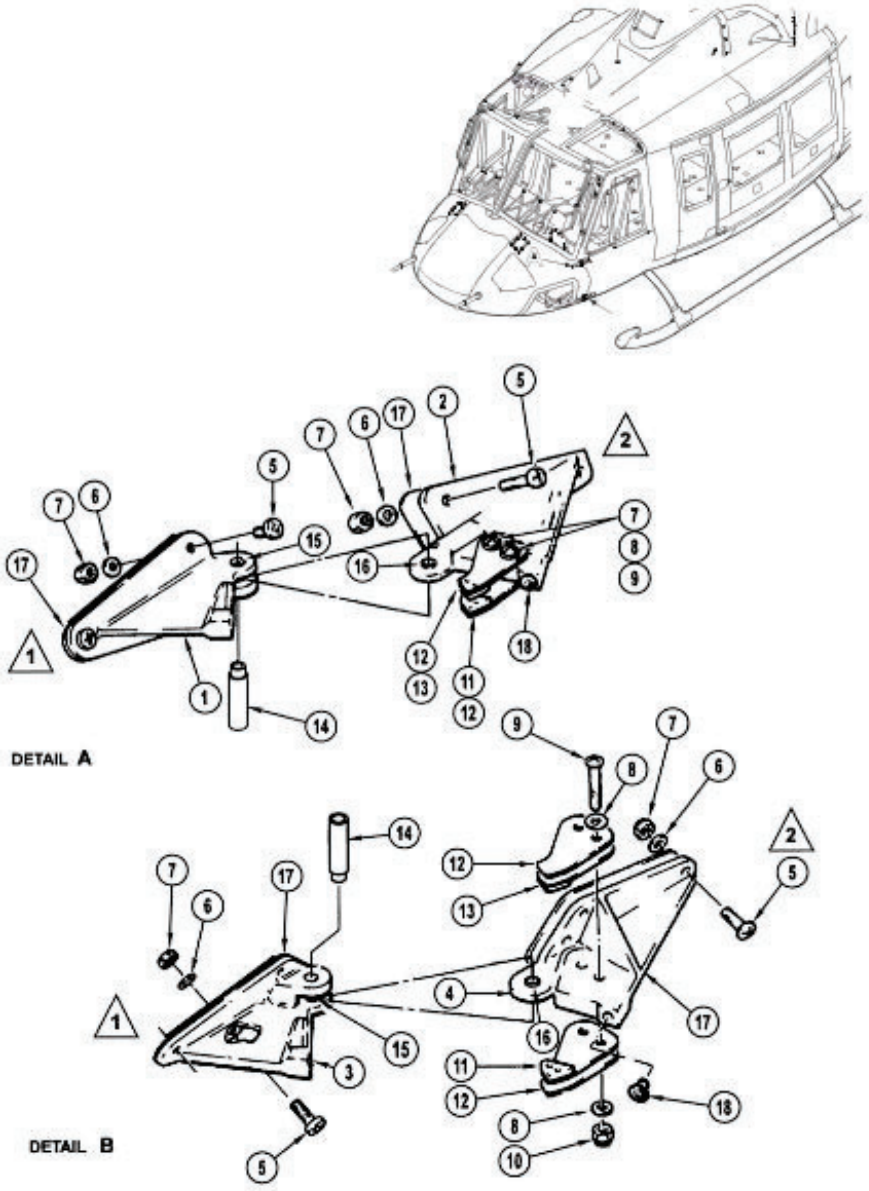
1. *Remove the crew door (412-A-52-10-00-00A-520B-A / 00778.1).*

2. *Carefully remove the sealant from around the upper fixed hinge-half-assembly (1) and the upper movable hinge-half-assembly (2).*

3. *Remove the screws (5), nuts (7), and washers (6).*

4. *Remove the upper fixed hinge-half-assembly (1), upper movable hinge-half-assembly (2), and shims (17) from the crew door.*

5. Carefully remove the remaining sealant with aliphatic naphtha (C-305).



ICN-412-A-521000-B-97499-00004-A-001-01

Figure 1. Crew Door Hinges - Removal/Installation (Sheet 1 of 2)

1. Hinge half assembly, upper fixed
2. Hinge half assembly, upper movable
3. Hinge half assembly, lower fixed
4. Hinge half assembly, lower movable
5. Screw
6. Washer
7. Nut
8. Washer
9. Screw
10. Nut
11. Spring assembly, lower
12. Spring
13. Spring assembly, upper
14. Bushing
15. Bushing
16. Bushing
17. Shim
18. Bumper

NOTES

1. If not previously accomplished, match drill holes to 0.190 to 0.196 inch (4.826 to 4.978 mm) in diameter.
2. If not previously accomplished, match drill holes to 0.190 to 0.218 inch (4.826 to 5.537 mm) in diameter.

ICN-412-A-521000-B-97499-00005-A-001-01

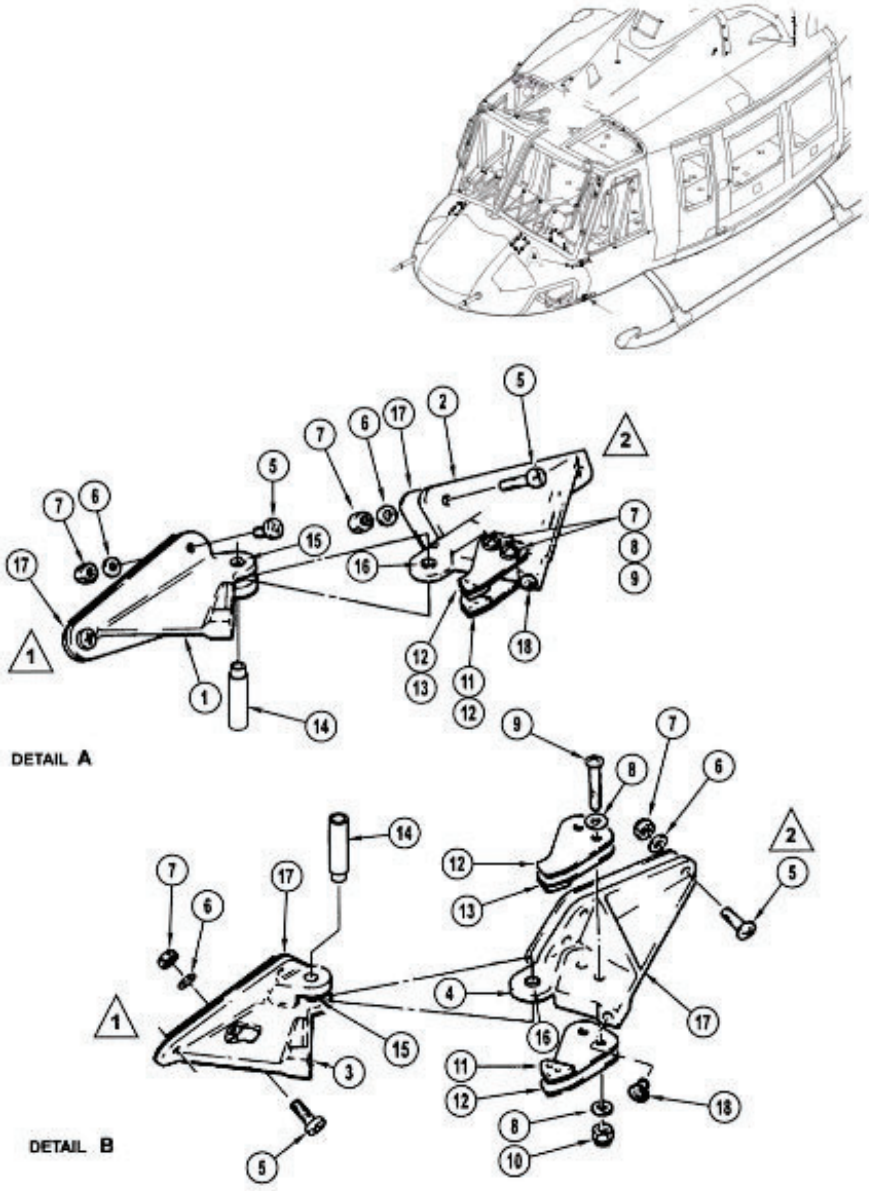
Figure 1. Crew Door Hinges - Removal/Installation (Sheet 2 of 2)

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Crew Door Hinges - Inspection

Procedure

1. Inspect the applicable fixed hinge half assembly (1 or 3, Figure 1) for cracks, wear, and holes with signs of deformation.
2. Inspect the applicable movable hinge half assembly (2 or 4) for cracks, wear, and holes with signs of deformation.
3. Inspect the bushings (15 and 16) for excessive wear or play.
4. Inspect the bushings (14) for excessive wear.
5. Inspect the spring assemblies (11 and 13) for excessive wear to the striker blocks.
6. Replace the spring assemblies (11 or 13) if the rivets of the striker blocks are visible.



ICN-412-A-521000-B-97499-00004-A-001-01

Figure 1. Crew Door Hinges (Sheet 1 of 2)

1. Hinge half assembly, upper fixed
2. Hinge half assembly, upper movable
3. Hinge half assembly, lower fixed
4. Hinge half assembly, lower movable
5. Screw
6. Washer
7. Nut
8. Washer
9. Screw
10. Nut
11. Spring assembly, lower
12. Spring
13. Spring assembly, upper
14. Bushing
15. Bushing
16. Bushing
17. Shim
18. Bumper

NOTES

1. If not previously accomplished, match drill holes to 0.190 to 0.196 inch (4.826 to 4.978 mm) in diameter.
2. If not previously accomplished, match drill holes to 0.190 to 0.218 inch (4.826 to 5.537 mm) in diameter.

ICN-412-A-521000-B-97499-00005-A-001-01

Figure 1. Crew Door Hinges (Sheet 2 of 2)

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Crew Door Hinges - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	
Sealant	C-308	AR	

Procedure

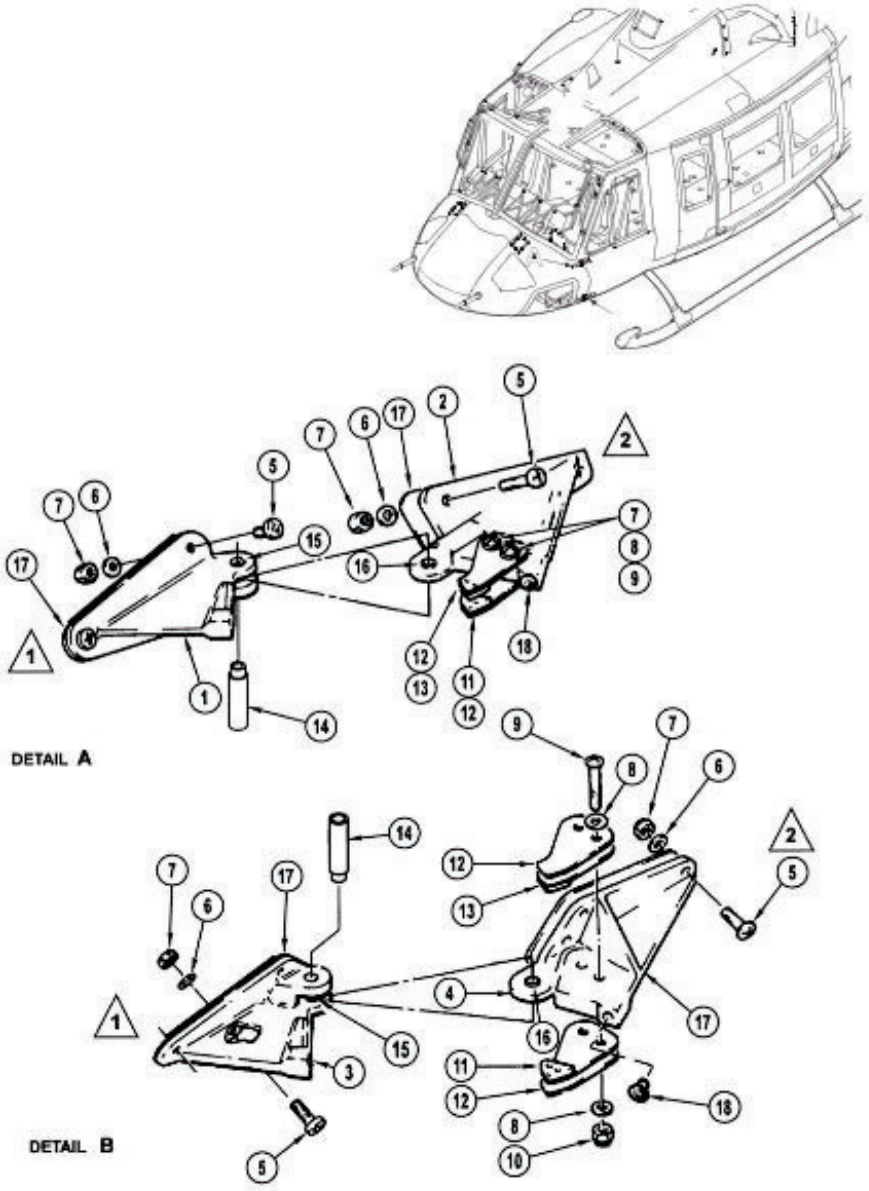
NOTE

This procedure is applicable to the upper and lower hinge half assemblies. The procedure that follows is for the upper hinge half assemblies (1 and 2, Figure 1) and the differences for the lower hinge half assemblies (3 and 4) are identified.

NOTE

New hinge half assemblies can be supplied without the three mounting holes.

1. *Install the upper fixed hinge-half-assembly (1) and the upper movable hinge-half-assembly (2) with shims (17), screws (5), washers (6) and nuts (7).*
2. *Temporarily install the crew door (412-A-52-10-00-00A-720B-A / 00781.1) to make sure the fit is correct.*
3. *Remove the crew door (412-A-52-10-00-00A-520B-A / 00778.1).*
4. If necessary, adjust the thickness of the shims (17) to get the correct fit.
5. *Apply sealant (C-308) around the edge of the upper fixed hinge-half-assembly (1) and the upper movable hinge-half-assembly (2).*
6. *Clean the unwanted sealant (C-308) with aliphatic naphtha (C-305).*
7. *Install the crew door (412-A-52-10-00-00A-720B-A / 00781.1).*



ICN-412-A-521000-B-97499-00004-A-001-01

Figure 1. Crew Door Hinges - Removal/Installation (Sheet 1 of 2)

1. Hinge half assembly, upper fixed
2. Hinge half assembly, upper movable
3. Hinge half assembly, lower fixed
4. Hinge half assembly, lower movable
5. Screw
6. Washer
7. Nut
8. Washer
9. Screw
10. Nut
11. Spring assembly, lower
12. Spring
13. Spring assembly, upper
14. Bushing
15. Bushing
16. Bushing
17. Shim
18. Bumper

NOTES

1. If not previously accomplished, match drill holes to 0.190 to 0.196 inch (4.826 to 4.978 mm) in diameter.
2. If not previously accomplished, match drill holes to 0.190 to 0.218 inch (4.826 to 5.537 mm) in diameter.

ICN-412-A-521000-B-97499-00005-A-001-01

Figure 1. Crew Door Hinges - Removal/Installation (Sheet 2 of 2)

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Crew Door Hinges - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Epoxy polyamide primer	C-204	AR	
Adhesive	C-317	AR	

Procedure

1. Replace the bushings (14 and 15, Figure 1) as follows:

1.1. Hold the legs that contains the bushings (14 or 15) and push them out.

1.2. Make sure that the holes are free of damage.

1.3. Measure the holes and make sure that the diameters are from 0.312 to 0.313 inch (7.925 to 7.950 mm).

NOTE

Make sure that the flanges of the bushings are in the hinge assembly slot.

1.4. Apply epoxy polyamide primer (C-204) to the holes.

1.5. Hold the legs of the fixed hinge half assemblies (1 and 3) and push the new bushings (15) into position.

NOTE

Install the bushings to be flush with the inner face of the fixed hinge half assemblies.

1.6. Apply epoxy polyamide primer (C-204) to the holes.

1.7. Hold the legs of the fixed hinge half assemblies (1 and 3) and push the new bushings (14) into position.

1.8. Ream bushings (14) to have a diameter from 0.1880 to 0.1885 inch (4.775 to 4.788 mm).

2. Replace bushings (16) as follows:

2.1. Hold the legs that contains the bushings (16) and push them out.

2.2. Make sure that the holes are free of damage.

2.3. Measure the holes and make sure that the diameters are from 0.312 to 0.313 inch (7.925 to 7.950 mm).

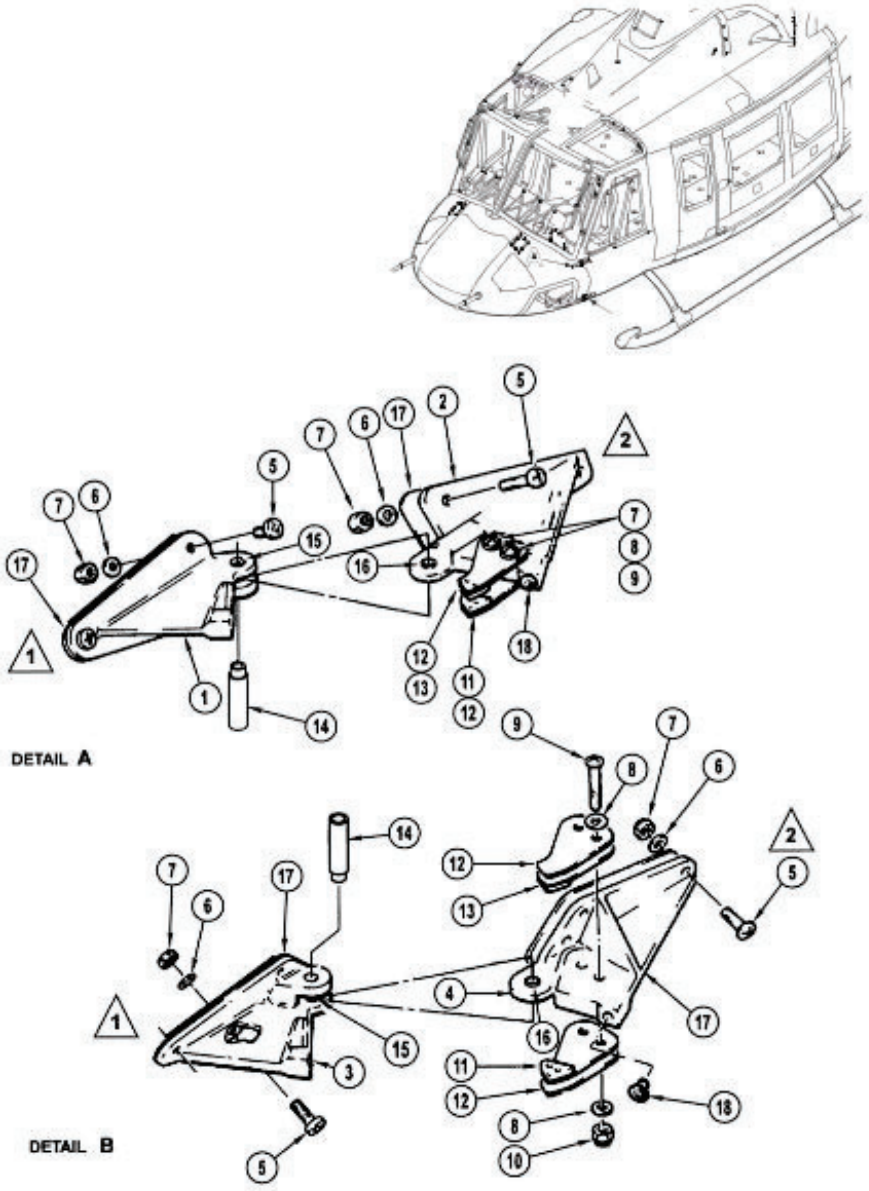
NOTE

Make sure that the ends of the bushings are flush or below the surface of the moveable hinge half assembly.

2.4. Apply epoxy polyamide primer (C-204) to the holes.

2.5. Ream bushings (16) to have a diameter from 0.1880 to 0.1885 inch (4.775 to 4.788 mm).

2.6. If removed, install a new bumper (18) with adhesive (C-317).



ICN-412-A-521000-B-97499-00004-A-001-01

Figure 1. Crew Door Hinges (Sheet 1 of 2)

1. Hinge half assembly, upper fixed
2. Hinge half assembly, upper movable
3. Hinge half assembly, lower fixed
4. Hinge half assembly, lower movable
5. Screw
6. Washer
7. Nut
8. Washer
9. Screw
10. Nut
11. Spring assembly, lower
12. Spring
13. Spring assembly, upper
14. Bushing
15. Bushing
16. Bushing
17. Shim
18. Bumper

NOTES

1. If not previously accomplished, match drill holes to 0.190 to 0.196 inch (4.826 to 4.978 mm) in diameter.
2. If not previously accomplished, match drill holes to 0.190 to 0.218 inch (4.826 to 5.537 mm) in diameter.

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Figure 1. Crew Door Hinges (Sheet 2 of 2)

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Composite Crew Doors - Description and Maintenance

The crew doors are made of composites. The doors are hinged on the forward side and are equipped with a bell crank assembly, which may be operated from either side of door, to secure the door in closed position. Each door incorporates the transparent acrylic plastic window. In an emergency, doors may be jettisoned by pulling the EMERGENCY RELEASE handle mounted inside cabin forward of each door.

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Crew Doors - Removal

Procedure

WARNING

USE TWO PERSONNEL MINIMUM TO HOLD THE DOOR BEFORE YOU REMOVE/INSTALL THE DOOR FROM/ON THE HELICOPTER. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR OR CAN CAUSE DAMAGE TO THE PART.

NOTE

The procedure is for the crew door on the left side. The procedure is same for the crew door on the right side.

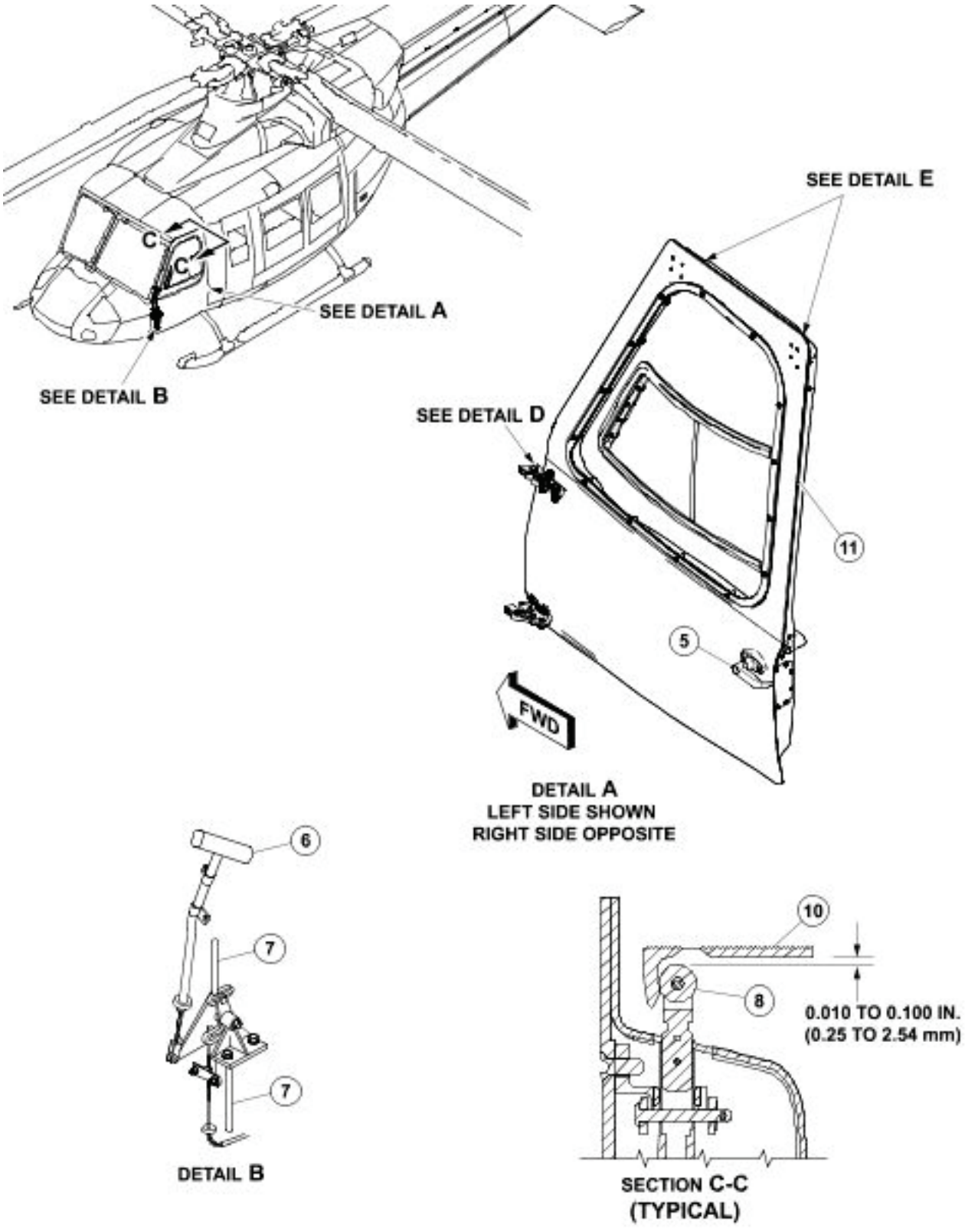
NOTE

The procedure is provided for the upper hinge assembly. The procedure is same for the lower hinge assembly.

NOTE

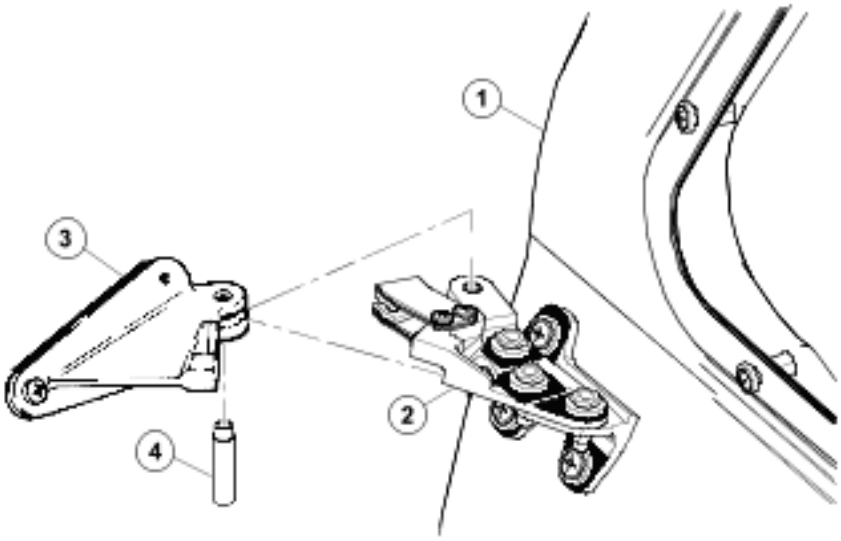
If the helicopter is equipped with the vertical reference operations kit (412-706-135), for procedures refer to (412-A-52-10-00-00A-520B-A / 00778.1).

1. Turn down the outer handle (5, Figure 1) to the open position.
2. Open the crew door (1).
3. *Pull the ejection handle (6) to retract the hinge pins (7) from the bushing (4). Refer to the jettison mechanism (412-A-52-10-00-01A-520B-A / 00781.2).*
4. Remove the bushing (4).
5. Remove the crew door (1) from the helicopter.

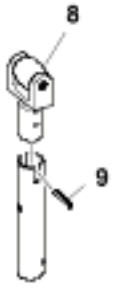


ICN-412-A-521100-B-97499-00001-A-001-01

Figure 1. Crew Doors (Sheet 1 of 2)



DETAIL D



DETAIL E

1. Door
2. Hinge assembly, upper
3. Hinge assembly, fuselage
4. Bushing
5. Handle, door
6. Ejection handle
7. Hinge pin
8. Roller
9. Pin
10. Striker
11. Extrusion

ICN-412-A-521100-B-97499-00002-A-001-01

Figure 1. Crew Doors (Sheet 2 of 2)

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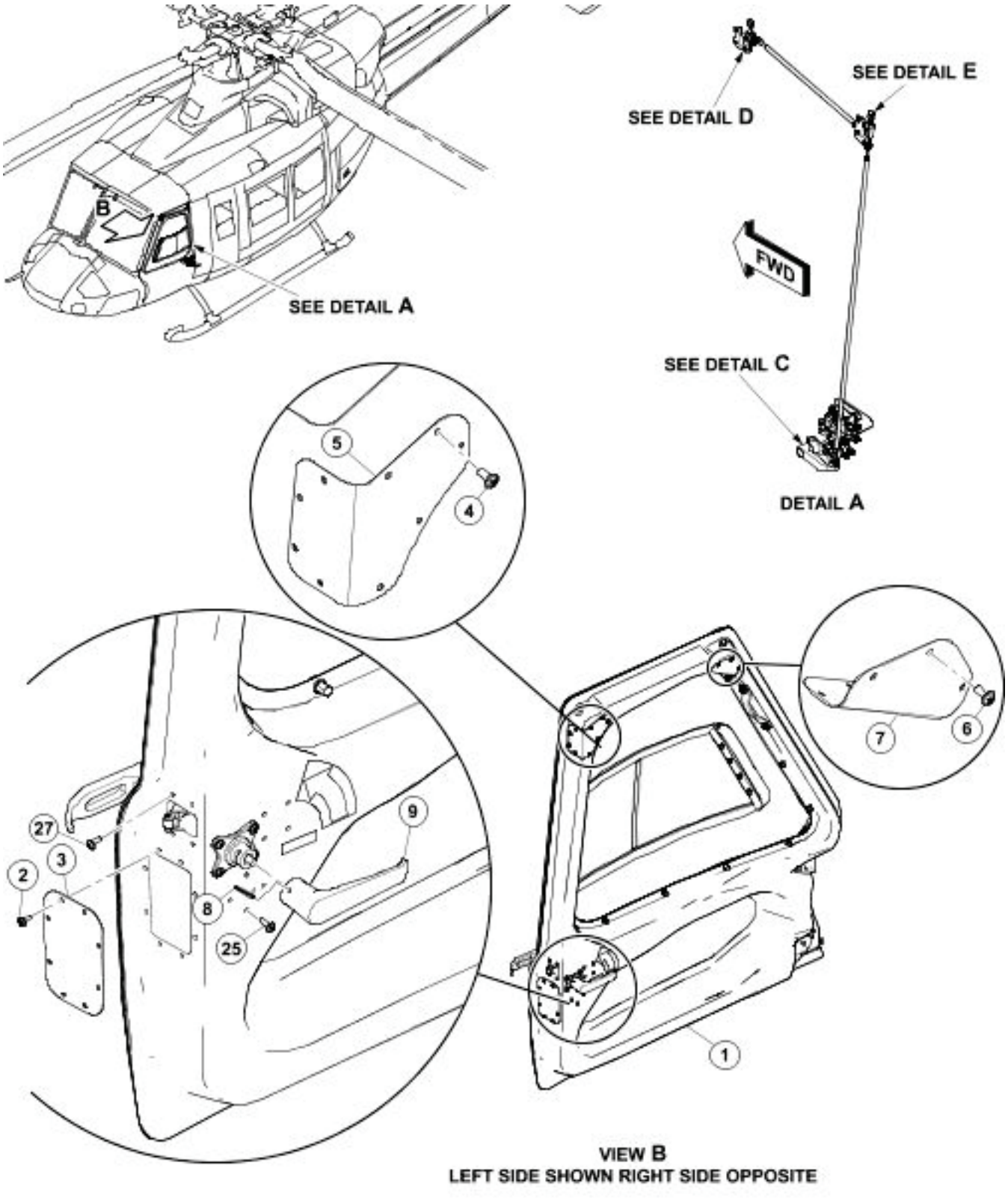
Crew Doors - Inspection

Procedure

NOTE

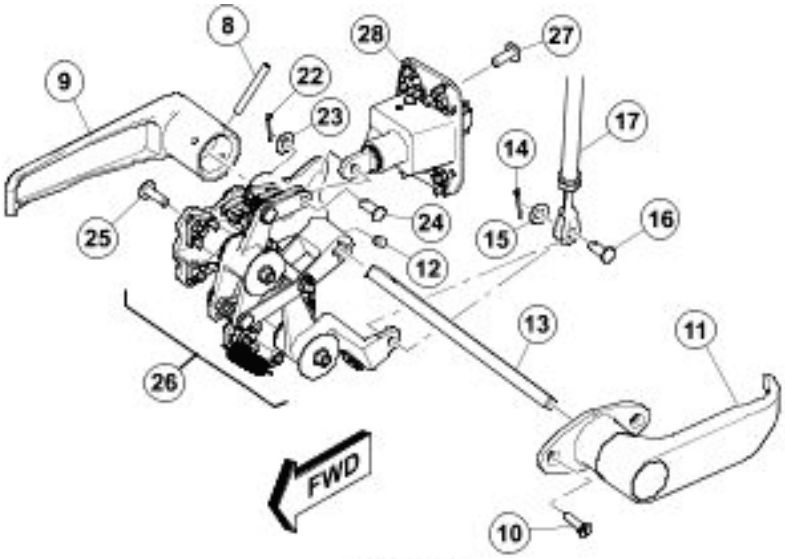
The inspection procedure is provided for the crew door on the left side. The inspection procedure is same for the crew door on the right side.

1. Inspect the visible portions of the door bell crank and plunger assembly (Figure 1) for secure installation, corrosion, and damage.
2. Inspect the visible portions of the crew door jettison mechanism for secure installation, corrosion, and damage.
3. Inspect the crew door structure and hinges for cracks and distortion.
4. Inspect the crew door window (412-A-52-01-00-10A-280A-A / 00882).
5. Inspect the crew door seals for secure bonding and for deterioration.

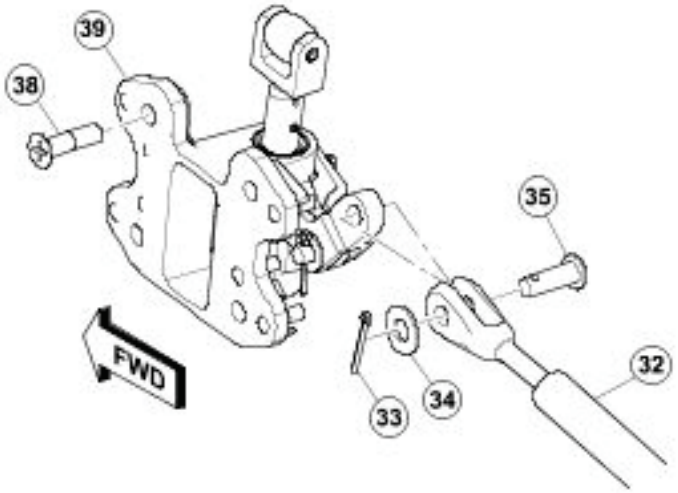


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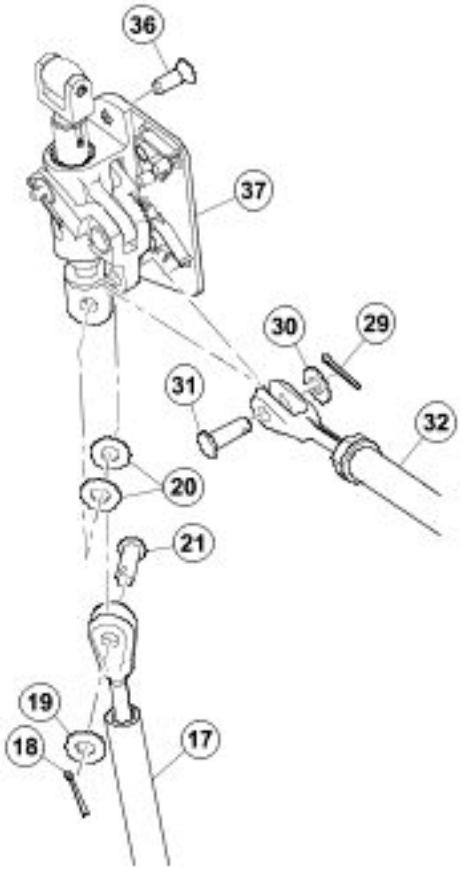
Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 1 of 2)



DETAIL C



DETAIL D



DETAIL E

1. Door
2. Screw
3. Cover, aft Lower
4. Screw
5. Cover, aft upper
6. Screw
7. Cover, fwd upper
8. Pin
9. Handle, inner
10. Screw
11. Handle, outer
12. Set screw
13. Rod, handle
14. Cotter pin
15. Washer
16. Pin
17. Rod assembly, vertical
18. Cotter pin
19. Washer
20. Washer
21. Pin
22. Cotter pin
23. Washer
24. Pin
25. Screw
26. Bell crank assembly
27. Screw
28. Plunger assembly, aft lower
29. Cotter Pin
30. Washer
31. Pin
32. Rod assembly, horizontal
33. Cotter pin
34. Washer
35. Pin
36. Screw
37. Plunger assembly, upper aft
38. Screw
39. Plunger assembly, upper fwd

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Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 2 of 2)

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Crew Doors - Repair

Procedure

NOTE

Repair of crew door bell crank assembly and jettison mechanism is restricted to replacement of unserviceable parts.

1. Replace the unserviceable parts in the crew door bell crank assembly (412-A-52-11-00-03A-520A-A / 00802, 412-A-52-11-00-03A-720A-A / 00803, 412-A-52-11-00-04A-520A-A / 00804, 412-A-52-11-00-04A-720A-A / 00805, 412-A-52-11-00-05A-520A-A / 00806 and 412-A-52-11-00-05A-720A-A / 00807).

2. Replace the unserviceable parts in the crew door jettison mechanism (412-A-52-11-00-02A-042A-A / 00797).

3. Repair the crew door structural damage. Refer to the BHT-MED-SRM-1 and to FAA Publication AC 43.13-1B Aircraft Inspection and Repair Manual for structural maintenance.

4. Repair the damaged crew door windows (412-A-52-01-00-10A-685A-A / 00883).

5. Install the new crew door seals to replace damaged or missing seals (412-A-52-11-00-01A-720A-A / 00796).

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Crew Doors - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	

Procedure



USE TWO PERSONNEL MINIMUM TO HOLD THE DOOR BEFORE YOU REMOVE/INSTALL THE DOOR FROM/ON THE HELICOPTER. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR OR CAN CAUSE DAMAGE TO THE PART.

NOTE

The procedure is for the crew door on the left side. The procedure is the same for the crew door on the right side.

NOTE

The procedure is provided for the upper hinge assembly. The procedure is same for the lower hinge assembly.

NOTE

If the helicopter is equipped with the vertical reference operations kit (412-706-135), for procedures refer to (412-A-52-10-00-00A-720B-A / 00781.1).

1. *If necessary, replace the extrusion (11, Figure 1) on the door (1) as follows:*

1.1. Remove the extrusion (11) from around the edges of the door (1).

1.2. Remove the old sealant from the edges of the door (1) with a plastic scraper.

1.3. Apply sealant (C-308) around the edges of the door (1).

1.4. Install the extrusion (11) around the edges of the door (1).

1.5. Clean the unwanted sealant.

2. Put the crew door (1) in its position on the helicopter.

3. Pull the ejection handle (6) to retract the upper hinge pin (7).

4. Align the upper hinge assembly (2) and the fuselage hinge assembly (3).

5. *Make sure that the lower and upper hinge pins (7) engage the hinge assemblies (2 and 3) correctly. Refer to jettison mechanism (412-A-52-10-00-00A-320A-A / 00785).*

6. *Close the door (1) slowly and monitor the movement of the bell crank assembly (26, Figure 2).*

7. *Slowly move the outer handle (11) to the lock position. Monitor the engagement of the two rollers (8, Figure 1) with the upper forward and aft strikers (10) and for a clearance of* 0.01 to 0.1 inches (0.3 to 2.5 mm) *above each roller (8) when fully extended.*

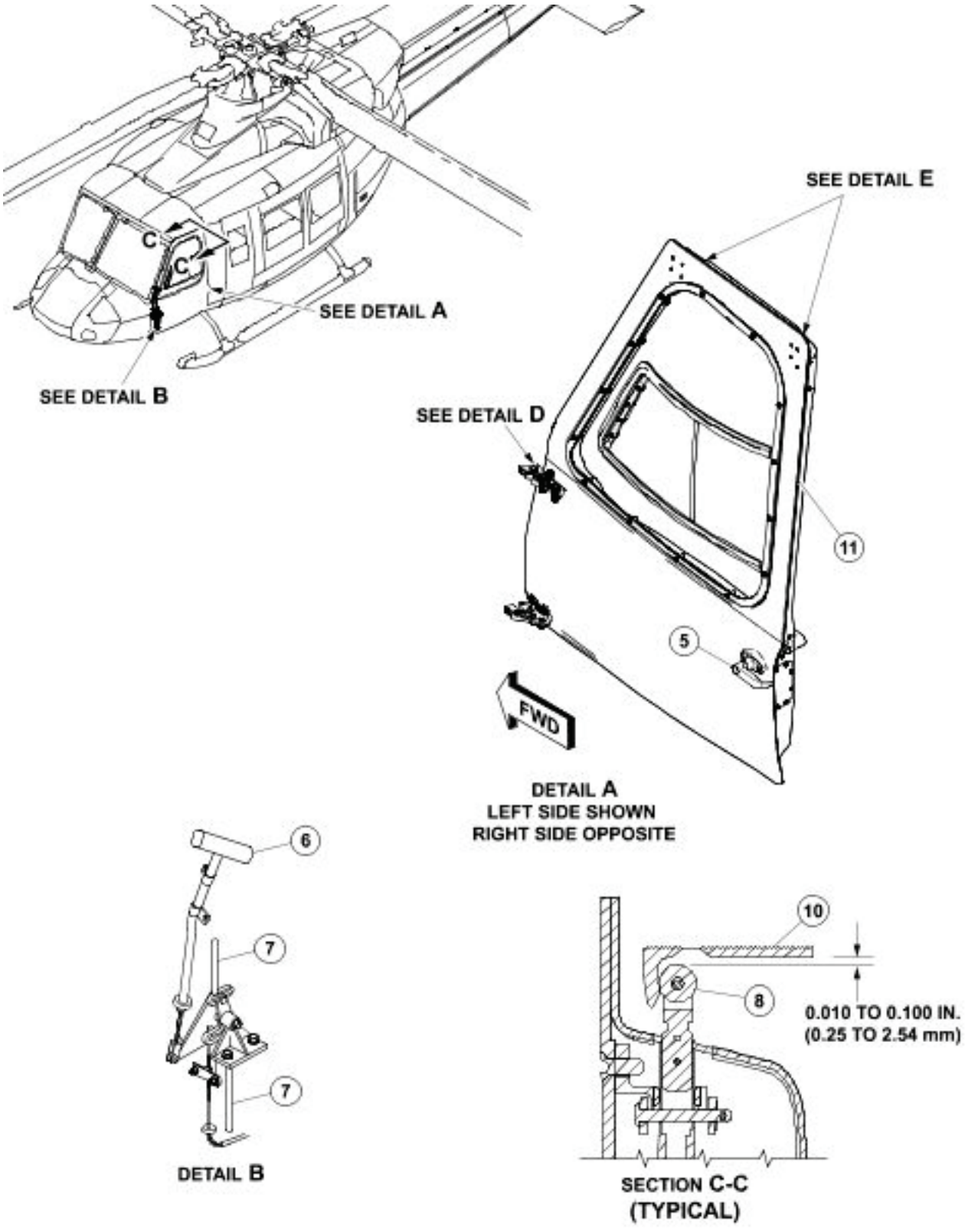
8. *Adjust the rollers (8) to get a clearance of* 0.01 to 0.1 inches (0.3 to 2.5 mm).

9. *Adjust the position of the strikers (10), if necessary.*

10. *When the adjustment is satisfactory, install the pin (9) in each roller (8).*

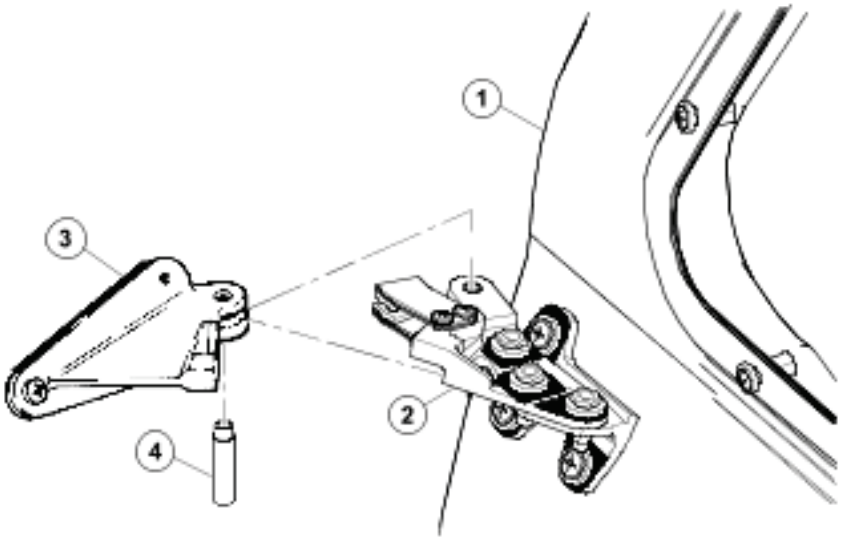
11. Install the bushing (4).

12. *Make sure that the door (1) opens and closes correctly.*

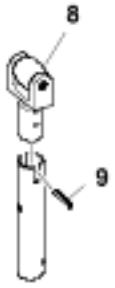


ICN-412-A-521100-B-97499-00001-A-001-01

Figure 1. Crew Doors (Sheet 1 of 2)



DETAIL D

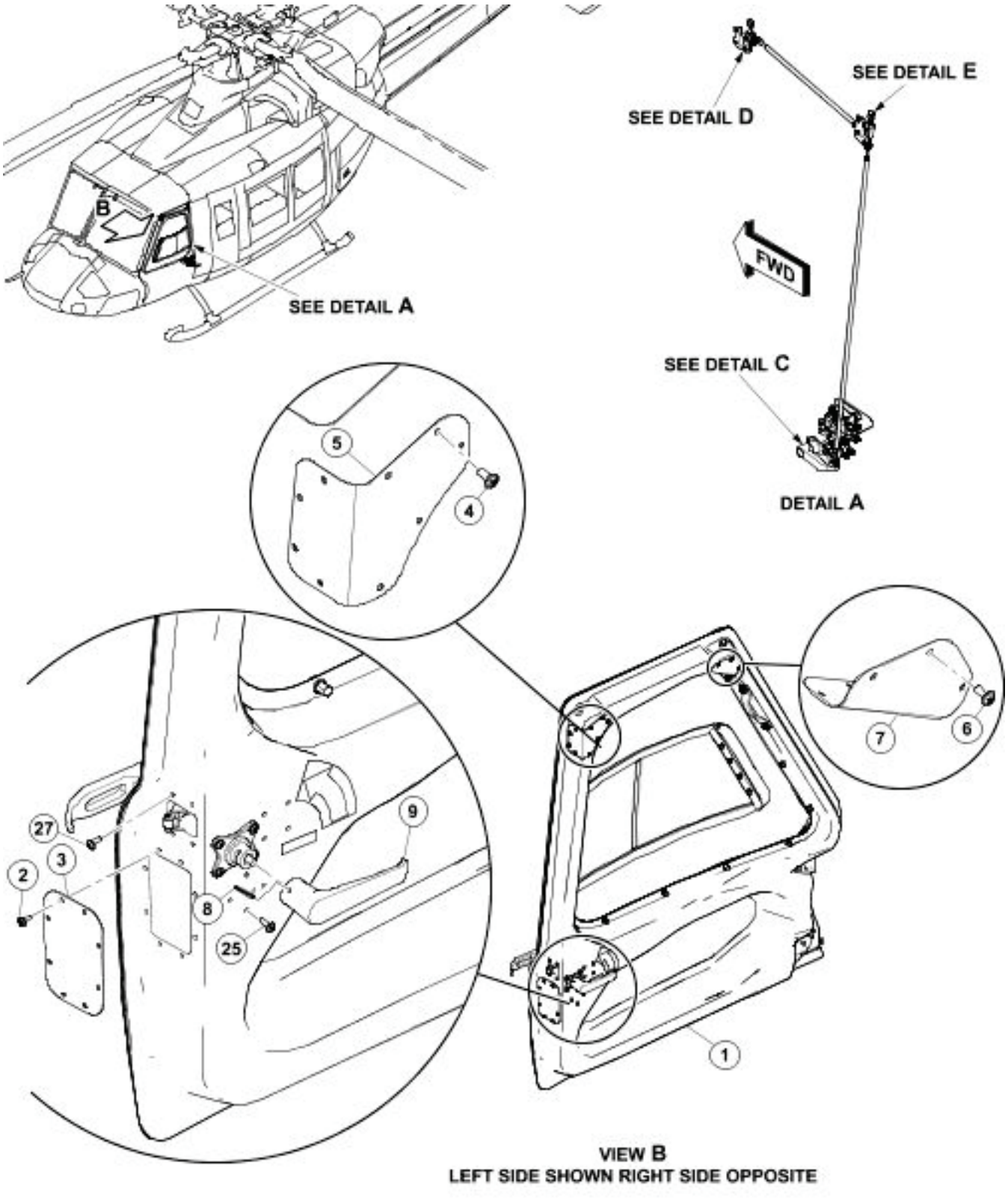


DETAIL E

1. Door
2. Hinge assembly, upper
3. Hinge assembly, fuselage
4. Bushing
5. Handle, door
6. Ejection handle
7. Hinge pin
8. Roller
9. Pin
10. Striker
11. Extrusion

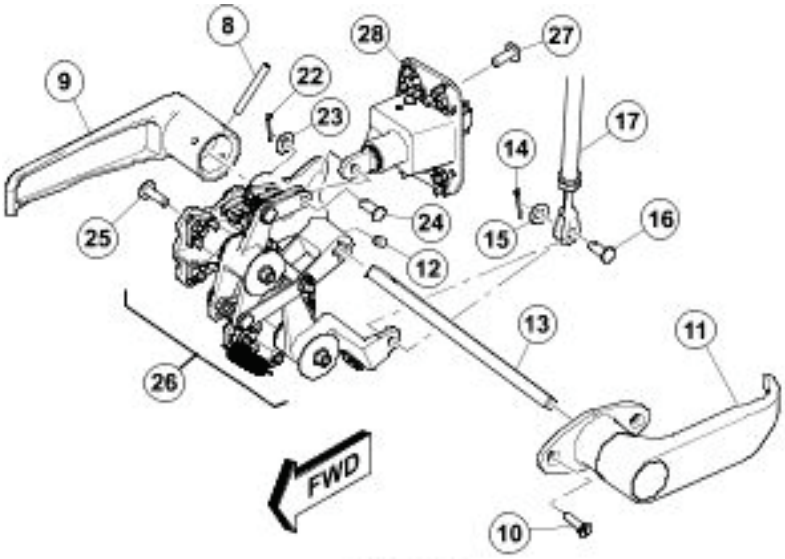
ICN-412-A-521100-B-97499-00002-A-001-01

Figure 1. Crew Doors (Sheet 2 of 2)

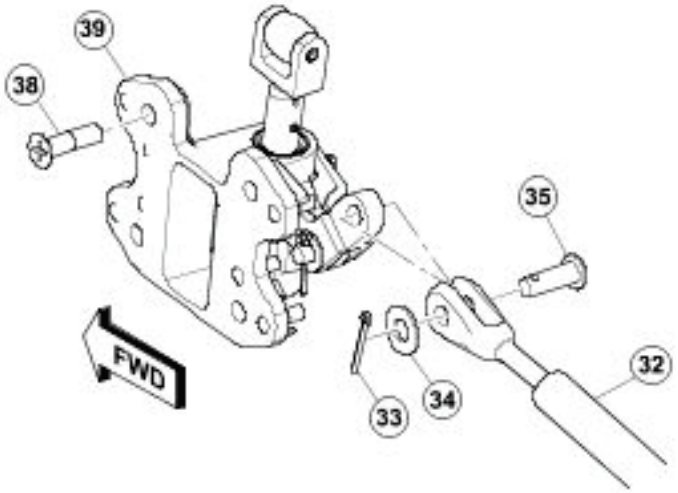


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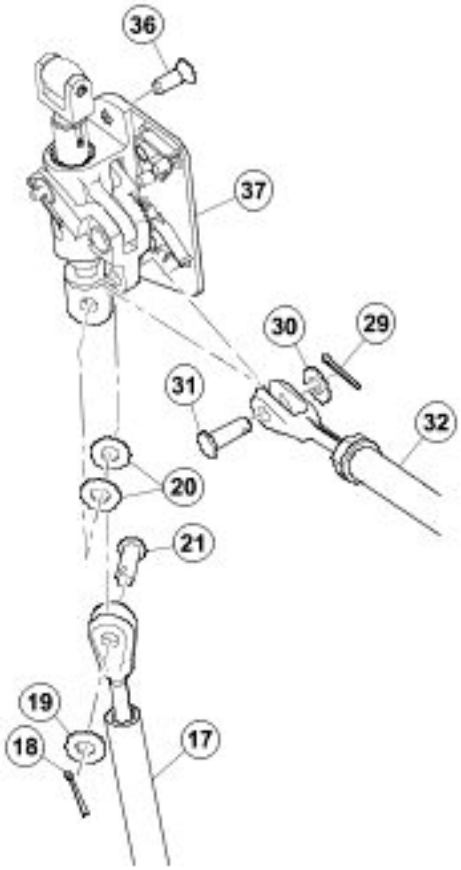
Figure 2. Crew Door Bell Crank Assembly and Plungers (Sheet 1 of 2)



DETAIL C



DETAIL D



DETAIL E

1. Door
2. Screw
3. Cover, aft Lower
4. Screw
5. Cover, aft upper
6. Screw
7. Cover, fwd upper
8. Pin
9. Handle, inner
10. Screw
11. Handle, outer
12. Set screw
13. Rod, handle
14. Cotter pin
15. Washer
16. Pin
17. Rod assembly, vertical
18. Cotter pin
19. Washer
20. Washer
21. Pin
22. Cotter pin
23. Washer
24. Pin
25. Screw
26. Bell crank assembly
27. Screw
28. Plunger assembly, aft lower
29. Cotter Pin
30. Washer
31. Pin
32. Rod assembly, horizontal
33. Cotter pin
34. Washer
35. Pin
36. Screw
37. Plunger assembly, upper aft
38. Screw
39. Plunger assembly, upper fwd

ICN-412-A-521100-B-97499-00005-A-001-01

Figure 2. Crew Door Bell Crank Assembly and Plungers (Sheet 2 of 2)

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Crew Door Jettison Mechanism - Description

*For the crew door jettison mechanism procedures refer to (*412-A-52-10-00-01A-520B-A / 00781.2 *and* 412-A-52-10-00-01A-720B-A / 00781.3*).*

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Crew Door Hinges - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	

Procedure

WARNING

SEALANT IS FLAMMABLE. DO NOT USE NEAR HEAT, OPEN FLAMES, OR SPARKS. OVEREXPOSURE CAN CAUSE IRRITATION OF SKIN AND EYES, HEADACHE, NAUSEA, VOMITING AND SYSTEMATIC PROBLEMS. THOROUGHLY WASH SKIN AREA WITH SOAP AND WATER AND IMMEDIATELY FLUSH EYES WITH WATER FOR 15 MINUTES. IN ALL CASES GET IMMEDIATE MEDICAL ATTENTION. WEAR APPROVED PROTECTIVE GLOVES AND GOGGLES. AVOID BREATHING OF VAPOURS AND PROLONGED OR REPEATED SKIN CONTACT.

NOTE

This procedure is provided for the hinge assemblies on the left door. The procedure is the same for the hinge assemblies on the right door.

NOTE

The procedure is provided for the upper hinge assembly and the procedure for the lower hinge assembly is same.

1. Remove the crew door (412-A-52-11-00-01A-520A-A / 00793).

2. Carefully remove the sealant from around the edges of the hinge assembly (2, Figure 1).

3. Remove the nut (14), washer (13) and the bolt (12) from the spring assembly (15) and arm assembly (3).

3.1. Remove the spring assembly (15) from the arm assembly (3).

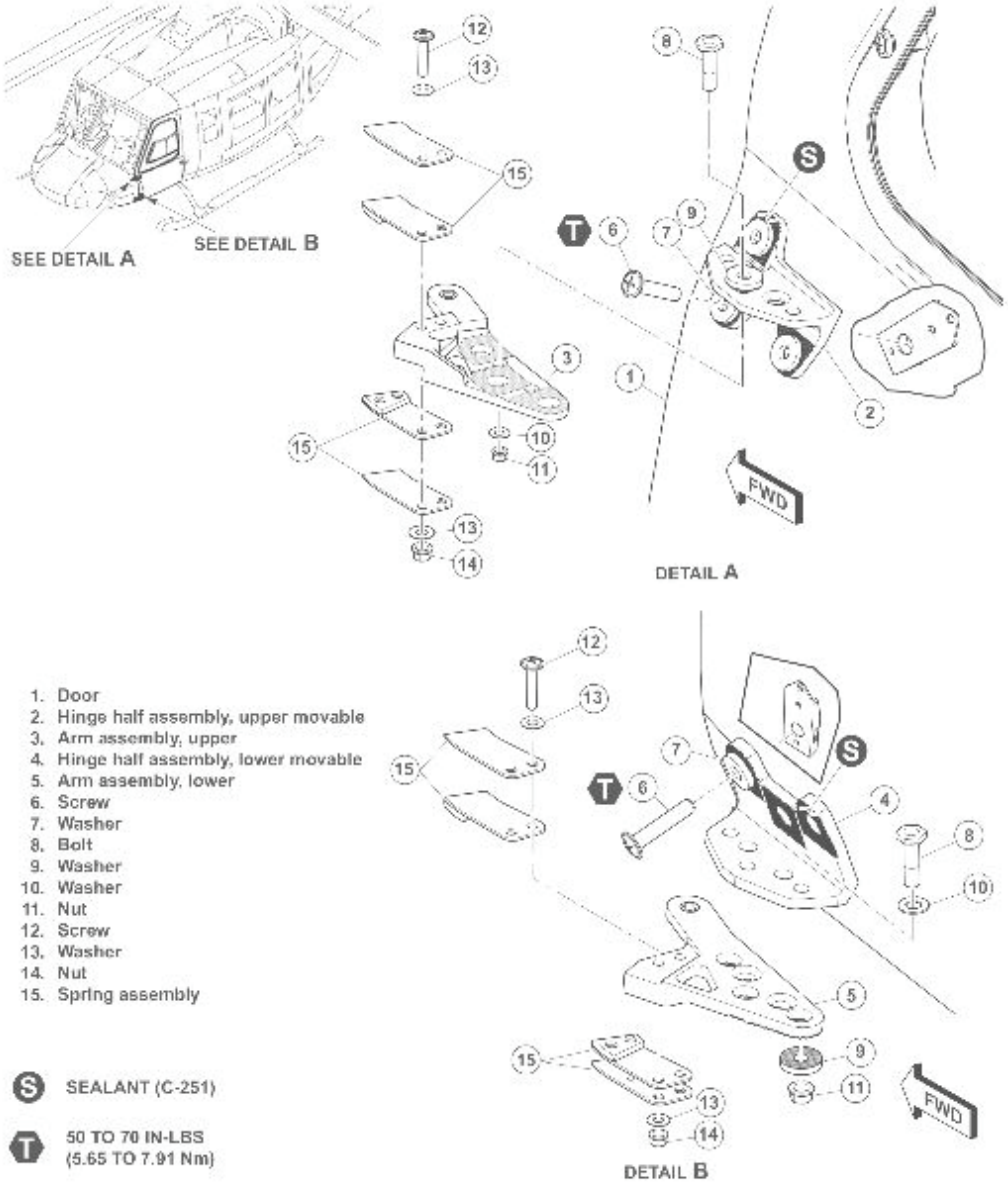
4. Remove the nut (11), washers (9 and 10) and the bolt (8) from the hinge assembly (2) and the arm assembly (3).

4.1. Remove the arm assembly (3).

5. Remove the screws (6) and washer (7) from the hinge assembly (2) and the door (1).

6. Remove the hinge assembly (2) from the door (1).

7. Carefully remove the remaining sealant with aliphatic naphtha (C-305).



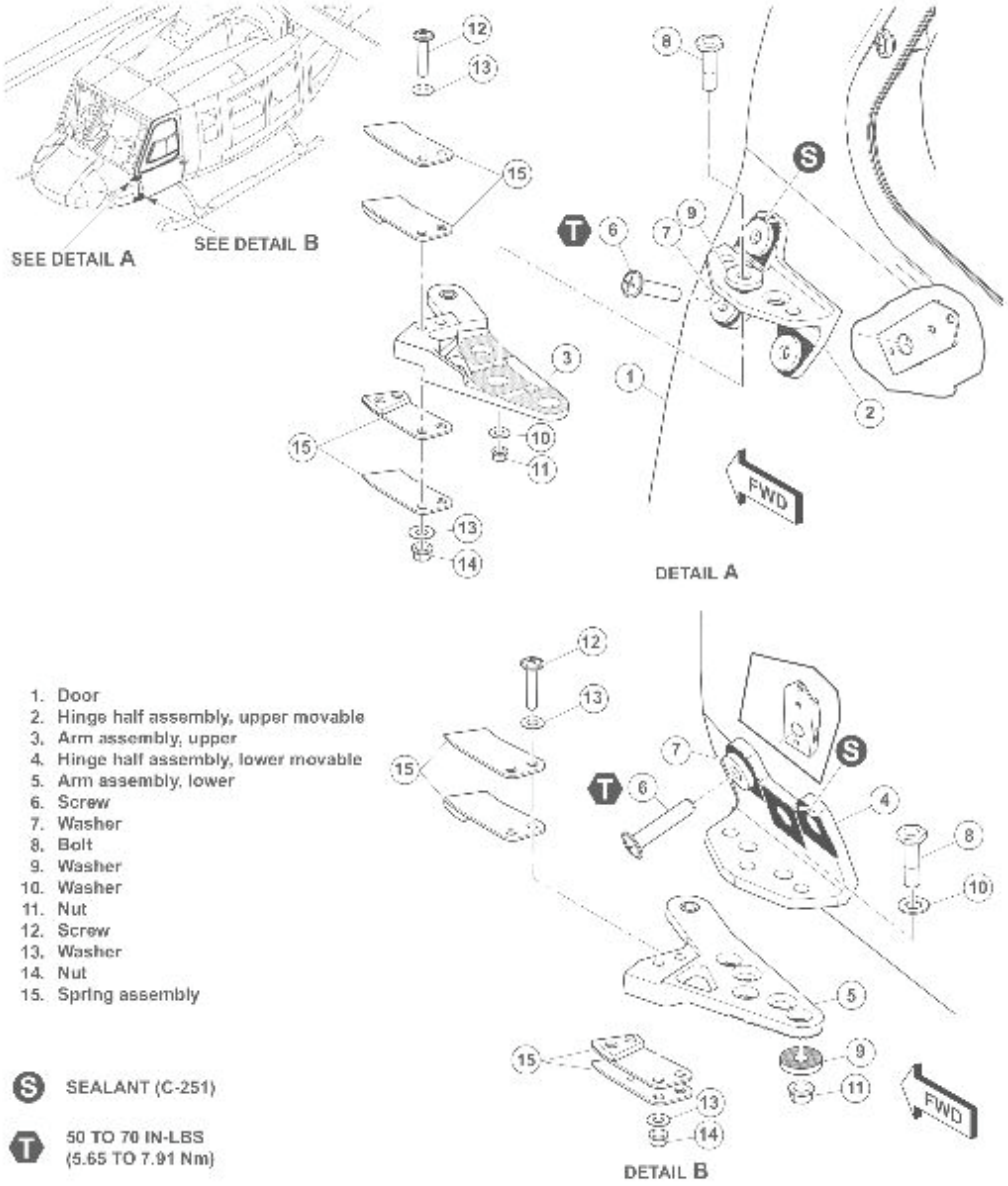
ICN-412-A-521100-B-97499-00003-A-001-01

Figure 1. Crew Door Hinges (Sheet 1 of 1)

Crew Door Hinges - Inspection

Procedure

1. Inspect the applicable movable hinge half assembly (2 or 4, Figure 1) for cracks, wear, and holes with signs of deformation.
2. Inspect the applicable arm assembly (3 and 5) for cracks, wear and the holes signs of deformation.
3. Inspect the bushings for excessive wear or play.
4. Inspect the spring assemblies (15) for excessive wear to the striker blocks.
5. Replace the spring assemblies (15) if the rivets of the striker blocks are visible.



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Figure 1. Crew Door Hinges (Sheet 1 of 1)

Crew Door Hinges - Repair

Procedure

1. For crew door hinges repair refer to the 412-A-52-10-00-03A-520A-A / 00788, 412-A-52-10-00-03A-280A-A / 00789, 412-A-52-10-00-03A-720A-A / 00790 and 412-A-52-10-00-03A-685A-A / 00791.

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Crew Door Hinges - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-251	AR	
Aliphatic naphtha	C-305	AR	

Procedure

NOTE

The procedure is provided for the upper hinge assembly and the procedure for the lower hinge assembly is same.

NOTE

This procedure is provided for the hinge assembly on the left door. The procedure is the same for the hinge assembly on the right door.

1. Put the hinge assembly (2, Figure 1) in its position.

2. Install the washers (7) and the screws (6).

3. Torque the screws 50 to 70 inch-pounds (5.65 to 7.91 Nm).

4. Put the arm assembly (3) in its position on the hinge assembly (2).

5. Install the bolt (8), washers (9 and 10) and the nut (11).

6. Put the spring assembly (15) in its position on the arm assembly (3).

7. Install the bolt (12), washer (13) and the nut (14).

8. Temporarily install the crew door (412-A-52-11-00-01A-720A-A / 00796).

9. Make sure that the fit is correct.

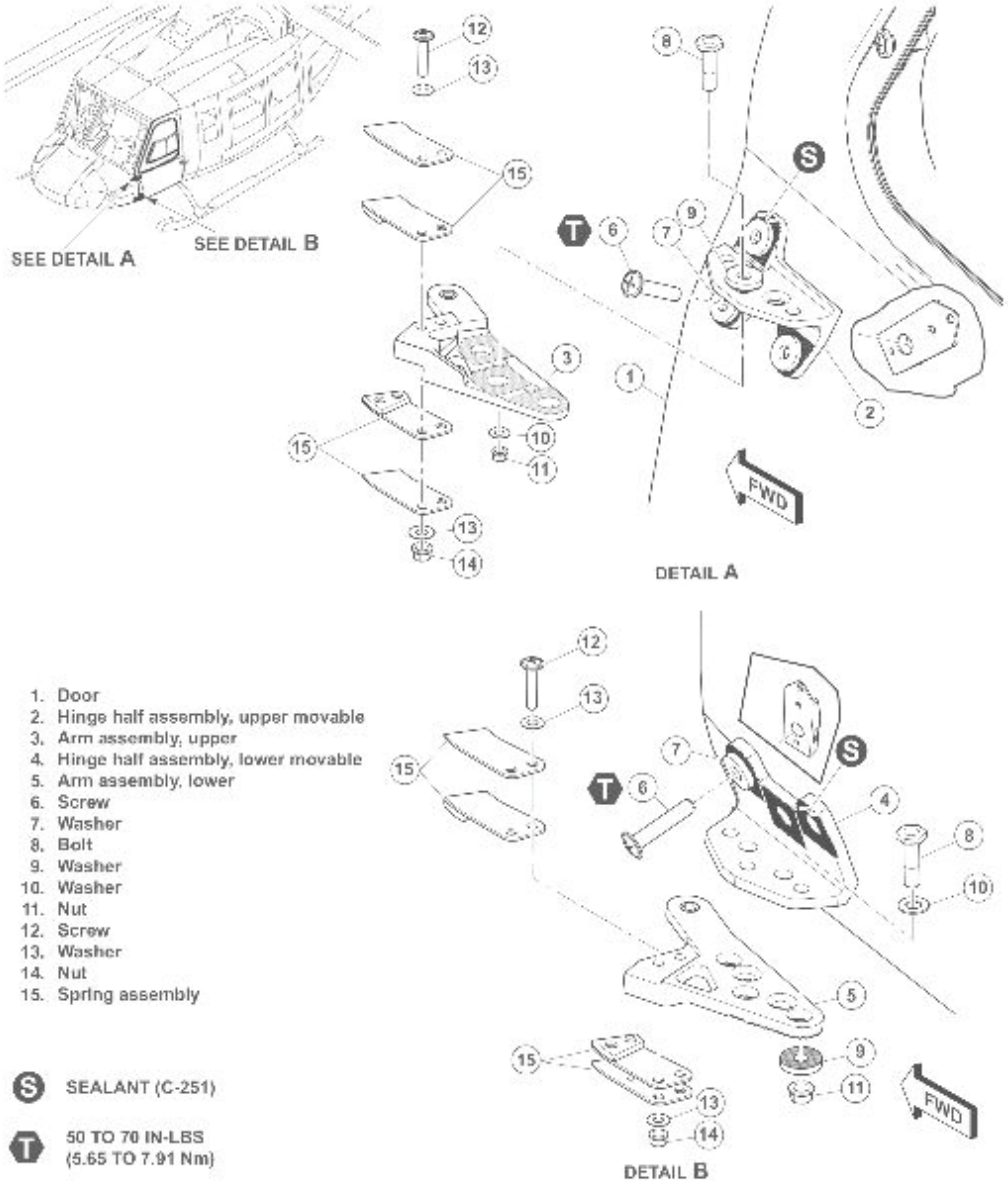
10. If necessary, adjust the fuselage hinge assembly (3) to get the correct fit.

11. Remove the crew door (412-A-52-11-00-01A-520A-A / 00793).

12. Apply sealant (C-251) around the edge of the hinge assembly (2).

13. Clean the excess sealant with aliphatic naphtha (C-305).

14. Install the crew door (412-A-52-11-00-01A-720A-A / 00796).



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Figure 1. Crew Door Hinges (Sheet 1 of 1)

Crew Door Cover - Removal

Procedure

NOTE

The procedure is for the crew door on the left side. The procedure is the same for the crew door on the right side.

1. Remove the lower aft cover (3, Figure 1) as follows:

1.1. Remove the screws (2).

1.2. Remove the cover (3) from the door (1).

2. Remove the upper aft cover (5) as follows:

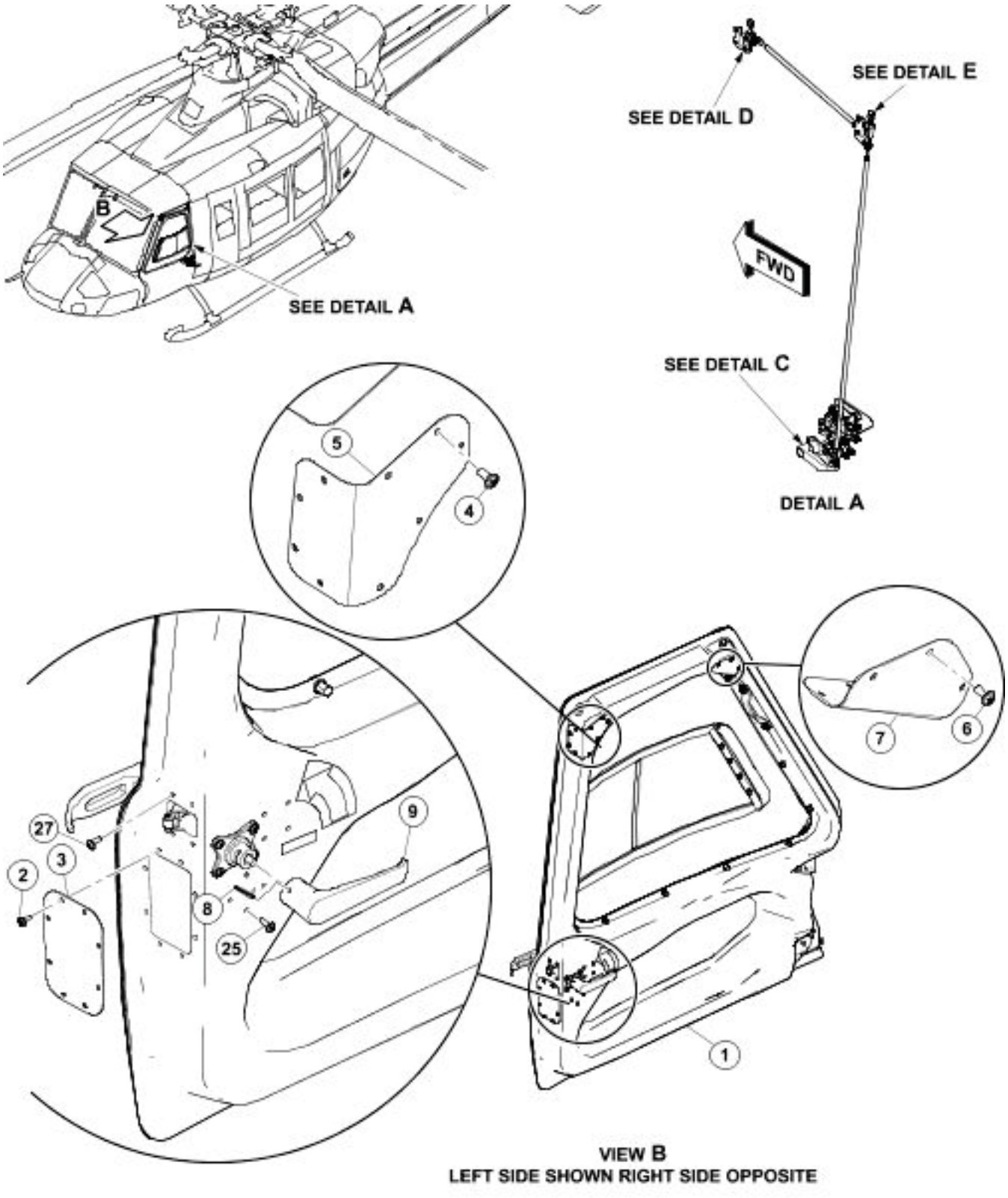
2.1. Remove the screws (4).

2.2. Remove the cover (5) from the door (1).

3. Remove the upper forward cover (7) as follows:

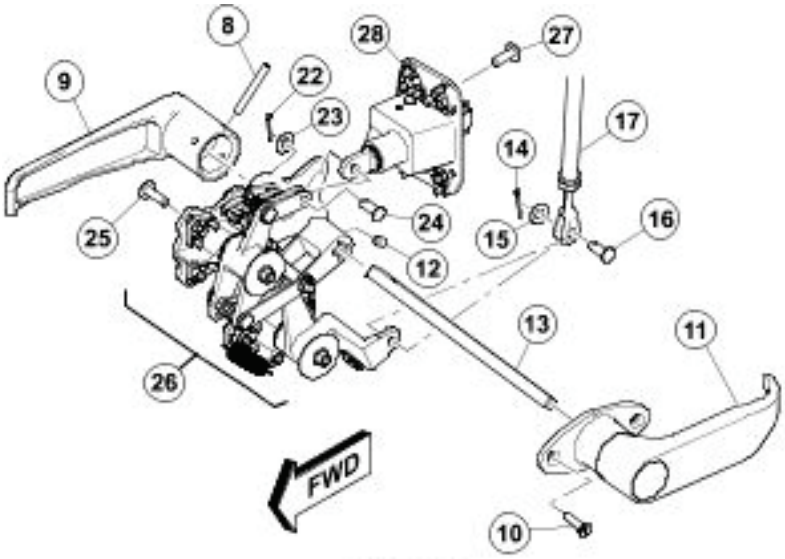
3.1. Remove the screws (6).

3.2. Remove the cover (7) from the door (1).

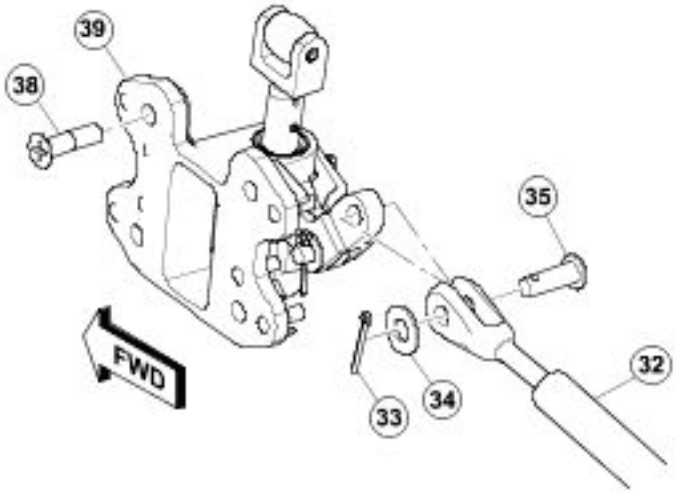


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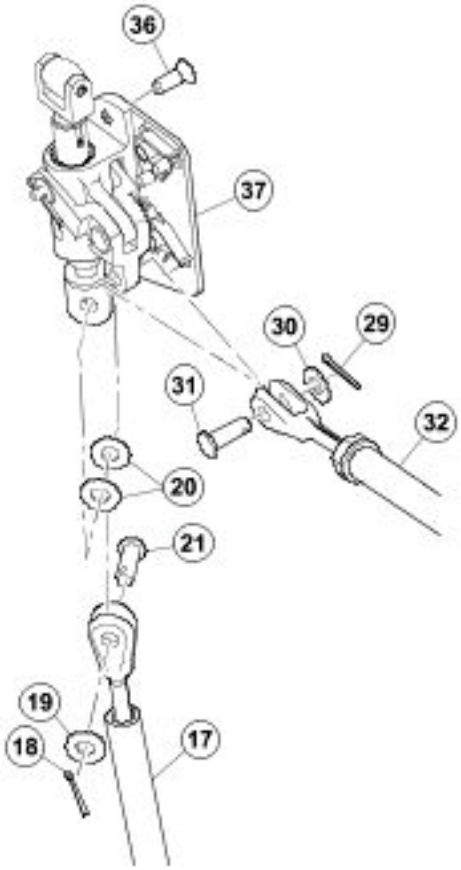
Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 1 of 2)



DETAIL C



DETAIL D



DETAIL E

1. Door
2. Screw
3. Cover, aft Lower
4. Screw
5. Cover, aft upper
6. Screw
7. Cover, fwd upper
8. Pin
9. Handle, inner
10. Screw
11. Handle, outer
12. Set screw
13. Rod, handle
14. Cotter pin
15. Washer
16. Pin
17. Rod assembly, vertical
18. Cotter pin
19. Washer
20. Washer
21. Pin
22. Cotter pin
23. Washer
24. Pin
25. Screw
26. Bell crank assembly
27. Screw
28. Plunger assembly, aft lower
29. Cotter Pin
30. Washer
31. Pin
32. Rod assembly, horizontal
33. Cotter pin
34. Washer
35. Pin
36. Screw
37. Plunger assembly, upper aft
38. Screw
39. Plunger assembly, upper fwd

ICN-412-A-521100-B-97499-00005-A-001-01

Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 2 of 2)

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Crew Door Cover - Installation

Procedure

NOTE

The procedure is for the crew door on the left side. The procedure is the same for the crew door on the right side.

1. Install the lower aft cover (3, Figure 1) as follows:

1.1. Put the cover (3) in its position on the door (1).

1.2. Install the screws (2).

2. Install the upper aft cover (5) as follows:

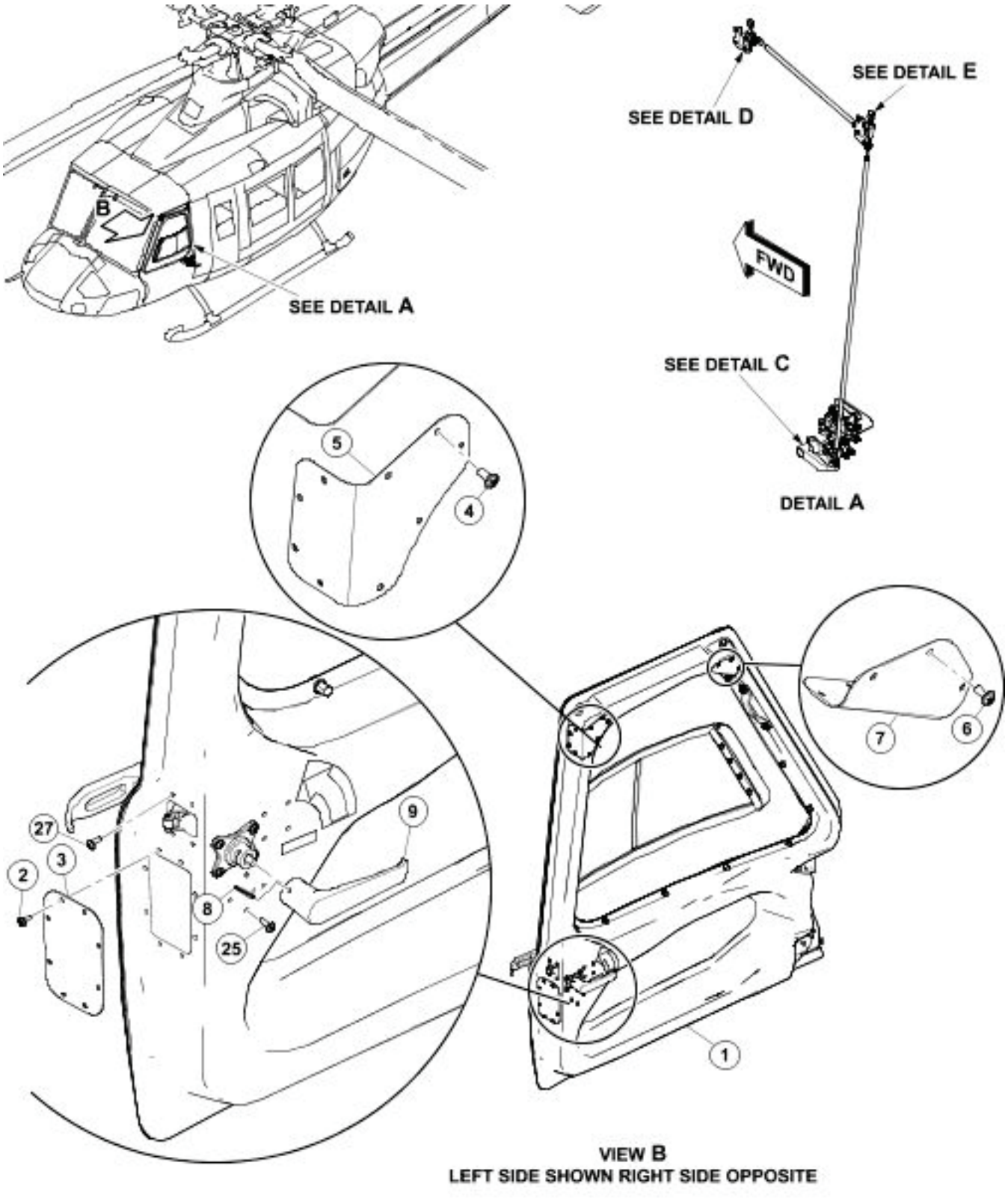
2.1. Put the cover (5) in its position on the door (1).

2.2. Install the screws (4).

3. Install the upper forward cover (7) as follows:

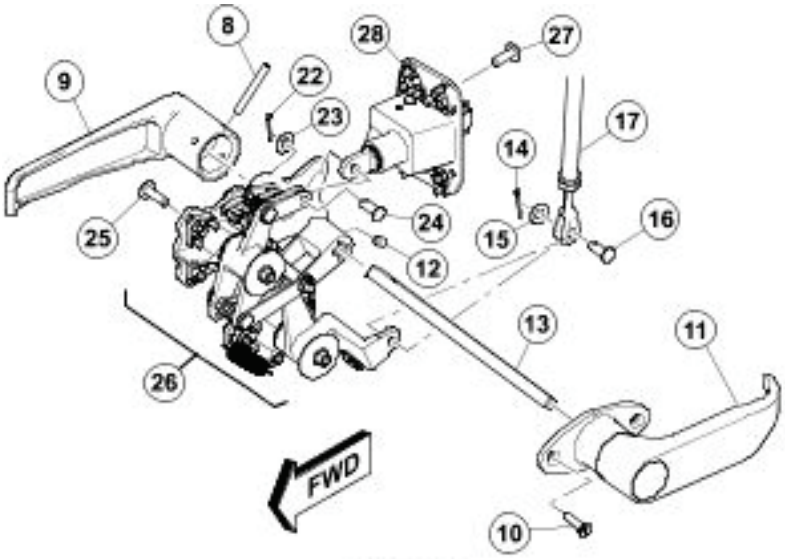
3.1. Put the cover (7) in its position on the door (1).

3.2. Install the screws (6).

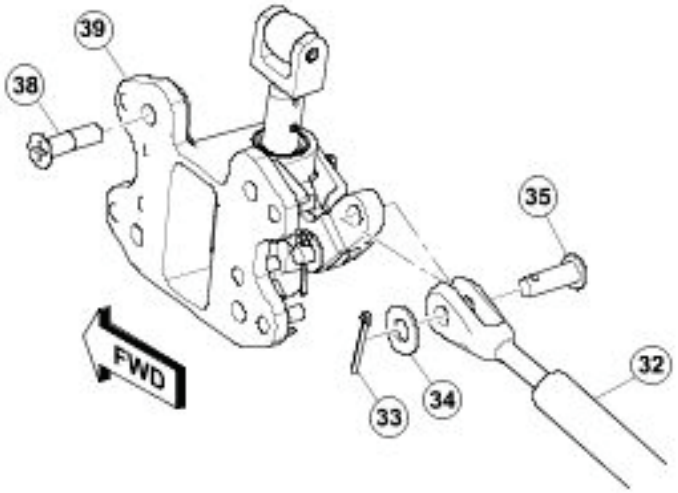


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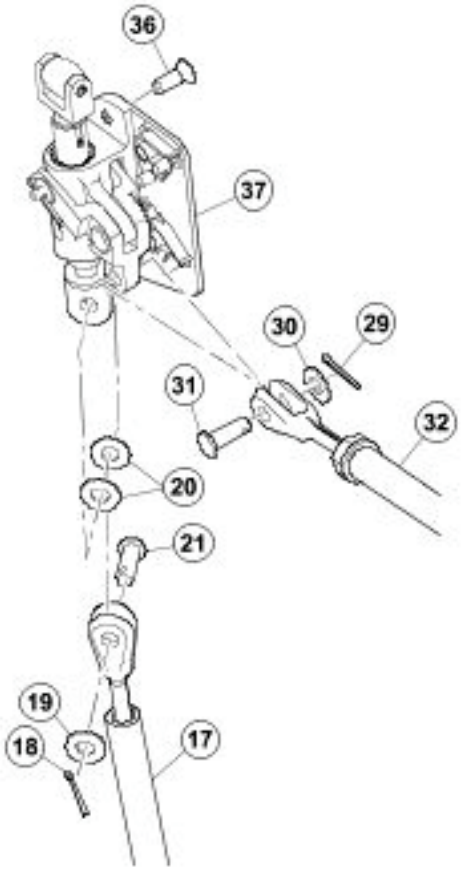
Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 1 of 2)



DETAIL C



DETAIL D



DETAIL E

1. Door
2. Screw
3. Cover, aft Lower
4. Screw
5. Cover, aft upper
6. Screw
7. Cover, fwd upper
8. Pin
9. Handle, inner
10. Screw
11. Handle, outer
12. Set screw
13. Rod, handle
14. Cotter pin
15. Washer
16. Pin
17. Rod assembly, vertical
18. Cotter pin
19. Washer
20. Washer
21. Pin
22. Cotter pin
23. Washer
24. Pin
25. Screw
26. Bell crank assembly
27. Screw
28. Plunger assembly, aft lower
29. Cotter Pin
30. Washer
31. Pin
32. Rod assembly, horizontal
33. Cotter pin
34. Washer
35. Pin
36. Screw
37. Plunger assembly, upper aft
38. Screw
39. Plunger assembly, upper fwd

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Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 2 of 2)

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Crew Door Handle - Removal

Procedure

NOTE

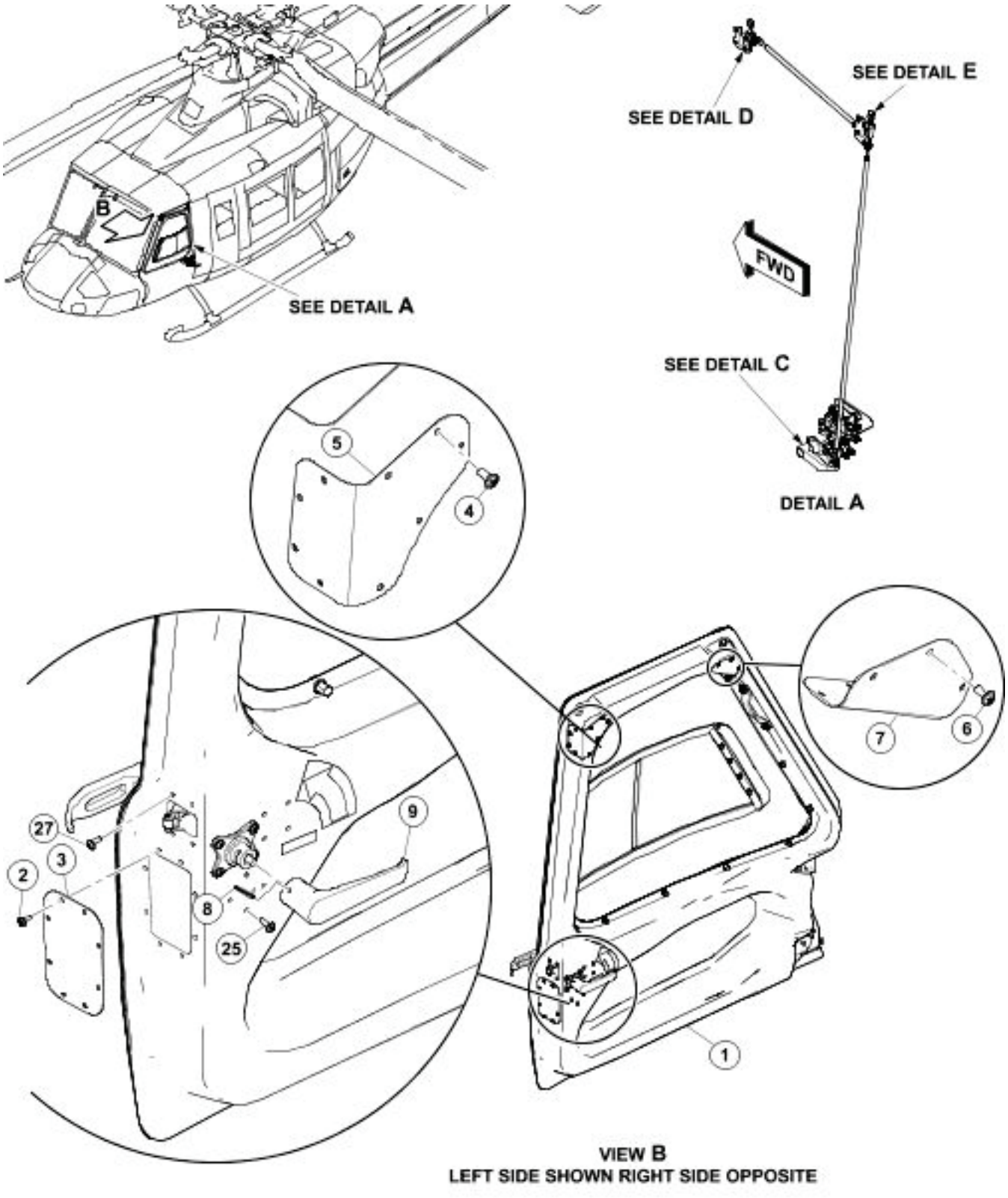
This procedure is for the left crew door. The procedure is same for the right crew door.

1. Remove the screws (10, Figure 1) from the outer handle (11).

2. Remove the outer handle (11).

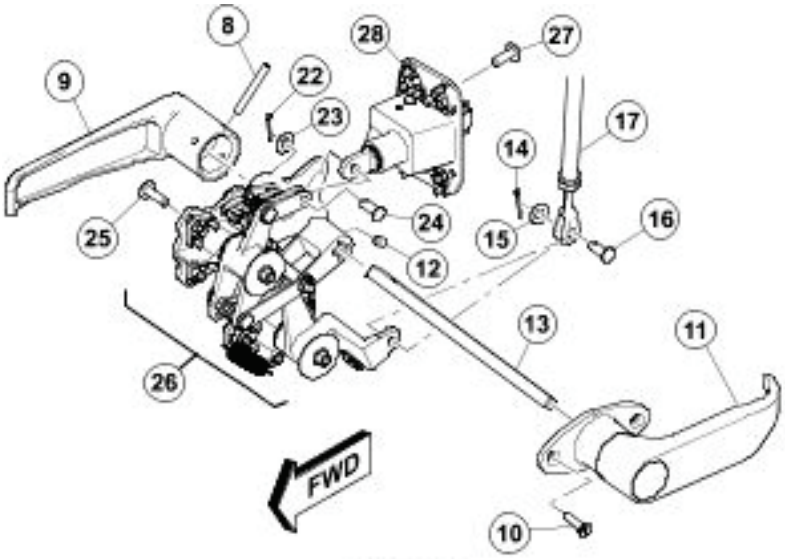
3. Remove the pin (8) from the inner handle (9).

4. Remove the inner handle (9).

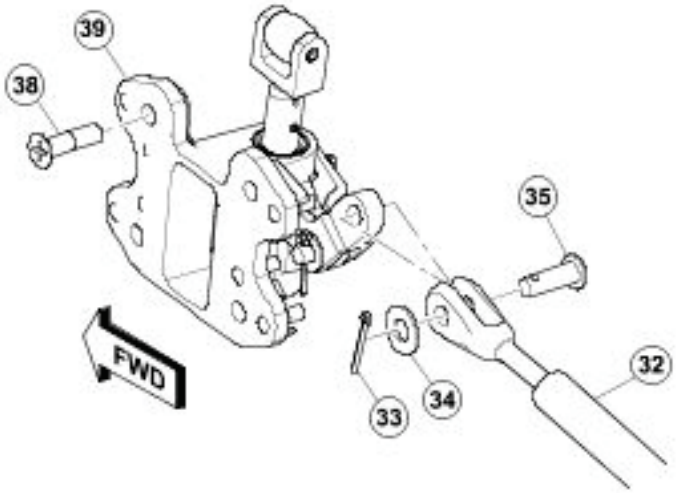


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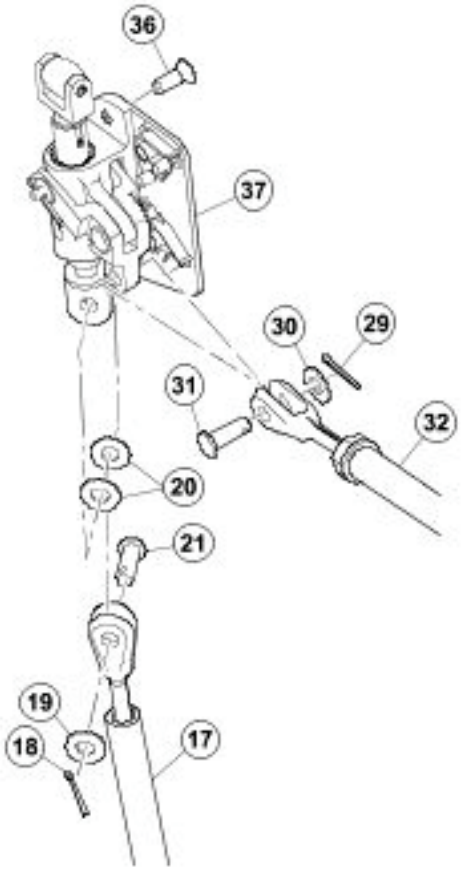
Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 1 of 2)



DETAIL C



DETAIL D



DETAIL E

1. Door
2. Screw
3. Cover, aft Lower
4. Screw
5. Cover, aft upper
6. Screw
7. Cover, fwd upper
8. Pin
9. Handle, inner
10. Screw
11. Handle, outer
12. Set screw
13. Rod, handle
14. Cotter pin
15. Washer
16. Pin
17. Rod assembly, vertical
18. Cotter pin
19. Washer
20. Washer
21. Pin
22. Cotter pin
23. Washer
24. Pin
25. Screw
26. Bell crank assembly
27. Screw
28. Plunger assembly, aft lower
29. Cotter Pin
30. Washer
31. Pin
32. Rod assembly, horizontal
33. Cotter pin
34. Washer
35. Pin
36. Screw
37. Plunger assembly, upper aft
38. Screw
39. Plunger assembly, upper fwd

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Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 2 of 2)

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Crew Door Handle - Installation

Procedure

NOTE

This procedure is for the left crew door. The procedure is same for the right crew door.

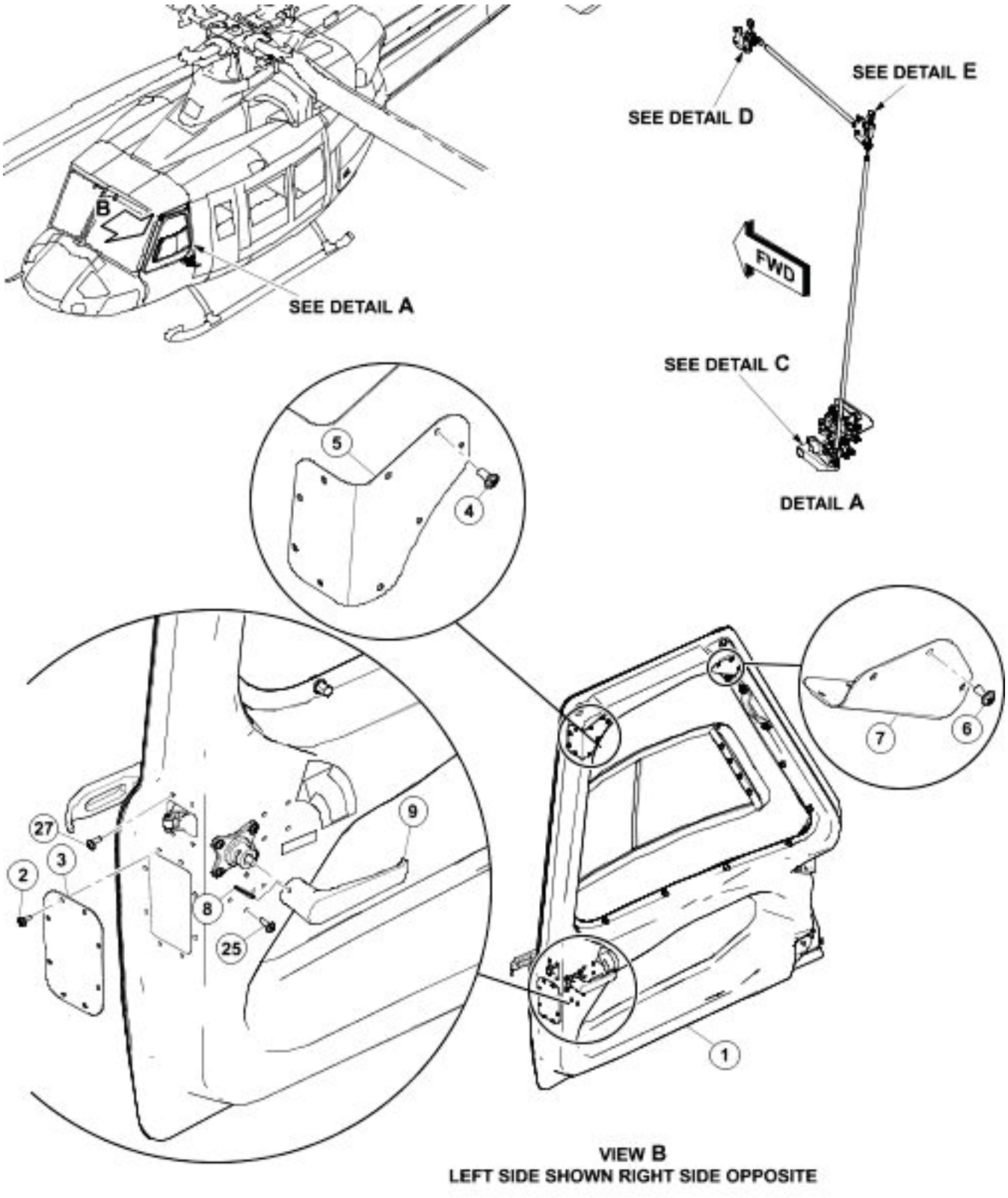
1. Put the handle (11, Figure 1) in its position on the door (1).

2. Install the screws (10).

3. Put the inner handle (9) in its position.

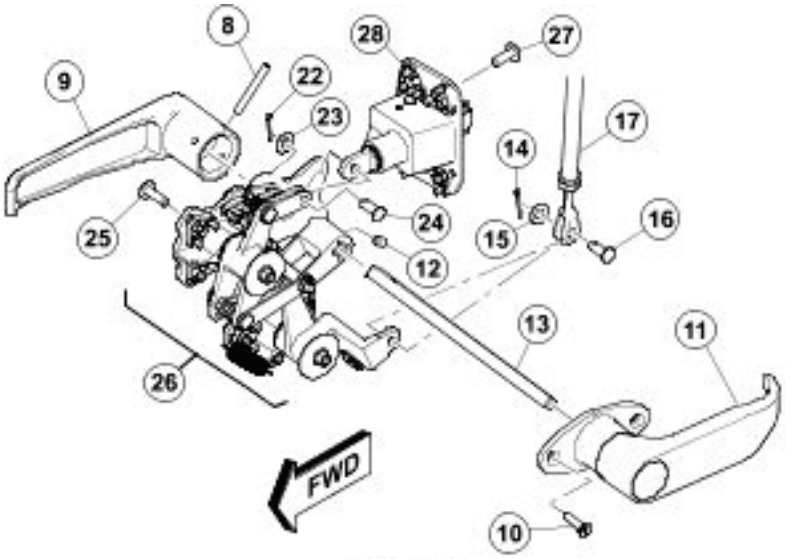
4. Install the pin (8).

5. Make sure that the inner and outer handles (9 and 11) operate correctly.

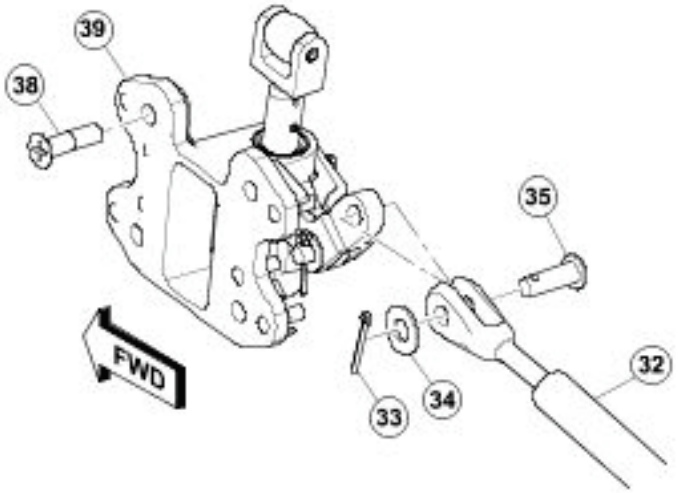


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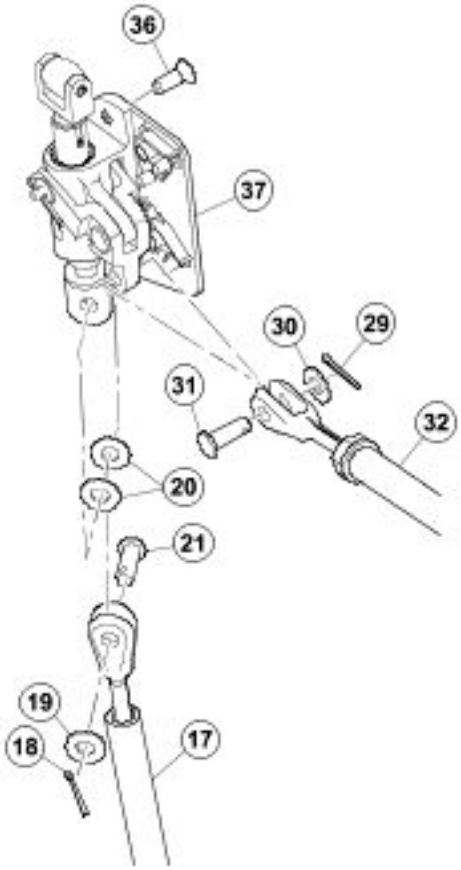
Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 1 of 2)



DETAIL C



DETAIL D



DETAIL E

1. Door
2. Screw
3. Cover, aft Lower
4. Screw
5. Cover, aft upper
6. Screw
7. Cover, fwd upper
8. Pin
9. Handle, inner
10. Screw
11. Handle, outer
12. Set screw
13. Rod, handle
14. Cotter pin
15. Washer
16. Pin
17. Rod assembly, vertical
18. Cotter pin
19. Washer
20. Washer
21. Pin
22. Cotter pin
23. Washer
24. Pin
25. Screw
26. Bell crank assembly
27. Screw
28. Plunger assembly, aft lower
29. Cotter Pin
30. Washer
31. Pin
32. Rod assembly, horizontal
33. Cotter pin
34. Washer
35. Pin
36. Screw
37. Plunger assembly, upper aft
38. Screw
39. Plunger assembly, upper fwd

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Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 2 of 2)

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Bell Crank Assembly - Removal

Procedure

NOTE

The procedure is for the crew door on the left side. The procedure is the same for the crew door on the right side.

1. Remove the covers (Figure 1) (412-A-52-11-00-03A-520A-A / 00802).

2. Remove the crew door handles (412-A-52-11-00-04A-520A-A / 00804).

3. Remove the vertical rod assembly (17) as follows:

3.1. Remove the cotter pin (14) from the pin (16).

3.2. Remove the washer (15) and pin (16).

3.3. Remove the cotter pin (18) from the pin (21).

3.4. Remove the washers (19 and 20) and pin (21) from the plunger (37).

3.5. Carefully remove the vertical rod assembly (17) thru the cover area.

4. Remove the bell crank assembly (26) as follows:

4.1. Remove the crew door handles (9 and 11) (412-A-52-11-00-04A-520A-A / 00804).

4.2. Remove the set screw(12).

4.3. Lightly tap the handle rod (13) with a hammer.

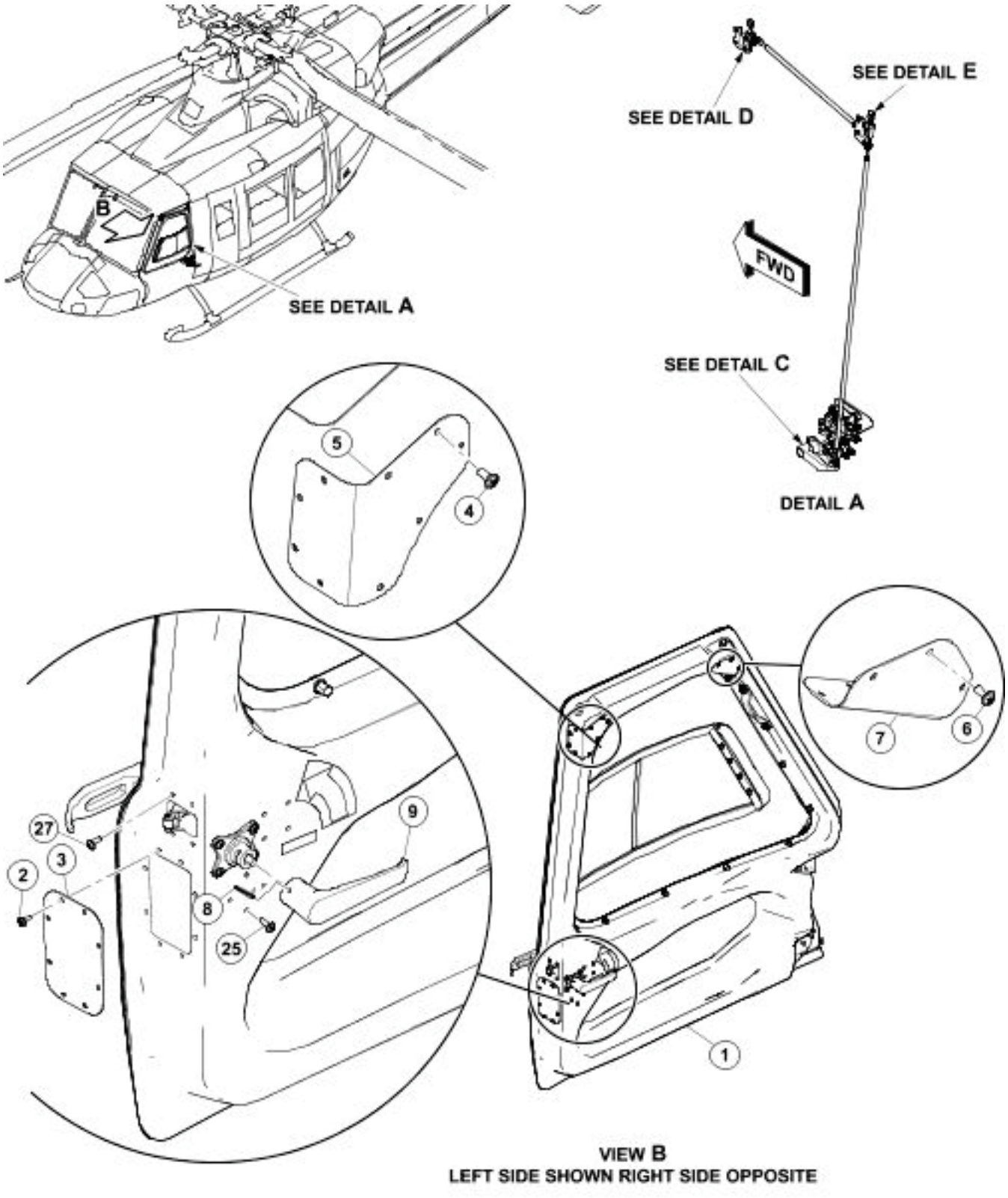
4.4. Remove the handle rod (13) from the bell crank lever arm.

4.5. Remove the cotter pin (22) and washer (23) from the pin (24).

4.6. Remove the pin (24) from the plunger assembly (28) and the bell crank assembly (26).

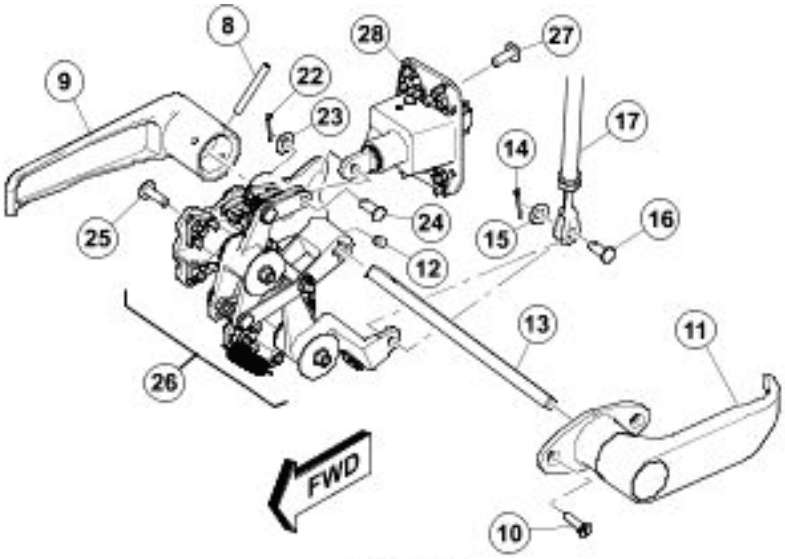
4.7. Remove the screws (25).

4.8. Remove the bell crank assembly (26).

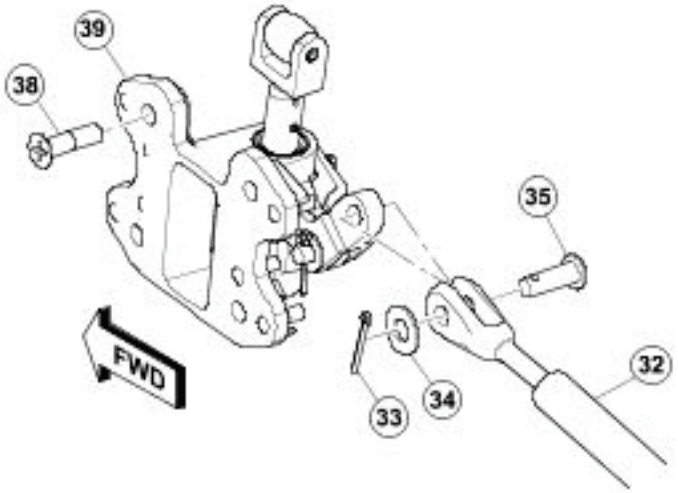


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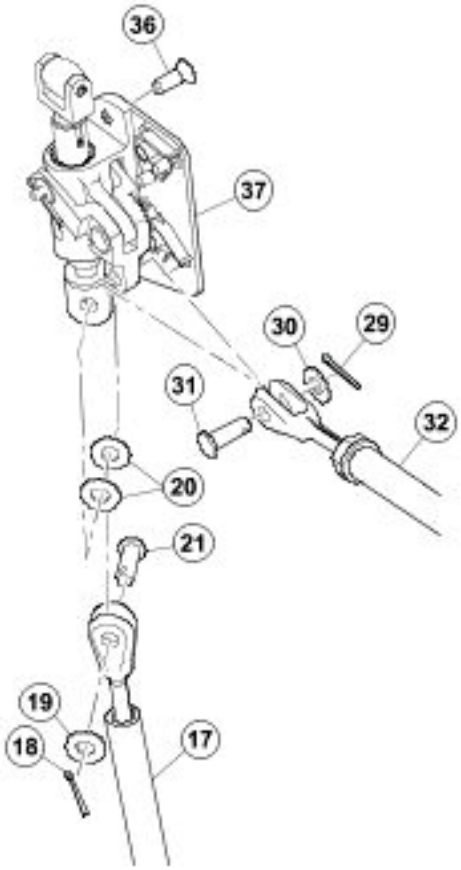
Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 1 of 2)



DETAIL C



DETAIL D



DETAIL E

1. Door
2. Screw
3. Cover, aft Lower
4. Screw
5. Cover, aft upper
6. Screw
7. Cover, fwd upper
8. Pin
9. Handle, inner
10. Screw
11. Handle, outer
12. Set screw
13. Rod, handle
14. Cotter pin
15. Washer
16. Pin
17. Rod assembly, vertical
18. Cotter pin
19. Washer
20. Washer
21. Pin
22. Cotter pin
23. Washer
24. Pin
25. Screw
26. Bell crank assembly
27. Screw
28. Plunger assembly, aft lower
29. Cotter Pin
30. Washer
31. Pin
32. Rod assembly, horizontal
33. Cotter pin
34. Washer
35. Pin
36. Screw
37. Plunger assembly, upper aft
38. Screw
39. Plunger assembly, upper fwd

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Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 2 of 2)

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Bell Crank Assembly - Installation

Procedure

NOTE

The procedure is for the crew door on the left side. The procedure is the same for the crew door on the right side.

1. Put the bell crank assembly (26, Figure 1) in its position.

2. Install the screws (25).

3. Install the pin (24) and washer (23).

4. Install the cotter pin (22).

5. Install the handle rod (13) thru the opening on the door and the bell crank lever arm.

6. Install the vertical rod assembly (17) as follows:

6.1. Put the vertical rod assembly (17) in its position.

6.2. Install the pin (21) and washers (19 and 20).

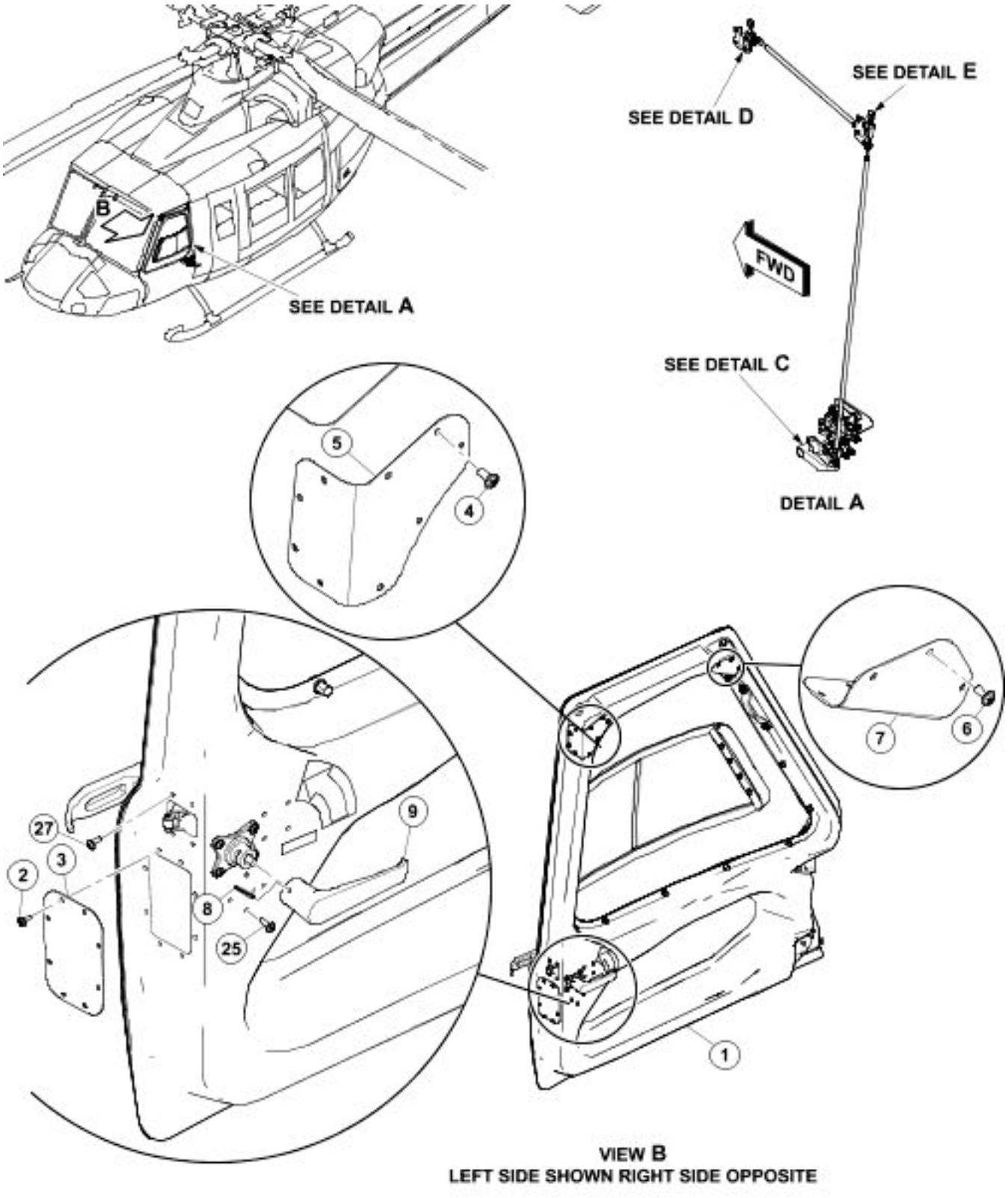
6.3. Install the cotter pin (18).

6.4. Install the pin (16) and washer (15).

6.5. Install the cotter pin (14).

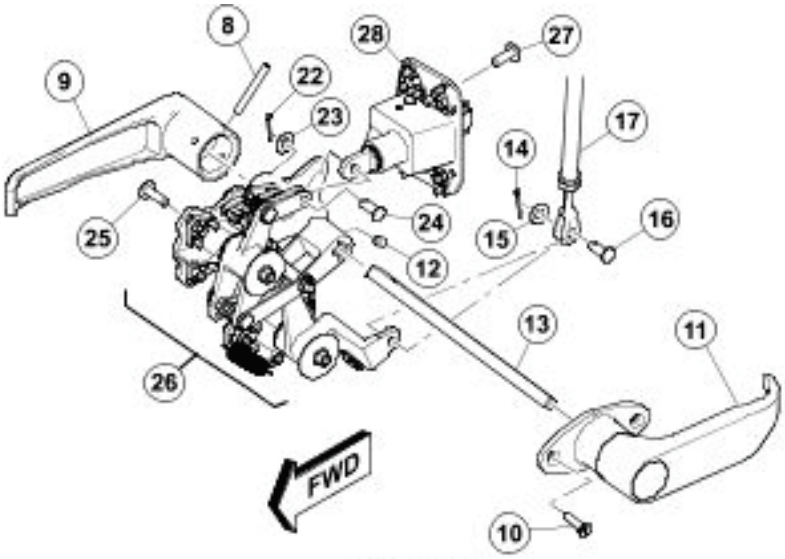
7. Install the crew door handles (412-A-52-11-00-04A-720A-A / 00805).

8. Install the covers (412-A-52-11-00-03A-720A-A / 00803).

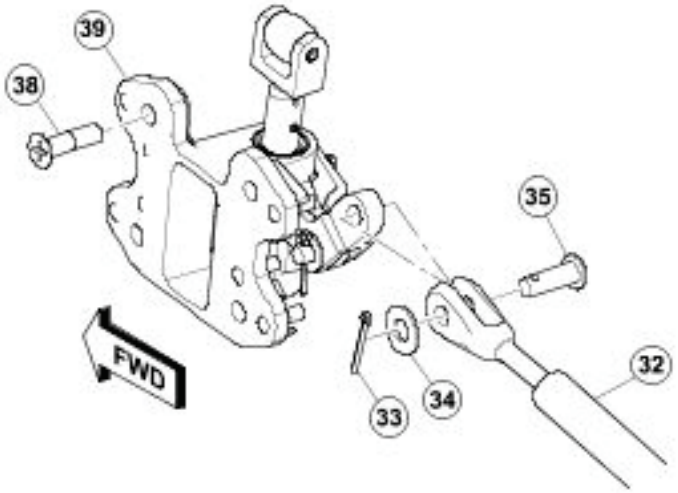


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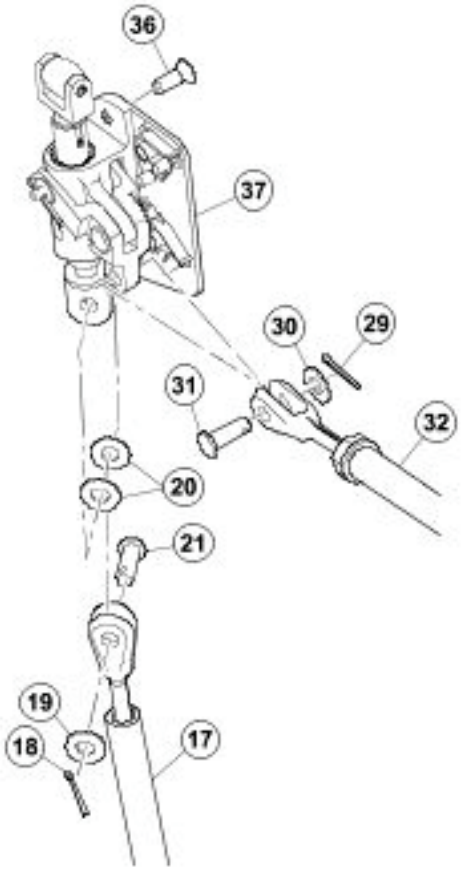
Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 1 of 2)



DETAIL C



DETAIL D



DETAIL E

1. Door
2. Screw
3. Cover, aft Lower
4. Screw
5. Cover, aft upper
6. Screw
7. Cover, fwd upper
8. Pin
9. Handle, inner
10. Screw
11. Handle, outer
12. Set screw
13. Rod, handle
14. Cotter pin
15. Washer
16. Pin
17. Rod assembly, vertical
18. Cotter pin
19. Washer
20. Washer
21. Pin
22. Cotter pin
23. Washer
24. Pin
25. Screw
26. Bell crank assembly
27. Screw
28. Plunger assembly, aft lower
29. Cotter Pin
30. Washer
31. Pin
32. Rod assembly, horizontal
33. Cotter pin
34. Washer
35. Pin
36. Screw
37. Plunger assembly, upper aft
38. Screw
39. Plunger assembly, upper fwd

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Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 2 of 2)

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Plunger Assembly - Removal

Procedure

NOTE

The procedure is provided for the left door. The procedure is same for the right door.

1. Remove the covers (Figure 1) (412-A-52-11-00-03A-520A-A / 00802).

2. Remove the vertical rod assembly (17) as follows:

2.1. Remove the cotter pin (14) from the pin (16).

2.2. Remove the washer (15) and pin (16).

2.3. Remove the cotter pin (18) from the pin (21).

2.4. Remove the washers (19 and 20) and pin (21) from the plunger assembly (37).

2.5. Carefully remove the vertical rod assembly (17) thru the cover area.

3. Remove the lower aft plunger assembly as follows:

3.1. Remove the cotter pin (22) and washer (23) from the pin (24).

3.2. Remove the pin (24) from the plunger assembly (28) and the bell crank assembly (26).

3.3. Remove the screws (27).

3.4. Remove the lower aft plunger assembly (28) thru the cover area.

4. Remove the horizontal rod assembly (17) from the plunger assembly (37 and 39) as follows:

4.1. Remove the cotter pin (29) from the pin (31).

4.2. Remove the washer (30) and pin (31).

4.3. Remove the cotter pin (33) from the pin (35).

4.4. Remove the washer (34) and pin (35) from the plunger assembly (39).

4.5. Carefully remove the horizontal rod assembly (32) thru the cover area.

5. Remove the upper aft plunger assembly (37) from the door (1) as follows:

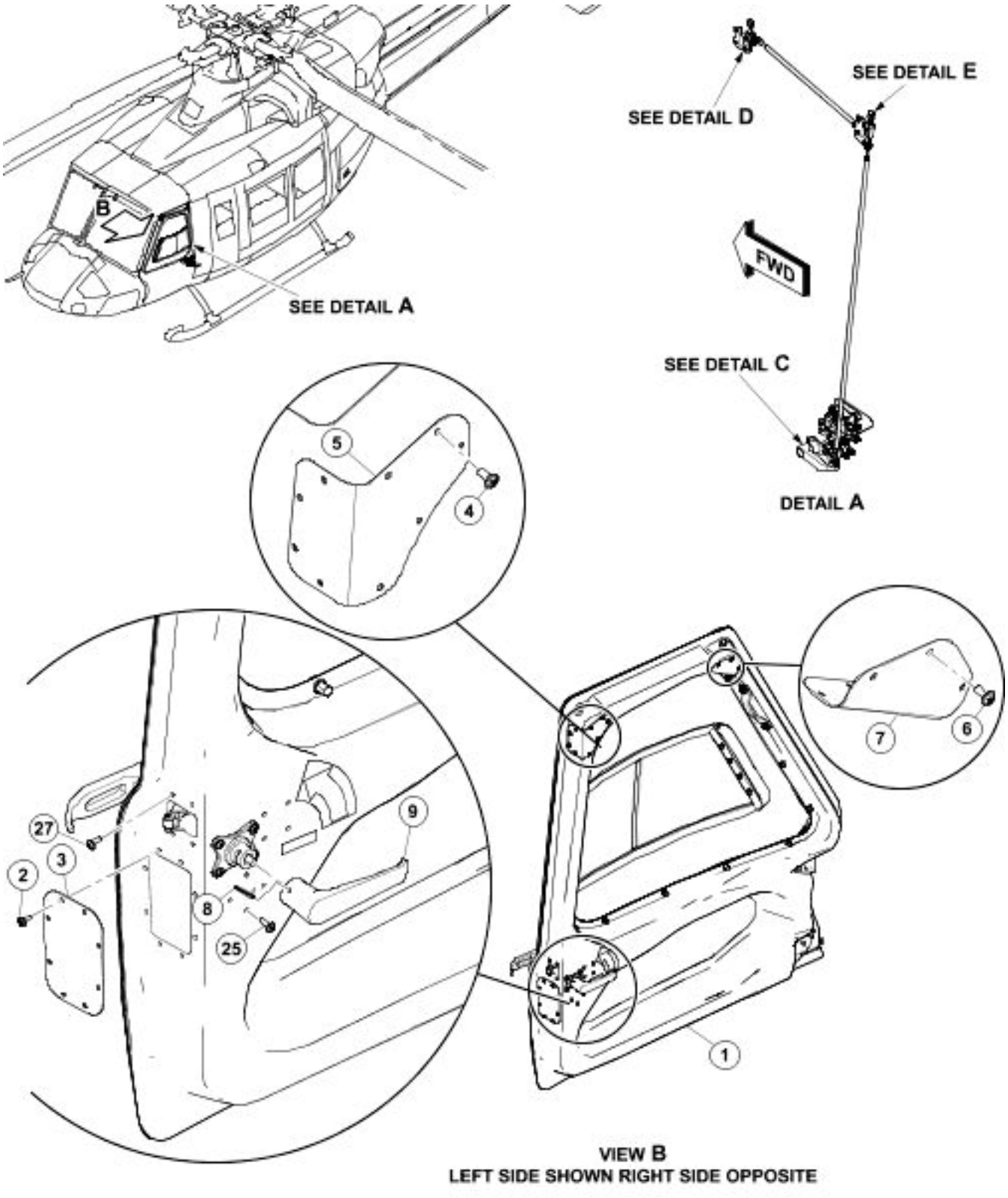
5.1. Remove the screws (36) from the door (1) and the plunger assembly (37).

5.2. Remove the plunger assembly (37) from the door (1).

6. Remove the upper forward plunger assembly (39) from the door (1) as follows:

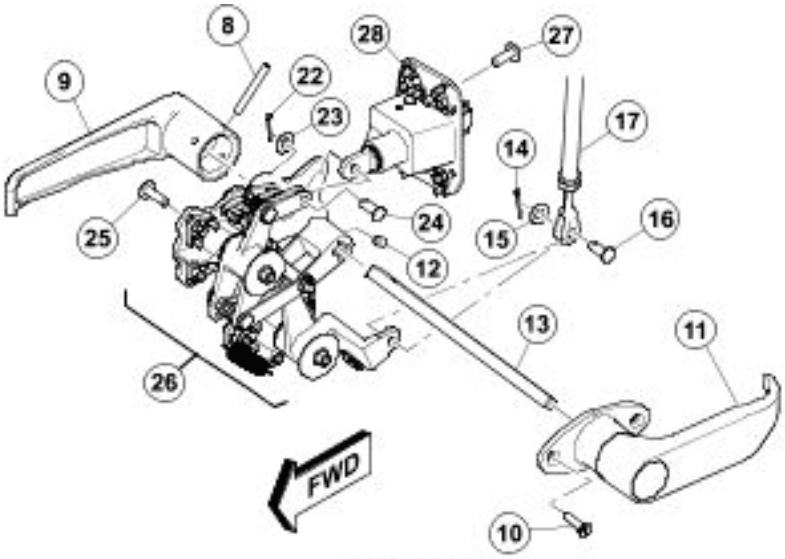
6.1. Remove the screws (38) from the door (1) and the plunger assembly (39).

6.2. Remove the plunger assembly (39) from the door (1).

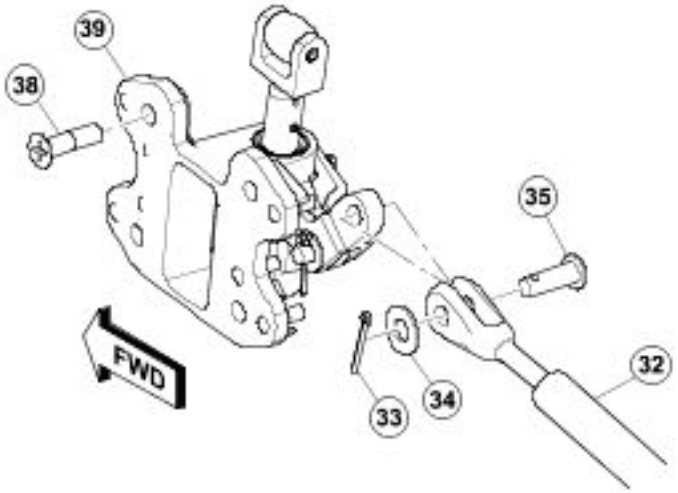


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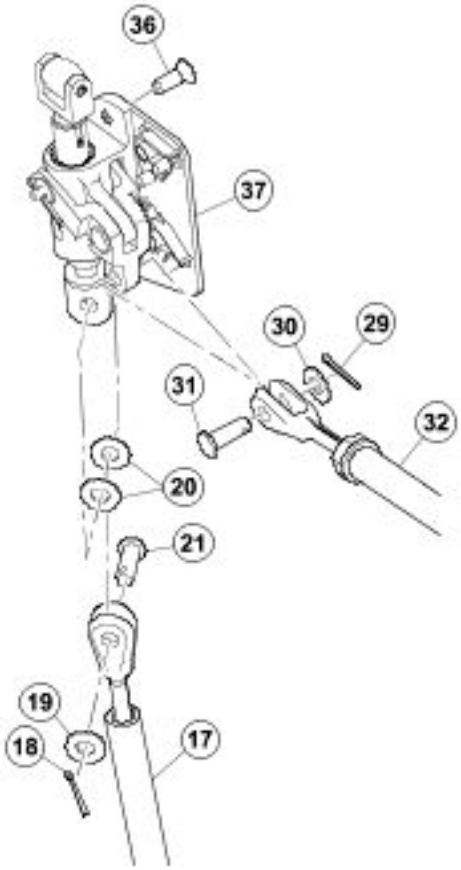
Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 1 of 2)



DETAIL C



DETAIL D



DETAIL E

1. Door
2. Screw
3. Cover, aft Lower
4. Screw
5. Cover, aft upper
6. Screw
7. Cover, fwd upper
8. Pin
9. Handle, inner
10. Screw
11. Handle, outer
12. Set screw
13. Rod, handle
14. Cotter pin
15. Washer
16. Pin
17. Rod assembly, vertical
18. Cotter pin
19. Washer
20. Washer
21. Pin
22. Cotter pin
23. Washer
24. Pin
25. Screw
26. Bell crank assembly
27. Screw
28. Plunger assembly, aft lower
29. Cotter Pin
30. Washer
31. Pin
32. Rod assembly, horizontal
33. Cotter pin
34. Washer
35. Pin
36. Screw
37. Plunger assembly, upper aft
38. Screw
39. Plunger assembly, upper fwd

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Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 2 of 2)

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Plunger Assembly - Installation

Procedure

NOTE

The procedure is provided for the left door. The procedure is same for the right door.

1. Install the upper forward plunger assembly (39, Figure 1) on the door (1) as follows:

1.1. Put the upper forward plunger assembly (39) in its position on the inner part of the door (1).

1.2. Install the screws (38) on the plunger assembly (39) from outer part of the door (1).

2. Install the upper aft plunger assembly (37) on the door (1) as follows:

2.1. Put the upper aft plunger assembly (37) in its position on the inner part of the door (1).

2.2. Install the screws (36) on the plunger assembly (37) from outer part of the door (1).

3. Install the horizontal rod assembly (32) on the upper aft and upper forward plunger assembly (37 and 39) as follows:

3.1. Put the horizontal rod assembly (32) in its position.

3.2. Install the pin (35) and the washer (34).

3.3. Install the cotter pin (33).

3.4. Install the pin (31) and washer (30).

3.5. Install the cotter pin (29).

4. Install the lower aft plunger assembly (28) on the door (1) as follows:

4.1. Put the lower aft plunger assembly (28) in its position on the inner part of the door (1).

4.2. Install the screws (27) on the plunger assembly (28) from outer part of the door (1).

4.3. Install the pin (24) thru the bell crank lever (26) and the plunger assembly (28).

4.4. Install the washer (23).

4.5. Install the cotter pin (22).

5. Install the vertical rod assembly (17) as follows:

5.1. Put the vertical rod assembly (17) in its position.

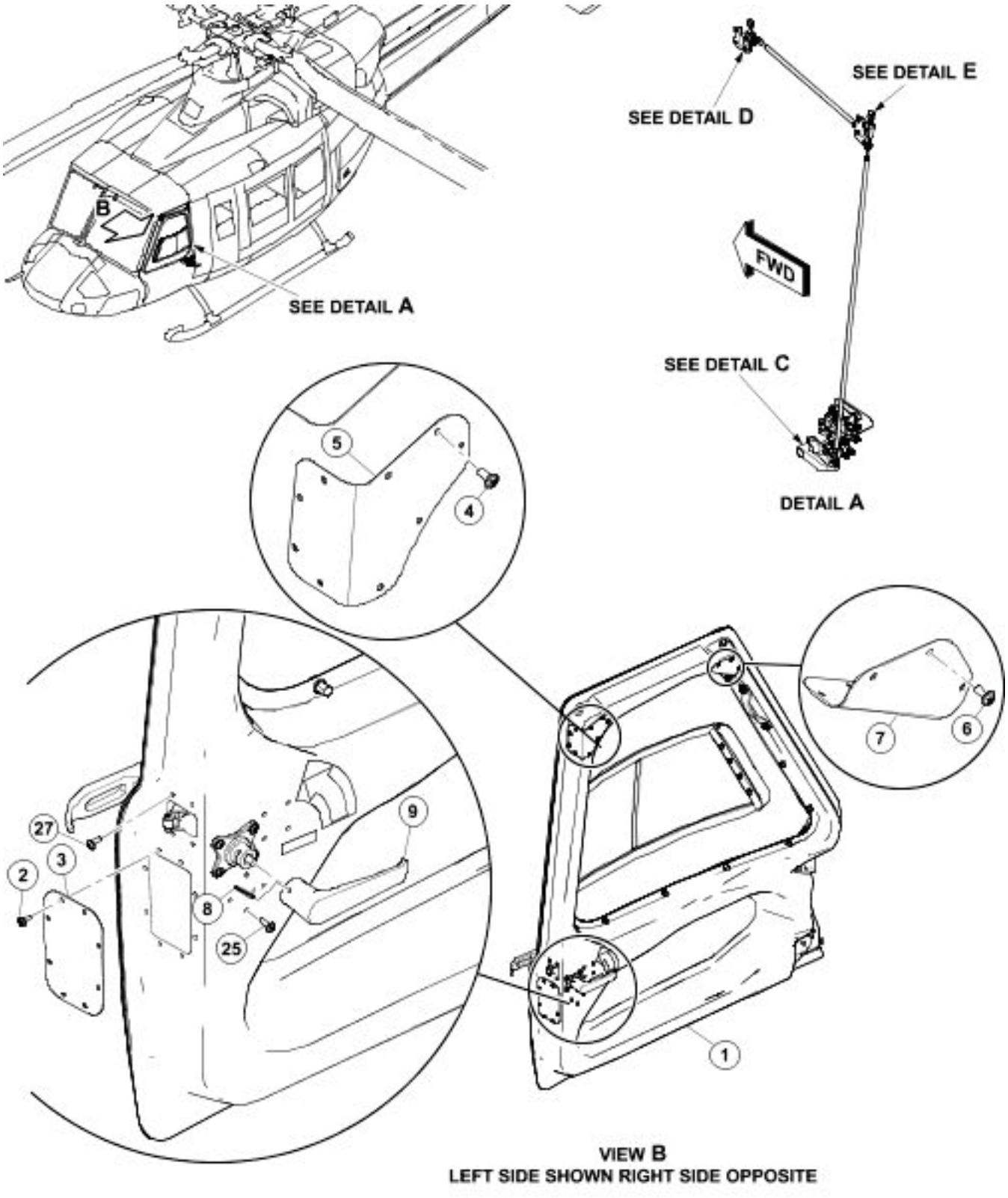
5.2. Install the pin (21) and washers (19 and 20).

5.3. Install the cotter pin (18).

5.4. Install the pin (16) and washer (15).

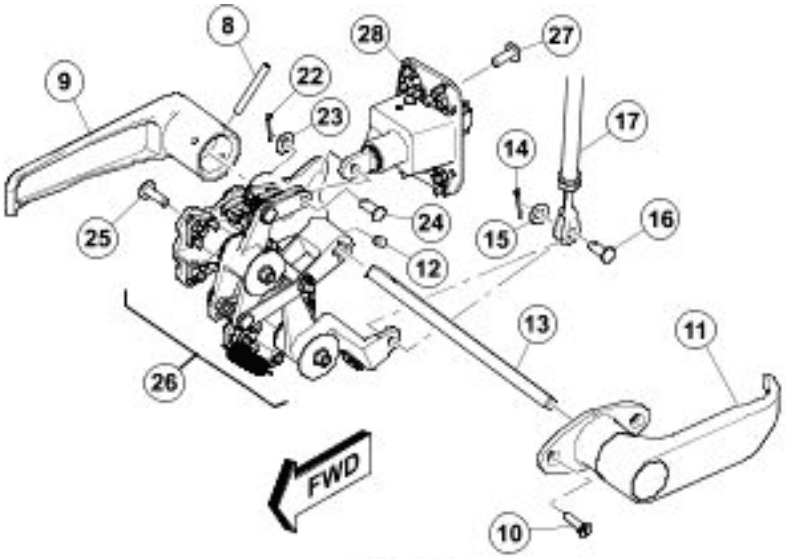
5.5. Install the cotter pin (14).

6. Install the covers (Figure 1) (412-A-52-11-00-03A-720A-A / 00803).

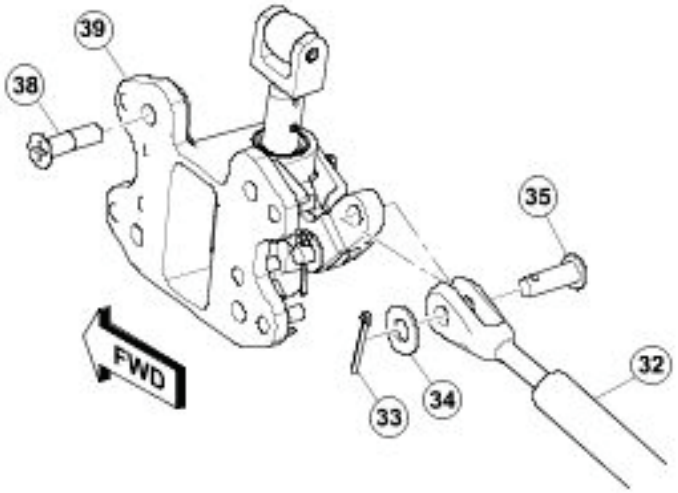


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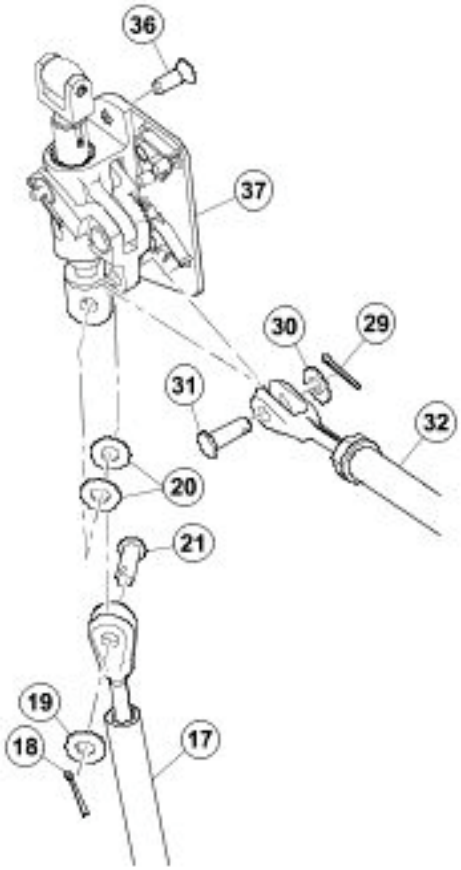
Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 1 of 2)



DETAIL C



DETAIL D



DETAIL E

1. Door
2. Screw
3. Cover, aft Lower
4. Screw
5. Cover, aft upper
6. Screw
7. Cover, fwd upper
8. Pin
9. Handle, inner
10. Screw
11. Handle, outer
12. Set screw
13. Rod, handle
14. Cotter pin
15. Washer
16. Pin
17. Rod assembly, vertical
18. Cotter pin
19. Washer
20. Washer
21. Pin
22. Cotter pin
23. Washer
24. Pin
25. Screw
26. Bell crank assembly
27. Screw
28. Plunger assembly, aft lower
29. Cotter Pin
30. Washer
31. Pin
32. Rod assembly, horizontal
33. Cotter pin
34. Washer
35. Pin
36. Screw
37. Plunger assembly, upper aft
38. Screw
39. Plunger assembly, upper fwd

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Figure 1. Crew Door Bell Crank Assembly and Plungers (Sheet 2 of 2)

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Passenger/Cargo Area Access Doors - General Description

The passenger/cargo area access doors consist of the sliding passenger/cargo door and the hinged panel door located on either side of the helicopter aft of the crew doors.

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Passenger/Cargo Door - Description

The passenger/cargo doors operate on rollers and tracks. These doors give access to passenger/cargo area. Each door has two emergency exit push-out windows. Earlier configuration passenger/cargo doors are constructed from aluminum. Later configuration passenger/cargo doors (Approximately S/N 36191 and Subsequent) are a composite construction and have larger push-out windows.

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Passenger/Cargo Door - Removal

Procedure

NOTE

The removal procedure for left passenger/cargo door is given. The removal procedure for right passenger/cargo door is the same.

1. Unlatch cargo door (38, Figure 1).

2. Retract open position stop spring (22).

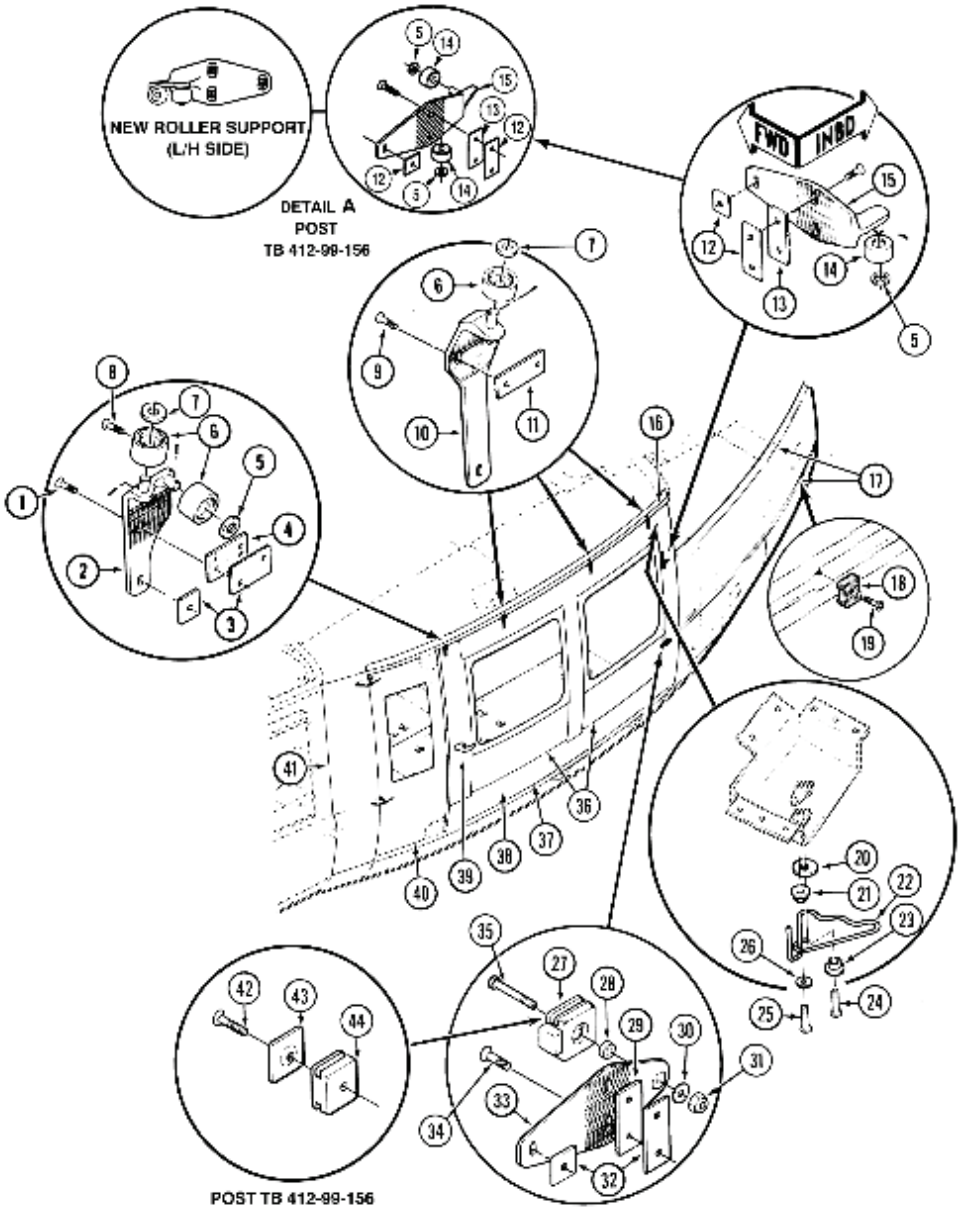
3. Remove screw (19) and stop (18) from aft end of lower door track.

4. Slide cargo door (38) aft out of tracks.

NOTE

Step 5. is applicable only if new door is to be installed.

5. Remove all attaching hardware from door. Mark position of serrated plates (4, 11, and 13) before removal of forward roller support (2), upper roller support (10), and aft upper roller support (15). This will provide a starting point during adjustment procedure.



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Figure 1. Passenger/Cargo Door (Sheet 1 of 2)

1. Screw
2. Forward roller support
3. Shims
4. Serrated plate
5. Washer
6. Roller
7. Washer
8. Screw
9. Screw
10. Upper roller support
11. Serrated plate
12. Shims
13. Serrated plate
14. Roller
15. Aft upper roller support
16. Upper track
17. Aft tracks
18. Stop
19. Screw
20. Washer
21. Spacer
22. Spring
23. Bushing
24. Screw
25. Screw
26. Washer
27. Slide
28. Spacer
29. Serrated plate
30. Washer
31. Nut
32. Shims
33. Slider support
34. Screw
35. Bolt
36. Escape panel
37. Lower track
38. Cargo door
39. Latch handle
40. Hinged panel
41. Door post
42. Screw
43. Retainer
44. Slider

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Figure 1. Passenger/Cargo Door (Sheet 2 of 2)

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Passenger/Cargo Door - Inspection

Procedure

1. Perform operational check for both passenger/cargo doors (412-A-52-12-00-01A-320A-A / 00816).
2. Inspect door latch for binding, wear, damage, and secure mounting.
3. Inspect door structure and hinges for cracks and/or distortion.
4. Inspect door windows (412-A-52-01-00-06A-280A-A / 00873).
5. Inspect door seals for secure bonding and deterioration.
6. Inspect door lock caution switch (412-A-96-03-01-14A-271A-A / 02061).

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Passenger/Cargo Door - Repair

Procedure

1. Replace unserviceable parts in passenger/cargo door latch mechanism (412-A-52-12-00-02A-520A-A / 00817 and 412-A-52-12-00-02A-720A-A / 00818).

2. Repair passenger/cargo door structural damage. Refer to the BHT-MED-SRM-1 and to FAA Publication AC 43.13-1B Aircraft Inspection and Repair Manual for structural maintenance.

3. Repair damaged passenger door windows (412-A-52-01-00-06A-042A-A / 00871).

4. Repair damaged passenger/cargo door seals (412-A-52-00-00-00A-720A-A / 00777).

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Passenger/Cargo Door - Installation

Procedure

NOTE

The installation for left passenger/cargo door is given. The installation procedure for right passenger cargo door is the same.

1. If new door is being installed, install all attaching hardware.

NOTE

To aid in adjustment procedures, note marked position of serrated plates (4, 11, and 13, Figure 1) on installation of forward roller support (2), upper roller support (10), and aft upper roller support (15).

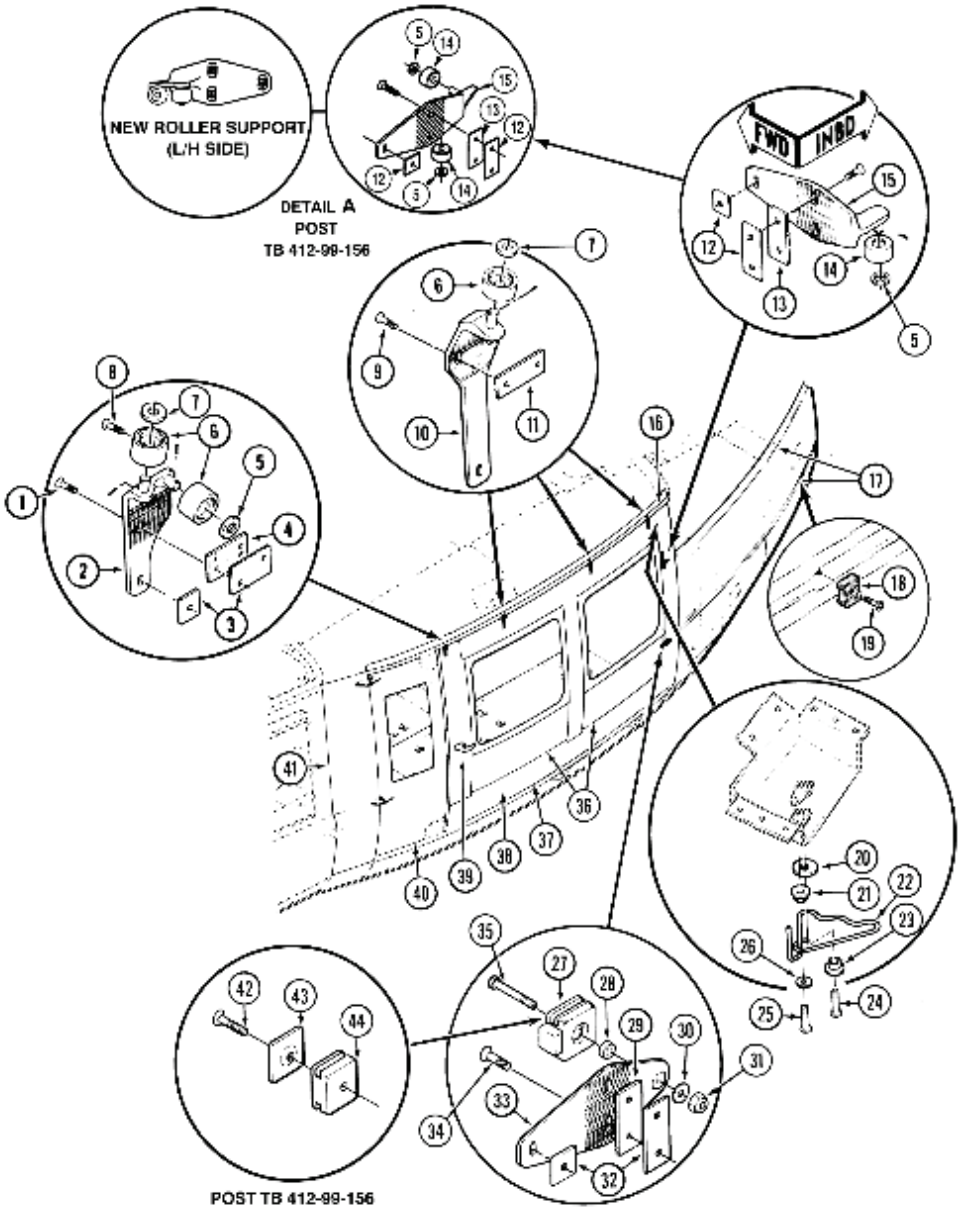
2. Position passenger/cargo door (38, Figure 1) on helicopter with forward edge in line with aft end of door tracks.

3. Start rollers and slider through cut outs at aft ends of tracks.

4. Push passenger/cargo door (38) forward.

5. With concave face of stop (18) facing forward, position stop at end of lower track on structure and install with screw (19).

6. Perform operational check (412-A-52-12-00-01A-320A-A / 00816).



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Figure 1. Passenger/Cargo Door (Sheet 1 of 2)

1. Screw
2. Forward roller support
3. Shims
4. Serrated plate
5. Washer
6. Roller
7. Washer
8. Screw
9. Screw
10. Upper roller support
11. Serrated plate
12. Shims
13. Serrated plate
14. Roller
15. Aft upper roller support
16. Upper track
17. Aft tracks
18. Stop
19. Screw
20. Washer
21. Spacer
22. Spring
23. Bushing
24. Screw
25. Screw
26. Washer
27. Slide
28. Spacer
29. Serrated plate
30. Washer
31. Nut
32. Shims
33. Slider support
34. Screw
35. Bolt
36. Escape panel
37. Lower track
38. Cargo door
39. Latch handle
40. Hinged panel
41. Door post
42. Screw
43. Retainer
44. Slider

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Figure 1. Passenger/Cargo Door (Sheet 2 of 2)

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Passenger/Cargo Door - Operational Check and Adjustment

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Make sure that the hinged panel (40, Figure 1) is closed and latched. Place door to fully closed and latched position. Verify upper edge of door is parallel to top of cabin door frames and upper track (16).

2. If door is out of alignment, loosen screws attaching aft upper roller support (15) and slider support (33) on rear edge of door. Adjust slider support. Tighten slider support screws.

3. Adjust aft upper roller support (15) so that roller is fully engaged in track. Tighten attaching screws.

4. Operate door through full travel while verifying all rollers on upper edge are fully engaged in upper track (16) at all positions. Adjust upper roller supports (10) as required.

5. With door fully closed and latched, make sure that lower door track (37) is engaged not less than 0.25 inch (6.35 mm) in cabin door channel without restricting door travel through full range from fully closed to fully open. Verify door track attaching screws are tightened after adjustment.

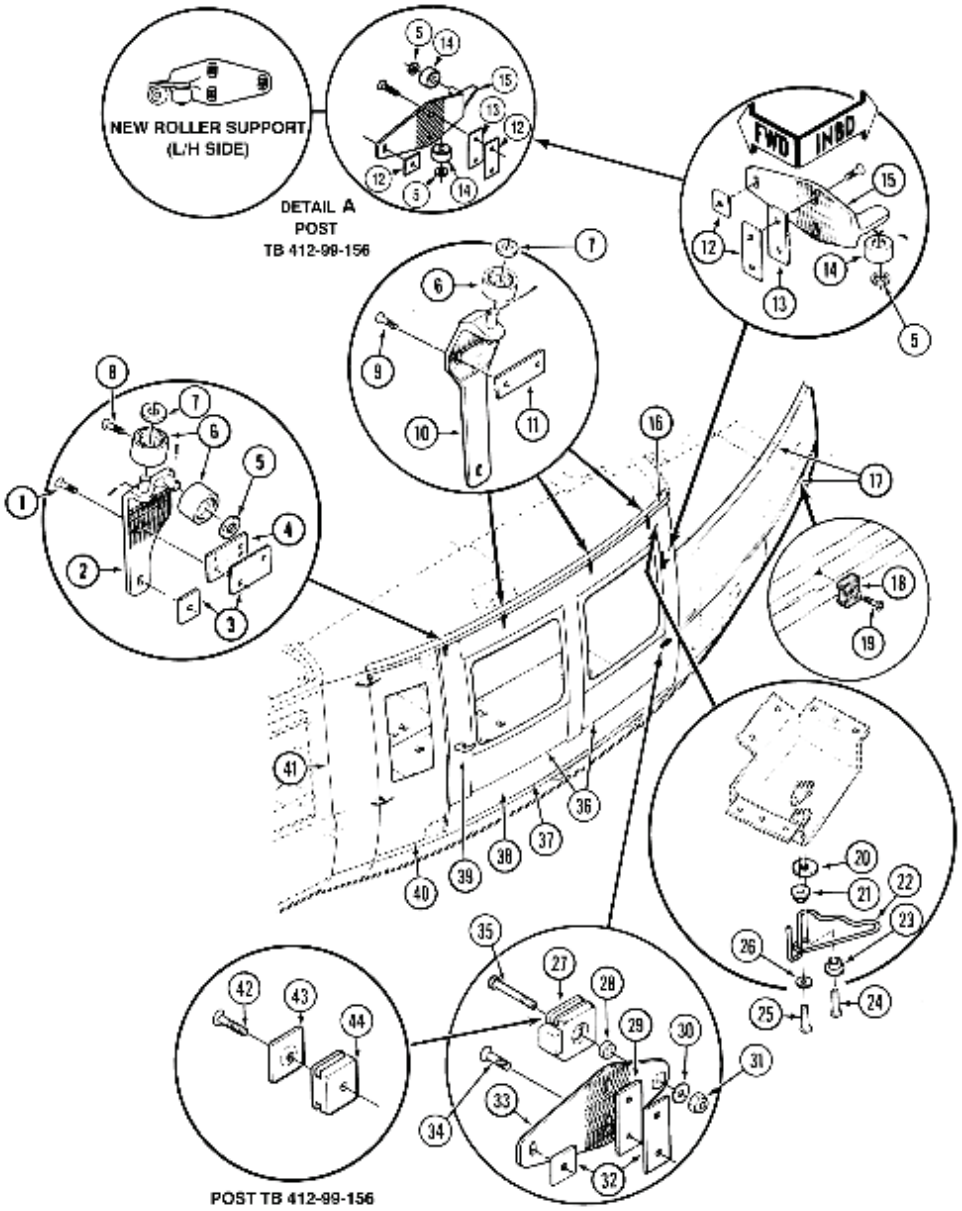
6. Operate door through full travel while checking for dragging or binding.

7. Roller supports (2 and 15) and slider support (33) are installed with peel shims (3, 12, and 32). To adjust, peel new shims to required thickness. Pay particular attention to slider (44) when checking for binding. Minimum friction on slider will improve operation of door.

8. Cut lockwire on adjustment screws (8, Figure 2) and adjust so upper and lower cams (9) will bottom simultaneously. After adjustment has been obtained, secure adjusting screws (8) with lockwire (C-405).

9. Move door to full open position and check for fit of hold-open spring (22, Figure 1) on striker bushing near upper edge of door.

10. Adjust lock caution light switch (412-A-96-03-01-14A-271A-A / 02061).



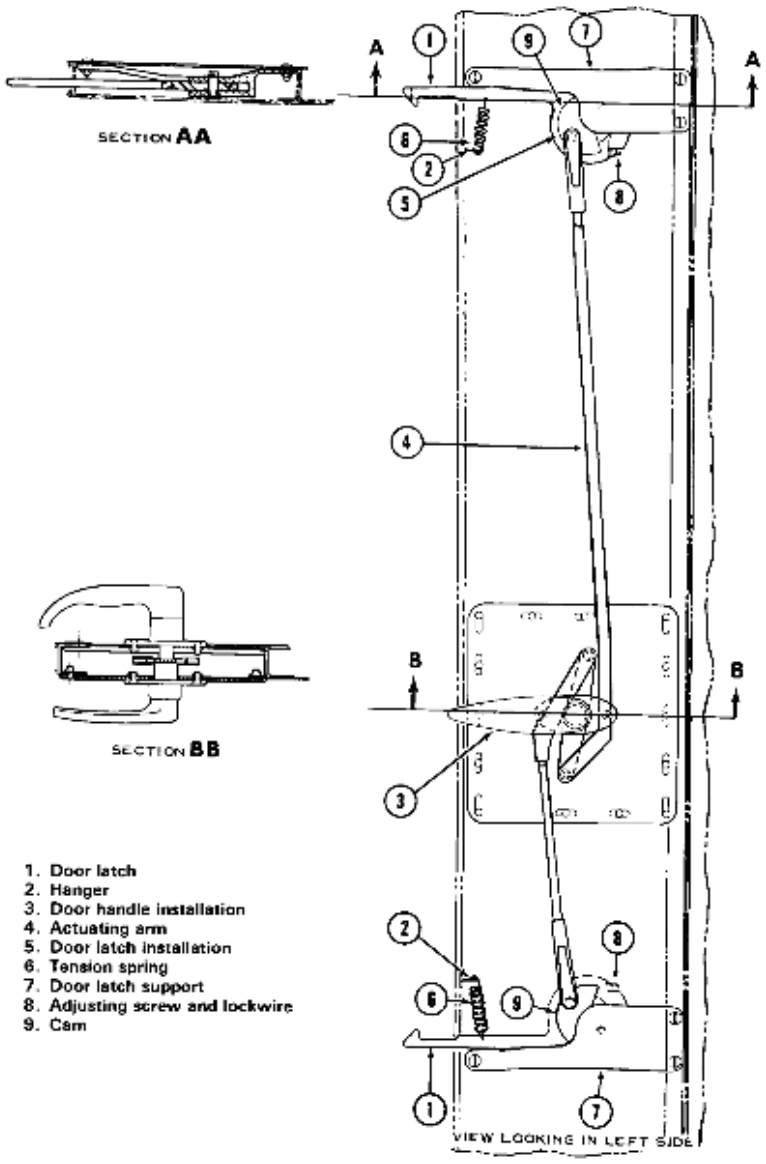
ICN-412-A-521200-B-97499-00001-A-001-01

Figure 1. Passenger/Cargo Door (Sheet 1 of 2)

1. Screw
2. Forward roller support
3. Shims
4. Serrated plate
5. Washer
6. Roller
7. Washer
8. Screw
9. Screw
10. Upper roller support
11. Serrated plate
12. Shims
13. Serrated plate
14. Roller
15. Aft upper roller support
16. Upper track
17. Aft tracks
18. Stop
19. Screw
20. Washer
21. Spacer
22. Spring
23. Bushing
24. Screw
25. Screw
26. Washer
27. Slide
28. Spacer
29. Serrated plate
30. Washer
31. Nut
32. Shims
33. Slider support
34. Screw
35. Bolt
36. Escape panel
37. Lower track
38. Cargo door
39. Latch handle
40. Hinged panel
41. Door post
42. Screw
43. Retainer
44. Slider

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Figure 1. Passenger/Cargo Door (Sheet 2 of 2)



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Figure 2. Passenger/Cargo Door Latch (Sheet 1 of 1)

Passenger/Cargo Door Latches - Removal

Procedure

NOTE

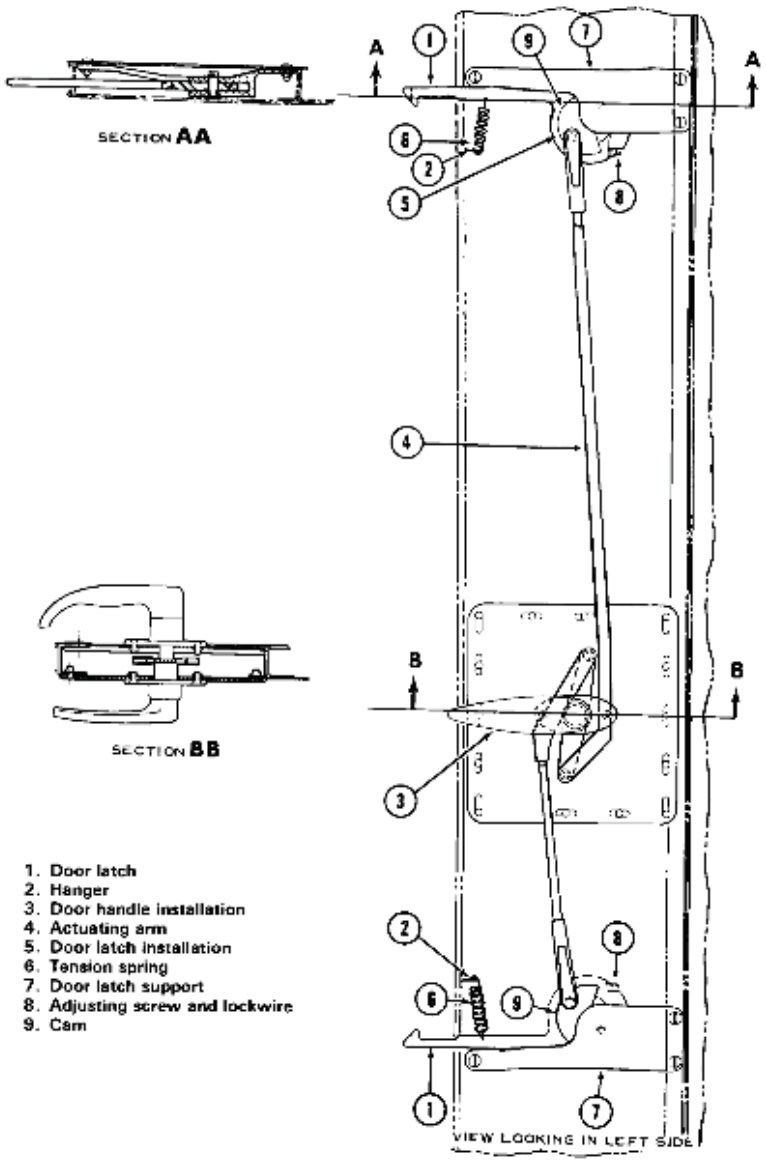
The following instructions are for removing the upper door latch. Procedure for removing lower door latch is the same.

1. Unhook tension springs (6, Figure 1) from door latches (1) and hangers (2) on door structure.

2. Remove cotter pin, washer, and guide pin from actuating arm (4) and door latch (1).

3. Remove cotter pin, washer, and guide pin from door latch (1) and door latch support (7).

4. Remove door latch (1).



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Figure 1. Passenger/Cargo Door Latch (Sheet 1 of 1)

Passenger/Cargo Door Latches - Installation

Procedure

NOTE

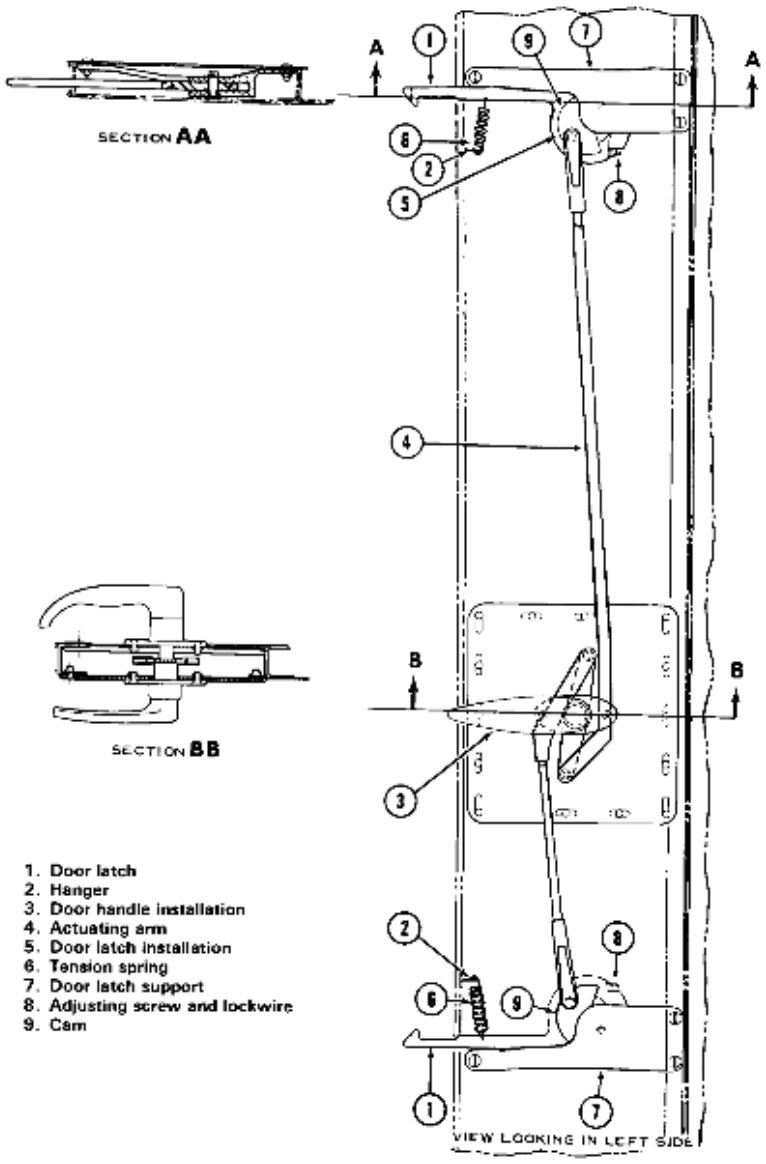
The following instructions are for installing the upper door latch. Procedure for installing the lower door latch is the same.

1. Insert door latch (1, Figure 1) in door latch support (7).

2. Align door latch (1) to door latch support (7) and insert guide pin. Secure guide pin with washer and cotter pin.

3. Align actuating arm (4) to door latch (1) and insert guide pin. Secure guide pin with washer and cotter pin.

4. Connect tension springs (6) to door latch (1) and hangers (2) on door structure.

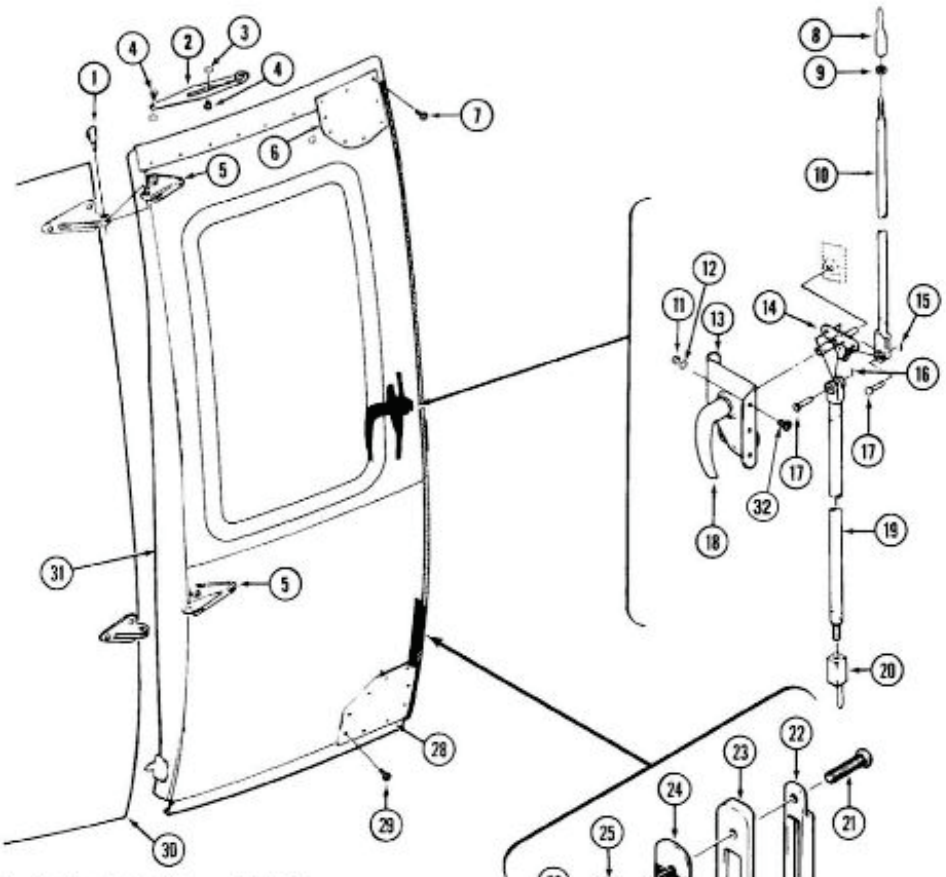


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Figure 1. Passenger/Cargo Door Latch (Sheet 1 of 1)

Cargo Hinged Panels - Description

The cargo hinged panels (Figure 1), located forward of passenger/cargo doors, provide a wider opening for cargo loading. Earlier configuration cargo hinged panels are constructed from aluminum. Later configuration (S/N 36191 and Subsequent) cargo hinged panels are a composite construction.



1. Quick release pin
2. Positioning spring
3. Nut plate
4. Pivot
5. Hinge
6. Cover
7. Screw
8. Pin
9. Nut
10. Tube
11. Nut
12. Washer
13. Escutcheon
14. Spindle assembly
15. Cotter pin
16. Cotter pin
17. Pin
18. Handle
19. Tube
20. Pin
21. Screw
22. Striker guide
23. Shim
24. Plate
25. Striker
26. Washer
27. Nut
28. Cover
29. Screw
30. Door post
31. Hinged panel
32. Screw

ICN-412-A-523000-B-97499-00001-A-001-01

Figure 1. Cargo Hinged Panel (Sheet 1 of 1)

Cargo Hinged Panels - Removal

Procedure

NOTE

The removal procedure of the left cargo hinged panel is given. The removal procedure for the right cargo hinged panel is the same.

1. Open passenger/cargo door.

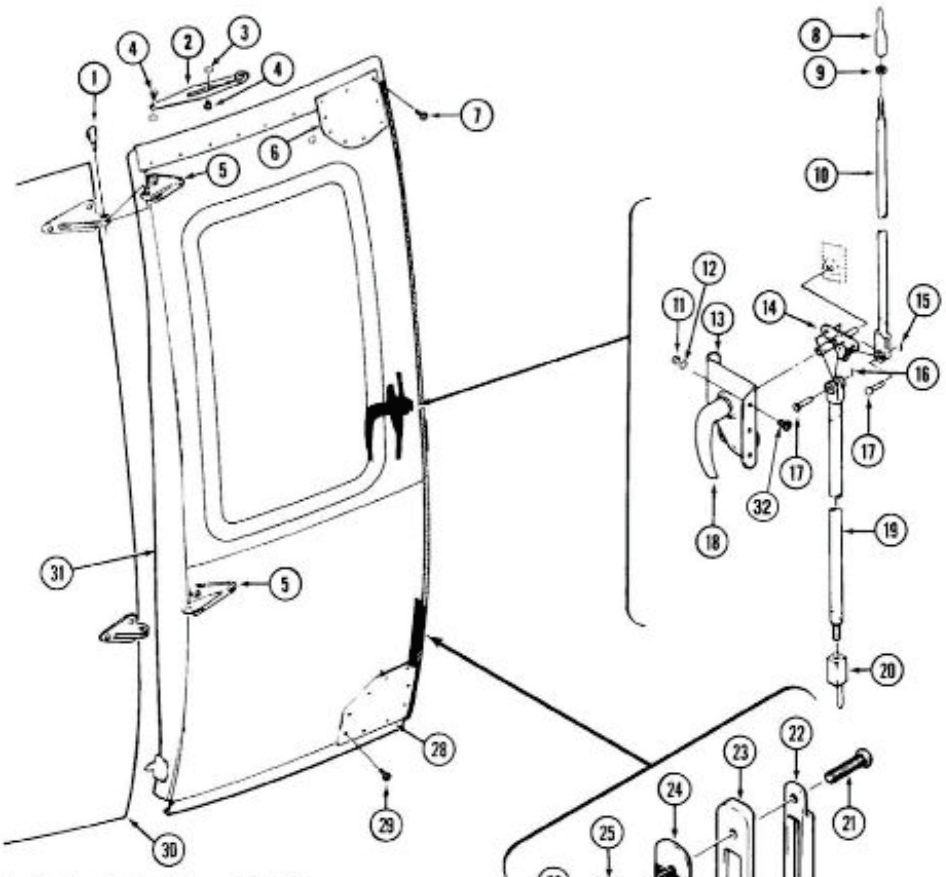
2. Operate latch handle (18, Figure 1) of panel door to release pins (8 and 20) from upper and lower channels of door opening.

3. Open cargo hinged panel (31).

4. Remove positioning spring (2) from stud at top of opening.

5. Remove two quick-release pins (1) from hinges (5).

6. Remove cargo hinged panel (31).



1. Quick release pin
2. Positioning spring
3. Nut plate
4. Pivot
5. Hinge
6. Cover
7. Screw
8. Pin
9. Nut
10. Tube
11. Nut
12. Washer
13. Escutcheon
14. Spindle assembly
15. Cotter pin
16. Cotter pin
17. Pin
18. Handle
19. Tube
20. Pin
21. Screw
22. Striker guide
23. Shim
24. Plate
25. Striker
26. Washer
27. Nut
28. Cover
29. Screw
30. Door post
31. Hinged panel
32. Screw

ICN-412-A-523000-B-97499-00001-A-001-01

Figure 1. Cargo Door Latch (Sheet 1 of 1)

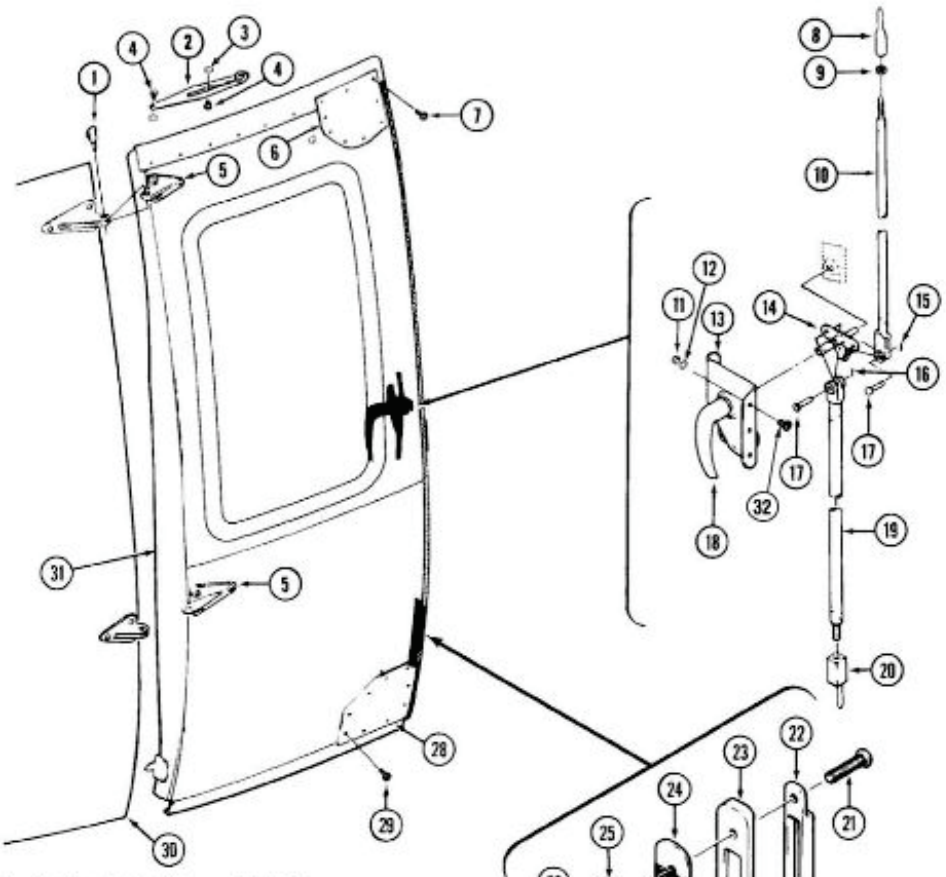
Cargo Hinged Panels - Inspection

Procedure

NOTE

The inspection procedure of the left cargo hinged panel is given. The inspection procedure for the right cargo hinged panel is the same.

1. Inspect visible portions of both hinge panel latching mechanism for binding, wear, damage, and secure mounting (Figure 1).
2. Inspect panel structure and hinge for cracks and/or distortion.
3. Inspect panel door windows for damage (412-A-52-01-00-05A-280A-A / 00868).
4. Inspect panel seals for secure bonding and deterioration.



1. Quick release pin
2. Positioning spring
3. Nut plate
4. Pivot
5. Hinge
6. Cover
7. Screw
8. Pin
9. Nut
10. Tube
11. Nut
12. Washer
13. Escutcheon
14. Spindle assembly
15. Cotter pin
16. Cotter pin
17. Pin
18. Handle
19. Tube
20. Pin
21. Screw
22. Striker guide
23. Shim
24. Plate
25. Striker
26. Washer
27. Nut
28. Cover
29. Screw
30. Door post
31. Hinged panel
32. Screw

ICN-412-A-523000-B-97499-00001-A-001-01

Figure 1. Cargo Door Latch (Sheet 1 of 1)

Cargo Hinged Panels - Repair

Procedure

NOTE

The repair procedure for the left cargo hinged panel is given. The repair procedure for the right cargo hinged panel is the same.

1. Replace unserviceable parts in hinged panel latch mechanism (412-A-52-30-00-01A-520A-A / 00824 and 412-A-52-30-00-01A-720A-A / 00825).

2. Repair cargo hinged panel structural damage. Refer to the BHT-MED-SRM-1 and to FAA Publication AC 43.13-1A Aircraft Inspection and Repair Manual for structural maintenance instructions.

3. Repair cargo hinged panel windows (412-A-52-01-00-05A-685A-A / 00869).

4. Repair damaged cargo hinged panel door seals (412-A-52-00-00-00A-720A-A / 00777).

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Cargo Hinged Panels - Installation

Procedure

NOTE

The installation procedure for the left cargo hinged panel is given. The installation for the right cargo hinged panel is the same.

1. Align cargo hinged panel (31, Figure 1) on hinge and install two quick release pins (1).

2. Swing panel partly closed and engage positioning spring (2) on stud at top of door opening.

3. Close door and operate handle (18) to extend latch pins (8 and 20) into holes in upper and lower structural channels of frame. Make sure that pins properly engage channels.

4. Adjust hinge panel as follows:

4.1. Verify pin on lower inside of panel is engaged in door post.

4.2. Remove screws (7) to remove cover (6) and screws (29) to remove cover (28).

4.3. Operate handle (18) to open hinged panel.

4.4. Verify upper and lower pins (8 and 20) retract to clear top and bottom of door opening.

4.5. If necessary to adjust either pin (8 or 20), loosen nut (9) and adjust pin to obtain secure engagement and proper release.

4.6. Tighten nut (9) against pin (8).

4.7. Install covers (6 and 28).

4.8. Check action of positioning spring (2) on pivot (4) as panel is opened. Detent in spring slot should catch and hold panel at approximately 90° to fuselage. Hook of spring should catch when panel is forced beyond detent.

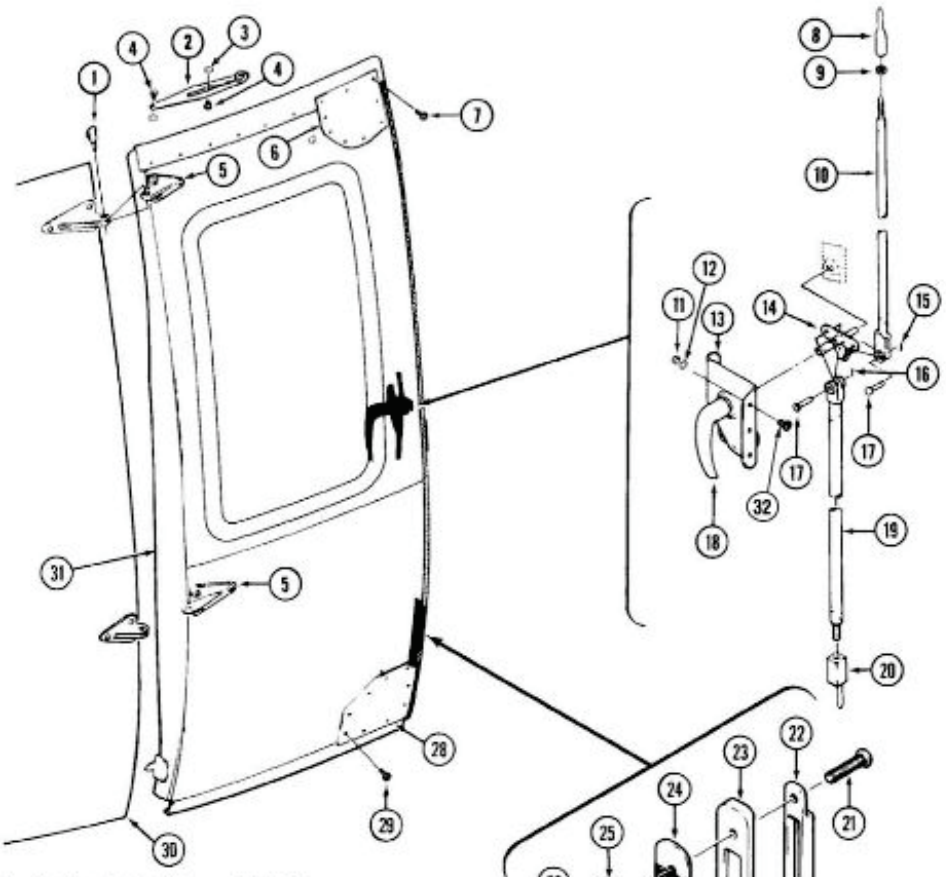
4.9. Close and latch hinged panel.

4.10. Slowly close passenger/cargo door.

4.11. Check proper engagement of latches on passenger/cargo door with striker (25).

4.12. When adjustment is necessary, remove cover (28) over striker (25). Adjust striker positioning by means of slotted mounting holes, or adjust thickness of laminated shim (23). Tighten screws (21), washers (26), and nuts (27).

4.13. Install cover (28) on striker (25).



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Figure 1. Cargo Hinged Panel (Sheet 1 of 1)

Cargo Hinged Panel Latching Mechanism - Removal

Procedure

NOTE

The maintenance procedures for the left cargo hinged panel latching mechanism are given. The procedures for the right cargo hinged panel latching mechanism are the same.

1. Remove screws (7, Figure 1) and remove cover (6).

2. Remove screws (29) and remove cover (28).

3. Remove screws (32), washers (12), and nuts (11) to detach escutcheon (13).

4. Remove escutcheon (13) and handle (18) as an assembly.

5. Remove cotter pins (15 and 16) and pins (17) to detach tubes (10 and 19) from spindle assembly (14).

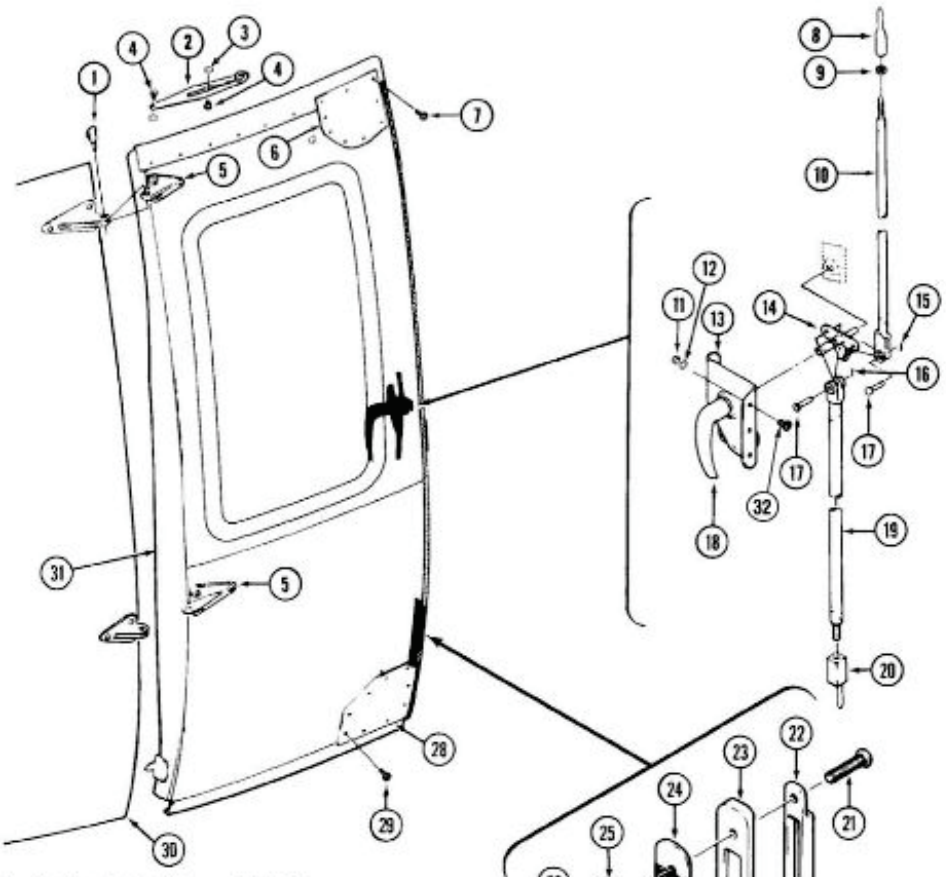
6. Pull tube (10) until pin (8) is free of guide in top of panel.

7. Pull tube (19) until pin (20) is free of guide in bottom of panel.

8. Remove striker mechanism from top or bottom of panel as follows:

8.1. Remove screw (21), washer (26), and nut (27).

8.2. Remove striker guide (22), shim (23), plate (24), and striker (25).



1. Quick release pin
2. Positioning spring
3. Nut plate
4. Pivot
5. Hinge
6. Cover
7. Screw
8. Pin
9. Nut
10. Tube
11. Nut
12. Washer
13. Escutcheon
14. Spindle assembly
15. Cotter pin
16. Cotter pin
17. Pin
18. Handle
19. Tube
20. Pin
21. Screw
22. Striker guide
23. Shim
24. Plate
25. Striker
26. Washer
27. Nut
28. Cover
29. Screw
30. Door post
31. Hinged panel
32. Screw

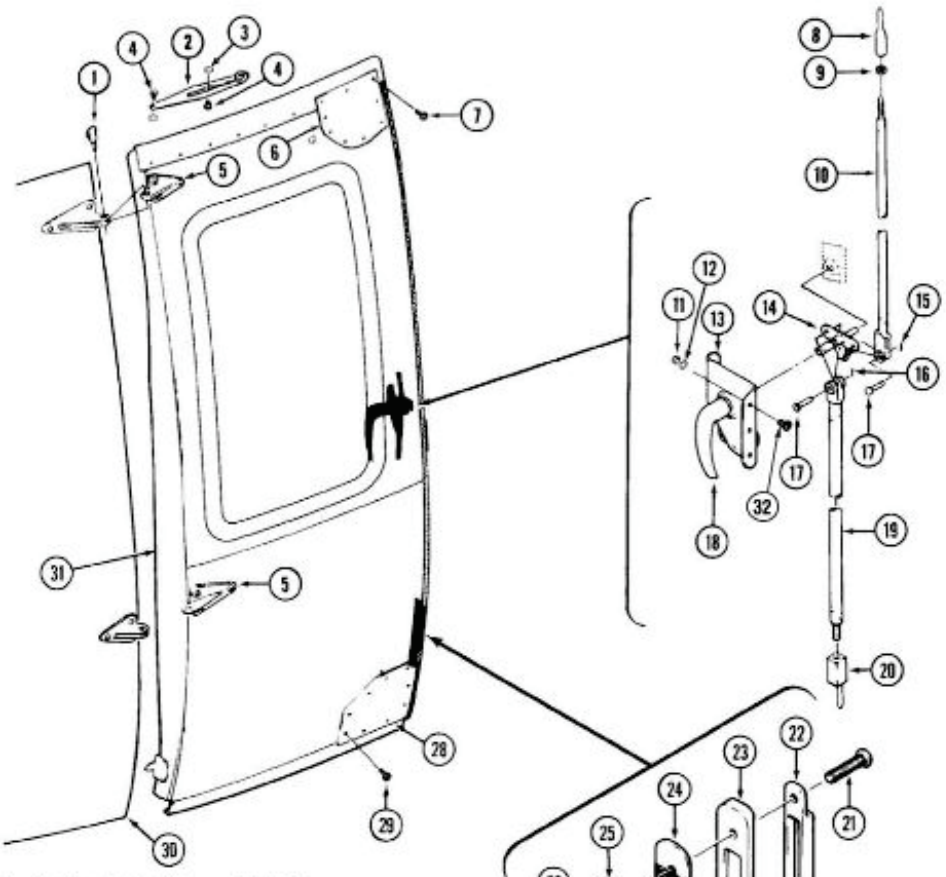
ICN-412-A-523000-B-97499-00001-A-001-01

Figure 1. Cargo Hinged Panel (Sheet 1 of 1)

Cargo Hinged Panel Latching Mechanism - Installation

Procedure

1. Position striker guide (22, Figure 1) on outside surface of panel and position shim (23), plate (24), and striker (25) on inside surface of panel.
2. Install screws (21) with washers (26) and nuts (27).
3. Insert tube (10) through access opening at top of panel.
4. Install nut (9) and pin (8) on tube (10).
5. Insert pin (8) through guide at top of panel.
6. Connect tube (10) to spindle assembly (14) with pin (17) and cotter pin (15).
7. Insert tube (19) through access opening at bottom of panel.
8. Install pin (20) on tube (19).
9. Insert pin (20) through guide at bottom of panel.
10. Connect tube (19) to spindle assembly (14) with pin (17) and cotter pin (16).
11. Install escutcheon (13) and handle (18) as an assembly. Secure with screws (32), washers (12), and nuts (11).
12. Install cover (6) with screws (7).
13. Install cover (28) with screws (29).



1. Quick release pin
2. Positioning spring
3. Nut plate
4. Pivot
5. Hinge
6. Cover
7. Screw
8. Pin
9. Nut
10. Tube
11. Nut
12. Washer
13. Escutcheon
14. Spindle assembly
15. Cotter pin
16. Cotter pin
17. Pin
18. Handle
19. Tube
20. Pin
21. Screw
22. Striker guide
23. Shim
24. Plate
25. Striker
26. Washer
27. Nut
28. Cover
29. Screw
30. Door post
31. Hinged panel
32. Screw

ICN-412-A-523000-B-97499-00001-A-001-01

Figure 1. Cargo Hinged Panel (Sheet 1 of 1)

Baggage Compartment Door - Description

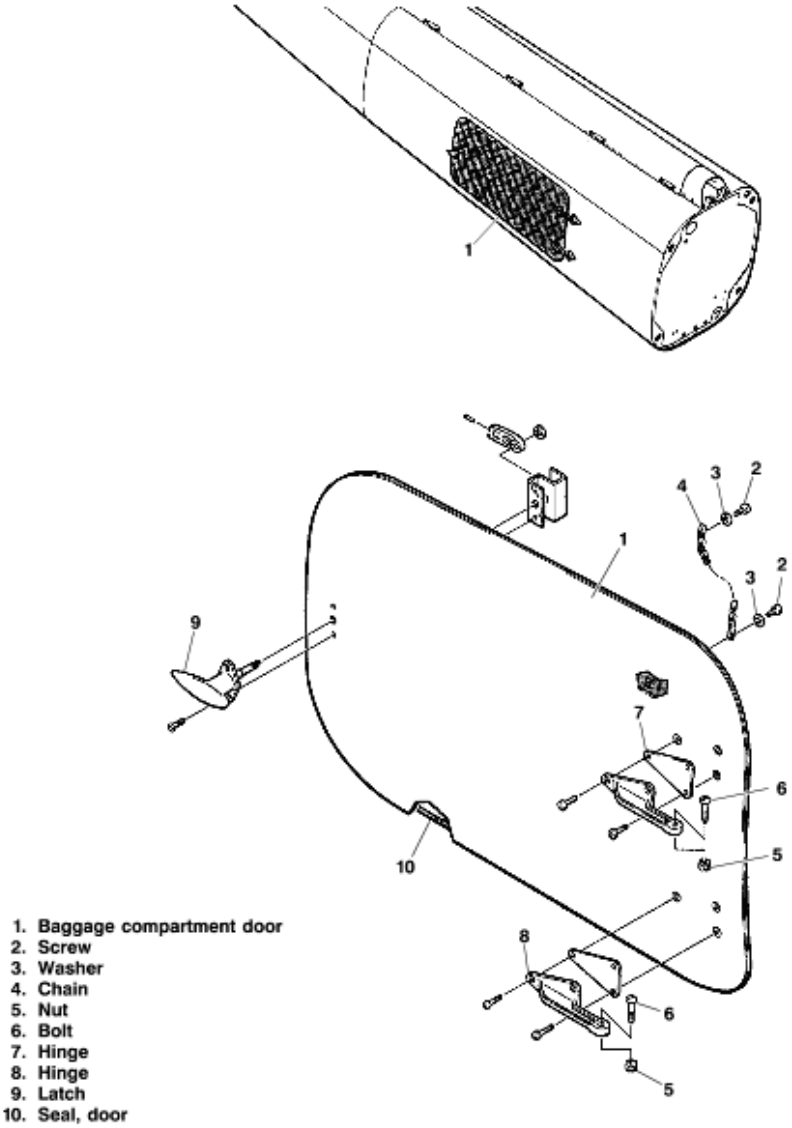
The baggage compartment door is located on forward right side of tailboom. The door is hinged on forward end and has a latch and lock on aft end. The inside of the door is honeycomb panel construction sandwiched between inner and outer panels of aluminum alloy. The baggage door latch includes a roller to actuate a switch when door is closed. The switch illuminates a caution panel worded segment DOOR LOCK when the door is open and activates an overhead light in the baggage compartment.

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Baggage Compartment Door - Removal

Procedure

1. Open baggage compartment door (1, Figure 1).
2. Remove screws (2) and washers (3) to disconnect chain (4) from baggage compartment.
3. Remove nuts (5) and bolts (6) from hinges (7 and 8). Remove baggage compartment door (1).



1. Baggage compartment door
2. Screw
3. Washer
4. Chain
5. Nut
6. Bolt
7. Hinge
8. Hinge
9. Latch
10. Seal, door

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Figure 1. Baggage Compartment Door (Sheet 1 of 1)

Baggage Compartment Door - Inspection

Procedure

1. Inspect chain (4) for security of installation.
2. Inspect hinge (7 and 8) for cracks and distortion.
3. Inspect latch (9) for proper operation and security of installation.
4. Inspect lock (not shown) for proper operation.
5. Inspect switch for operation of baggage compartment light and caution panel segment DOOR LOCK when baggage compartment door is opened. Inspect visually and refer to 412-A-96-03-01-14A-271A-A / 02061.
6. Inspect door for cracks and distortion.
7. Inspect door seals (10) for secure bonding and for deterioration.

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Baggage Compartment Door - Repair

Procedure

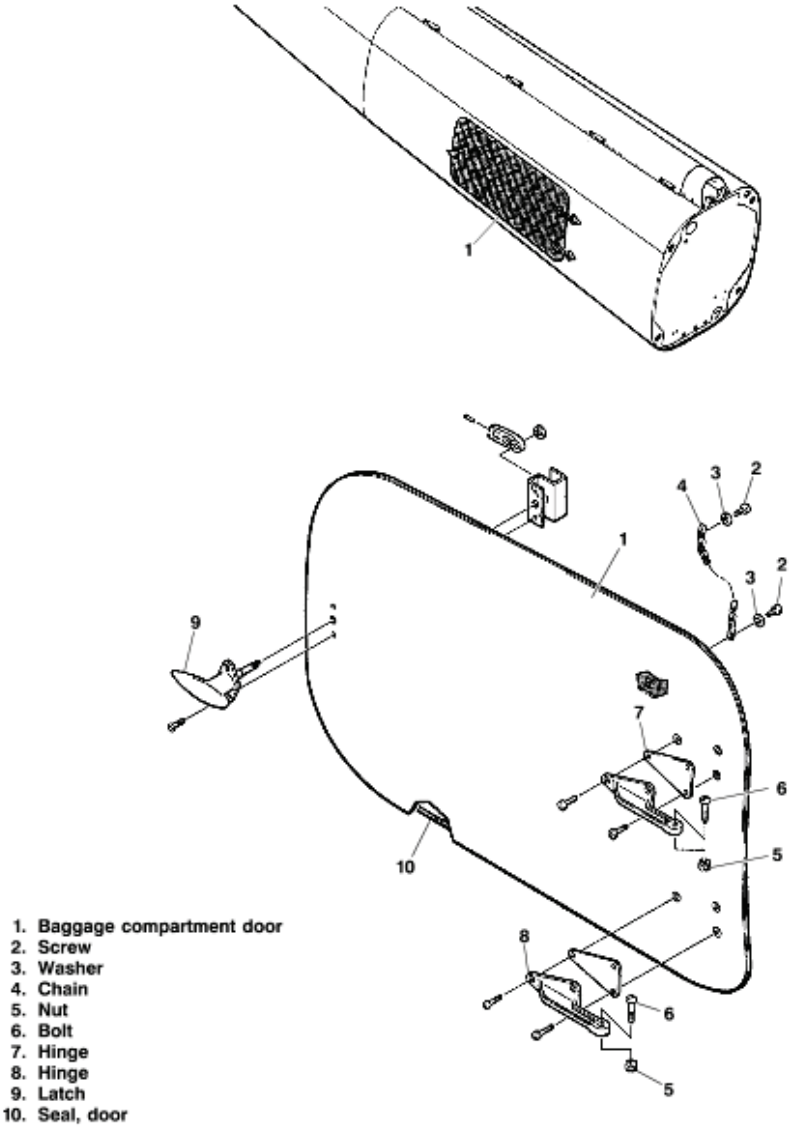
1. Replace unserviceable parts in baggage compartment door latch mechanism.
2. Repair baggage compartment door structural damage. Refer to the BHT-MED-SRM-1 and to FAA Publication AC 43.13-1B Aircraft Inspection and Repair Manual for repair of honeycomb panels.
3. Replace damaged baggage compartment door seals (412-A-52-00-00-00A-720A-A / 00777).

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Baggage Compartment Door - Installation

Procedure

1. Position baggage door (1, Figure 1) over hinges on tailboom.
2. Install bolts (6) and nuts (5).
3. Connect chain (4) to door using screws (2) and washers (3).
4. Check operation of door lock caution switch (412-A-96-03-01-14A-271A-A / 02061).



1. Baggage compartment door
2. Screw
3. Washer
4. Chain
5. Nut
6. Bolt
7. Hinge
8. Hinge
9. Latch
10. Seal, door

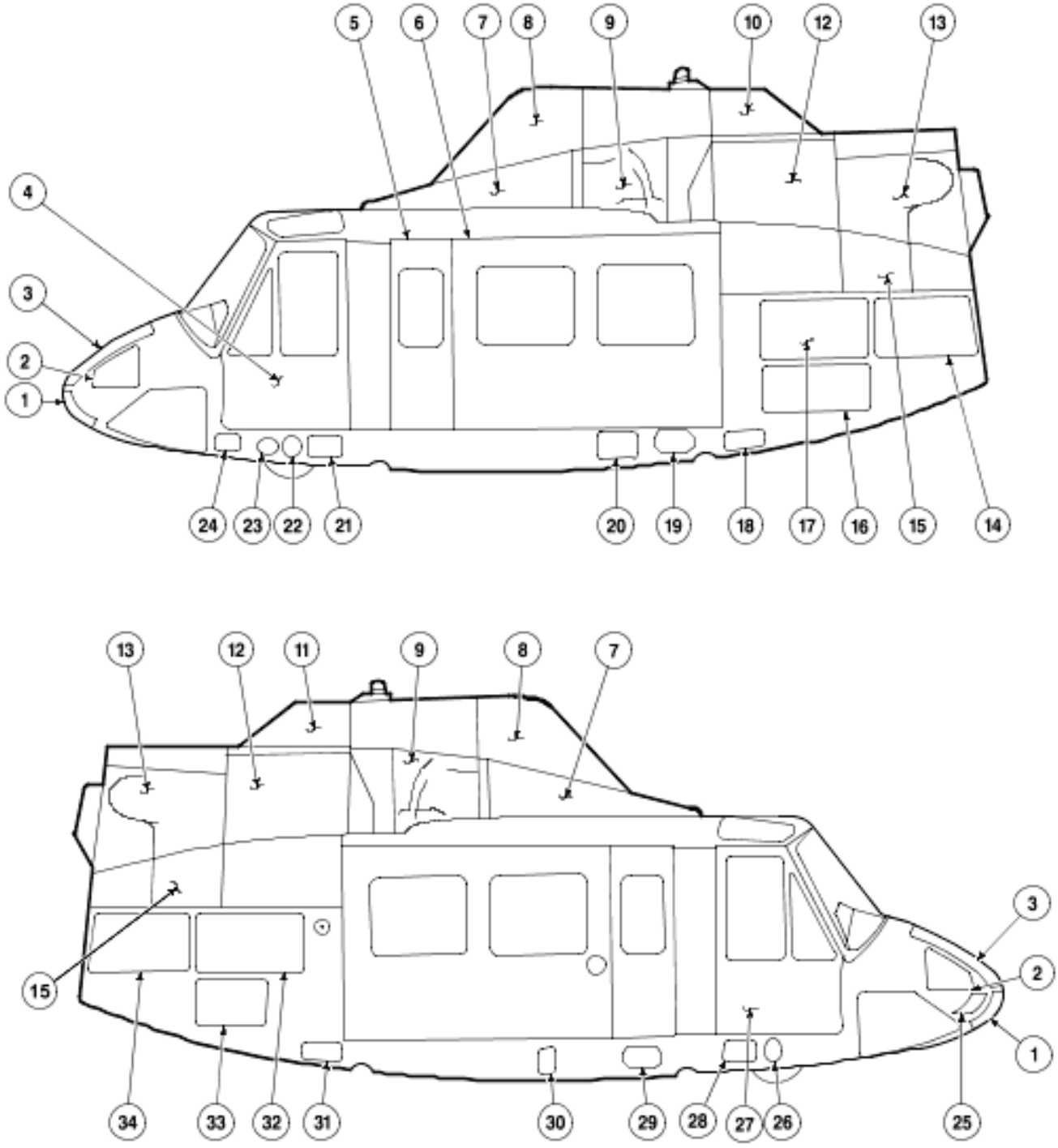
ICN-412-A-523000-B-97499-00002-A-001-01

Figure 1. Baggage Compartment Door (Sheet 1 of 1)

Nose Access Doors - Description

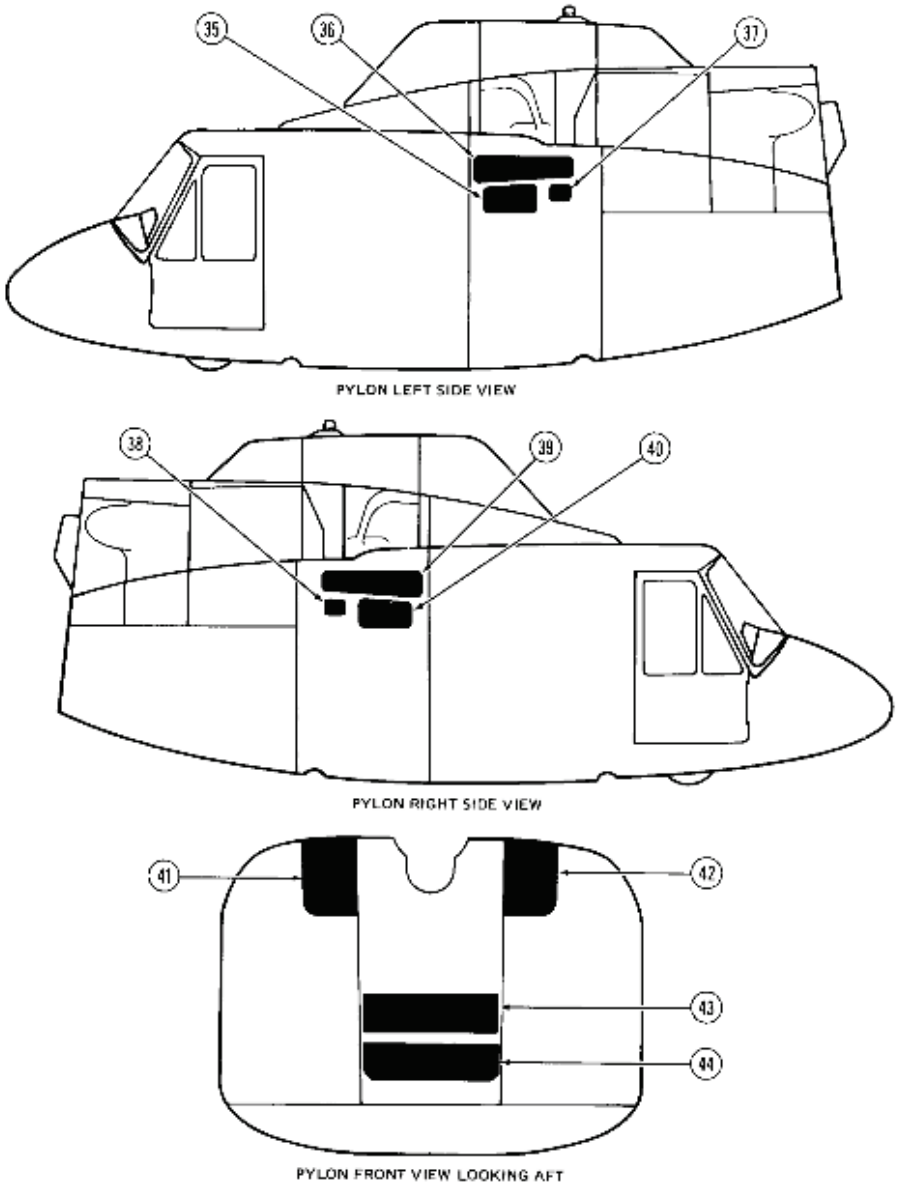
Two doors give access to the nose compartment. On helicopters prior to S/N 33001 through 33213 and S/N 36001 through 36257 (Pre TB 412-03-192), lower nose door (1, Figure 1) is of aluminum alloy construction and swings down on hinges after release of attaching studs.

On helicopters S/N 36258 and Subsequent, lower nose door is a one-piece composite construction secured with studs and screws. Upper nose door (3) is of fiberglass faced honeycomb construction and swings up after release of either six latches or six fasteners.



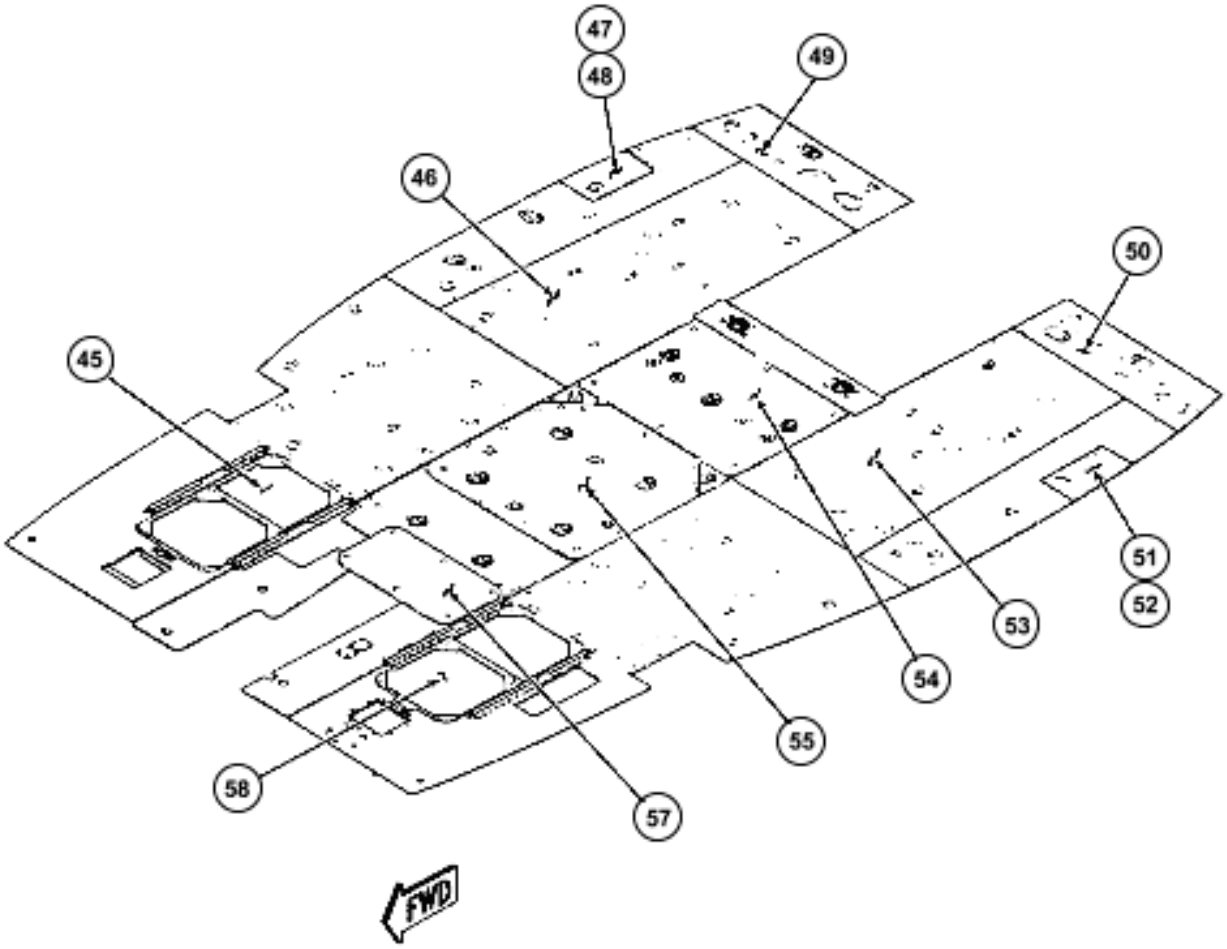
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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 1 of 6)



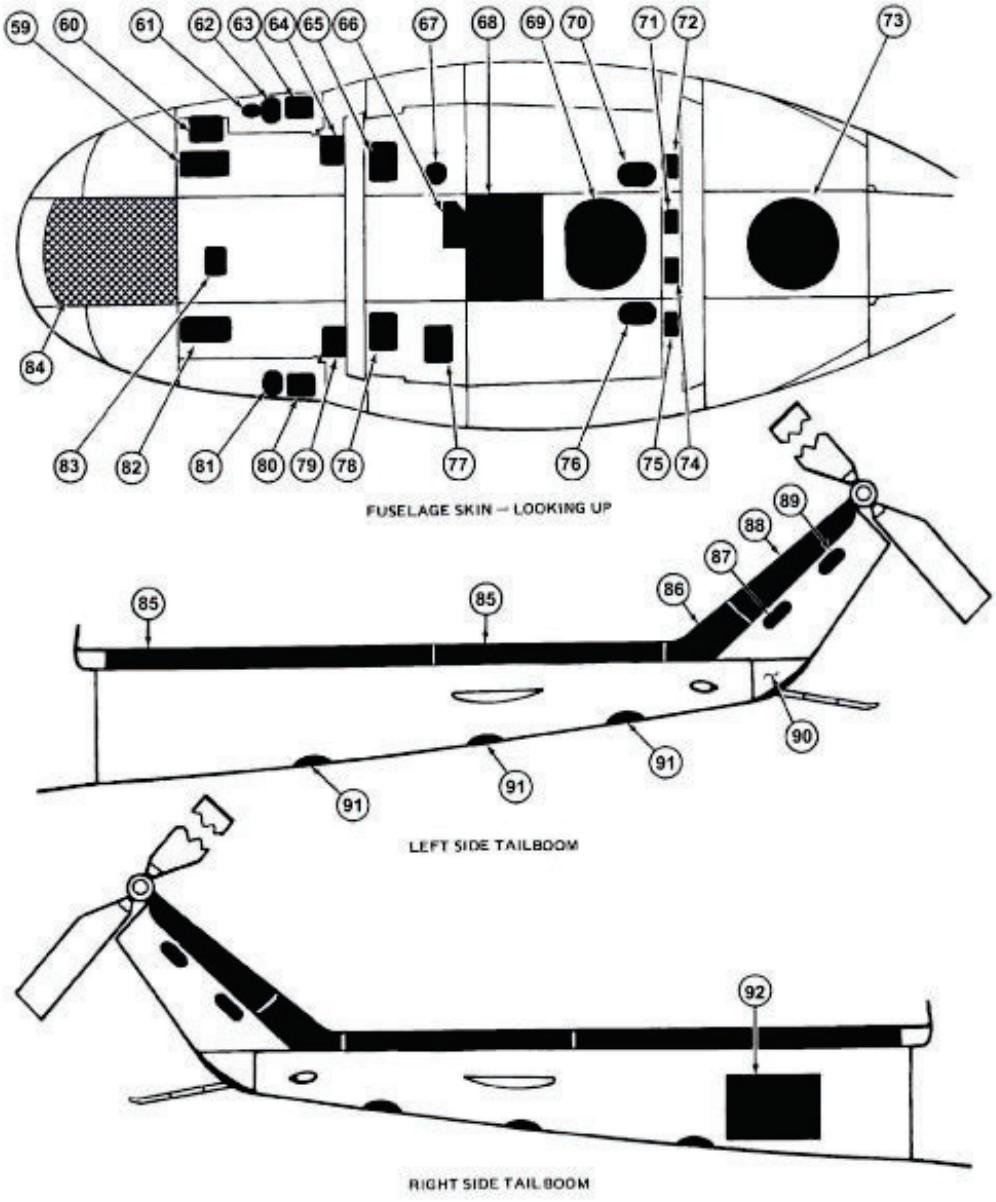
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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 2 of 6)



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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 3 of 6)



ICN-412-A-520000-B-97499-00004-A-001-01

Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 4 of 6)

INDEX NO.	NAME	ACCESS TO
1.	Lower nose door	Battery
2.	Cover	Automatic direction finder
3.	Upper nose door	Avionics equipment
4.	Crew door (copilot)	Copilot station
5.	Cargo hinged panel	Cabin
6.	Passenger cargo door	Cabin
7.	Transmission fairing	Hydraulic reservoirs, rotating controls
8.	Forward pylon fairing assembly	Transmission, swashplate, hydraulic reservoirs, hyd system 2 pump
9.	Fairing, engine inlet	Engine inlet, fuel controls, starter generator
10.	Aft pylon fairing assembly	Transmission, rotating controls
11.	Door	LH/RH engine
12.	Door	LH/RH engine power section
13.	Engine oil cooler support cowling assembly	Engine reduction gearbox
14.	Door	Radar altimeter
15.	Lower engine cowl assembly	LH/RH engine power section, fire extinguisher container
16.	Door	Electrical compartment
17.	Door	Navigation equipment
18.	Door	Avionics and electrical wiring
19.	Door	Hydraulic system 1 test connectors
20.	Door	Hydraulic system 2 test connectors
21.	Door	Cyclic flight controls
22.	Door	Flight controls
23.	Door	Flight controls
24.	Door	Flight controls
25.	Door	Auxiliary power unit receptacle
26.	Door	
27.	Crew door (pilot)	Pilot station
28.	Door	
29.	Door	
30.	Door	
31.	Door	Heater duct
32.	Door	
33.	Door	Heater variable control mixing valve
34.	Door	Tail rotor servo actuator, heater duct noise suppressor
35.	Door	Transmission, flight controls, lift link
36.	Door, pylon bulkhead	Transmission mounts
37.	Door	Transmission oil lines
38.	Door	Transmission oil line
39.	Door, pylon bulkhead	Transmission mounts
40.	Door	Hydraulic system 1 pump
41.	Cover	Engine accessory gearbox and drain valve, electric chip detector
42.	Cover	Engine accessory gearbox and drain valve, electric chip detector
43.	Cover	Flight controls hydraulic actuators
44.	Cover	Hydraulic integrated modules and filters

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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 5 of 6)

INDEX NO.	NAME	ACCESS TO
45.	Panel	Fore and aft force gradient, magnetic brake, flight controls
46.	Panel	Fuel lines and fuel cell
47.	Panel	
48.	Door	
49.	Panel	Fuel lines
50.	Panel	Fuel lines
51.	Panel	
52.	Door	
53.	Panel	Fuel cell
54.	Panel	Flight controls
55.	Panel	AFCS actuators, fwd fuel crossover valve
56.	Panel	Collective jackshaft
57.	Door	Flight controls
58.	Panel	Force gradient, flight controls
59.	Panel	Flight controls
60.	Door	Flight controls
61.	Door	Flight controls
62.	Door	Flight controls
63.	Door	Flight controls
64.	Door	Flight controls
65.	Door	Fuel lines, electrical wiring
66.	Door	Forward fuel crossover valve
67.	Door	Fuel line
68.	Door	VHF antenna
69.	Opening, cargo hook	Cargo hook, hydraulic lines, transmission sump
70.	Door	Fuel boost pump, fuel cell drain
71.	Door	Fuel lines, aft crossover and main
72.	Door	Fuel lines, aft crossover and main
73.	Door	Droop compensator control, Hydraulic lines, aft center fuel cell
74.	Door	Fuel lines, main and aft crossover
75.	Door	Fuel lines, main and aft crossover
76.	Door	Fuel boost pump, fuel cell drain
77.	Door	Fuel lines, heater controls, duct, mixing valve
78.	Door	AFCS, flight controls
79.	Door	Flight controls, collective
80.	Door	Flight controls, cyclic
81.	Door	Flight controls
82.	Door	Flight controls, searchlight
83.	Door	Flight controls, jackshaft
84.	Lower nose access panel	Electrical equipment
85.	Cover	Driveshaft, tail rotor
86.	Cover	Intermediate gearbox
87.	Door	Flight controls
88.	Cover	Driveshaft, tail rotor
89.	Door	Flight controls
90.	Fairing	Tail skid
91.	Door	Flight controls
92.	Door	Baggage compartment

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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 6 of 6)

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Nose Access Doors - Removal

Procedure

NOTE

To remove lower nose door (1, Figure 1), upper nose door (3) must be open.

Applicable: SN: 33001-33107:

1. Release latches or attaching studs from upper nose door (3).

2. Open upper nose door (3).

3. To remove upper nose door (3), disconnect hinge assemblies from brackets on door by removing screws, washers, and nuts.

Applicable: SN: 33001-33213 or 36001-36257 or TB 412-03-192: Pre:

4. To remove lower nose door (1) on helicopters:

4.1. Release attaching studs from lower nose door (1) by turning them counterclockwise.

4.2. Open lower nose door (1).

4.3. Disconnect hinge assemblies from fittings on structure supporting lower nose door (1) by removing screws, nuts, and washers.

4.4. Remove lower nose door (1).

Applicable: SN: 36258-Subsequent TB 412-03-192: Post:

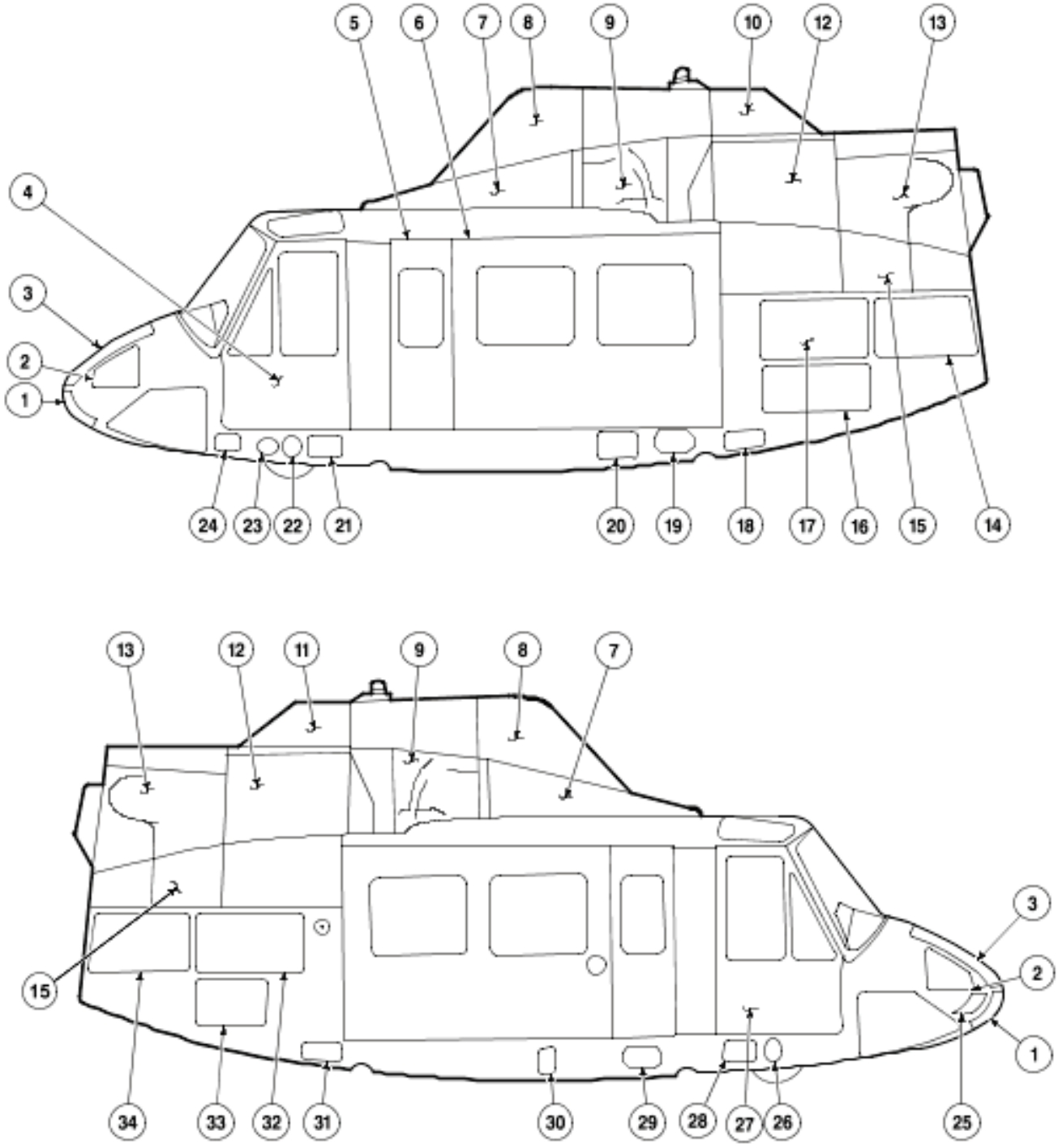
5. To remove lower nose door (1) on helicopters:

5.1. Support lower nose access panel (85) and release attaching studs by turning them counterclockwise.

5.2. Remove lower nose access panel (85).

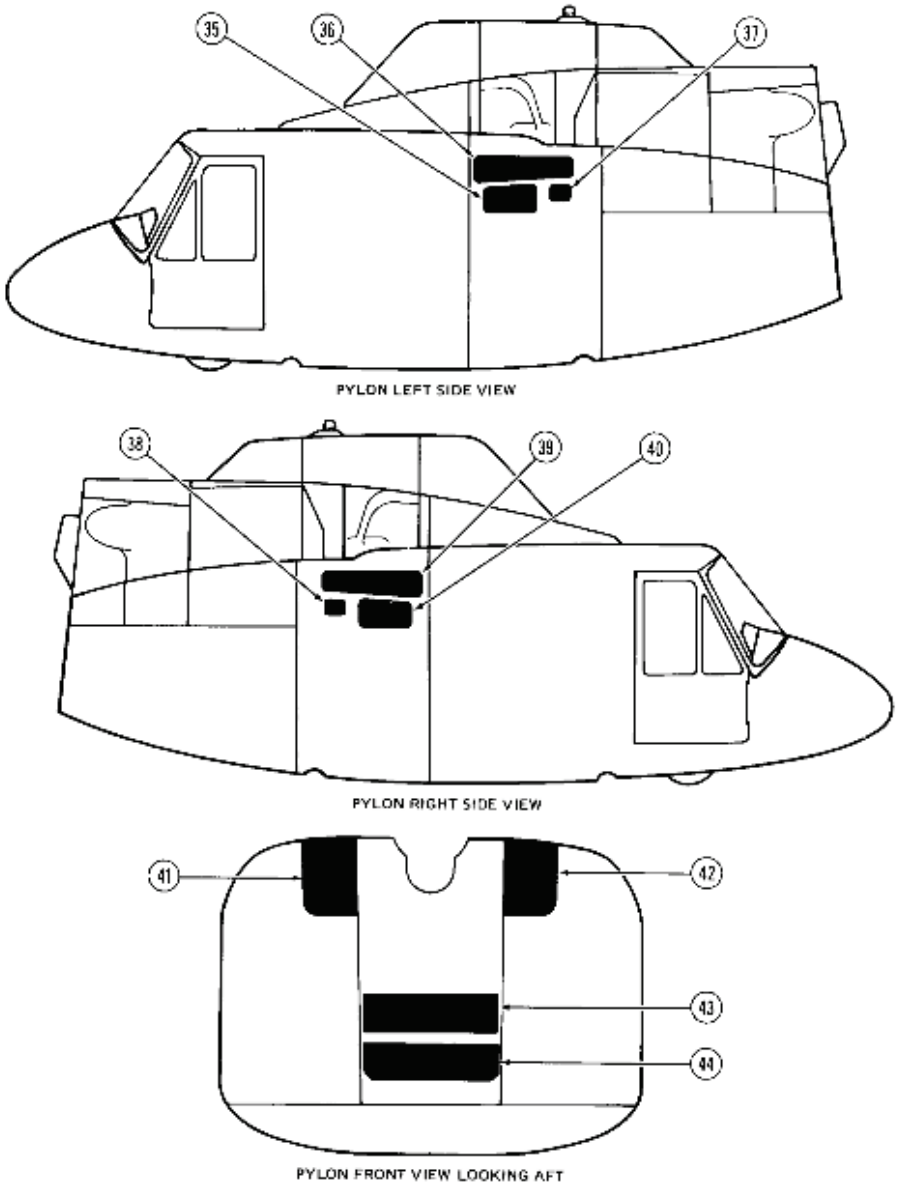
5.3. Support lower nose door (1), remove screws, and release attaching studs.

5.4. Remove lower nose door (1).



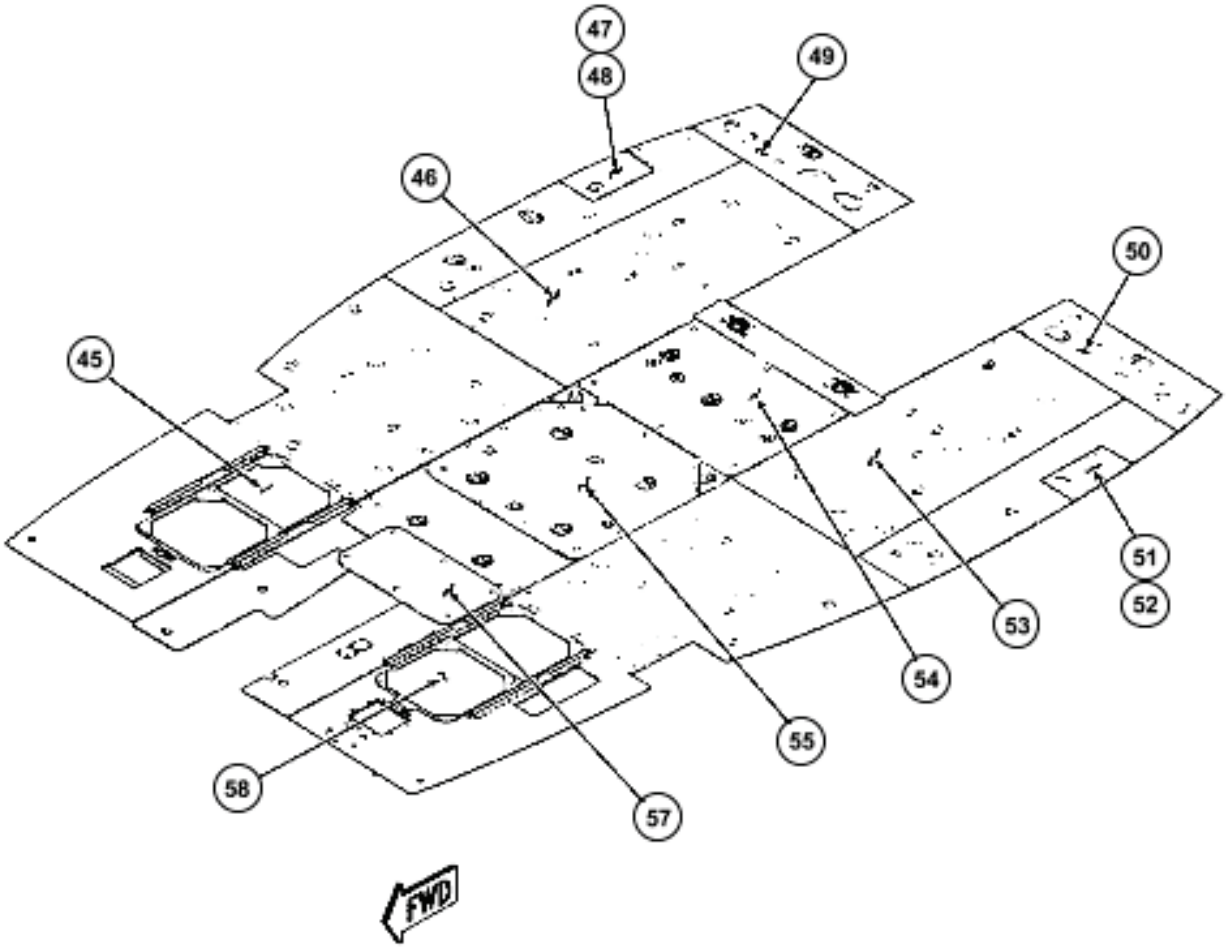
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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 1 of 6)



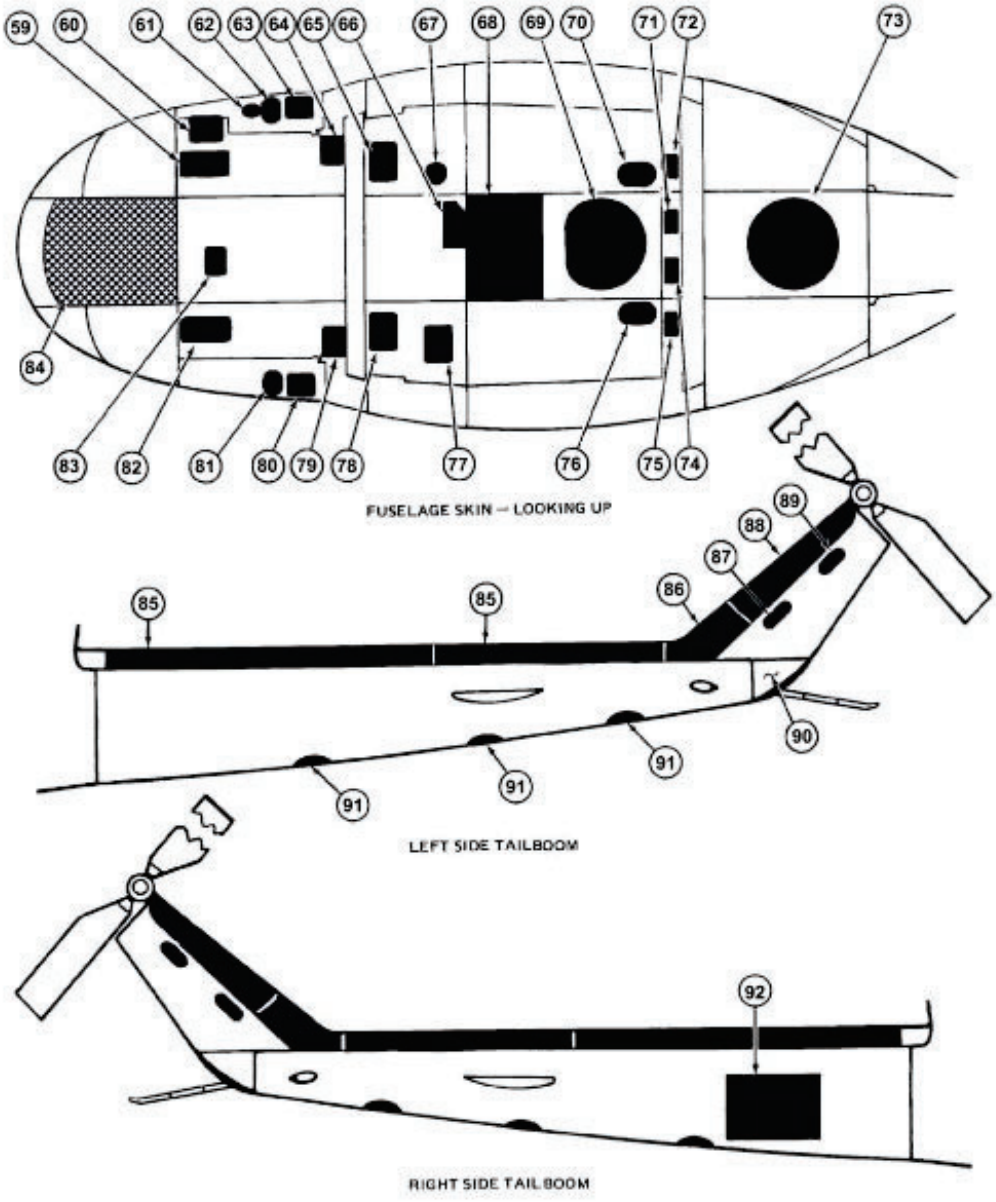
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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 2 of 6)



ICN-412-A-520000-B-97499-00003-A-001-01

Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 3 of 6)



ICN-412-A-520000-B-97499-00004-A-001-01

Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 4 of 6)

INDEX NO.	NAME	ACCESS TO
1.	Lower nose door	Battery
2.	Cover	Automatic direction finder
3.	Upper nose door	Avionics equipment
4.	Crew door (copilot)	Copilot station
5.	Cargo hinged panel	Cabin
6.	Passenger cargo door	Cabin
7.	Transmission fairing	Hydraulic reservoirs, rotating controls
8.	Forward pylon fairing assembly	Transmission, swashplate, hydraulic reservoirs, hyd system 2 pump
9.	Fairing, engine inlet	Engine inlet, fuel controls, starter generator
10.	Aft pylon fairing assembly	Transmission, rotating controls
11.	Door	LH/RH engine
12.	Door	LH/RH engine power section
13.	Engine oil cooler support cowling assembly	Engine reduction gearbox
14.	Door	Radar altimeter
15.	Lower engine cowl assembly	LH/RH engine power section, fire extinguisher container
16.	Door	Electrical compartment
17.	Door	Navigation equipment
18.	Door	Avionics and electrical wiring
19.	Door	Hydraulic system 1 test connectors
20.	Door	Hydraulic system 2 test connectors
21.	Door	Cyclic flight controls
22.	Door	Flight controls
23.	Door	Flight controls
24.	Door	Flight controls
25.	Door	Auxiliary power unit receptacle
26.	Door	
27.	Crew door (pilot)	Pilot station
28.	Door	
29.	Door	
30.	Door	
31.	Door	Heater duct
32.	Door	
33.	Door	Heater variable control mixing valve
34.	Door	Tail rotor servo actuator, heater duct noise suppressor
35.	Door	Transmission, flight controls, lift link
36.	Door, pylon bulkhead	Transmission mounts
37.	Door	Transmission oil lines
38.	Door	Transmission oil line
39.	Door, pylon bulkhead	Transmission mounts
40.	Door	Hydraulic system 1 pump
41.	Cover	Engine accessory gearbox and drain valve, electric chip detector
42.	Cover	Engine accessory gearbox and drain valve, electric chip detector
43.	Cover	Flight controls hydraulic actuators
44.	Cover	Hydraulic integrated modules and filters

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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 5 of 6)

INDEX NO.	NAME	ACCESS TO
45.	Panel	Fore and aft force gradient, magnetic brake, flight controls
46.	Panel	Fuel lines and fuel cell
47.	Panel	
48.	Door	
49.	Panel	Fuel lines
50.	Panel	Fuel lines
51.	Panel	
52.	Door	
53.	Panel	Fuel cell
54.	Panel	Flight controls
55.	Panel	AFCS actuators, fwd fuel crossover valve
56.	Panel	Collective jackshaft
57.	Door	Flight controls
58.	Panel	Force gradient, flight controls
59.	Panel	Flight controls
60.	Door	Flight controls
61.	Door	Flight controls
62.	Door	Flight controls
63.	Door	Flight controls
64.	Door	Flight controls
65.	Door	Fuel lines, electrical wiring
66.	Door	Forward fuel crossover valve
67.	Door	Fuel line
68.	Door	VHF antenna
69.	Opening, cargo hook	Cargo hook, hydraulic lines, transmission sump
70.	Door	Fuel boost pump, fuel cell drain
71.	Door	Fuel lines, aft crossover and main
72.	Door	Fuel lines, aft crossover and main
73.	Door	Droop compensator control, Hydraulic lines, aft center fuel cell
74.	Door	Fuel lines, main and aft crossover
75.	Door	Fuel lines, main and aft crossover
76.	Door	Fuel boost pump, fuel cell drain
77.	Door	Fuel lines, heater controls, duct, mixing valve
78.	Door	AFCS, flight controls
79.	Door	Flight controls, collective
80.	Door	Flight controls, cyclic
81.	Door	Flight controls
82.	Door	Flight controls, searchlight
83.	Door	Flight controls, jackshaft
84.	Lower nose access panel	Electrical equipment
85.	Cover	Driveshaft, tail rotor
86.	Cover	Intermediate gearbox
87.	Door	Flight controls
88.	Cover	Driveshaft, tail rotor
89.	Door	Flight controls
90.	Fairing	Tail skid
91.	Door	Flight controls
92.	Door	Baggage compartment

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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 6 of 6)

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Nose Access Doors - Inspection

Procedure

1. Inspect hinges for cracks and distortion.
2. Inspect latches for operation and security of installation.
3. Inspect doors for structural damage.
4. Inspect door seals for secure bonding and for deterioration.

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Nose Access Doors - Repair

Procedure

1. Replace unserviceable parts in nose access doors.
2. Repair nose access door structural damage. Refer to the BHT-MED-SRM-1 and to FAA Publication AC 43.13-1B Aircraft Inspection and Repair Manual for structural maintenance instructions.
3. Replace damaged nose access door seals (412-A-52-00-00-00A-720A-A / 00777).

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Nose Access Doors - Installation

Procedure

NOTE

To install or close lower nose door (1, Figure 1), upper nose door (3) must be open.

Applicable: SN: 33001-33213 or 36001-36257 TB 412-03-192: Pre:

1. To install lower nose door (1) on helicopters:

1.1. Support lower nose door (1) and connect assemblies to fittings on structure with screws, nuts, and washers.

1.2. Close lower nose door (1).

1.3. Secure lower nose door (1) with attaching studs by turning them clockwise.

Applicable: SN: 36258-Subsequent or TB 412-03-192: Post:

2. To install lower nose door (1) on helicopters prior to:

2.1. Hold lower nose door (1) in place. Secure with screws and attaching studs.

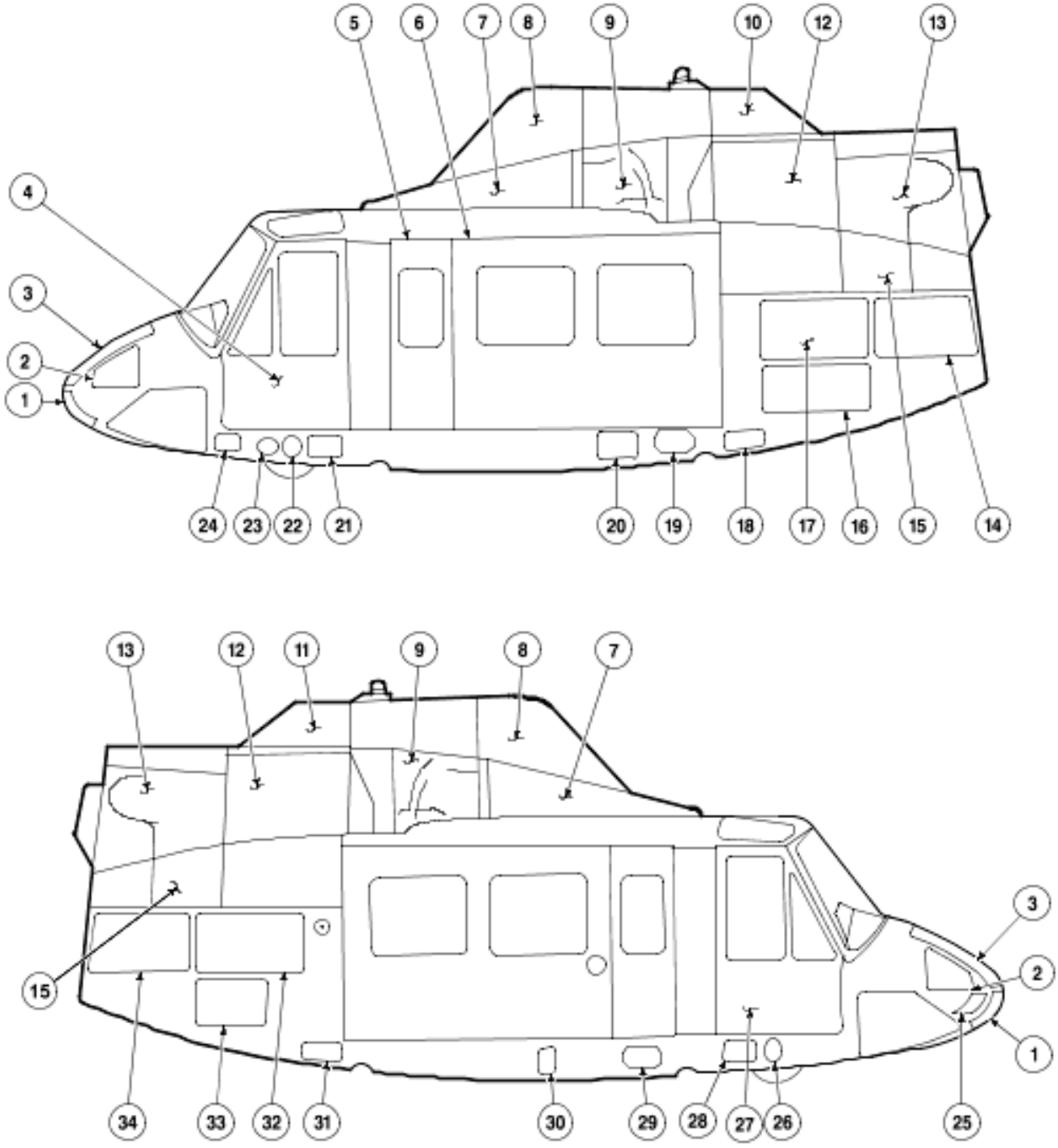
2.2. Support lower nose access panel (85). Secure with attaching studs.

3. To install upper nose door (3) and connect hinge assemblies to brackets with screws, washers, and nuts.

4. Close upper nose door (3).

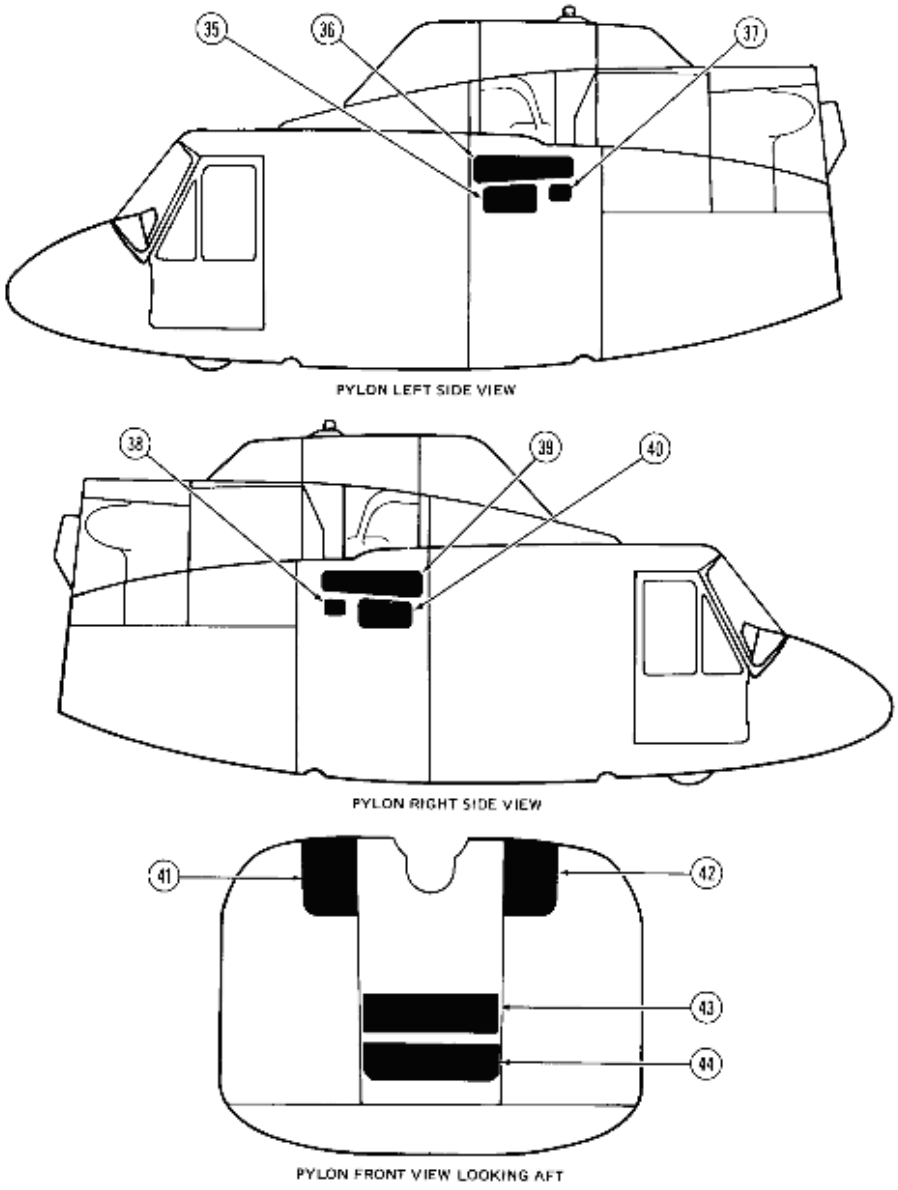
Applicable: SN: 33001-33107:

5. Secure upper nose door (3) with latches or attaching studs.



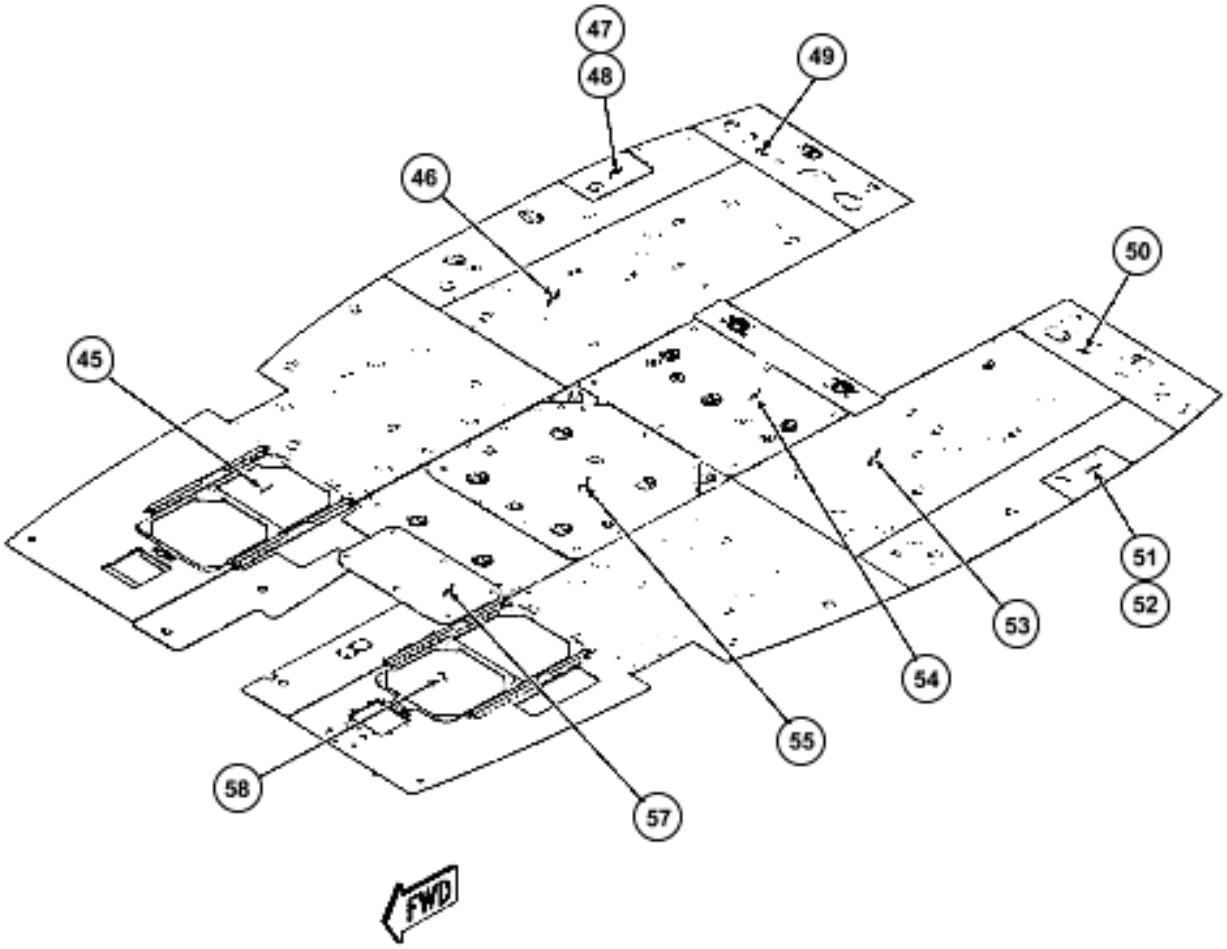
ICN-412-A-520000-B-97499-00001-A-001-01

Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 1 of 6)



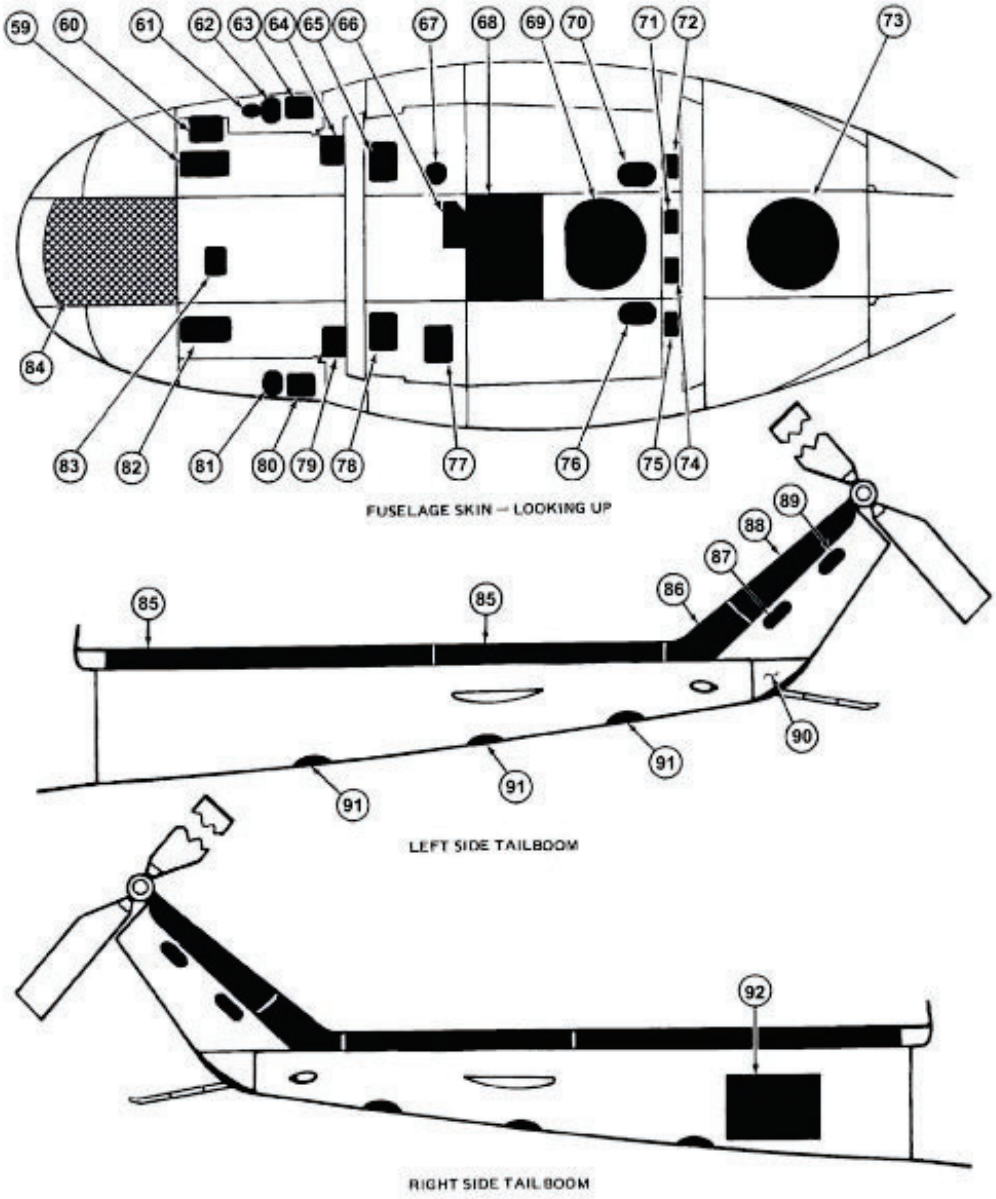
ICN-412-A-520000-B-97499-00002-A-001-01

Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 2 of 6)



ICN-412-A-520000-B-97499-00003-A-001-01

Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 3 of 6)



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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 4 of 6)

INDEX NO.	NAME	ACCESS TO
1.	Lower nose door	Battery
2.	Cover	Automatic direction finder
3.	Upper nose door	Avionics equipment
4.	Crew door (copilot)	Copilot station
5.	Cargo hinged panel	Cabin
6.	Passenger cargo door	Cabin
7.	Transmission fairing	Hydraulic reservoirs, rotating controls
8.	Forward pylon fairing assembly	Transmission, swashplate, hydraulic reservoirs, hyd system 2 pump
9.	Fairing, engine inlet	Engine inlet, fuel controls, starter generator
10.	Aft pylon fairing assembly	Transmission, rotating controls
11.	Door	LH/RH engine
12.	Door	LH/RH engine power section
13.	Engine oil cooler support cowling assembly	Engine reduction gearbox
14.	Door	Radar altimeter
15.	Lower engine cowl assembly	LH/RH engine power section, fire extinguisher container
16.	Door	Electrical compartment
17.	Door	Navigation equipment
18.	Door	Avionics and electrical wiring
19.	Door	Hydraulic system 1 test connectors
20.	Door	Hydraulic system 2 test connectors
21.	Door	Cyclic flight controls
22.	Door	Flight controls
23.	Door	Flight controls
24.	Door	Flight controls
25.	Door	Auxiliary power unit receptacle
26.	Door	
27.	Crew door (pilot)	Pilot station
28.	Door	
29.	Door	
30.	Door	
31.	Door	Heater duct
32.	Door	
33.	Door	Heater variable control mixing valve
34.	Door	Tail rotor servo actuator, heater duct noise suppressor
35.	Door	Transmission, flight controls, lift link
36.	Door, pylon bulkhead	Transmission mounts
37.	Door	Transmission oil lines
38.	Door	Transmission oil line
39.	Door, pylon bulkhead	Transmission mounts
40.	Door	Hydraulic system 1 pump
41.	Cover	Engine accessory gearbox and drain valve, electric chip detector
42.	Cover	Engine accessory gearbox and drain valve, electric chip detector
43.	Cover	Flight controls hydraulic actuators
44.	Cover	Hydraulic integrated modules and filters

ICN-412-A-520000-B-97499-00005-A-001-01

Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 5 of 6)

INDEX NO.	NAME	ACCESS TO
45.	Panel	Fore and aft force gradient, magnetic brake, flight controls
46.	Panel	Fuel lines and fuel cell
47.	Panel	
48.	Door	
49.	Panel	Fuel lines
50.	Panel	Fuel lines
51.	Panel	
52.	Door	
53.	Panel	Fuel cell
54.	Panel	Flight controls
55.	Panel	AFCS actuators, fwd fuel crossover valve
56.	Panel	Collective jackshaft
57.	Door	Flight controls
58.	Panel	Force gradient, flight controls
59.	Panel	Flight controls
60.	Door	Flight controls
61.	Door	Flight controls
62.	Door	Flight controls
63.	Door	Flight controls
64.	Door	Flight controls
65.	Door	Fuel lines, electrical wiring
66.	Door	Forward fuel crossover valve
67.	Door	Fuel line
68.	Door	VHF antenna
69.	Opening, cargo hook	Cargo hook, hydraulic lines, transmission sump
70.	Door	Fuel boost pump, fuel cell drain
71.	Door	Fuel lines, aft crossover and main
72.	Door	Fuel lines, aft crossover and main
73.	Door	Droop compensator control, Hydraulic lines, aft center fuel cell
74.	Door	Fuel lines, main and aft crossover
75.	Door	Fuel lines, main and aft crossover
76.	Door	Fuel boost pump, fuel cell drain
77.	Door	Fuel lines, heater controls, duct, mixing valve
78.	Door	AFCS, flight controls
79.	Door	Flight controls, collective
80.	Door	Flight controls, cyclic
81.	Door	Flight controls
82.	Door	Flight controls, searchlight
83.	Door	Flight controls, jackshaft
84.	Lower nose access panel	Electrical equipment
85.	Cover	Driveshaft, tail rotor
86.	Cover	Intermediate gearbox
87.	Door	Flight controls
88.	Cover	Driveshaft, tail rotor
89.	Door	Flight controls
90.	Fairing	Tail skid
91.	Door	Flight controls
92.	Door	Baggage compartment

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Figure 1. Access Doors, Panels, Cowlings, Fairings, and Covers (Sheet 6 of 6)

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Miscellaneous Access Doors and Panels - Description

Doors and panels are provided at various locations on the fuselage for access to interior areas. Three types of construction are used: Honeycomb with sheet metal and fiberglass faces, honeycomb with both faces of fiberglass, and contoured flat sheet metal. Screws or twist type studs retain doors in place.

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Miscellaneous Access Doors and Panels - Removal

Procedure

1. Remove screws or disengage studs as applicable and remove door, panel, or fairing.

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Miscellaneous Access Doors and Panels - Inspection

Procedure

1. Inspect doors, panels, and fairings for damaged and/or missing twist-type studs if applicable.
2. Inspect doors, panels, and fairings for mechanical and corrosion damage.

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Miscellaneous Access Doors and Panels - Repair

Procedure

1. Replace unserviceable and/or missing twist-type studs.

2. Repair structural and corrosion damage. Refer to the BHT-MED-SRM-1 and to FAA Publication AC 43.13-1B Aircraft Inspection and Repair Manual for structural repair instructions.

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Miscellaneous Access Doors and Panels - Installation

Procedure

1. Position door, panel, or fairing on helicopter and fasten twist-type stud or install screws as applicable.

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Tail Rotor Driveshaft Access Covers - Description

The tail rotor driveshaft access cover installation consists of three formed aluminum alloy parts. Each cover is hinged on right side and is secured on left side by twist-type studs through support angles between BS 18 and 200.

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Tail Rotor Driveshaft Access Covers - Removal

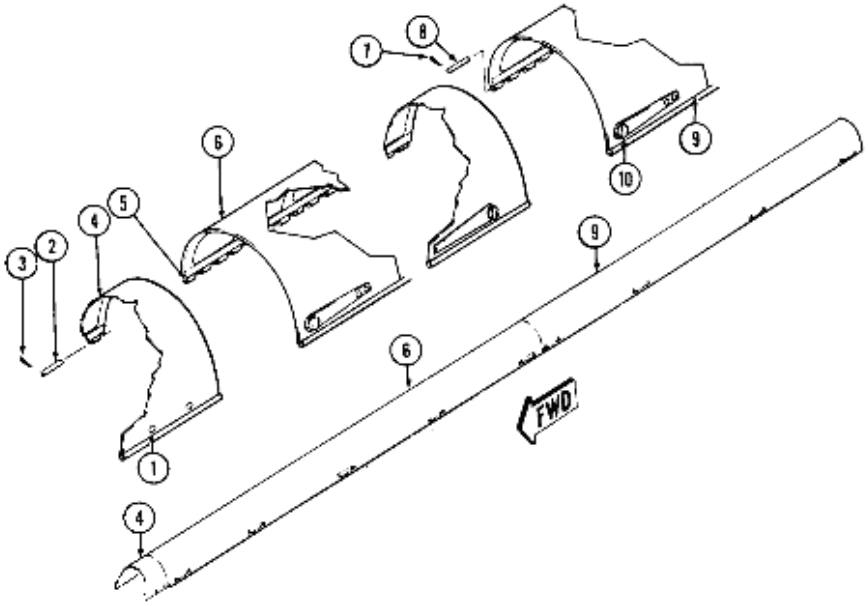
Procedure

1. Loosen stud (1, Figure 1) in two places and hinge access cover (4).

2. Loosen studs (10) and open hinged access covers (6 and 9).

3. Remove cotter pin (3) and hinge pin (2). Remove access covers (4 and 6).

4. Remove cotter pin (7), hinge pin (8), and aft access cover (9).



1. Stud (2 places)
2. Hinge pin
3. Cotter pin
4. Access cover
5. Hinge half (typical)
6. Forward access cover
7. Cotter pin
8. Hinge pin
9. Aft access cover
10. Stud with ejecting blade (9 places)

ICN-412-A-524000-B-97499-00001-A-001-01

Figure 1. Tail Rotor Driveshaft Access Cover (Sheet 1 of 1)

Tail Rotor Driveshaft Access Covers - Inspection

Procedure

1. Inspect access covers (4, 6, and 9, Figure 1) for missing, damaged, or loose studs with ejecting blades (10).

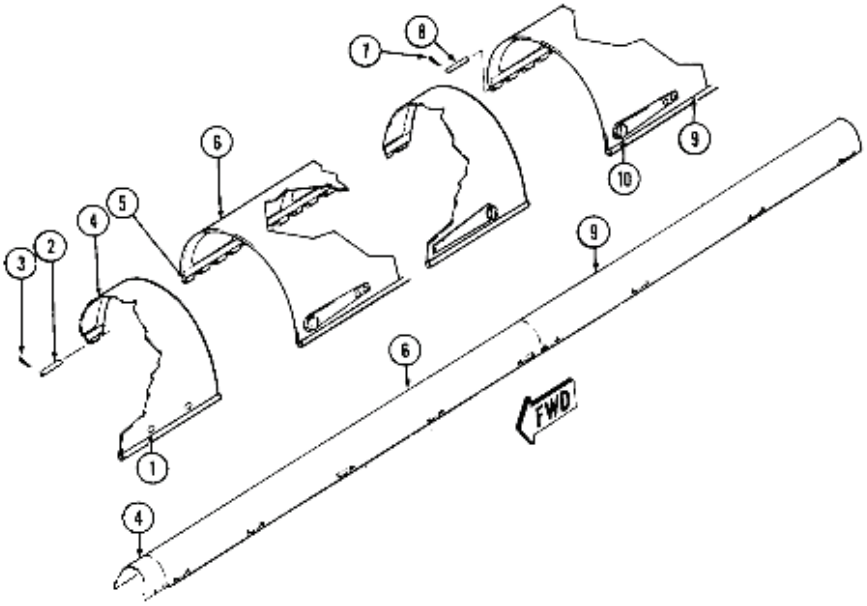
2. Inspect support angles on left side for missing, damaged, or loose grommets.

3. Inspect support angles and hinge on right side for corrosion and mechanical damage.

4. Inspect the hinges for the following:

- Corrosion and mechanical damage
- Damage to the upper end and lower end loops. No damage is permitted
- For missing loop, it is acceptable to have a missing loop in any 8 inches (230 mm) length
- For worn loops, it is acceptable to have a maximum quantity 3 consecutive worn loops as long as there is a maximum of quantity 4 undamaged loops on each side (except at the top and bottom as mentioned above), and as long as the limitations given above are also met

5. Inspect access covers (4,6 and 9) for corrosion and mechanical damage.



1. Stud (2 places)
2. Hinge pin
3. Cotter pin
4. Access cover
5. Hinge half (typical)
6. Forward access cover
7. Cotter pin
8. Hinge pin
9. Aft access cover
10. Stud with ejecting blade (9 places)

ICN-412-A-524000-B-97499-00001-A-001-01

Figure 1. Tail Rotor Driveshaft Access Cover (Sheet 1 of 1)

Tail Rotor Driveshaft Access Covers - Repair

Procedure

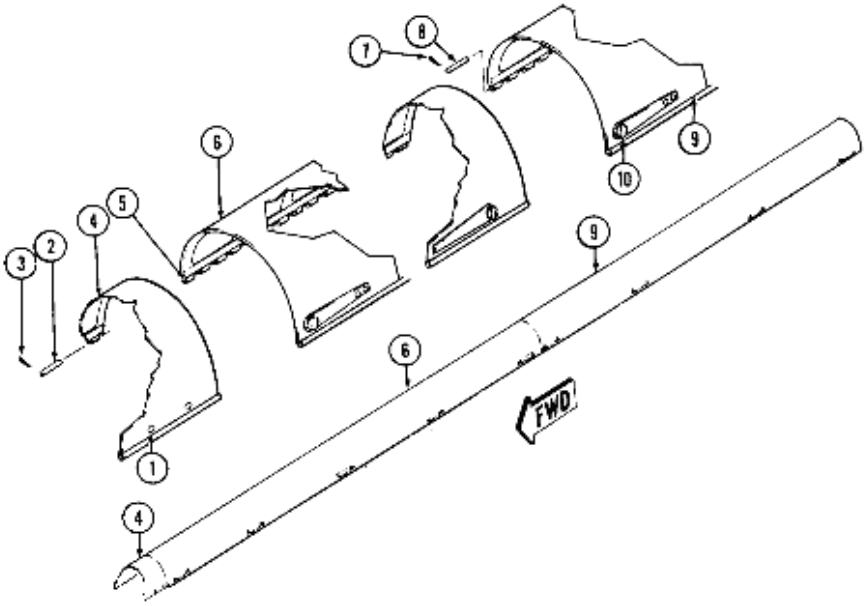
1. Repair structural and corrosion damage. Refer to the BHT-MED-SRM-1 and to FAA Publication AC 43.13-1B Aircraft Inspection and Repair Manual for structural repair instructions.

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Tail Rotor Driveshaft Access Covers - Installation

Procedure

1. Install aft access cover (9, Figure 1) with hinge pin (8) and secure with cotter pin (7).
2. Align forward access cover (6) with hinge half (5). Align access cover (4) with hinge half (5).
3. Install hinge pin (2) forward to aft through access cover (4) then through forward access cover (6), and secure with cotter pin (3).
4. Close access cover (4) and fasten two studs (1). Close access covers (6 and 9) and latch studs (10).



1. Stud (2 places)
2. Hinge pin
3. Cotter pin
4. Access cover
5. Hinge half (typical)
6. Forward access cover
7. Cotter pin
8. Hinge pin
9. Aft access cover
10. Stud with ejecting blade (9 places)

ICN-412-A-524000-B-97499-00001-A-001-01

Figure 1. Tail Rotor Driveshaft Access Cover (Sheet 1 of 1)

Cabin Roof Windows - Description

The cabin roof windows are made of tinted acrylic plastic.

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Cabin Roof Windows - Removal

Procedure

1. Remove free air temperature gauge (412-A-95-63-03-00A-042A-A / 01866).
2. Remove nuts, washers, and screws attaching windows to roof structure.
3. Separate window from sealant using a non-metallic spatula, putty knife, or similar tool (e.g., Snap-on P/N YA109A).
4. Carefully remove window from helicopter.

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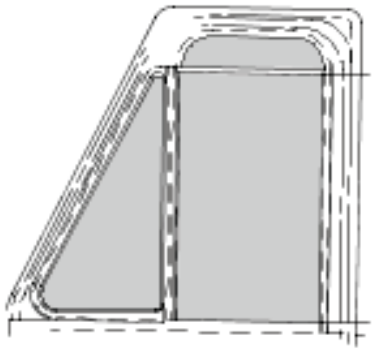
Cabin Roof Windows - Inspection

Procedure

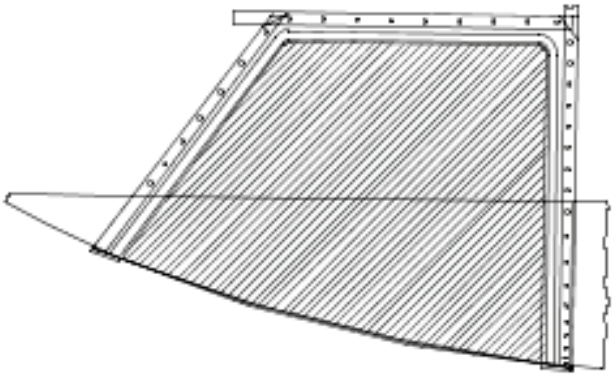
1. Inspect windows in accordance with Figure 1.



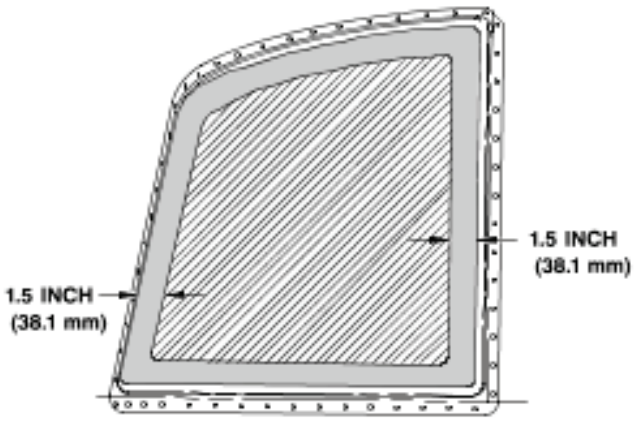
PASSENGER DOOR WINDOWS



PILOT AND COPILOT DOOR WINDOWS



LOWER CABIN NOSE WINDOWS



CABIN ROOF WINDOWS

Area "A": Scratches and pits are permitted in this area provided they are not so numerous or form such a pattern as to be objectionable to the viewer. Cracks, holes, or other damage may be temporarily repaired by stop drilling, patching or other approved methods (refer to FAA Advisory Circular 43.13-1, Aircraft Inspection and Repair), but window must be replaced at the earliest opportunity.

Area "B": Scratches and pits are permitted in this area, providing the structural integrity of the window is not impaired. Cracks, holes, or other damage may be repaired by stop drilling, patching or other approved methods provided structural integrity is not impaired (refer to FAA Advisory Circular 43.13-1, Aircraft Inspection and Repair).

ICN-412-A-520100-B-97499-00001-A-001-01

Figure 1. Windows - Damage Limits (Sheet 1 of 1)

Cabin Roof Windows - Repair

Procedure

1. Repair damage in accordance with the BHT-MED-SRM-1 and FAA Publication AC 43.13-1B Aircraft Inspection and Repair Manual.

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Cabin Roof Windows - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	
Sealant	C-308	AR	

Procedure

1. Prior to installation of cabin roof windows, wash windows with a mild detergent and water solution. Dry with a soft dry cloth.

2. Remove all sealing compound from mounting flange with a non-metallic putty knife, spatula, or similar tool.



WHEN CLEANING WINDOWS AND MOUNTING FLANGE, USE ONLY TYPE II aliphatic naphtha (C-305). ALIPHATIC NAPHTHA TYPE I WILL CLOUD WINDOWS.

3. Wipe clean mounting flange with cloth dampened with aliphatic naphtha (C-305).

4. If a new window is being installed, position window over opening. Trim only enough surplus material from edge of window to permit proper alignment of window against mounting flange.

NOTE

Do not trim window to final size until all mounting holes have been drilled.

5. Position window against mounting flange and, using a No. 4 drill, back drill two holes on each edge of window. Use holes in mounting flange as template.

6. Secure window to mounting flange with four existing screws, washer, and nuts. Lightly tighten. Finish drilling holes in windows.

7. Measure window and mark proper edge distance required for installation. Remove window and trim window edge to 0.80 inch (20.32 mm).

8. Remove all dust and dirt from windows and mounting flange with a clean cloth dampened with aliphatic naphtha (C-305).

9. Apply a 0.125 inch (3.18 mm) bead of sealant (C-308).

10. Position window to mounting flange and align holes. Install retaining screws, washers, and nuts.

11. Remove excess sealant (C-308) from around window with a clean cloth dampened with aliphatic naphtha (C-305).

12. Install free air temperature gauge (412-A-95-63-03-00A-042A-A / 01866).

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Upper Crew Door Windows - Description

The upper crew door windows are constructed of clear acrylic material.

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Upper Crew Door Windows - Removal

Procedure

1. Remove nuts, washers, and screws attaching window to door.
2. Separate window from sealing compound.
3. Remove window from door.

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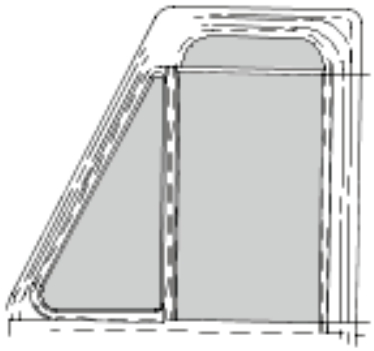
Upper Crew Door Windows - Inspection

Procedure

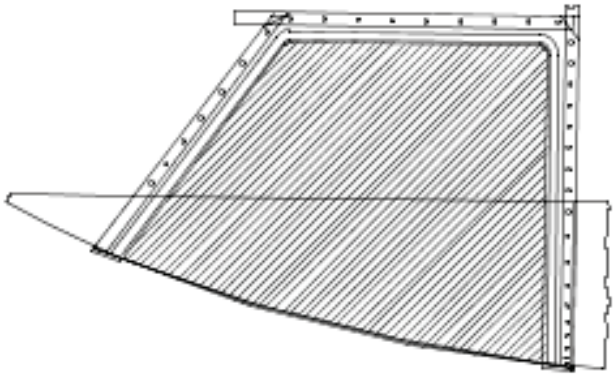
1. Inspect windows in accordance with Figure 1.



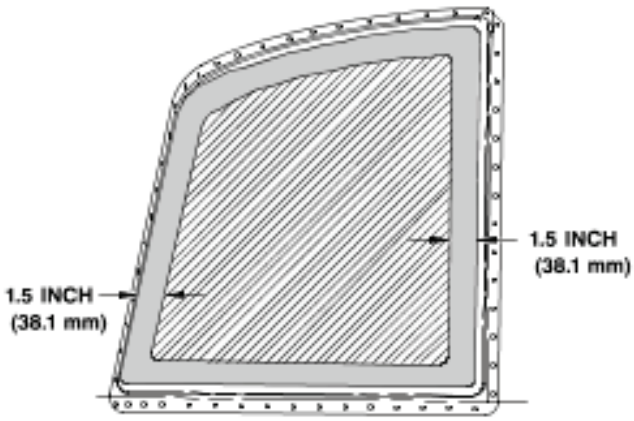
PASSENGER DOOR WINDOWS



PILOT AND COPILOT DOOR WINDOWS



LOWER CABIN NOSE WINDOWS



CABIN ROOF WINDOWS

Area "A": Scratches and pits are permitted in this area provided they are not so numerous or form such a pattern as to be objectionable to the viewer. Cracks, holes, or other damage may be temporarily repaired by stop drilling, patching or other approved methods (refer to FAA Advisory Circular 43.13-1, Aircraft Inspection and Repair), but window must be replaced at the earliest opportunity.

Area "B": Scratches and pits are permitted in this area, providing the structural integrity of the window is not impaired. Cracks, holes, or other damage may be repaired by stop drilling, patching or other approved methods provided structural integrity is not impaired (refer to FAA Advisory Circular 43.13-1, Aircraft Inspection and Repair).

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Figure 1. Windows - Damage Limits (Sheet 1 of 1)

Upper Crew Door Windows - Repair

Procedure

1. Refer to FAA Publication AC 43.13-1B Aircraft Inspection and Repair Manual for repair procedures.

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Upper Crew Door Windows - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	
Sealant	C-308	AR	

Procedure

1. Prior to installation of upper crew door windows, wash windows with a mild detergent and water solution. Dry with a soft dry cloth.

2. Remove old sealing compound from mounting flange with a non-metallic putty knife, spatula, or similar tool.



WHEN CLEANING WINDOWS AND MOUNTING FLANGE, USE ONLY TYPE II aliphatic naphtha (C-305). ALIPHATIC NAPHTHA TYPE I WILL CLOUD WINDOWS.

3. Wipe and clean mounting flange with cloth dampened with aliphatic naphtha (C-305).

4. If a new window is being installed, position window over opening. Trim only enough surplus material from edge of window to permit proper alignment of window against mounting flange.

NOTE

Do not trim window to final size until all mounting holes have been drilled.

5. Position window against mounting flange. Using a No. 4 drill, back drill two holes on each edge of window. Use holes in mounting flange as template.

6. Secure window to mounting flange with four existing screws, washer, and nuts. Lightly tighten. Finish drilling holes in windows.

7. Measure window and mark proper edge distance required for installation.

8. Remove window and trim window edge per measurements from Step 7..

9. Remove all dust and dirt from windows and mounting flange with a clean cloth moistened in aliphatic naphtha (C-305).

10. Apply a 0.125 inch (3.18 mm) bead of sealant (C-308).

11. Position window to mounting flange, align holes, and install screws, washers, and nuts.

12. Remove excess sealant (C-308) from around window with a clean cloth dampened with aliphatic naphtha (C-305).

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Forward Crew Door Windows - Description

The forward crew door windows are made of clear acrylic material.

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Forward Crew Door Windows - Removal

Procedure

1. Drill out rivets using a No. 30 drill.
2. Separate window from sealant using a non-metallic putty knife, spatula, or similar tool.
3. Remove window from helicopter.

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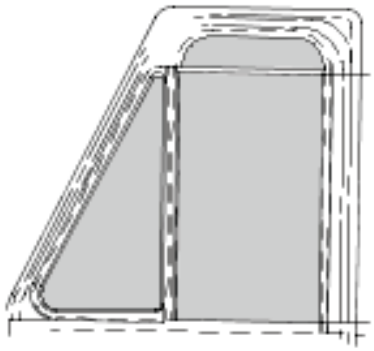
Forward Crew Door Windows - Inspection

Procedure

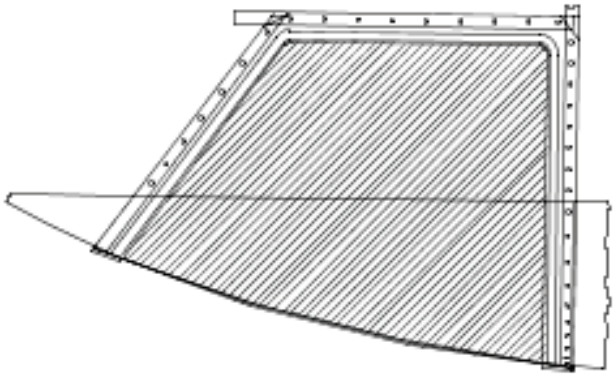
1. Inspect window in accordance with Figure 1.



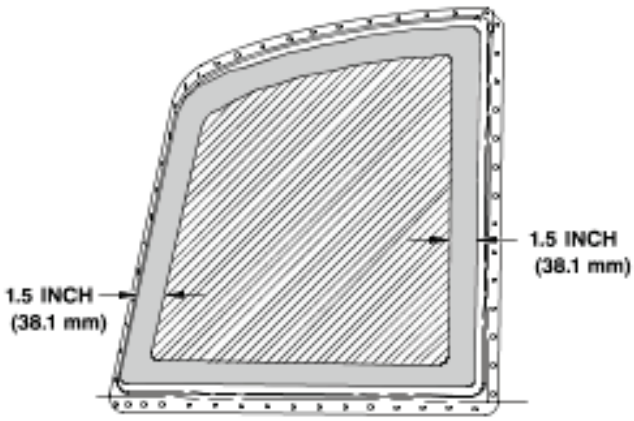
PASSENGER DOOR WINDOWS



PILOT AND COPILOT DOOR WINDOWS



LOWER CABIN NOSE WINDOWS



CABIN ROOF WINDOWS

Area "A": Scratches and pits are permitted in this area provided they are not so numerous or form such a pattern as to be objectionable to the viewer. Cracks, holes, or other damage may be temporarily repaired by stop drilling, patching or other approved methods (refer to FAA Advisory Circular 43.13-1, Aircraft Inspection and Repair), but window must be replaced at the earliest opportunity.

Area "B": Scratches and pits are permitted in this area, providing the structural integrity of the window is not impaired. Cracks, holes, or other damage may be repaired by stop drilling, patching or other approved methods provided structural integrity is not impaired (refer to FAA Advisory Circular 43.13-1, Aircraft Inspection and Repair).

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Figure 1. Windows - Damage Limits (Sheet 1 of 1)

Forward Crew Door Windows - Repair

Procedure

1. Refer to FAA Publication AC 43.13-1B Aircraft Inspection and Repair Manual for repair procedures.

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Forward Crew Door Windows - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	
Sealant	C-308	AR	

Procedure

1. Remove old sealant from mounting flange with a non-metallic putty knife, spatula, or similar tool.



WHEN CLEANING MOUNTING FLANGE AND WINDOWS, USE ONLY TYPE II aliphatic naphtha (C-305). ALIPHATIC NAPHTHA TYPE I WILL CLOUD WINDOWS.

2. Wipe and clean edging and window with aliphatic naphtha (C-305).

3. When installing new window, trim window to final size. Position window into mounting flange and drill mounting holes.

4. Apply sealant (C-308) in a bead around window edge.

5. Install rivets to secure window to crew door mounting flange.

6. Remove excess sealant (C-308) from around edge of window using clean cloth dampened with aliphatic naphtha (C-305).

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Lower Crew Door Windows - Description

The lower crew door windows are made of clear acrylic material.

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Lower Crew Door Windows - Removal

Procedure

1. Remove access door from lower edge of door.
2. Remove screws attaching handle to lower crew door window.
3. Loosen adjustable stop and slide window down and remove through access door opening.

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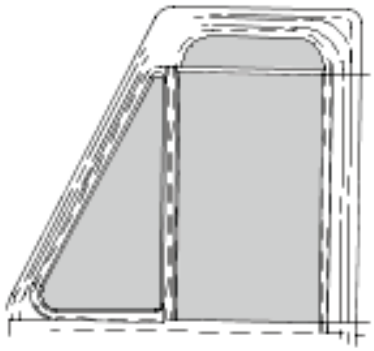
Lower Crew Door Windows - Inspection

Procedure

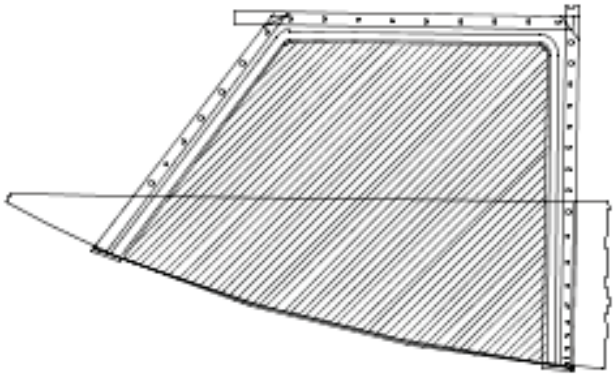
1. Inspect window in accordance with Figure 1.



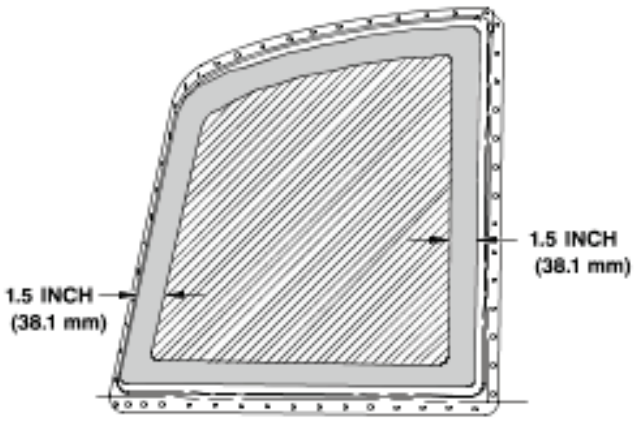
PASSENGER DOOR WINDOWS



PILOT AND COPILOT DOOR WINDOWS



LOWER CABIN NOSE WINDOWS



CABIN ROOF WINDOWS

Area "A": Scratches and pits are permitted in this area provided they are not so numerous or form such a pattern as to be objectionable to the viewer. Cracks, holes, or other damage may be temporarily repaired by stop drilling, patching or other approved methods (refer to FAA Advisory Circular 43.13-1, Aircraft Inspection and Repair), but window must be replaced at the earliest opportunity.

Area "B": Scratches and pits are permitted in this area, providing the structural integrity of the window is not impaired. Cracks, holes, or other damage may be repaired by stop drilling, patching or other approved methods provided structural integrity is not impaired (refer to FAA Advisory Circular 43.13-1, Aircraft Inspection and Repair).

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Figure 1. Windows - Damage Limits (Sheet 1 of 1)

Lower Crew Door Windows - Repair

Procedure

1. Refer to FAA Publication AC 43.13-1B Aircraft Inspection and Repair Manual for repair procedures.

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Lower Crew Door Windows - Installation

Procedure

1. Guide window up through slot in bottom of door and into channels.
2. Check progress of window through opening in aft edge of door.
3. With window in partially open position, engage adjustable stop.
4. Install handle with three screws.
5. Install lower access door.

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Cargo Hinged Panel Door Windows - Description

The cargo hinged panel door windows are made of clear acrylic material.

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Cargo Hinged Panel Door Windows - Removal

Procedure

1. Remove cargo hinged panel door from helicopter (412-A-52-30-00-00A-520A-A / 00820).
2. Remove hardware securing door latch Plexiglas covers.

NOTE

Earlier configuration cargo hinged panels are constructed from aluminum. Later configuration cargo (S/N 36191 and Subsequent) hinged panels are a composite construction.

3. For aluminum configuration cargo hinged panel, drill out rivets using a No. 30 drill.
4. For composite configuration cargo hinged panel, remove remaining screws, washers, and nuts around the periphery of window. Retain hardware for reuse.
5. Separate window from sealant using a non-metallic putty knife, spatula, or similar tool.
6. Carefully remove window from door panel.

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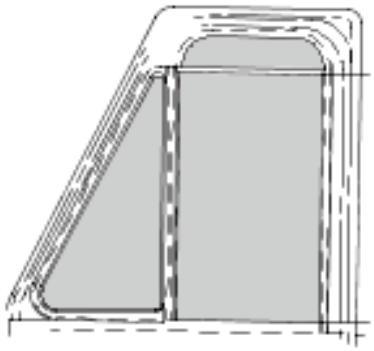
Cargo Hinged Panel Door Windows - Inspection

Procedure

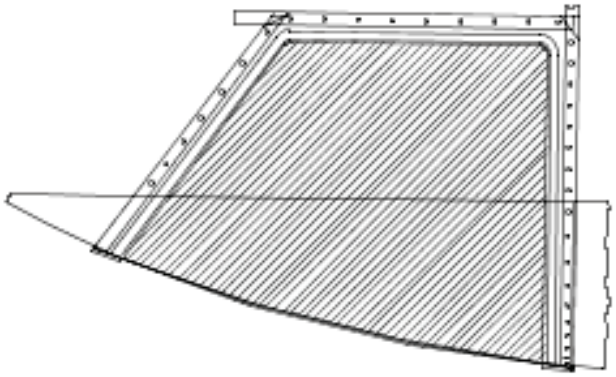
1. Inspect window in accordance with Figure 1.



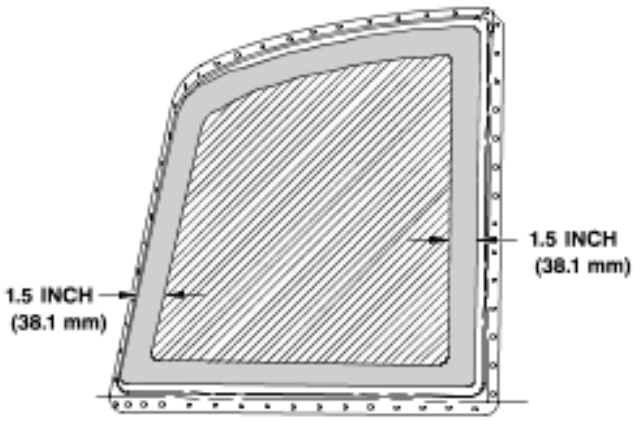
PASSENGER DOOR WINDOWS



PILOT AND COPILOT DOOR WINDOWS



LOWER CABIN NOSE WINDOWS



CABIN ROOF WINDOWS

Area "A": Scratches and pits are permitted in this area provided they are not so numerous or form such a pattern as to be objectionable to the viewer. Cracks, holes, or other damage may be temporarily repaired by stop drilling, patching or other approved methods (refer to FAA Advisory Circular 43.13-1, Aircraft Inspection and Repair), but window must be replaced at the earliest opportunity.

Area "B": Scratches and pits are permitted in this area, providing the structural integrity of the window is not impaired. Cracks, holes, or other damage may be repaired by stop drilling, patching or other approved methods provided structural integrity is not impaired (refer to FAA Advisory Circular 43.13-1, Aircraft Inspection and Repair).

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Figure 1. Windows - Damage Limits (Sheet 1 of 1)

Cargo Hinged Panel Door Windows - Repair

Procedure

1. Refer to FAA Publication AC 43.13-1B Aircraft Inspection and Repair Manual for repair procedures.

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Cargo Hinged Panel Door Windows - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	
Sealant	C-308	AR	

Procedure

1. Prior to installation, clean windows with a mild detergent and water solution. Dry with a soft dry cloth.



WHEN CLEANING WINDOWS, USE ONLY TYPE II aliphatic naphtha (C-305). ALIPHATIC NAPHTHA TYPE I WILL CLOUD WINDOWS.

2. Clean windows and mounting flange with aliphatic naphtha (C-305).

NOTE

Earlier configuration cargo hinged panels are constructed from aluminum. Later configuration (S/N 36191 and Subsequent) cargo hinged panels are a composite construction.

3. For aluminum configuration cargo hinged panel, proceed as follows:

3.1. If a new window is being installed, drill rivet mounting holes.

3.2. Apply a bead of sealant (C-308) on window edges.

3.3. Mount window in panel and secure using rivets.

4. For composite configuration cargo hinged panel, proceed as follows:

4.1. Center new window in opening within 0.030 inch.

4.2. Transfer screw holes to replacement window and enlarge holes in window only to 0.312 to 0.318 inch. Deburr and clean debris from window and frame.

4.3. Install window and secure with screws, washers, and nuts. Torque nuts to 5 to 8 inch-pounds.

4.4. Apply a bead of sealant (C-308) on window edges.

5. Install screws securing door latch Plexiglas cover to window.

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Passenger Door Windows - Description

The passenger door windows are made of clear acrylic material. The windows are installed in metal frames with silicone rubber retainers and are designed for emergency egress. EMERGENCY PUSH HERE decals are installed in lower corners.

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Passenger Door Windows - Removal

Procedure

NOTE

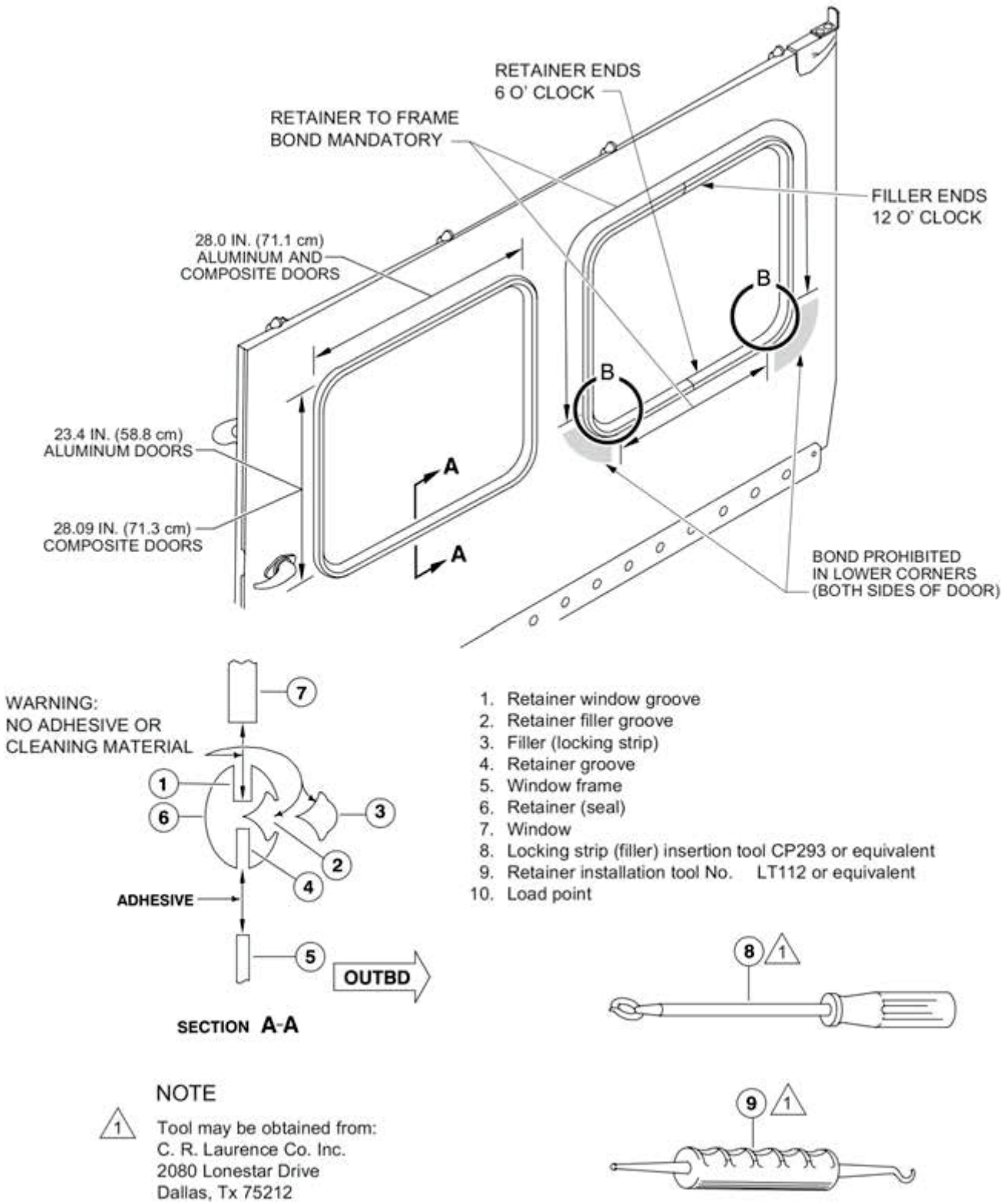
Removal is not recommended unless required by window or retainer damage. Procedures are same for all Bell Helicopter Textron and Aeronautical Accessories push-out windows..

1. Remove interior decor/trim as necessary to gain access to window supporting structure.
2. Using mylar, tedlar, or similar tape, mask off structure adjacent to window retainer. The mask edge should be pressed firmly against, but not on, retainer.
3. Remove filler (locking strip) (3, Figure 1) from retainer (6). Discard filler.



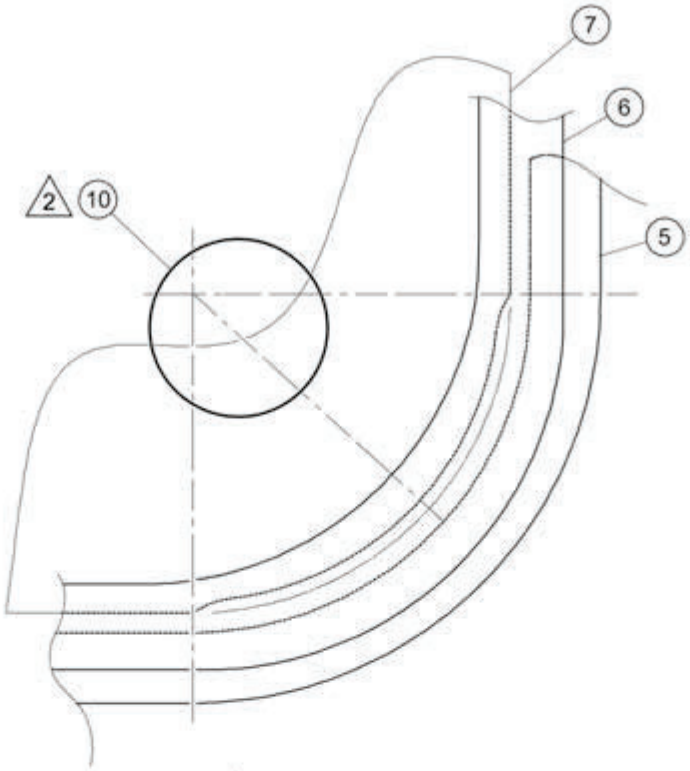
DO NOT ALLOW WINDOW TO FALL. STORE WINDOW ON EDGE TO PREVENT CONTOUR CHANGE.

4. Push window out by pressing at corner as directed by decal.
5. Pull retainer (6) from structure starting at unbonded area at lower corners. Discard retainer.



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Figure 1. Passenger Door Window (Sheet 1 of 2)



DETAIL B
TYPICAL



Application of the
Force Measuring Tool

NOTE

2 Pressure to be applied at this location.

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Figure 1. Passenger Door Window (Sheet 2 of 2)

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Passenger Door Windows - Inspection

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	
Abrasive cloth or paper	C-423	AR	
Tape	C-439	AR	
Clean cloth	C-516	AR	

Procedure

1. Inspection - windows installed.



NO REPAIRS ARE ALLOWED ON RETAINER (6) AND FILLER (3).

NOTE

The inspection criteria that follow are applicable to the Bell Textron and Aeronautical Accessories push-out windows.

1.1. Visually inspect retainer (6, Figure 1) and filler (3) for cuts, cracks, or signs of deterioration (swelling). Replace if damage is found.

1.2. Verify retainer (6) is securely bonded to both sides of window frame (5) except in radius area of two lower corners. Replace if any bond separation is found.

1.3. Inspect window (7) for cracking, crazing, or reduced transparency. Replace damaged windows.

NOTE

Retainers (6) and fillers (3) shall be replaced when window is removed from retainer.

2. Inspection - windows removed.

2.1. Mark both lower corners of window frame as shown in Figure 1. Mask these lower corner areas where the bond is prohibited with tape (C-439). Apply tape to both sides of the door.

2.2. Remove all old adhesive and finish from unmasked portion of window frame (5) using stiff fiber brush and a clean cloth (C-516) dampened with aliphatic naphtha (C-305).

2.3. For aluminum doors, examine the primer on the unmasked area of window frame (5). If necessary, do the touch up repair of the primer.

2.4. For composite doors, use 360 grit abrasive cloth or paper (C-423) and lightly abrade surface to be in contact with the retainer (seal) (6) on both sides of the door.

2.5. Remove the tape from the lower corners of the window frame.

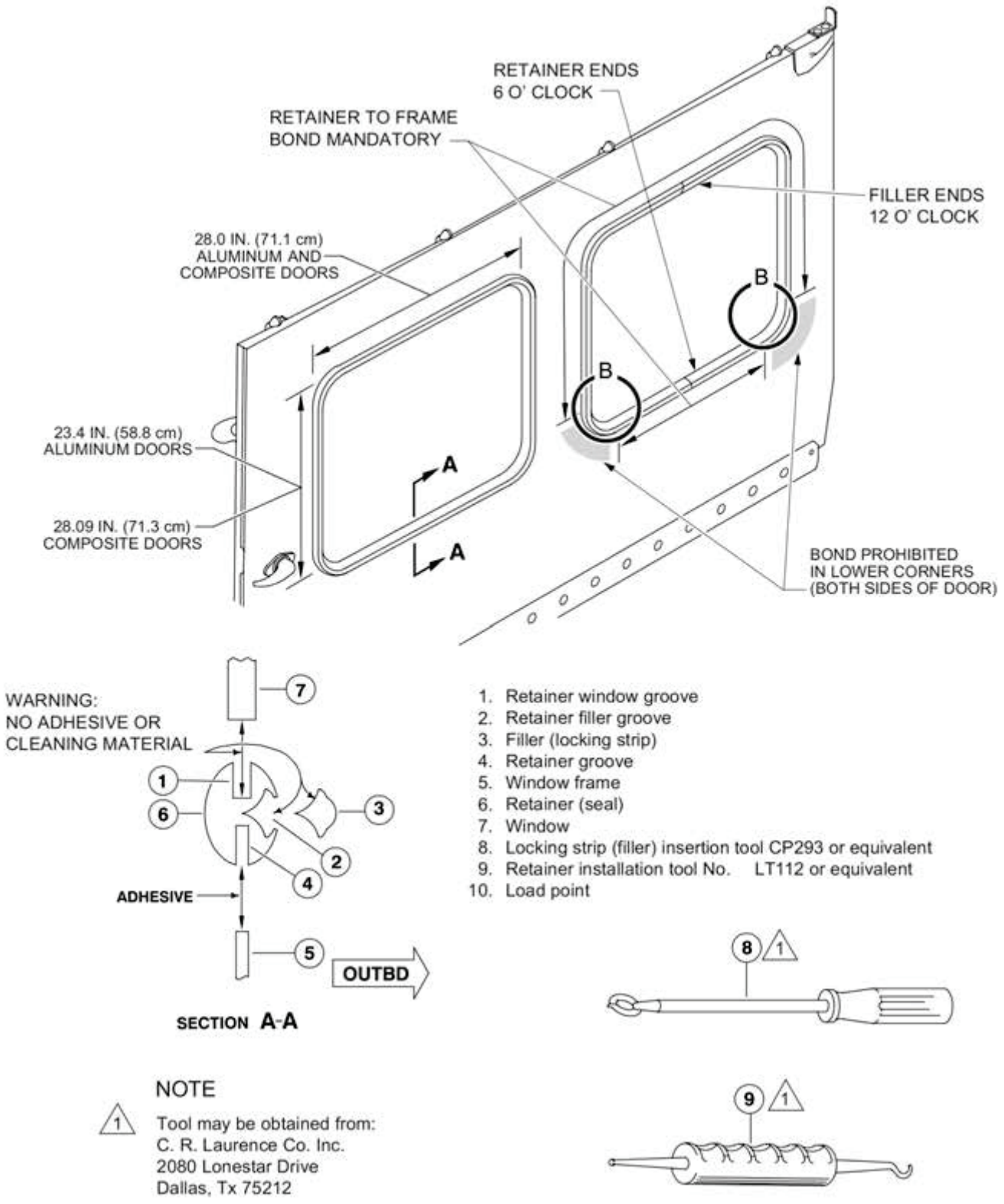
2.6. Inspect edge of frame (5) for scratches, cracks, or other damage. Any scratches less than 0.004 inch (0.10 mm) deep and 1.00 inch (25.40 mm) long may be polished out. Repairs must not be closer than 3.0 inches (76.2 mm) from adjacent repairs. An area may only be repaired once.



WHEN CLEANING WINDOWS, USE ONLY TYPE II aliphatic naphtha (C-305). ALIPHATIC NAPHTHA TYPE I WILL CLOUD WINDOWS.

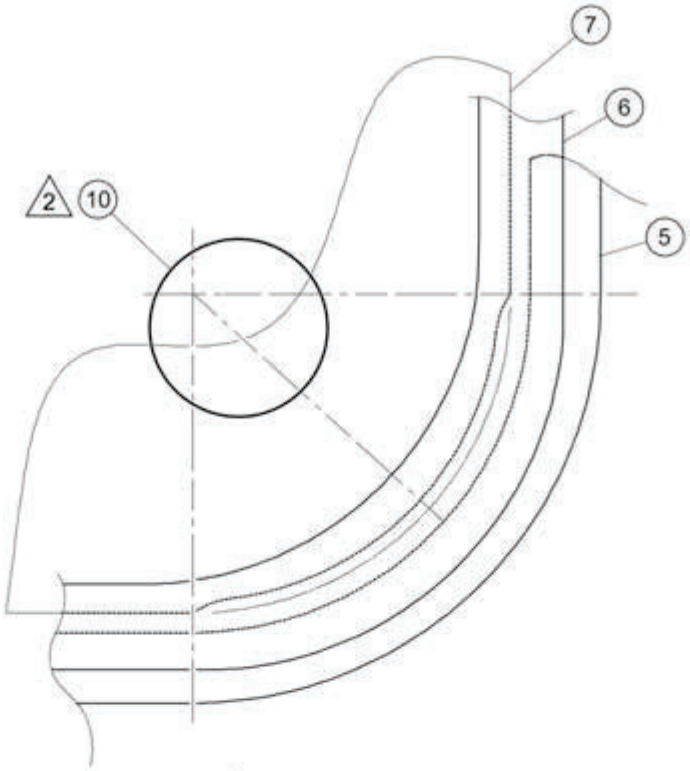
2.7. Inspect window (7) for chamfered, nicked or notched edge, cracking, crazing, or reduced transparency. Replace damaged windows.

3. Repair window (7) in accordance with Figure 2.



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Figure 1. Passenger Door Window (Sheet 1 of 2)



DETAIL B
TYPICAL



Application of the
Force Measuring Tool

NOTE

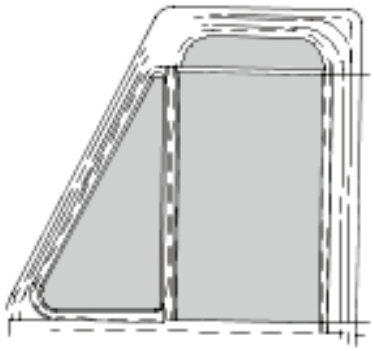
2 Pressure to be applied at this location.

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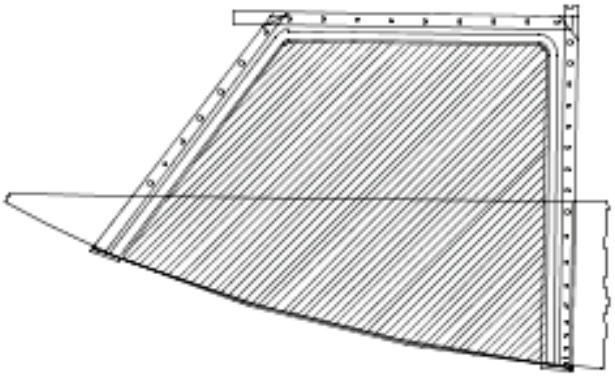
Figure 1. Passenger Door Window (Sheet 2 of 2)



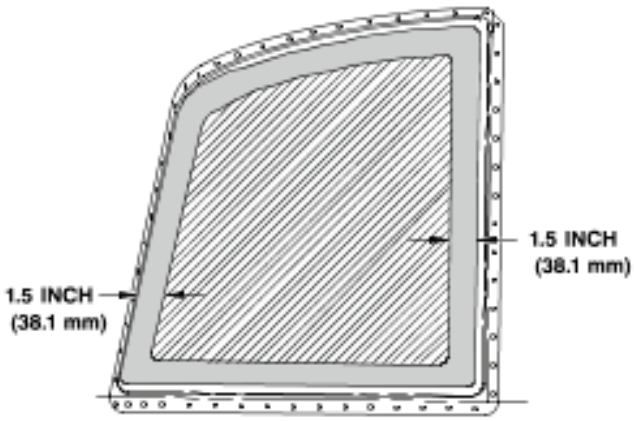
PASSENGER DOOR WINDOWS



PILOT AND COPILOT DOOR WINDOWS



LOWER CABIN NOSE WINDOWS



CABIN ROOF WINDOWS

Area "A": Scratches and pits are permitted in this area provided they are not so numerous or form such a pattern as to be objectionable to the viewer. Cracks, holes, or other damage may be temporarily repaired by stop drilling, patching or other approved methods (refer to FAA Advisory Circular 43.13-1, Aircraft Inspection and Repair), but window must be replaced at the earliest opportunity.

Area "B": Scratches and pits are permitted in this area, providing the structural integrity of the window is not impaired. Cracks, holes, or other damage may be repaired by stop drilling, patching or other approved methods provided structural integrity is not impaired (refer to FAA Advisory Circular 43.13-1, Aircraft Inspection and Repair).

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Figure 2. Windows - Damage Limits (Sheet 1 of 1)

Passenger Door Windows - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Locking Strip (Filler) Insertion Tool	CP 293	1	
Retainer Installation Tool	LT112	1	
Force Measuring Tool	512-230-002-101 or DG-200 or equivalent	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Adhesive	C-300	AR	
Aliphatic naphtha	C-305	AR	
Toluene	C-306	AR	
Adhesive	C-307	AR	
Acetone	C-316	AR	
Primer	C-337	AR	
	C-347	AR	
Adhesive	C-363	AR	
Abrasive cloth or paper	C-423	AR	
Masking tape	C-426	AR	
Clean cloth	C-516	AR	

Procedure



STRICT ADHERENCE TO THE INSTALLATION INSTRUCTIONS IS CRITICAL AS FAILURE TO DO SO MAY RESULT IN IN-FLIGHT LOSS OF WINDOW.

NOTE

The installation instructions that follow are applicable to the Bell Textron and Aeronautical Accessories push-out windows.

1. If the window was removed to perform a structural repair that may have affected the window opening, measure window opening. Trim opening as required to the following dimensions (±0.03 inch (±0.8 mm)):

- For aluminum doors, window opening is 28.00 inches (71.1 cm) wide by 23.14 inches (58.8 cm) inches high (Figure 1).
- For composite doors, window opening is 28.00 inches (71.1 cm) wide by 28.09 inches (71.3 cm) inches high (Figure 1).



ENSURE WINDOW FRAME (5) IS CLEAN AND COMPLETELY FREE OF ANY PAINT FINISHING MATERIAL.

NOTE

Helicopter finishes will not adhere to surfaces where primer and adhesive are applied.

NOTE

Window opening must be in a protected area, shielded from wind, rain, or other adverse environmental conditions during installation.

2. Following applicable trimming or initial preparation, remove all residue using a clean cloth (C-516) moistened with toluene (C-306), (C-347), or acetone (C-316). Wipe dry with a clean cloth (C-516).



A NEW FILLER (LOCKING STRIP) (3) AND RETAINER (SEAL) (6) MUST BE USED FOR EACH WINDOW INSTALLATION.

3. Temporarily position new retainer (seal) (6) in window frame (5) opening.

4. With the new retainer (seal) (6) positioned in place, apply masking tape (C-426) around its periphery to the surface of both sides of the door. Remove the retainer.

5. For composite doors, apply adhesive (C-363) to the trimmed edge of the window frame (5) opening. Allow to dry for 6 hours at room temperature before installing retainer (seal) (6).

6. For aluminum doors, apply the primer (C-337) on the unmasked area of the window frame (5). Allow to dry 30 minutes before installing retainer (seal) (6).

7. Prepare and fit new retainer (seal) (6) in window frame (5) as follows:

7.1. Abrade retainer groove (4) with 80 grit abrasive cloth or paper (C-423).

7.2. Clean retainer groove (4) with a clean cloth (C-516) dampened with toluene (C-306), (C-347), or acetone (C-316). Wipe dry with a clean cloth (C-516).

NOTE

Retainer filler groove (2) shall be positioned outboard.

7.3. Position and hold retainer (seal) (6) in place with masking tape (C-426) so that it nests in all four corners of the opening. Position ends at approximately the 6 o'clock position and trim to length.

7.4. Mix adhesive (C-300) and apply to both trimmed ends of retainer (seal) (6). Allow to air dry for 15 to 30 minutes to a strong tack and press the ends together.



WHEN CLEANING WINDOWS, USE TYPE II aliphatic naphtha (C-305). ALIPHATIC NAPHTHA TYPE I WILL CLOUD WINDOWS.

NOTE

The height and width are very similar on the square window. The window is normally oriented correctly when the part number is at the bottom.

8. Place window (7) in retainer window groove (1).



IF PROCURED AS A PRE-TRIMMED ITEM, DO NOT CUT OR STRETCH THE FILLER (LOCKING STRIP) (3) TO LENGTH. THE FULL LENGTH MUST BE USED.

IF FILLER (LOCKING STRIP) (3) IS PROCURED AS RAW MATERIAL BY THE FOOT, TEMPORARILY PLACE FILLER (LOCKING STRIP) (3) ON TOP OF RETAINER FILLER GROOVE (2) AND CUT TO LENGTH USING A SHARP KNIFE. AFTER TRIMMING, THE FULL LENGTH MUST BE USED.

9. Install filler (locking strip) (3) in retainer filler groove (2), using locking strip insertion tools (8 and 9). Ends of filler shall meet approximately at the 12 o'clock position in retainer (seal) (6).

10. To avoid adhesive contact with the painted surfaces of the door, verify installation of masking tape (C-426) on both sides of the door around the periphery of the retainer (seal) (6). Mark lower corner areas not to be bonded on both sides of the door, as shown in Figure 1.

CAUTION

DO NOT APPLY ADHESIVE TO THE RETAINER WINDOW GROOVE (1), FILLER (LOCKING STRIP) (3), OR LOWER CORNER AREAS SHOWN IN Figure 1.

11. Using a clean wooden or plastic stick or the AT 515 tool, locally lift the retainer (seal) (6) and inject adhesive (C-307) in a small bead between the retainer and window frame (5), all around the retainer except in lower corner areas as shown in Figure 1. A small amount of adhesive squeeze-out around the edges is desirable.

CAUTION

DO NOT ALLOW TOLUENEtoluene (C-306) OR (C-347) TO COME INTO CONTACT WITH WINDOW (7).

12. Remove excess silicone adhesive using a clean cloth (C-516) dampened with toluene (C-306), (C-347), or acetone (C-316). Wipe dry with a clean cloth (C-516).

13. Turn the door over and apply adhesive (C-307) to the other side of the retainer (seal) (6) and window frame (5) per Step 11. and Step 12..

14. Remove all masking tape from around retainer (seal) (6).

15. Inspect periphery of window (7) and retainer (seal) (6) for gaps between the retainer and window and between the retainer and window frame (5).

CAUTION

CURE THE adhesive (C-307) FOR A MINIMUM OF 24 HOURS AT ROOM TEMPERATURE BEFORE THE WINDOW REMOVAL TEST AND THE RELEASE OF HELICOPTER FOR FLIGHT.

NOTE

adhesive (C-307) curing is obtained from moisture in the air. If window installation is performed in a very dry environment, curing may be accelerated by covering the window with a sealed plastic bag and placing a cloth wet with water in a sealed compartment on each side of the window. Make sure no water makes direct contact with the retainer.

16. Placard the window (7) with a “DO NOT DISTURB” placard and allow to dry. Do not remove the placard or release the helicopter for flight until the silicone adhesive has cured for a minimum of 24 hours at room temperature.

17. Perform the window removal test or push test as follows:

17.1. Make sure that the Force Measuring Tool DG-200 or equivalent is calibrated.

CAUTION

MAKE SURE NOT TO PUSH THE WINDOW COMPLETELY OUT WHILE YOU APPLY THE LOAD, OTHERWISE, THE WINDOW INSTALLATION WILL HAVE TO BE RE-DONE. MAKE SURE THAT THE WINDOW IS ADEQUATELY PROTECTED TO PREVENT THE DAMAGE WHILE YOU APPLY THE FORCE WITH THE TOOL.

NOTE

The procedure is typical for all the four windows.

17.2. Hold the loading pad of the Force Measuring Tool at one of the load points (10,Figure 1) in the center of the corner radius, perpendicular to the inner surface of the window (7).

NOTE

Make sure to apply a center of pressure not more than 1.5 inches (38.1 mm) from the interface of the seal (6) and the window (7).

17.3. Carefully apply the load to the window (7).

NOTE

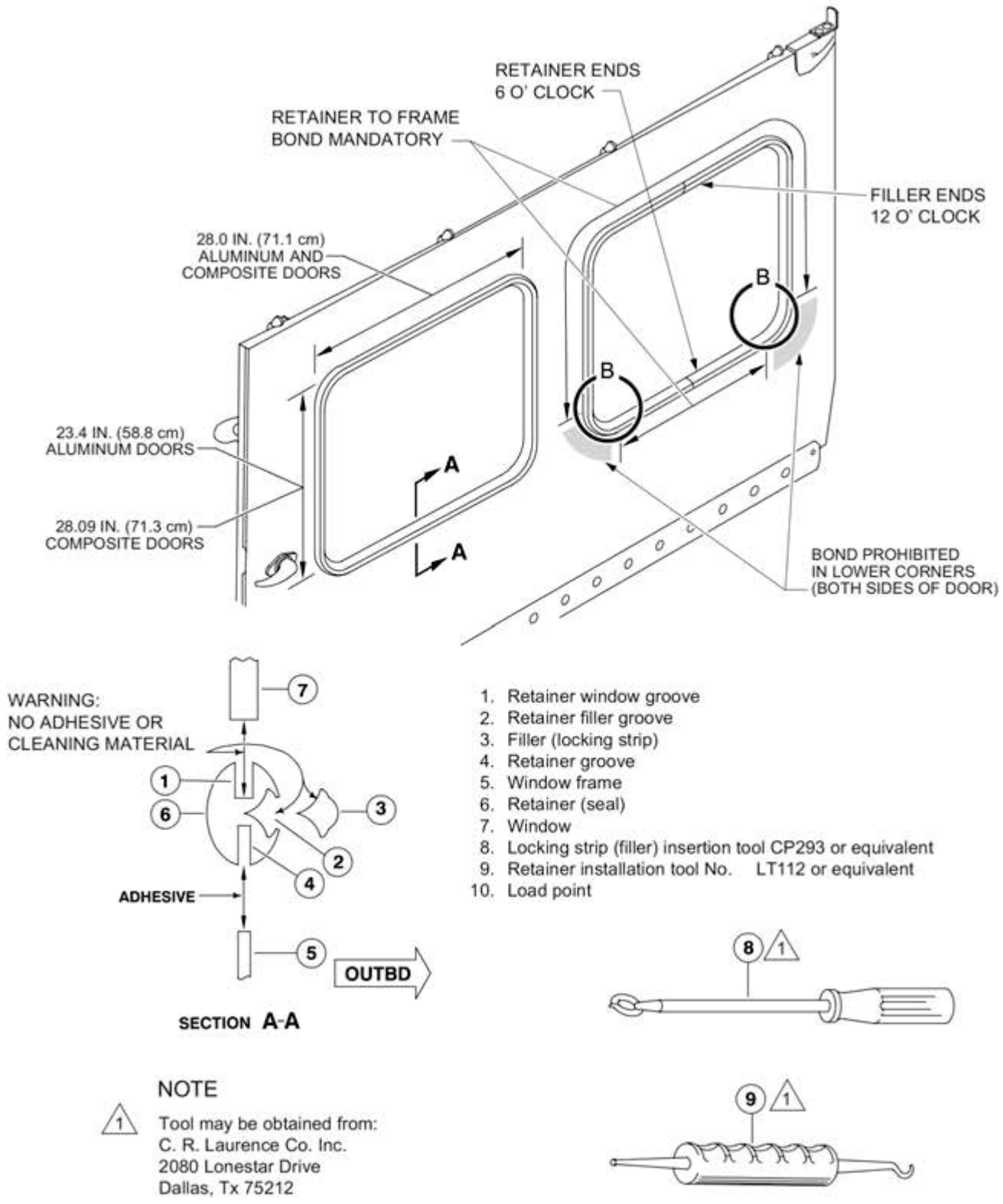
Monitor the relationship between the window (7) and seal (6) and the seal (6) and window frame (5).

17.4. Make sure that at one point of the seal (6) starts to curl away (Figure 1, Detail B) from the window (7) or the window frame (5). Record the load applied.

17.5. Make sure that the load recorded in above step is between 40 to 55 lbs (178 to 245 N) for all the windows (7).

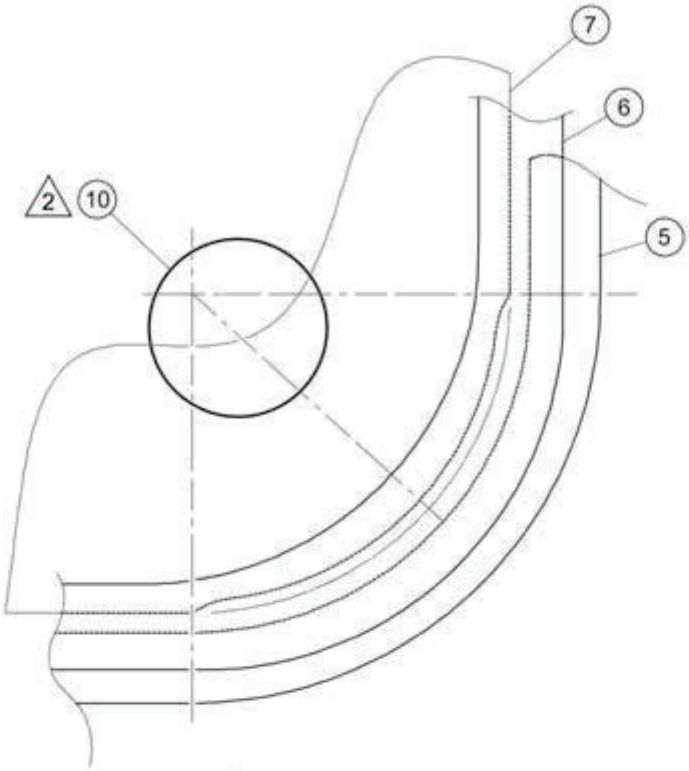
18. If not already installed, apply “EMERGENCY PUSH HERE” decals on lower corners of window designated for emergency egress (412-A-11-00-00-00A-067A-A / 00124).

19. Install interior decor/trim (if applicable).



ICN-412-A-520100-B-97499-00002-A-001-01

Figure 1. Passenger Door Window (Sheet 1 of 2)



DETAIL B
TYPICAL



Application of the
Force Measuring Tool

NOTE

2 Pressure to be applied at this location.

ICN-412-A-520100-B-97499-01031-A-001-01

Figure 1. Passenger Door Window (Sheet 2 of 2)

Lower Nose Section Windows - Description

The lower nose section windows are made of clear acrylic material.

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Lower Nose Section Windows - Removal

Procedure

1. Remove nuts, washers, and screws attaching window.
2. Separate window from sealant.
3. Remove window from helicopter.

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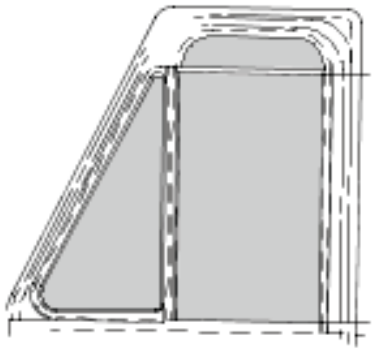
Lower Nose Section Windows - Inspection

Procedure

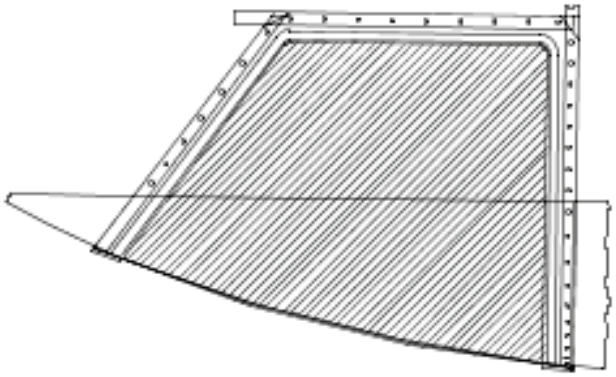
1. Inspect window in accordance with Figure 1.



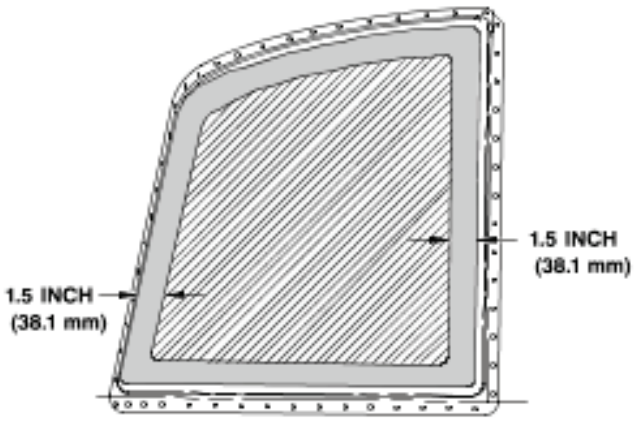
PASSENGER DOOR WINDOWS



PILOT AND COPILOT DOOR WINDOWS



LOWER CABIN NOSE WINDOWS



CABIN ROOF WINDOWS

Area "A": Scratches and pits are permitted in this area provided they are not so numerous or form such a pattern as to be objectionable to the viewer. Cracks, holes, or other damage may be temporarily repaired by stop drilling, patching or other approved methods (refer to FAA Advisory Circular 43.13-1, Aircraft Inspection and Repair), but window must be replaced at the earliest opportunity.

Area "B": Scratches and pits are permitted in this area, providing the structural integrity of the window is not impaired. Cracks, holes, or other damage may be repaired by stop drilling, patching or other approved methods provided structural integrity is not impaired (refer to FAA Advisory Circular 43.13-1, Aircraft Inspection and Repair).

ICN-412-A-520100-B-97499-00001-A-001-01

Figure 1. Windows - Damage Limits (Sheet 1 of 1)

Lower Nose Section Windows - Repair

Procedure

1. Refer to FAA Publication AC 43.13-1B Aircraft Inspection and Repair Manual for repair procedures.

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Lower Nose Section Windows - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	
Sealant	C-308	AR	

Procedure

1. Prior to installation of lower nose section windows, wash windows with a mild detergent and water solution. Dry with a soft, dry cloth.

2. Remove old sealant from mounting flange with a putty knife, spatula, or other suitable tool.



WHEN CLEANING WINDOWS, USE ONLY TYPE II aliphatic naphtha (C-305). ALIPHATIC NAPHTHA TYPE I WILL CLOUD WINDOWS.

3. Wipe and clean mounting flange with cloth dampened with aliphatic naphtha (C-305). If an old window is being installed, clean edges of window using aliphatic naphtha (C-305).

4. Position a new window over opening. Trim surplus edge to permit window to be in mounting position against flange.

NOTE

Do not trim window to final size until all mounting holes have been drilled.

5. Position window against mounting flange and, using a No. 4 drill, back drill two holes on each edge of window.

6. Secure window to mounting flange with four existing screws, washers, and nuts. Lightly tighten screws. Finish drilling holes in window.

7. Determine edge distance and mark window. Remove window and trim to 1.04 inches (29 mm) edge distance.

8. Remove all dust and foreign matter from window mating area from window mounting flange using aliphatic naphtha (C-305).

9. Apply a bead of sealant (C-308) around windows.

10. Position window in mounting flange, align holes, and install screws, washers, and nuts.

11. Remove excess sealant (C-308) from around window.

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Composite Crew Door Windows - Description

412-A-52-01-00-10A-520A-A / 00881, 412-A-52-01-00-10A-280A-A / 00882, 412-A-52-01-00-10A-685A-A / 00883 and 412-A-52-01-00-10A-720A-A / 00884 gives the maintenance information for the composite crew door windows.

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Composite Crew Door Windows - Removal

Procedure

NOTE

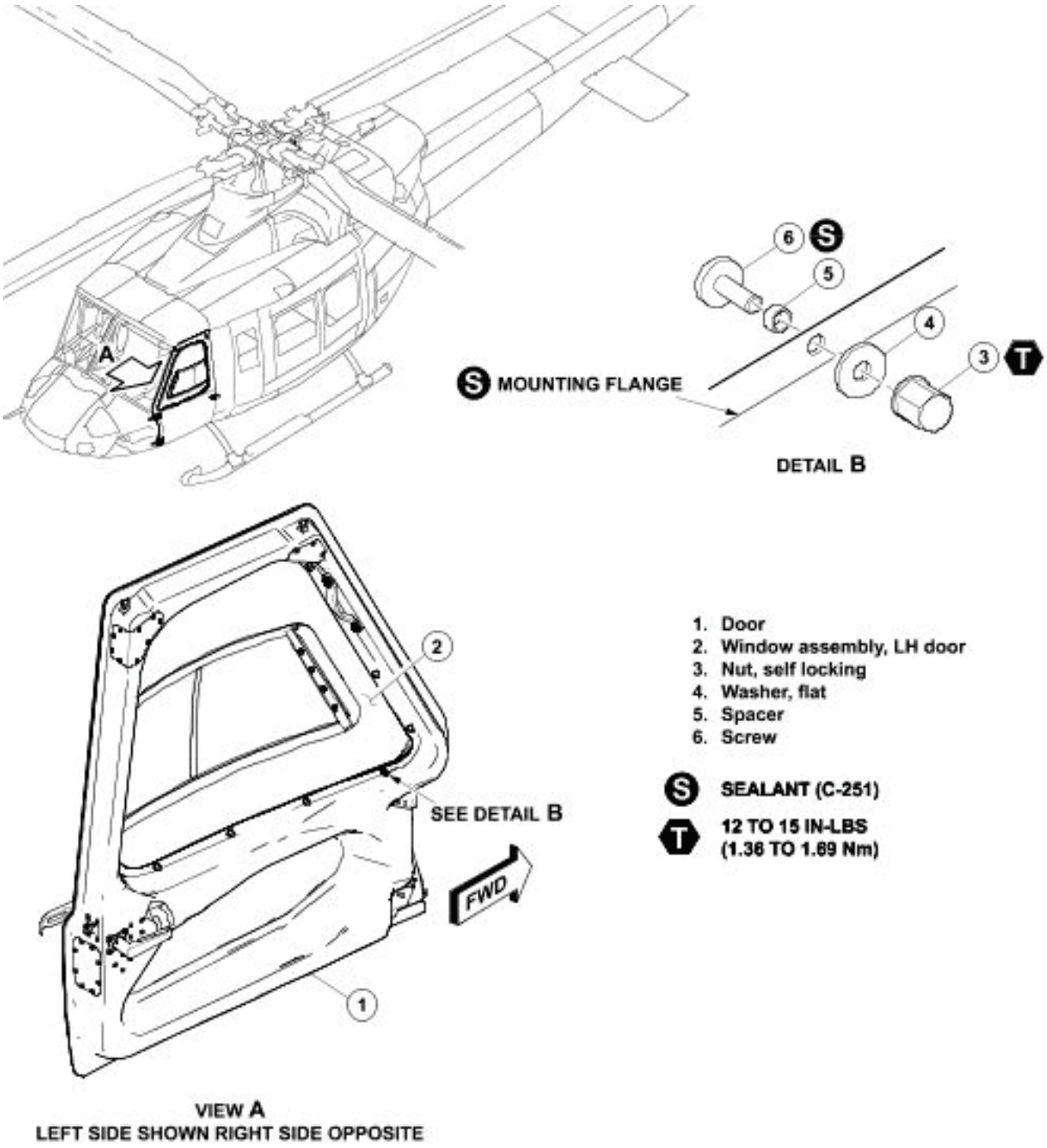
The following procedure is for the left side crew door window. The procedure is same for the right side crew door window.

1. With a plastic scraper remove the edge seal around the window (2, Figure 1).

2. Remove the nuts (3), washers (4), spacers (5) and the screws (6).

3. Separate the window (2) from the sealing tape.

4. Remove the window (2) from the door (1).



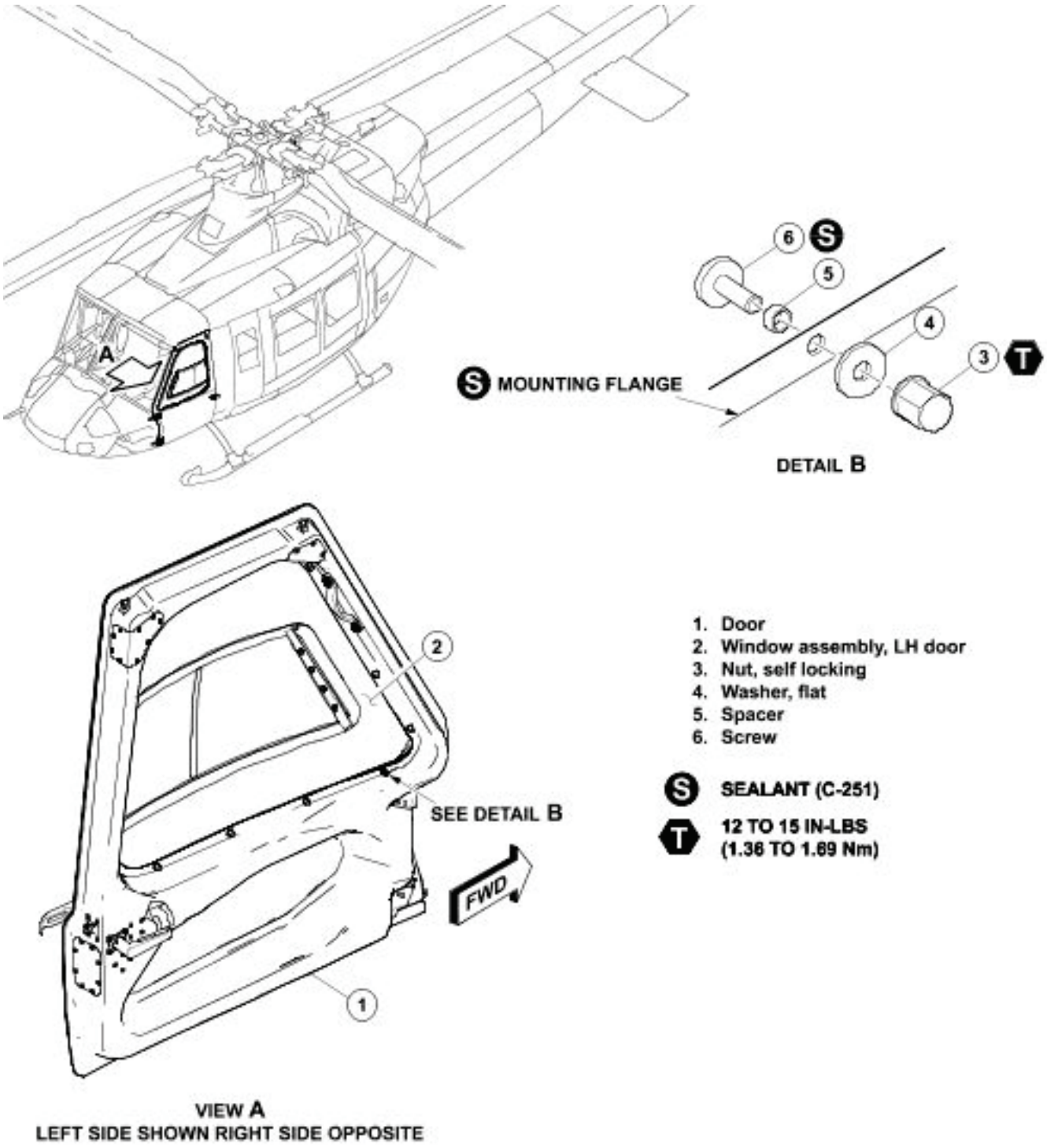
ICN-412-A-520100-B-97499-00005-A-001-01

Figure 1. Crew Door Windows (Sheet 1 of 1)

Composite Crew Door Windows - Inspection

Procedure

1. Visually inspect window assembly (2, Figure 1) for cuts, cracks, or signs of deterioration (swelling). Replace if damage is found.



ICN-412-A-520100-B-97499-00005-A-001-01

Figure 1. Crew Door Windows (Sheet 1 of 1)

Composite Crew Door Windows - Repair

Procedure

1. Refer to FAA Publication AC 43.13-1B Aircraft Inspection and Repair Manual for repair procedures.

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Composite Crew Door Windows - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Tape	C-067	AR	
Sealant	C-251	AR	
Aliphatic naphtha	C-305	AR	

Procedure

NOTE

The following procedure is for the left side crew door window. The procedure is same for the right side crew door window.

1. Prior to the installation, wash window (2, Figure 1) with a mild detergent and water solution. Dry with a soft dry cloth.

2. Remove and discard the old adhesive tape.



WHEN CLEANING WINDOWS AND MOUNTING FLANGE, USE ONLY TYPE II aliphatic naphtha (C-305). ALIPHATIC NAPHTHA TYPE I WILL CLOUD WINDOWS.

3. Wipe and clean the mounting flange with cloth dampened with aliphatic naphtha (C-305).

4. If a new window (2) is being installed, put the window (2) over opening. Trim only enough surplus material from edge of the window (2) to permit proper alignment of window (2) against mounting flange.

NOTE

Do not trim the window to final size until all mounting holes have been drilled.

5. Position the window (2) against mounting flange. Using a No. 4 drill, back drill two holes on each edge of the window (2). Use holes in mounting flange as template.

6. Secure the window (2) to mounting flange with four screws (6), spacers (5), washers (4) and nuts (3). Lightly tighten. Finish drilling holes in the window (2).

7. Measure the window (2) and mark correct edge distance required for installation.

8. Position the window (2) against mounting flange. Using a No. 4 drill, back drill two holes on each edge of the window (2). Use holes in mounting flange as template.

9. Secure the window (2) to mounting flange with four existing screws (6), spacers (5), washers (4) and nuts (3). Lightly tighten. Finish drilling holes in the window (2).

10. Measure the window (2) and mark correct edge distance required for installation.

11. Remove the window (2) and trim window (2) edge per measurements from Step 7..

12. Remove all the dust and dirt from window (2) and mounting flange with a clean cloth moistened in aliphatic naphtha (C-305).

13. Put a new tape (C-067) in its position around the full perimeter of the mounting flange. Refer to BHT-ALL-SPM, Standard Practices Manual.

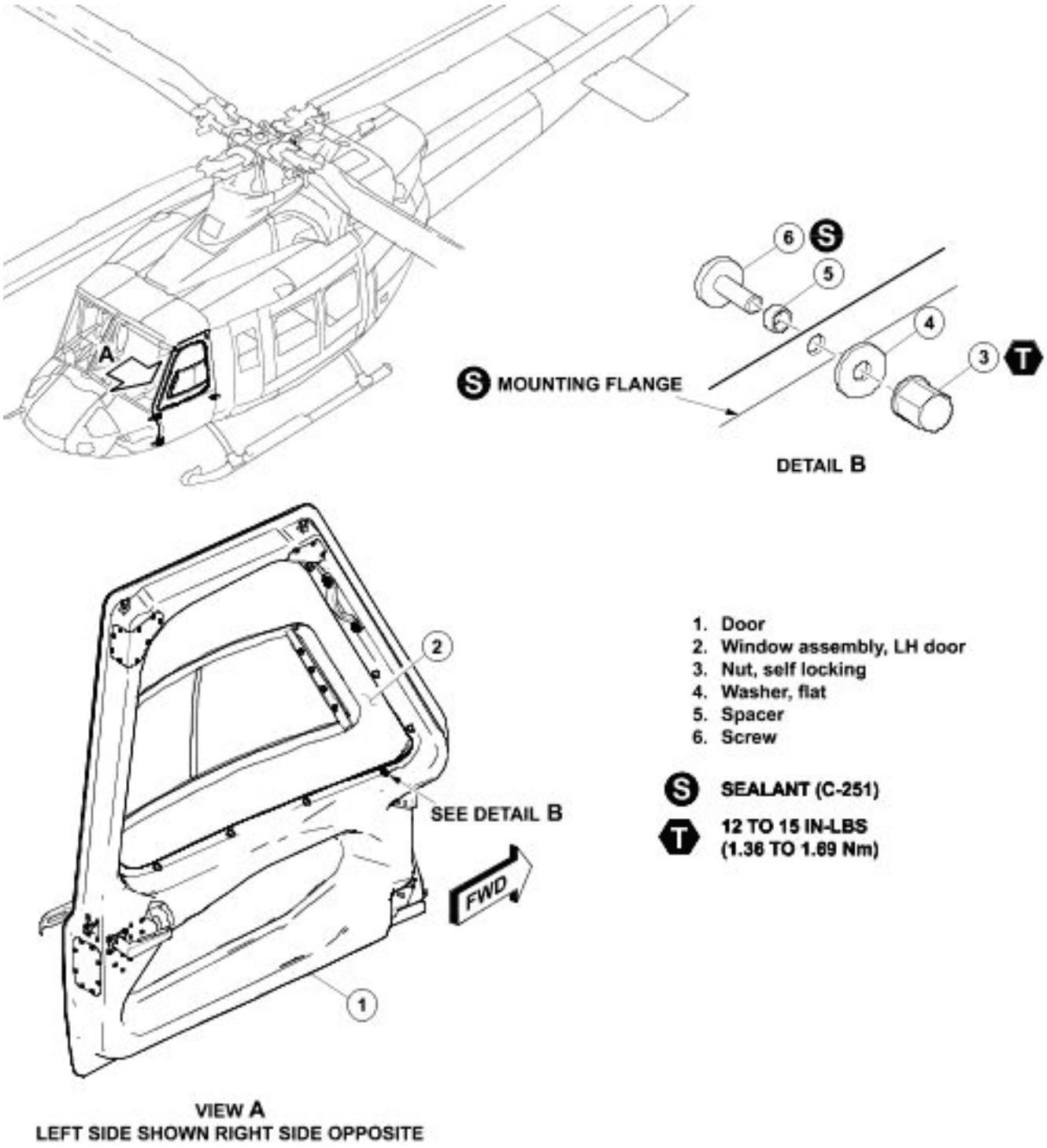
14. Position the window (2) on the mounting flange and align holes.

15. Wet install the screws (6), spacers (5), washers (4) and the nuts (3) on the window (2) with sealant (C-251).

16. Torque the nuts from 12 to 15 inch-pounds (1.36 to 1.69 Nm) before the sealant dries.

17. Apply sealant (C-251) around the edges of the window and the mounting flange.

18. Remove the unwanted sealant (C-251) from around the window (2) with a clean cloth dampened with aliphatic naphtha (C-305).



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Figure 1. Crew Door Windows (Sheet 1 of 1)

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Windshields - Description

The windshield is of three-layer, sandwich-type construction. The outer layer is made of tempered glass. The middle layer is made of applicable laminated glass. The inner layer may be made of glass or plastic. The windshields are set in weather-tight sealer and are mounted to the cabin structure with screws, washers, and nuts.

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Windshields - Removal

Procedure

1. Mark location of wiper stop (approximately BL 7.0). Remove windshield wiper arm (412-A-30-40-00-02A-520A-A / 00687) and blade (412-A-30-40-00-00A-520A-A / 00683).

2. Noting location of hardware, remove nuts, washers, and screws attaching windshield to fuselage.

NOTE

Shims may have been installed to reduce any gaps between windshield frame and mounting flange. If same windshield is to be reinstalled, mark location of any shims installed.

3. Separate windshield from sealing compound using a non-metallic spatula, putty knife, or other suitable tool (e.g., Snap-on P/N YA109C).

4. Carefully remove windshield from helicopter.

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Windshields - Inspection

Procedure

1. Inspect the windshield frame for nicks, dents, scratches, distortion, and holes. Minor scratches, nicks, and non-puncturing dents are acceptable without repair if the damage does not adversely affect the windshield or windshield security.

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Windshields - Cleaning

Procedure

1. Wash the windshields with a mild detergent and water solution. Dry with a soft, clean cloth.

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Windshields - Repair

Procedure

1. Glass windshields cannot be repaired. Glass windshields must be replaced if there are cracks.

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Windshields - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	
Sealant	C-328	AR	

Procedure

1. Remove all sealing compound from the mounting flange with a non-metallic putty knife, spatula, or similar tool.



WHEN CLEANING WINDOWS, USE ONLY TYPE II aliphatic naphtha (C-305). ALIPHATIC NAPHTHA TYPE I WILL CLOUD WINDOWS.

2. Wipe and clean the mounting flange with a cloth dampened with aliphatic naphtha (C-305).

3. Position the windshield in the mounting position so that it mates with the contour and that no portion of the bevel edge on the inside surface contacts the frame (minimum 0.010 inch (0.25 mm) clearance).

NOTE

Do not trim the windshield to the final size until all the mounting holes have been drilled.

4. Trim the windshield only as required to allow temporary installation. Temporarily secure with heavy tape.

5. Mark the location of all the screw holes on the windshield flange.

6. Remove the windshield and confirm a minimum edge distance of 0.38 inch (9.7 mm) for all the holes. Readjust the position as required to obtain proper edge distance.

7. Position the windshield in place with all the marked holes aligned.

8. Starting at the center of the vertical row (center post) and progressing towards the top and bottom, transfer the screw holes into the windshield flange with a 0.190 to 0.196 inch drill. Temporarily attach with Cleco fasteners.

9. Transfer the remaining screw holes starting at the inboard corners, progressing towards the outboard edge. Temporarily attach with Cleco fasteners.

NOTE

If a gap greater than 0.015 inch (0.38 mm) exists due to variations in the windshield contour or the mounting flange, a shim may be installed.

10. Check for an excessive gap between the windshield frame and the mounting flange over a distance of three screw holes. If a gap is greater than 0.015 inch (0.38 mm), install shim material as follows:

10.1. Prepare shim material from aluminum alloy 2024 T3 with a maximum thickness of 0.063 inch (1.60 mm) to reduce any gaps to a maximum of 0.015 inch (0.38 mm).

10.2. Install a shim between the windshield frame and the mounting flange. Shim material must be secured with a minimum of two fasteners.

10.3. Mark the location of the shim for proper installation in later steps.

11. Backdrill two holes on each edge of the windshield with a No. 4 drill. Use the holes in the mounting flange as a template.

12. Secure the windshield to the mounting flange with the four screws, washers, and nuts. Lightly tighten. Finish drilling the holes in the windshield.

13. Confirm the proper edge distance, mark the windshield and the shims, if required, and remove. Trim the windshield edge and the shims, if required, to the proper size.

14. Remove all dust and foreign matter from the windshield mating area and from the windshield mounting flange.

15. Apply a small even bead of sealant (C-328) to the window frame.

16. Position the windshield, align the holes, and install the shims, if required, screws, washers, and nuts.

17. Increase the torque slowly and evenly on all the screws or nuts to a maximum of 23 inch-pounds (2.6 Nm). Verify the torque before the sealant dries.

18. Remove excess sealant (C-328) from around the windshield with aliphatic naphtha (C-305). Touch up the paint if bare metal is evident.

19. Install the windshield wiper assemblies (412-A-30-40-00-01A-720A-A / 00686).

Heated Windshield Kit (412-706-010/-138) - Description

The heated windshield kit can be installed on the helicopter as an option. The windshield is of three-layer, sandwich-type construction. The outer layer is made of tempered glass. The middle layer is made of applicable laminated glass. The inner layer may be made of glass or plastic. It is installed on the windshield frame, outboard cabin post, frame assembly, and center post. The windshields are set in weather-tight sealer and are mounted to the cabin structure with screws, washers, and nuts.

The heated windshield kit is connected to the DC power and has a defrost device activated by a switch.

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Heated Windshield Kit (412-706-010/-138) - Removal

Procedure

WARNING

OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT (412-A-96-00-01-02A-042A-A / 01874).

NOTE

The following procedure is for the heated windshield kit (8, Figure 1) on the left side. The procedure is the same for the heated windshield on the right side.

NOTE

Remove the components to the extent necessary to do maintenance.

1. Open the nose doors and/or panels, as applicable (412-A-52-40-00-01A-520A-A / 00832).

2. Disconnect the battery and/or the external power from the helicopter.

3. Remove the windshield wiper arm (412-A-30-40-00-02A-520A-A / 00687) and blade (412-A-30-40-00-00A-520A-A / 00683).

4. Remove the center post cover.

5. With a blunt edge spatula, carefully remove the bead of sealant from around the heated windshield (8, Figure 1).

6. Remove the cover assembly (17) from the terminal board assembly (18).

7. Tag and disconnect the terminal lugs from the terminal board assembly (18).

8. Put protective tape on the exposed ends of all the electrical leads.

9. Mark the location of the wiper stop (3) (approximately BL 7.0) with a nonpermanent marker.

10. Remove the nuts (6), screws (4), washers (5), and wiper stop (3) from the heated windshield (8) and the windshield frame (2).

11. *Remove the nuts (21), screws (19, 22, and 23), and washers (20) from the heated windshield (8), solid shims (7 and 24), as applicable, and windshield frame (2).*

12. Remove the screws (15) from the heated windshield (8), fitting (14), and frame assembly (16).

13. Remove the screws (13) from the heated windshield (8) and the frame assembly (16).

14. Remove the nuts (11), screws (12), and washers (10) from the heated windshield (8), windshield frame (2), outboard cabin post (9), and center post (1).

NOTE

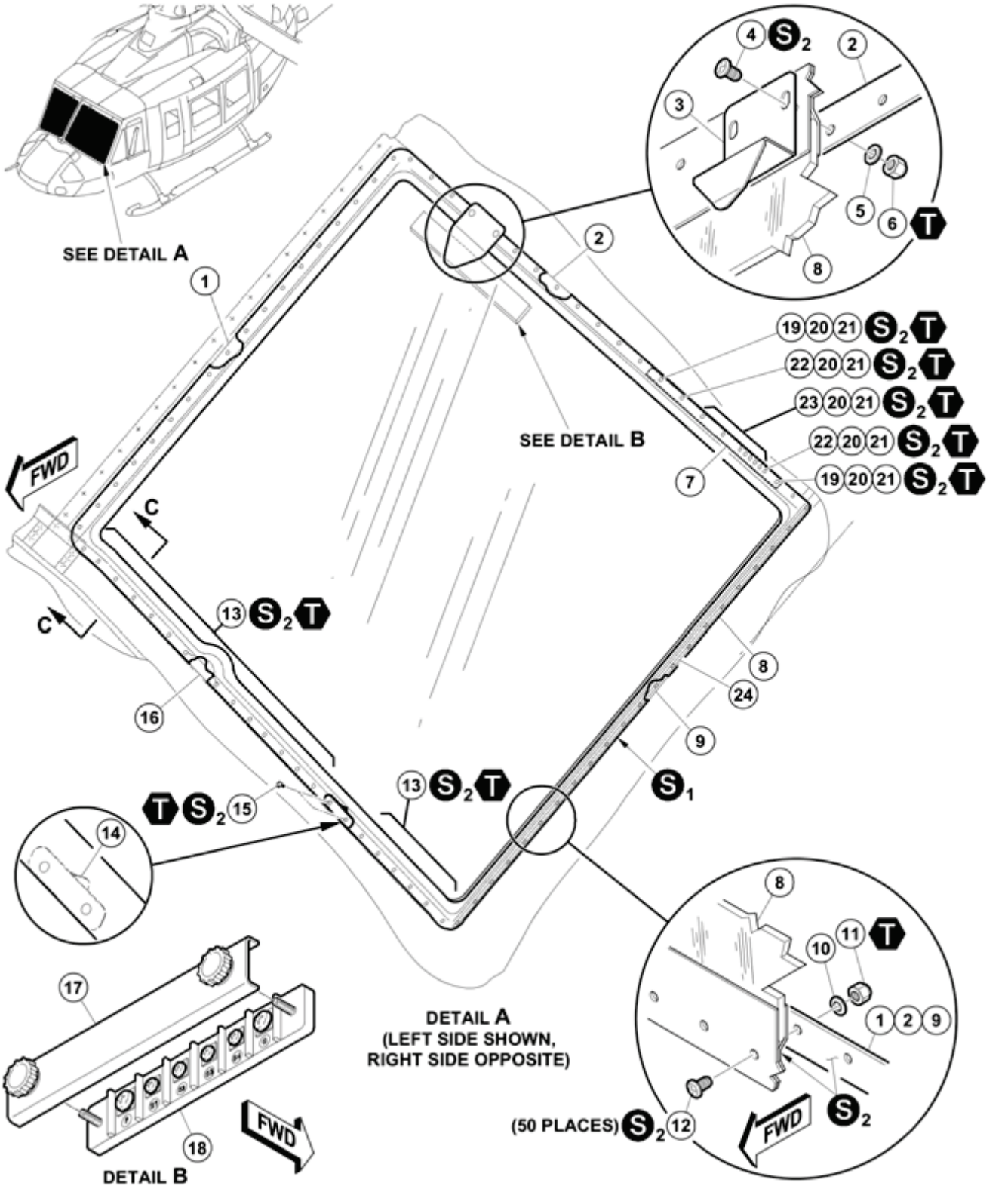
Solid shim may have been installed at the top area with sealant to reduce any gaps between the windshield frame (2) and the mounting flange. If the same windshield is to be installed, mark the location of any shim installed.

15. Carefully remove the heated windshield (8) from the windshield frame (2), outboard cabin post (9), center post (1), and frame assembly (16) with a non-metallic spatula, putty knife, or other suitable tool (e.g., Snap-on P/N YA109C).

16. If required, remove the solid shim (7) from the windshield frame (2).

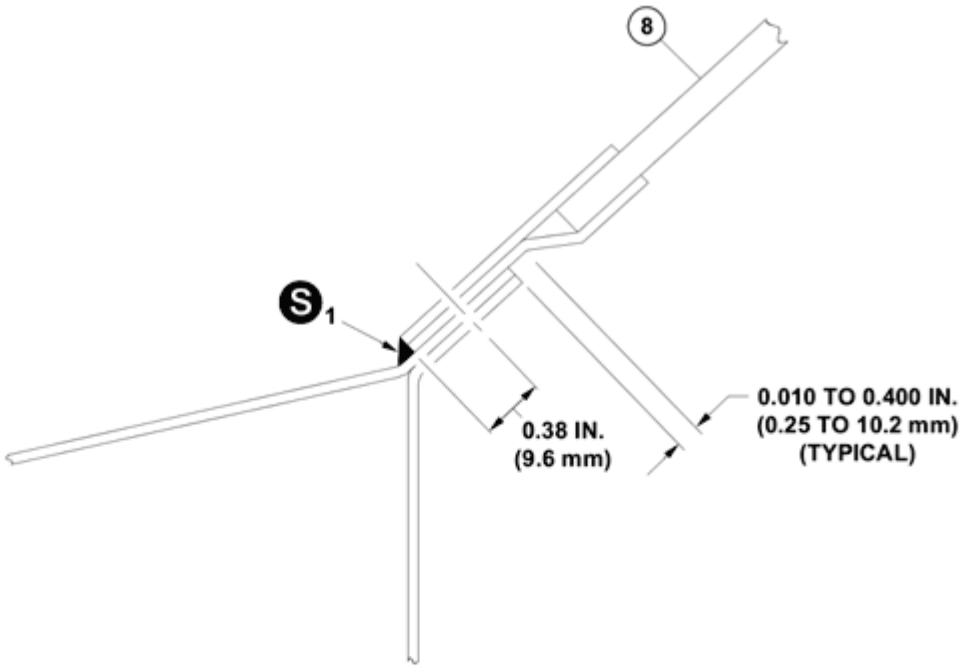
Applicable: Heated Windshield Kit (412-706-138): Installed:

17. *If required, remove the solid shim (24) from the outboard cabin post (9).*



ICN-412-A-520100-B-97499-00003-A-002-01

Figure 1. Heated Windshield - Removal and Installation (Sheet 1 of 2)



SECTION C-C
(TYPICAL)

1. Center post
2. Windshield frame
3. Stop
4. Screw
5. Washer
6. Nut
7. Solid shim
8. Heated windshield
9. Outboard cabin post
10. Washer
11. Nut
12. Screw
13. Screw
14. Fitting
15. Screw
16. Frame assembly
17. Cover assembly
18. Terminal board assembly
19. Screw
20. Washer
21. Nut
22. Screw
23. Screw
24. Solid shim

S1 SEALANT (C-151)

S2 SEALANT (C-328)

T 20 TO 25 IN-LBS
(2.3 TO 2.8 Nm)

ICN-412-A-520100-B-97499-00004-A-002-01

Figure 1. Heated Windshield - Removal and Installation (Sheet 2 of 2)

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Heated Windshield Kit (412-706-010/-138) - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	

Procedure



WHEN CLEANING WINDOWS, USE ONLY TYPE II aliphatic naphtha (C-305). ALIPHATIC NAPHTHA TYPE I WILL CLOUD THE WINDOWS.

1. For the cleaning of the windshield, refer to the instructions for the cleaning of plastic and glass in the BHT-ALL-SPM, Chapter 5, Standard Practices Manual.

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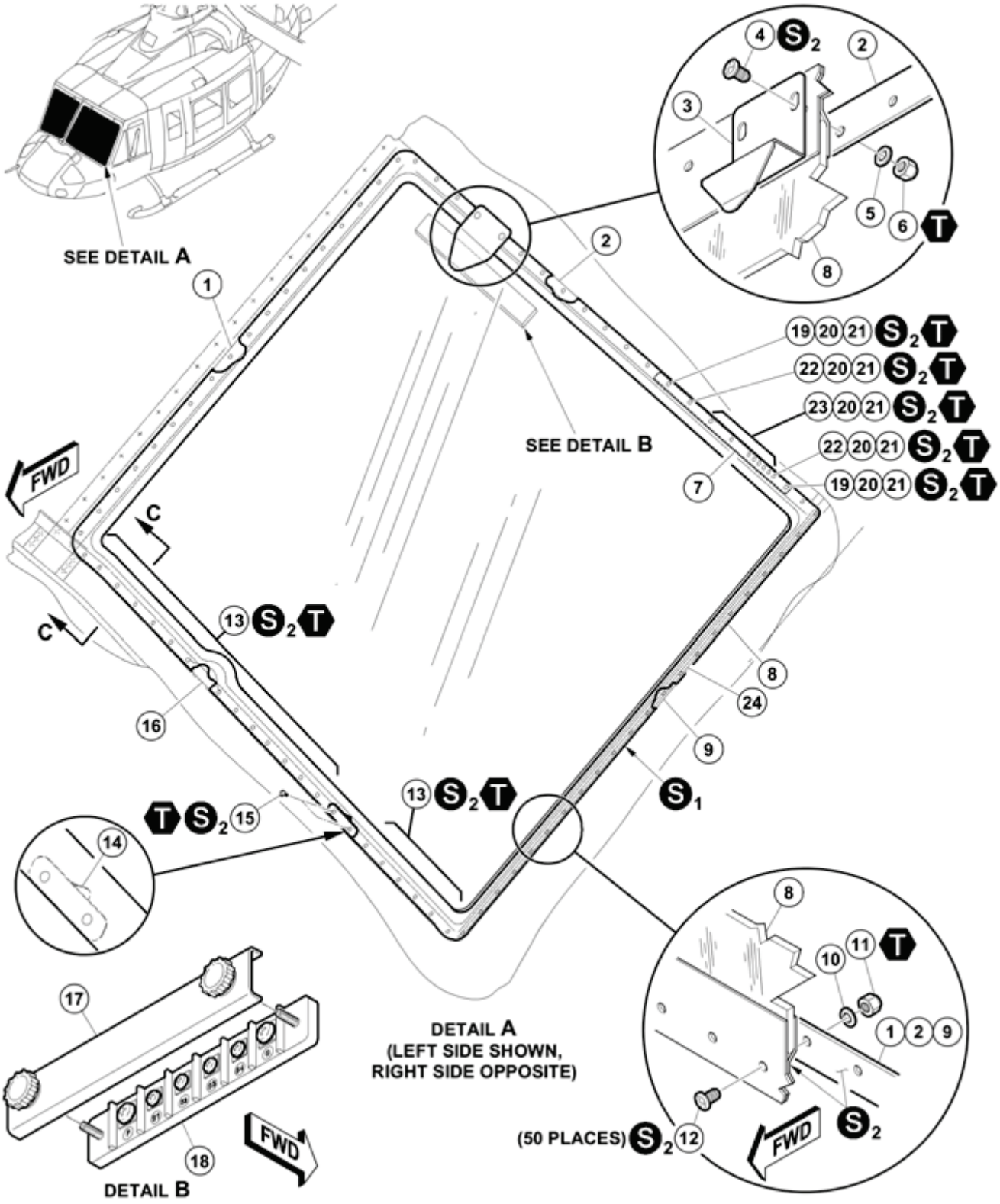
Heated Windshield Kit (412-706-010/-138) - Inspection

Procedure

NOTE

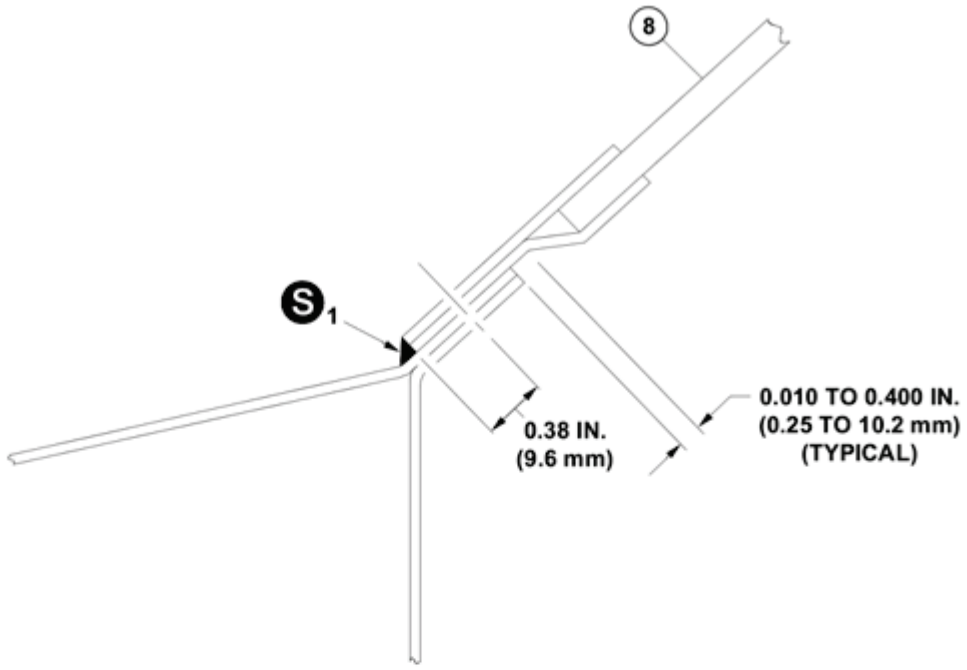
The following inspection criteria are primarily intended for when the part or component has been removed from the helicopter. To accommodate all inspection requirements, applicable criteria may also be used when the part or component is installed on the helicopter.

1. Inspect the heated windshield (8, Figure 1) frame for nicks, dents, scratches, distortion, and holes. Minor scratches, nicks, and non-puncturing dents are acceptable without repair if the damage does not adversely affect the windshield or windshield security.



ICN-412-A-520100-B-97499-00003-A-002-01

Figure 1. Heated Windshield - Removal and Installation (Sheet 1 of 2)



SECTION C-C
(TYPICAL)

1. Center post
2. Windshield frame
3. Stop
4. Screw
5. Washer
6. Nut
7. Solid shim
8. Heated windshield
9. Outboard cabin post
10. Washer
11. Nut
12. Screw
13. Screw
14. Fitting
15. Screw
16. Frame assembly
17. Cover assembly
18. Terminal board assembly
19. Screw
20. Washer
21. Nut
22. Screw
23. Screw
24. Solid shim

S1 SEALANT (C-151)

S2 SEALANT (C-328)

T 20 TO 25 IN-LBS
(2.3 TO 2.8 Nm)

ICN-412-A-520100-B-97499-00004-A-002-01

Figure 1. Heated Windshield - Removal and Installation (Sheet 2 of 2)

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Heated Windshield Kit (412-706-010/-138) - Repair

Procedure

1. Glass windshields cannot be repaired. Glass windshields must be replaced if there are cracks, or if there is distortion or low visibility due to cracks.

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Heated Windshield Kit (412-706-010/-138) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Chemical film material	C-100	AR	
Sealant	C-151	AR	
Epoxy polyamide primer	C-204	AR	
Aliphatic naphtha	C-305	AR	
Adhesive	*C-317*	*AR*	
Sealant	C-328	AR	
Adhesive tape	*C-460*	*AR*	
Clean cloth	C-516	AR	

Procedure



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT (412-A-96-00-01-02A-042A-A / 01874).

NOTE

The following procedure is for the heated windshield kit (8, Figure 1) on the left side. The procedure is the same for the heated windshield on the right side.

1. *Remove all sealant from the windshield frame (2), outboard cabin post (9), center post (1), frame assembly (16), and solid shims (7 and 24), as applicable with a non-metallic putty knife, spatula, or equivalent tool.*



WHEN CLEANING WINDOWS, USE ONLY TYPE II aliphatic naphtha (C-305). ALIPHATIC NAPHTHA TYPE I WILL CLOUD WINDOWS.

2. *Clean the windshield frame (2), outboard cabin post (9), center post (1), solid shims (7 and 24), as applicable, and frame assembly (16) with a clean cloth (C-516) moistened with aliphatic naphtha (C-305).*

NOTE

If the same heated windshield is installed go to Step 5.:

3. If a new heated windshield (8) is installed, do as follows:

3.1. If installed, remove the solid shim (7) from the windshield frame (2).

3.2. Remove the sealant from the solid shim (7).

Applicable: Heated Windshield Kit (412-706-138): Installed:

3.3. *If installed, remove the solid shim (24) from the outboard cabin post (9).*

Applicable: Heated Windshield Kit (412-706-138): Installed:

3.4. *Remove the sealant from the solid shim (24) and adhesive from the solid shim (24) and the outboard cabin post (9).*

3.5. Put the heated windshield (8) in position on the windshield frame (2), outboard cabin post (9), center post (1), and frame assembly (16) so that it mates with the contour.

NOTE

Do not trim the windshield to final size until all mounting holes have been drilled.

3.6. It is permitted to trim the heated windshield (8) only as required for temporary installation. Temporarily secure with heavy tape.

3.7. Make sure that a gap of 0.010 to 0.400 inch (0.25 to 10.2 mm) exists between the bevel edge on the inside surface of the heated windshield (8) and the windshield frame (2), outboard cabin post (9), center post (1), and frame assembly (16). Adjust, as required, until satisfactory.

3.8. Mark the location of all screw holes on the flange of the heated windshield (8) with a nonpermanent marker.

3.9. Remove the heated windshield (8) from the windshield frame (2), outboard cabin post (9), center post (1), and frame assembly (16).

3.10. Make sure the minimum edge distance is 0.38 inch (9.6 mm) for all holes. If not, adjust the location of holes to get the required edge distance.

3.11. At the center post location of the heated windshield (8), drill holes 0.190 to 0.196 inch (4.82 to 4.98 mm) in diameter through the flange of the windshield. Start at the middle area and continue to drill alternately above and below to get to the top and the bottom of the windshield to transfer the screw holes with a No. 10 drill.

3.12. At the top area of the heated windshield (8), start from the inboard edge and drill the holes 0.190 to 0.196 inch (4.82 to 4.98 mm) in diameter through the flange of the windshield to transfer the screw holes with a No. 10 drill.

3.13. At the bottom area of the heated windshield (8), start from the inboard edge and drill the holes 0.190 to 0.196 inch (4.82 to 4.98 mm) in diameter through the flange windshield to transfer the screw holes with a No. 10 drill.

3.14. At the outboard area of the heated windshield (8), drill holes 0.190 to 0.196 inch (4.82 to 4.98 mm) in diameter through the flange of the windshield to transfer the screw holes with a No. 10 drill.

3.15. Put the heated windshield (8) in position and align all the drilled holes with the holes in the windshield frame (2), outboard cabin post (9), center post (1), and frame assembly (16). Temporarily secure in place with Clecos fasteners.

3.16. Measure and record the gap between the heated windshield (8) and the windshield frame (2).

3.17. Verify that the edge distance is 0.38 inch (9.6 mm) for all the holes.

3.18. If required, mark the flange of the heated windshield (8) with a nonpermanent marker for the final trim adjustment.

3.19. Remove the heated windshield (8) from the windshield frame (2), outboard cabin post (9), center post (1), and frame assembly (16).

3.20. Trim the flange of the heated windshield (8) as required.

3.21. If you install a new heated windshield (8), do the shimming between the heated windshield and the windshield frame (2) as follows:

NOTE

The solid shim (7) is 0.250 inch (6.35 mm) thick at the middle area and is tapered to 0.010 inch (0.25mm) at each end. It is made of aluminum.

3.21.1. Trim the solid shim (7) to fill the gap between the heated windshield (8) and the windshield frame (2).

3.21.2. Apply chemical film material (C-100) to the shim (BHT-ALL-SPM, Chapter 3, Standard Practices Manual).

3.21.3. Apply a coat of epoxy polyamide primer (C-204) to the solid shim (7).

3.21.4. Put the solid shim (7) in position on the inboard side of the flange of the heated windshield (8), centered with the holes in the windshield.

3.21.5. Drill the holes 0.190 to 0.196 inch (4.82 to 4.98 mm) in diameter from the flange of the heated windshield (8) through the solid shim (7) with a No. 10 drill.

3.22. Open all the holes 0.203 to 0.208 inch (5.16 to 5.28 mm) in diameter from the flange of the heated windshield (8) through the solid shim (7), as required, with a No. 5 drill.

3.23. Deburr all the holes on the flange of the heated windshield (8) and the solid shim (7), as required.

3.24. If required, install the solid shim (7) on the windshield frame (2) with sealant (C-328).

3.25. *DELETED*

Applicable: Heated Windshield Kit (412-706-138): Installed:

4. *If a new heated windshield (8) is installed, do the shimming procedure as follows:*

4.1. *Determine the shimming requirement between the new heated windshield (8) and outboard cabin post (9) as follows:*

4.1.1. *Put the new heated windshield (8) in position on the windshield frame (2), outboard cabin post (9), center post (1), and frame assembly (16) so that it mates with the contour.*

4.1.2. *Temporarily install the fasteners (screw, washer and nut, as applicable) on the heated windshield (8) with a minimum gap of 4 holes (one screw out of five).*

4.1.3. *Hand torque the screws or nuts as applicable from 2 to 5 inch pounds (0.225 to 0.565 Nm) and make sure that the clamp load is reduced to a minimum.*

4.1.4. *With the heated windshield (8) positioned in place, measure and record the gaps under the screw heads on the outboard cabin post (9).*

4.1.5. *Remove the heated windshield (8) from the windshield frame (2), outboard cabin post (9), center post (1), and frame assembly (16).*

4.1.6. *If the gaps are within 0.01 inch (0.254 mm), go to* Step 5..

NOTE

During the final installation of the heated windshield (8), if required apply excessive sealant to seal the gaps between the new heated windshield (8) and outboard cabin post (9).

4.1.7. *Do the shimming procedure as per the condition below based on recorded gaps:*

- *If the gaps are between 0.01 inch (0.254 mm) and 0.03 inch (0.762 mm), apply shimming compound as per* Step 4.2.
- *If the gaps greater than 0.03 inch (0.762 mm), install the solid shim (24) made of aluminum as per* Step 4.3.

4.2. *Apply shimming compound on the outboard cabin post (9) surface that mates with the heated windshield (8) as follows:*

4.2.1. *On the outboard cabin post (9), lightly abrade the area to be shimmed with 180 grit or finer and make sure not to remove the primer.*

4.2.2. *Clean the outboard cabin post (9) with a* clean cloth (C-516) *moistened with* aliphatic naphtha (C-305).

4.2.3. *Apply* adhesive (C-317) *on the mating surface (to be shimmed) of the outboard cabin post (9) in a sheet form.*

NOTE

If additional sheets of shimming compound are required to obtain thickness, apply additional sheets of shimming compound directly over the original lay-up until the desired thickness is obtained. Make sure that there is no air entrapment.

NOTE

Make sure that the holes on the flange of the outboard cabin post (9) is not filled with adhesive.

4.2.4. *Go to* Step 4.4.

4.3. *Install the solid shim (24) on the outboard cabin post (9) surface that mates with the heated windshield (8) as follows:*

NOTE

Sold shim (24) is supplied with two different thickness 0.063 inch (1.6 mm) and 0.1 inch (2.54 mm). The adhesive (C-317) *bond line thickness must be considered when you select the solid shim thickness.*

4.3.1. *Select the solid shim (24) in accordance with the dimensions of the gap to be shimmed.*

4.3.2. *Trim the solid shim (24) as required to the fill the gap between heated windshield (8) and outboard cabin post (9).*

4.3.3. *Put the shim (24) in its position on the flange of the outboard cabin post (9).*

4.3.4. *Use the holes of the outboard cabin post (9) as a template, mark the holes location on the shim (24) with a temporary marker.*

4.3.5. *Drill the holes 0.190 to 0.196 inch (4.82 to 4.98 mm) in diameter on the solid shim (24), as required with a No. 10 drill.*

4.3.6. *Open all the holes 0.203 to 0.208 inch (5.16 to 5.28 mm) in diameter on the solid shim (24), as required, with a No. 5 drill.*

4.3.7. *Deburr all the holes on the solid shim (24), as required.*

4.3.8. *Apply* chemical film material (C-100) *to the shim (24) (BHT-ALL-SPM, Chapter 3, Standard Practices Manual).*

4.3.9. *Apply a coat of* epoxy polyamide primer (C-204) *to the solid shim (24).*

4.3.10. *Clean the surface of the solid shim (24) with a* clean cloth (C-516) *moistened with* aliphatic naphtha (C-305).

4.3.11. *On the outboard cabin post (9), lightly abrade the area to be shimmed with 180 grit or finer and make sure not to remove the primer.*

4.3.12. *Clean the surface of the outboard cabin post (9) with a* clean cloth (C-516) *moistened with* aliphatic naphtha (C-305).

4.3.13. *Apply a thin continuous film of* adhesive (C-317) *over the entire length of solid shim (24) and bond with outboard cabin post (9) structure. Make sure there is a smooth transition between the adhesive and the adjacent solid shims.*

4.4. *Apply* adhesive tape (C-460) *around the periphery of the heated windshield (8) to avoid adhesive from bonding to heated windshield (8).*

4.5. *Install the heated windshield (8) as per the* Step 4.1.1. *thru* Step 4.1.3..

4.6. *Allow the adhesive to cure for 8 to 12 hours.*

4.7. *Trim the cured adhesive as required.*

4.8. *Remove the heated windshield (8) from the windshield frame (2), outboard cabin post (9), center post (1), and frame assembly (16).*



WHEN CLEANING WINDOWS, USE ONLY TYPE II aliphatic naphtha (C-305). ALIPHATIC NAPHTHA TYPE I WILL CLOUD THE WINDOWS.

5. *Clean the heated windshield (8) and the windshield frame (2), outboard cabin post (9), center post (1), solid shims (7 and 24), as required, and frame assembly (16) with a* clean cloth (C-516) *moistened with* aliphatic naphtha (C-305).

6. *Apply* sealant (C-328) *to the faying surfaces of the heated windshield (8), solid shims (7 and 24), as required, windshield frame (2), outboard cabin post (9), center post (1), and frame assembly (16).*

7. *Put the heated windshield (8) in position on the windshield frame (2), outboard cabin post (9), center post (1), solid shims (7 and 24), as required, and frame assembly (16). Temporarily secure with Cleco fasteners.*

NOTE

The total quantity of screws for the left heated windshield (8) is 84. The total quantity of screws on the right windshield is 83.

8. Wet install the screws (15) on the heated windshield (8), fitting (14), and frame assembly (16) with sealant (C-328).

9. Wet install the screws (13) on the heated windshield (8) and the frame assembly (16) with sealant (C-328).

10. Apply sealant (C-328) to the shanks of the screws (4).

11. Put the wiper stop (3) in position at the fourth and fifth hole on the heated windshield (8) and windshield frame (2) and secure with the screws (4), washers (5), and nuts (6).

12. At two locations, wet install the screw (19), washer (20), and nut (21) on the heated windshield (8), solid shim (7), as required, and windshield frame (2) with sealant (C-328).

13. At two locations, wet install the screw (22), washer (20), and nut (21) on the heated windshield (8), solid shim (7), as required, and windshield frame (2) with sealant (C-328).

14. At seven locations, wet install the screw (23), washer (20), and nut (21) on the heated windshield (8), solid shim (7), as required, and windshield frame (2) with sealant (C-328).

15. At 50 locations, wet install the screw (12), washer (10), and nut (11) on the heated windshield (8), center post (1), windshield frame (2), and outboard cabin post (9) with sealant (C-328).

16. Increase torque slowly and equally on the screws (13 and 15) and the nuts (6, 11, and 21). Torque the screws and nuts before the sealant cures .



WHEN CLEANING WINDOWS, USE ONLY TYPE II aliphatic naphtha (C-305). ALIPHATIC NAPHTHA TYPE I WILL CLOUD THE WINDOWS.

17. Clean excess sealant from around the heated windshield (8) with a clean cloth (C-516) moistened with aliphatic naphtha (C-305).

18. Apply a bead of sealant (C-151) to seal the edges of the heated windshield (8).

18.1. Tag and disconnect the terminal lugs from the terminal board assembly (18).



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT (412-A-96-00-01-02A-042A-A / 01874).

NOTE

Refer to 412-A-98-00-00-41B-051A-A / 02601 for the correct installation of the wires and contacts.

19. Remove the tag and connect the terminal to the terminal board assembly (18).

20. Install the cover assembly (17) on the terminal board assembly (18). Tighten the nuts.

21. Install the center post cover.

22. Install the windshield wiper arm (412-A-30-40-00-02A-720A-A / 00688) and blade (412-A-30-40-00-01A-720A-A / 00686).

23. Connect the battery and/or the external power to the helicopter (412-A-96-00-00-00A-009A-A / 01870).

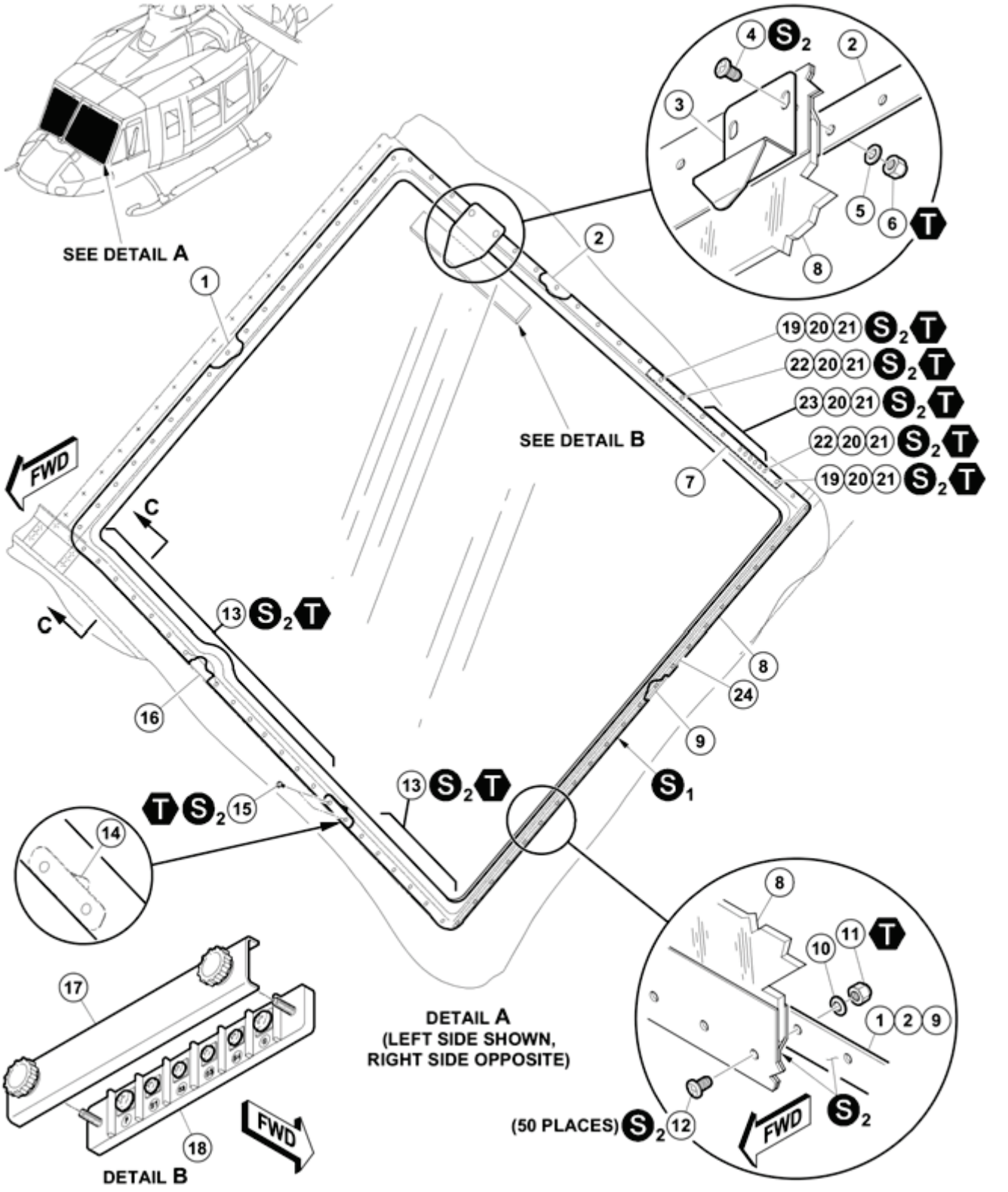
24. Close the nose doors and/or panels, as applicable (412-A-52-40-00-01A-720A-A / 00835).

Applicable: SN: 36087-36999:

25. Do an operational check of the heated windshield (8) (BHT-412-SI-49, Chapter 2).

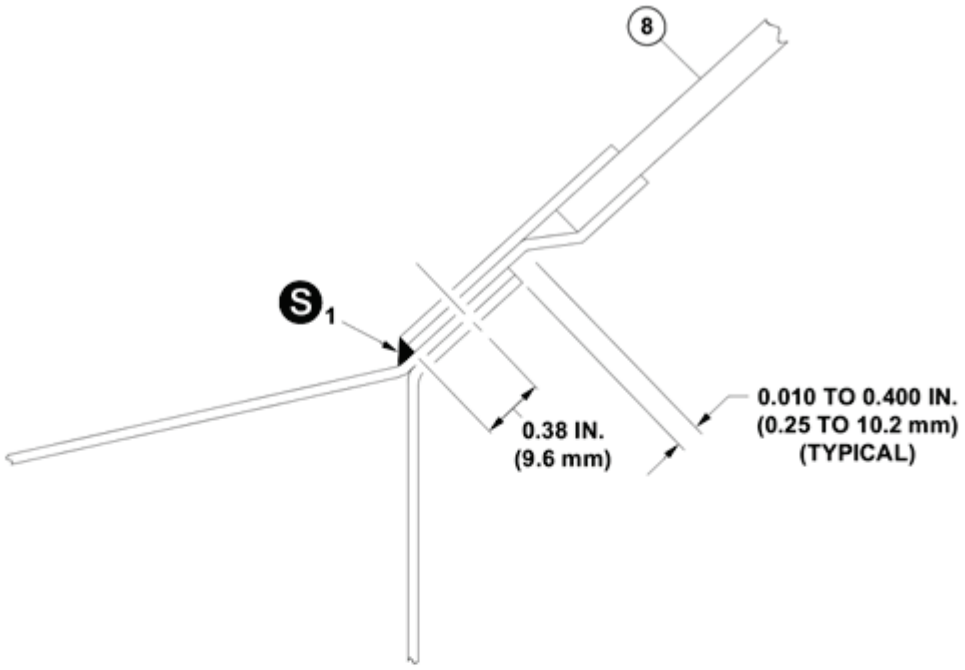
Applicable: SN: 37002-37999:

26. Do an operational check of the heated windshield (8) (BHT-412-MMS-EPI, Chapter 96).



ICN-412-A-520100-B-97499-00003-A-002-01

Figure 1. Heated Windshield - Removal and Installation (Sheet 1 of 2)



SECTION C-C
(TYPICAL)

1. Center post
2. Windshield frame
3. Stop
4. Screw
5. Washer
6. Nut
7. Solid shim
8. Heated windshield
9. Outboard cabin post
10. Washer
11. Nut
12. Screw
13. Screw
14. Fitting
15. Screw
16. Frame assembly
17. Cover assembly
18. Terminal board assembly
19. Screw
20. Washer
21. Nut
22. Screw
23. Screw
24. Solid shim

S1 SEALANT (C-151)

S2 SEALANT (C-328)

T 20 TO 25 IN-LBS
(2.3 TO 2.8 Nm)

ICN-412-A-520100-B-97499-00004-A-002-01

Figure 1. Heated Windshield - Removal and Installation (Sheet 2 of 2)

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CHAPTER 53
FUSELAGE

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

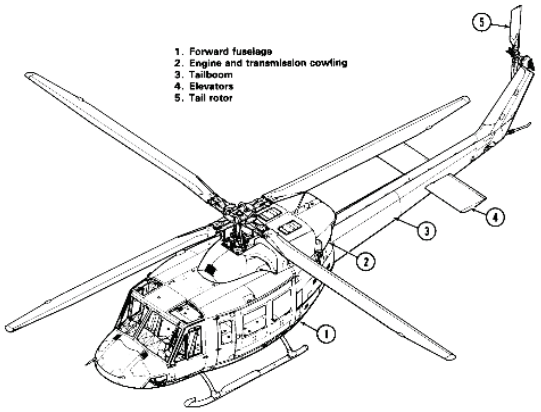
C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Fuselage - Description	DMC-412-A-53-00-00-00A-028A-A / 00898	2023-05-31	ALL
Generic Sheet Metal Part - Replacement Procedure	DMC-412-A-53-00-00-00A-921A-A / 00899	2023-05-31	ALL
Forward Fuselage - Description	DMC-412-A-53-10-00-00A-042A-A / 00900	2023-05-31	ALL
Fuel Cell Cover - Removal	DMC-412-A-53-10-00-01A-520A-A / 00901	2023-05-31	ALL
Fuel Cell Cover - Inspection	DMC-412-A-53-10-00-01A-280A-A / 00902	2023-05-31	ALL
Fuel Cell Cover - Installation	DMC-412-A-53-10-00-01A-720A-A / 00903	2023-05-31	ALL
Crew Seat Tracks - Removal	DMC-412-A-53-10-00-02A-520A-A / 00904	2023-05-31	SN: Prior to 36420
Crew Seat Tracks - Inspection	DMC-412-A-53-10-00-02A-280A-A / 00905	2023-05-31	SN: Prior to 36420
Crew Seat Tracks - Installation	DMC-412-A-53-10-00-02A-720A-A / 00906	2023-05-31	SN: Prior to 36420
Crew Seat Tracks - Description	DMC-412-A-53-10-00-02B-042A-A / 00907	2023-05-31	SN: 36420-Subsequent
Tension Rod Assembly - Removal	DMC-412-A-53-10-00-03A-520A-A / 00908	2023-05-31	ALL
Tension Rod Assembly - Inspection	DMC-412-A-53-10-00-03A-280A-A / 00909	2023-05-31	ALL
Tension Rod Assembly - Installation	DMC-412-A-53-10-00-03A-720A-A / 00910	2023-05-31	ALL
Tailboom - Description	DMC-412-A-53-50-00-00A-042A-A / 00911	2023-05-31	ALL
Tailboom - Removal	DMC-412-A-53-50-00-00A-520A-A / 00912	2023-05-31	ALL
Tailboom - Inspection	DMC-412-A-53-50-00-00A-280A-A / 00913	2023-05-31	ALL
Tailboom - Repair	DMC-412-A-53-50-00-00A-685A-A / 00914	2023-05-31	ALL
Tailboom - Installation	DMC-412-A-53-50-00-00A-720A-A / 00915	2023-05-31	ALL
Upper Tail Fairing - Description	DMC-412-A-53-50-00-01A-042A-A / 00916	2023-05-31	ALL
Upper Tail Fairing - Removal	DMC-412-A-53-50-00-01A-520A-A / 00917	2023-05-31	ALL
Upper Tail Fairing - Cleaning	DMC-412-A-53-50-00-01A-251A-A / 00918	2023-05-31	ALL
Upper Tail Fairing - Inspection	DMC-412-A-53-50-00-01A-280A-A / 00919	2023-05-31	ALL
Upper Tail Fairing - Repair	DMC-412-A-53-50-00-01A-685A-A / 00920	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Upper Tail Fairing - Installation	DMC-412-A-53-50-00-01A-720A-A / 00921	2023-05-31	ALL
Lower Tail Fairing - Description	DMC-412-A-53-50-00-02A-042A-A / 00922	2023-05-31	ALL
Lower Tail Fairing - Removal	DMC-412-A-53-50-00-02A-520A-A / 00923	2023-05-31	ALL
Lower Tail Fairing - Cleaning	DMC-412-A-53-50-00-02A-251A-A / 00924	2023-05-31	ALL
Lower Tail Fairing - Inspection	DMC-412-A-53-50-00-02A-280A-A / 00925	2023-05-31	ALL
Lower Tail Fairing - Repair	DMC-412-A-53-50-00-02A-685A-A / 00926	2023-05-31	ALL
Lower Tail Fairing - Installation	DMC-412-A-53-50-00-02A-720A-A / 00927	2023-05-31	ALL
Strakes - Description	DMC-412-A-53-50-00-03A-042A-A / 00928	2023-05-31	ALL
Strakes - Inspection	DMC-412-A-53-50-00-03A-280A-A / 00929	2023-05-31	ALL
Strakes - Repair	DMC-412-A-53-50-00-03A-685A-A / 00930	2023-05-31	ALL
Strakes - Replacement	DMC-412-A-53-50-00-03A-921A-A / 00931	2023-05-31	ALL

Fuselage - Description

The fuselage (Figure 1) consists of the forward fuselage and tailboom. The forward fuselage is a combined semi-monocoque and reinforced shell construction. The tailboom is a semi-monocoque construction.



ICN-412-A-530000-B-97499-00001-A-001-01

Figure 1. Fuselage Assembly (Sheet 1 of 1)

Generic Sheet Metal Part - Replacement Procedure

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-251	AR	
Sealant	C-308	AR	
Sealant	C-328	AR	

Procedure

NOTE

As no procedure can cover all sheet metal part replacements, the following guidelines can be used for most applications, but must be adapted as required. Refer to the BHT-MED-SRM-1 for specific applications. For general practices, refer to FAA Publication AC 43.13-1B.

1. Once the part to be replaced has been determined, provide bracing and support as required. Refer to BHT-MED-SRM-1 (Table 4-1) for suggestions.

2. Identify the type, size, and location of fasteners by looking at the manufactured head of the fasteners and referring to information in the BHT-MED-SRM-1.

3. Remove part while making a note which portion of the part was installed with sealant (if applicable).

4. Inspect supporting structure for cracks and damaged fastener holes. Repair or replace as required.

5. Trim and drill new part to fit in place. Maintain edge distance of 2D for all regular holes and 2.5D for all countersunk holes. Deburr.

6. Apply sealant on faying surfaces, as determined previously (if applicable). Unless otherwise stated, use sealant (C-251) or sealant (C-308) for general applications, or sealant (C-328) for screwed-on access panels (low adhesion).

7. Install fasteners as identified previously. Apply a small bead of sealant (C-251) or sealant (C-308) around the part (if required).

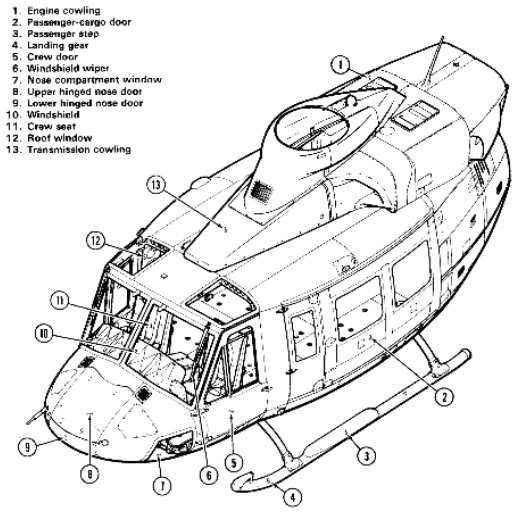
8. Refinish as required.

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Forward Fuselage - Description

The forward fuselage (Figure 1) is built up on two main longitudinal beams and is a combined semi-monocoque and reinforced-shell construction with transverse bulkheads and metal covering. The longitudinal beams act as a primary supporting structure for the cabin section, landing gear, fuel cells, power plant, transmission, tailboom, and useful load. Work and engine decks, some main beam panels, as well as cabin floor, cabin roof panels, are of aluminum alloy honeycomb construction. Avionics and electrical equipment is contained primarily in the nose compartment. Lower and upper hinged nose doors (8 and 9) provide access to the nose compartment. A windshield wiper (6) is located on each of the two windshields (10). Crew seats (11) are located in the forward cabin area. Crew doors (5), passenger-cargo doors (2), and hinged panel doors on both sides of the helicopter permit entrance and exit. Acrylic windows in the lower nose compartment, roof window (12), and doors permit a wide range of visibility. Transmission cowling (13) and engine cowling may be opened or removed for access. The cargo-passenger area is equipped with tie-down rings and studs.

Repair of structural components of the forward fuselage is provided in the BHT-MED-SRM-1. Repair of components installed on the fuselage is contained within appropriate chapters of this manual.



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Figure 1. Forward Fuselage (Sheet 1 of 1)

Fuel Cell Cover - Removal

Procedure

1. Remove all screws attaching fuel cell panel, noting the type, size, and location of screws.

2. Remove fuel cell panel while paying attention not to damage underlying structure. Do not use a sharp object to pry cover off.

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Fuel Cell Cover - Inspection

Procedure

1. Inspect supporting structure for mechanical damage, cracks, and damaged fastener holes. Repair or replace as required and refinish.

2. Inspect cover panel for delamination, corrosion, and mechanical damage. Repair or replace as required.

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Fuel Cell Cover - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-251	AR	
Sealant	C-308	AR	
Sealant	C-328	AR	

Procedure

1. Apply a continuous bead (approximately 1/8 to 1/4-inch diameter) of sealant on one of the faying surfaces. Do not spread. Unless otherwise stated, use sealant (C-328) (low adhesion).

2. Install cover.

3. Install fasteners as identified previously. Remove excess sealant. Apply a small bead of sealant (C-251) or sealant (C-308) around the edge of cover (if required). Remove excess sealant.

4. Refinish as required.

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Crew Seat Tracks - Removal

Procedure

1. Remove the applicable crew seat (412-A-25-10-02-00A-520A-A / 00248).

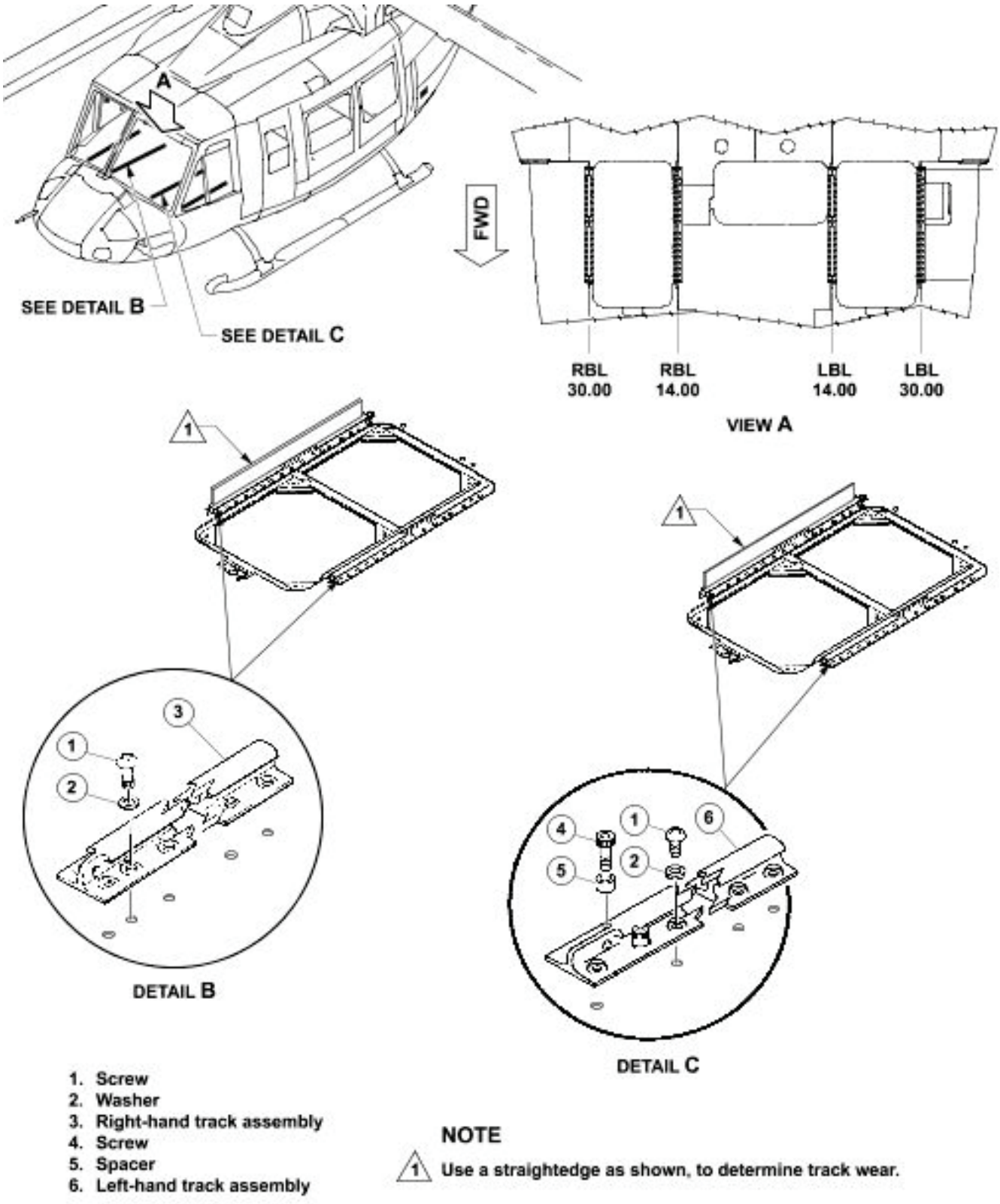
2. Remove the right-hand track assemblies (3, Figure 1) as follows:

2.1. Remove the screws (1), washers (2), and the right-hand track assembly (3).

3. Remove left-hand track assemblies (6) as follows:

3.1. Remove the two screws (4) and spacers (5).

3.2. Remove the screws (1), washers (2), and the left-hand track assembly (6).



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Figure 1. Crew Seat Track Assemblies - Removal and Installation (Sheet 1 of 1)

Crew Seat Tracks - Inspection

Procedure

NOTE

It is not necessary to remove a track assembly (3 or 6, Figure 1) to inspect it for wear; however, the seat assembly must be removed.

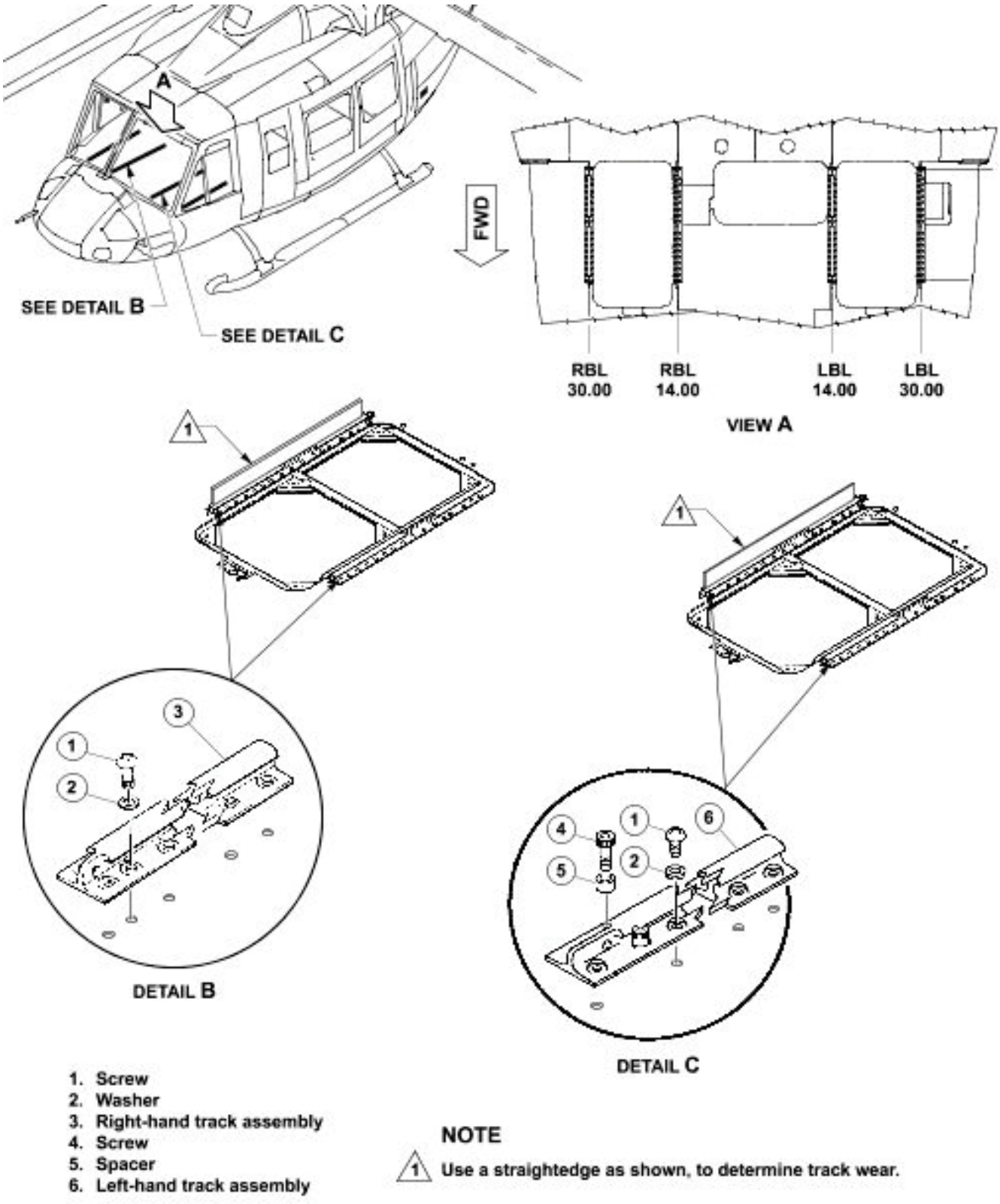
1. Hold a straightedge against the crown of the central rail of the right-hand or left-hand track assembly (3 or 6) as shown. The straightedge must be at least as long as the track assembly.

2. Look for gaps between the bottom of the straightedge and the rail, particularly near the adjustment holes at the forward end.

3. If you find a gap that is more than 0.050 inch (1.27 mm), remove and replace the right-hand or left-hand track assembly (3 or 6) with a serviceable part (412-A-53-10-00-02A-520A-A / 00904 and 412-A-53-10-00-02A-720A-A / 00906).

4. Visually examine the exposed areas of the right-hand or left-hand track assembly (3 or 6) for cracks, mechanical damage, and corrosion damage. Replace as required.

5. Visually examine the exposed areas of the attaching hardware for wear, mechanical damage, and corrosion damage. Replace as required.



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Figure 1. Crew Seat Track Assemblies - Removal and Installation (Sheet 1 of 1)

Crew Seat Tracks - Installation

Procedure

1. Install right-hand track assemblies (3, Figure 1) as follows:

1.1. Make sure the mating surfaces are clean.

1.2. Install the right-hand track assembly (3) with the screws (1) and the washers (2).



MAKE SURE YOU INSTALL THE SPACERS (5) ON THE LEFT-HAND TRACK ASSEMBLY (6) AS SHOWN. IF THE SPACERS ARE OMITTED, THE SEAT CAN INTERFERE WITH COLLECTIVE STICK TRAVEL.

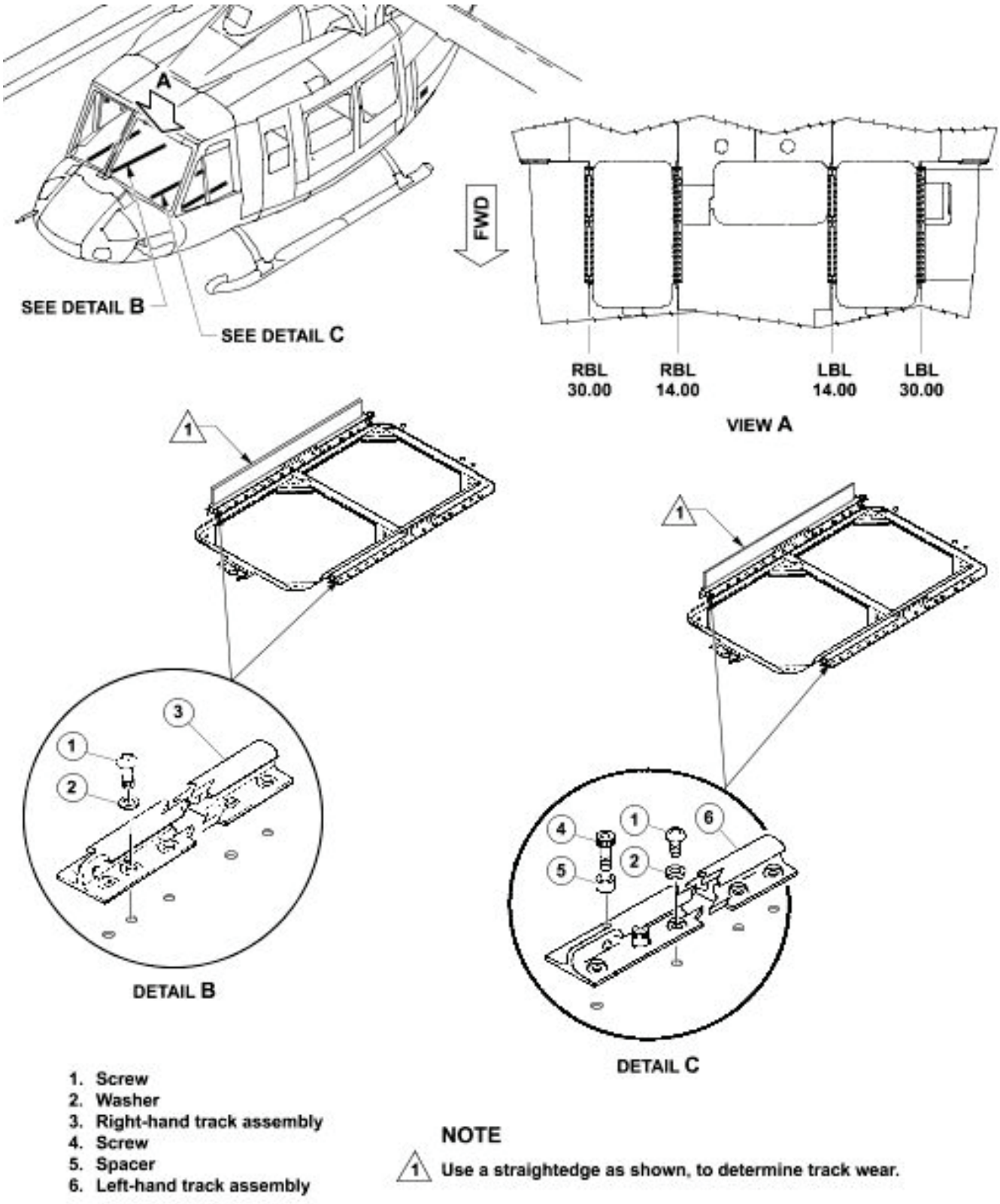
2. Install left-hand track assemblies (6) as follows:

2.1. Make sure the mating surfaces are clean.

2.2. Install the left-hand track assembly (6) with the two screws (4) and spacers (5) in the second hole position from the front of the track.

2.3. Install the screws (1) and the washers (2) in the remaining holes.

3. Install the applicable crew seat (412-A-25-10-02-00A-720A-A / 00251).



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Figure 1. Crew Seat Track Assemblies - Removal and Installation (Sheet 1 of 1)

Crew Seat Tracks - Description

For the helicopters with crash attenuating crew seats installed, refer to BHT-412-SI-75 for the maintenance procedures of crew seat tracks.

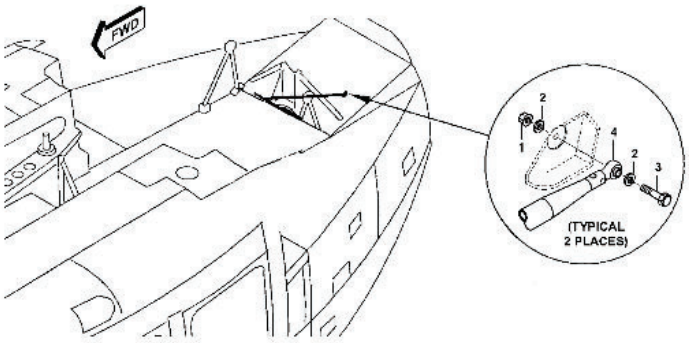
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Tension Rod Assembly - Removal

Procedure

1. Remove the nut (1, Figure 1), bolt (3), and washers (2) from the aft end of the tension rod assembly (4).

2. Remove the nut (1), bolt (3), washers (2), and tension rod assembly (4) from the bulkhead attachment point.



1. Nut
2. Washer
3. Bolt
4. Tension rod assembly

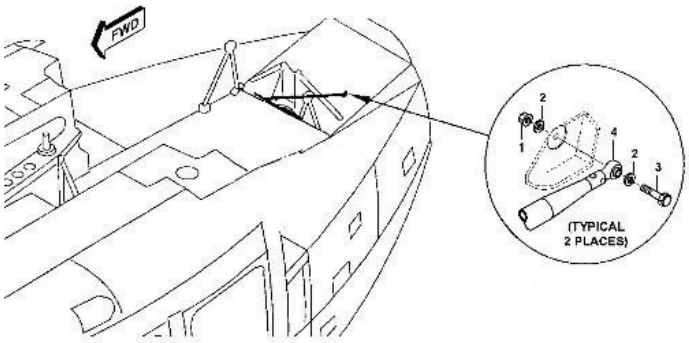
ICN-412-A-531000-B-97499-00003-A-001-01

Figure 1. Tension Rod Assembly - Removal and Installation (Sheet 1 of 1)

Tension Rod Assembly - Inspection

Procedure

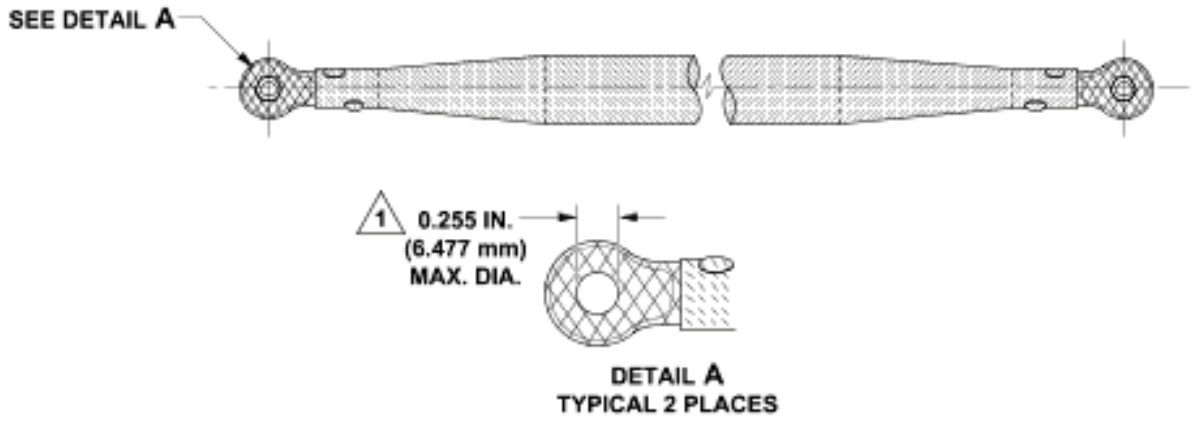
1. Examine the tension rod assembly (4, Figure 1) for cracks, wear, mechanical damage, and corrosion damage (Figure 2). Repair or replace as applicable.



1. Nut
2. Washer
3. Bolt
4. Tension rod assembly

ICN-412-A-531000-B-97499-00003-A-001-01

Figure 1. Tension Rod Assembly - Removal and Installation (Sheet 1 of 1)



TENSION ROD (212-030-170)

TYPE OF DAMAGE	DAMAGE AREA REPAIR SYMBOLS MAXIMUM DEPTH AND REPAIR AREAS ALLOWED	
SCRATCHES AND NICKS	0.010 In. (0.254 mm) after cleanup	
SCRATCHES		0.005 In. (0.127 mm) after cleanup 2
DENTS		Depth: 0.005 In. (0.127 mm) Length: 1.00 In. (25.40 mm) maximum 2
CORROSION	0.005 In. (0.127 mm) to 0.010 In. (0.254 mm) after cleanup	0.0025 In. (0.0635 mm) before and 0.005 In. (0.127 mm) after cleanup 3

NOTES

1. Bushings 20-006J-8 or 20-006C-8 can be installed in the holes. Wet install with epoxy polyamide primer (C-204). Ream bushings from 0.249 to 0.251 inch (6.325 to 6.375 mm) after installation.
2. Cleanup in longitudinal direction only and must not be more than 25% of the circumference at each location.
3. Dent must be to a minimum of 3.00 inches (76.20 mm) apart and no dent is permitted in the neck area.

ICN-412-A-531000-B-97499-00004-A-001-01

Figure 2. Tension Rod Assembly - Damage Limits (Sheet 1 of 1)

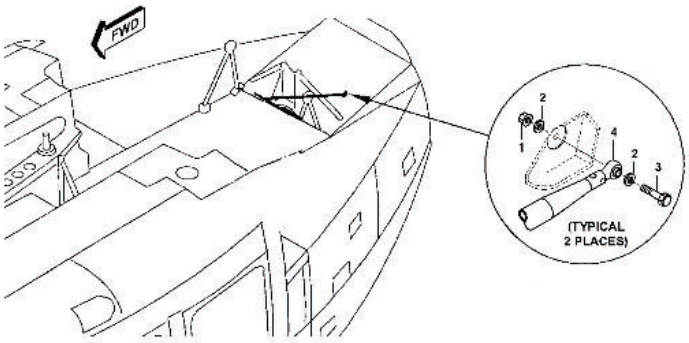
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Tension Rod Assembly - Installation

Procedure

1. Install the tension rod assembly (4, Figure 1) on the bulkhead attachment point with the bolt (3), washers (2), and nut (1).

2. Install the aft end of the tension rod assembly (4) with the bolt (3), washers (2), and nut (1). Tighten the nuts.



1. Nut
2. Washer
3. Bolt
4. Tension rod assembly

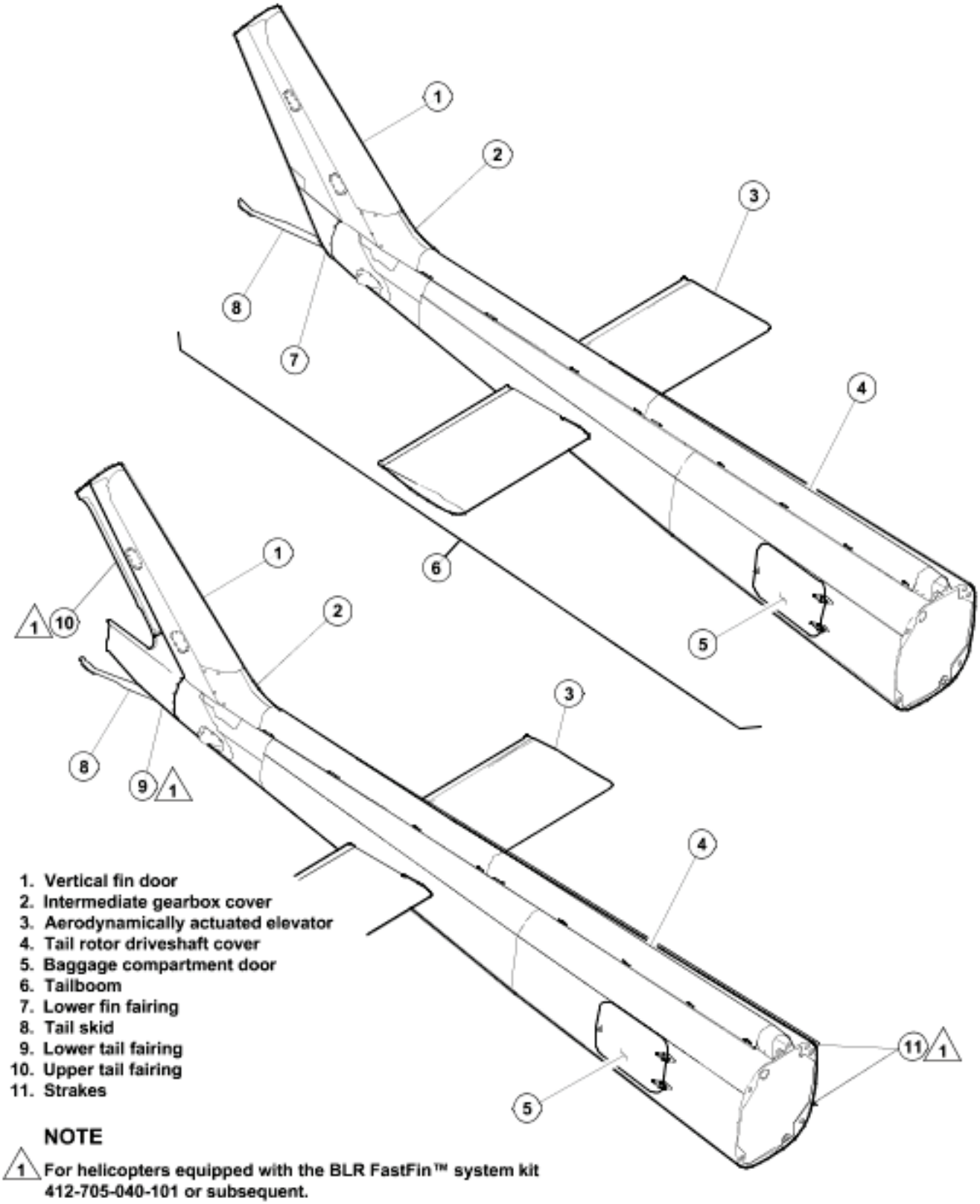
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Figure 1. Tension Rod Assembly - Removal and Installation (Sheet 1 of 1)

Tailboom - Description

The tailboom (Figure 1) is of semi-monocoque construction. External components consist of formed aluminum alloy skins and tail rotor driveshaft covers (4), a fiberglass fairing, and one aluminum alloy honeycomb cover. Internal components of the tailboom and aerodynamically actuated elevator are of aluminum alloy. The tailboom supports the tail rotor driveshafts, gearboxes, and driveshaft hangers. The baggage compartment is located in the forward end with the door (5) on the right side.

On helicopters equipped with the BLR FastFin™ system kit 412-705-040-101 (or Subsequent), an upper and lower tail fairing (10 and 9) provide the vertical fin with a new shape to increase thrust from the tail rotor. As part of this kit, the tailboom includes two aerodynamic strakes (11) to control the down-wash from the main rotor around the tailboom.



1. Vertical fin door
2. Intermediate gearbox cover
3. Aerodynamically actuated elevator
4. Tail rotor driveshaft cover
5. Baggage compartment door
6. Tailboom
7. Lower fin fairing
8. Tail skid
9. Lower tail fairing
10. Upper tail fairing
11. Strakes

NOTE

1 For helicopters equipped with the BLR FastFin™ system kit 412-705-040-101 or subsequent.

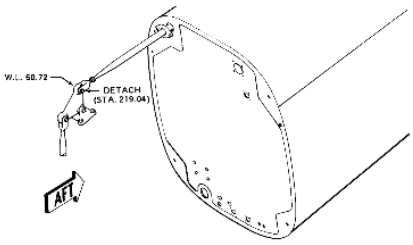
ICN-412-A-535000-B-97499-00001-A-001-01

Figure 1. Tailboom (Sheet 1 of 1)

Tailboom - Removal

Procedure

1. Open combining gearbox side panels.
2. Remove clamps from No. 2 driveshaft (412-A-63-21-02-07A-520A-A / 01171). Remove driveshaft section.
3. Open access door on right aft side of forward fuselage. Disconnect electrical harnesses and antenna cables.
4. Disconnect tail rotor control tube (Figure 1).
5. Remove plug bottoms from four tailboom attaching points.
6. Support tailboom. Remove bolts attaching tailboom to fuselage.
7. Remove tailboom. Place in a suitable cradle.



ICN-412-A-535000-B-97499-00002-A-001-01

Figure 1. Disconnect Points for Tailboom (Sheet 1 of 1)

Tailboom - Inspection

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	
MEK	C-309	AR	

Procedure

1. Inspect doors, covers, and access panels for misalignment, damage, and security of mounting.

2. Inspect tailboom skid for dents, corrosion, scratches, or other damage.

3. Inspect structure for cracks, distortion, corrosion, or loose rivets.

4. Inspect forward side of vertical fin spar caps and web from upper tailboom skin to approximately 4.0 inches (101.6 mm) below upper tailboom skin as follows:

4.1. Remove aft tailboom flight control inspection door (412-A-52-40-00-03A-520A-A / 00842).

NOTE

If necessary, clean area to be inspected with cloth dampened with MEK (C-309).

4.2. Face aft from inside of tailboom and use bright light and small mirror to inspect area for cracks. Pay particular attention to areas near rivet holes, especially cold worked holes.

5. Inspect tailboom upper longerons as follows:

5.1. Remove plug button at BS 99.0.

NOTE

Paint fissures are common in the splice area, cracks will be evident by black powder emanating from cracked area, and corrosion will leave evidence of white powder.

5.2. Using a borescope or suitable means, inspect inside of longeron, paying particular attention to cracked and/or corrosion in area of splice.

5.3. Report any cracks to Product Support Engineering.

5.4. Install plug button with sealant (C-308).

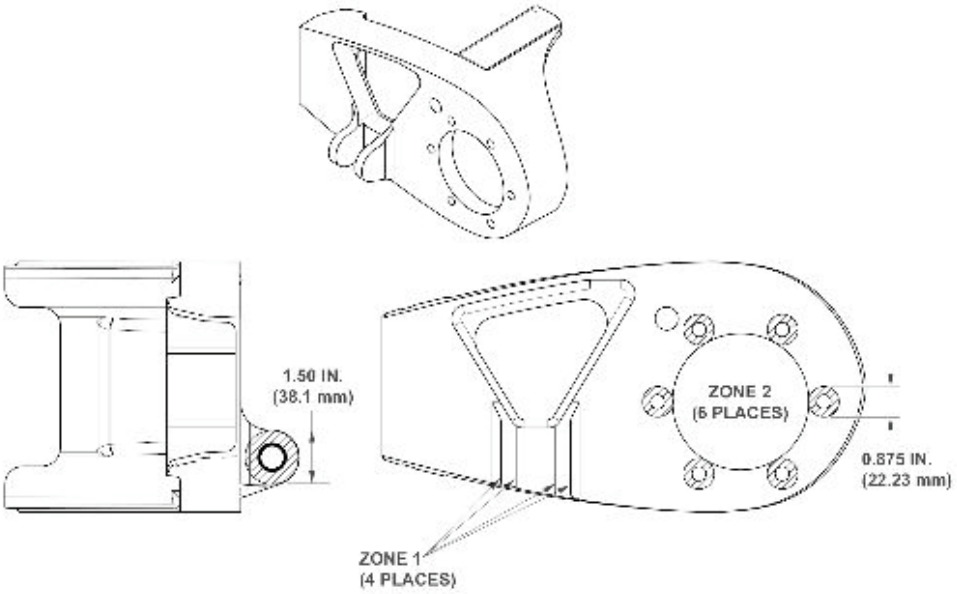
6. Inspect tailboom bonded panels for dents, scratches, delamination and corrosion.

7. Inspect diameter of tailboom to fuselage attaching holes. The allowable limits are as follows:

LOCATION	MAXIMUM DIAMETER
Upper right	0.516 inch (13.1 mm)
Upper left	0.578 inch (14.7 mm)
Lower left	0.391 inch (9.9 mm)
Lower right	0.391 inch (9.9 mm)

8. Inspect corresponding four tailboom-to-fuselage attaching bolt holes in fuselage fittings for elongation.

9. Inspect tail rotor gearbox support fitting in accordance with Figure 1.



SUPPORT FITTING (212-030-103-001/-007)
MATERIAL: ALUMINUM ALLOY

NO.	CHARACTERISTIC	INSPECTION METHOD	LIMIT
	ZONE 1		
1.	Corrosion on one lug face	Measure	For a corrosion depth of <=0.020 inch (0.51 mm), remove twice the depth of the corrosion. Do not remove more than 0.040 inch (1.02 mm) from the cross-hatched area.
2.	Corrosion of both lug faces	Measure	For a combined corrosion depth of <=0.020 inch (0.51 mm), remove twice the depth of the corrosion (the sum of the removed material on both sides of the cross-hatched area must not exceed 0.040 inch (1.02 mm).
3.	Corrosion damage to outer edge of the lug	Measure	For a corrosion depth of <=0.020 inch (0.51 mm), remove twice the depth of the corrosion. Do not remove more than 0.040 inch (1.02 mm) from the outer edge of the lug.
4.	Corrosion damage to the lug hole (not the bushing hole)	Measure	For a corrosion depth of <=0.010 inch (0.25 mm), uniformly remove twice the depth of the corrosion around the periphery of the hole. Do not remove more than 0.020 inch (0.51 mm) from any part of the lug hole surface. Maintain original lug hole centerline. Rebush hole.

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Figure 1. Tail Rotor Gearbox Support Fitting - Damage and Corrosion Limits (Sheet 1 of 2)

NO.	CHARACTERISTIC	INSPECTION METHOD	LIMIT
5.	Corrosion damage to the bushing hole	Measure	Not allowed. Replace bushing.
6.	Mechanical damage on one lug face	Measure	For a mechanical damage of <=0.040 inch (1.02 mm), remove the depth of the damage. Do not remove more than 0.040 inch (1.02 mm) from the cross-hatched area.
7.	Mechanical damage on both lug faces	Measure	For a combined mechanical damage of <= 0.040 inch (1.02 mm), remove the depth of the damage (the sum of the removed material on both sides of the lug must not exceed 0.040 inch (1.02 mm).
8.	Mechanical damage to outer edge of the lug	Measure	For a mechanical damage depth of <=0.020 inch (0.51 mm), remove the depth of the damage. Do not remove more than 0.020 inch (0.51 mm) from the edge of the lug.
9.	Mechanical damage to the lug hole (not the bushing hole)	Measure	For a damage depth of <=0.020 inch (0.51 mm), uniformly remove the depth of the damage around the periphery of the hole. Do not remove more than 0.020 inch (0.51 mm) from any part of the lug. Maintain original lug hole centerline. Rebush hole.
10.	Mechanical damage to the bushing hole		Not allowed. Replace bushing.
	ZONE 2		
1.	Corrosion on one spot face	Measure	For a corrosion depth of <=0.020 inch (0.51 mm), remove twice the depth of the corrosion. Do not remove more than 0.040 inch (1.02 mm) from the cross-hatched area.
2.	Corrosion damage to the lug hole (not the bushing hole)	Measure	For a corrosion depth of <=0.010 inch (0.25 mm), uniformly remove twice the depth of the corrosion around the periphery of the hole. Do not remove more than 0.020 inch (0.51 mm) from any part of the lug hole. Maintain original lug hole centerline. Rebush hole.
3.	Corrosion damage to the bushing hole	Measure	Not allowed. Replace bushing.
4.	Mechanical damage on one spot face	Measure	For a mechanical damage of <=0.040 inch (1.02 mm), remove the depth of the damage. Do not remove more than 0.040 inch (1.02 mm) from the cross-hatched area.
5.	Mechanical damage to the lug hole (not the bushing hole)	Measure	For a mechanical damage depth of <=0.020 inch (0.51 mm), uniformly remove the depth of the damage around the periphery of the hole. Do not remove more than 0.020 inch (0.51 mm) from any part of the lug hole surface. Maintain original lug hole centerline. Rebush hole.
6.	Mechanical damage to the bushing hole		Not allowed. Replace bushing.

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Figure 1. Tail Rotor Gearbox Support Fitting - Damage and Corrosion Limits (Sheet 2 of 2)

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Tailboom - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Catalyzed epoxy primer	C-222	AR	
Polyurethane enamel	C-233	AR	
MEK	C-309	AR	

Procedure

1. Replace damaged or unserviceable doors, covers, and access panels.

2. Touch up paint on mating surfaces of tail rotor driveshaft covers as follows:

2.1. Clean area with MEK (C-309). Allow to air dry.

2.2. Apply one coat of catalyzed epoxy primer (C-222). Allow to dry 1 hour. Apply one coat of polyurethane enamel (C-233).

3. Perform necessary structural repairs in accordance with the BHT-MED-SRM-1.

4. Replace damaged or unserviceable fasteners and hinges on tail rotor driveshaft covers.

5. Replace tailboom if attaching holes in tailboom fittings exceed maximum allowable diameter.

NOTE

If a new tailboom is installed, confirmation of tail rotor driveshaft alignment is recommended (412-A-65-00-00-00A-272A-A / 01269).

6. When replacing tailboom, remove the following and install on replacement tailboom:

- Tail rotor controls (412-A-64-30-00-00A-042A-A / 01233)
- Electrical and avionics (412-A-96-00-00-00A-009A-A / 01870 and 412-A-97-00-00-00A-009A-A / 02235)
- Aerodynamically actuated elevator (412-A-67-02-00-01A-520A-A / 01540)
- Tail rotor driveshaft and hanger and bearings (412-A-65-10-02-00A-520A-A / 01270)
- Intermediate gearbox (412-A-65-20-00-00A-520A-A / 01280)
- Tail rotor gearbox (412-A-65-20-01-00A-520A-A / 01302)
- Tail rotor driveshaft covers and vertical fin cover
- Access doors and inspection panels

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Tailboom - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-104	AR	

Procedure

NOTE

If a replacement tailboom is installed or misalignment is suspected, carry out tail rotor driveshaft alignment procedure (412-A-65-00-00-00A-272A-A / 01269).

NOTE

If the tailboom was removed as part of the 5000 hour/ 5 year inspection, install a new upper left tailboom attachment bolt and inspect the three remaining bolts for mechanical or corrosion damage. Any evidence of corrosion is cause for rejection.

1. Position tailboom adjacent to forward fuselage. Line up bolt holes (Figure 1).

2. Apply a coating of corrosion preventive compound (C-104) to all bolt shanks prior to installation. Do not apply corrosion preventive compound to bolt threads.

3. Place countersunk washer on each bolt with countersunk side of washer toward bolt head. Install bolts using plain washers, as required, between countersunk washer and fitting so not less than one thread or more than two threads are showing. Torque bolts as follows:

3.1. Upper left - 1300 to 1600 inch-pounds (147 to 181 Nm).

3.2. Upper right - 1000 to 1200 inch-pounds (113 to 136 Nm).

3.3. Both bottom bolts - 400 to 430 inch-pounds (45.2 to 48.6 Nm).

4. Install plug buttons in four tailboom attaching points.

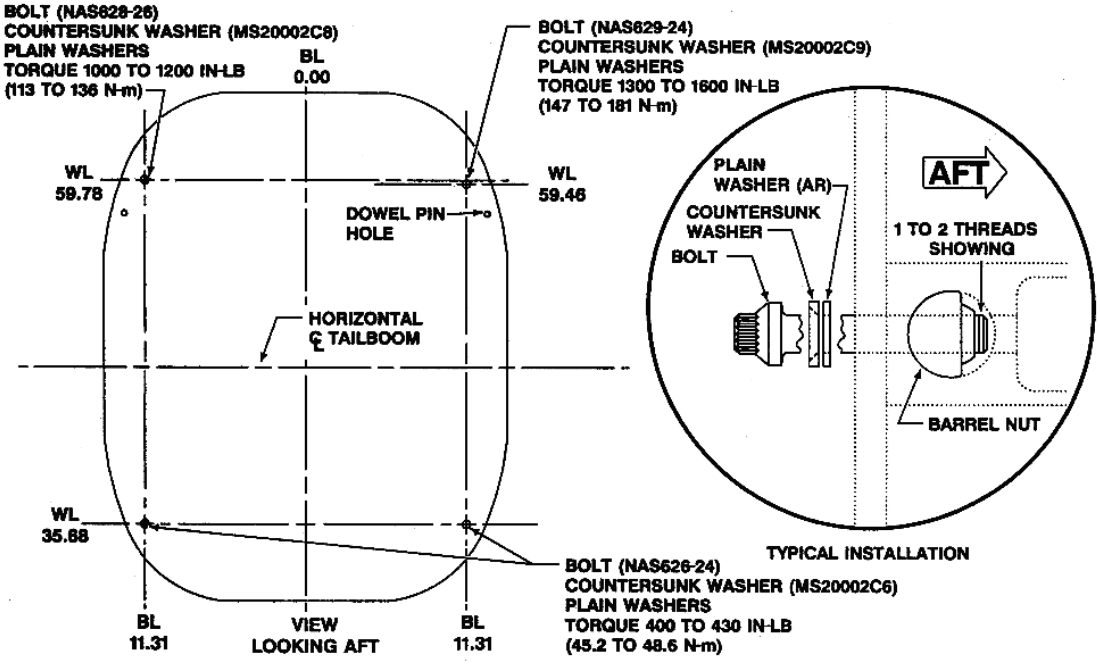
5. Connect tail rotor control tube (Figure 2).

6. Connect electrical harnesses and antenna cables.

7. Install No. 2 tail rotor driveshaft section (412-A-63-21-02-07A-720A-A / 01175).

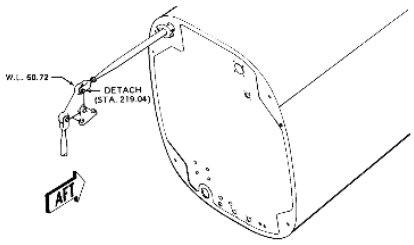
8. Install matched clamp sets on couplings, with bolt heads in direction of rotation and clamp joints indexed 90° to those of adjacent clamps. Tighten clamp bolts evenly, torque 30 to 35 inch-pounds (3.39 to 3.95 Nm) above nut friction.

9. Close combining gearbox side panels.



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Figure 1. Tailboom Attaching Bolts (Sheet 1 of 1)



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Figure 2. Disconnect Points for Tailboom (Sheet 1 of 1)

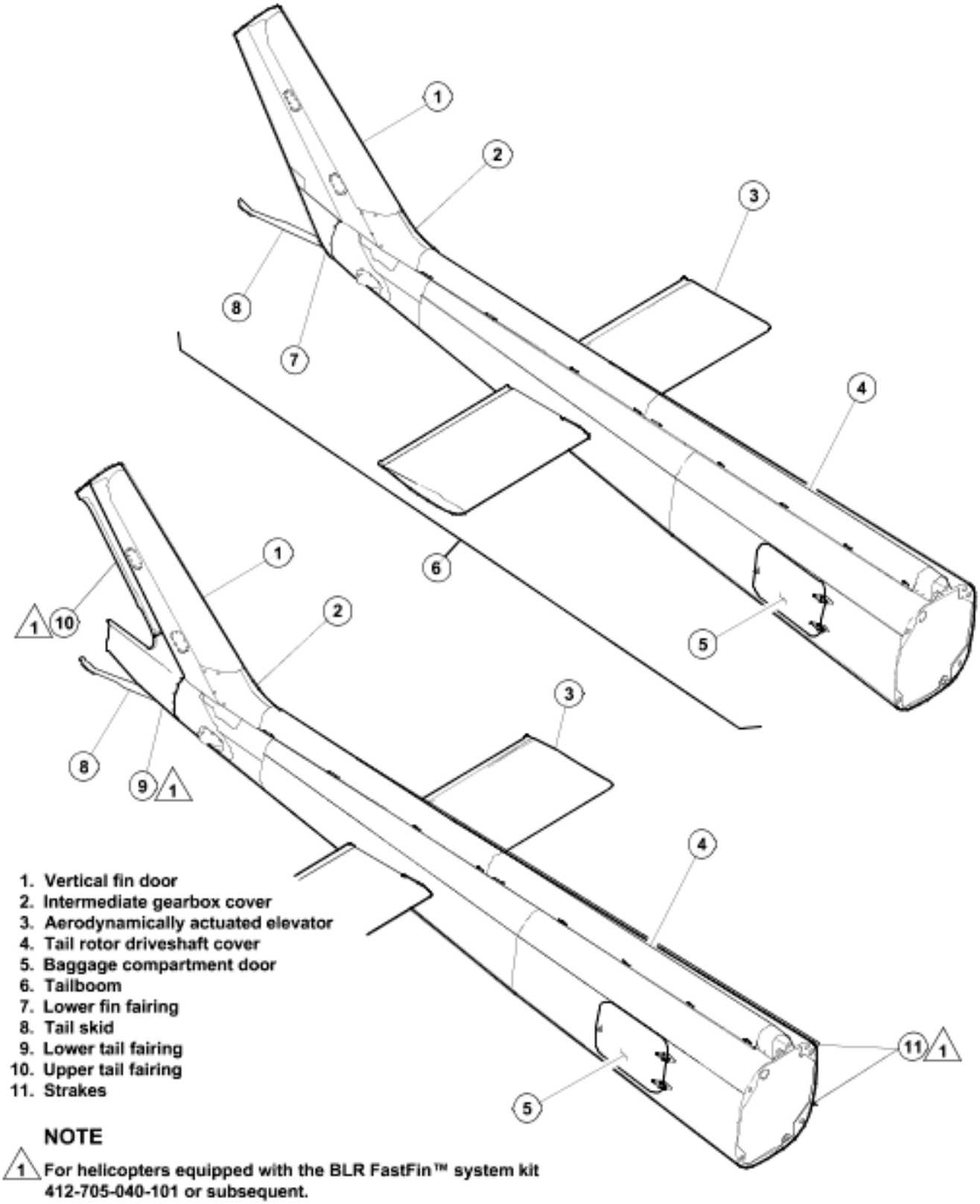
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Upper Tail Fairing - Description

NOTE

Instructions for the upper tail fairing (10, Figure 1) apply to helicopters equipped with the BLR FastFin™ system kit 412-705-040-101 (or Subsequent). For helicopters equipped with the Dual Tailboom Strakes and FastFin™ system Supplemental Type Certificate (STC) , refer to the applicable BLR Aerospace documents.

The upper tail fairing (10) is made of graphite and installed on the vertical fin with screws. The fairing replaces the aluminum trailing edge of the vertical fin on helicopters equipped with the BLR FastFin™ system kit 412-705-040-101 (or Subsequent).



1. Vertical fin door
2. Intermediate gearbox cover
3. Aerodynamically actuated elevator
4. Tail rotor driveshaft cover
5. Baggage compartment door
6. Tailboom
7. Lower fin fairing
8. Tail skid
9. Lower tail fairing
10. Upper tail fairing
11. Strakes

NOTE

1 For helicopters equipped with the BLR FastFin™ system kit 412-705-040-101 or subsequent.

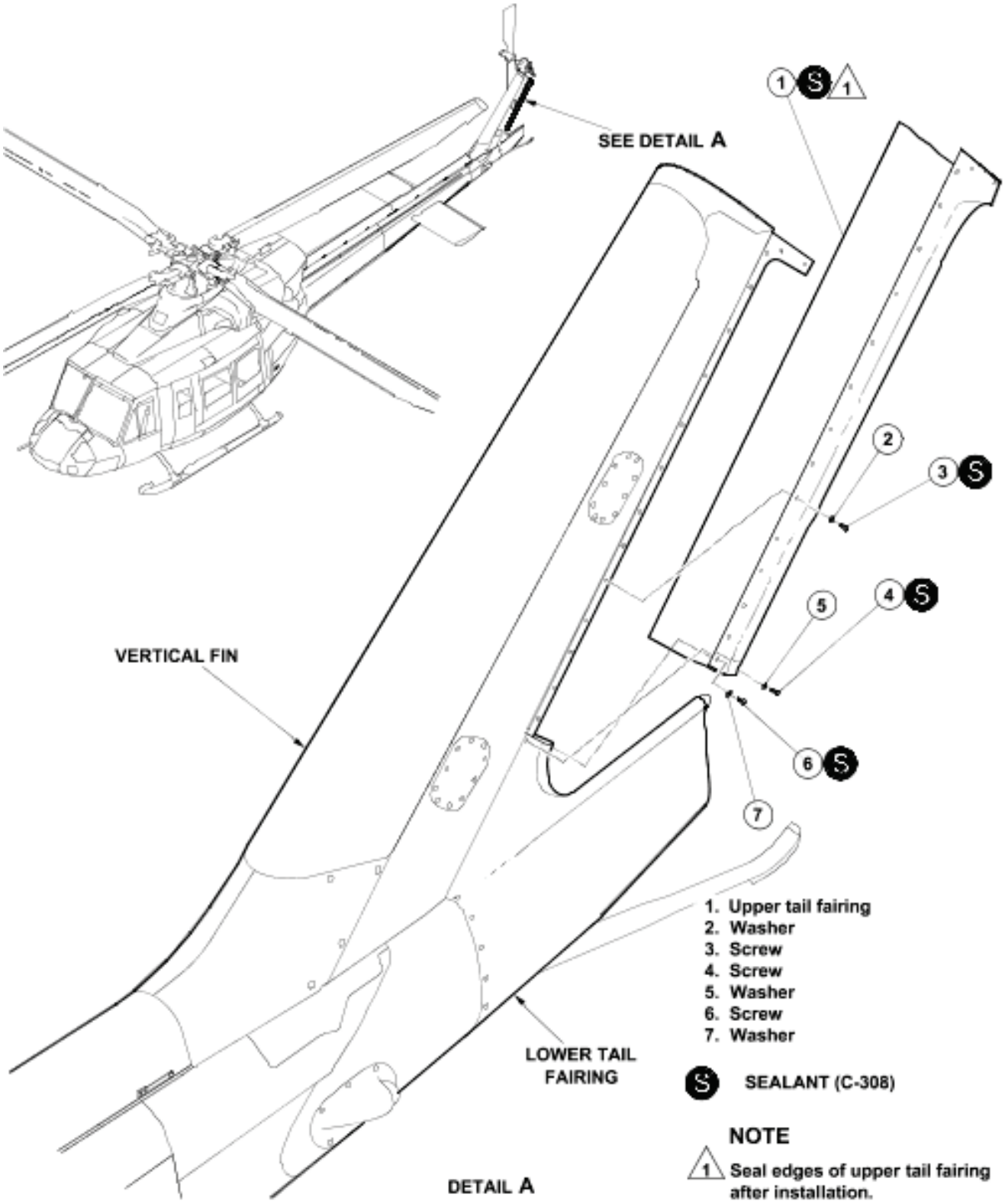
ICN-412-A-535000-B-97499-00001-A-001-01

Figure 1. Tailboom (Sheet 1 of 1)

Upper Tail Fairing - Removal

Procedure

1. Remove sealant from the edges of the upper tail fairing (1, Figure 1) with a plastic scraper.
2. Remove the screws (4 and 6) and the washers (5 and 7) from the overlap of the upper tail fairing (1) and the lower tail fairing.
3. Remove the screws (3) and the washers (2) from the upper tail fairing (1).
4. Remove the upper tail fairing (1) from the vertical fin.



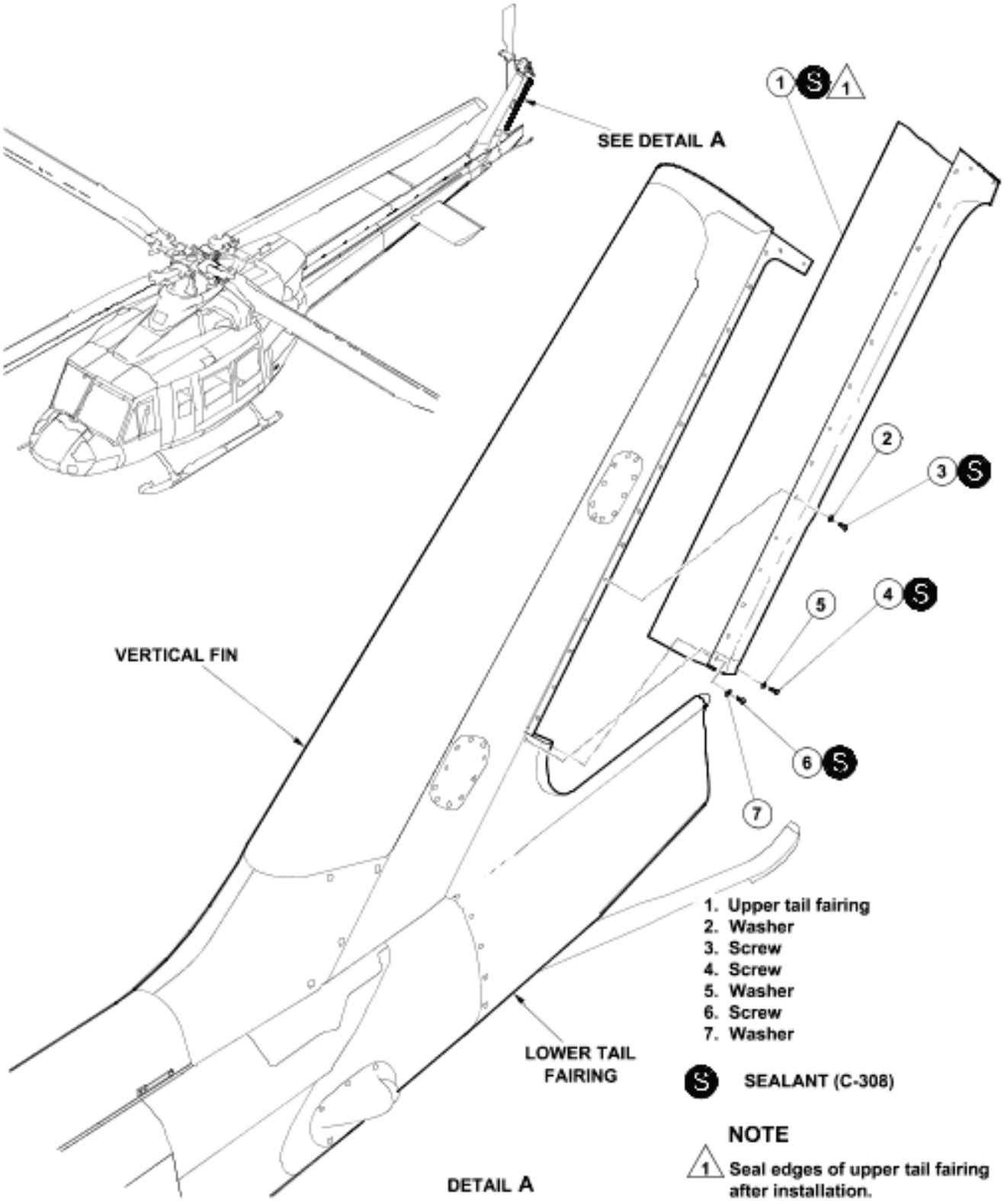
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Figure 1. Upper Tail Fairing - Removal and Installation (Sheet 1 of 1)

Upper Tail Fairing - Cleaning

Procedure

1. Clean the upper tail fairing (1, Figure 1) in accordance with the instructions for the cleaning of plastic and glass in the BHT-ALL-SPM, Chapter 5, Standard Practices Manual.



ICN-412-A-535000-B-97499-00006-A-001-01

Figure 1. Upper Tail Fairing - Removal and Installation (Sheet 1 of 1)

Upper Tail Fairing - Inspection

Preliminary requirements

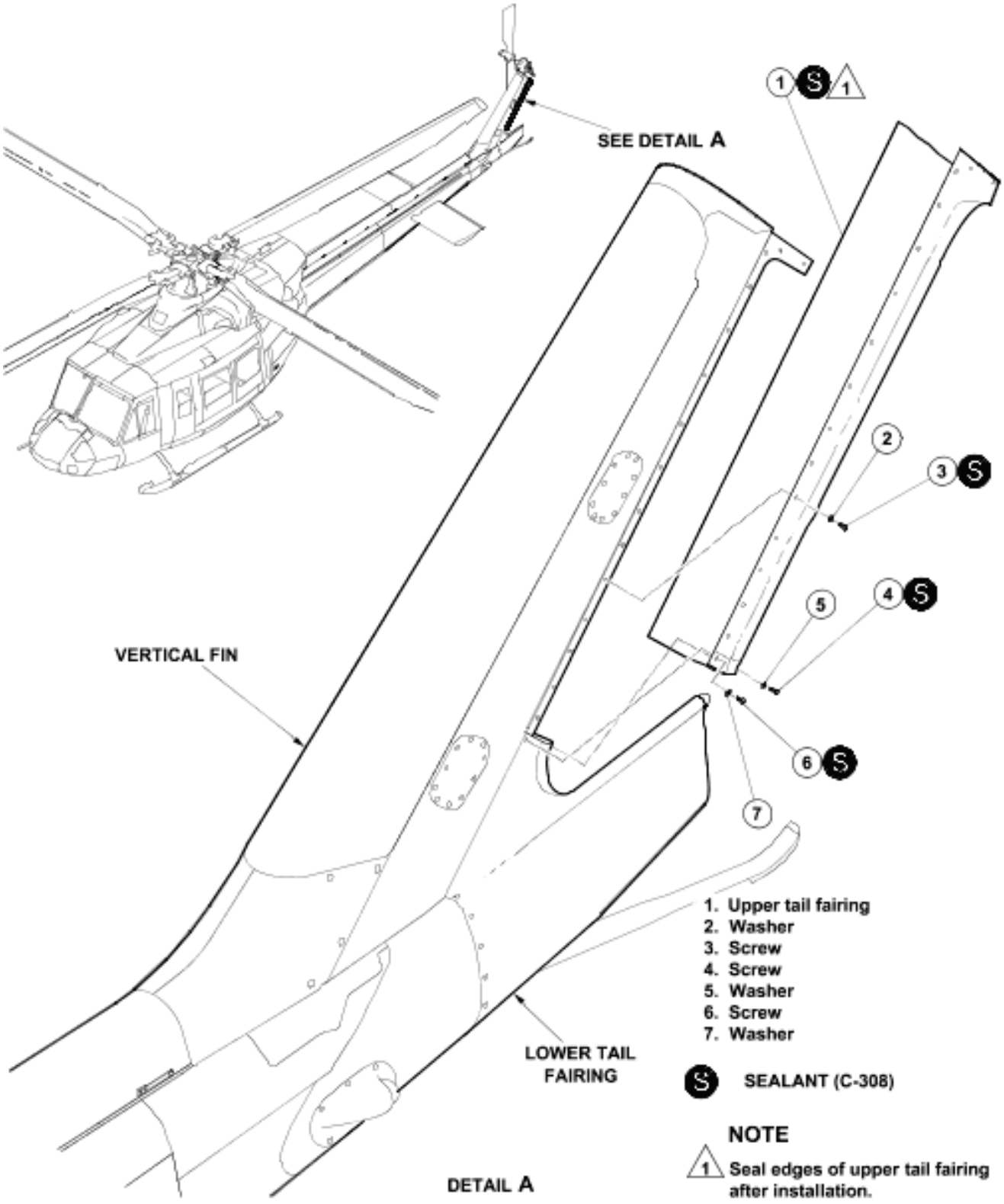
Support equipment

Nomenclature	Identification No.	Qty	Remarks
Tap Hammer	Commercial	1	

Procedure

1. Examine the upper tail faring (1, Figure 1) for cracks, damage, delamination, and voids. As required, repair damage (412-A-53-50-00-01A-685A-A / 00920) or replace the part.

2. Visually examine the attaching hardware for condition. Replace as required.



ICN-412-A-535000-B-97499-00006-A-001-01

Figure 1. Upper Tail Fairing - Removal and Installation (Sheet 1 of 1)

Upper Tail Fairing - Repair

Preliminary requirements

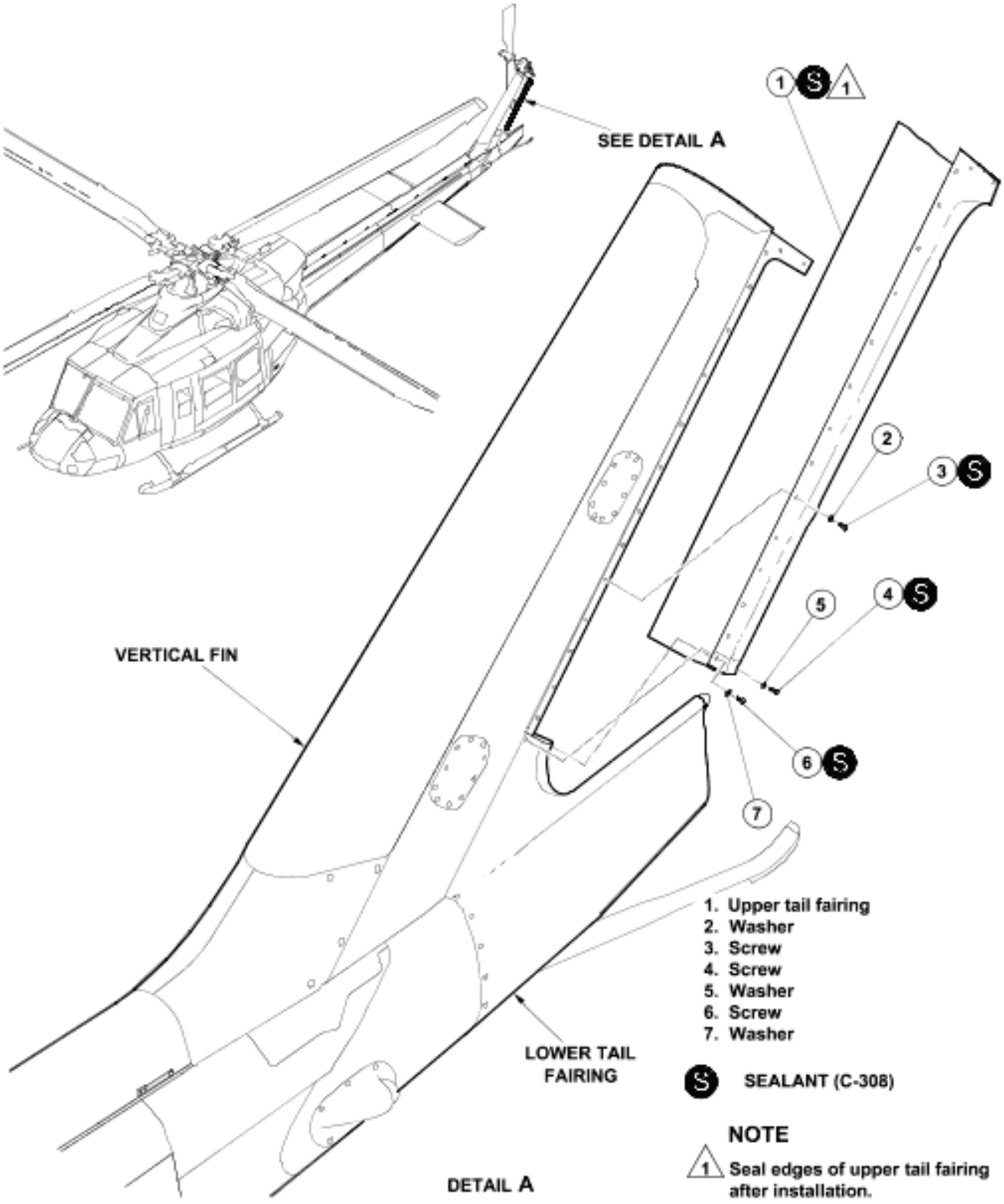
Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Fairing compound	C-323	AR	

Procedure

1. For information on damage repair on the upper tail faring (1, Figure 1) other than those specified in Step 2., contact Product Support Engineering.

2. Fill and fair scratches, nicks, or any equivalent superficial damage on the upper tail faring (1) with fairing compound (C-323). Do a finishing of the area after repair.



ICN-412-A-535000-B-97499-00006-A-001-01

Figure 1. Upper Tail Fairing - Removal and Installation (Sheet 1 of 1)

Upper Tail Fairing - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	

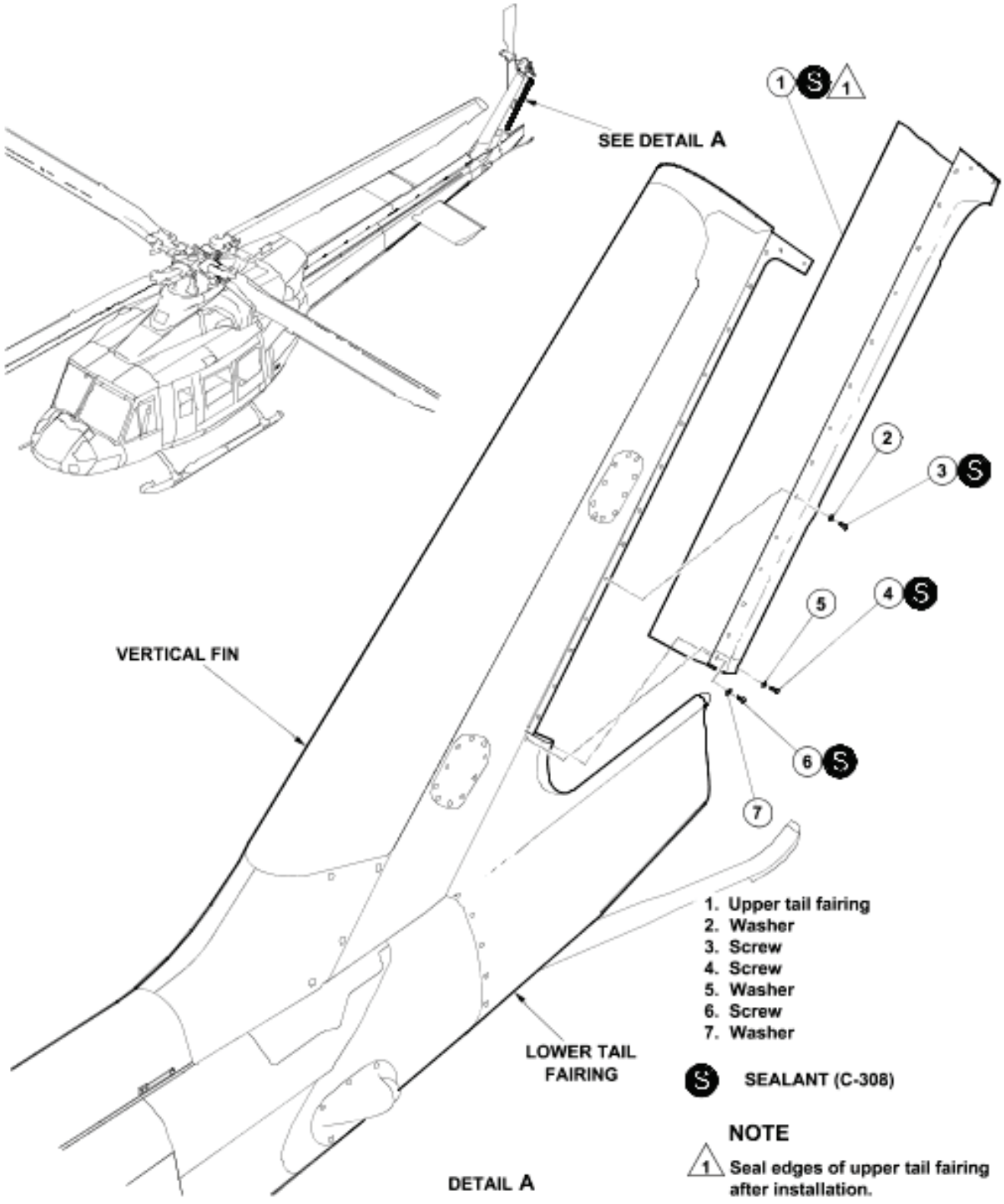
Procedure

1. Put the upper tail fairing (1, Figure 1) in position on the vertical fin.

2. Wet install the screws (3) and the washers (2) on the upper tail fairing (1) with sealant (C-308). Tighten the screws.

3. Wet install the screws (4 and 6) and the washers (5 and 7) on the overlap of the upper tail fairing (1) and lower tail fairing with sealant (C-308). Tighten the screws.

4. Seal the edges of the upper tail fairing (1) with sealant (C-308).



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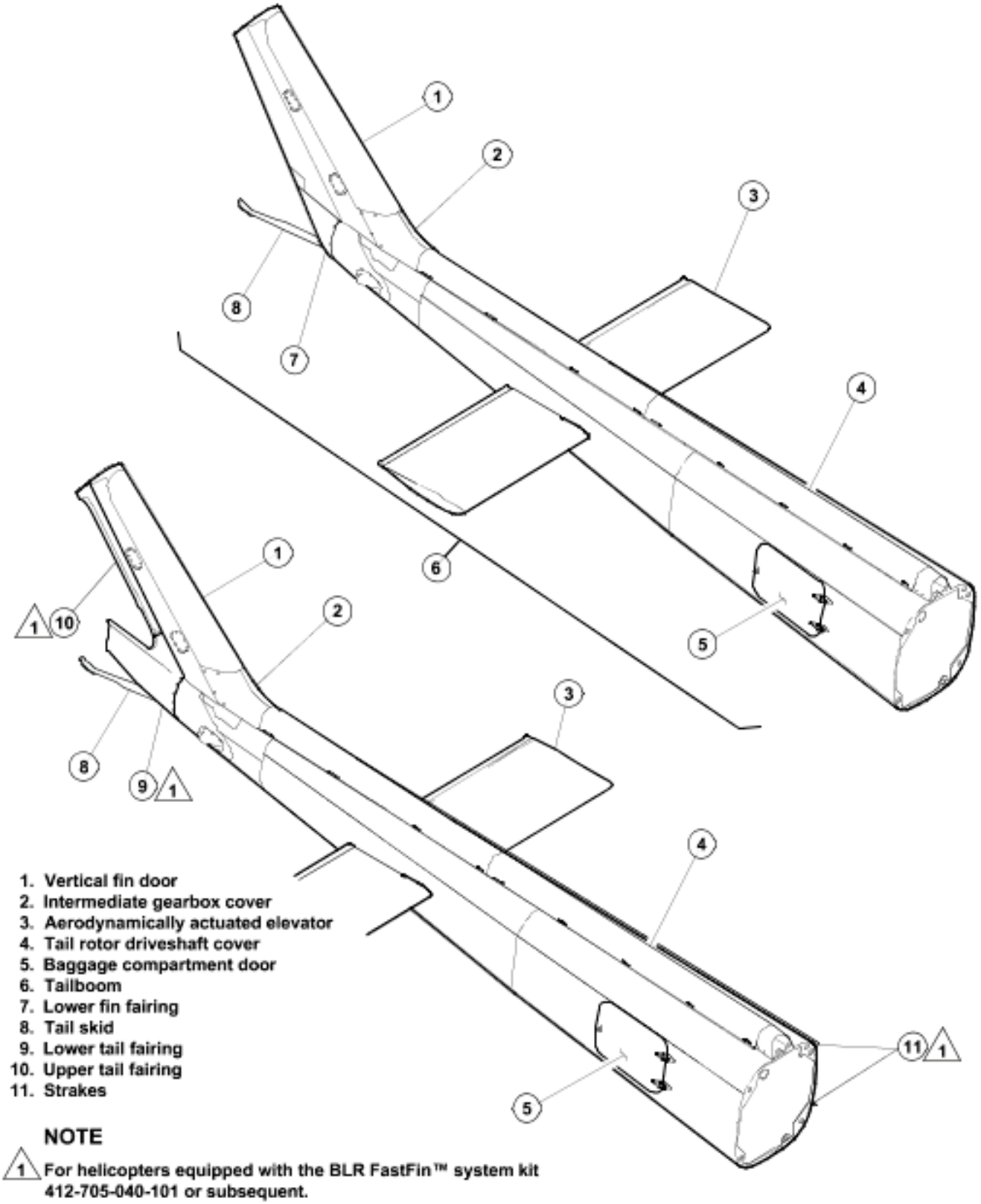
Figure 1. Upper Tail Fairing - Removal and Installation (Sheet 1 of 1)

Lower Tail Fairing - Description

NOTE

Instructions for the lower tail fairing (9, Figure 1) apply to helicopters equipped with the BLR FastFin™ system kit 412-705-040-101 (or Subsequent). For helicopters equipped with the Dual Tailboom Strakes and FastFin™ system Supplemental Type Certificate (STC) , refer to the applicable BLR Aerospace documents.

The lower tail fairing (9) is made of graphite and installed on the tailboom, vertical fin, and upper tail fairing (10) with screws. The fairing replaces the lower fin fairing (7) on helicopters equipped with the BLR FastFin™ system kit 412-705-040-101 (or Subsequent).



1. Vertical fin door
2. Intermediate gearbox cover
3. Aerodynamically actuated elevator
4. Tail rotor driveshaft cover
5. Baggage compartment door
6. Tailboom
7. Lower fin fairing
8. Tail skid
9. Lower tail fairing
10. Upper tail fairing
11. Strakes

NOTE

1 For helicopters equipped with the BLR FastFin™ system kit 412-705-040-101 or subsequent.

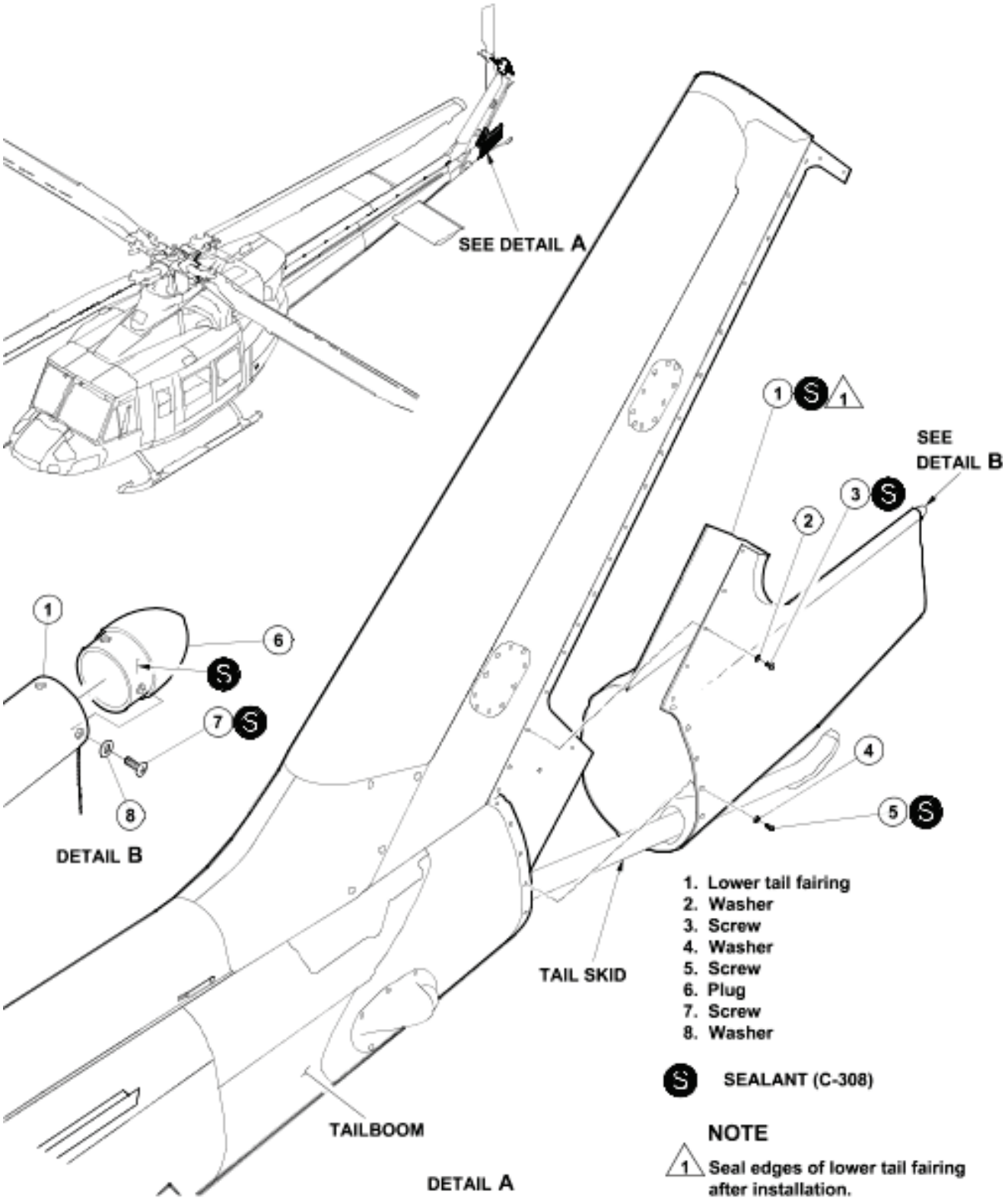
ICN-412-A-535000-B-97499-00001-A-001-01

Figure 1. Tailboom (Sheet 1 of 1)

Lower Tail Fairing - Removal

Procedure

1. Remove sealant from the edges of the lower tail fairing (1, Figure 1) with a plastic scraper.
2. Remove the upper tail fairing (412-A-53-50-00-01A-520A-A / 00917).
3. Remove the screws (3 and 5) and the washers (2 and 4) from the lower tail fairing (1).
4. Remove the lower tail fairing (1) from the tail skid and the tailboom.
5. If necessary, remove the screws (7), washers (8), and plug (6) from the lower tail fairing (1).



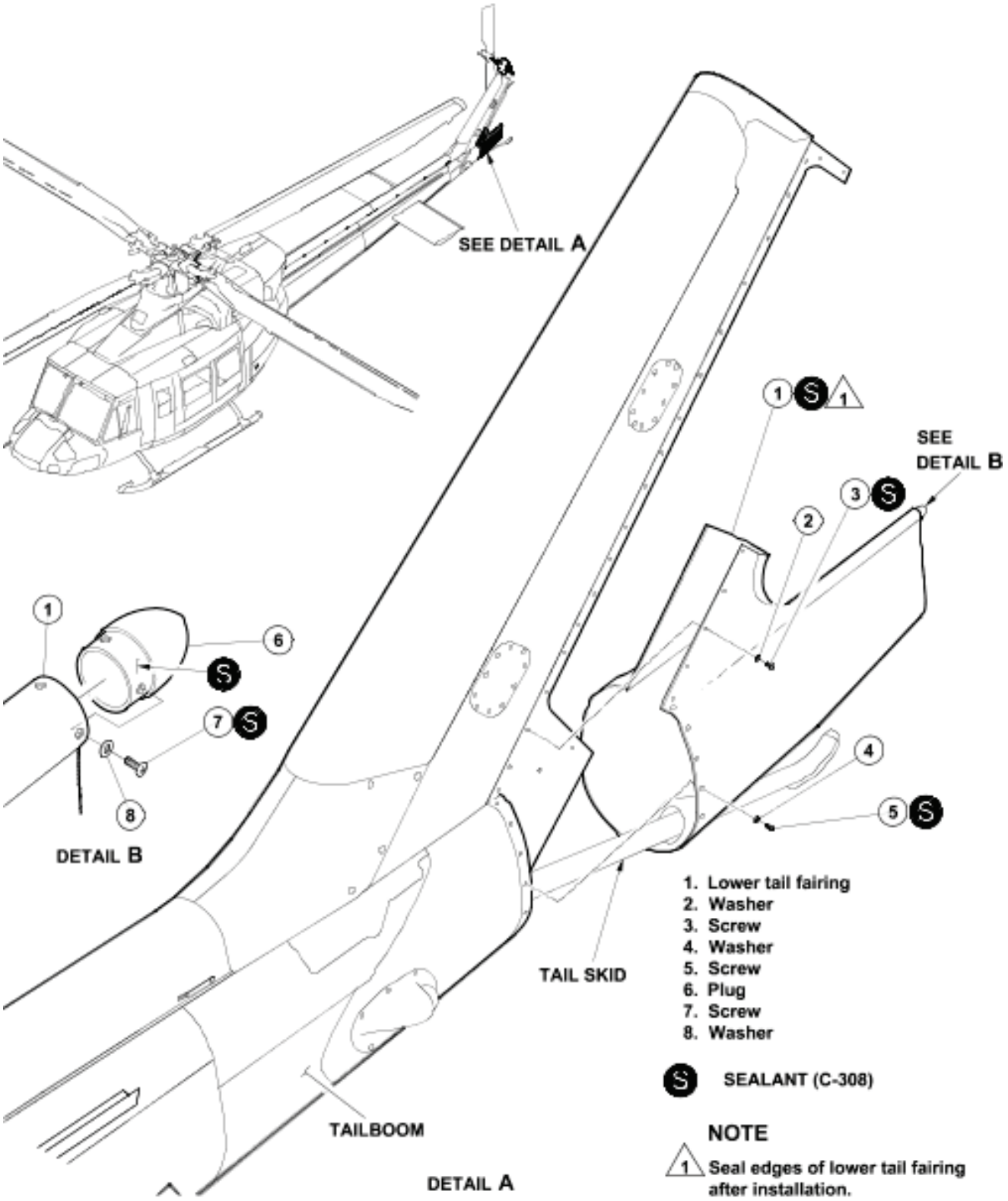
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Figure 1. Lower Tail Fairing - Removal and Installation (Sheet 1 of 1)

Lower Tail Fairing - Cleaning

Procedure

1. Clean the lower tail fairing (1, Figure 1) in accordance with the instructions for the cleaning of plastic and glass in the BHT-ALL-SPM, Chapter 5, Standard Practices Manual.



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Figure 1. Lower Tail Fairing - Removal and Installation (Sheet 1 of 1)

Lower Tail Fairing - Inspection

Preliminary requirements

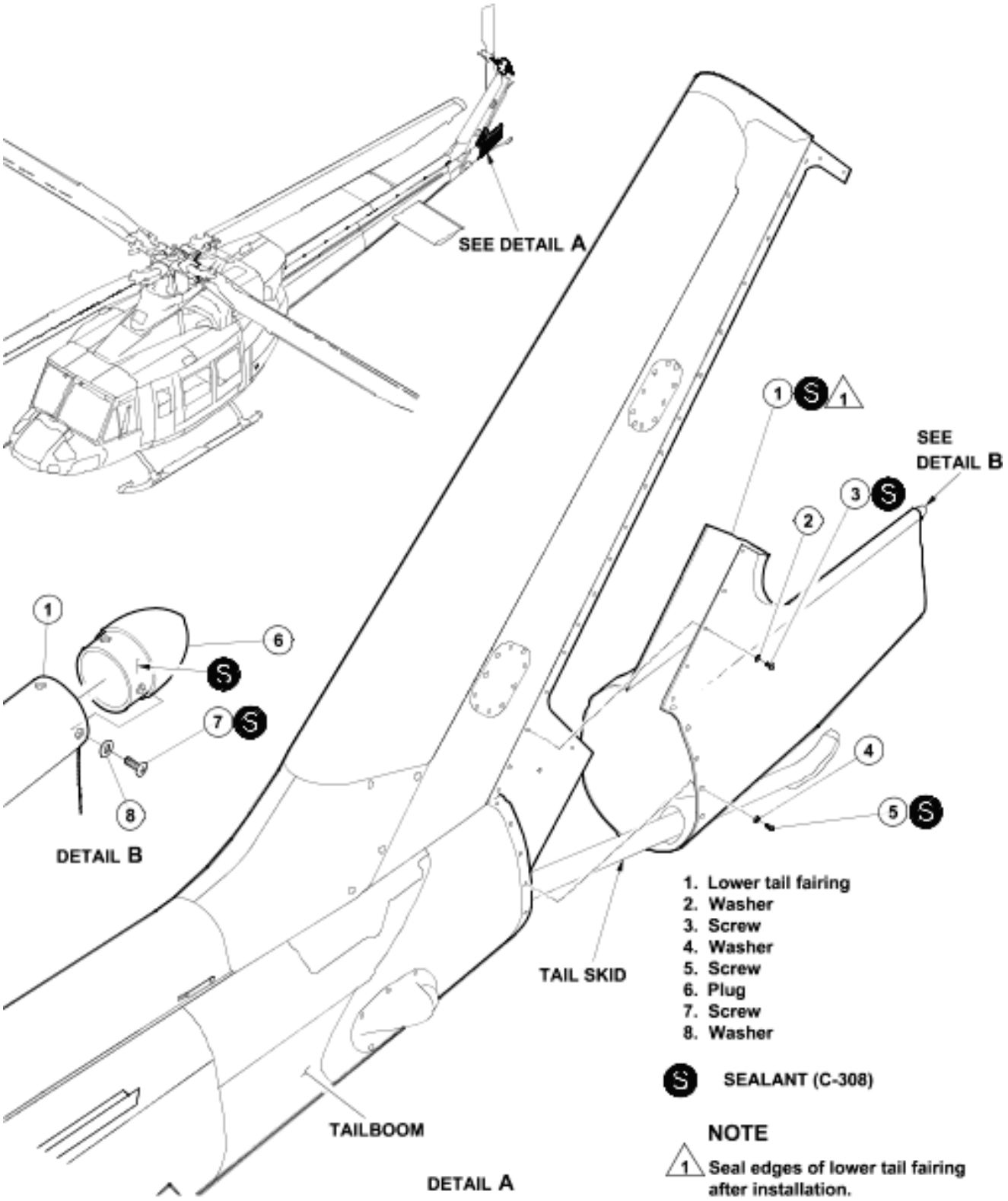
Support equipment

Nomenclature	Identification No.	Qty	Remarks
Tap Hammer	Commercial	1	

Procedure

1. Examine the lower tail faring (1, Figure 1) for cracks, damage, delamination, and voids. As required, repair damage (412-A-53-50-00-02A-685A-A / 00926) or replace the part.

2. Visually examine the attaching hardware for condition. Replace as required.



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Figure 1. Lower Tail Fairing - Removal and Installation (Sheet 1 of 1)

Lower Tail Fairing - Repair

Preliminary requirements

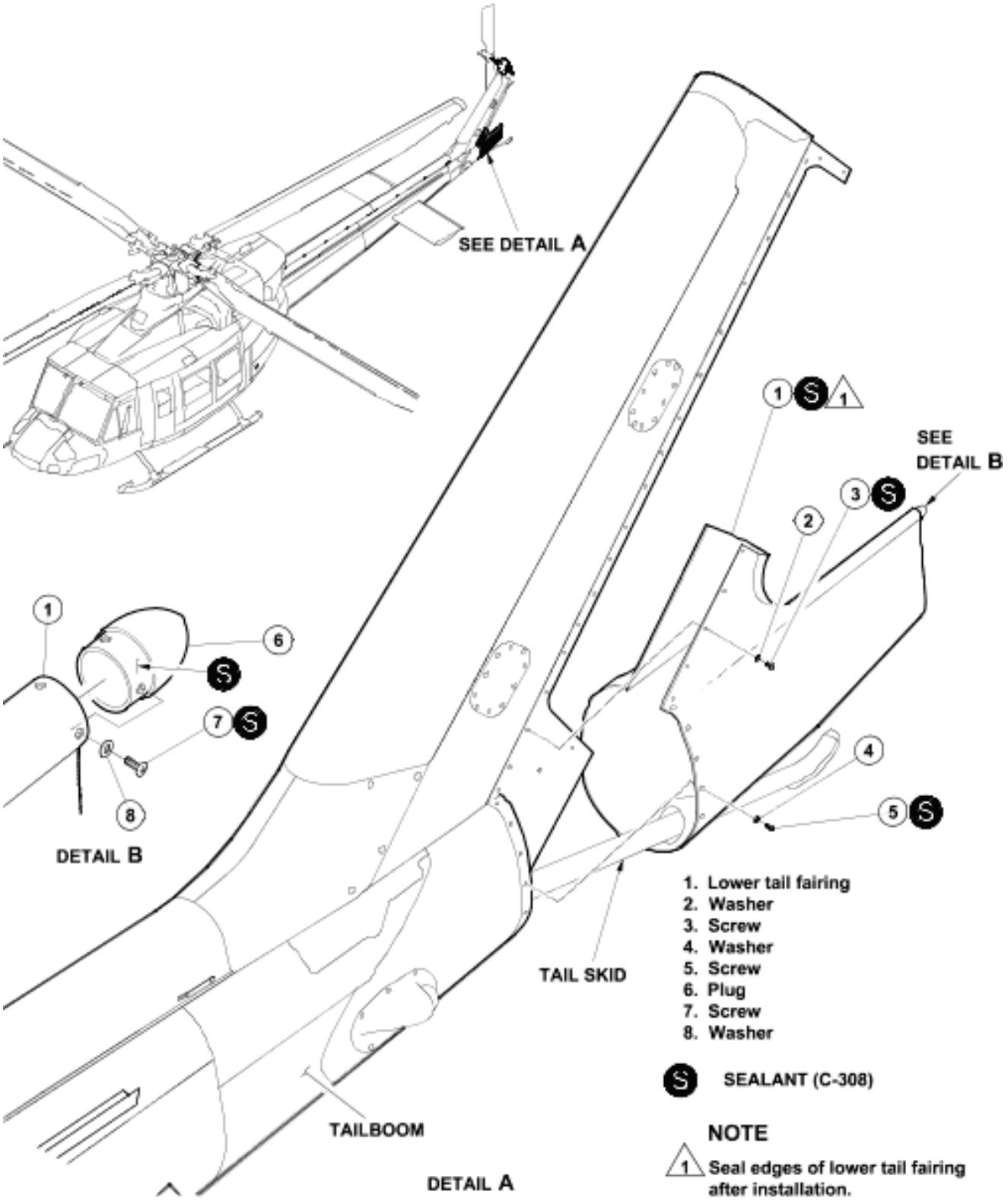
Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Fairing compound	C-323	AR	

Procedure

1. For information on damage repair on the lower tail faring (1, Figure 1) other than those specified in Step 2., contact Product Support Engineering.

2. Fill and fair scratches, nicks, or any equivalent superficial damage on the lower tail faring (1) with fairing compound (C-323). Do a finishing of the area after repair.



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Figure 1. Lower Tail Fairing - Removal and Installation (Sheet 1 of 1)

Lower Tail Fairing - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	

Procedure

1. If previously removed, install the plug (6, Figure 1) on the lower tail fairing (1) as follows:

1.1. Apply sealant (C-308) on the outer diameter of the plug (6).

1.2. While sealant is wet, install the plug (6) on the lower tail fairing (1).

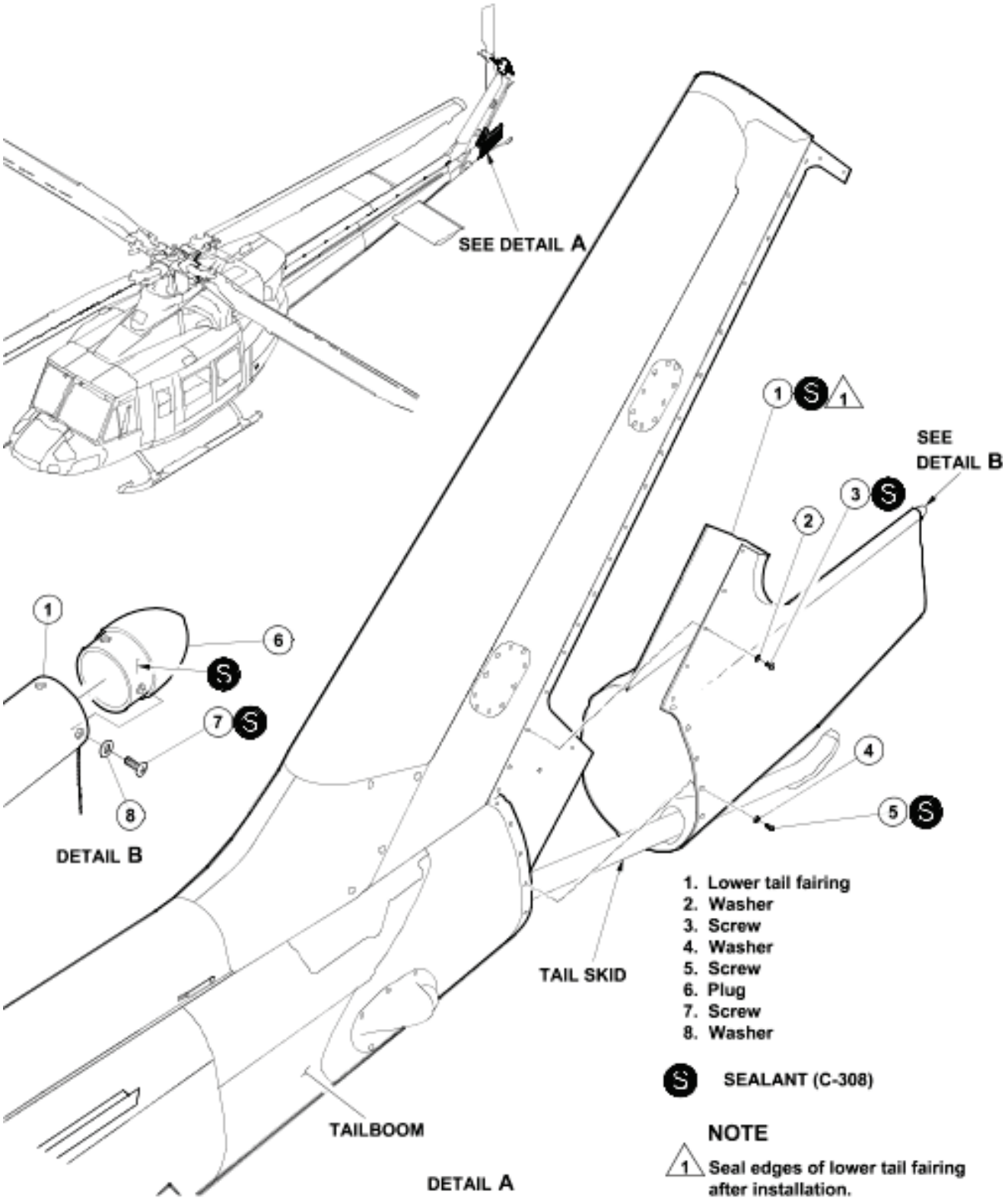
1.3. Wet install the screws (7) and the washers (8) on the plug (6). Tighten the screws.

2. Move the lower tail fairing (1) over the tail skid and put it in position on the tailboom.

3. Wet install the screws (3 and 5) and the washers (2 and 4) on the lower tail fairing (1) with sealant (C-308). Tighten the screws.

4. Install the upper tail fairing (412-A-53-50-00-01A-720A-A / 00921).

5. Seal the edges of the lower tail fairing (1) with sealant (C-308).



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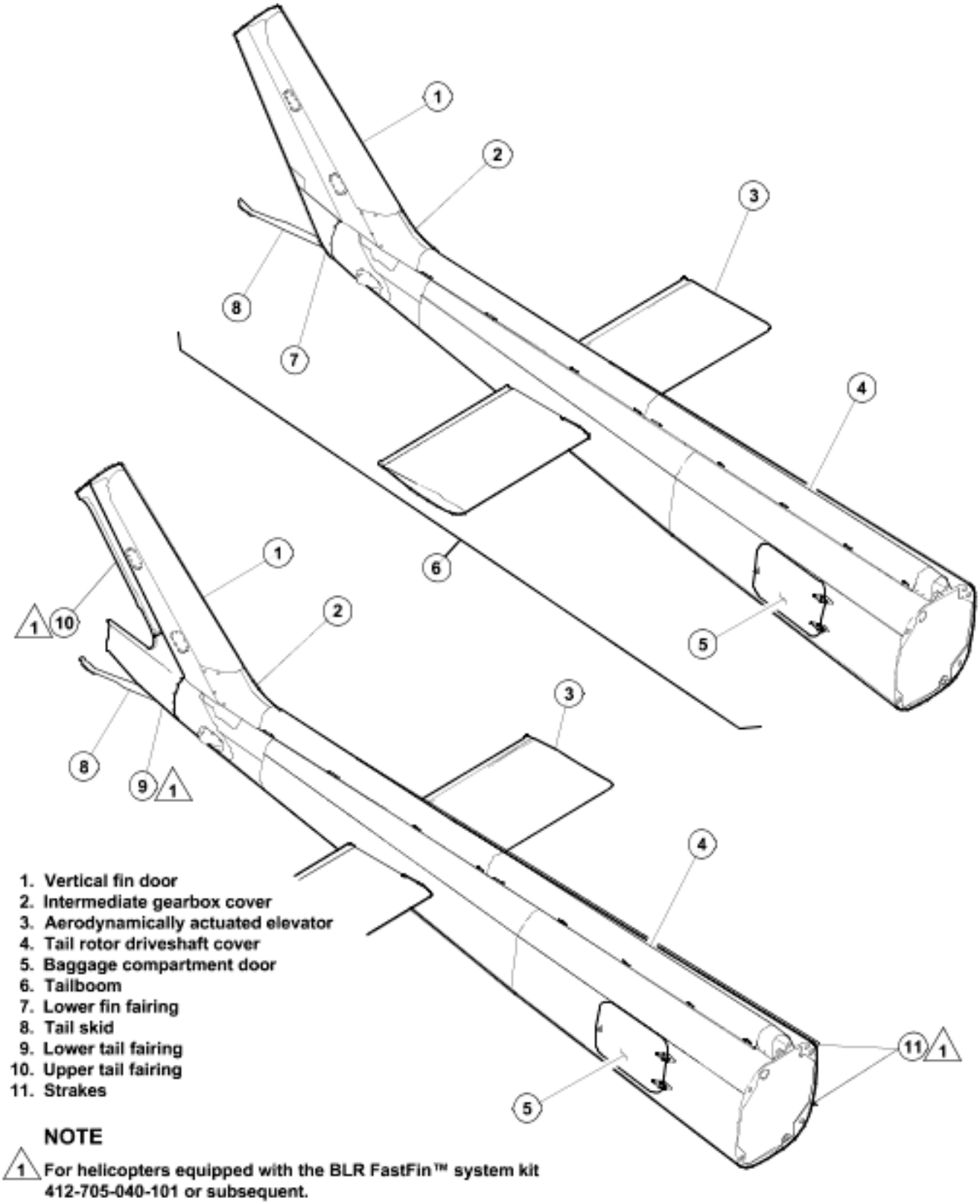
Figure 1. Lower Tail Fairing - Removal and Installation (Sheet 1 of 1)

Strakes - Description

NOTE

Instructions for the strakes (11, Figure 1) apply to helicopters equipped with the BLR FastFin™ system kit 412-705-040-101 (or Subsequent). For helicopters equipped with the Dual Tailboom Strakes and FastFin™ system Supplemental Type Certificate (STC) , refer to the applicable BLR Aerospace documents.

The strakes (11) consist of an upper and a lower strake riveted along the left side of the tailboom, between Boom Station (BS) 17.42 and BS 185.15. Each strake is made of four aluminum segments or angles. The free flange of the angles is the aerodynamic blade, the other is the mounting flange for attachment to the boom skin.



1. Vertical fin door
2. Intermediate gearbox cover
3. Aerodynamically actuated elevator
4. Tail rotor driveshaft cover
5. Baggage compartment door
6. Tailboom
7. Lower fin fairing
8. Tail skid
9. Lower tail fairing
10. Upper tail fairing
11. Strakes

NOTE

1 For helicopters equipped with the BLR FastFin™ system kit 412-705-040-101 or subsequent.

ICN-412-A-535000-B-97499-00001-A-001-01

Figure 1. Tailboom (Sheet 1 of 1)

Strakes - Inspection

Procedure

1. Visually examine the upper and lower strakes for cracks, damage, corrosion, deformation, and loose or missing rivets. As required, repair damage (412-A-53-50-00-03A-685A-A / 00930) or replace the part (412-A-53-50-00-03A-921A-A / 00931).

2. Visually examine sealant around the upper and lower strakes for condition. As applicable, repair voids or replace sealant in case of deterioration (412-A-53-50-00-03A-685A-A / 00930).

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Strakes - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	
Sealant	C-559	AR	Use CS 5500 Class B2.

Procedure

1. For information on damage repair and corrosion removal on segments of the upper or lower strake other than those specified in Step 2., contact Product Support Engineering.

2. Lightly sand to remove scratches, nicks, or any equivalent superficial damage from the strakes. Do a finishing of the area after repair.

NOTE

sealant (C-308) may have been used as an alternate to sealant (C-559) for installation of the lower strake.

3. Touch up repair or replace fillet of sealant around the upper or lower strake with sealant (C-559) or sealant (C-308), as applicable.

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Strakes - Replacement

Procedure

1. For instructions to replace one or more segments of the upper or lower strake, contact Product Support Engineering.

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CHAPTER 62
MAIN ROTOR

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Main Rotor Hub and Blades - Description	DMC-412-A-62-00-00-00A-028A-A / 00933	2023-05-31	ALL
Main Rotor Blades - Description	DMC-412-A-62-10-00-00A-042A-A / 00934	2023-05-31	ALL
Main Rotor Blades - Removal	DMC-412-A-62-10-00-00A-520A-A / 00935	2023-05-31	ALL
Main Rotor Blades - Cleaning	DMC-412-A-62-10-00-00A-251A-A / 00936	2023-05-31	ALL
Main Rotor Blades - Inspection	DMC-412-A-62-10-00-00A-280A-A / 00937	2023-05-31	ALL
Main Rotor Blades - Repair	DMC-412-A-62-10-00-00A-685A-A / 00938	2023-05-31	ALL
Main Rotor Blades - Installation	DMC-412-A-62-10-00-00A-720A-A / 00946	2023-05-31	ALL
Polyurethane Protective Tape - Description	DMC-412-A-62-20-00-01A-042A-A / 00939	2023-05-31	ALL
Polyurethane Protective Tape - Cleaning and Surface Preparation	DMC-412-A-62-20-00-01A-251A-A / 00940	2023-05-31	ALL
Polyurethane Protective Tape - Wet Application Method	DMC-412-A-62-20-00-01A-252A-A / 00941	2023-05-31	ALL
Polyurethane Protective Tape - Dry Application Method	DMC-412-A-62-20-00-01A-252B-A / 00942	2023-05-31	ALL
Polyurethane Protective Tape - Inspection	DMC-412-A-62-20-00-01A-280A-A / 00943	2023-05-31	ALL
Polyurethane Protective Tape - Edge Sealing (Optional)	DMC-412-A-62-20-00-01A-621A-A / 00944	2023-05-31	ALL
Polyurethane Protective Tape - Removal	DMC-412-A-62-20-00-01A-520A-A / 00945	2023-05-31	ALL
Main Rotor Hub - Description	DMC-412-A-62-20-00-02A-042A-A / 00947	2023-05-31	ALL
Main Rotor Hub - Removal	DMC-412-A-62-20-00-02A-520A-A / 00948	2023-05-31	ALL
Main Rotor Hub - Cleaning	DMC-412-A-62-20-00-02A-251A-A / 00949	2023-05-31	ALL
Main Rotor Hub - Inspection	DMC-412-A-62-20-00-02A-280A-A / 00950	2023-05-31	ALL
Main Rotor Hub - Repair	DMC-412-A-62-20-00-02A-685A-A / 00951	2023-05-31	ALL
Main Rotor Hub - Installation	DMC-412-A-62-20-00-02A-720A-A / 00952	2023-05-31	ALL
Main Rotor Controls - Description	DMC-412-A-62-30-00-00A-028A-A / 00953	2023-05-31	ALL
Swashplate and Support Assembly - Description	DMC-412-A-62-30-00-01A-042A-A / 00954	2023-05-31	ALL
Swashplate and Support Assembly - Removal	DMC-412-A-62-30-00-01A-520A-A / 00955	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Swashplate and Support Assembly - Cleaning	DMC-412-A-62-30-00-01A-251A-A / 00956	2023-05-31	ALL
Swashplate and Support Assembly - Inspection	DMC-412-A-62-30-00-01A-280A-A / 00957	2023-05-31	ALL
Swashplate and Support Assembly - Repair	DMC-412-A-62-30-00-01A-685A-A / 00958	2023-05-31	ALL
Swashplate and Support Assembly - Installation	DMC-412-A-62-30-00-01A-720A-A / 00959	2023-05-31	ALL
Main Rotor Pitch Link Assemblies - Description	DMC-412-A-62-30-00-02A-042A-A / 00960	2023-05-31	ALL
Main Rotor Pitch Link Assemblies - Removal	DMC-412-A-62-30-00-02A-520A-A / 00961	2023-05-31	ALL
Main Rotor Pitch Link Assemblies - Cleaning	DMC-412-A-62-30-00-02A-251A-A / 00962	2023-05-31	ALL
Main Rotor Pitch Link Assemblies - Inspection	DMC-412-A-62-30-00-02A-280A-A / 00963	2023-05-31	ALL
Main Rotor Pitch Link Assemblies - Repair	DMC-412-A-62-30-00-02A-685A-A / 00964	2023-05-31	ALL
Main Rotor Pitch Link Assemblies - Installation	DMC-412-A-62-30-00-02A-720A-A / 00965	2023-05-31	ALL
Hub and Sleeve Assembly - Description	DMC-412-A-62-30-00-03A-042A-A / 00966	2023-05-31	ALL
Teflon Lined Bearings (Installed) - Inspection	DMC-412-A-62-30-00-03A-280A-A / 00967	2023-05-31	ALL
Hub and Sleeve Assembly - Removal	DMC-412-A-62-30-00-03A-520A-A / 00968	2023-05-31	ALL
Hub and Sleeve Assembly - Cleaning	DMC-412-A-62-30-00-03A-251A-A / 00969	2023-05-31	ALL
Hub and Sleeve Assembly - Inspection	DMC-412-A-62-30-00-03A-280B-A / 00970	2023-05-31	ALL
Hub and Sleeve Assembly - Repair	DMC-412-A-62-30-00-03A-685A-A / 00971	2023-05-31	ALL
Hub and Sleeve Assembly - Installation	DMC-412-A-62-30-00-03A-720A-A / 00972	2023-05-31	ALL
Rephasing Levers - Removal	DMC-412-A-62-30-00-04A-520A-A / 00973	2023-05-31	ALL
Rephasing Levers - Installation	DMC-412-A-62-30-00-04A-720A-A / 00974	2023-05-31	ALL
Drive Link Assembly - Removal	DMC-412-A-62-30-00-05A-520A-A / 00975	2025-05-13	ALL
Drive Link Assembly - Installation	DMC-412-A-62-30-00-05A-720A-A / 00976	2025-05-13	ALL
Swashplate Link Assembly - Removal	DMC-412-A-62-30-00-06A-520A-A / 00977	2023-05-31	ALL
Swashplate Link Assembly - Installation	DMC-412-A-62-30-00-06A-720A-A / 00978	2023-05-31	ALL
Collective Pitch Drive Plate Set - Removal	DMC-412-A-62-30-00-07A-520A-A / 00979	2024-05-09	ALL
Collective Pitch Drive Plate Set - Installation	DMC-412-A-62-30-00-07A-720A-A / 00980	2024-05-09	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Collective Lever Assembly - Removal	DMC-412-A-62-30-00-08A-520A-A / 00981	2023-05-31	ALL
Collective Lever Assembly - Cleaning	DMC-412-A-62-30-00-08A-251A-A / 00982	2023-05-31	ALL
Collective Lever Assembly - Inspection	DMC-412-A-62-30-00-08A-280A-A / 00983	2023-05-31	ALL
Collective Lever Assembly - Repair	DMC-412-A-62-30-00-08A-685A-A / 00984	2023-05-31	ALL
Collective Lever Assembly - Installation	DMC-412-A-62-30-00-08A-720A-A / 00985	2023-05-31	SN: 33001-33167 TB 412-88-68: Pre
Collective Lever Assembly - Installation	DMC-412-A-62-30-00-08A-720B-A / 00986	2023-05-31	SN: 33168-Subsequent TB 412-88-68: Post

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Main Rotor Hub and Blades - Description

NOTE

Use of expandable rotor blade bolt instead of standard rotor blade retention bolt is optional.

The main rotor hub and blade consists of the hub, four main rotor blades and four pitch horns. Each main rotor blade is connected to hub by either two fixed diameter bolts or one fixed diameter bolt on leading edge side of rotor blade and an expandable bolt on trailing edge side of rotor blade. The purpose of expandable bolts (when used) is to facilitate removal of bolts when rotor blades are folded. The main rotor droop restraint assembly attaches to lower damper bridges and to attach lugs of lower cone seat, (four places) and will minimize yoke flexing during static rotor blade droop.

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Main Rotor Blades - Description

The main rotor blades are constructed of fiberglass with a honeycomb core, stainless steel tip cap and outboard abrasion strip, and a titanium inboard abrasion strip. The glass fiber spar is made up of four spar caps and a nose block, constructed of spanwise oriented unidirectional glass fibers, all bonded between inner and outer glass fiber torque wraps. The rotor blades are covered with fiberglass skins.

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Main Rotor Blades - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Blade Bolt Tool	T101266	1	
Sling	T101626	1	

Procedure

NOTE

Removal procedure for main rotor blade (3, Figure 1) is given. Removal procedure for the other three rotor blades is the same.

1. Inspect rotor blade (3, Figure 1) and spindle (1) for color code markings. If color code markings are missing, apply new marking prior to removal (412-A-11-00-00-00A-067A-A / 00124).



USE HOIST WITH MINIMUM CAPACITY OF 1000 POUNDS (454 KILOGRAMS).

2. Attach sling (4) to hoist with minimum capacity of 1000 pounds (454 kilograms). Position hoist directly over center of gravity of rotor blade to be removed. Attach sling straps around rotor blade. Take up slack with hoist.



MAIN ROTOR BLADES AND HUB COMPONENTS ARE COLOR CODED. AFTER EXPANDABLE BOLT (10, Figure 1) IS ADJUSTED AT INITIAL INSTALLATION, IT SHALL BE KEPT WITH SAME ROTOR BLADE AND HUB SPINDLE, OR EXPANDABLE BOLT SHALL BE READJUSTED.

NOTE

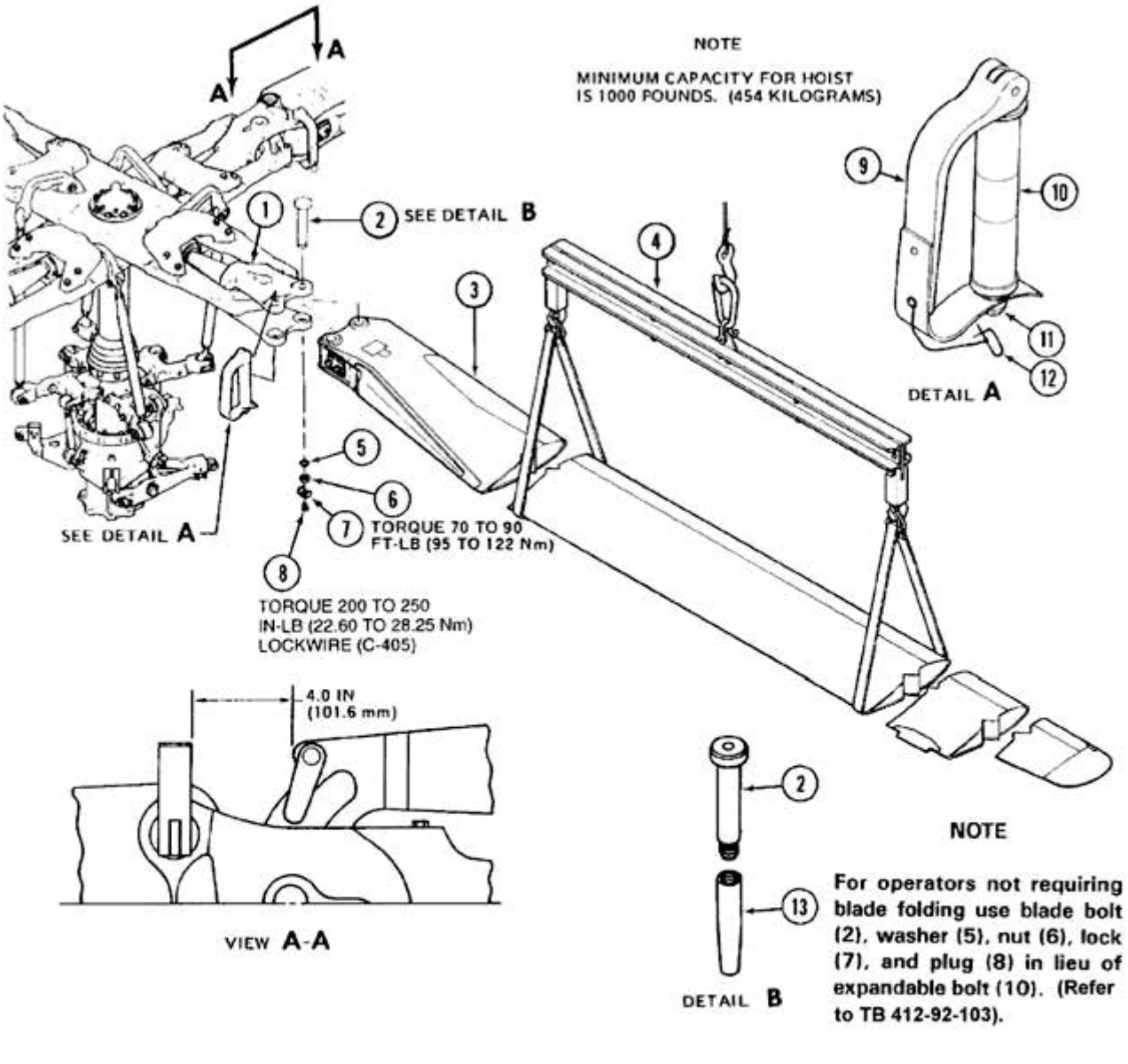
For helicopters with expandable rotor blade bolts installed, accomplish following Step 3. and Step 4.. For helicopters with standard rotor blade bolts installed in both rotor blade spindle holes, accomplish Step 5. for both bolts.

3. Remove safety lock hair pin (12) from lower end of expandable bolt (10).

4. Raise bolt handle (9) to reduce diameter of bolt (10). Remove bolt (10).

5. Cut and remove lockwire. Remove plug (8) and lock (7) from lower end of rotor blade bolt (2). Remove nut (6) and washer (5). Remove rotor blade bolt (2) using rotor blade bolt tool (13).

6. Slide the rotor blade (3) from the spindle (1). Lower the rotor blade and place in a suitable blade rack.



1. Spindle
2. Blade bolt
3. Blade
4. Sling (T101626)
5. Washer (206-010-154-3)
6. Nut
7. Lock
8. Plug (AN814-4L)
9. Bolt handle
10. Expandable bolt
11. Adjustment nut
12. Safety lock hair pin
13. Blade bolt tool (T101266)

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Figure 1. Main Rotor Blade Installation (Sheet 1 of 1)

Main Rotor Blades - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Cleaning compound	C-318	AR	
Wax	C-412	AR	

Procedure

1. Wash the rotor blades with a mild solution of cleaning compound (C-318) and water. Rinse the rotor blades with clean water and wipe dry with clean cloths.



DO NOT USE SILICONE TYPE WAXES. SILICONE WAXES MAY INTERFERE WITH FUTURE REPAIRS OR REFINISHING.

NOTE

Waxing is not required for blades 412-015-300-109 and subsequent or for any blade finished with polyurethane paint.

2. Coat the rotor blades with wax (C-412).

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Main Rotor Blades - Inspection

Procedure

1. Inspect the leading edge abrasion strip (3, Figure 1) and the splice covers (4 and 10) as follows:

1.1. Return the main rotor blade to Bell Helicopter Textron for replacement of leading edge abrasion strip under the following conditions:

1.1.1. Leading edge deformation due to abrasion.

1.1.2. Edge voids deeper than 0.50 inch (12.7 mm).

1.1.3. Holes, cracks, or tears.

1.2. Nicks, scratches, or sharp dents not exceeding 0.012 inch (0.31 mm) are permitted if polished smooth. Non-sharp dents not exceeding 0.015 inch (0.38 mm) are permitted. If damage exceeds the inspection requirements, replace the abrasion strip (3) and/or the splice covers (4 or 10).

2. Inspect the fretting pads (9) for cracks and adhesive voids. If cracks are found or edge voids are in excess of 0.25 inch (6.4 mm), notify Product Support Engineering.

NOTE

Some main rotor blades are equipped with only one trim tab (outboard), or two trim tabs (inboard and outboard), or three trim tabs (inboard, mid, and outboard).

3. Inspect the inboard, mid, and outboard trim tabs (6, 7, and 8) and the tip assembly (5) for damage, debonding, or corrosion (BHT-412-CR&O).

4. Inspect the rotor blade bolt retention bushings (1) (BHT-412-CR&O).

5. Inspect the main rotor blade bonded assembly (2) as follows:

5.1. Inspect for cracks. If cracks are found, notify Product Support Engineering.

5.2. Nicks, scratches, gouges, or dents on fiberglass surfaces are permitted without repair provided the fibers are not broken or cause delamination. If fibers are broken or delaminations develop, refer to the BHT-412-CR&O.

5.3. Scratches, gouges, or nicks that penetrate one or more layers of the upper and lower skins (12 and 14), but do not damage the core (15), may be repaired (BHT-412-CR&O).

5.4. Holes in the skin outboard of station 90.0 may be repaired (BHT-412-CR&O).

5.5. Nicks or notches in extreme trailing edge strip (13) of blade less than 0.120 inch (3.05 mm) in depth are acceptable if polished out over distance of at least 2.0 inches (51 mm) each side of nick or notch.

5.6. Nicks or scratches in erosion shield (11, detail A) are acceptable provided there is no penetration. Excessive damage requires erosion shield replacement at BHT authorized repair station.

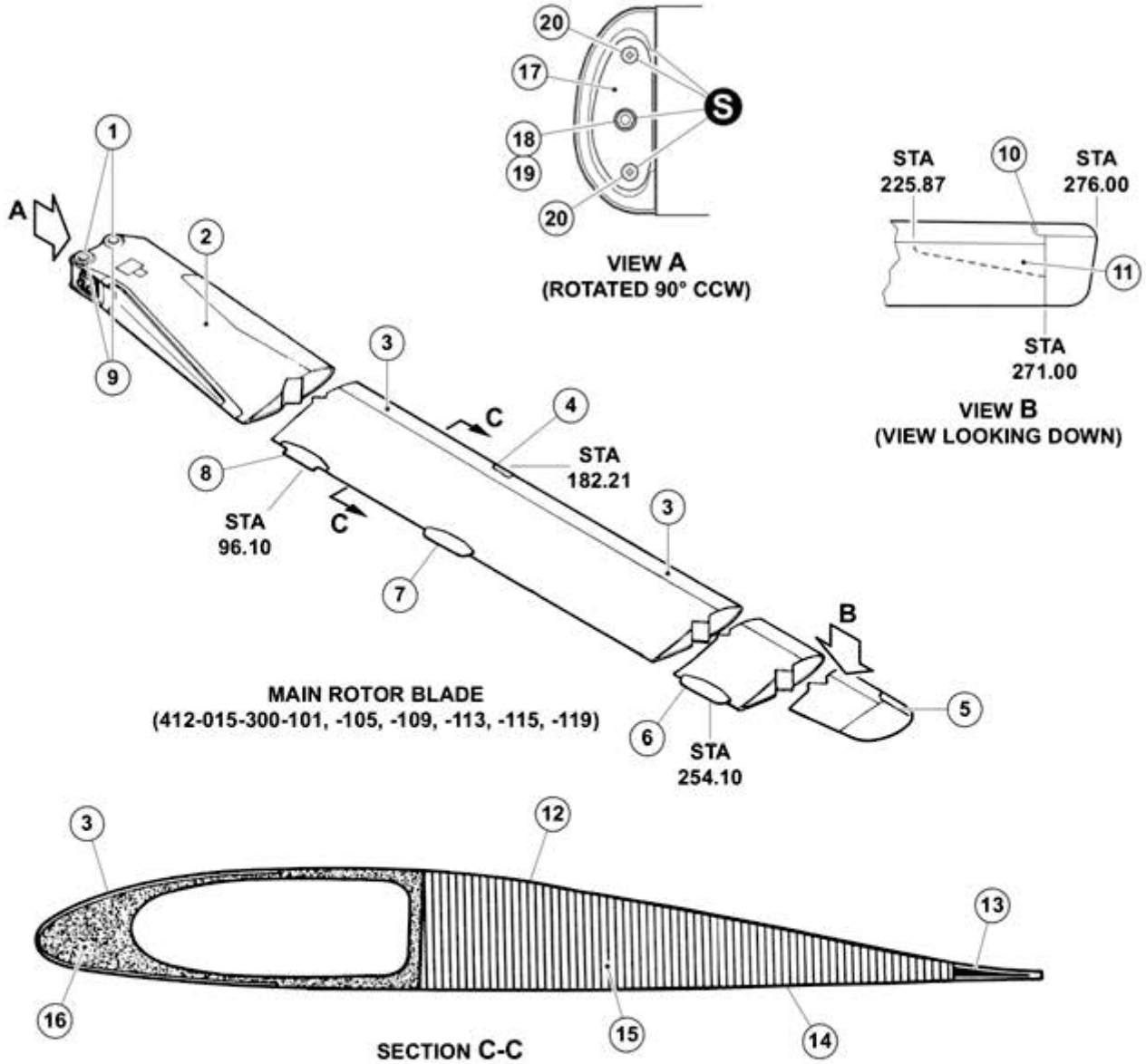
NOTE

A void is defined as any unbonded area. Many sub-definitions of voids are often used, such as lack of adhesive, gas pocket, misfit, etc. This manual makes no distinction among these sub-definitions, but groups all such unbonded areas under the general term, void. Inspect for voids using a small steel tapping hammer similar to BHT tool number T-75449-2 on all surfaces.

6. Inspect for voids under leading edge abrasion strips (3) and skin of main rotor blade bonded assembly (2). If any voids are found, refer to BHT-412-CR&O for limitation and authorized repair.

7. Inspect paint condition. Refer to BHT-412-CR&O for refinish procedures.

8. Inspect sealant applied in balance pocket area, on root end weight cover (17, Figure 1, View A) and it's attaching hardware (18, 19 and 20) for gaps, cracks, voids, deterioration, erosion and signs of water ingression. If sealant is damaged and/or there is presence of rust stains, indicating water ingress and corrosion of weights, refer to BHT-412-CR&O manual for limitation and authorized repair.



1. Blade bolt retention bushings
2. Main rotor blade bonded assembly
3. Abrasion strip
4. Splice cover (inboard)
5. Tip assembly
6. Outboard trim tab
7. Mid trim tab
8. Inboard trim tab
9. Fretting pad
10. Splice cover (outboard)
11. Erosion shield
12. Upper skin
13. Trailing edge strip
14. Lower skin
15. Core
16. Spar
17. Cover (412-015-045-101 or 412-015-045-103)
18. Nut
19. Washer
20. Screw

S SEALANT (C-308)

412_MM_62_0001_c01

ICN-412-A-621000-B-97499-00008-A-001-01

Figure 1. Main Rotor Blade (Sheet 1 of 1)

Main Rotor Blades - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Steel wool	C-411	AR	
Abrasive cloth or paper	C-423	AR	

Procedure

1. Polish out all acceptable nicks and scratches on leading edge abrasion strip (3, Figure 1), using 180 grit or finer abrasive cloth or paper (C-423). Polish to a surface finish, removing only enough material to remove the nicks or scratches. The use of steel wool (C-411) on abrasion strips to polish out defects is acceptable. Acceptable dents need not be filled or faired.

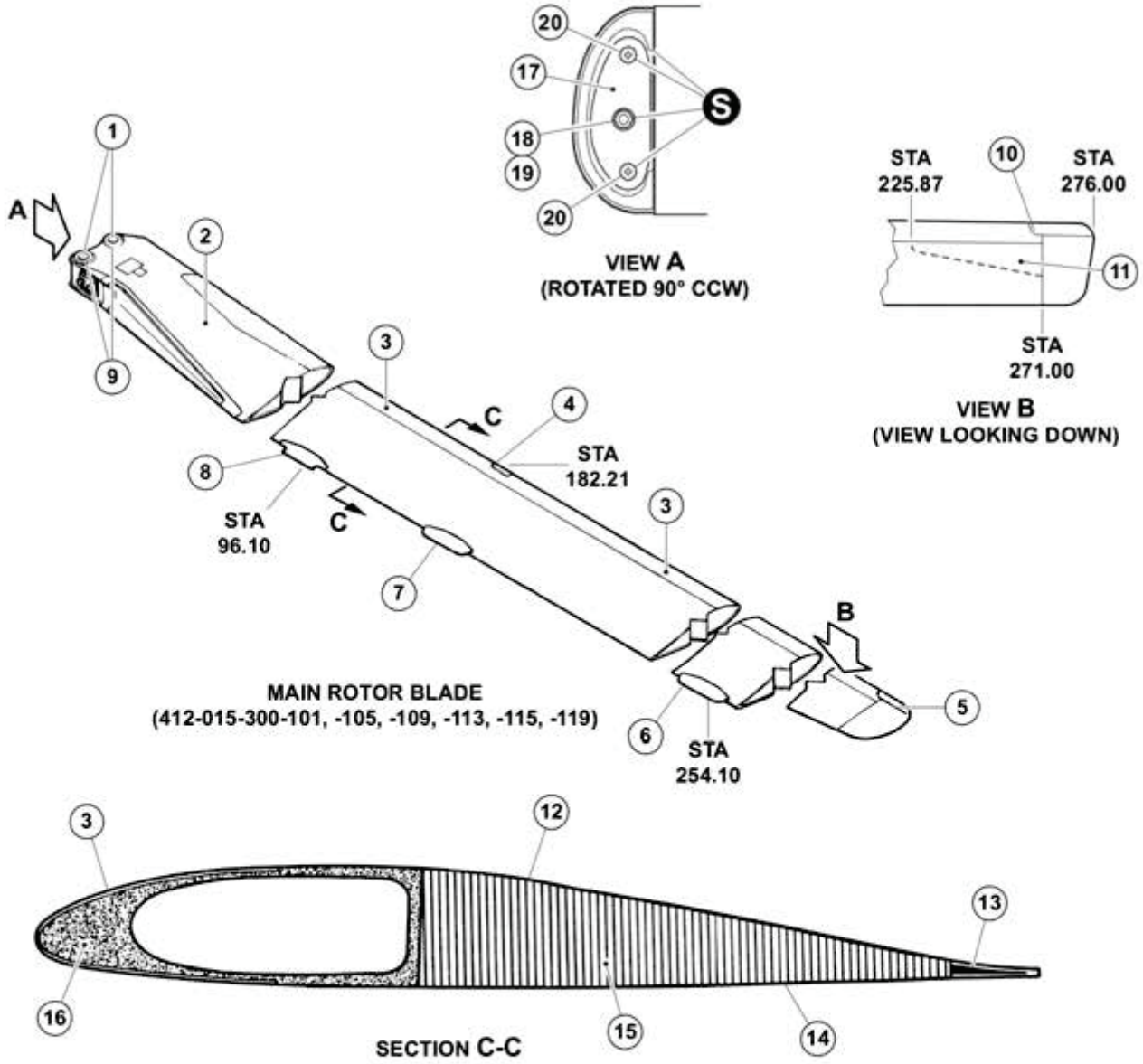
2. Repair rotor blade bolt retention bushings (1) (BHT-412-CR&O).

3. Polish out all acceptable nicks and scratches on fiberglass surfaces and erosion shield (11) with 200 grit abrasive cloth or paper (C-423). Lightly abrade painted surfaces immediately surrounding polished areas.

4. Polish out nicks, scratches and notches in trailing edge strip (13) which do not extend forward more than 0.10 inch (2.5 mm) with 200 grit or finer abrasive cloth or paper (C-423). Fair repair out a minimum of 2.0 inches (51 mm) each side.

5. Fill all acceptable edge voids (BHT-412-CR&O).

6. Touch up finish of repaired areas (BHT-412-CR&O).



1. Blade bolt retention bushings
2. Main rotor blade bonded assembly
3. Abrasion strip
4. Splice cover (inboard)
5. Tip assembly
6. Outboard trim tab
7. Mid trim tab
8. Inboard trim tab
9. Fretting pad
10. Splice cover (outboard)
11. Erosion shield
12. Upper skin
13. Trailing edge strip
14. Lower skin
15. Core
16. Spar
17. Cover (412-015-045-101 or 412-015-045-103)
18. Nut
19. Washer
20. Screw

S SEALANT (C-308)

412_MM_62_0001_c01

ICN-412-A-621000-B-97499-00008-A-001-01

Figure 1. Main Rotor Blade (Sheet 1 of 1)

Main Rotor Blades - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Blade Bolt Tool	T101266	1	
Sling	T101626	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Grease	C-561	AR	
Lockwire	C-405	AR	

Procedure



USE HOIST WITH A MINIMUM CAPACITY OF 1000 POUNDS (454 KG).

MAIN ROTOR BLADES AND HUB COMPONENTS ARE COLOR CODED. AFTER EXPANDABLE BOLT (10, Figure 1) IS ADJUSTED AT INITIAL INSTALLATION, IT SHALL BE KEPT WITH THE SAME ROTOR BLADE AND HUB SPINDLE, OR EXPANDABLE BOLT SHALL BE READJUSTED.

NOTE

When replacing main rotor blade assembly 412-015-200-101 with main rotor blade assembly 412-015-300-101, blades shall be installed in pairs and in same yoke and spindle of hub and blade assembly.

1. Attach sling (4, Figure 1) to a suitable hoist and position directly over center of blade (3).

2. Attach straps of sling (4) around blade (3). Hoist and align blade with hub.

3. Guide blade (3) into spindle (1) per color code and align bolt holes.

4. Apply grease (C-561) to mating surfaces of blade bolts (2) and (if being used) expandable bolts (10).



STANDARD BLADE BOLTS MAY BE INSTALLED IN LIEU OF EXPANDABLE BOLTS IN BLADE TRAILING EDGE BOLT HOLE. HOWEVER, STANDARD AND EXPANDABLE BLADE BOLTS SHALL NOT BE INTERMIXED ON SAME HUB.

5. Install rotor blade bolt (2) through spindle (1) and rotor blade (3) in leading edge bolt hole. Use blade bolt tool (T101266) to install bolt (2). Do not install nut (6) at this time. If standard bolt is being installed in aft hole, repeat procedure for trailing edge bolt.



EXPANDABLE BOLT (10), BLADE (3) AND SPINDLE ARE COLOR CODED. EXPANDABLE BOLT (10) AND ROTOR BLADE (3) SHALL BE INSTALLED IN SAME SPINDLE FROM WHICH REMOVED. IF FOR ANY REASON EXPANDABLE BOLT (10), ROTOR BLADE (3) AND SPINDLE COMBINATION ARE CHANGED, EXPANDABLE BOLT (10) SHALL BE ADJUSTED AS DESCRIBED BELOW.

6. For helicopters with expandable bolt installed in aft blade hole, accomplish the following:

6.1. Raise bolt handle (9) on expandable bolt (10).

6.2. Install expandable bolt (10) in spindle (1) and blade (3) in trailing edge hole with handle positioned as shown in View A-A.

6.3. Push down on bolt handle (9). Force required to push handle down should be 50 to 75 pounds (222 to 334 N) measured at end of handle. If force is within limits, latch spring clip of bolt handle (9) on core pin of expandable bolt (10).

6.4. Install safety lock hair pin (12) to hold bolt handle (9) in down position. If force required to push handle down is not within limits, accomplish the following step.

6.5. Adjust expandable bolt (10) only if force required to latch bolt handle (9) is not within 50 to 75 pounds (222 to 334 N) measured at end of handle. Install expandable bolt (10) in correct spindle (1) and blade (3) combination. Adjust adjustment nut (11) in increments and push bolt handle (9) down until force required is within limits noted above. When expandable bolt (10) is adjusted within limits, install safety lock hair pin (12) to hold handle (9) in down position.

7. For standard blade bolts, accomplish the following:

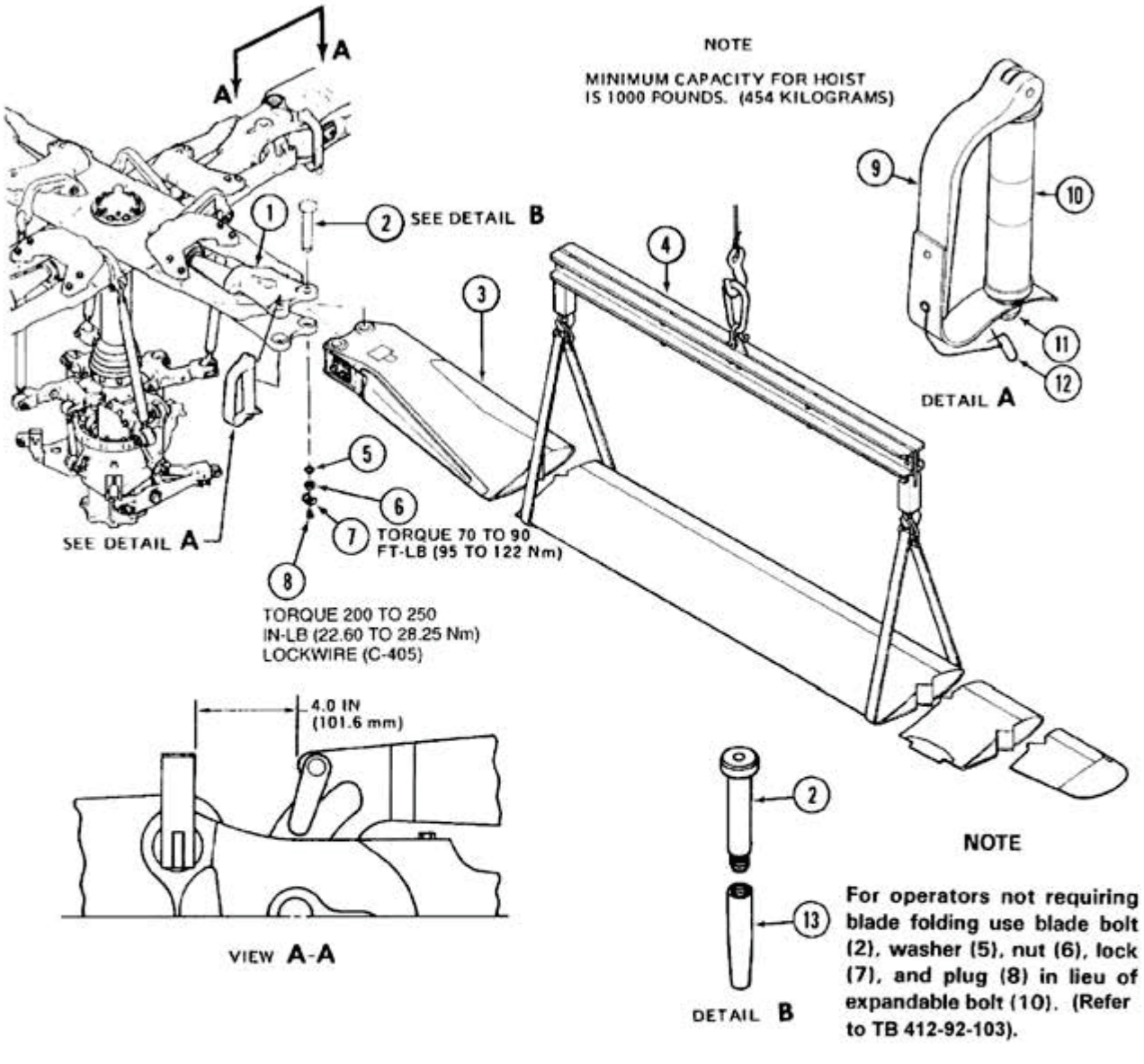
7.1. Install washer (5) and nut (6).

7.2. Torque nut (6) 70 to 90 foot-pounds (95 to 122 Nm).

7.3. Install lock (7) on nut (6) and install plug (8) to secure lock (7) on nut (6).

7.4. Torque plug (8) 200 to 250 inch-pounds (22.60 to 28.25 Nm). Secure plug (8) to lock (7) using lockwire (C-405).

8. After all four main rotor blades are installed, track main rotor blades (412-A-18-11-01-00A-373A-A / 00177) and perform functional check flight BHT-412-FM.



1. Spindle
2. Blade bolt
3. Blade
4. Sling (T101626)
5. Washer (206-010-154-3)
6. Nut
7. Lock
8. Plug (AN814-4L)
9. Bolt handle
10. Expandable bolt
11. Adjustment nut
12. Safety lock hair pin
13. Blade bolt tool (T101266)

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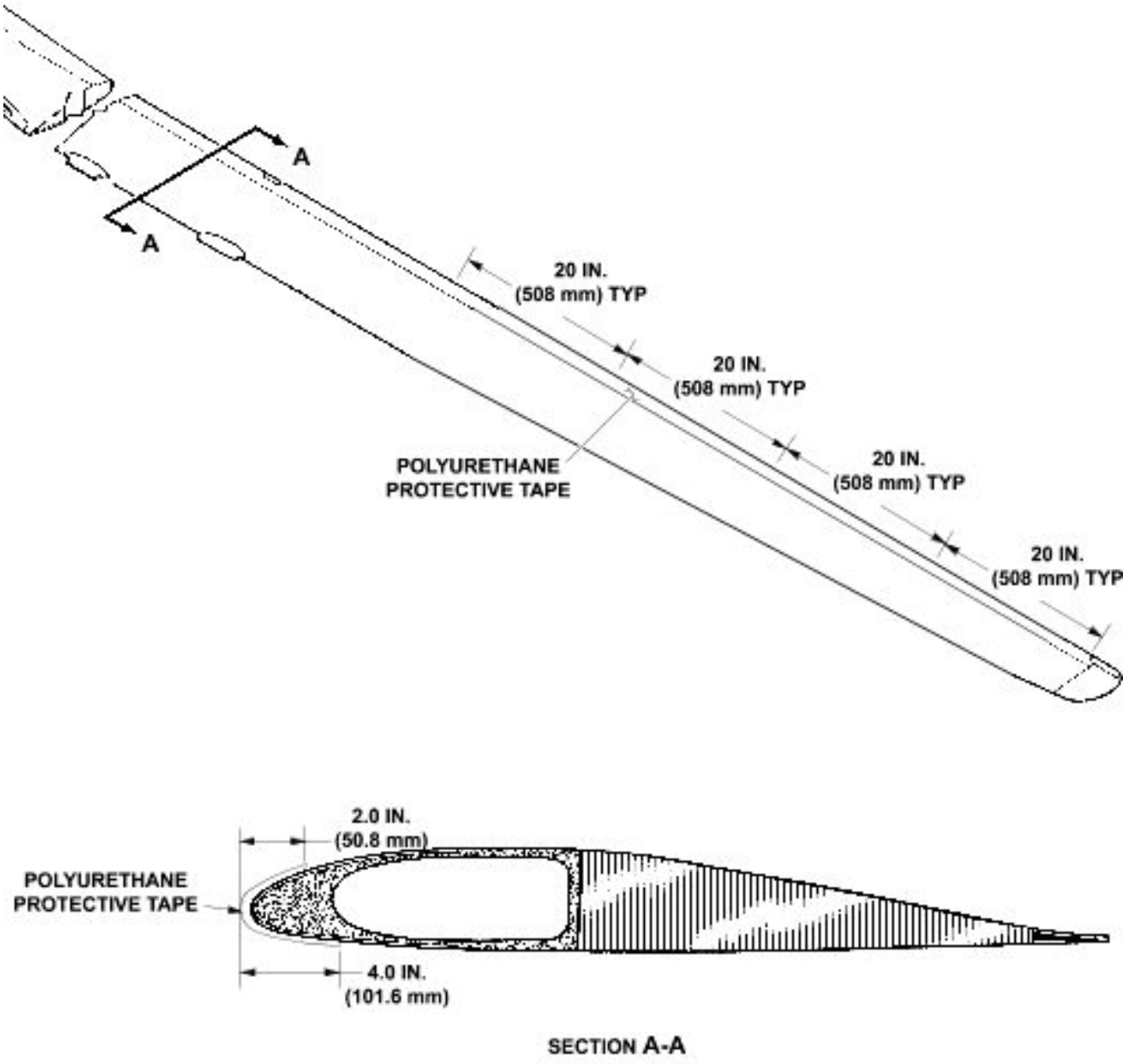
Figure 1. Main Rotor Blade Installation (Sheet 1 of 1)

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Polyurethane Protective Tape - Description

Use of polyurethane tape is optional with individual operator. Its use is recommended for protection against abrasion of leading edge of blade in areas of high sand and dust conditions. The tape affords medium protection against rain. It may be removed and replaced with new tape as often as necessary.

Refer to Figure 1 for tape size and location.



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Figure 1. Polyurethane Protective Tape - Size and Location (Sheet 1 of 1)

Polyurethane Protective Tape - Cleaning and Surface Preparation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Isopropyl alcohol	C-385	AR	
Abrasive cloth or paper	C-423	AR	
Tape	C-434	AR	
Clean cloth	C-516	AR	
Adhesive promoter	C-558	AR	

Procedure

1. Dampen a clean cloth (C-516) with isopropyl alcohol (C-385) and wipe the blade surface.

2. Wipe dry with a clean cloth (C-516) before the isopropyl alcohol (C-385) evaporates from the surface.

3. Mask application area with 1-inch masking tape (C-434).

4. Sand surface in a spanwise direction only with 220 or 240 grit abrasive cloth or paper (C-423).

5. Thoroughly wipe with clean cloth (C-516) and isopropyl alcohol (C-385) until there is no trace of residue and wipe dry with clean cloth (C-516).

6. Apply a thin, uniform coat of adhesive promoter (C-558) to the blade surface, using the minimum amount that will fully coat the surface. Allow to dry thoroughly (5 minutes minimum).

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Polyurethane Protective Tape - Wet Application Method

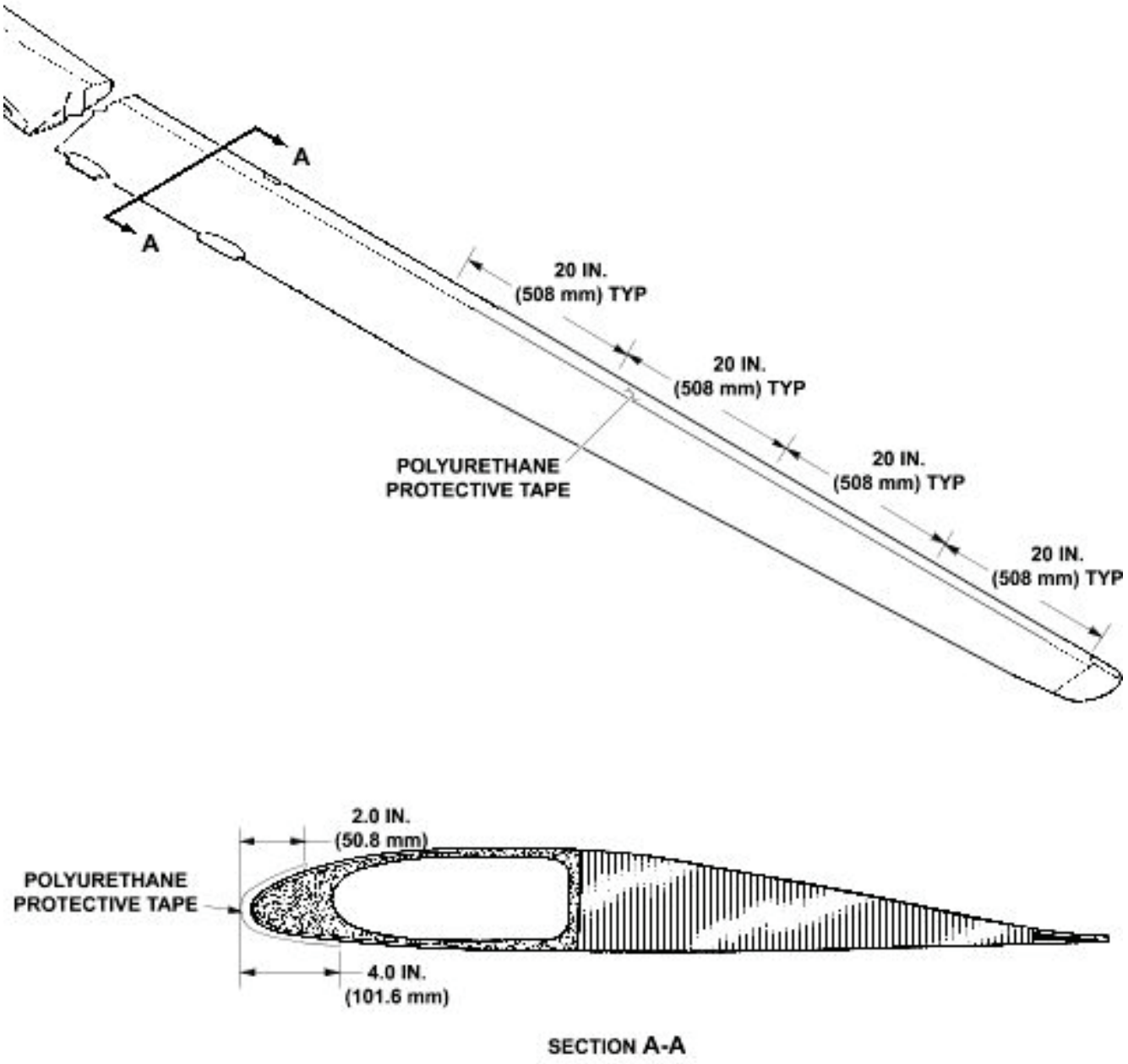
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Isopropyl alcohol	C-385	AR	
Erosion tape	C-557	AR	

Procedure

1. Mix the wetting solution as follows: thoroughly mix four drops of liquid detergent (not soap) in a pint (1/2 liter) solution of 25% isopropyl alcohol (C-385) and 75% water.
2. Mark correct position of tape on blade surface (Figure 1).
3. Remove liner and flood adhesive side with wetting solution.
4. Thoroughly spray blade surface with wetting solution.
5. Position erosion tape (C-557) on blade surface and spray wetting solution on film surface (prevents squeegee from sticking).
6. Squeegee from leading edge to trailing edge of blade using firm, overlapping strokes to smooth out film. Dry entire area with cloth. Work out blisters with the plastic squeegee.
7. To apply next section of erosion tape (C-557), repeat Step 3. through Step 6..
8. Allow applied erosion tape (C-557) to dwell overnight (12 hours).
9. If blisters are present after the application, puncture tape at one end of blister with a pin or needle and press out with squeegee or thumb toward puncture.



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Figure 1. Polyurethane Protective Tape - Size and Location (Sheet 1 of 1)

Polyurethane Protective Tape - Dry Application Method

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Erosion tape	C-557	AR	

Procedure

1. Remove half of liner from adhesive side of erosion tape (C-557).
2. Align erosion tape (C-557) and press one edge to blade surface with finger.
3. Squeegee from leading edge to trailing edge of blade using firm overlapping strokes to smooth out tape.
4. Remove remaining half of liner and repeat Step 3..

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Polyurethane Protective Tape - Inspection

Procedure

1. After each mission, when practical, inspect the tape on the leading edge of main rotor blades. If any of the following conditions exist, replace tape.

- Wear-through of tape
- Void or blistering of tape

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Polyurethane Protective Tape - Edge Sealing (Optional)

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Edge sealer	C-149	AR	
Tape	C-434	AR	
Erosion tape	C-557	AR	

Procedure

1. Prepare erosion tape (C-557) for edge sealing by applying a strip of tape (C-434) across the top of the erosion tape (C-557) 1/32 inch back from the edge. Apply a second strip of tape (C-434) 1/8 inch forward of the edge of the erosion tape (C-557). This will provide a neat, defined area for the application of the epoxy edge sealant.

2. Apply a bead of edge sealer (C-149) between both strips of tape (C-434). With a fitted rubber glove, use a finger to spread and smooth the sealant bead between the strips of tape.

3. Allow the adhesive to partially cure, approximately 30 minutes. Remove the tape (C-434) to allow the edges of the adhesive to flow smoothly on the surface and the erosion tape (C-557) to make a tapered edge. The 30-minute cure time can be adjusted up or down depending if the ambient temperature is higher or lower than 72°F (22°C).

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Polyurethane Protective Tape - Removal

Procedure

1. Pull on tape by keeping the angle of removal perpendicular (90°) to the surface and pull at a moderate speed.

- If the angle is more or less, there is greater chance of adhesive transferring from the tape to the surface.
- The same problem is possible if the tape is removed too slowly. If removed too quickly, the tape may tear or sliver.
- At 60 to 120°F (15 to 49°C), the adhesive is firm enough for clean removal without tearing or slivering.
- Removal at hotter temperature increases the potential for adhesive transfer.
- At colder temperature, the tape is more brittle with increased chance for slivering.
- For minor adhesive transfer rub the adhesive with your finger into a ball and wipe it off. Or, dab up the adhesive with another piece of tape.
- For major adhesive transfer that is tacky, use common cleaning agents such as 3M citrus base cleaner, mineral spirits, or industrial solvents. Refer to BHT-ALL-SPM, Standard Practices Manual for cleaning of stainless steal leading edge abrasion strip.
- For hard adhesive on hard surfaces, scrape with a plastic/Plexiglas scraper or sand with fine grit paper. Use caution to prevent surface damage.

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Main Rotor Hub - Description

The main rotor hub consists of two titanium yokes, each of which mounts two steel spindles in elastomeric bearings and elastomeric dampers. Each spindle has two spindle lugs for mounting a rotor blade. Rotor blade pitch control is achieved by means of a pitch horn attached to inboard end of each spindle. The opposite end of each pitch horn is attached to a pitch link. The pitch links are attached to the rotating controls. The main rotor hub is attached to mast by a procedure which does not require application of large torque values to hub retaining nut. The cap assembly, which secures main rotor hub to mast, is threaded onto mast and is also bolted to hub with eight bolts.

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Main Rotor Hub - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Support Assembly	214-782-003-1	1	
Shaft Weldment	T101356-3	1	(Part of T101263-101)
Bolt	NAS6208-9	1	(Part of T101263-101)
Transportation Stand	SWE 13875	1	
Adapter	SWE 13875-30	1	
Adapter	T101263-105	1	(Part of T101263-101)
Jackscrew	T101308	1	
Sling Assembly	T101626	1	

Procedure

1. Inspect main rotor hub, main rotor blades, and pitch links for color code markings. If any color code markings (dots) are missing, apply new markings (412-A-11-00-00-00A-067A-A / 00124).

2. Remove main rotor blades (412-A-62-10-00-00A-520A-A / 00935).

3. Remove bolt (13, Figure 1) and washers (14) securing pitch link (19) to pitch horn (15). Secure pitch link (19) to main rotor mast (8). Disconnect remaining three pitch links (19). Secure pitch links (19) to main rotor mast (8).

4. Cut and remove lockwire from bolts (9).

5. Remove bolts (9) and washers (10).



WHEN REMOVING SEALANT FROM HUB, USE PLASTIC OR BRASS TOOLS ONLY.

6. Remove sealant from threaded holes in lower cone (11) of main rotor hub (5) adjacent to where bolts (9) were removed. Remove sealant with plastic or brass tools to prevent damage to rotor hub.

7. Install three jackscrew (T101308) in lower cone (11) of main rotor hub (5) where sealant was removed. If jackscrew (T101308) is not available, use three 1/4 X 28 UNF threaded bolts approximately 2.5 inches (64 mm) long. Tighten jackscrews (or bolts) evenly and push lower cone (11) out of lower cone seat of main rotor hub (5).

8. Remove sealant from joint between cap assembly (3) and main rotor hub (5) with sharp plastic scraper.

9. Cut and remove lockwire from bolts (1).

10. Remove bolts (1) and washers (2).

11. Remove cap assembly (3).

NOTE

Cap assembly (3) consists of two major parts: upper cap and nut which threads onto mast. After eight bolts (1) and bead of sealant are removed, cap assembly can be removed with a wrench attached to flat area at top of cap assembly.

12. Place sling assembly (T101626) (2, Figure 2) on a hoist with minimum capacity of 1000 pounds (454 kg). Hoist sling into position above main rotor hub assembly (1). Position sling straps around main rotor hub assembly, as shown in Figure 2.

13. Check that adapter (6) with support assembly (5), adapter (4), and bolt (3) are installed on maintenance transportation stand (7).

NOTE

If adapter (6) with support assembly (5), adapter (4), and bolt (3) are installed on maintenance transportation stand (7), proceed to Step 14.. If not installed, perform Step 13. and Step 14..

14. Install adapter (6) on maintenance transportation stand (7).

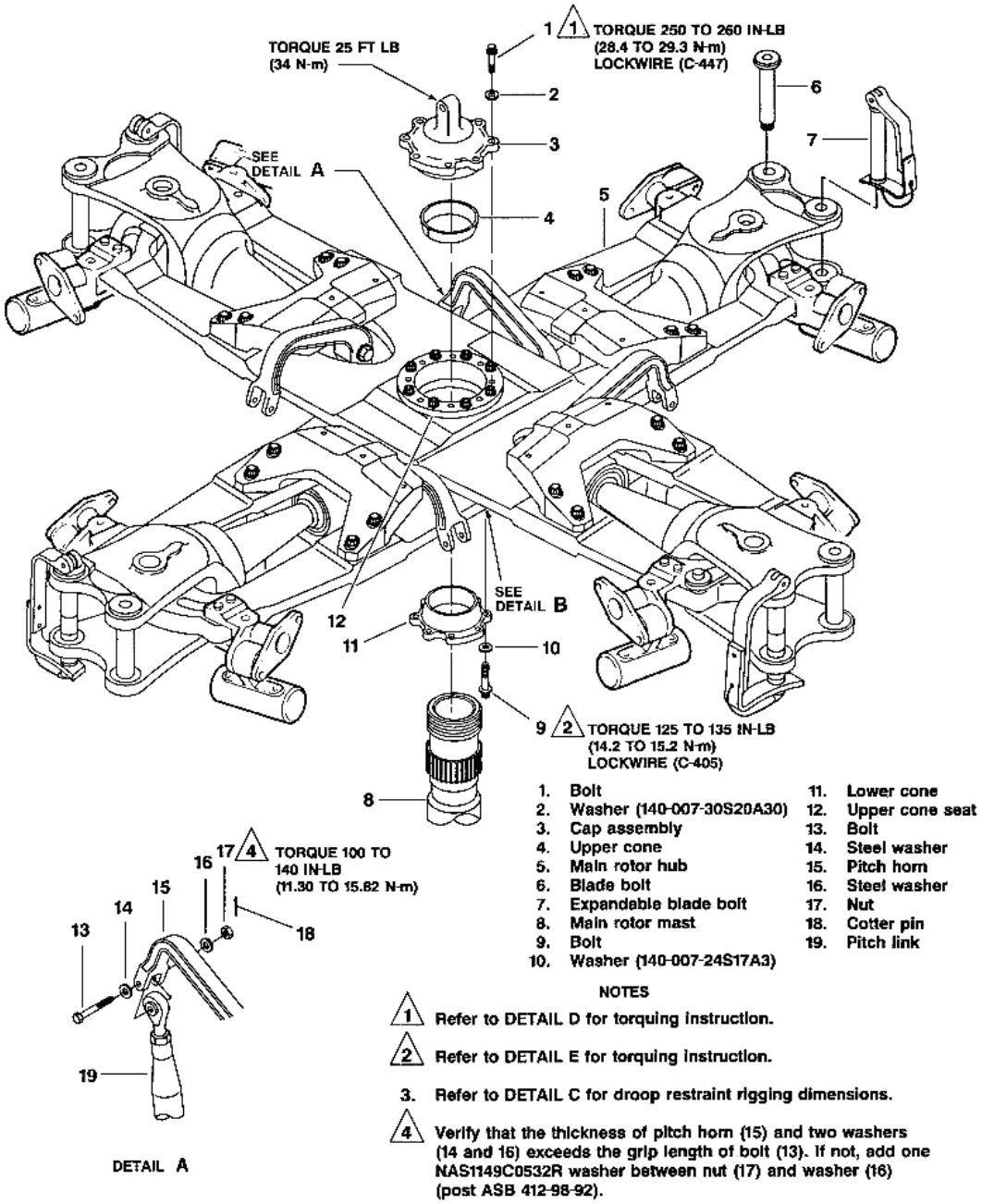
15. Install adapter (4) on support assembly (5) and secure with bolt (3).

16. Hoist main rotor hub assembly (1) approximately 2.0 inches (51 mm) and remove upper cone (4, Figure 1).

17. Hoist main rotor hub (5) clear of main rotor mast (8).

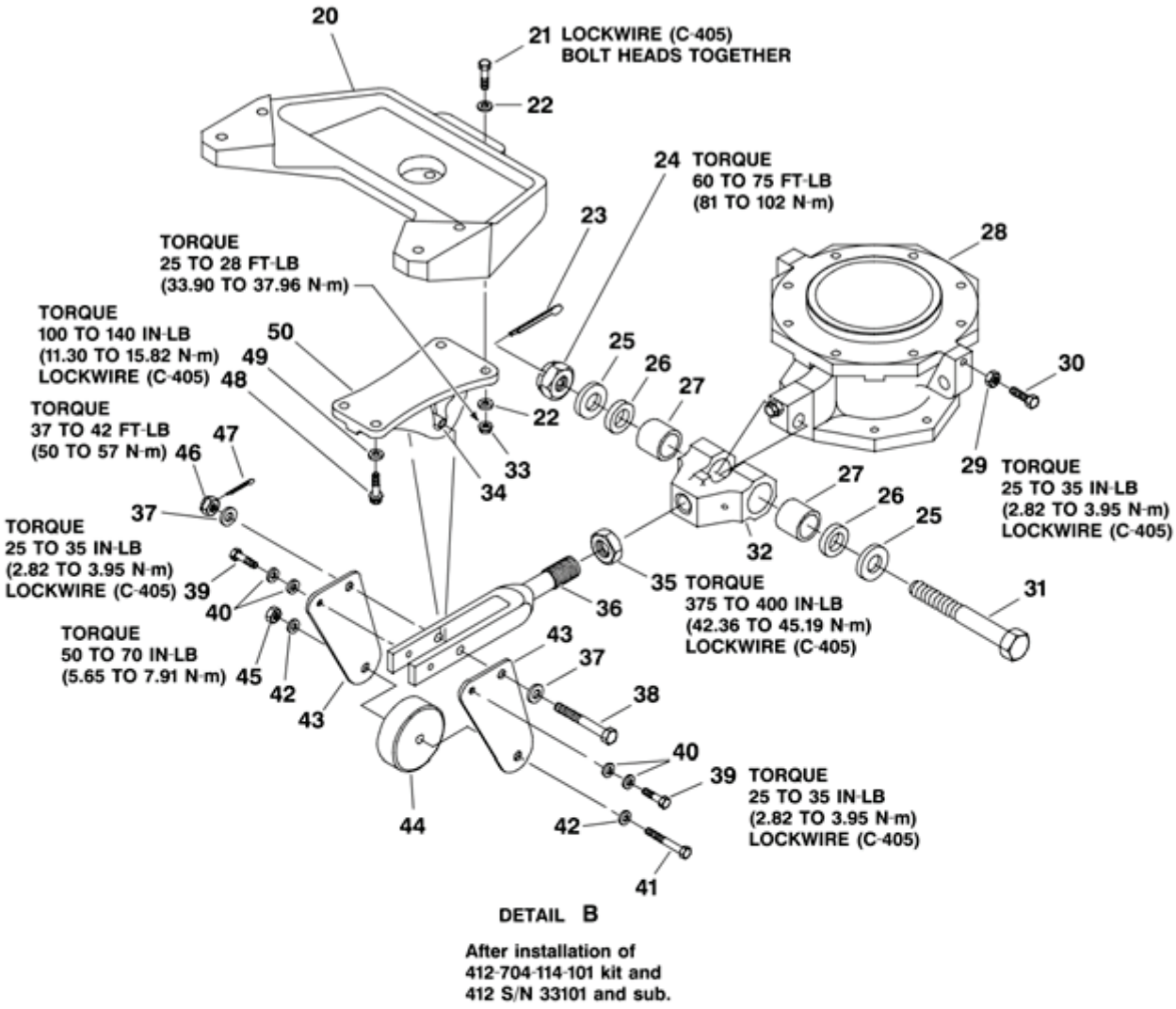
18. Align main rotor hub assembly (1, Figure 2) with adapter (4) and lower main rotor hub assembly into position on maintenance transportation stand (7).

19. Remove sling (2) from main rotor hub assembly (1).



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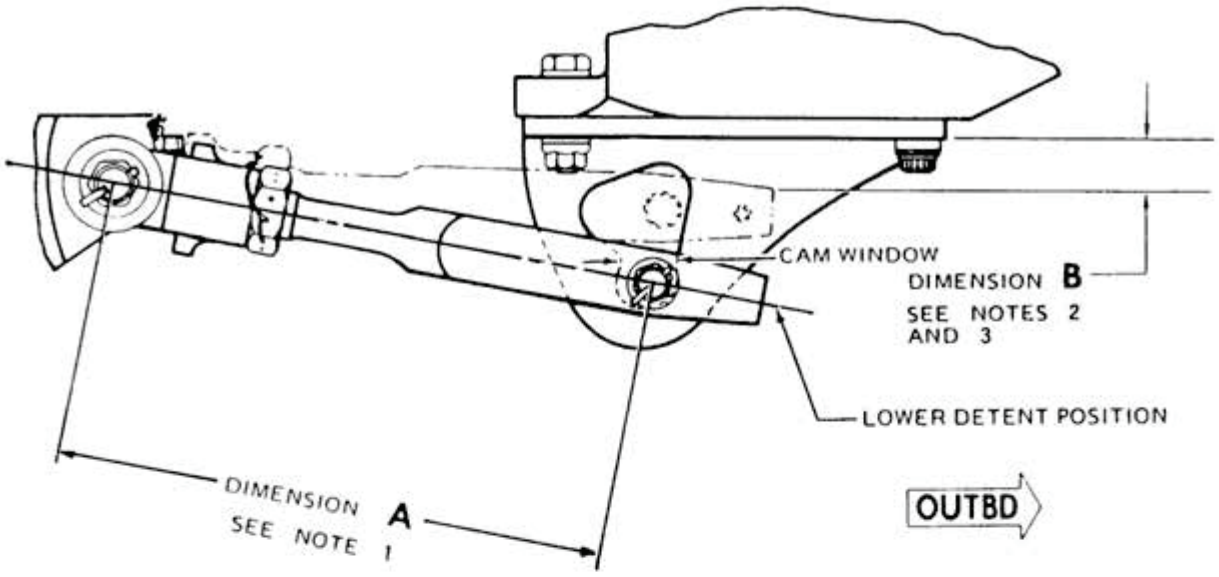
Figure 1. Main Rotor Hub (Sheet 1 of 6)



20. Damper bridge
21. Bolt
22. Washer (AN960-616)
23. Cotter pin
24. Nut
25. Washer (412-018-064-101)
26. Washer (140-00733S40C4)
27. Race
28. Lower cone seat
29. Nut
30. Bolt
31. Bolt
32. Clevis
33. Nut
34. Bushing (412-018-062-109)
35. Nut
36. Clevis
37. Washer (AN960-716)
38. Bolt
39. Bolt
40. Washer (MS9482-10)
41. Bolt
42. Washer (AN960-516)
43. Arm
44. Weight
45. Nut
46. Nut
47. Cotter pin
48. Bolt
49. Washer (MS9482-11)
50. Cam

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Figure 1. Main Rotor Hub (Sheet 2 of 6)



FLEXURE SHOWN IN STATIC DROOP POSITION WITH DROOP RESTRAINT ENGAGED FOUR PLACES (ARM AND WEIGHT REMOVED FOR CLARITY)

DETAIL C

NOTES

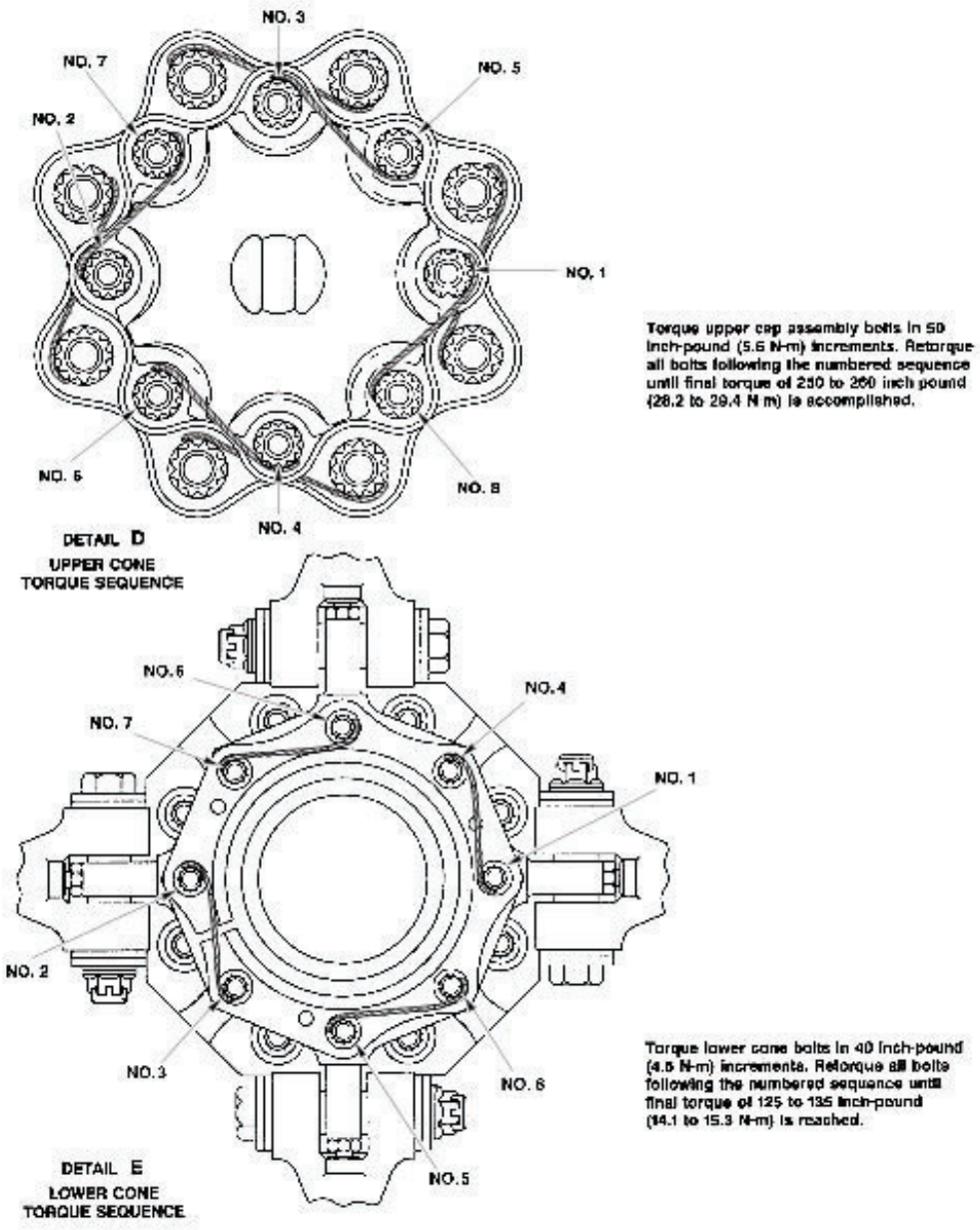
1. **Droop restraint arm length (dimension A) to be adjusted with weight, brackets and associated hardware removed per rigging procedures.**

2. **Dimension B is to be set per rigging instructions following setting of arm length (dimension A) and prior to reinstallation of weight, brackets and associated hardware. For each restraint arm, dimension B is to be measured on both leading and trailing edge clevis tangs and measured difference is to be 0.030 inch (0.762 mm) maximum.**

3. **Pivot stop rigging dimension to be set after adjusting (dimension A) per rigging procedures.**

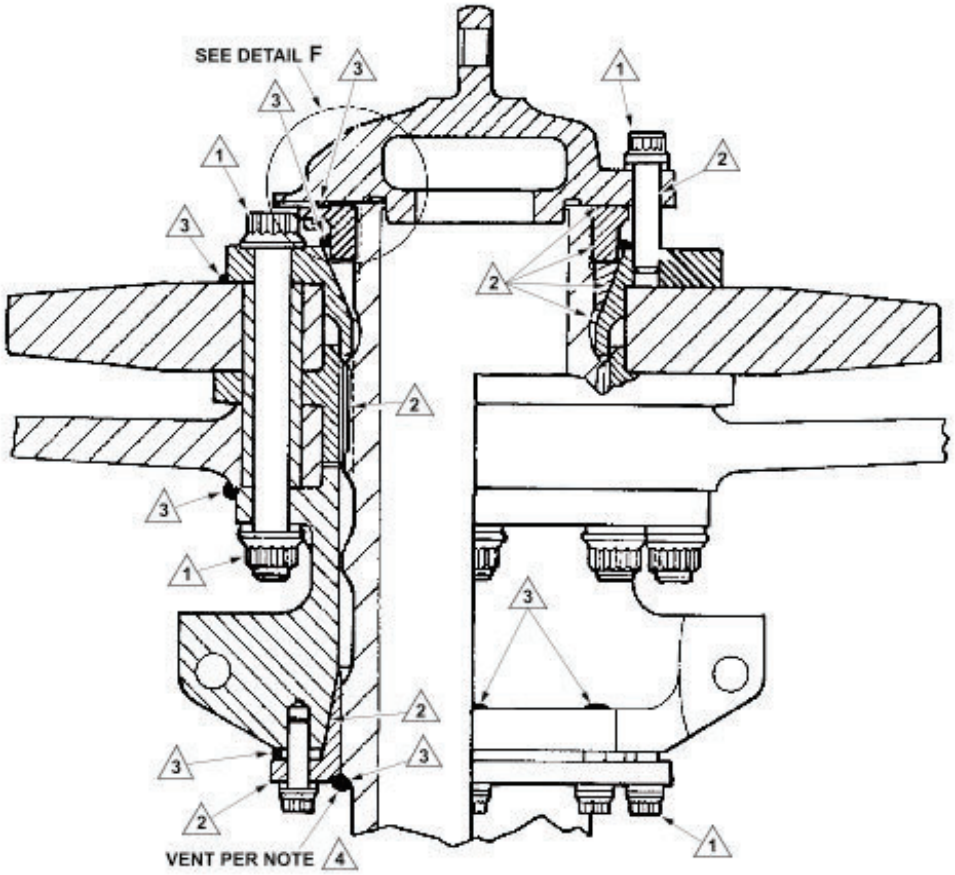
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Figure 1. Main Rotor Hub (Sheet 3 of 6)



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Figure 1. Main Rotor Hub (Sheet 4 of 6)

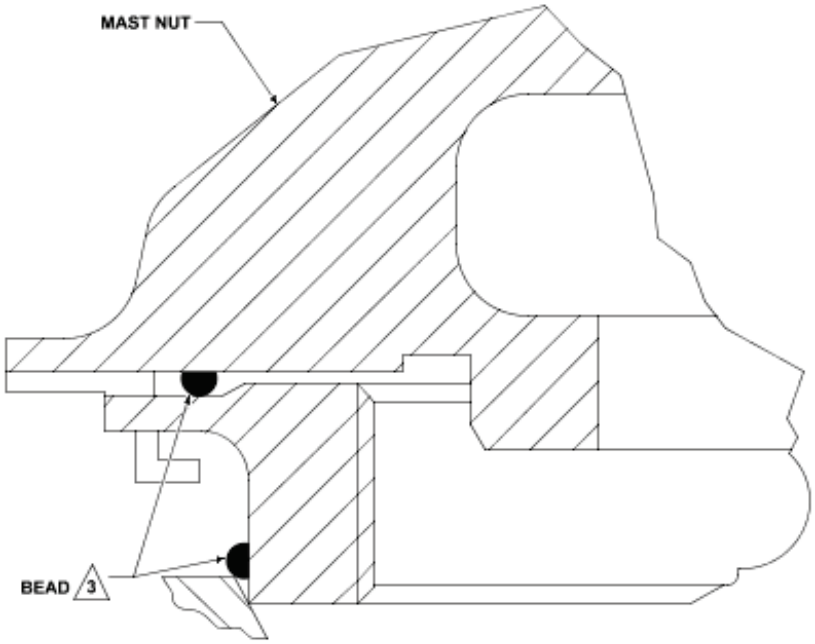


NOTES

1. Apply corrosion preventive compound (C-101).
2. Apply Thixogrease (C-561) on faying surfaces.
3. Seal all around using sealant (C-158).
4. Small passageway must remain at lower cone split to vent space between mast and hub.

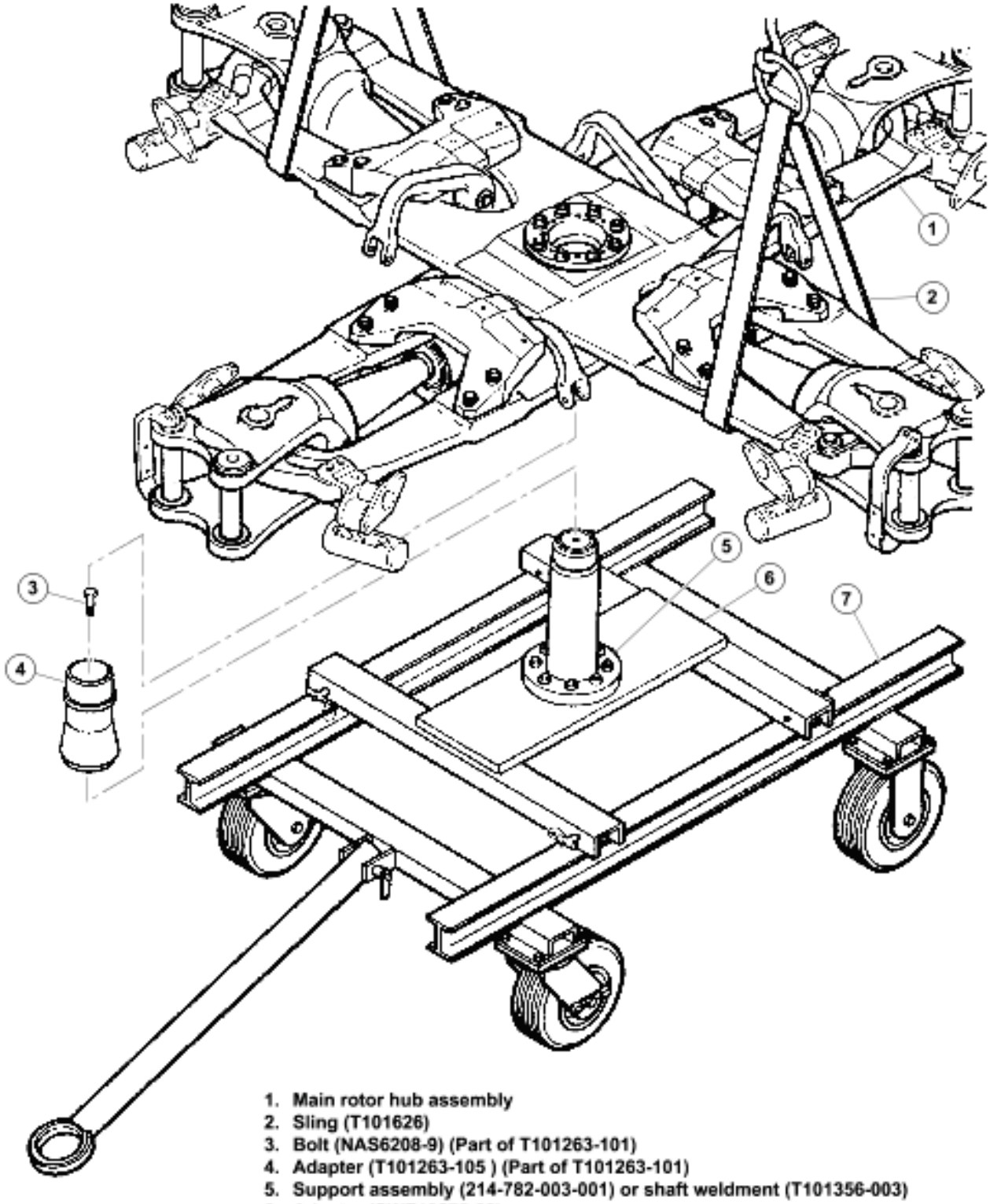
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Figure 1. Main Rotor Hub (Sheet 5 of 6)



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Figure 1. Main Rotor Hub (Sheet 6 of 6)



1. Main rotor hub assembly
2. Sling (T101626)
3. Bolt (NAS6208-9) (Part of T101263-101)
4. Adapter (T101263-105) (Part of T101263-101)
5. Support assembly (214-782-003-001) or shaft weldment (T101356-003)
6. Adapter (SWE13875-30)
7. Maintenance transportation stand (SWE13875)

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Figure 2. Main Rotor Hub Stand - Tool Application (Sheet 1 of 1)

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Main Rotor Hub - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Clean main rotor hub assembly with cloths dampened with drycleaning solvent (C-304).

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Main Rotor Hub - Inspection

Procedure

1. For detailed inspection of main rotor hub assembly components, refer to the BHT-412-CR&O.

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Main Rotor Hub - Repair

Procedure

1. For detailed repair procedures of main rotor hub assembly components, refer to the BHT-412-CR&O.

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Main Rotor Hub - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Sling Assembly	T101626	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Corrosion preventive compound	C-104	AR	
Sealant	C-158	AR	
Lockwire	C-405	AR	
Lockwire	C-447	AR	
Grease	C-561	AR	

Procedure

1. Place sling assembly (T101626) (2, Figure 1) on a hoist with minimum capacity of 1000 pounds (454 kg) or more.

2. Hoist sling (2) into position above main rotor hub assembly (1).

3. Position sling (2) straps around hub as shown on Figure 1.

4. Hoist main rotor hub assembly (1) clear of adapter (4).



MAKE SURE GREASE DOES NOT COME IN CONTACT WITH TEFLON OF CONE OR MATING SURFACES WITH TEFLON.

5. Apply a thin coat of grease (C-561) to mating surfaces of main rotor mast (8, Figure 2) lower cone (11), main rotor hub (5), upper cone (4), and cap assembly (3).

6. Apply grease (C-561) to main rotor mast (8) and splined plate splines.

7. Position lower cone (11) on main rotor mast (8) with beveled side up.

8. Hoist main rotor hub (5) into position above main rotor mast (8).

9. Align master tooth of splined plate in main rotor hub (5) with master tooth of main rotor mast (8) splines.

10. Carefully lower main rotor hub (5) onto main rotor mast (8) until the upper cone seat (12) is supported by splines on main rotor mast.

11. Remove sling (2, Figure 1) from main rotor hub (5, Figure 2).

12. Install upper cone (4) which was previously coated with grease (C-561), on main rotor mast (8) with beveled side down.

13. Apply a thin coat of grease (C-561) under the heads and to the shanks of bolts (9).

14. Slide lower cone (11) into position and install bolts (9) with washers (10) under bolt heads. Do not torque bolts (9) at this time.

15. Install cap assembly (3) on main rotor mast (8), and torque cap assembly to approximately 25 foot-pounds (34 Nm) to ensure cone (4) is seated.

16. Loosen cap assembly until holes in cap assembly flange are aligned with threaded holes in upper cone seat (12). If cap assembly has been loosened less than half the distance between holes, loosen one additional hole for installation of bolts (1).

17. Apply a bead of sealant (C-158) around cap assembly (3) where cap assembly joins main rotor hub (5).

18. Apply a thin coat of grease (C-561) under the heads and to shanks of eight bolts (1).

19. Install eight bolts (1) with washers (2) under bolt heads, and torque bolts in sequence as per Detail D.

20. Secure bolts (1) in pairs using lockwire (C-447).

21. Apply a thin coat of corrosion preventive compound (C-101) to head of bolts (1).

22. Check torque of bolts (1) after every flight during rotor smoothing process until torque is maintained.

23. Torque bolts (9) in sequence as per Detail E, and secure bolt heads in pairs using lockwire (C-405).

24. Apply a thin coat of corrosion preventive compound (C-101) to head of bolts (9) and washers.

25. Check torque of bolts (9) after every flight during rotor smoothing process until torque is maintained.

26. Fill threaded holes in lower cone seat of main rotor hub (5) in area above bolts (9), area at lower cone (11) to main rotor mast (8) interface, and gap between lower cone and cone seat with sealant (C-158).

NOTE

Small passageway must remain open at lower cone split to vent space between mast and hub.

27. Position pitch link (19) in pitch horn (15).

28. Coat bolt (13) with corrosion preventive compound (C-104).

29. Install bolt (13), washer (14), washer (16), and nut (17) with head of bolt (13) in direction of rotation.

NOTE

Verify that the thickness of the pitch horn (15) and two washers (14 and 16) exceeds the spindle length of the bolt (13). If not, add one NAS1149C0532R washer between nut (17) and washer (16).

30. Torque nut (17) 100 to 140 inch-pounds (11.30 to 15.82 Nm). Install cotter pin (18).

31. Repeat Step 27. through Step 29. for installation of remaining pitch links.

NOTE

Main rotor hub can remain on maintenance transportation stand (1, Figure 1) or be installed on helicopter for accomplishment of rigging procedures.

32. Install main rotor blades (412-A-62-10-00-00A-720A-A / 00946).



PROPER RIGGING IS EXTREMELY IMPORTANT AND MUST BE PERFORMED UNDER IDEAL CONDITIONS. MAIN ROTOR BLADE MOVEMENT, SUCH AS CAUSED BY WIND MUST BE MINIMIZED. IT IS RECOMMENDED THE FOLLOWING RIGGING PROCEDURES BE PERFORMED IN A HANGER WITH MAIN ROTOR HUB AND BLADE ASSEMBLY EITHER ON HELICOPTER OR INSTALLED ON MAINTENANCE TRANSPORTATION STAND (1, Figure 1) WITH BLADE TIPS UNSUPPORTED. IF MAINTENANCE TRANSPORTATION STAND (1) IS USED, DISREGARD REFERENCES IN RIGGING PROCEDURE REGARDING BLADE POSITION RELATIVE TO HELICOPTER. OUTDOOR RIGGING MAY BE PERFORMED PROVIDING WIND VELOCITY IS FIVE KNOTS OR LESS.

33. Manually rotate rotor until blade corresponding to first droop restraint arm to be rigged is pointing forward over nose of helicopter. Rigging procedure is to be performed while blade is at normal static droop.

34. Cut and remove lockwire from bolt (30, Figure 2).

35. Loosen nut (29) on pivot adjusting bolt (30) and turn bolt inward until it bottoms; do not torque nut.

36. Cut and remove lockwire from nut (35).

37. Adjust droop restraint arm length (Dimension A, Figure 2, Detail C) as follows:

NOTE

Dimension A is established using bolt (38) without bushing (34) installed to obtain length baseline.

37.1. Manually hold droop restraint arm to position 7/16-inch holes of outboard clevis (36) into outboard radius of cam (50) lower detent. Using bolt, verify bolt will install freely through both clevis tangs while passing through lower detent of cam window. If bolt does not pass through, turn clevis as necessary, until bolt freely installs.

37.2. Manually stabilize forward blade at tip during Step 37.3. and Step 37.4.. Do not support blade.

37.3. Lengthen arm by turning clevis (36) outward in 1/2-turn increments (installing bolt each time) until first instance occurs where bolt will not freely install. This establishes “length baseline”.

37.4. Turn clevis inward 1/2 turn from “length baseline”. Bolt should install freely. With bolt loosely in place, attempt to swing droop restraint arm upward into upper portion of cam window. If bolt hits cam window surface and arm is restrained in cam lower detent, length adjustment is correct. If bolt clears cam window surface and arm swings up freely, turn clevis outward 1/2 turn.

37.5. Square up clevis (36) with cam (50) by pushing clevis upward until both clevis tangs are in contact with the cam mounting flange.

37.6. Torque nut (35) 375 to 400 inch-pounds (42.36 to 45.19 Nm).

37.7. Make sure that clearance exists between clevis (36) and cam (50).

38. Adjust pivot adjusting bolt (30) to achieve dimension B, 0.60 to 0.70 inch (15.2 to 17.8 mm) (measured from face of cam plate mounting flange to outboard upper corner of each clevis tang) (Figure 2, Detail C). Dimension B on same clevis may vary 0.030 inch (0.76 mm) maximum leading edge side to trailing edge side.

39. Torque nut (29) 25 to 35 inch-pounds (2.82 to 3.95 Nm).

40. Recheck Dimension B. Repeat Step 37., if necessary.

41. Secure nut (29) to lower cone seat (28) and nut (35) to clevis (32) with lockwire (C-405).

42. Install weight (44) to arms (43); secure with bolt (41), washers (42) and nut (45). Bolt head must be installed in direction of rotation.

NOTE

It may be necessary to raise blade slightly to allow installation of bolt (38) through bushing (34).

43. Install bushing (34) inside cam (50).

44. Position arms (43) on clevis (36) and align arms with bolt holes on clevis.

45. Position clevis (36) to align bolt holes in flange of clevis with bushing (34) in cam (50).

NOTE

Bolt head of bolts (38) should be installed in direction of rotation.

46. Install bolt (38) with washer (37) under head of bolt through arm (43), clevis (36), bushing (34) in cam (50), and through arm (43).

47. Install washers (37) and nuts (46) on bolt (38).

48. Install bolts (39) with two washers (40) under each head of bolts through arm (43) into clevis (36).

49. Torque nut (46) 37 to 42 foot-pounds (50 to 57 Nm). Install cotter pin (47).

50. Torque bolts (39) 25 to 35 inch-pounds (2.82 to 3.95 Nm). Secure with lockwire (C-405) to arm (43).

51. Torque nut (45) 50 to 70 inch-pounds (5.65 to 7.91 Nm).

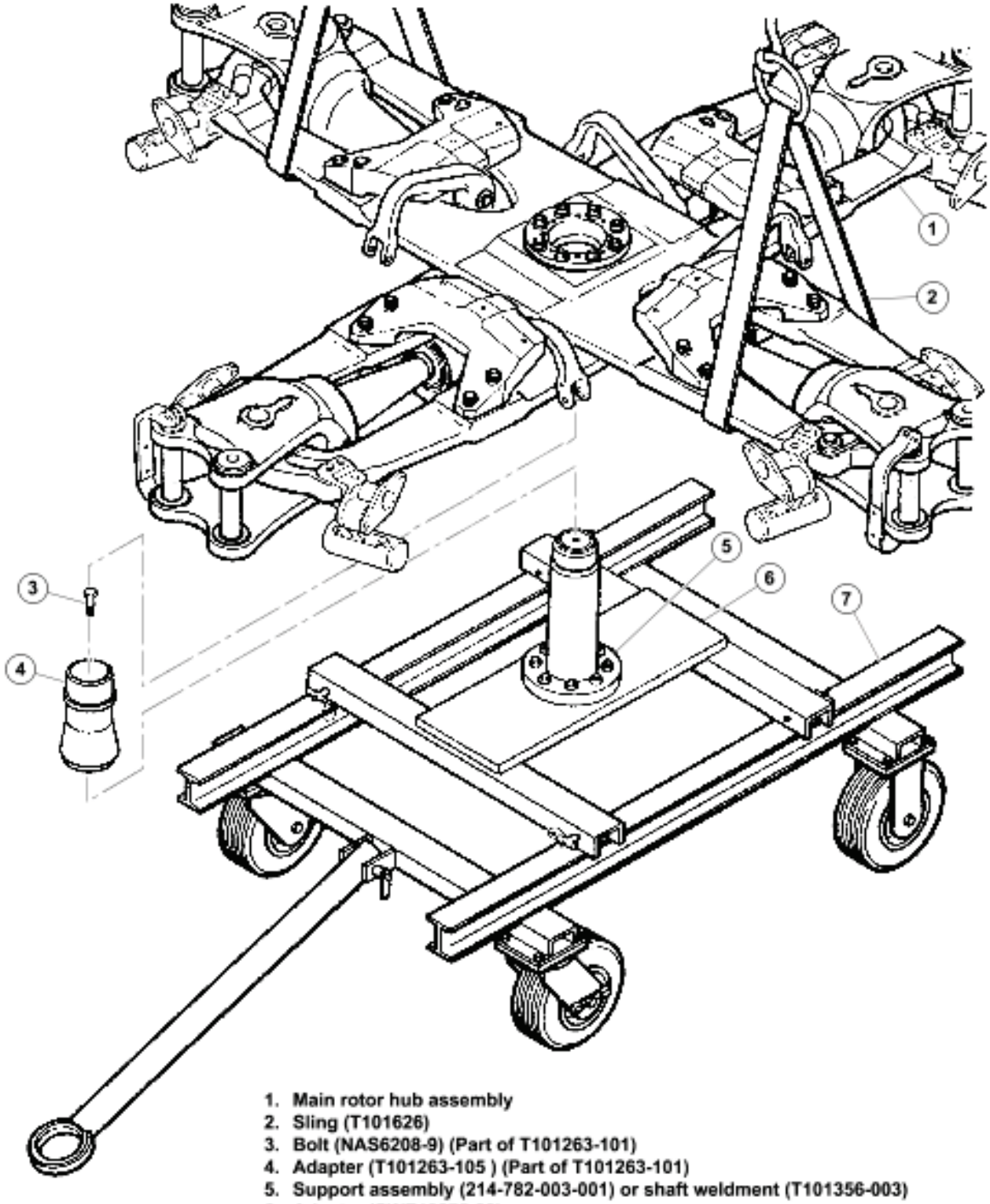
52. Manually raise blade and allow droop restraint arm to fall into lower detent of cam window.

53. Repeat Step 33. through Step 52. for remaining droop restraint arms.

NOTE

During first rotor run up following installation of main rotor droop restraint, have an observer visually verify all droop restraint arms disengage prior to RPM reaching idle. Normal disengagement occurs at 80 to 100 RPM (25 to 30% N_R). Arm disengagement may be observed by using a fixed reference point (cowling surface, etc.) to gage droop restraint arm position as RPM increases, or a strobe light of sufficient intensity may be used. A noticeable vibration will be detectable at idle if arm disengagement does not occur during application of collective with neutral cyclic. If vibration occurs, shut down helicopter and verify droop restraint arm rigging by repeating Step 33. through Step 52..

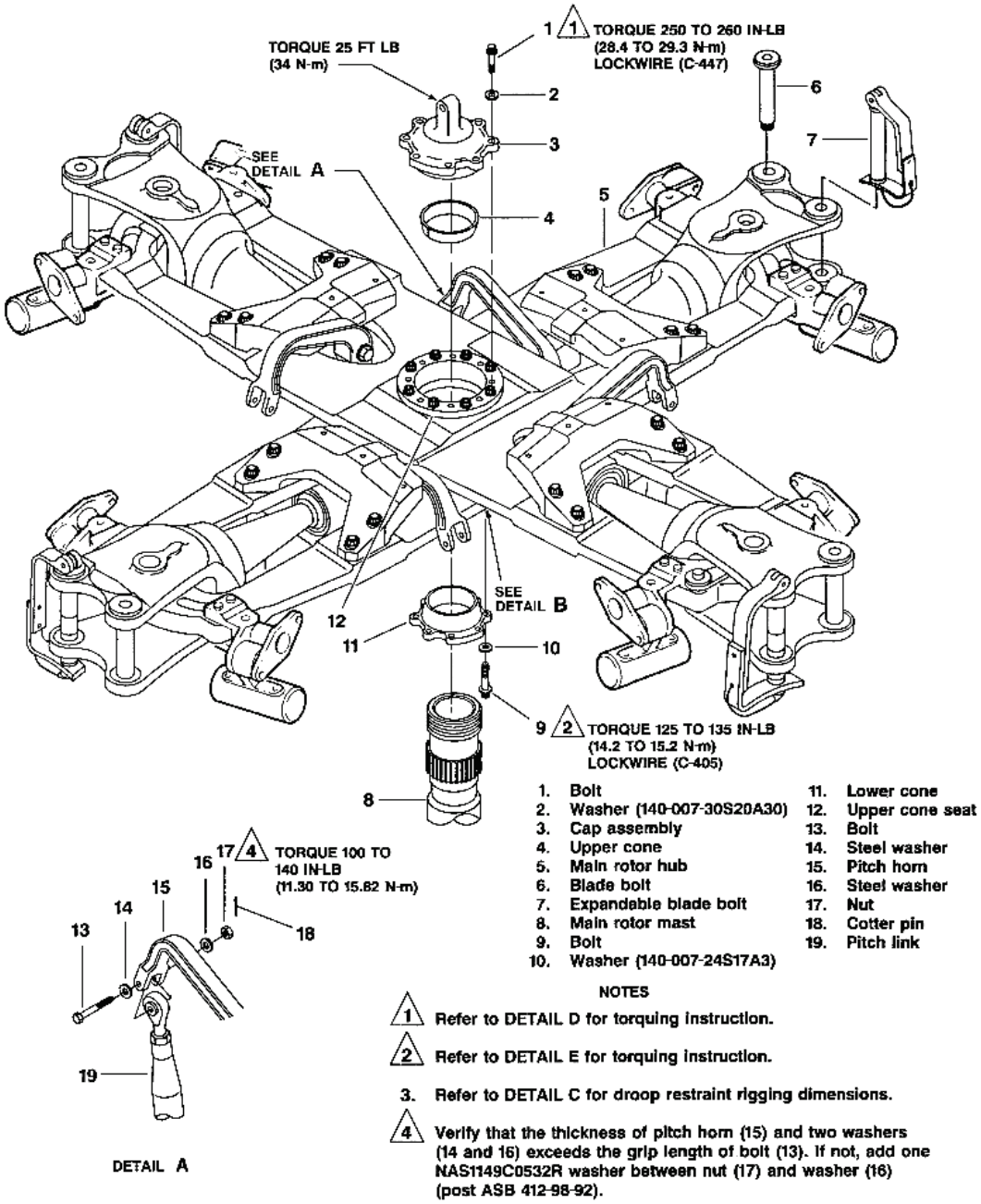
54. Make sure main rotor rigging is within limits (412-A-62-20-00-02A-280A-A / 00950).



1. Main rotor hub assembly
2. Sling (T101626)
3. Bolt (NAS6208-9) (Part of T101263-101)
4. Adapter (T101263-105) (Part of T101263-101)
5. Support assembly (214-782-003-001) or shaft weldment (T101356-003)
6. Adapter (SWE13875-30)
7. Maintenance transportation stand (SWE13875)

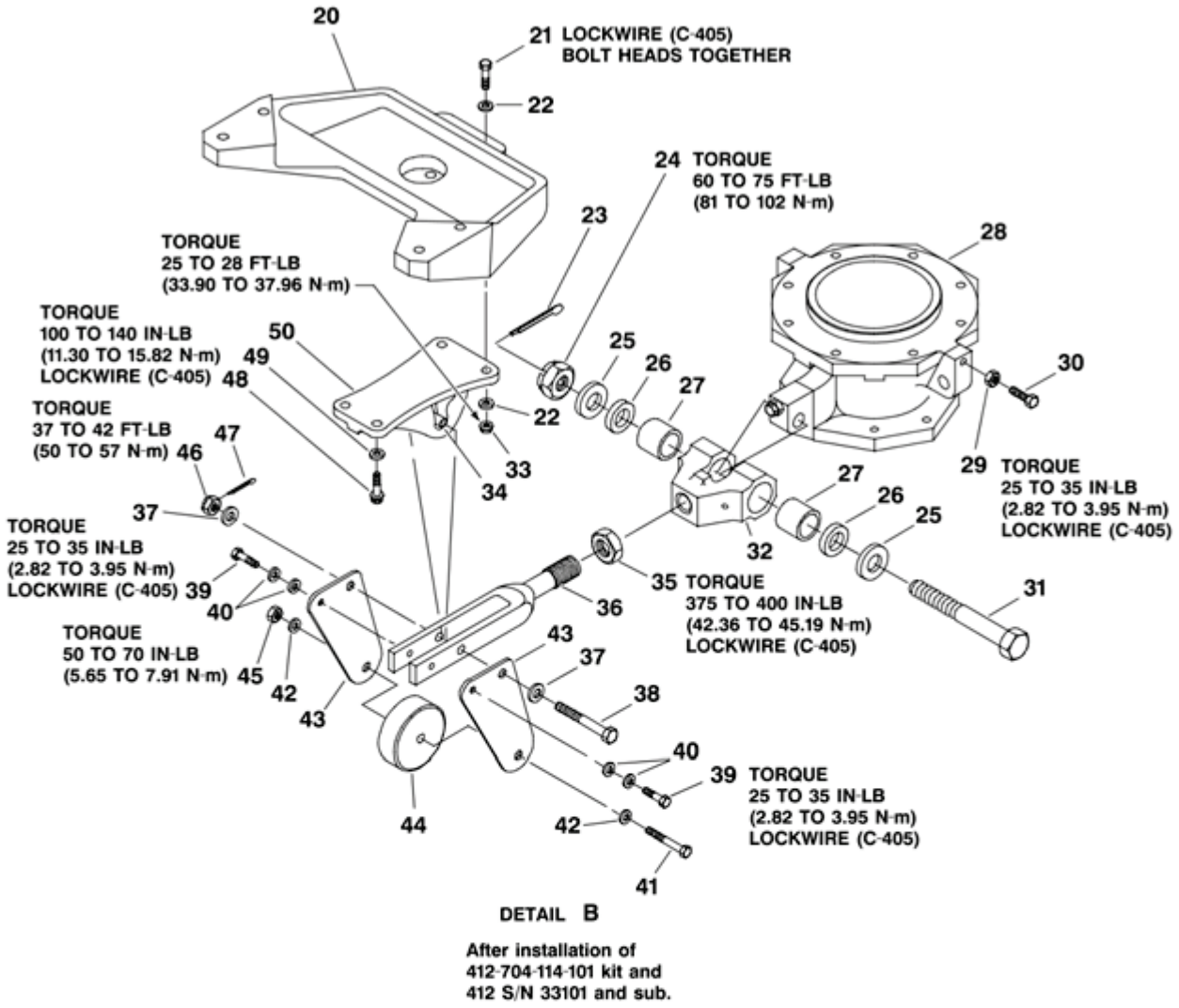
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Figure 1. Main Rotor Hub Stand - Tool Application (Sheet 1 of 1)



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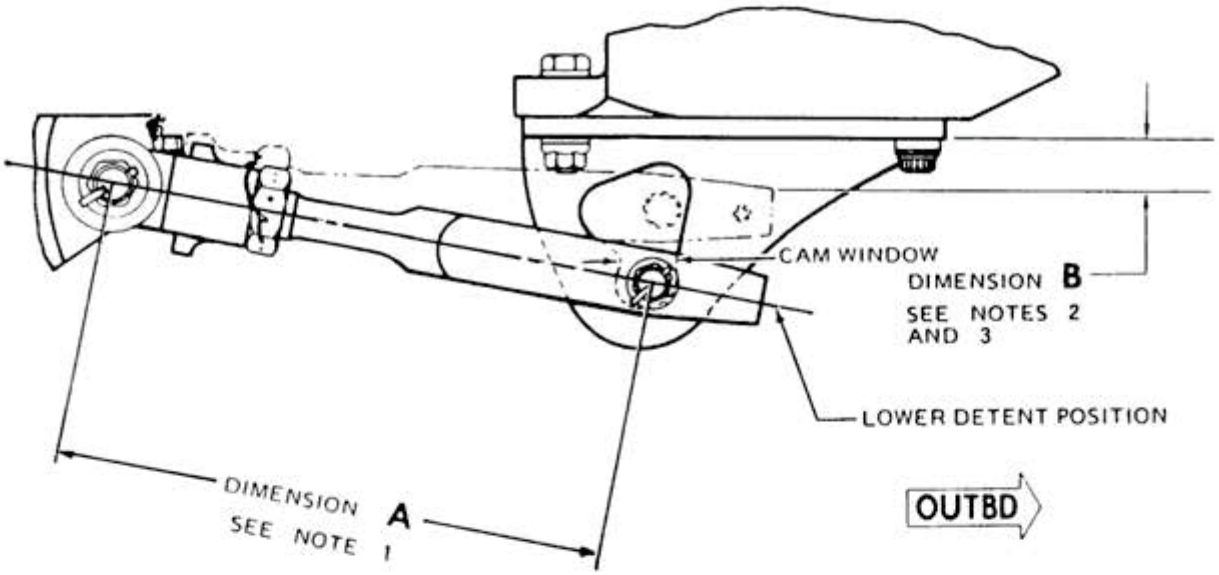
Figure 2. Main Rotor Hub (Sheet 1 of 6)



20. Damper bridge
21. Bolt
22. Washer (AN960-616)
23. Cotter pin
24. Nut
25. Washer (412-018-064-101)
26. Washer (140-00733S40C4)
27. Race
28. Lower cone seat
29. Nut
30. Bolt
31. Bolt
32. Clevis
33. Nut
34. Bushing (412-018-062-109)
35. Nut
36. Clevis
37. Washer (AN960-716)
38. Bolt
39. Bolt
40. Washer (MS9482-10)
41. Bolt
42. Washer (AN960-516)
43. Arm
44. Weight
45. Nut
46. Nut
47. Cotter pin
48. Bolt
49. Washer (MS9482-11)
50. Cam

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Figure 2. Main Rotor Hub (Sheet 2 of 6)



FLEXURE SHOWN IN STATIC DROOP POSITION WITH DROOP RESTRAINT ENGAGED FOUR PLACES (ARM AND WEIGHT REMOVED FOR CLARITY)

DETAIL C

NOTES

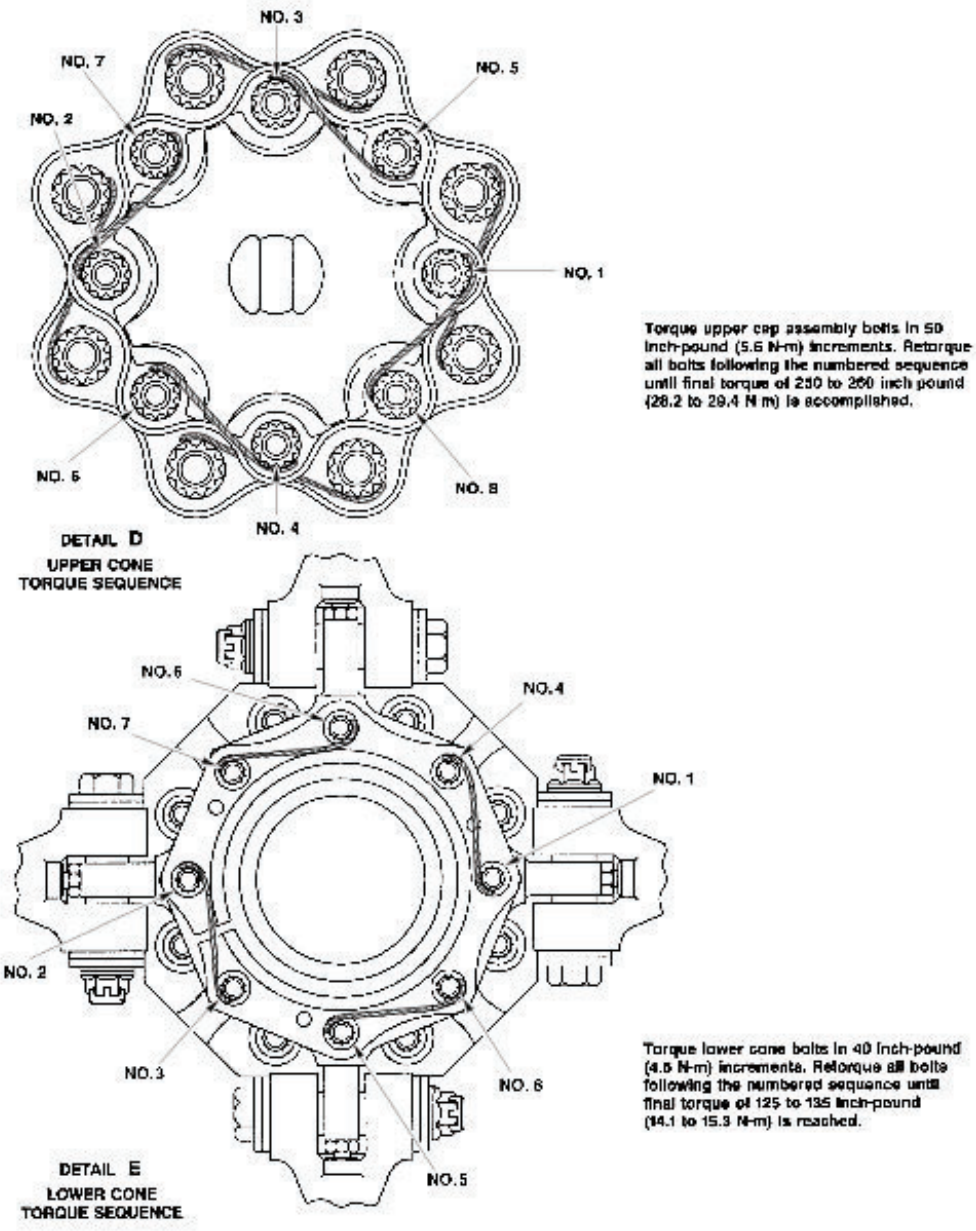
1. **Droop restraint arm length (dimension A) to be adjusted with weight, brackets and associated hardware removed per rigging procedures.**

2. **Dimension B is to be set per rigging instructions following setting of arm length (dimension A) and prior to reinstallation of weight, brackets and associated hardware. For each restraint arm, dimension B is to be measured on both leading and trailing edge clevis tangs and measured difference is to be 0.030 inch (0.762 mm) maximum.**

3. **Pivot stop rigging dimension to be set after adjusting (dimension A) per rigging procedures.**

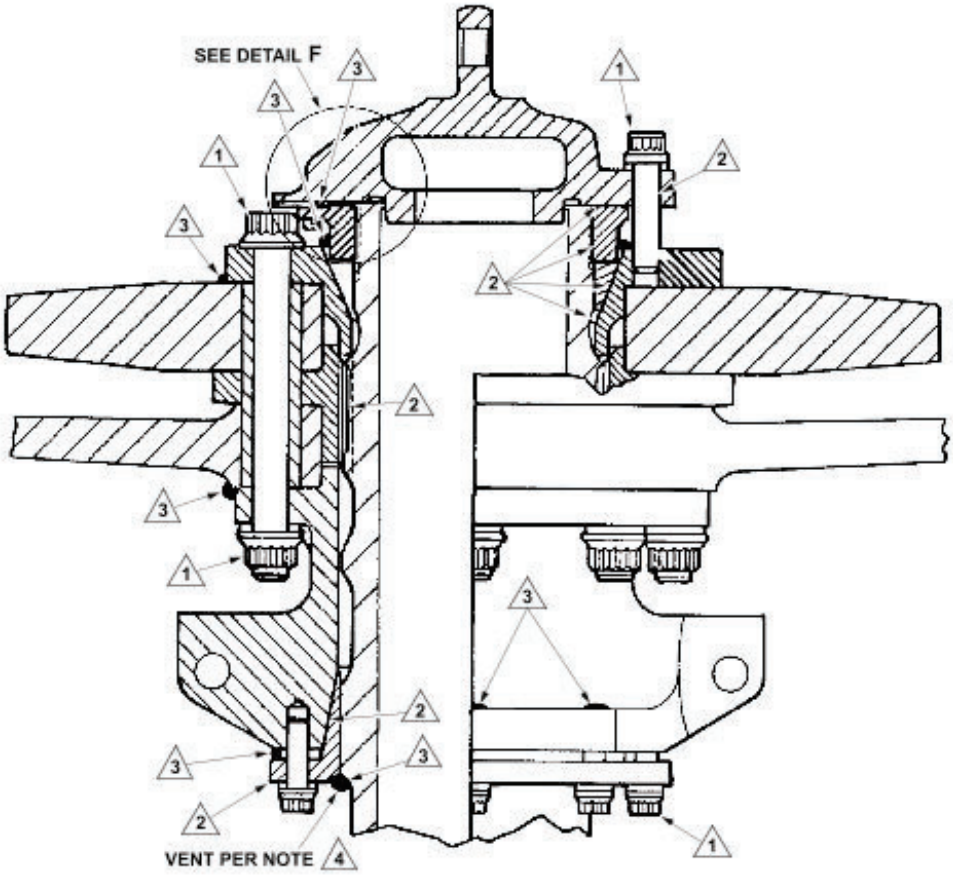
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Figure 2. Main Rotor Hub (Sheet 3 of 6)



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Figure 2. Main Rotor Hub (Sheet 4 of 6)

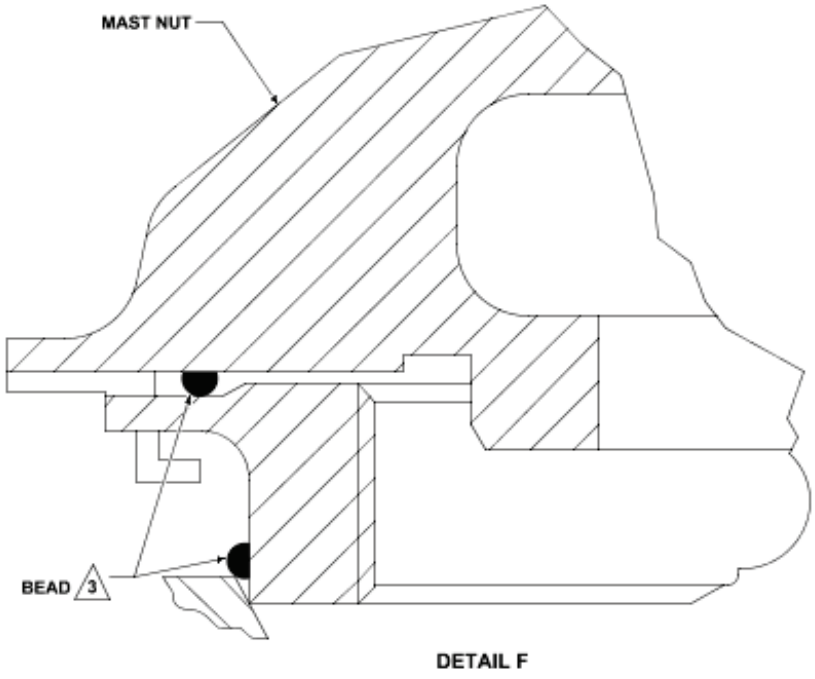


NOTES

1. Apply corrosion preventive compound (C-101).
2. Apply Thixogrease (C-561) on faying surfaces.
3. Seal all around using sealant (C-158).
4. Small passageway must remain at lower cone split to vent space between mast and hub.

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Figure 2. Main Rotor Hub (Sheet 5 of 6)



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Figure 2. Main Rotor Hub (Sheet 6 of 6)

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Main Rotor Controls - Description

The two major assemblies of main rotor controls are hub and sleeve assembly and swashplate and support assembly. The four pitch links, four rephasing levers, one drive link, three swashplate links and collective lever, are also included in main rotor control installation. The function of main rotor controls is to transmit cyclic and collective input forces to the main rotor hub and blade assembly by means of the hub and sleeve, swashplate and support, and interconnecting links and tubes.

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Swashplate and Support Assembly - Description

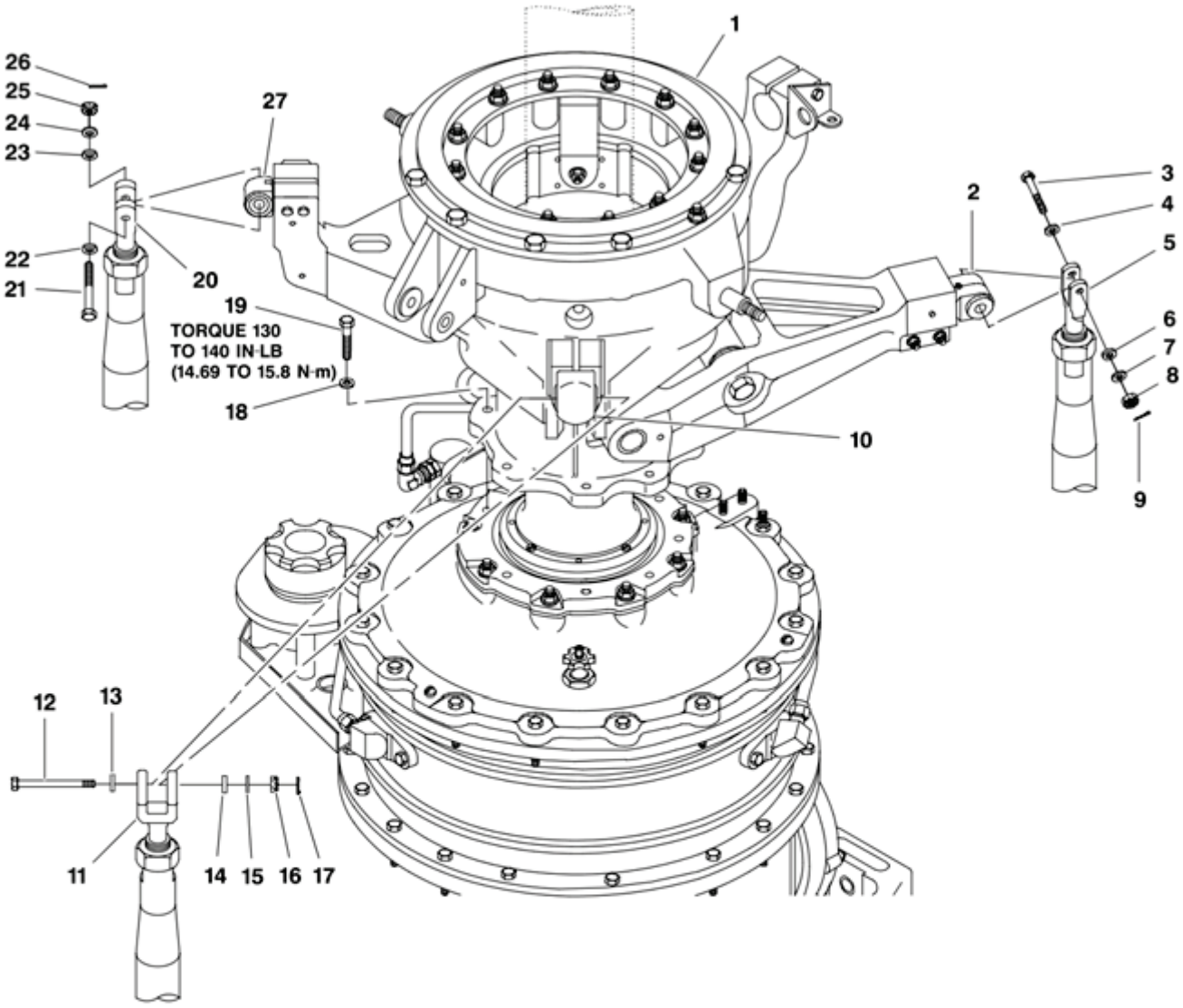
The swashplate and support assembly consists of the outer rotating ring mounted through a duplex bearing set on the inner non-rotating ring. The inner ring is connected to the swashplate support by a gimbal ring assembly. Servo actuator control tubes connect to two trunnions on the inner ring assembly to tilt the ring in the desired direction. The outer ring assembly tilts with the inner ring assembly, but rotates independently with the rotating controls.

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Swashplate and Support Assembly - Removal

Procedure

1. Remove main rotor hub and blade assemblies (412-A-62-10-00-00A-520A-A / 00935).
2. Remove hub and sleeve assembly (412-A-62-30-00-03A-520A-A / 00968).
3. Remove cotter pin (9, Figure 1), nut (8), washers (7 and 6), bolt (3) with washer (4) from clevis (5) and separate clevis (5) from trunnion (2).
4. Remove cotter pin (17) and remove nut (16) and washers (14 and 15) from bolt (12).
5. Remove bolt (12) and washer (13) from clevis (11) and separate clevis (11) from trunnion (10).
6. Remove cotter pin (26) and remove nut (25) and washers (24 and 23) from bolt (21).
7. Remove bolt (21) and washer (22) from clevis (20) and separate clevis (20) from trunnion (27).
8. Remove lockwire from bolts (19) and remove bolts (19) and washers (18).
9. Carefully remove swashplate and support assembly (1) from transmission.



1. Swashplate and support
2. Trunnion
3. Bolt
4. Washer (thick)
5. Clevis
6. Washer (thick)
7. Washer (thin)
8. Nut
9. Cotter pin
10. Trunion
11. Clevis
12. Bolt
13. Washer (thick)
14. Washer (thick)
15. Washer (thin)
16. Nut
17. Cotter pin
18. Washer (thick)
19. Bolt
20. Clevis
21. Bolt
22. Washer (thick)
23. Washer (thick)
24. Washer (thin)
25. Nut
26. Cotter pin
27. Trunnion

ICN-412-A-623000-B-97499-00023-A-001-01

Figure 1. Swashplate and Support (Sheet 1 of 1)

Swashplate and Support Assembly - Cleaning

Preliminary requirements

Consumables, materials and expendables

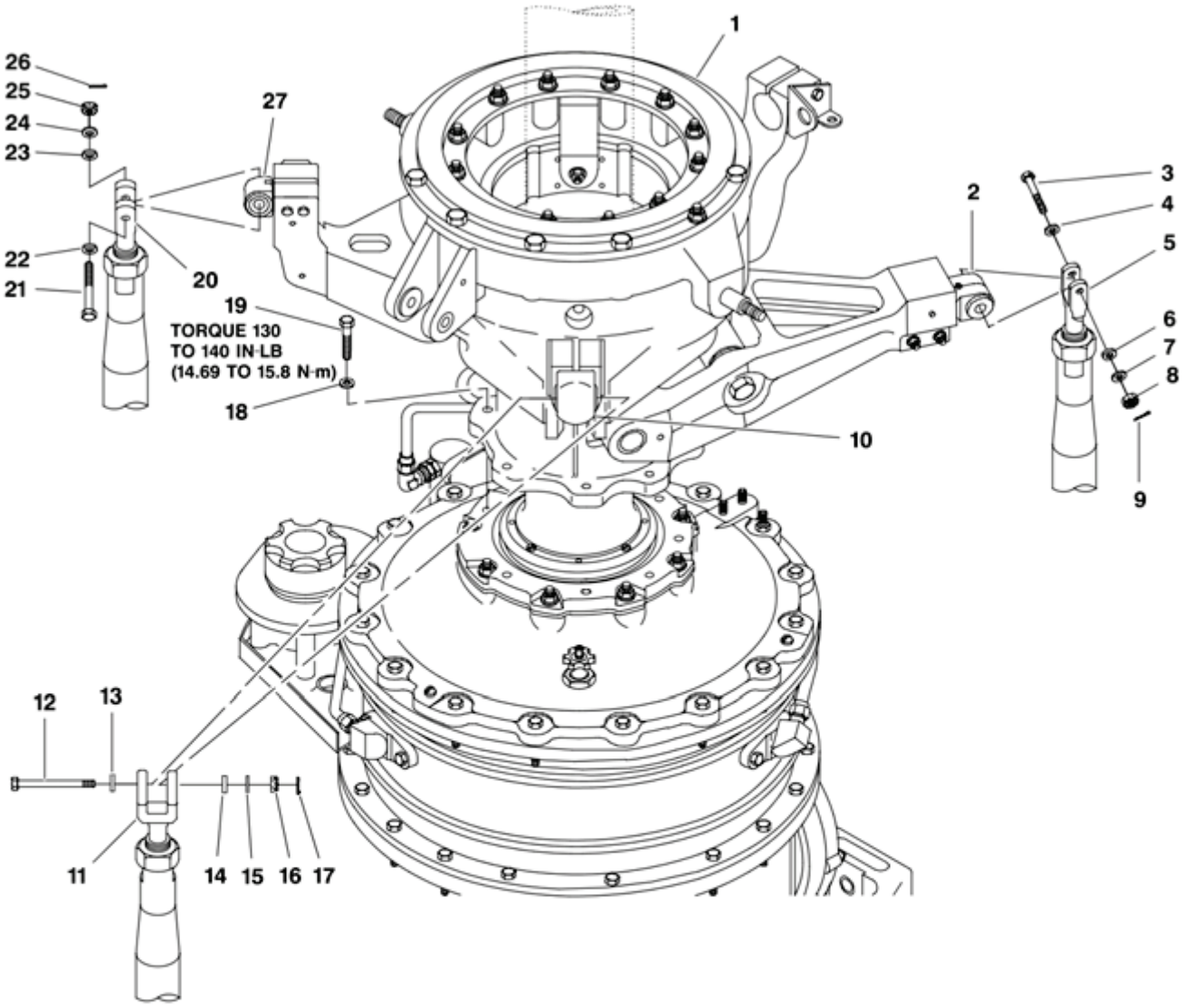
Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

NOTE

Do not allow solvent to saturate trunnion bearings, gimbal ring bearings or thrust bearing in swashplate inner ring.

1. Clean swashplate and support assembly (1, Figure 1) with clean cloth damp with drycleaning solvent (C-304).



1. Swashplate and support
2. Trunnion
3. Bolt
4. Washer (thick)
5. Clevis
6. Washer (thick)
7. Washer (thin)
8. Nut
9. Cotter pin
10. Trunion
11. Clevis
12. Bolt
13. Washer (thick)
14. Washer (thick)
15. Washer (thin)
16. Nut
17. Cotter pin
18. Washer (thick)
19. Bolt
20. Clevis
21. Bolt
22. Washer (thick)
23. Washer (thick)
24. Washer (thin)
25. Nut
26. Cotter pin
27. Trunnion

ICN-412-A-623000-B-97499-00023-A-001-01

Figure 1. Swashplate and Support (Sheet 1 of 1)

Swashplate and Support Assembly - Inspection

Procedure

NOTE

If records or physical appearance indicate swashplate and support assembly has been involved in an accident or incident, refer to 412-A-05-50-00-00A-284A-A / 00063 for inspection requirements.

1. Refer to 412-CR&O, Chapter 62 for instructions to inspect swashplate and support assembly.

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Swashplate and Support Assembly - Repair

Procedure

1. Refer to BHT-412-CR&O for repair limits and procedures.

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Swashplate and Support Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Corrosion preventive compound	C-104	AR	
Lockwire	C-405	AR	

Procedure

NOTE

A thin coat of corrosion preventive compound (C-104) should be applied to shanks and under bolt heads of all rotating control bolts prior to installation.

A thin coat of corrosion preventive compound (C-101) should be applied to all rotating control nuts, washers, and bolt heads after installation.

1. Carefully lower swashplate and support assembly (1, Figure 1) over mast and position on mast bearing plate on top of transmission.

2. Apply a thin coat of corrosion preventive compound (C-104) under the heads and to shanks of bolts (19).

3. Install bolts (19) with washers (18) under bolt heads. Torque bolts (19) 130 to 140 inch-pounds (14.69 to 15.82 Nm).

4. Secure bolts (19) in pairs using lockwire (C-405).

5. Apply a thin coat of corrosion preventive compound (C-101) to the heads of bolts (19) and washers (18).

6. Install hub and sleeve assembly (412-A-62-30-00-03A-720A-A / 00972).

7. Position clevis (5) on trunnion (2).

8. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (3).

9. Install bolt (3) with washer (4) under bolt head.

10. Install nut (8) with washers (6 and 7) as required to attain proper thread engagement for cotter pin (9). Install cotter pin (9).

11. Apply a thin coat of corrosion preventive compound (C-101) to the head of bolt (3), washer (4), nut (8) with washers (6 and 7), and cotter pin (9).

12. Position clevis (11) on trunnion (10).

13. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (13).

14. Install bolt (12) with washer (13) under bolt head.

15. Install nut (16) with washers (14 and 15) as required to attain proper thread engagement for cotter pin (17). Install cotter pin (17).

16. Apply a thin coat of corrosion preventive compound (C-101) to the head of bolt (12), washer (13), nut (16) with washers (14 and 15), and cotter pin (17).

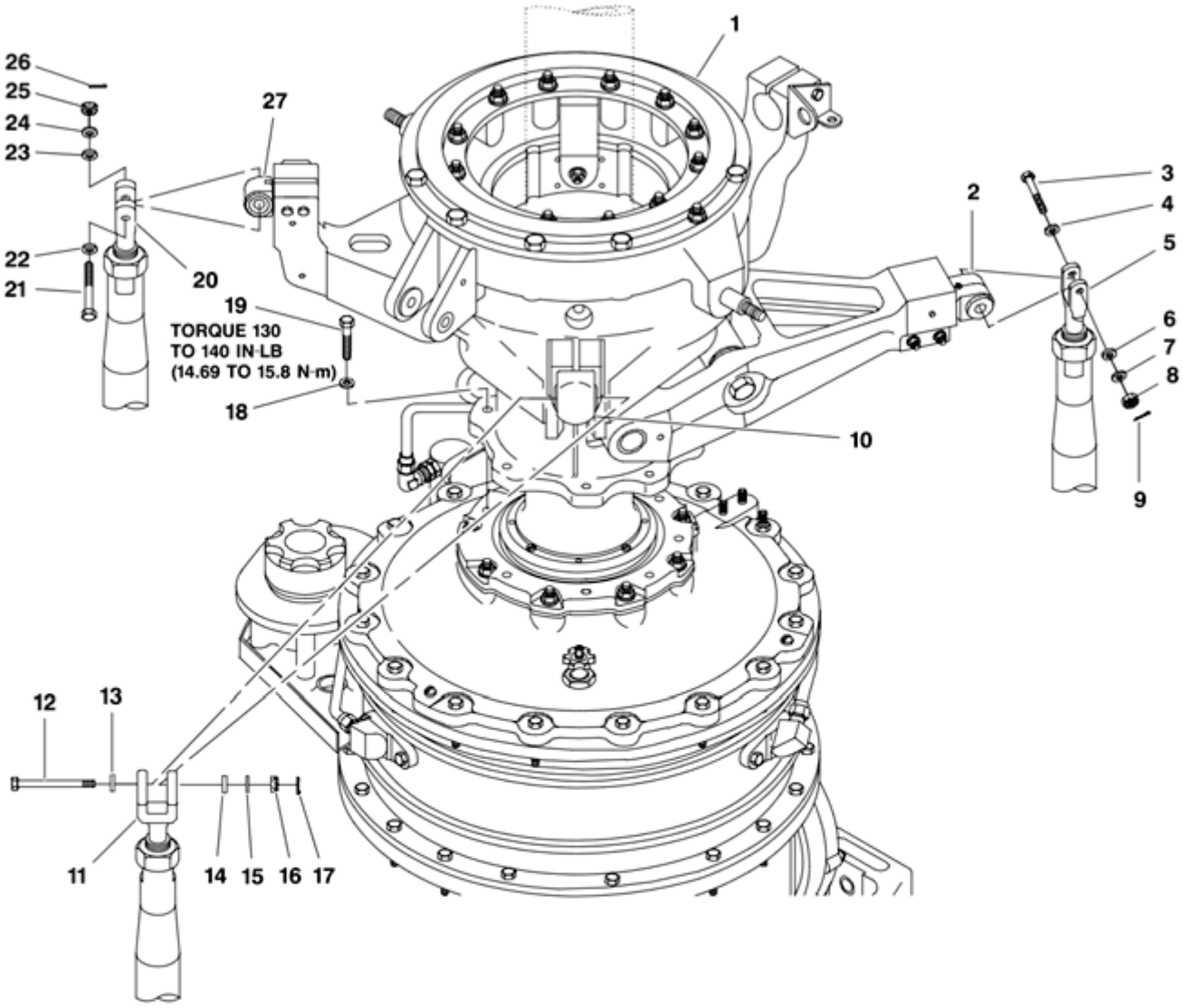
17. Position clevis (20) on trunnion (27).

18. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (21).

19. Install bolt (21) with washer (22) under bolt head.

20. Install nut (25) with washers (23 and 24) as required to attain proper thread engagement for cotter pin (26). Install cotter pin (26).

21. Apply a thin coat of corrosion preventive compound (C-101) to the head of bolt (21), washer (22), nuts (25) with washers (23) and (24), and cotter pin (26).



1. Swashplate and support
2. Trunnion
3. Bolt
4. Washer (thick)
5. Clevis
6. Washer (thick)
7. Washer (thin)
8. Nut
9. Cotter pin
10. Trunion
11. Clevis
12. Bolt
13. Washer (thick)
14. Washer (thick)
15. Washer (thin)
16. Nut
17. Cotter pin
18. Washer (thick)
19. Bolt
20. Clevis
21. Bolt
22. Washer (thick)
23. Washer (thick)
24. Washer (thin)
25. Nut
26. Cotter pin
27. Trunnion

ICN-412-A-623000-B-97499-00023-A-001-01

Figure 1. Swashplate and Support (Sheet 1 of 1)

Main Rotor Pitch Link Assemblies - Description

The pitch links are composed of a tube, two swaged inserts, two adjustable rod end bearings with lock nuts, a nameplate and decal. The pitch link assemblies connect the semi-circular pitch horn on the blade spindle to the rephasing levers on hub and sleeve.

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Main Rotor Pitch Link Assemblies - Removal

Procedure

1. Removal of pitch link with elastomeric lower rod end bearing.

1.1. With boost off, center cyclic and pull collective up as high as possible.

1.2. Remove lockwire from bolt (23, Figure 1) and remove bolt with washer (24) from rephasing lever (25).

1.3. Remove cotter pin (22), nut (21), washer (20), and bushing (19) from bolt (17).

1.4. Remove bolt (17) with washer (18) and bushing (19) from rephrasing lever (25).

1.5. Remove cotter pin (22), nut (10), and washer (9) from bolt (6).

1.6. Remove bolt (6) with washer (7) and separate pitch link (13) from pitch horn (5) and rephasing lever (25).

1.7. Repeat Step 1.1. through Step 1.4. to remove remaining pitch links.

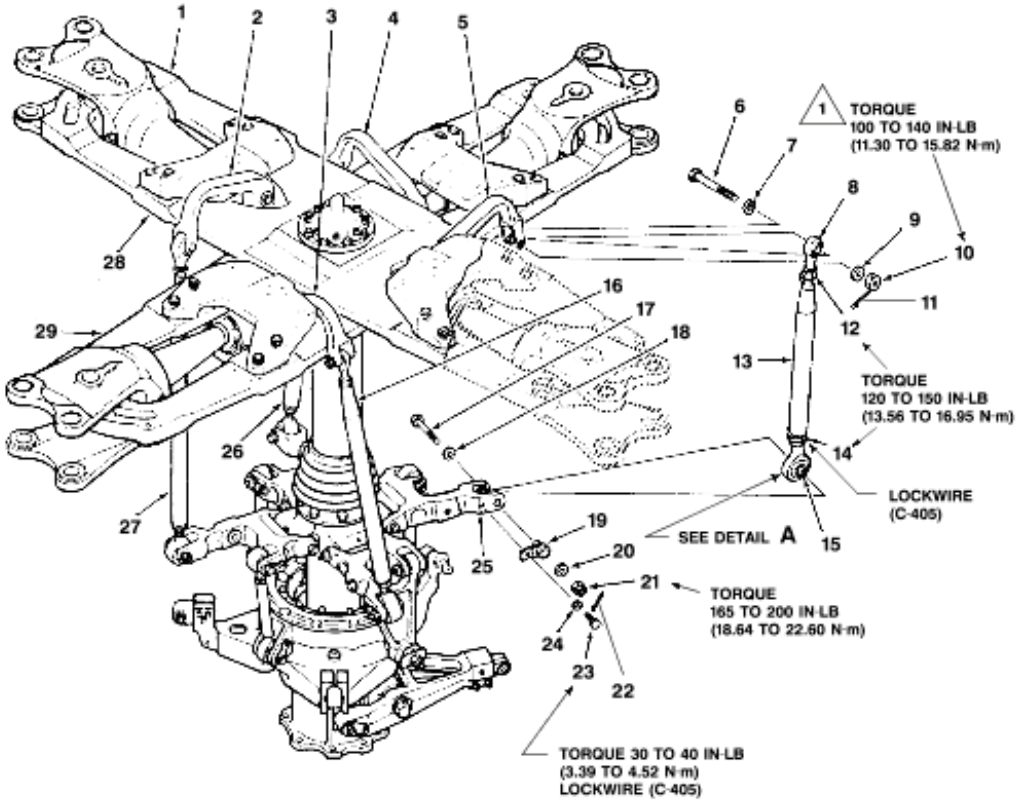
2. Removal of pitch link with uniball Teflon lined bearing.

2.1. With boost off, center cyclic and pull collective up as high as possible.

2.2. Remove cotter pin (30, detail B), nut (31), bolt (17), washer (34), long bushing (32), bracket (33), and short bushing (35) from rephasing lever (25).

2.3. Remove cotter pin (11), nut (10), washers (9 and 7), bolt (6), and separate pitch link (13) from pitch horn (5) and rephasing lever (25).

2.4. Repeat Step 2.1. through Step 2.3. to remove the remaining pitch links.



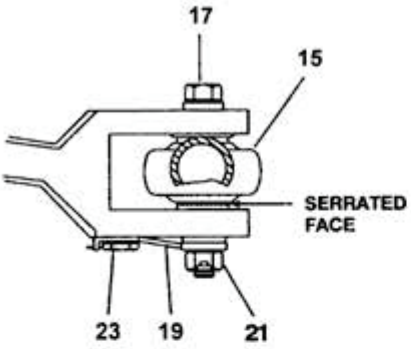
1. Main rotor hub
2. Pitch horn (upper yoke)
3. Pitch horn (lower yoke)
4. Pitch horn (lower yoke)
5. Pitch horn (upper yoke)
6. Bolt
7. Steel washer
8. Rod end bearing
9. Steel washer
10. Nut
11. Cotter pin
12. Nut
13. Pitch link
14. Nut
15. Rod end bearing
16. Pitch link
17. Bolt
18. Washer (140-007-25S-20C4)
19. Bushing
20. Thin steel washer
21. Nut
22. Cotter pin
23. Bolt
24. Steel washer
25. Rephasing lever
26. Pitch link
27. Pitch link
28. Upper yoke
29. Lower yoke

NOTE:

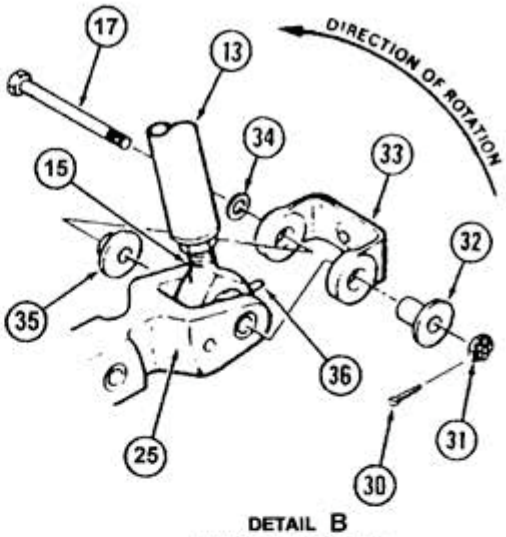
1 Verify that the thickness of pitch horn (5) and two washers (7 and 9) exceeds the grip length of the bolt (6). If not, add one NAS1149C0532R washer between nut (10) and washer (9) (post ASB 412-98-92).

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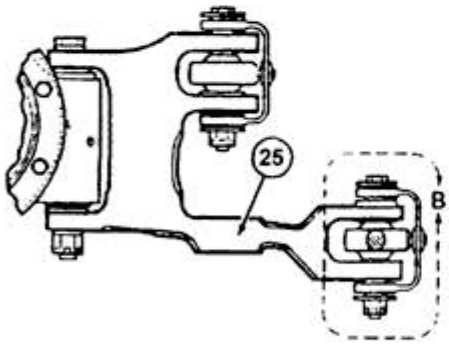
Figure 1. Main Rotor Pitch Link (Sheet 1 of 2)



DETAIL A
ELASTOMERIC ROD END BEARING



DETAIL B
UNI-BALL ROD END BEARING
(4 PLACES)



LOOKING DOWN ON REPHASING LEVER

30. Cotter pin
31. Nut
32. Long bushing
33. Bracket
34. Washer
35. Short bushing
36. Anti-rotation pin

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Figure 1. Main Rotor Pitch Link (Sheet 2 of 2)

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Main Rotor Pitch Link Assemblies - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

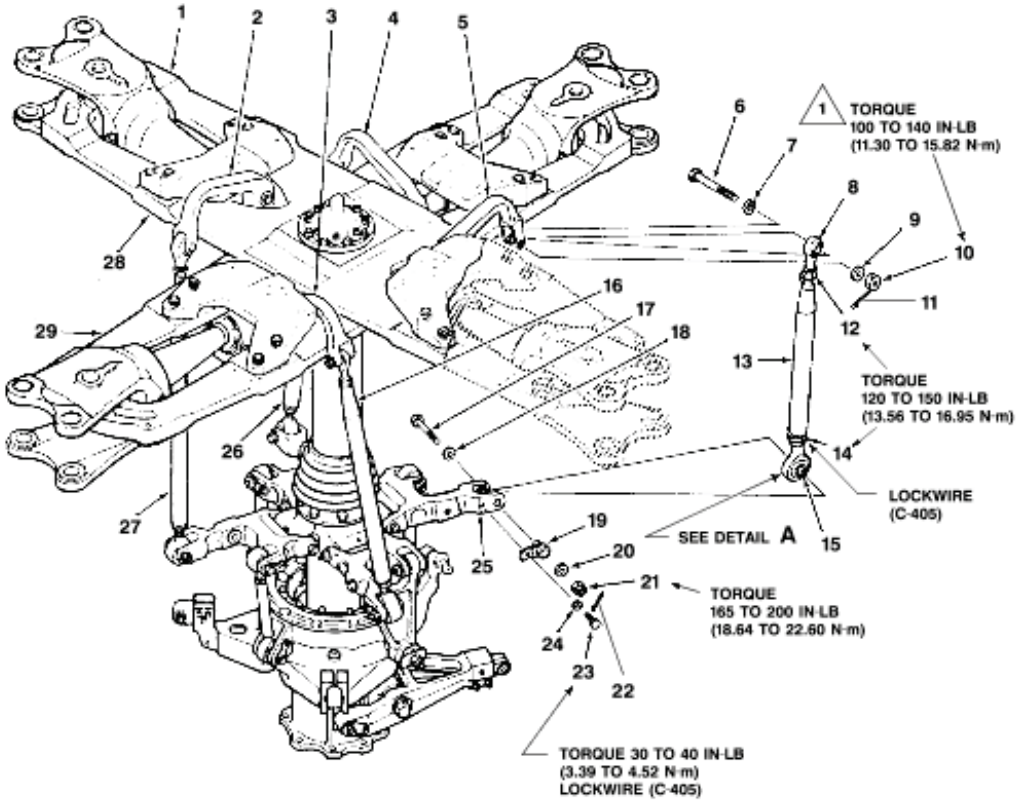
Procedure

NOTE

Do not immerse rod end bearing (8 and 15, Figure 1) in drycleaning solvent (C-304).

1. Wipe pitch link (13) with clean cloth damp with drycleaning solvent (C-304).

2. Dry with filtered, compressed air.



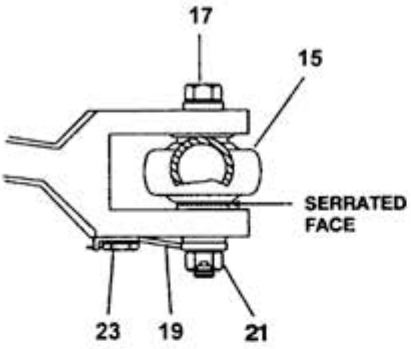
1. Main rotor hub
2. Pitch horn (upper yoke)
3. Pitch horn (lower yoke)
4. Pitch horn (lower yoke)
5. Pitch horn (upper yoke)
6. Bolt
7. Steel washer
8. Rod end bearing
9. Steel washer
10. Nut
11. Cotter pin
12. Nut
13. Pitch link
14. Nut
15. Rod end bearing
16. Pitch link
17. Bolt
18. Washer (140-007-25S-20C4)
19. Bushing
20. Thin steel washer
21. Nut
22. Cotter pin
23. Bolt
24. Steel washer
25. Rephasing lever
26. Pitch link
27. Pitch link
28. Upper yoke
29. Lower yoke

NOTE:

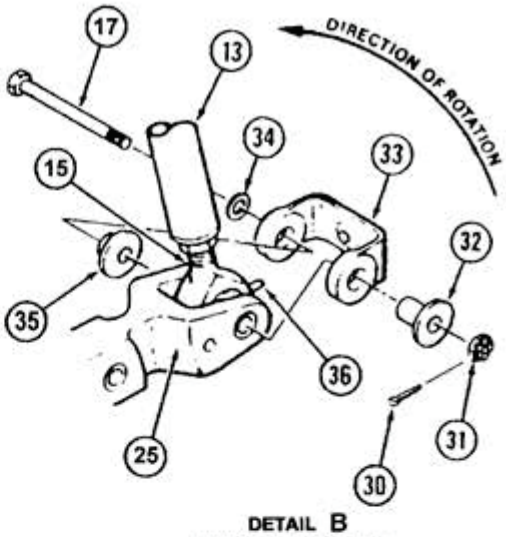
1 Verify that the thickness of pitch horn (5) and two washers (7 and 9) exceeds the grip length of the bolt (6). If not, add one NAS1149C0532R washer between nut (10) and washer (9) (post ASB 412-98-92).

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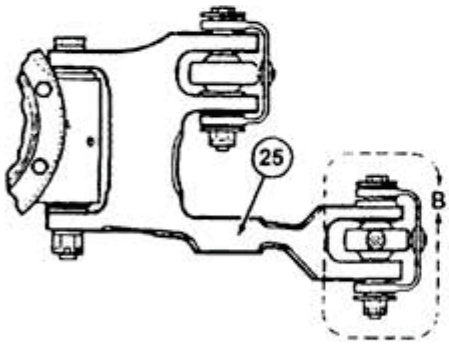
Figure 1. Main Rotor Pitch Link (Sheet 1 of 2)



DETAIL A
ELASTOMERIC ROD END BEARING



DETAIL B
UNI-BALL ROD END BEARING
(4 PLACES)



LOOKING DOWN ON REPHASING LEVER

30. Cotter pin
31. Nut
32. Long bushing
33. Bracket
34. Washer
35. Short bushing
36. Anti-rotation pin

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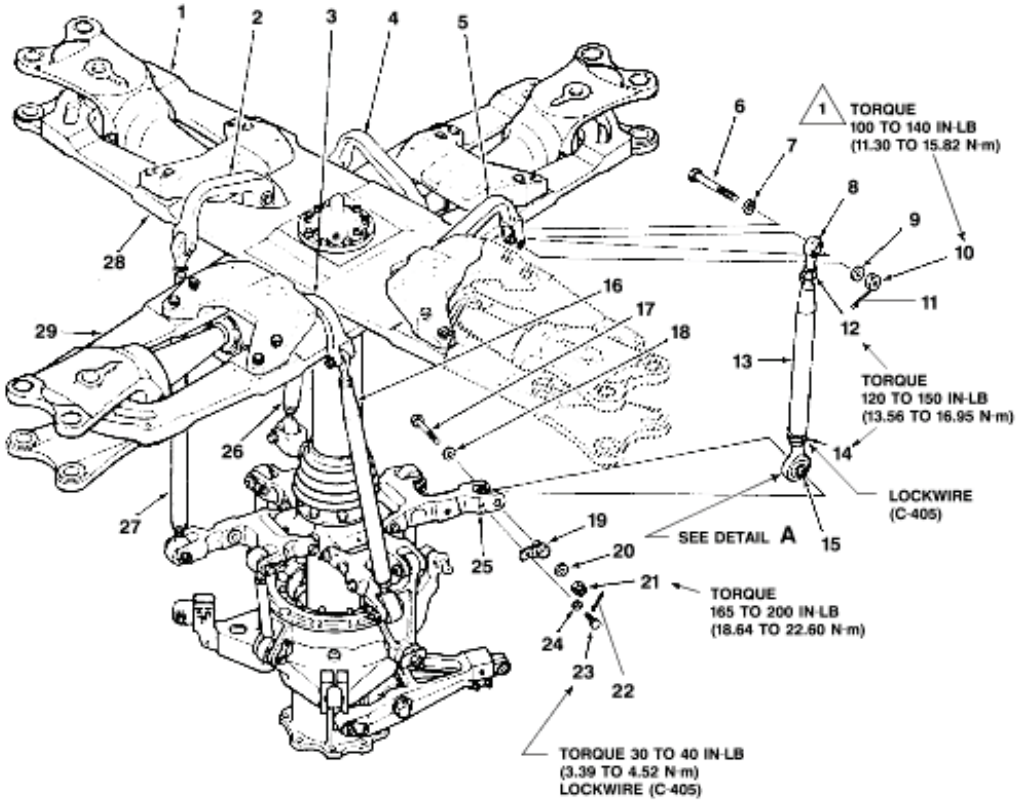
Figure 1. Main Rotor Pitch Link (Sheet 2 of 2)

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Main Rotor Pitch Link Assemblies - Inspection

Procedure

1. Inspect pitch link assemblies (13, 16, 26, and 27, Figure 1) for damage in accordance with BHT-412-CR&O.
2. Inspect bushings (19) for cracks and damage (BHT-412-CR&O).
3. Inspect bolts (6, 17, and 23) for thread damage distortion and corrosion.
4. Inspect rod end bearings (8 and 15) for wear (BHT-412-CR&O).



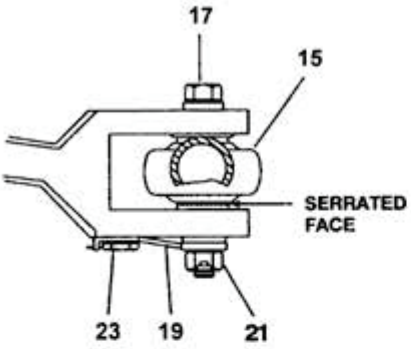
1. Main rotor hub
2. Pitch horn (upper yoke)
3. Pitch horn (lower yoke)
4. Pitch horn (lower yoke)
5. Pitch horn (upper yoke)
6. Bolt
7. Steel washer
8. Rod end bearing
9. Steel washer
10. Nut
11. Cotter pin
12. Nut
13. Pitch link
14. Nut
15. Rod end bearing
16. Pitch link
17. Bolt
18. Washer (140-007-25S-20C4)
19. Bushing
20. Thin steel washer
21. Nut
22. Cotter pin
23. Bolt
24. Steel washer
25. Rephasing lever
26. Pitch link
27. Pitch link
28. Upper yoke
29. Lower yoke

NOTE:

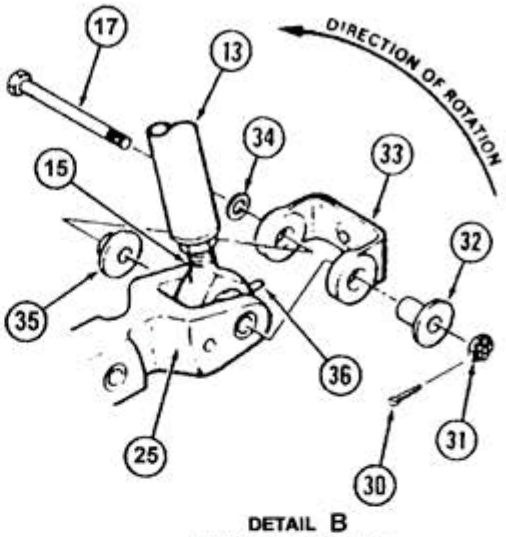
1 Verify that the thickness of pitch horn (5) and two washers (7 and 9) exceeds the grip length of the bolt (6). If not, add one NAS1149C0532R washer between nut (10) and washer (9) (post ASB 412-98-92).

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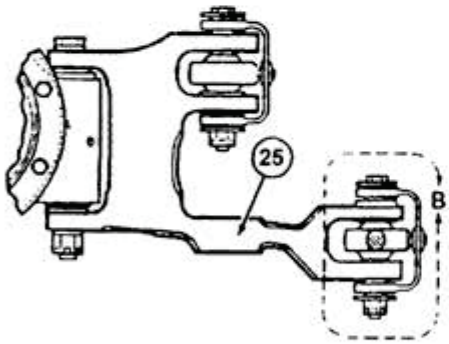
Figure 1. Main Rotor Pitch Link (Sheet 1 of 2)



DETAIL A
ELASTOMERIC ROD END BEARING



DETAIL B
UNI-BALL ROD END BEARING
(4 PLACES)



LOOKING DOWN ON REPHASING LEVER

30. Cotter pin
31. Nut
32. Long bushing
33. Bracket
34. Washer
35. Short bushing
36. Anti-rotation pin

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Figure 1. Main Rotor Pitch Link (Sheet 2 of 2)

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Main Rotor Pitch Link Assemblies - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Chemical film material	C-100	AR	
Corrosion preventive compound	C-104	AR	
Epoxy coating	C-210	AR	

Procedure

NOTE

Repair is restricted to polishing out damage and replacing rod end bearings, decal, and nameplates.

1. Polish out mechanical and corrosion damage within limits (BHT-412-CR&O).

2. Apply chemical film material (C-100) to all bare aluminum areas (BHT-ALL-SPM, Standard Practices Manual).

3. Apply epoxy coating (C-210) to areas where metal is exposed (BHT-ALL-SPM, Standard Practices Manual).

4. Touch up paint to match existing finish (BHT-ALL-SPM, Standard Practices Manual).

5. Replace rod end bearing (1 and 8, Figure 1) if rod end bearings are worn beyond limits (BHT-412-CR&O) as follows:

5.1. Loosen nut (2) and remove rod end bearing (1) from tube (9).

5.2. Remove nut (2) from rod end bearing (1).

5.3. Apply light coat of corrosion preventive compound (C-104) to threads of new rod end bearing.

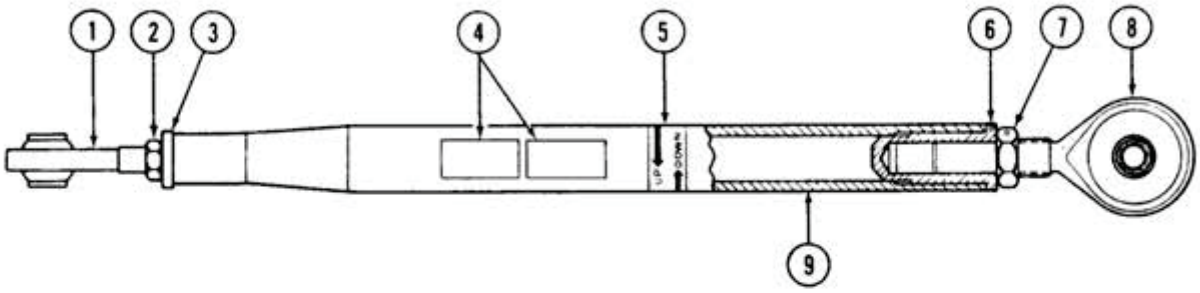
5.4. Install nut (2) on rod end bearing (1).

5.5. Install rod end bearing (1) on tube (9).

5.6. Repeat Step 5.1. through Step 5.5. to replace opposite rod end bearing (8).

5.7. Adjust pitch link to initial setting dimension (Figure 2).

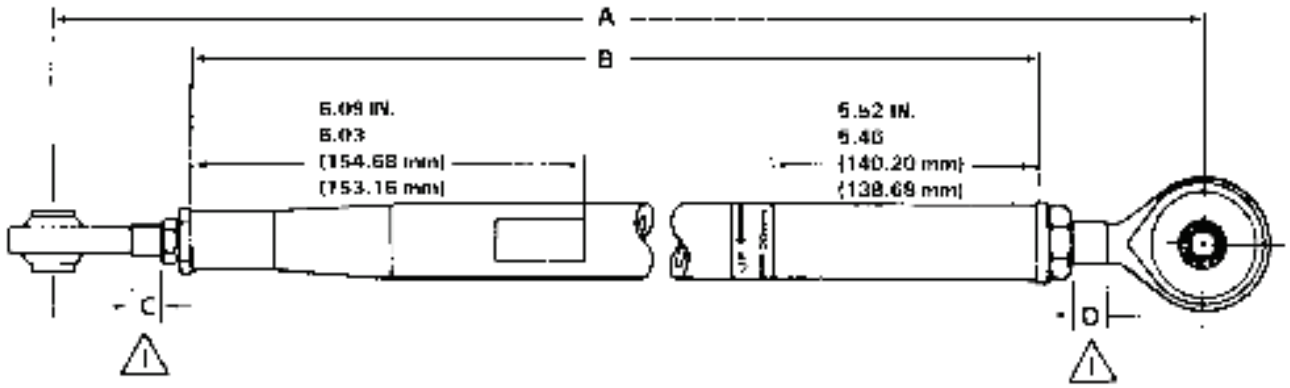
5.8. Tighten nut (2, Figure 1) enough to hold set dimensions. Do not torque nut at this time.



1. Rod end bearing
2. Nut
3. Insert
4. Name plate
5. Decal
6. Insert
7. Nut
8. Rod end bearing
9. Tube

ICN-412-A-623000-B-97499-00027-A-001-01

Figure 1. Main Rotor Pitch Link Assembly (Sheet 1 of 1)



PITCH LINK FOR	DIMENSIONS		DIMENSIONS METRIC	
	A	B	A	B
UPPER YOKE	20.770 INCH 20.750 INCH	16.64 INCH 16.61 INCH	527.558 mm 527.050 mm	422.65 mm 421.89 mm
LOWER YOKE	19.025 INCH 19.005 INCH	14.985 INCH 15.015 INCH	483.235 mm 482.727 mm	380.619 mm 381.381 mm

NOTE

1 **After initial setting: Exposed threads on rod end bearings (dimensions "C" and "D") must be equal within two threads.**

After installation and rigging: Maximum exposed threads shall not exceed eleven at dimensions "C" and eight at dimension "D".

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Figure 2. Main Rotor Pitch Link Assembly Initial Dimensions (Sheet 1 of 1)

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Main Rotor Pitch Link Assemblies - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Corrosion preventive compound	C-104	AR	
Lockwire	C-405	AR	

Procedure

NOTE

The two shorter pitch links (16 and 26, Figure 1) are installed on pitch horns (3 and 4) of lower yoke (29). The longer pitch links (13 and 27) are installed on pitch horns (2 and 5) and can be identified for installation in correct position by color code markings or pitch link tube dimensions (Figure 2).

A thin coat of corrosion preventive compound (C-104) should be applied to shanks and under bolt heads of all rotating control bolts prior to installation.

A thin coat of corrosion preventive compound (C-101) should be applied to all rotating control nuts, washers, and bolt heads after their installation.

1. If not previously accomplished, adjust pitch link to initial setting dimensions (Figure 2).

NOTE

Step 2.1. through Step 2.3. are required for pitch links with lower elastomeric rod ends only.

2. Select pitch link to be installed by color code or pitch link tube dimension. Refer to note preceding Step 1.. During installation of pitch links, ensure links are installed such that any elastomeric bearing has zero torsional preload at flat pitch when swashplate is level with respect to mast. This may be accomplished as follows:

2.1. Center cyclic to maintain swashplate level in respect to mast.

2.2. Lower collective lever until down stop is contacted.

2.3. Rotate rotor until lower link of control system in which link (swashplate link or pitch link) is being installed is directly forward of mast and over center line of helicopter (within ±10°). All links of control system shall be installed in this manner to ensure any cyclic input to swashplate will not be a significant factor.

3. To install pitch link with elastomeric bearing proceed as follows:

3.1. Install link so elastomeric bearing is within one serration of being parallel to mast.

NOTE

All bolts installed in rotating controls are to be installed with heads in direction of rotation.

3.2. Position upper end of pitch link (13, Figure 1) in pitch horn (5).

3.3. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (6).

3.4. Install bolt (6), washer (7 and 9), and nut (10).

NOTE

Verify that the thickness of the pitch horn (5) and two washers (7 and 9) exceeds the spindle length of the bolt (6). If not, add one NAS1149C0532R washer between nut (10) and washer (9).

3.5. Torque nut (10) 100 to 140 inch-pounds (11.30 to 15.82 Nm). Install cotter pin (11).

3.6. Apply a thin coat of corrosion preventive compound (C-101) to the head of bolt (6) and washer (7), washer (9), nut (10), and cotter pin (11).

3.7. Position rod end bearing (15) on rephasing lever (25) with serrated face of lower rod end bearing facing bushing (19) as shown on detail A.

3.8. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (17).

3.9. Install bolt (17) with washer (18) under bolt head.

3.10. Install bushing (19), washer (20) and nut (21). Do not torque nut at this time.

3.11. Ensure serrations of bushing (19) and rod end bearing (15) are properly engaged.

3.12. Install bolt (23) and washer (24), and torque bolt 30 to 40 inch-pounds (3.39 to 4.52 Nm).

3.13. Secure bolt (23) to bushing (19) with lockwire (C-405).

3.14. Torque nut (21) 165 to 200 inch-pounds (18.64 to 22.60 Nm). Install cotter pin (22).

3.15. Apply a thin coat of corrosion preventive compound (C-101) to head of bolt (17), washer (18), washer (20), nut (21), cotter pin (22), washer (24), and bolt (23).

4. To install pitch link with uniball Teflon lined bearing.

NOTE

All bolts in rotating controls are to be installed with heads in direction of rotation.

4.1. Position upper end of pitch link (13, Figure 1) in pitch horn (5).

4.2. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (6).

4.3. Install bolt (6), washer (7 and 9), and nut (10).

4.4. Torque nut (10) 100 to 140 inch-pounds (11.30 to 15.82 Nm). Install cotter pin (11).

4.5. Apply a thin coat of corrosion preventive compound (C-101) to the head of bolt (6) and washer (7), washer (9), nut (10), and cotter pin (11).

4.6. Install short bushing (35) into inside flange of bracket (33). Detail B (pitch link lower end).

4.7. Position bracket (33) at end of rephasing lever (25) arm in alignment with rod end (15) and anti-rotation pin (36).

4.8. Install long bushing (32) into side of bracket (33) with head of bushing opposite direction of rotation as shown in detail B.

4.9. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (17).

4.10. Install washer (34) under head of bolt (17). Push bolt through bushings in bracket (33), rod end bearing (15), and rephasing lever (25).

4.11. Install nut (31).

4.12. Torque nut (31) 165 to 200 inch-pounds (18.64 to 22.60 Nm). Install cotter pin (30).

5. Repeat Step 3. or Step 4., as applicable, to install remaining pitch links.

6. Center rod end bearing (8) in pitch horn clevis (5).

7. Torque nut (12) 120 to 150 inch-pounds (13.56 to 16.95 Nm).

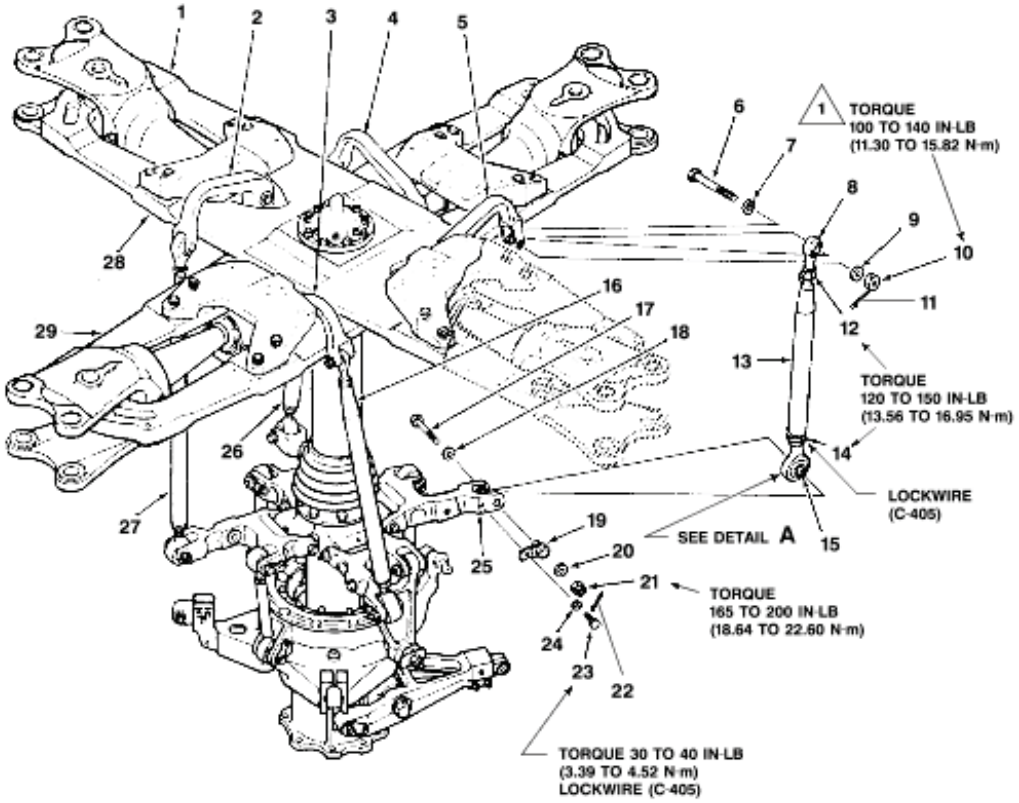
8. Secure nut (12) to insert (3, Figure 3) with lockwire (C-405).

9. Center rod end bearing (15, Figure 1) in rephasing lever clevis (25).

10. Torque nut (14) 120 to 150 inch-pounds (13.56 to 16.95 Nm).

11. Secure nut (14) to insert (6, Figure 3) with lockwire (C-405).

12. Repeat Step 1. through Step 10. to install remaining pitch links.



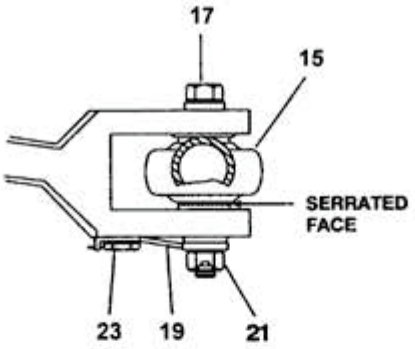
1. Main rotor hub
2. Pitch horn (upper yoke)
3. Pitch horn (lower yoke)
4. Pitch horn (lower yoke)
5. Pitch horn (upper yoke)
6. Bolt
7. Steel washer
8. Rod end bearing
9. Steel washer
10. Nut
11. Cotter pin
12. Nut
13. Pitch link
14. Nut
15. Rod end bearing
16. Pitch link
17. Bolt
18. Washer (140-007-25S-20C4)
19. Bushing
20. Thin steel washer
21. Nut
22. Cotter pin
23. Bolt
24. Steel washer
25. Rephasing lever
26. Pitch link
27. Pitch link
28. Upper yoke
29. Lower yoke

NOTE:

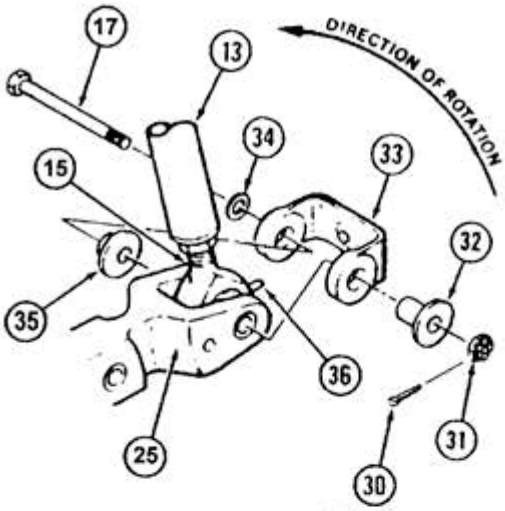
1 Verify that the thickness of pitch horn (5) and two washers (7 and 9) exceeds the grip length of the bolt (6). If not, add one NAS1149C0532R washer between nut (10) and washer (9) (post ASB 412-98-92).

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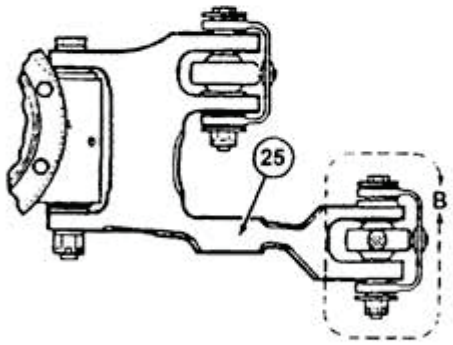
Figure 1. Main Rotor Pitch Link (Sheet 1 of 2)



DETAIL A
ELASTOMERIC ROD END BEARING



DETAIL B
UNI-BALL ROD END BEARING
(4 PLACES)

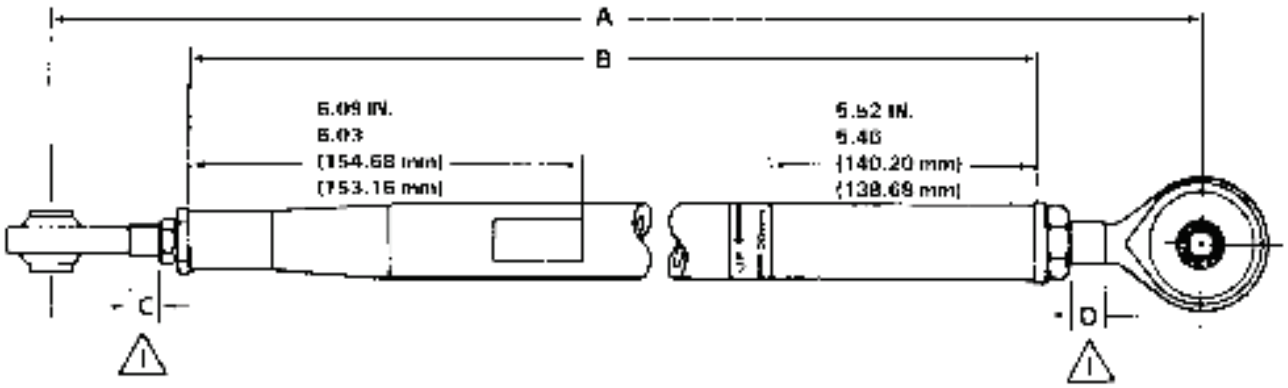


LOOKING DOWN ON REPHASING LEVER

30. Cotter pin
31. Nut
32. Long bushing
33. Bracket
34. Washer
35. Short bushing
36. Anti-rotation pin

ICN-412-A-623000-B-97499-00025-A-001-01

Figure 1. Main Rotor Pitch Link (Sheet 2 of 2)



PITCH LINK FOR	DIMENSIONS		DIMENSIONS METRIC	
	A	B	A	B
UPPER YOKE	20.770 INCH 20.750 INCH	16.64 INCH 16.61 INCH	527.558 mm 527.050 mm	422.65 mm 421.89 mm
LOWER YOKE	19.025 INCH 19.005 INCH	14.985 INCH 15.015 INCH	483.235 mm 482.727 mm	380.619 mm 381.381 mm

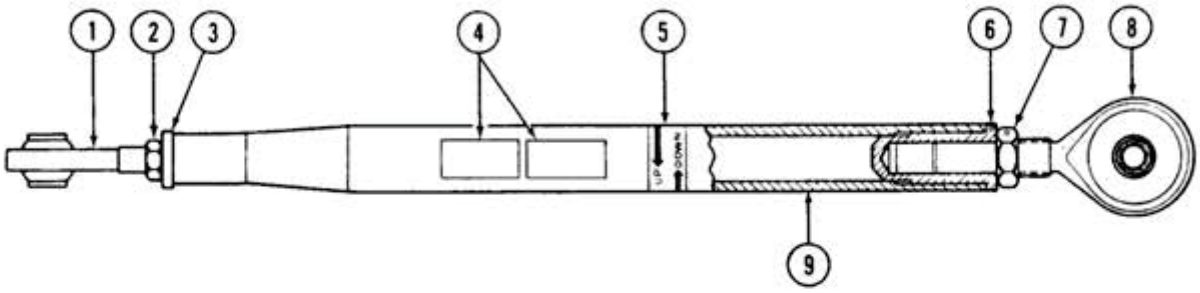
NOTE

1 **After initial setting: Exposed threads on rod end bearings (dimensions "C" and "D") must be equal within two threads.**

After installation and rigging: Maximum exposed threads shall not exceed eleven at dimensions "C" and eight at dimension "D".

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Figure 2. Main Rotor Pitch Link Assembly Initial Dimensions (Sheet 1 of 1)



1. Rod end bearing
2. Nut
3. Insert
4. Name plate
5. Decal
6. Insert
7. Nut
8. Rod end bearing
9. Tube

ICN-412-A-623000-B-97499-00027-A-001-01

Figure 3. Main Rotor Pitch Link Assembly (Sheet 1 of 1)

Hub and Sleeve Assembly - Description

Applicable: SN: 33001-33107:

The hub and sleeve assembly consists of four rephasing levers and a rotating hub splined to mast and mounted through ball bearings on upper end of non-rotating collective sleeve. The collective hub and sleeve operates between the swashplate and support assembly and mast, and is actuated by collective lever attached to lower end. The inboard ends of rephasing levers are connected to the hub. One rephasing lever is connected to the drive link and three rephasing levers are connected to swashplate links.

Applicable: SN: 33108-Subsequent or (SN: 33001-33107 TB 412-86-52: Post):

The rephasing levers are connected to two drive links and two swashplate links. The outboard ends of rephasing levers are connected to pitch horns by four pitch links which transmit collective and cyclic control motions to the main rotor hub and blade assembly.

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Teflon Lined Bearings (Installed) - Inspection

Procedure

1. Fabricate work aid as shown on Figure 1, Detail A.

NOTE

Inspection procedure is typical for all four pitch change links.

2. Hook work aid (2, Figure 2) on pitch horn (3) and pull down to relieve loading. Pitch change link loading occurs from main rotor elastomeric bearings.

3. Grasp pitch link (4) and move in a direction to permit a check for axial looseness of the pitch link rod end bearings (Teflon lined) (1 and 9). Maximum acceptable axial wear in pitch link rod end bearings (Teflon lined) (1 and 9) is 0.015 inch (0.38 mm).

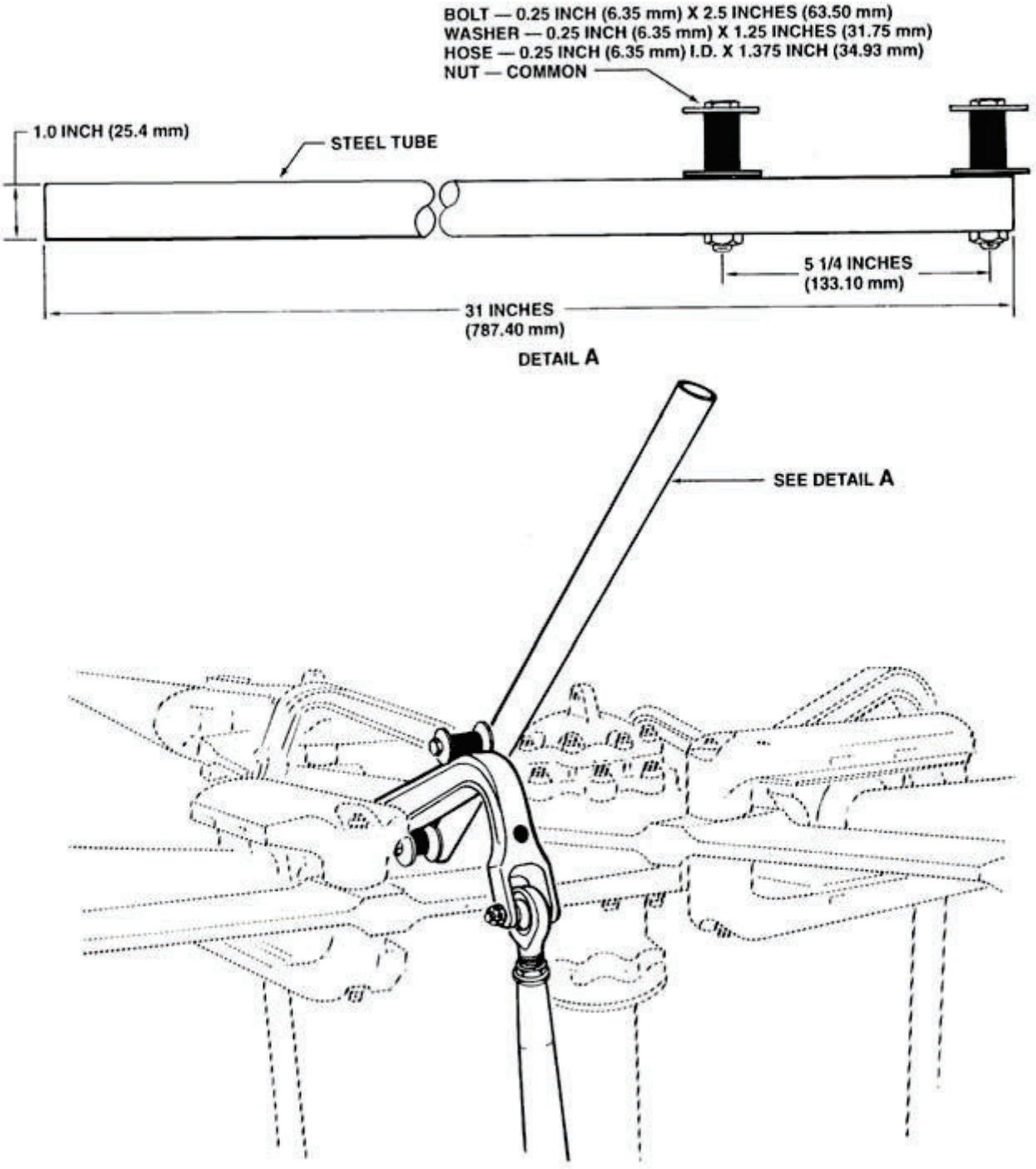
4. Grasp swashplate link (8) and attempt to move axially. Maximum acceptable axial wear in swashplate link rod end bearing (Teflon lined) (7) is 0.015 inch (0.38 mm).

5. Grasp drive link (5) and attempt to move axially. Maximum acceptable axial wear in drive link rod end bearing (Teflon lined) (6) is 0.015 inch (0.38 mm).

6. Replace pitch link rod end bearings (1 and 9) exceeding maximum acceptable wear (BHT-412-CR&O).

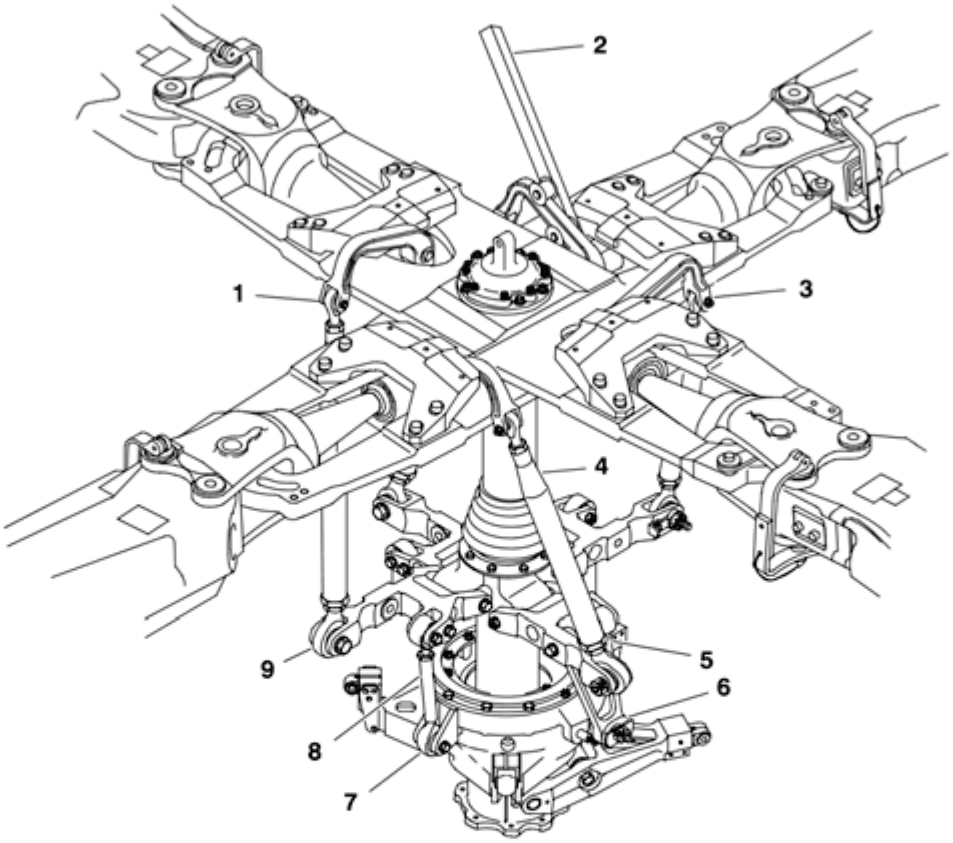
7. Replace swashplate link rod end bearing (7) exceeding maximum acceptable in swashplate link rod end bearing (BHT-412-CR&O).

8. Replace drive link rod end bearing (6) exceeding maximum acceptable wear (BHT-412-CR&O).



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Figure 1. Swashplate Link and Drive Link Bearings - Work Aid (Sheet 1 of 1)



1. Pitch link rod end bearing (teflon lined)
2. Work aid
3. Pitch horn
4. Pitch link
5. Drive link
6. Drive link bearing (teflon lined)
7. Swashplate link rod end bearing (teflon lined) (on lower end of swashplate link)
8. Swashplate link
9. Pitch link rod end bearing (teflon lined)

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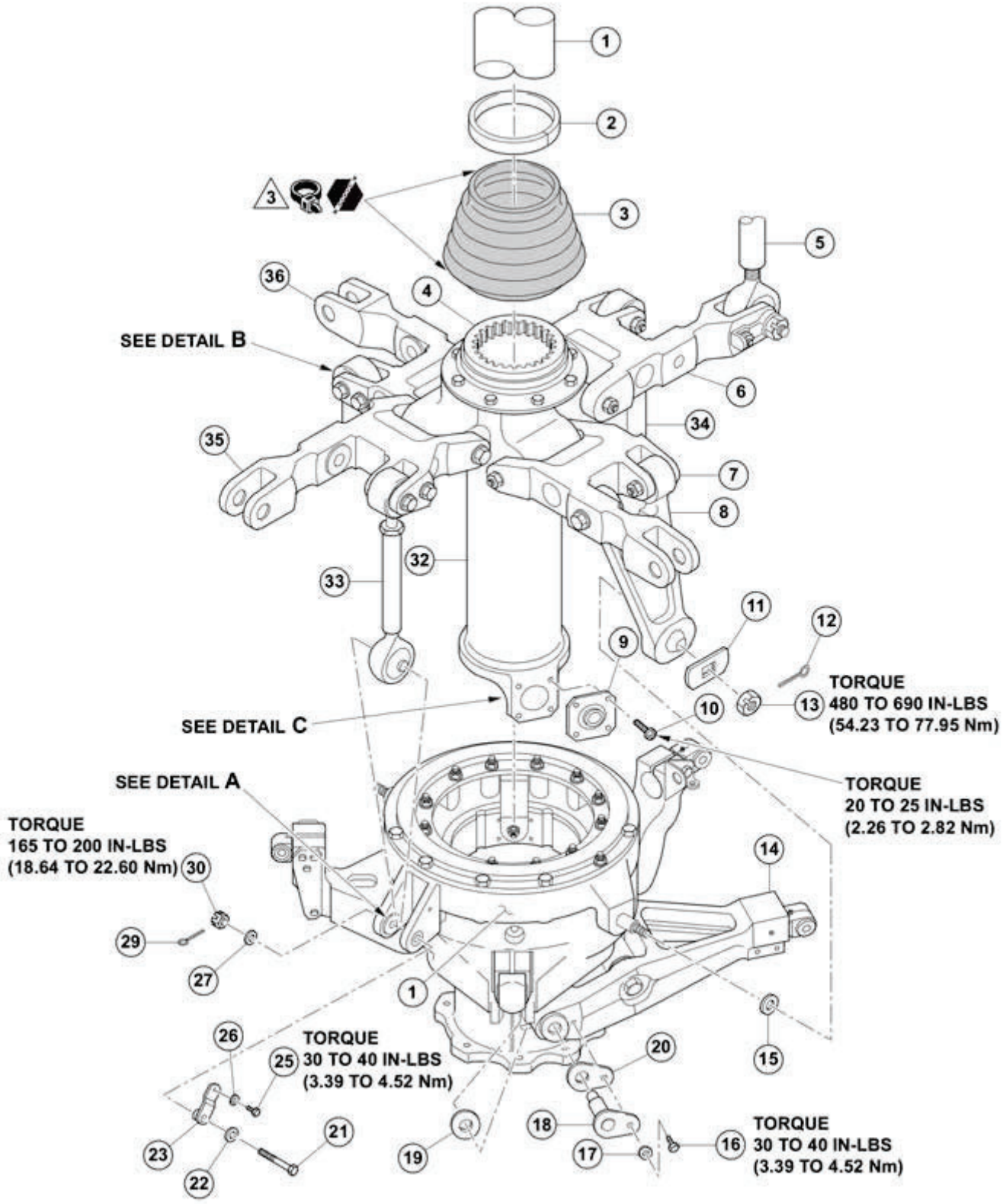
Figure 2. Hub and Sleeve Assembly Teflon Bearing Inspection (Sheet 1 of 1)

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Hub and Sleeve Assembly - Removal

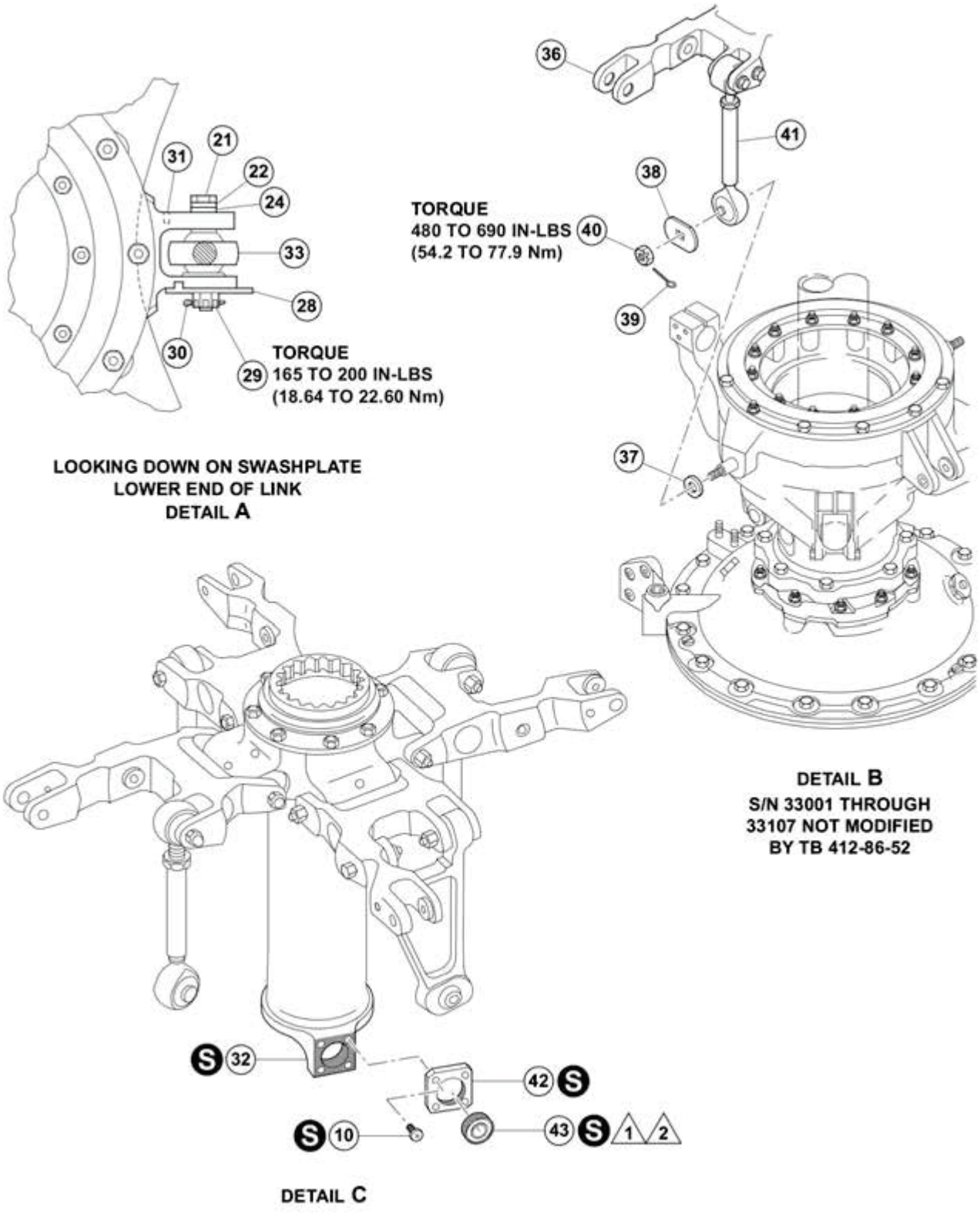
Procedure

1. Remove main rotor hub and blade assemblies (412-A-62-10-00-00A-520A-A / 00935 and 412-A-62-20-00-02A-520A-A / 00948).
2. Cut and remove lockwire or tie wrap, as applicable, from dust boot (3, Figure 1). Remove dust boot (3) and spacer (2) from mast (1).
3. Prior to removal of hub and sleeve assembly, measure spline wear on collective pitch drive plate set (4) (412-A-62-30-00-03A-280B-A / 00970).
4. Remove four pitch link assemblies (5) from rephasing levers (6, 7, 35, and 36) (412-A-62-30-00-02A-520A-A / 00961).
5. Cut and remove lockwire from bolt (25) and remove bolt with washer (26).
6. Remove cotter pin (29), nut (30), washer (27), bolt (21), washer (22) and anti-torque bushing (23).
7. Separate swashplate link (33) from outer ring (31).
8. Repeat Step 5. through Step 7. to disconnect opposite swashplate link (34) and (41), if installed.
9. Remove cotter pin (12), nut (13), and safety washer (11). Separate rephrasing lever drive link (8) from outer ring (31).
10. Remove washer (15) from stud on outer ring (31).
11. Remove cotter pin (39, detail B), nut (40) and safety washer (38). Separate swashplate link (41) from outer ring (31). Repeat for opposite swashplate link, if installed.
12. Remove washer (37) from stud on outer ring (31).
13. Cut and remove lockwire from bolt (16) and remove bolt, washer (17), collective lever pin (18), shim (19), and if installed, shim (20).
14. Index shim (19) and if installed, shim (20), for reinstallation in same location.
15. Repeat Step 13. and Step 14. to remove opposite collective lever pin (18) on collective levers.
16. Cut and remove lockwire from four screws (10). Remove four screws and bearing and liner (9).
17. Repeat Step 16. to remove opposite bearing and liner (9).
18. Remove collective pitch drive plate set (4) (412-A-62-30-00-07A-520A-A / 00979 and 412-A-62-30-00-07A-720A-A / 00980).
19. Carefully lift hub and sleeve assembly from swashplate.



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Figure 1. Hub and Sleeve (Sheet 1 of 3)



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Figure 1. Hub and Sleeve (Sheet 2 of 3)

1. Mast
2. Spacer
3. Dust boot
4. Collective pitch drive plate set
5. Pitch link
6. Rephasing lever
7. Rephasing lever
8. Rephasing lever drive link
9. Bearing and liner assembly
10. Screw
11. Safety washer
12. Cotter pin
13. Nut
14. Collective lever
15. Washer (140-009D41S60)
16. Bolt
17. Steel washer
18. Collective lever pin
19. Shim (pre TB 412-88-68)
20. Shim (post TB 412-88-68)
21. Bolt
22. Washer (140-007-25S-20C4)
23. Anti-torque bushing
24. Bushing
25. Bolt
26. Steel washer
27. Steel washer
28. Interrupter bracket
29. Cotter pin
30. Nut
31. Outer ring
32. Collective sleeve
33. Swashplate link (412-010-406-111)
34. Swashplate link
35. Rephasing lever
36. Rephasing lever
37. Washer (140-009D41S60)
38. Safety washer
39. Cotter pin
40. Nut
41. Swashplate link (412-010-406-113)
42. Liner
43. Bearing

LOCKWIRE (C-405)

S SEALANT (C-352)

TIE WRAP (C-592)

NOTE

1. Apply sealant MIL-R-46082, TYPE I (C-352) to faying surfaces as indicated by shading.
2. Ensure that no sealant is allowed to enter the bearing.
3. It is permitted, as an alternate to lockwire to install tie wrap (C-592) at the location shown with a plastic strap tool P/N 90387-2 with tension set between 5 and 8.

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Figure 1. Hub and Sleeve (Sheet 3 of 3)

Hub and Sleeve Assembly - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure



DO NOT IMMERSE HUB AND SLEEVE ASSEMBLY IN SOLVENT.

1. Clean hub and sleeve assembly with clean cloth damp with drycleaning solvent (C-304).
2. Dry with filtered, compressed air.

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Hub and Sleeve Assembly - Inspection

Procedure

NOTE

If records or physical appearance indicate hub and sleeve assembly has been involved in an accident or incident, refer to 412-A-05-50-00-00A-284A-A / 00063 for inspection requirements.

1. Inspect hub and sleeve assembly Teflon lined rod end bearings (installed) as required by 412-A-05-50-00-03A-284A-A / 00066 and Figure 1.

2. Inspect boot (3, Figure 2) for tears, cracks, and deterioration.

NOTE

The hub and sleeve assembly must be installed on main rotor mast to accomplish the following step.

3. Inspect spacer (2) for damage and deterioration.

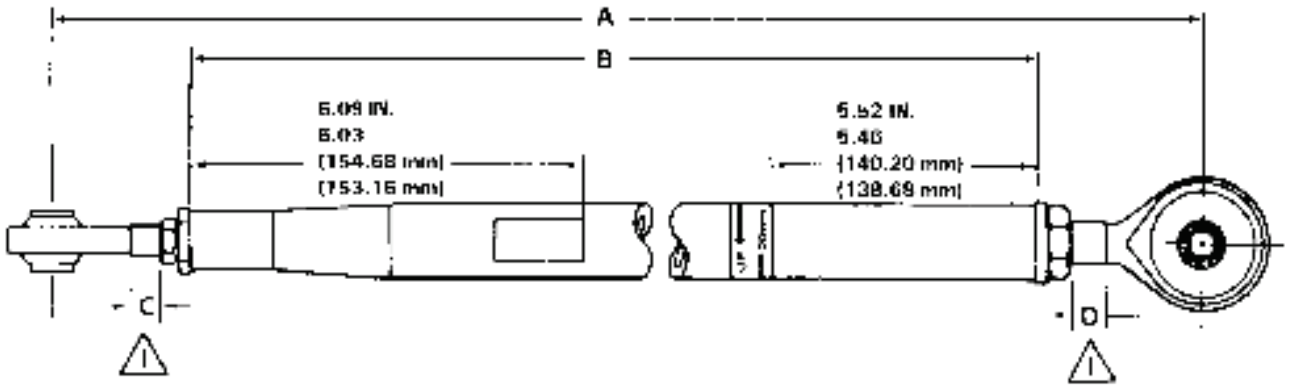
4. Measure spline wear on collective drive plate set (7, Figure 3).

5. Inspect safety washers (11 and 38, Figure 2) for distortion and cracks. Superficial damage is acceptable without repair.

6. Hold collective sleeve (32) and rotate hub. Check bearing set by feel for looseness and/or binding. If looseness and/or binding is detected, inspect in accordance with procedures contained in 412-CR&O, Chapter 62.

7. Inspect anti-torque bushing (23) for damage or distortion and wear which would prevent locking rod end bearing.

8. For remainder of hub and sleeve components, inspect in accordance with procedures contained in 412-CR&O, Chapter 62.



PITCH LINK FOR	DIMENSIONS		DIMENSIONS METRIC	
	A	B	A	B
UPPER YOKE	20.770 INCH 20.750 INCH	16.64 INCH 16.61 INCH	527.558 mm 527.050 mm	422.65 mm 421.89 mm
LOWER YOKE	19.025 INCH 19.005 INCH	14.985 INCH 15.015 INCH	483.235 mm 482.727 mm	380.619 mm 381.381 mm

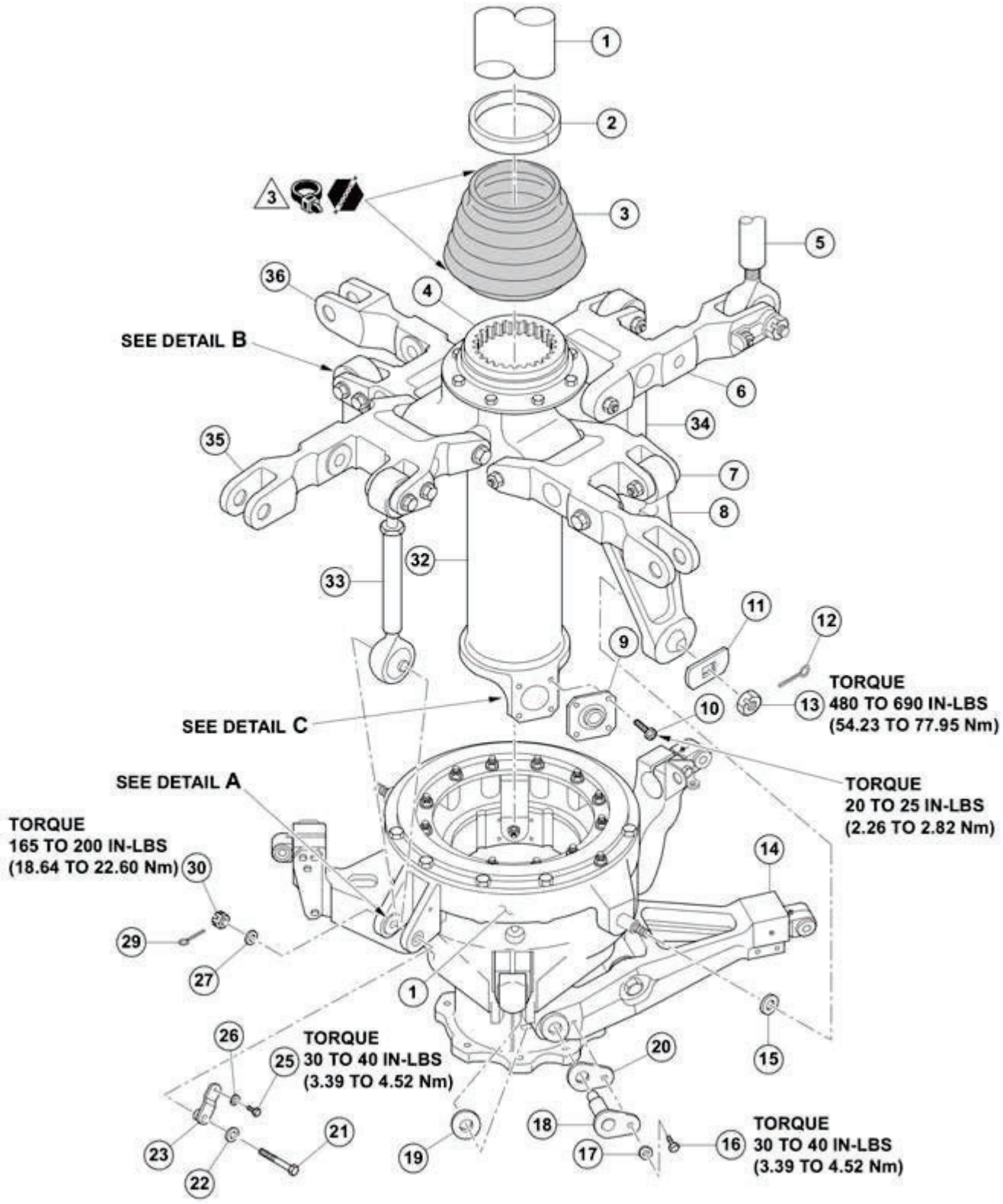
NOTE

1 After initial setting: Exposed threads on rod end bearings (dimensions "C" and "D") must be equal within two threads.

After installation and rigging: Maximum exposed threads shall not exceed eleven at dimensions "C" and eight at dimension "D".

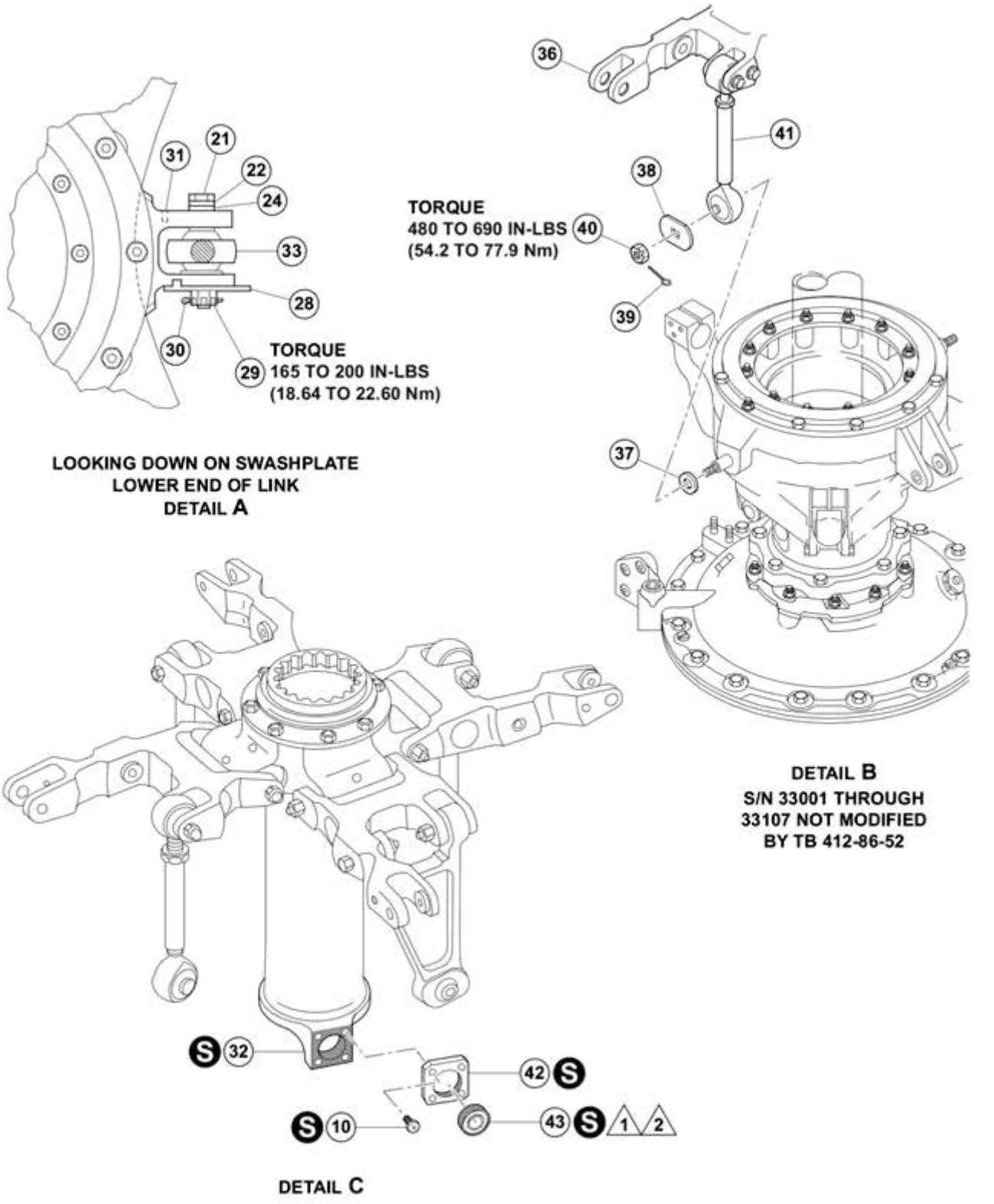
ICN-412-A-623000-B-97499-00026-A-001-01

Figure 1. Main Rotor Pitch Link Assembly Initial Dimensions (Sheet 1 of 1)



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Figure 2. Hub and Sleeve (Sheet 1 of 3)



ICN-412-A-623000-B-97499-00031-A-001-01

Figure 2. Hub and Sleeve (Sheet 2 of 3)

1. Mast
2. Spacer
3. Dust boot
4. Collective pitch drive plate set
5. Pitch link
6. Rephasing lever
7. Rephasing lever
8. Rephasing lever drive link
9. Bearing and liner assembly
10. Screw
11. Safety washer
12. Cotter pin
13. Nut
14. Collective lever
15. Washer (140-009D41S60)
16. Bolt
17. Steel washer
18. Collective lever pin
19. Shim (pre TB 412-88-68)
20. Shim (post TB 412-88-68)
21. Bolt
22. Washer (140-007-25S-20C4)
23. Anti-torque bushing
24. Bushing
25. Bolt
26. Steel washer
27. Steel washer
28. Interrupter bracket
29. Cotter pin
30. Nut
31. Outer ring
32. Collective sleeve
33. Swashplate link (412-010-406-111)
34. Swashplate link
35. Rephasing lever
36. Rephasing lever
37. Washer (140-009D41S60)
38. Safety washer
39. Cotter pin
40. Nut
41. Swashplate link (412-010-406-113)
42. Liner
43. Bearing

LOCKWIRE (C-405)

SEALANT (C-352)

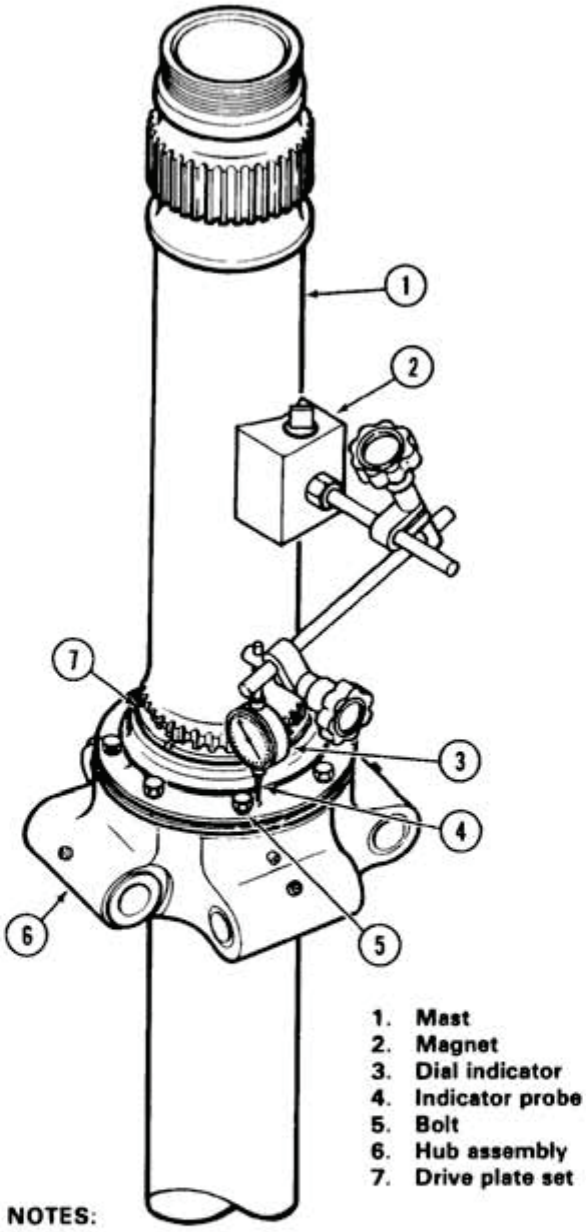
TIE WRAP (C-592)

NOTE

1. Apply sealant MIL-R-46082, TYPE I (C-352) to faying surfaces as indicated by shading.
2. Ensure that no sealant is allowed to enter the bearing.
3. It is permitted, as an alternate to lockwire to install tie wrap (C-592) at the location shown with a plastic strap tool P/N 90387-2 with tension set between 5 and 8.

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Figure 2. Hub and Sleeve (Sheet 3 of 3)



1. Mast
2. Magnet
3. Dial indicator
4. Indicator probe
5. Bolt
6. Hub assembly
7. Drive plate set

NOTES:

Attach dial indicator (3) to mast (1) with magnet (2). Position indicator probe (4) in contact with a flat of bolt (5) (Flange attaching bolt). Move hub (6) back and forth, in plane of rotation and measure amount of play.

Maximum acceptable radial play allowed between mast (1) and plate set drive (7) is 0.040 inch (1.01 mm).

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Figure 3. Collective Drive Plate Set - Damage Limits (Sheet 1 of 1)

Hub and Sleeve Assembly - Repair

Procedure

1.

NOTE

Repair of hub and sleeve assembly is restricted to polishing out mechanical and corrosion damage and installation of new parts such as bearing and liner assembly, drive link assembly, swashplate link assembly, and rephasing lever drive assembly. Refer to BHT-412-CR&O for mechanical and corrosion damage limits and repair procedures and limits.

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Hub and Sleeve Assembly - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Plastic Strap Tool	MS90387-2	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Corrosion preventive compound	C-104	AR	
Sealant	C-308	AR	
Adhesive	C-325	AR	
Denatured alcohol	C-326	AR	
Sealant	C-352	AR	
Lockwire	C-405	AR	
Grease	C-561	AR	
Tie wrap	C-592	AR	Use tie wrap MS3367-6-9.

Procedure

NOTE

A thin coat of corrosion preventive compound (C-104) should be applied to shanks and under bolt heads of all rotating control bolts prior to installation.

A thin coat of corrosion preventive compound (C-101) should be applied to all rotating control nuts, washers, and bolt heads after there installation.

1. Apply grease (C-561) to splines of mast and collective pitch drive plate set (4, Figure 1).

2. Remove collective pitch drive plate set (4) from hub and sleeve, if not previously accomplished (412-A-62-30-00-07A-520A-A / 00979).

3. Carefully position hub and sleeve assembly on mast (1) and lower hub and sleeve assembly into swashplate.



MAKE SURE THAT NO SEALANT IS ALLOWED TO ENTER BEARINGS.

4. If replacing bearing (43), proceed as follows:

4.1. Clean all faying surfaces with denatured alcohol (C-326).

4.2. Refer to Detail C. Apply sealant (C-352) sparingly to outside diameter of bearing (43) and inside diameter of liner (42).

4.3. While sealant is still wet, install bearing (42) in liner (43).

5. Refer to Detail C. Clean all faying surfaces with denatured alcohol (C-326). Apply sealant (C-352) sparingly to faying surfaces of liner (42), collective sleeve (32) and screws (10).

6. While sealant is still wet, position bearing and liner assembly (9) on collective sleeve (32) and install four screws (10).

7. Torque screws (10) evenly 20 to 25 inch-pounds (2.26 to 2.82 Nm).

8. Secure screws (10) together in pairs using lockwire (C-405).

9. Repeat Step 4. through Step 8. to install opposite bearing and liner (9).

10. Install pin (18) (412-A-62-30-00-08A-720A-A / 00985 or 412-A-62-30-00-08A-720B-A / 00986).

11. Install washer (15) on stud of outer ring (31) and position rephrasing lever drive link (8) on swashplate outer ring.

12. Install safety washer (11), with raised lip toward drive link bearing, and nut (13).

13. Torque nut (13) 480 to 690 inch-pounds (54.2 to 77.9 Nm). Install cotter pin (12).

14. If second drive link is installed, repeat Step 11. through Step 13., then proceed to Step 19. or Step 20., as required.

15. Install washer (37) on stud of outer ring (31).

16. Position swashplate link (41) on outer ring (31).

17. Install safety washer (38), with raised lip toward swashplate link bearing, and nut (40).

18. Torque nut (40) 480 to 690 inch-pounds (54.2 to 77.9 Nm). Install cotter pin (39).

19. To install swashplate link assembly with elastomeric bearings, proceed as follows:

19.1. Position swashplate link (33) on outer ring (31) with serrated face of rod end bearing toward anti-torque bushing (23) (Detail A).

19.2. Install anti-torque bushing (23) with serrated face of bushing in contact with serrated face of rod end bearing (Detail A).

19.3. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (21).

19.4. Install bolt (21) with washer (22) under head of bolt.

19.5. Install washer (27) and nut (30) on bolt (21). Do not torque nut at this time.

19.6. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (25).

19.7. Install bolt (25) with washer (26) under head of bolt.

19.8. Torque bolt (25) 30 to 40 inch-pounds (3.39 to 4.52 Nm).

19.9. Secure bolt (25) to bushing (23) using lockwire (C-405).

19.10. Torque nut (30) 165 to 200 inch-pounds (18.64 to 22.60 Nm). Install cotter pin (29).

19.11. Apply a thin coat of corrosion preventive compound (C-101) to nut (30), to head of bolt (21) and washer (22), and to head of bolt (25) and washer (26).

20. To install swashplate link assembly with uniball Teflon lined bearing, proceed as follows:

20.1. Install lower end of swashplate link (33) with head of bolt (21) in direction of rotation.

20.2. Apply a thin coat of corrosion preventive compound (C-104) under the head and to the shank of bolt (21).

20.3. Install washer (22) under bolt head (21).

20.4. Slide bushing (24) into swashplate outer ring (31).

20.5. Install bolt (21) with washer (10) through bushing (24), swashplate link (33), and outer ring (31).

20.6. Install interrupter bracket (28) and nut (30).

20.7. Torque nut (30) 165 to 200 inch-pounds (18.64 to 22.60 Nm) and secure with cotter pin (29).

21. Repeat Step 19.1. through Step 19.11. to install remaining swashplate link.

22. Install collective pitch drive plate set (412-A-62-30-00-07A-720A-A / 00980).

23. Install dust boot (3) and spacer (2).

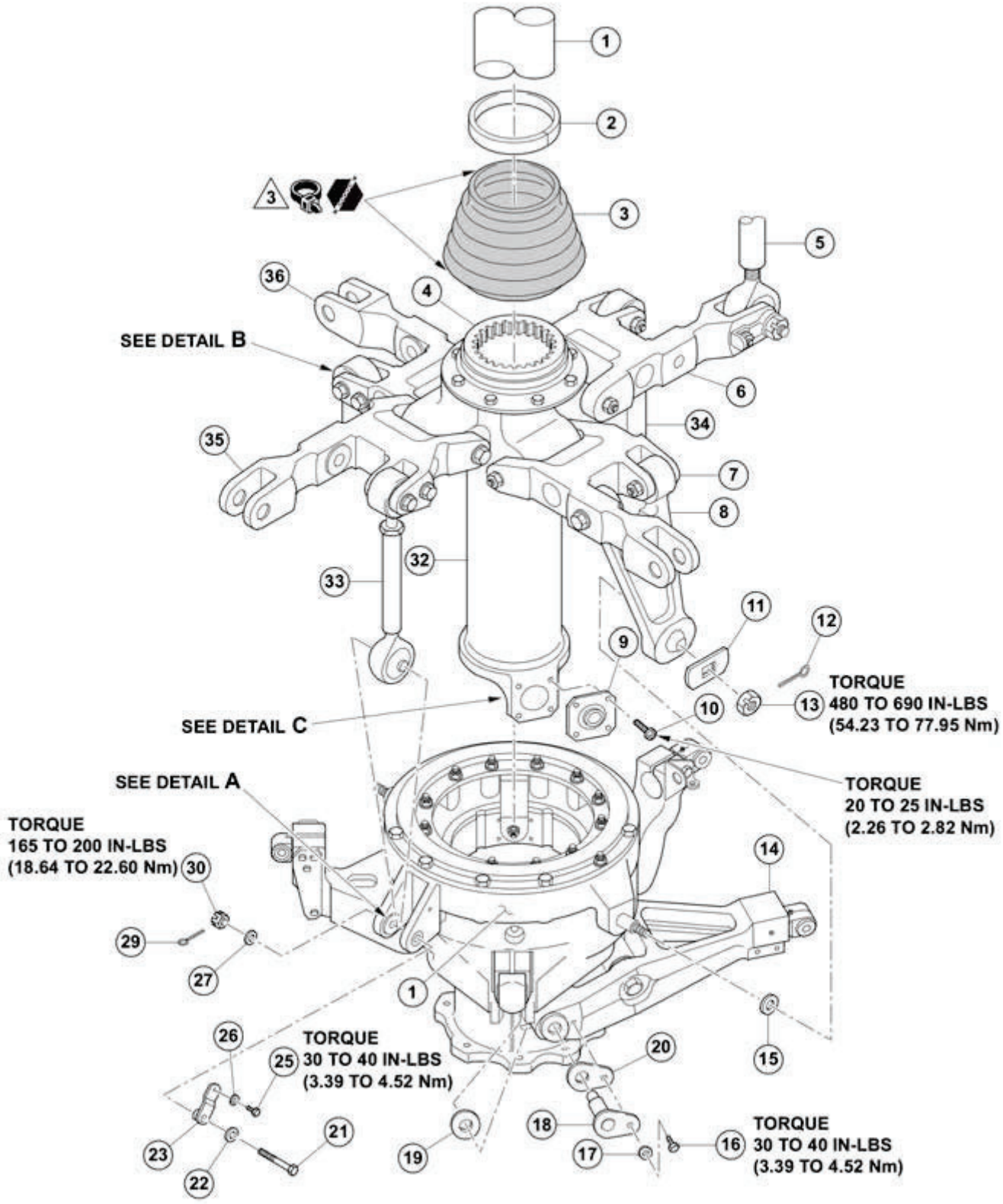
24. Bond flap and closure of boot (3) together at junction using adhesive (C-325).

25. Seal spacer (2) where ends come together with sealant (C-308)

26. Secure boot to mast (1) and collective pitch drive plate set (4) using lockwire (C-405) or tie wrap (C-592) as applicable, with the plastic strap tool (MS90387-2) with the tension adjusted between 5 and 8.

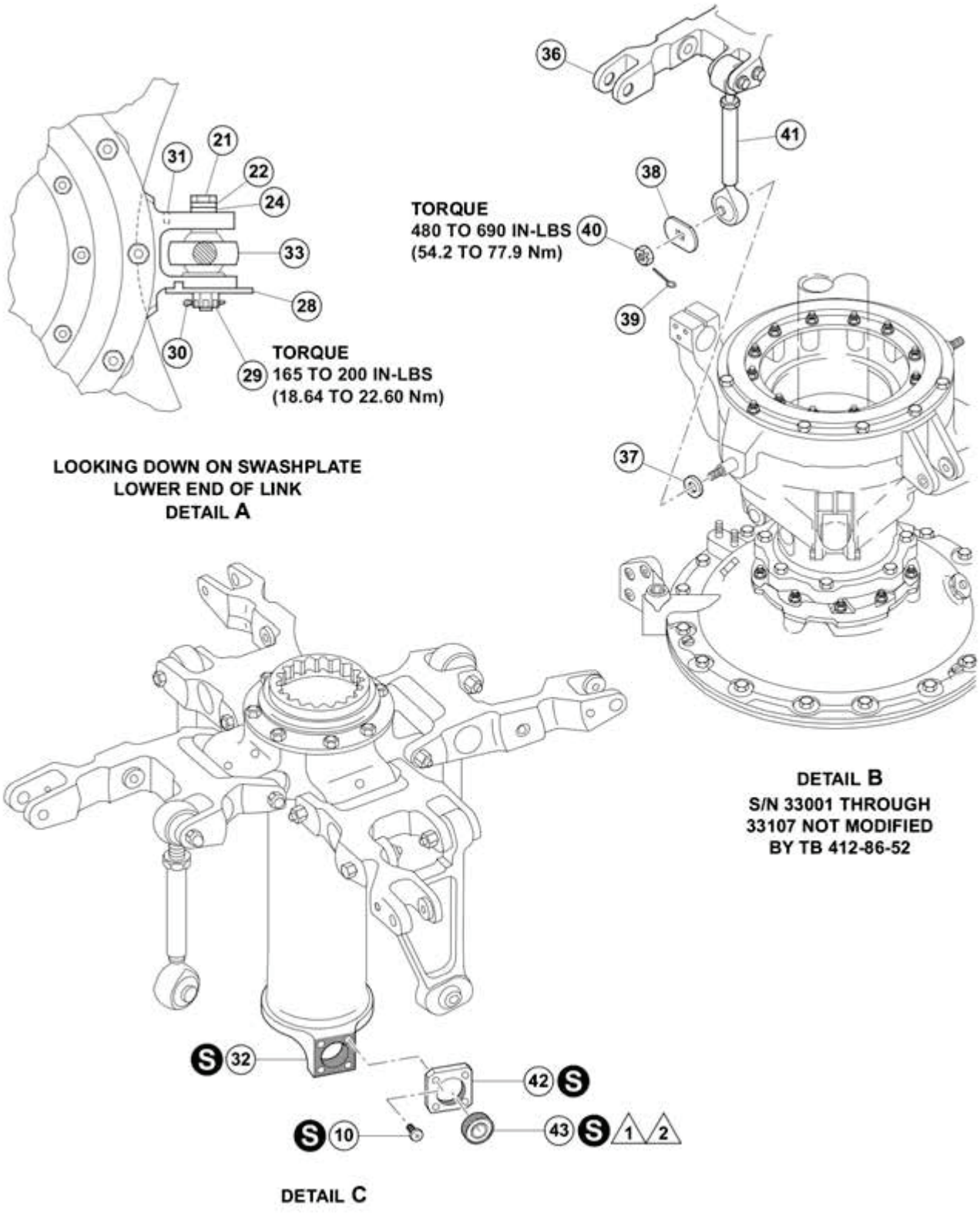
27. Lubricate hub and sleeve assembly (412-A-12-01-00-00A-040A-A / 00141).

28. Confirm control rigging is within limits (412-A-67-01-00-08A-276A-A / 01548).



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Figure 1. Hub and Sleeve (Sheet 1 of 3)



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Figure 1. Hub and Sleeve (Sheet 2 of 3)

1. Mast
2. Spacer
3. Dust boot
4. Collective pitch drive plate set
5. Pitch link
6. Rephasing lever
7. Rephasing lever
8. Rephasing lever drive link
9. Bearing and liner assembly
10. Screw
11. Safety washer
12. Cotter pin
13. Nut
14. Collective lever
15. Washer (140-009D41S60)
16. Bolt
17. Steel washer
18. Collective lever pin
19. Shim (pre TB 412-88-68)
20. Shim (post TB 412-88-68)
21. Bolt
22. Washer (140-007-25S-20C4)
23. Anti-torque bushing
24. Bushing
25. Bolt
26. Steel washer
27. Steel washer
28. Interrupter bracket
29. Cotter pin
30. Nut
31. Outer ring
32. Collective sleeve
33. Swashplate link (412-010-406-111)
34. Swashplate link
35. Rephasing lever
36. Rephasing lever
37. Washer (140-009D41S60)
38. Safety washer
39. Cotter pin
40. Nut
41. Swashplate link (412-010-406-113)
42. Liner
43. Bearing

LOCKWIRE (C-405)

SEALANT (C-352)

TIE WRAP (C-592)

NOTE

1. Apply sealant MIL-R-46082, TYPE I (C-352) to faying surfaces as indicated by shading.
2. Ensure that no sealant is allowed to enter the bearing.
3. It is permitted, as an alternate to lockwire to install tie wrap (C-592) at the location shown with a plastic strap tool P/N 90387-2 with tension set between 5 and 8.

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Figure 1. Hub and Sleeve (Sheet 3 of 3)

Rephasing Levers - Removal

Procedure

NOTE

SN: 33001-33107 or TB 412-86-52: Pre:

One rephasing lever is connected to the drive link and three rephasing levers are connected to swashplate links.

NOTE

(SN: 33001-33107 TB 412-86-52: Post) or SN: 33108-Subsequent:

The rephasing levers are connected to two drive links and two swashplate links.

1. Remove rephasing lever (2, Figure 1) with swashplate (3) link attached, as follows:

1.1. Remove pitch link (1) with uniball or elastomeric rod end bearings from rephasing lever (2) (412-A-62-30-00-02A-520A-A / 00961).

1.2. Remove swashplate link (3) with uniball or elastomeric rod ends from rephasing lever (2) (412-A-62-30-00-06A-520A-A / 00977).

1.3. Remove cotter pin (4), nut (5), washer (6), bolt (7), washer (8), and washers (9 and 10). Discard cotter pin (4).

1.4. Lift rephasing lever (2) from hub (11).

1.5. Repeat Step 1. for remaining rephasing levers with swashplate links attached.

2. Remove rephasing lever (12) with drive link (13) attached, as follows:

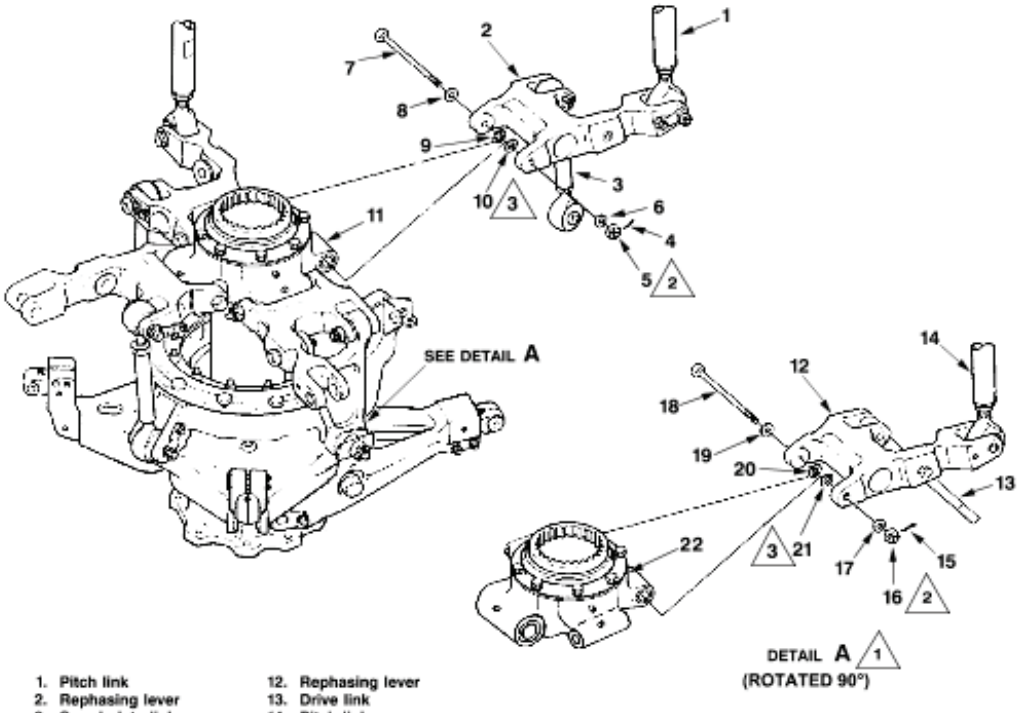
2.1. Remove pitch link (14) with uniball or elastomeric rod end bearings from rephasing lever (12) (412-A-62-30-00-02A-520A-A / 00961).

2.2. Remove drive link (13) from rephasing lever (12) (412-A-62-30-00-05A-520A-A / 00975).

2.3. Remove cotter pin (15), nut (16), washer (17), bolt (18), washer (19), and washers (20 and 21).

2.4. Lift rephasing lever (12) from hub (22).

2.5. Repeat Step 2. for remaining drive link, if installed.



1. Pitch link
2. Rephasing lever
3. Swashplate link
4. Cotter pin
5. Nut
6. Washer
7. Bolt
8. Washer
9. Washer
10. Washer
11. Hub
12. Rephasing lever
13. Drive link
14. Pitch link
15. Cotter pin
16. Nut
17. Washer
18. Bolt
19. Washer
20. Washer
21. Washer
22. Hub

NOTES:

1. Helicopters S/N 33108 and subsequent and helicopters modified by TB 412-86-52, two drive links are installed.

2. Torque 1/2 inch bolts 24 to 34 ft·lb (33 to 46 N·m). Torque 3/8 inch bolts 165 to 200 in·lb (18.64 to 22.60 N·m).

3. Helicopters not modified by TB 412-03-189, with hub 412-010-445-101 installed, use washer 412-010-423-103.

Helicopters modified by TB 412-03-189, with hub 412-010-445-101FM installed, use washer 412-010-423-105; with hub 412-010-445-105FM installed, use washer 412-010-423-107.

S/N 36240 and subsequent and helicopters with hub 412-010-445-105 installed use washer 412-010-423-107.

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Figure 1. Rephasing Lever Installation (Sheet 1 of 1)

Rephasing Levers - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Corrosion preventive compound	C-104	AR	

Procedure

WARNING

IT IS POSSIBLE TO INSTALL REPHASING LEVER (2) INVERTED. ENSURE REPHASING LEVER (2) IS INSTALLED ON HUB (10) WITH SHORT ARM FOR ATTACHMENT OF SWASHPLATE LINK (3) ON SIDE IN DIRECTION OF ROTATION OF MAST WHICH WILL AFFECT FLIGHT CONTROL RIGGING.

NOTE

SN: 33001-33107 or TB 412-86-52: Pre:

One rephasing lever is connected to the drive link and three rephasing levers are connected to swashplate links.

NOTE

(SN: 33001-33107 TB 412-86-52: Post) or SN: 33108-Subsequent:

The rephasing levers are connected to two drive links and two swashplate links.

1. Install rephasing lever (2) with swashplate link as follows:

NOTE

Refer to Figure 1 for correct usage of washer (10).

1.1. Position rephasing lever (2, Figure 1) on hub (11) with washers (9 and 10).

1.2. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (7).

1.3. Install bolt (7) with washer (8) under bolt head.

1.4. Install washer (6) and nut (5).

1.5. Torque nut (5) 24 to 34 foot-pounds (32 to 45 Nm). Install cotter pin (4).

1.6. Apply a thin coat of corrosion preventive compound (C-101) on nut (5) and washer (6) and to head of bolt (7) and washer (8).

1.7. Install swashplate link (3) on rephasing lever (2) (412-A-62-30-00-06A-720A-A / 00978).

1.8. Install pitch link (1) on rephasing lever (2) (412-A-62-30-00-02A-720A-A / 00965).

1.9. Repeat Step 1. for remaining rephasing levers (11) with swashplate links (3) installed.

WARNING

IT IS POSSIBLE TO INSTALL REPHASING LEVER (11) INVERTED. ENSURE REPHASING LEVER (11) IS INSTALLED ON HUB (20) WITH SHORT ARM FOR ATTACHMENT OF DRIVE LINK (12) ON SIDE IN DIRECTION OF ROTATION OF MAST.

2. Install rephasing lever (12) with drive link (13) as follows:

NOTE

Refer to Figure 1 for correct usage of washer (21).

2.1. Position rephasing lever (12, Figure 1) on hub (22) with washers (20 and 21).

2.2. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (18).

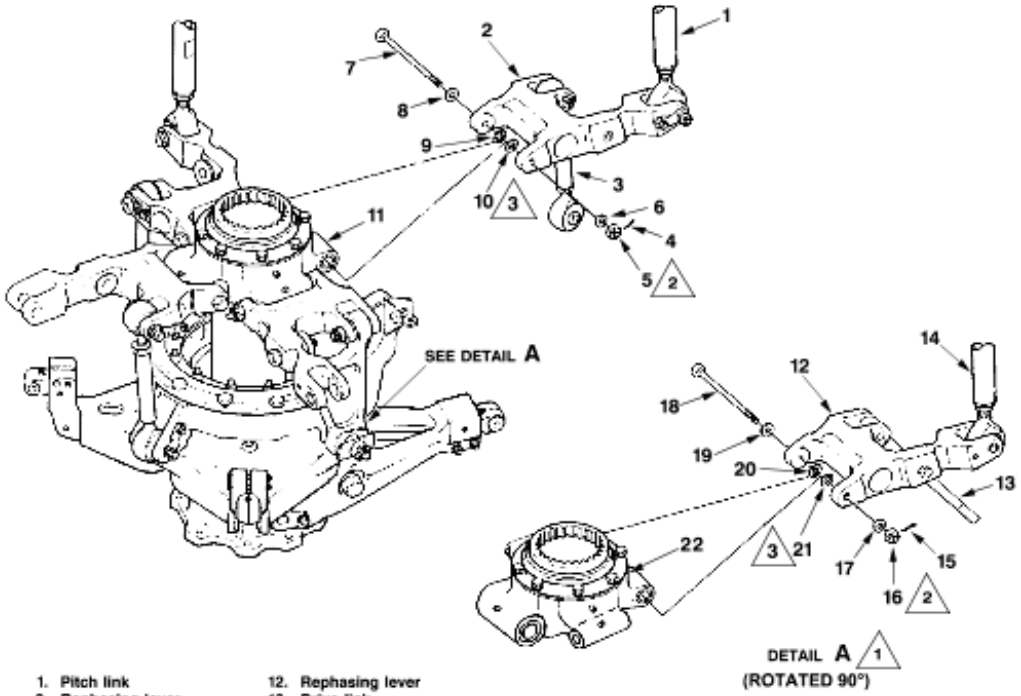
2.3. Install bolt (18) with washer (19) under bolt head.

2.4. Install washer (17) and nut (16).

2.5. Torque nut 24 to 34 foot-pounds (33 to 46 Nm). Install cotter pin (15).

2.6. Apply a thin coat of corrosion preventive compound (C-101) on nut (16) and washer (17) and to head of bolt (18) and washer (19).

3. Confirm control rigging is within limits (412-A-67-01-00-08A-276A-A / 01548).



1. Pitch link
2. Rephasing lever
3. Swashplate link
4. Cotter pin
5. Nut
6. Washer
7. Bolt
8. Washer
9. Washer
10. Washer
11. Hub
12. Rephasing lever
13. Drive link
14. Pitch link
15. Cotter pin
16. Nut
17. Washer
18. Bolt
19. Washer
20. Washer
21. Washer
22. Hub

NOTES:

1. Helicopters S/N 33108 and subsequent and helicopters modified by TB 412-86-52, two drive links are installed.

2. Torque 1/2 inch bolts 24 to 34 ft·lb (33 to 46 N·m). Torque 3/8 inch bolts 165 to 200 in·lb (18.64 to 22.60 N·m).

3. Helicopters not modified by TB 412-03-189, with hub 412-010-445-101 installed, use washer 412-010-423-103.

Helicopters modified by TB 412-03-189, with hub 412-010-445-101FM installed, use washer 412-010-423-105; with hub 412-010-445-105FM installed, use washer 412-010-423-107.

S/N 36240 and subsequent and helicopters with hub 412-010-445-105 installed use washer 412-010-423-107.

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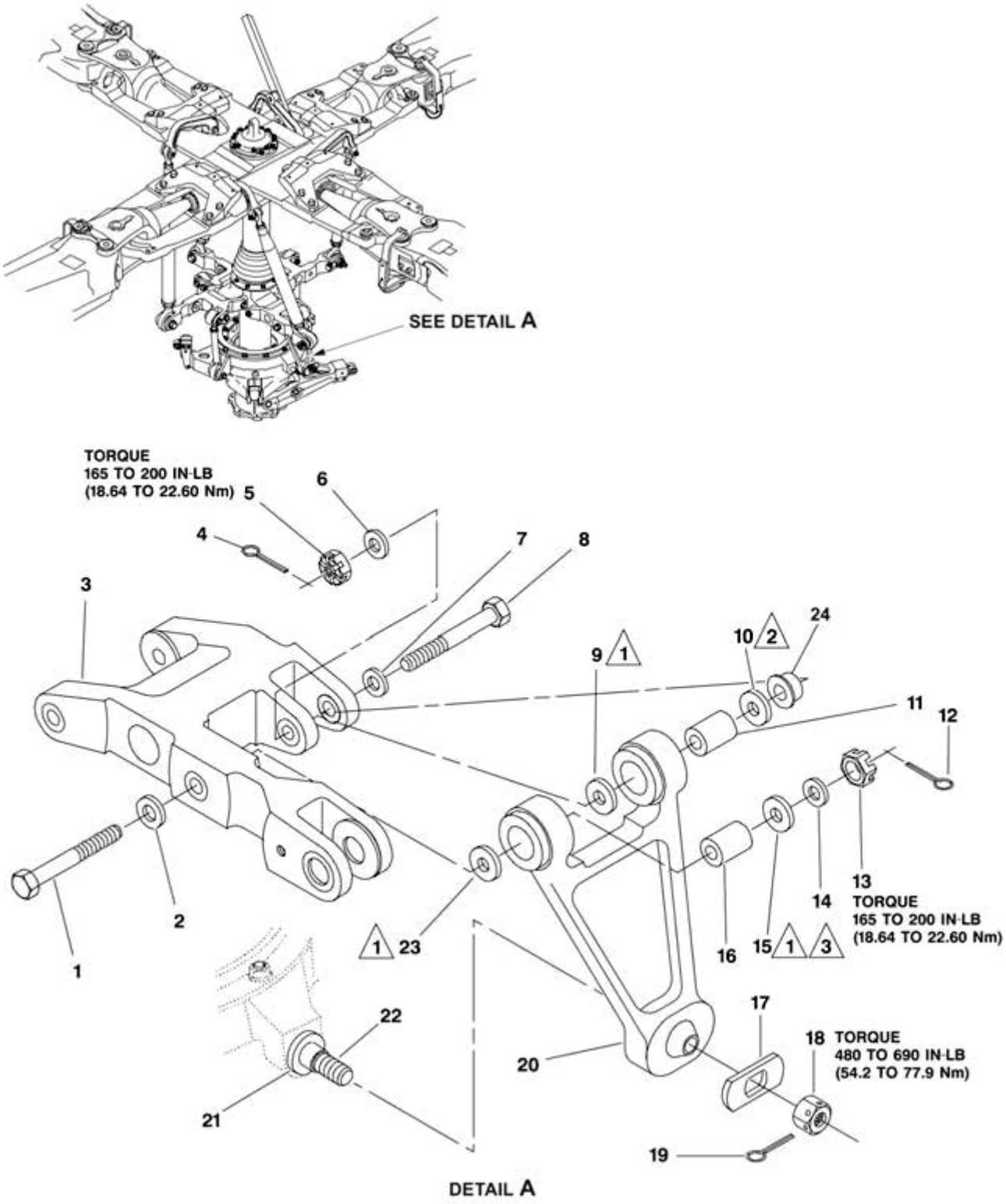
Figure 1. Rephasing Lever Installation (Sheet 1 of 1)

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Drive Link Assembly - Removal

Procedure

1. Remove cotter pin (19, Figure 1), nut (18), and safety washer (17).
2. Remove cotter pin (12), nut (13), washers (14, 15, 23, and 2), spacer (16), and bolt (1).
3. Remove cotter pin (4), nut (5), washers (6, 9, 10, and 7), spacer (11) and bolt (8).
4. Remove drive link (20).
5. *If required, remove the bushing (24) from the rephasing lever (3).*



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Figure 1. Drive Link (Sheet 1 of 2)

NOTES

1. Helicopters S/N 33108 and subsequent and helicopters modified by TB412-86-52 use washer P/N 412-010-423-103.
2. Not used on helicopters S/N 33108 and subsequent and helicopters modified by TB412-86-52.
3. Helicopters modified by TB 412-03-189, with drive link 412-010-405-107FM/-109FM installed, use washer 412-010-423-105; with drive link 412-010-445-111FM installed, use washer 412-010-423-107. S/N 36240 and subsequent and helicopters with drive link 412-010-445-111 installed use washer 412-010-423-107.

1. Bolt
2. Washer (140-007-25S20C4)
3. Rephasing lever
4. Cotter pin
5. Nut
6. Washer (140-009D-25S40)
7. Washer (140-007-25S36C4)
8. Bolt
9. Drive washer (412-010-423-101)
10. Drive washer (412-010-423-101)
11. Spacer (412-010-418-101)
12. Cotter pin
13. Nut
14. Washer (140-009D25S72)
15. Drive washer (412-010-423-101)
16. Spacer
17. Safety washer
18. Nut
19. Cotter pin
20. Drive link
21. Washer (140-009D-41S60)
22. Stud
23. Drive washer (412-010-423-101)
24. Bushing

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Figure 1. Drive Link (Sheet 2 of 2)

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Drive Link Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Corrosion preventive compound	C-104	AR	

Procedure

NOTE

A thin coat of corrosion preventive compound (C-104) should be applied to shanks and under bolt heads of all rotating control bolts prior to installation.

A thin coat of corrosion preventive compound (C-101) should be applied to all rotating control nuts, washers, and bolt heads after their installation.

1. Position washer (21, Figure 1) on stud (22).

2. Position drive link (20) on stud (22).

3. Install washer (17), with raised lip toward drive link (20). Install nut (18). Do not torque nut at this time.

4. *If removed, install the bushing (24) in its position on the rephasing lever (3).*

5. Apply a thin coat corrosion preventive compound (C-104) under the head and to shank of bolt (8).

6. Assemble spacer (11), washer (9), washer (10) (if applicable) on drive link bearing and install bolt (8) with washer (7) under bolt head.

7. Install washer (6) and nut (5). Do not torque nut at this time.

8. Apply a thin coat corrosion preventive compound (C-104) under the head and to shank of bolt (1).

NOTE

Refer to Figure 1 for correct usage of washer (15).

9. Assemble spacer (16) and washer (23) on drive link bearing and install bolt (1) with washer (2) under bolt head.

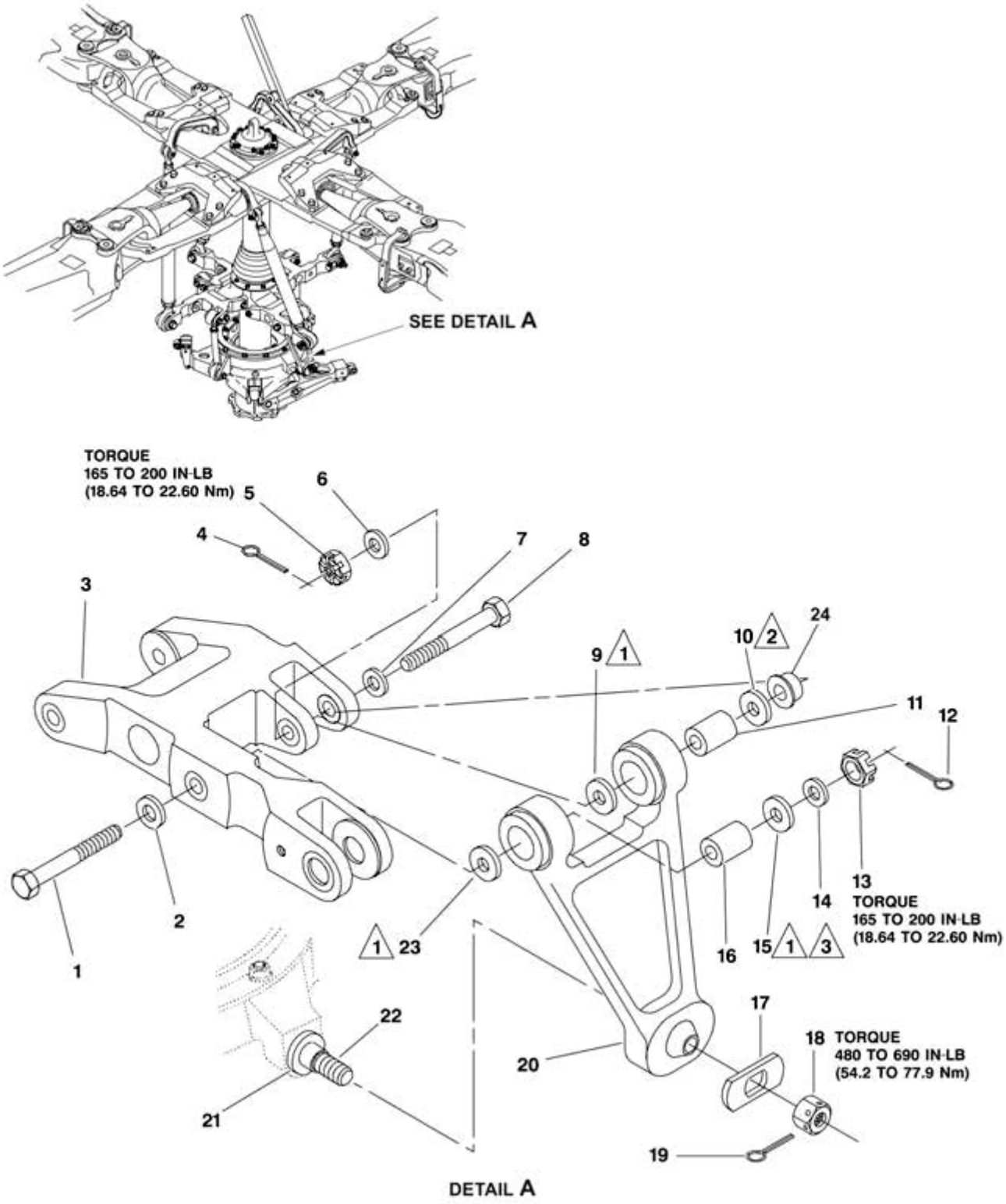
10. Install washers (15 and 14) and nut (13). Torque nut (13) 165 to 200 inch-pounds (18.64 to 22.60 Nm). Install cotter pin (12).

11. Torque nut (5) 165 to 200 inch-pounds (18.64 to 22.60 Nm). Install cotter pin (4).

12. Torque nut (18) 480 to 690 inch-pounds (54.2 to 77.9 Nm). Install cotter pin (19).

13. Apply a thin coat of corrosion preventive compound (C-101) on head of bolt (8 and 1), nut (13) and washer (14), nut (5) and washer (6), and nut (18) and washer (17).

14. Confirm control rigging is within limits (412-A-67-01-00-08A-276A-A / 01548).



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Figure 1. Drive Link (Sheet 1 of 2)

NOTES

1. Helicopters S/N 33108 and subsequent and helicopters modified by TB412-86-52 use washer P/N 412-010-423-103.
2. Not used on helicopters S/N 33108 and subsequent and helicopters modified by TB412-86-52.
3. Helicopters modified by TB 412-03-189, with drive link 412-010-405-107FM/-109FM installed, use washer 412-010-423-105; with drive link 412-010-445-111FM installed, use washer 412-010-423-107. S/N 36240 and subsequent and helicopters with drive link 412-010-445-111 installed use washer 412-010-423-107.

1. Bolt
2. Washer (140-007-25S20C4)
3. Rephasing lever
4. Cotter pin
5. Nut
6. Washer (140-009D-25S40)
7. Washer (140-007-25S36C4)
8. Bolt
9. Drive washer (412-010-423-101)
10. Drive washer (412-010-423-101)
11. Spacer (412-010-418-101)
12. Cotter pin
13. Nut
14. Washer (140-009D25S72)
15. Drive washer (412-010-423-101)
16. Spacer
17. Safety washer
18. Nut
19. Cotter pin
20. Drive link
21. Washer (140-009D-41S60)
22. Stud
23. Drive washer (412-010-423-101)
24. Bushing

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Figure 1. Drive Link (Sheet 2 of 2)

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Swashplate Link Assembly - Removal

Procedure

NOTE

SN: 33001-33107 TB 412-86-52: Pre:

Three rephasing levers are connected to swashplate links.

NOTE

SN: 33108-Subsequent:

Only two swashplate links are used.

1. Remove swashplate link with elastomeric rod end bearings as follows:

1.1. Cut and remove lockwire from bolt (7, Figure 1) and remove bolt (7) and washer (6).

1.2. Cut and remove lockwire from bolt (12) and remove bolt (12) and washer (13).

1.3. Remove cotter pin (16), nut (15), washer (14), bolt (9), washer (10) and anti-torque bushing (11).

1.4. Remove cotter pin (2), nut (3), washer (1), bolt (18), washer (21), and anti-torque bushing (5).

1.5. Remove swashplate link assembly (19).

1.6. Repeat Step 1.1. through Step 1.5. to remove opposite swashplate link assembly.

2. Remove swashplate link with uniball Teflon lined bearing as follows:

2.1. Remove cotter pin (50, detail E), nut (49), interrupter bracket (48), bolt (9), bushing (47), and washer (10). Discard cotter pin (50).

2.2. Remove cotter pin (43, detail D), nut (42), washer (41), short bushing (44), long bushing (39), and bracket (40).

2.3. Remove swashplate link assembly (45).

2.4. Repeat Step 2.1. through Step 2.3. to remove opposite swashplate link assembly.

Applicable: SN: 33001-33107:

3. Remove third swashplate link as follows:

NOTE

Step 3.1. and Step 3.2. are for swashplate links with elastomeric rod end bearings installed. For swashplate links with uniball rod end bearings, perform Step 2.2. from above in lieu of Step 1.1. and Step 1.2..

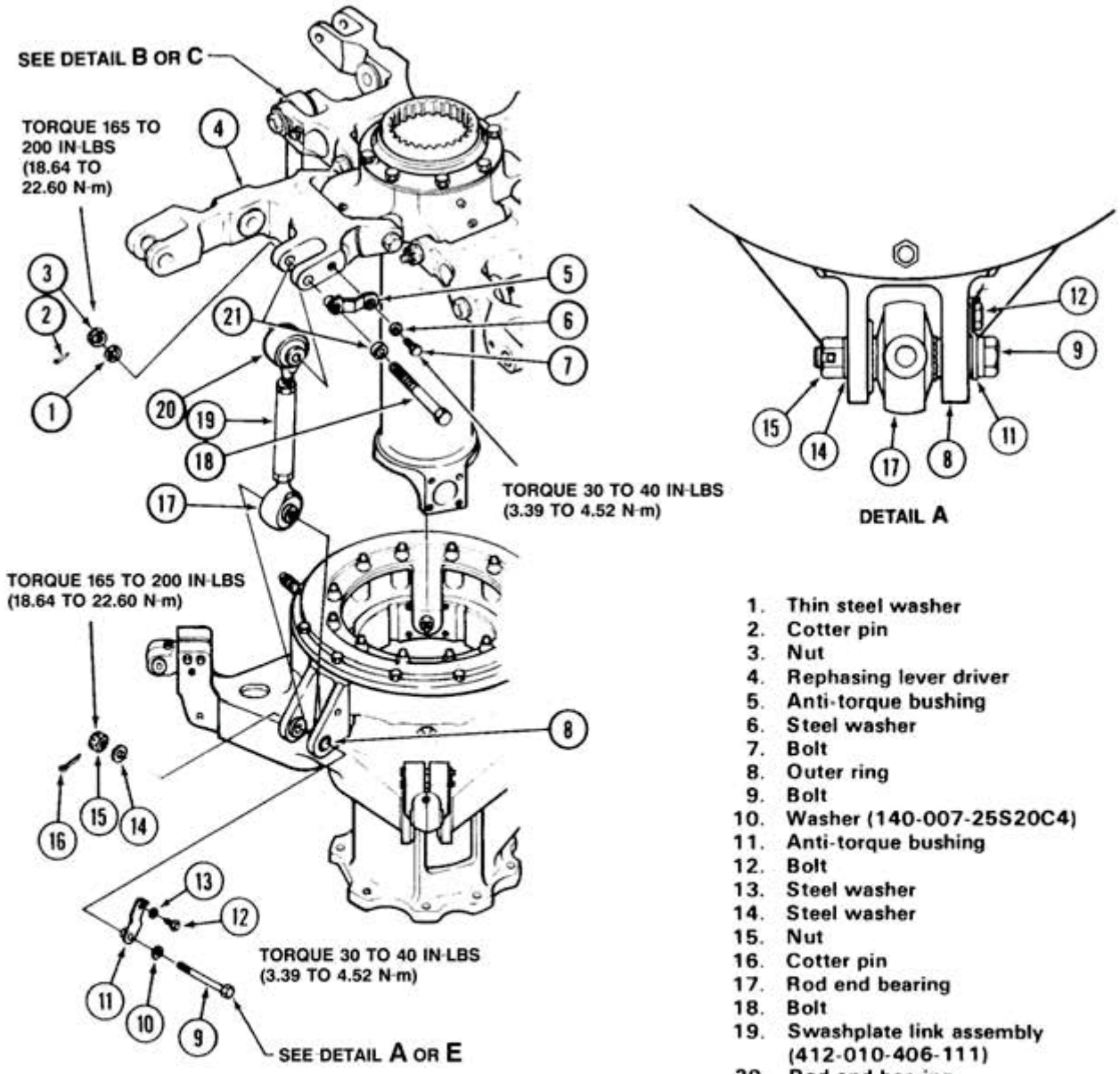
3.1. Cut and remove lockwire from bolt (25) and remove bolt (25) and washer (24).

3.2. Remove cotter pin (36), nut (37), washer (35), bolt (31) with washer (34) under bolt head, and anti-torque bushing (23).

3.3. Remove cotter pin (29) from nut (30) and remove nut (30) with safety washer (28).

3.4. Remove swashplate link assembly (32).

3.5. Remove washer (27) from stud of outer ring (26).



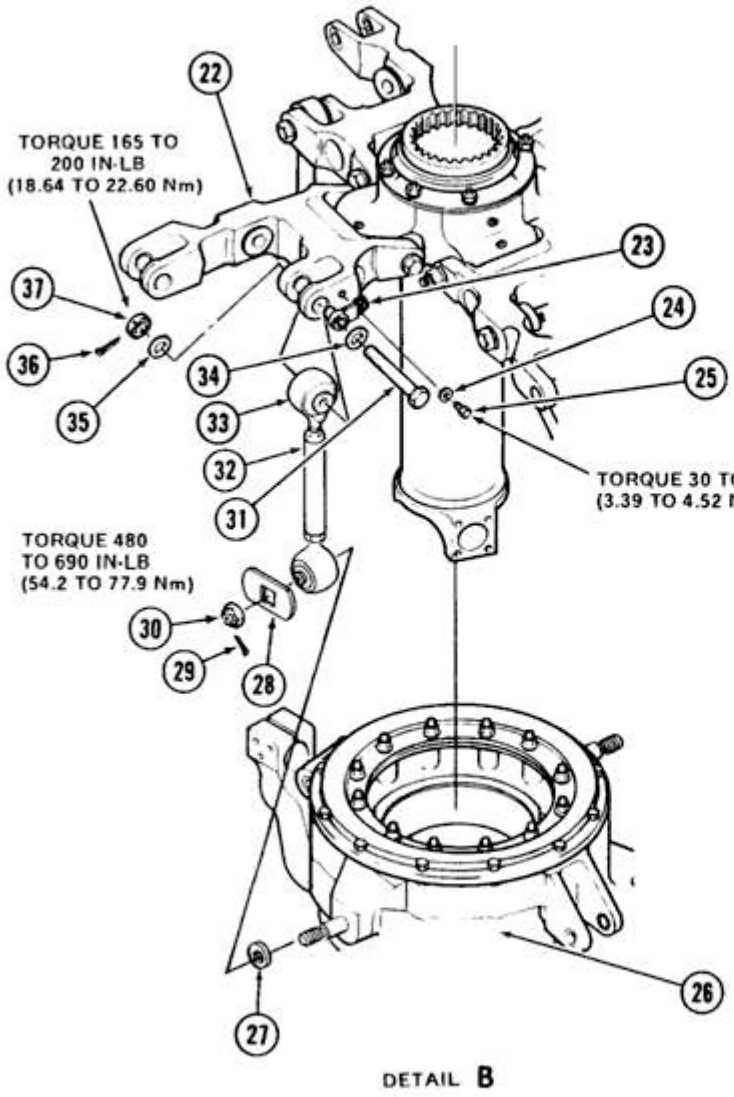
1. Thin steel washer
2. Cotter pin
3. Nut
4. Rephasing lever driver
5. Anti-torque bushing
6. Steel washer
7. Bolt
8. Outer ring
9. Bolt
10. Washer (140-007-25S20C4)
11. Anti-torque bushing
12. Bolt
13. Steel washer
14. Steel washer
15. Nut
16. Cotter pin
17. Rod end bearing
18. Bolt
19. Swashplate link assembly (412-010-406-111)
20. Rod end bearing
21. Washer (140-007-25S20C4)

NOTES:

Install rod end bearings (17, 20 and 33) with serrated faces toward bushings (5, 11 and 23) as shown in detail A.

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Figure 1. Swashplate Link (Sheet 1 of 3)

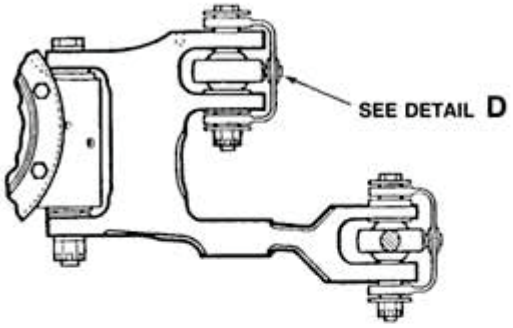


DETAIL B

22. Rephasing lever driver
23. Anti-torque bushing
24. Steel washer
25. Bolt
26. Outer ring
27. Washer (140-009D41S60)
28. Safety washer
29. Cotter pin
30. Nut
31. Bolt
32. Swashplate link (412-010-406-113)
33. Rod end bearing
34. Washer (140-007-25S20C4)
35. Steel washer
36. Cotter pin
37. Nut

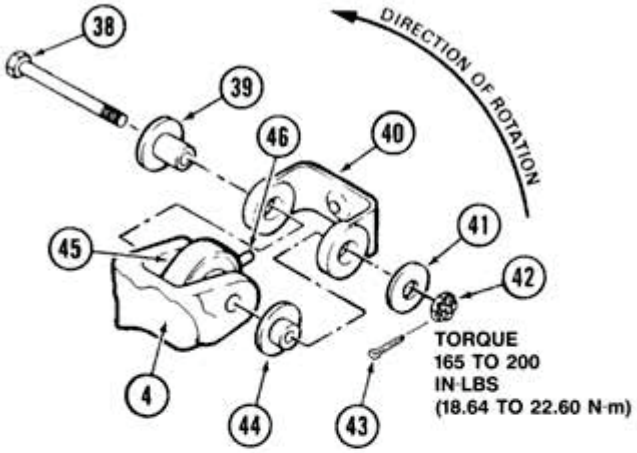
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Figure 1. Swashplate Link (Sheet 2 of 3)

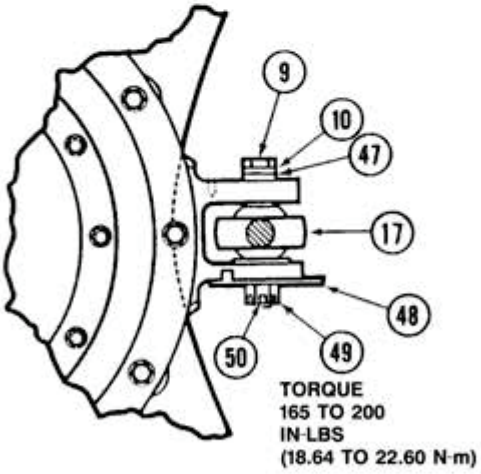


LOOKING DOWN ON
REPHASING LEVER
UPPER END OF LINK
(2 PLACES)

DETAIL C



DETAIL D
(2 PLACES)



LOOKING DOWN ON SWASHPLATE
LOWER END OF LINK
(2 PLACES)

DETAIL E

38. Bolt, 20-057-6-52D
39. Long bushing, 412-010-461-105
40. Bracket, 412-310-401-101
41. Washer, 140-007-25S40C4
42. Nut, MS14144L6
43. Cotter pin, MS24665-302
44. Short bushing, 412-010-461-101
45. Swashplate link, 412-010-406-115
46. Anti-rotation pin (ref)
47. Bushing, 412-010-462-101
48. Interrupter bracket (reuse)
49. Nut, MS14144L6
50. Cotter pin, MS24665-302

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Figure 1. Swashplate Link (Sheet 3 of 3)

Swashplate Link Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Corrosion preventive compound	C-104	AR	
Lockwire	C-405	AR	

Procedure

NOTE

A thin coat of corrosion preventive compound (C-104) should be applied to shanks and under bolt heads of all rotating control bolts prior to installation.

A thin coat of corrosion preventive compound (C-101) should be applied to all rotating control nuts, washers, and bolt heads after installation.

1. If swashplate link assembly (19 or 32, Figure 1) is to be installed in same position it was removed from without changing dimensions, proceed to Step 3..



IF NEW OR REPLACEMENT LINK ASSEMBLY IS BEING INSTALLED, COMPLIANCE WITH INSTRUCTIONS IN Step 2.. IS MANDATORY TO ENSURE ADEQUATE ROD END BEARING THREAD ENGAGEMENT IS MAINTAINED.

2. If new or replacement swashplate link assembly (19 or 32) is to be installed, adjust link to dimensions outlined in BHT-412-CR&O.

3. To install swashplate link assembly with elastomeric bearing proceed as follows:

3.1. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt

3.2. Position bushing (5) in rephrasing lever driver (4) and loosely install bolt (7) with washer (6).

3.3. Position swashplate link (19) 90° ± 5° to centerline of rephasing lever driver (4).

3.4. Ensure serrations of anti-torque bushing (5) and swashplate link assembly (19) are properly engaged.

3.5. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (18).

3.6. Install bolt (18) with washer (21) under bolt head and washer (1) and nut (3). Do not torque at this time.

3.7. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (12).

3.8. Position anti-torque bushing (11) in outer ring (8) and loosely install bolt (12) and washer (13).

3.9. Position swashplate link driver (19) in outer ring (8).

3.10. Ensure serrations of bushing (11) and link (19) are properly engaged.

3.11. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (9).

3.12. Install bolt (9) with washer (10) under bolt head, washer (14), and nut (15).

3.13. Torque nut (3) 165 to 200 inch-pounds (18.64 to 22.60 Nm) and install cotter pin (2).

3.14. Torque bolt (7) 30 to 40 inch-pounds (3.39 to 4.52 Nm). Secure bolt (7) to bushing (5) using lockwire (C-405).

3.15. Torque nut (15) 165 to 200 inch-pounds (18.64 to 22.60 Nm), and install cotter pin (16).

3.16. Torque bolt (12) 30 to 40 inch-pounds (3.39 to 4.60 Nm). Secure bolt (12) to bushing (11) using lockwire (C-405).

3.17. Apply a thin coat of corrosion preventive compound (C-101) to nut (3) and washer (1), head of bolt (7) and washer (6), nut (15) and washer (14), and head of bolt (12) and washer (13).

3.18. Repeat Step 3.1. through Step 3.17. to install opposite swashplate link assembly.

4. To install swashplate link with uniball Teflon lined bearing proceed as follows:

NOTE

The upper end of swashplate link assembly (45) incorporating anti-rotation pin (46) rod end must be installed first.

4.1. Position short bushing (44) in bracket (40).

4.2. Position bracket (40) at end of rephasing lever driver (4) arm in alignment with swashplate link (45) anti-rotation pin (46).

NOTE

Long bushing (39) passes through bracket (40), rephasing lever driver (4) bushings, and bottoms against swashplate link (45) upper rod end.

4.3. Install long bushing (39) into side of bracket (40) with head of bushing in direction of rotation.

4.4. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (38).

4.5. Install bolt (38) with head in direction of rotation through long bushing (39), swashplate link (45) and short bushing (44). Install washer (41) and nut (42) finger tight.

4.6. Install lower end of swashplate link (45) with head of bolt (9) in direction of rotation.

4.7. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (9).

4.8. Install washer (10) under bolt head (9).

4.9. Slide bushing (47) into swashplate outer ring (8).

4.10. Apply a thin coat of corrosion preventive compound (C-104) under the head and to shank of bolt (9).

4.11. Install bolt (9) with washer (10) through bushing (47), rod end (17), and swashplate outer ring (8).

4.12. Install interrupter bracket (48) and nut (49).

4.13. Torque nut (49) 165 to 200 inch-pounds (18.64 to 22.60 Nm) and secure with cotter pin (50).

4.14. Torque nut (42) 165 to 200 inch-pounds (18.64 to 22.60 Nm) and secure with cotter pin (43).

4.15. Apply a thin coat of corrosion preventive compound (C-101) to nut (49), head of bolt (9) and washer (10), and nut (42) and washer (41).

Applicable: SN: 33001-33107:

5. Install third swashplate link as follows:

5.1. Install washer (27) on stud of outer ring (26).

5.2. Position swashplate link assembly (32) on outer ring (26) and install safety washer (28) and nut (30).

5.3. Torque nut (30) 480 to 690 inch-pounds (54.2 to 77.9 Nm) and install cotter pin (29).

NOTE

Step 5.4. through Step 5.9. are for swashplate links with elastomeric rod end bearings. For swashplate links with uniball rod end bearings, perform Step 4.2. through Step 4.5. in lieu of Step 5.4. through Step 5.9..

5.4. Position anti-torque bushing (23) in rephasing lever driver (22) and loosely install bolt (25) with washer (24).

5.5. Position swashplate link assembly (32) 90° ± 5° to centerline of rephasing lever driver (22).

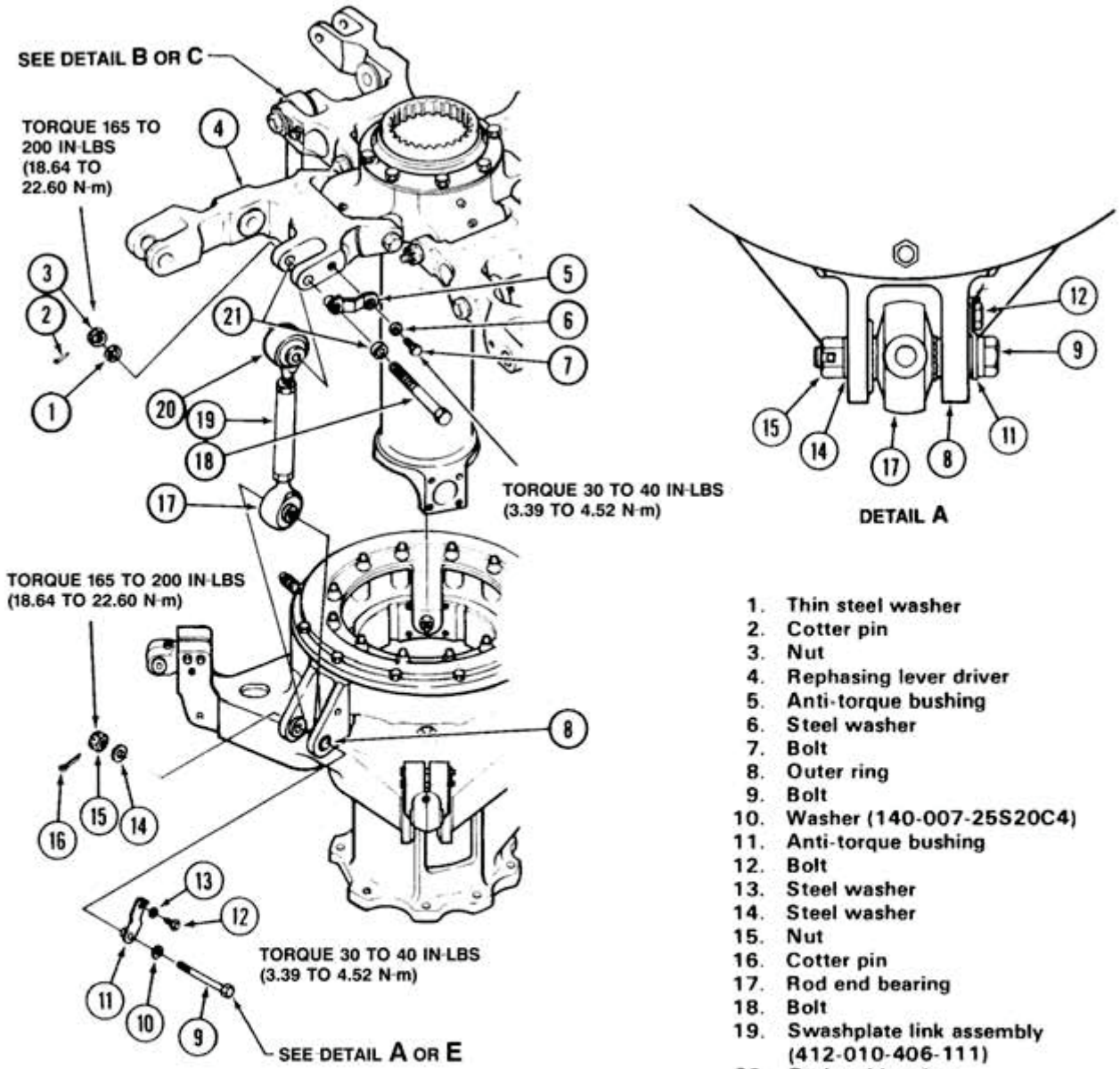
5.6. Ensure serrations of bushing (23) and swashplate link assembly (32) are properly engaged.

5.7. Install bolt (31) with washer (34) through rod end bearing (33).

5.8. Install washer (35) and nut (37) on bolt (31).

5.9. Torque bolt (25) 30 to 40 inch-pounds (3.39 to 4.52 Nm), and secure bolt (25) to bushing (23) using lockwire (C-405).

5.10. Torque nut (37) 165 to 200 inch-pounds (18.64 to 22.60 Nm) and install cotter pin (36).



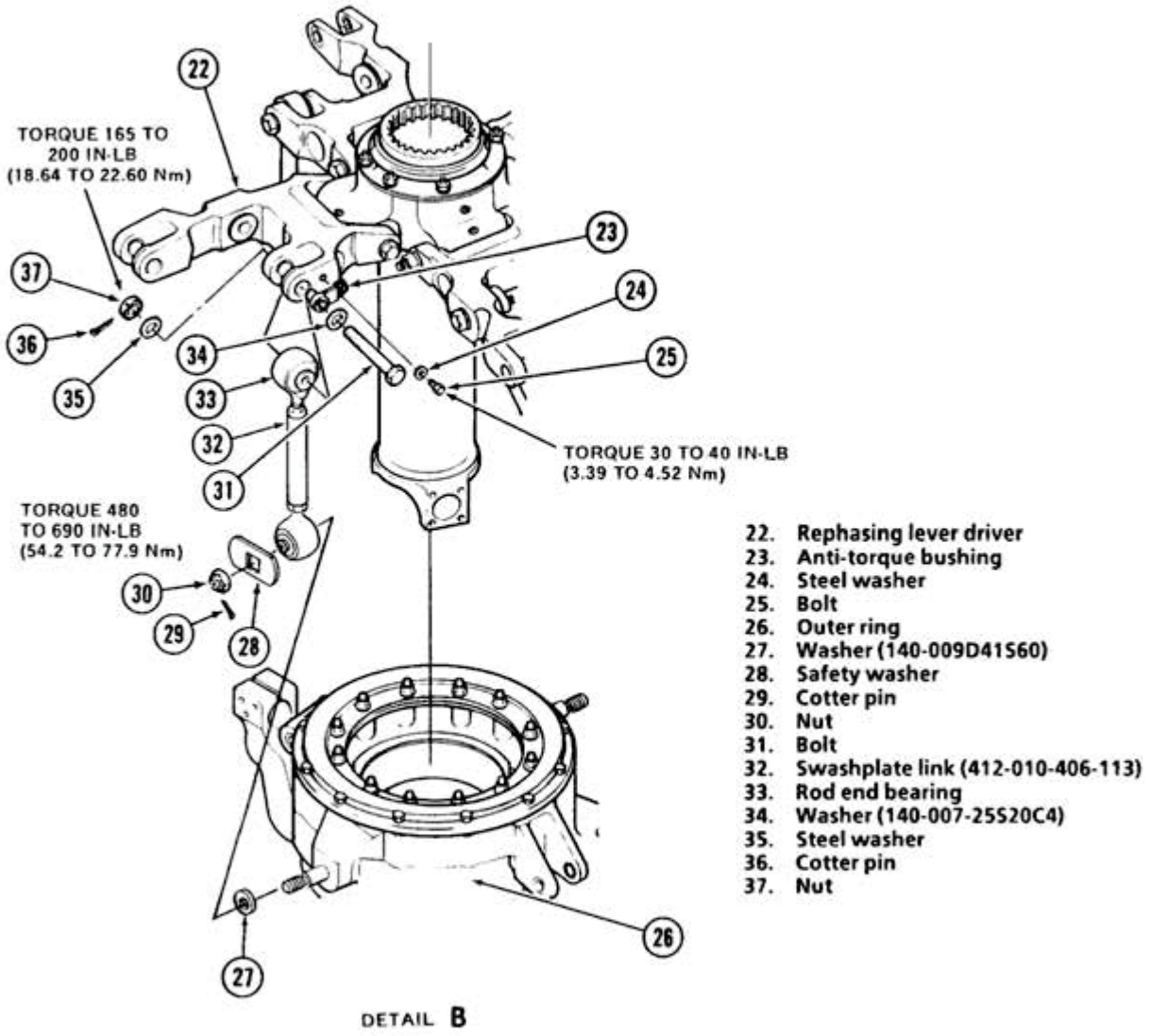
1. Thin steel washer
2. Cotter pin
3. Nut
4. Rephasing lever driver
5. Anti-torque bushing
6. Steel washer
7. Bolt
8. Outer ring
9. Bolt
10. Washer (140-007-25S20C4)
11. Anti-torque bushing
12. Bolt
13. Steel washer
14. Steel washer
15. Nut
16. Cotter pin
17. Rod end bearing
18. Bolt
19. Swashplate link assembly (412-010-406-111)
20. Rod end bearing
21. Washer (140-007-25S20C4)

NOTES:

Install rod end bearings (17, 20 and 33) with serrated faces toward bushings (5, 11 and 23) as shown in detail A.

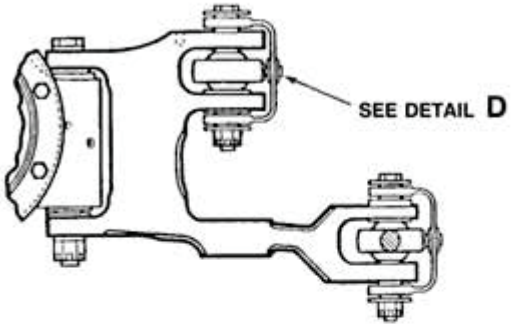
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Figure 1. Swashplate Link (Sheet 1 of 3)



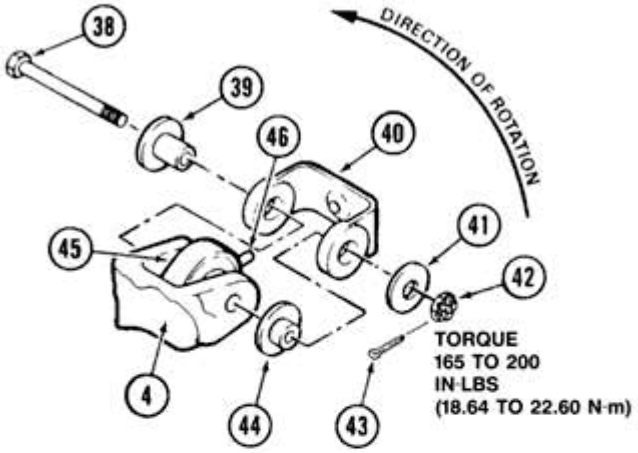
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Figure 1. Swashplate Link (Sheet 2 of 3)

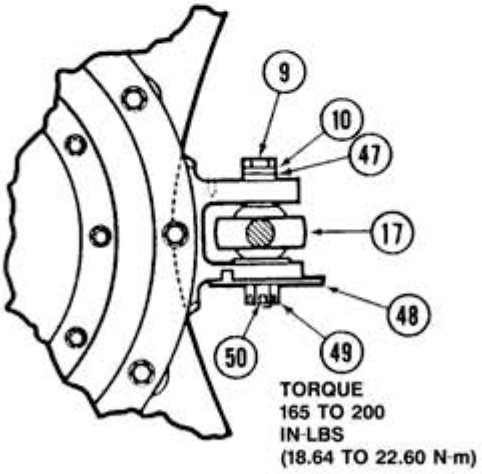


LOOKING DOWN ON
REPHASING LEVER
UPPER END OF LINK
(2 PLACES)

DETAIL C



DETAIL D
(2 PLACES)



LOOKING DOWN ON SWASHPLATE
LOWER END OF LINK
(2 PLACES)

DETAIL E

38. Bolt, 20-057-6-52D
39. Long bushing, 412-010-461-105
40. Bracket, 412-310-401-101
41. Washer, 140-007-25S40C4
42. Nut, MS14144L6
43. Cotter pin, MS24665-302
44. Short bushing, 412-010-461-101
45. Swashplate link, 412-010-406-115
46. Anti-rotation pin (ref)
47. Bushing, 412-010-462-101
48. Interrupter bracket (reuse)
49. Nut, MS14144L6
50. Cotter pin, MS24665-302

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Figure 1. Swashplate Link (Sheet 3 of 3)

Collective Pitch Drive Plate Set - Removal

Procedure

NOTE

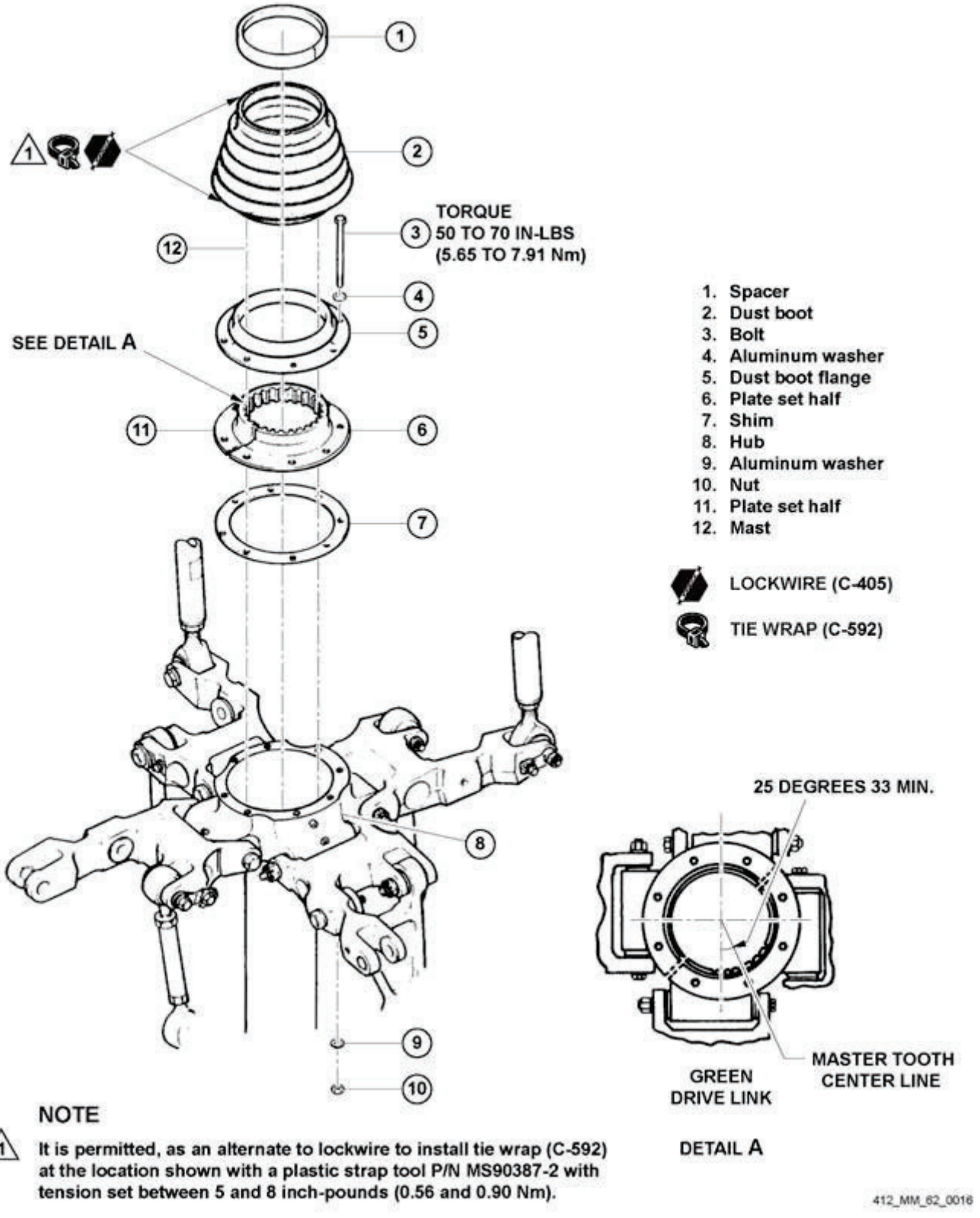
The collective pitch drive plate set is a serial numbered, matched set.

1. Cut and remove lockwire or tie wrap, as applicable, from dust boot (2, Figure 1) and remove dust boot (2) and spacer (1).
2. Remove eight nuts (10), aluminum washers (9), bolts (3), and aluminum washers (4).
3. Remove dust boot flange (5).



EXERCISE CARE WHEN REMOVING PLATE SET HALVES (6 AND 11) OR SHIM (7) TO PREVENT DAMAGE.

4. Remove plate set halves (6 and 11).
5. If collective pitch drive plate set is not to be installed immediately, secure shim (7) to hub (8).



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Figure 1. Collective Pitch Drive Plate Set (Sheet 1 of 1)

Collective Pitch Drive Plate Set - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Plastic Strap Tool	MS90387-2	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	
Corrosion preventive compound	C-104	AR	
Lockwire	C-405	AR	
Tie wrap	C-592	AR	Use tie wrap MS3367-6-9.

Procedure



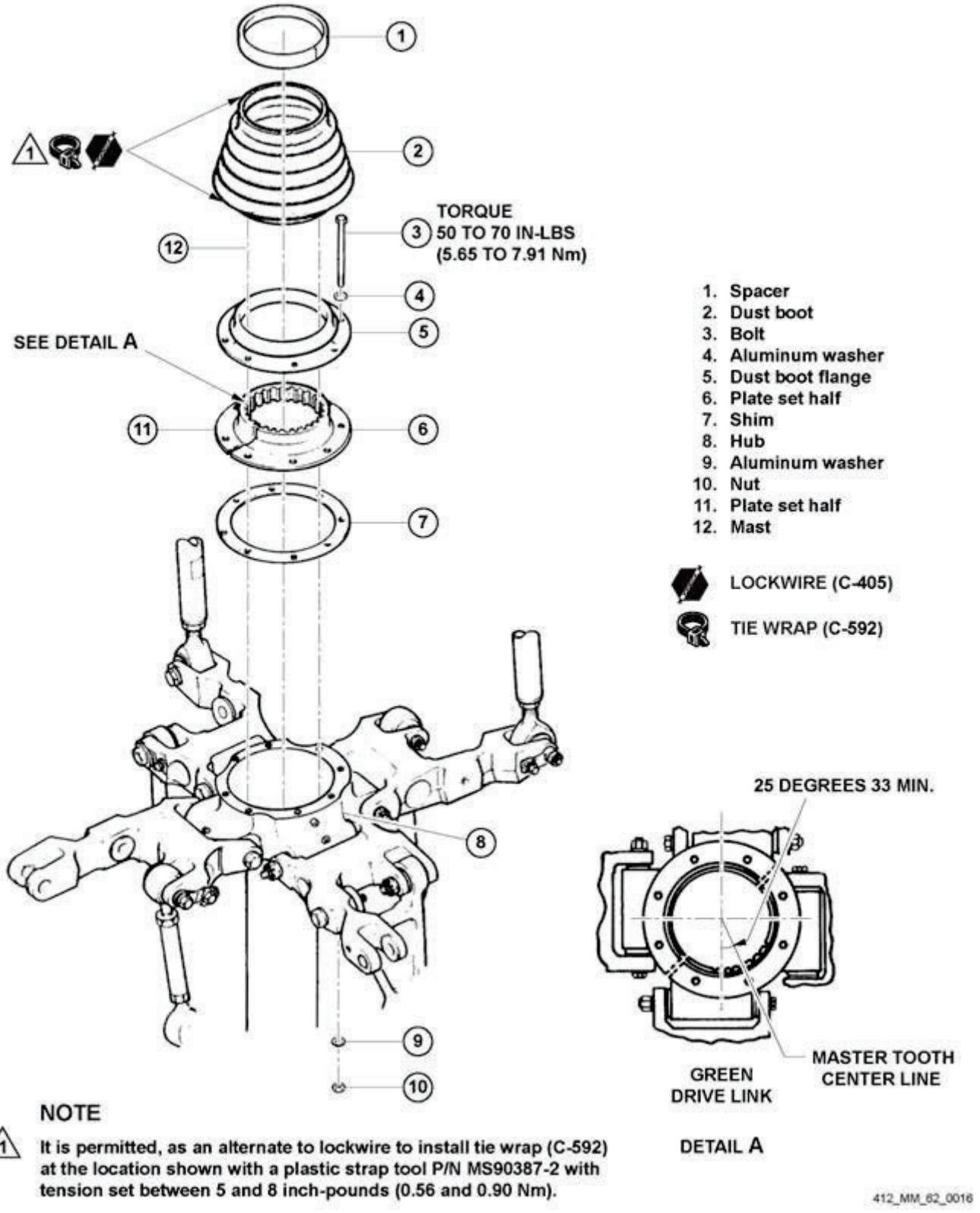
SHIM (7, Figure 1) SHALL BE OF CORRECT THICKNESS TO GIVE 0.003 TO 0.005 INCH (0.08 TO 0.13 MM) PINCH ON HUB BEARINGS. IF IT IS NOT KNOWN WHETHER SHIM IS OF CORRECT THICKNESS, REFER TO BHT-412-CR&O TO ESTABLISH SHIM THICKNESS.

NOTE

A thin coat of corrosion preventive compound (C-104) should be applied to shanks and under bolt heads of all rotating control bolts prior to installation.

A thin coat of corrosion preventive compound (C-101) should be applied to all rotating control nuts, washers, and bolt heads after there installation.

1. Position plate set halves (6 and 11) on hub (8) and align as shown on Detail A.
2. Position dust boot flange (5) on plate set halves.
3. Apply a thin coat of corrosion preventive compound (C-104) under the heads and to shanks of bolts (3).
4. Install bolts (3) with washers (4) under the heads of bolts, washers (9), and nuts (10). Torque nuts 50 to 70 inch-pounds (5.65 to 7.91 Nm).
5. Apply a thin coat of corrosion preventive compound (C-101) to nut (49), head of bolts (3) and washers (4), and nuts (10) and washers (9).
6. Install boot (2) and spacer (1). Secure boot to mast (12) and boot flange (5) with lockwire (C-405) or tie wrap (C-592), as applicable, with a plastic strap tool (MS90387-2) with tension adjusted between 5 and 8.



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Figure 1. Collective Pitch Drive Plate Set (Sheet 1 of 1)

Collective Lever Assembly - Removal

Procedure

1. Remove cotter pin (12, Figure 1), nut (13), washer (10), washer (11), bolt (7), washer (8) and separate clevis (9) from trunnion (6). Discard cotter pin (12).

2. Cut and remove lockwire from bolt (17) and remove bolt and washer (18).

NOTE

When removing shims (20 or 23), reinstall shims in the same location and order of removal.

3. Remove collective lever pin (19) and shims (20 or 23).

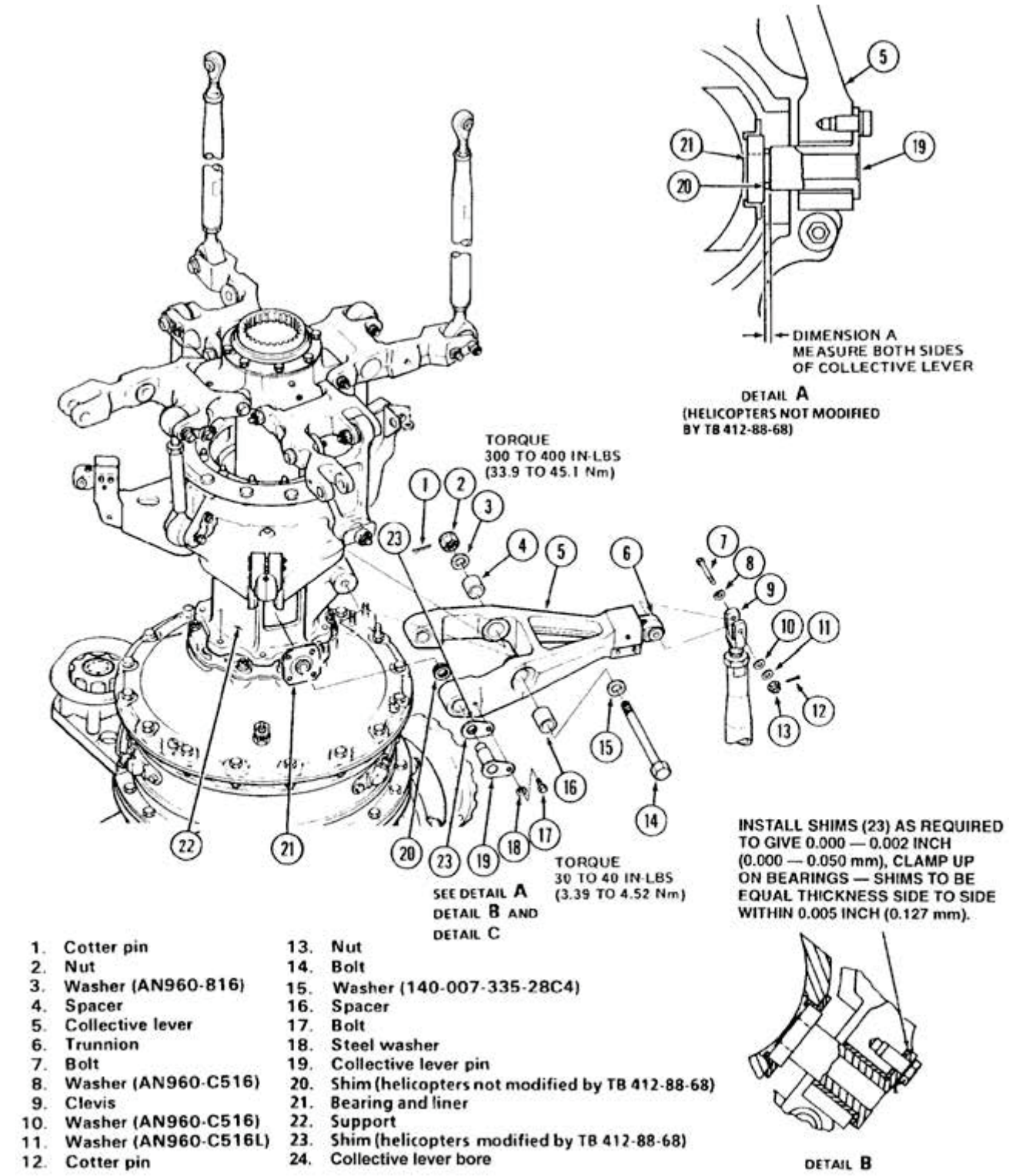
4. Repeat Step 2. and Step 3. to remove opposite collective lever pin (19).

NOTE

When removing washers (3 and 15) and spacers (4 and 16), reinstall in the same location and order of removal.

5. Remove cotter pin (1), nut (2), washer (3), bolt (14), washer (15), spacers (4 and 16). Discard cotter pin.

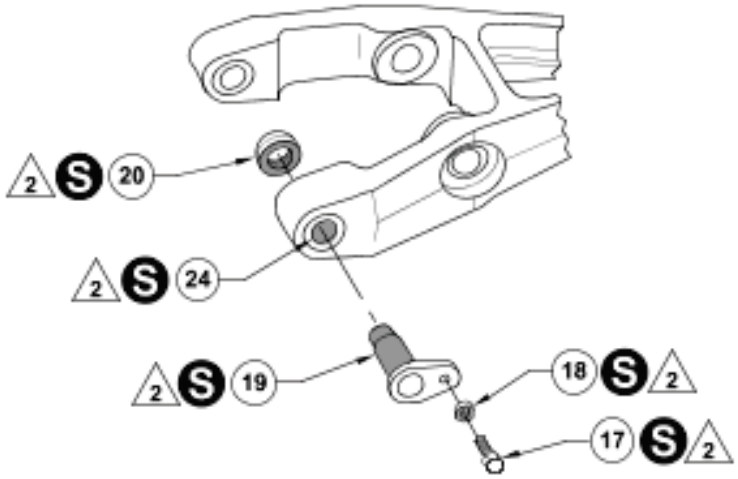
6. Carefully remove collective lever assembly (5) from support (22).



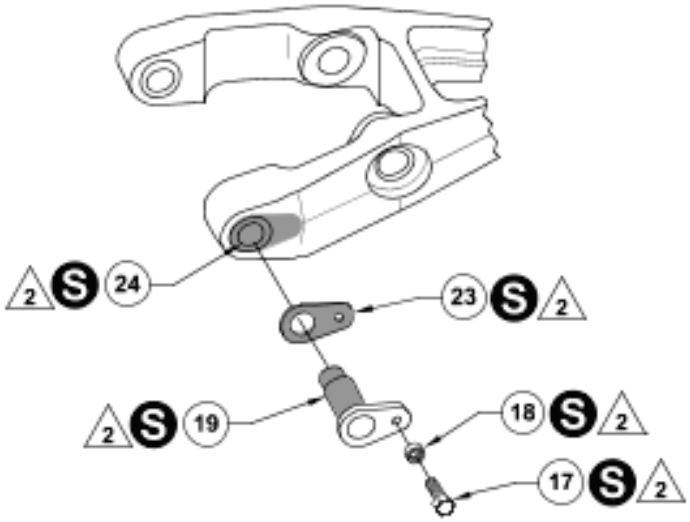
1. Cotter pin
2. Nut
3. Washer (AN960-816)
4. Spacer
5. Collective lever
6. Trunnion
7. Bolt
8. Washer (AN960-C516)
9. Clevis
10. Washer (AN960-C516)
11. Washer (AN960-C516L)
12. Cotter pin
13. Nut
14. Bolt
15. Washer (140-007-335-28C4)
16. Spacer
17. Bolt
18. Steel washer
19. Collective lever pin
20. Shim (helicopters not modified by TB 412-88-68)
21. Bearing and liner
22. Support
23. Shim (helicopters modified by TB 412-88-68)
24. Collective lever bore

ICN-412-A-623000-B-97499-00040-A-001-01

Figure 1. Collective Lever (Sheet 1 of 2)



PRE TB 412-88-68
S/N 33001 THROUGH 33167
S/N 34001 THROUGH 34024



POST TB 412-88-68
S/N 33168 AND SUBSEQUENT
S/N 34025 AND SUBSEQUENT

DETAIL C

S SEALANT (C-352)

NOTES

1. Ensure that no sealant is allowed to enter the bearing.
2. Apply sealant MIL-R-46082, type I (C-352) to faying surfaces as indicated by shading.

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Figure 1. Collective Lever (Sheet 2 of 2)

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Collective Lever Assembly - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

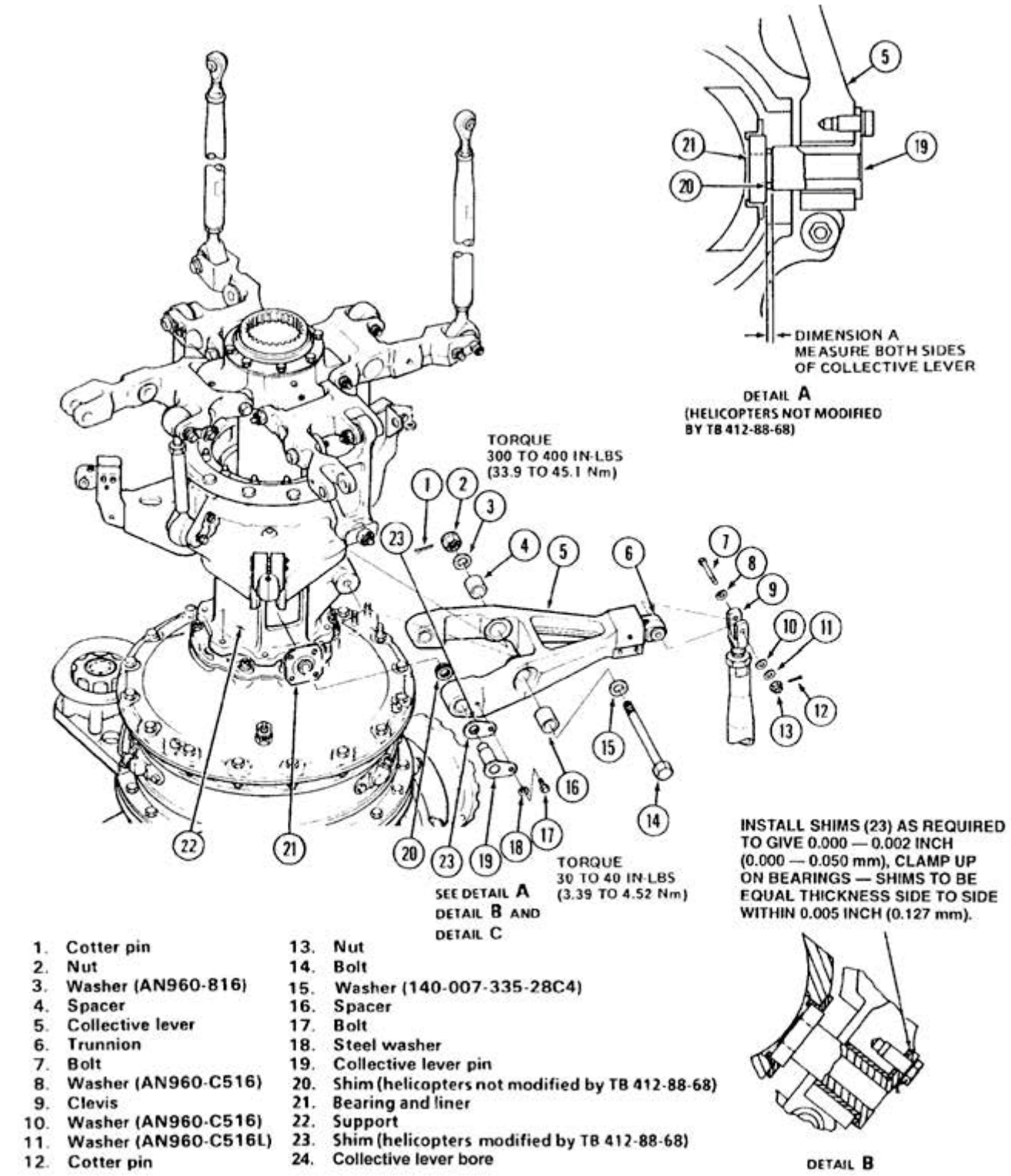
Procedure



DO NOT IMMERSE COLLECTIVE LEVER ASSEMBLY IN SOLVENT.

1. Carefully clean collective lever (5, Figure 1) with clean cloth damp in drycleaning solvent (C-304).

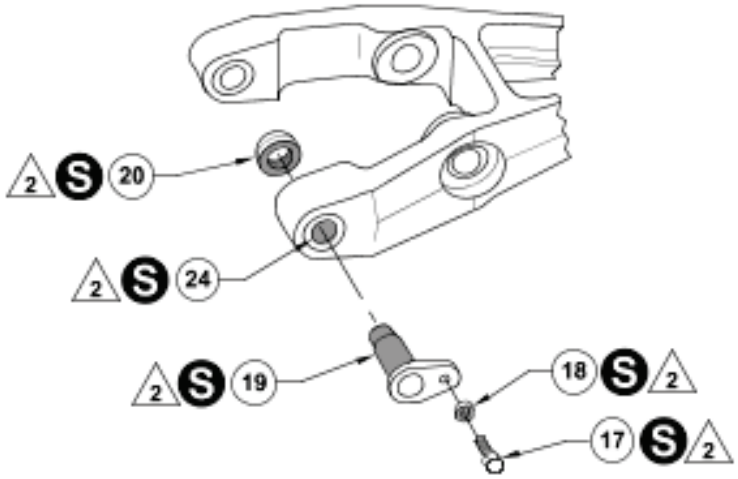
2. Dry collective lever (5) with filtered, compressed air.



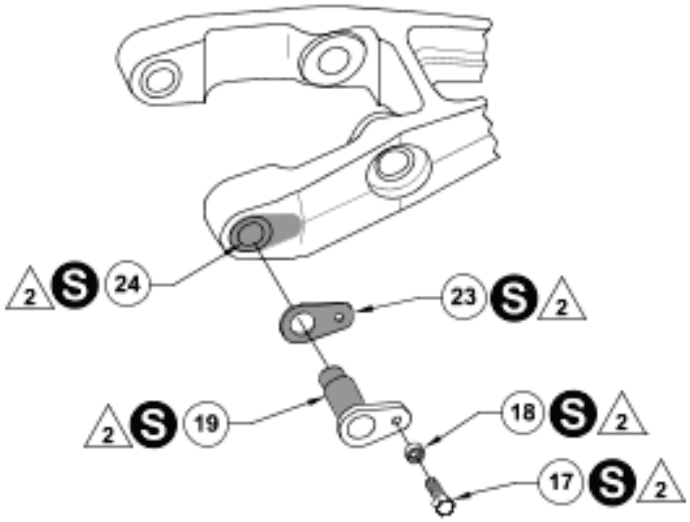
1. Cotter pin
2. Nut
3. Washer (AN960-816)
4. Spacer
5. Collective lever
6. Trunnion
7. Bolt
8. Washer (AN960-C516)
9. Clevis
10. Washer (AN960-C516)
11. Washer (AN960-C516L)
12. Cotter pin
13. Nut
14. Bolt
15. Washer (140-007-335-28C4)
16. Spacer
17. Bolt
18. Steel washer
19. Collective lever pin
20. Shim (helicopters not modified by TB 412-88-68)
21. Bearing and liner
22. Support
23. Shim (helicopters modified by TB 412-88-68)
24. Collective lever bore

ICN-412-A-623000-B-97499-00040-A-001-01

Figure 1. Collective Lever (Sheet 1 of 2)



PRE TB 412-88-68
S/N 33001 THROUGH 33167
S/N 34001 THROUGH 34024



POST TB 412-88-68
S/N 33168 AND SUBSEQUENT
S/N 34025 AND SUBSEQUENT

DETAIL C

S SEALANT (C-352)

NOTES

1. Ensure that no sealant is allowed to enter the bearing.
2. Apply sealant MIL-R-46082, type I (C-352) to faying surfaces as indicated by shading.

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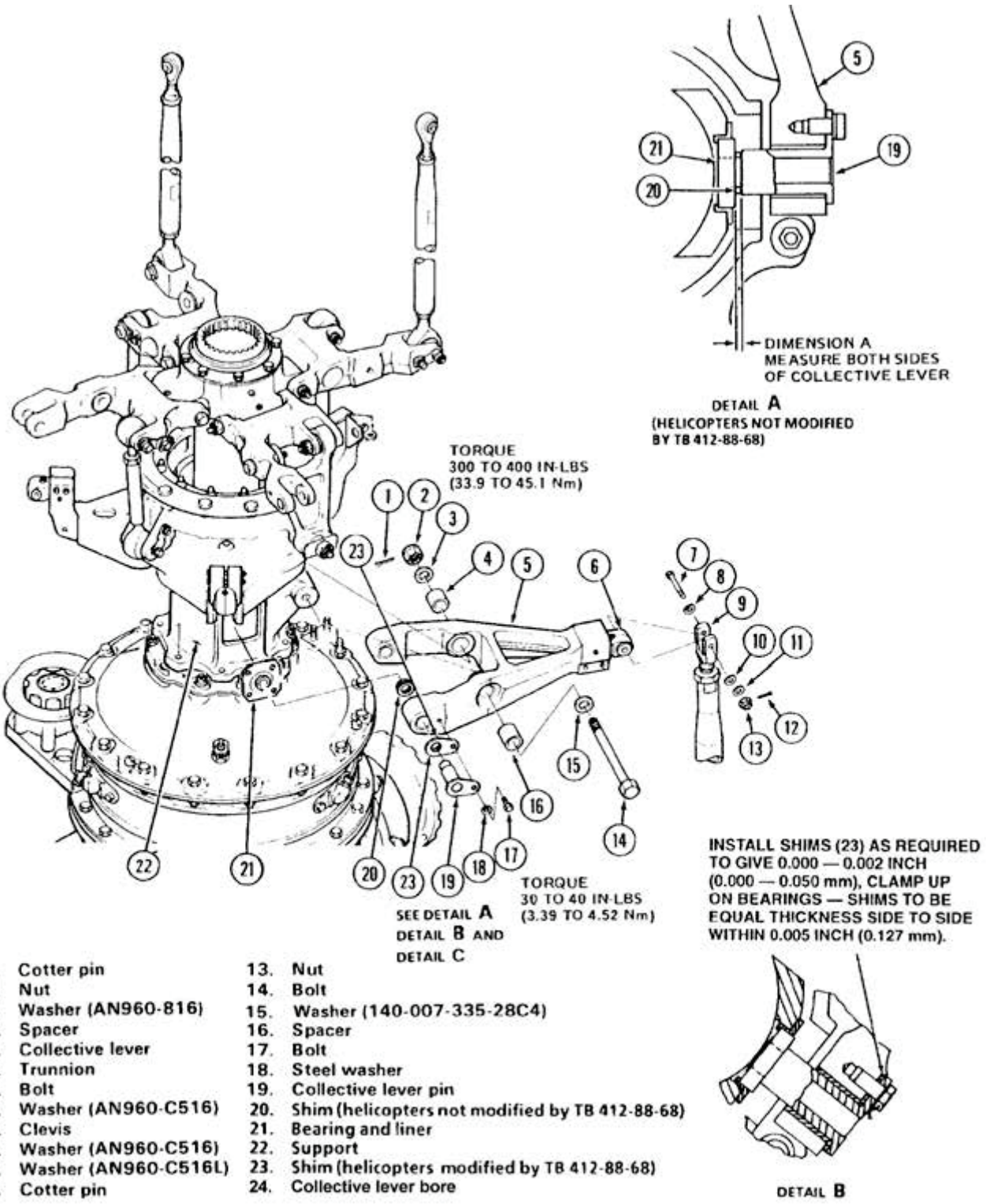
Figure 1. Collective Lever (Sheet 2 of 2)

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Collective Lever Assembly - Inspection

Procedure

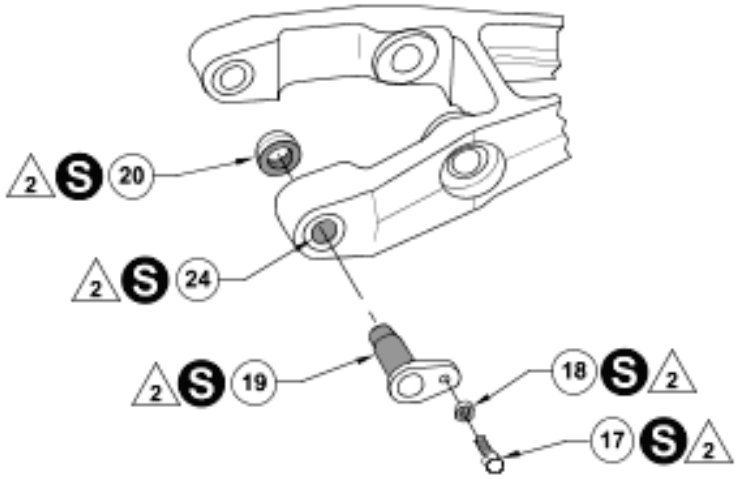
1. Inspect collective lever pins (19, Figure 1) for damage, cracks, and distortion. Damage in excess of superficial marks are not acceptable.



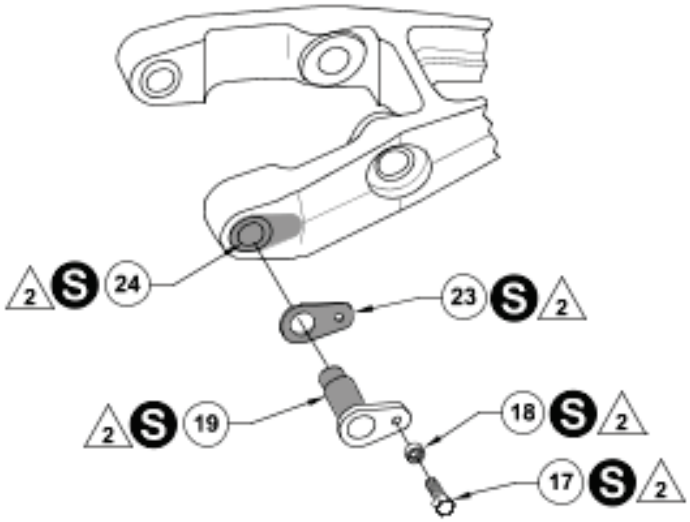
1. Cotter pin
2. Nut
3. Washer (AN960-816)
4. Spacer
5. Collective lever
6. Trunnion
7. Bolt
8. Washer (AN960-C516)
9. Clevis
10. Washer (AN960-C516)
11. Washer (AN960-C516L)
12. Cotter pin
13. Nut
14. Bolt
15. Washer (140-007-335-28C4)
16. Spacer
17. Bolt
18. Steel washer
19. Collective lever pin
20. Shim (helicopters not modified by TB 412-88-68)
21. Bearing and liner
22. Support
23. Shim (helicopters modified by TB 412-88-68)
24. Collective lever bore

ICN-412-A-623000-B-97499-00040-A-001-01

Figure 1. Collective Lever (Sheet 1 of 2)



PRE TB 412-88-68
S/N 33001 THROUGH 33167
S/N 34001 THROUGH 34024



POST TB 412-88-68
S/N 33168 AND SUBSEQUENT
S/N 34025 AND SUBSEQUENT

DETAIL C

S SEALANT (C-352)

NOTES

1. Ensure that no sealant is allowed to enter the bearing.
2. Apply sealant MIL-R-46082, type I (C-352) to faying surfaces as indicated by shading.

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Figure 1. Collective Lever (Sheet 2 of 2)

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Collective Lever Assembly - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Epoxy coating	C-210	AR	
Abrasive cloth or paper	C-423	AR	

Procedure

1. Polish out mechanical and corrosion damage with fine to medium grades of abrasive cloth or paper (C-423). Do not exceed limits defined in BHT-412-CR&O.
2. Apply chemical film material to all bare aluminum areas.
3. Apply epoxy coating (C-210) to areas where metal is exposed.
4. Touch up paint to match existing finish.
5. Replace trunnion which does not meet requirements defined in BHT-412-CR&O.

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Collective Lever Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Denatured alcohol	C-326	AR	
Sealant	C-352	AR	
Lockwire	C-405	AR	

Procedure

NOTE

It is possible to install collective lever (5, Figure 1) in inverted position. Ensure collective lever (5) is installed with slotted area for installation of trunnion (6) facing down as illustrated.

1. Install collective lever (5, Figure 1) on support (22) as follows:

2. Install bolt (14) with washer (15) under bolt head, and then install bushing (16).

3. Install bushing (4), washer (3), and nut (2) on bolt (14).

4. Torque nut (2) 300 to 400 inch-pounds (33.9 to 45.1 N-m). Install cotter pin (1).

NOTE

If shims (20, Figure 1) have been set aside and are to be installed in same location as removed, proceed to Step 11.. If new shims (20) are to be installed, accomplish Step 6. through Step 10..

5. Position collective lever pins (19), without shims (23), through either side of collective lever (5) into bearing (21).

6. Temporally install bolts (17) and washers (18).

7. Push collective lever (5) to one side and hold pressure.

8. Measure gap between bearing and liner (21) and shoulder of collective liver pin.

9. Prepare two shims (20) of equal thickness to within 0.005 inch (0.13 mm). Total thickness of two shims shall be equal to value recorded in Step 8., add 0 to 0.004 inch (0.1 mm). Maintaining these dimensions will result in a 0.000 to 0.002 inch (0.00 to 0.05 mm) clamp up on each bearing when pins are installed.

10. Remove collective lever pin (19).

11. Clean lever bores (24), inside diameter of bearing (21) and outside diameter of pins (19) with denatured alcohol (C-326).



ENSURE THAT NO SEALANT IS ALLOWED TO ENTER BEARINGS.

12. Refer to detail C. Coat all faying surfaces of shims (20), pins (19), lever bores (24), washers (18) and bolts (17) with sealant (C-352).

NOTE

Accomplish following steps while sealant (C-352) is still wet.

13. Position collective lever pin (19) in each side of collective lever (5). Position shim (20), indexed at time of removal, or new shim prepared at Step 9., on each pin (19).

14. Carefully position collective lever (5), collective lever pins (19) and shims (20) on swashplate support assembly (22).

15. Ensure shim (20) is in position and insert pin (19) into bearing (21).

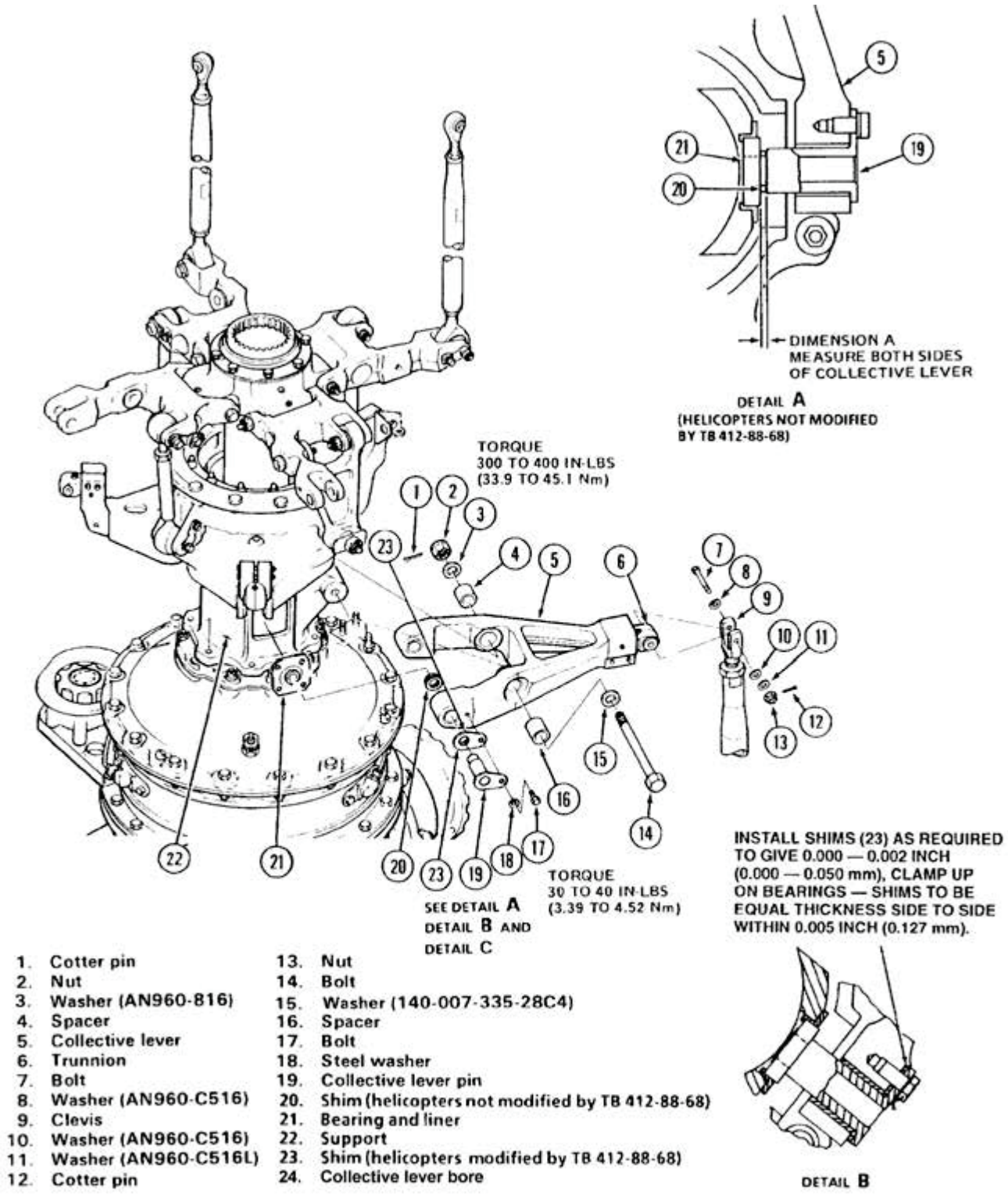
16. Install bolt (17) with washer (18), and torque bolt (17) 30 to 40 inch-pounds (3.39 to 4.52 Nm).

17. Secure bolt (17) to pin (19) using lockwire (C-405).

18. Install opposite collective lever pin (19) in same manner.

19. Position clevis (9) on trunnion (6), and install bolt (7) with washer (8) under bolt head.

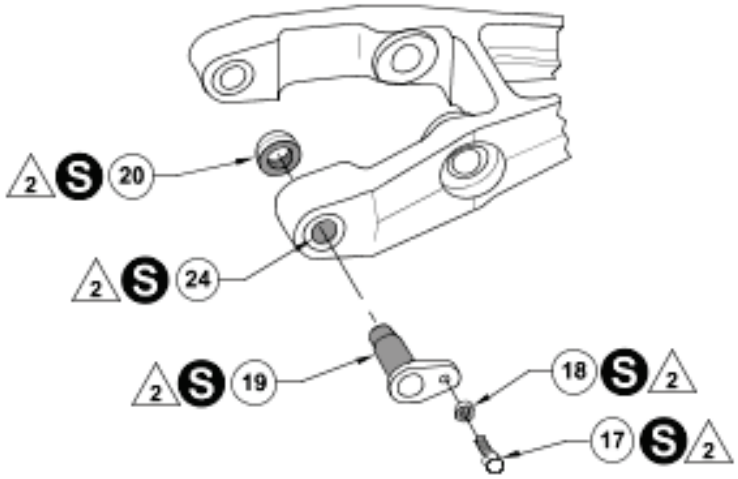
20. Install nut (13) with washers (10 and 11) as required to attain proper thread engagement for cotter pin (12), and install cotter pin (12).



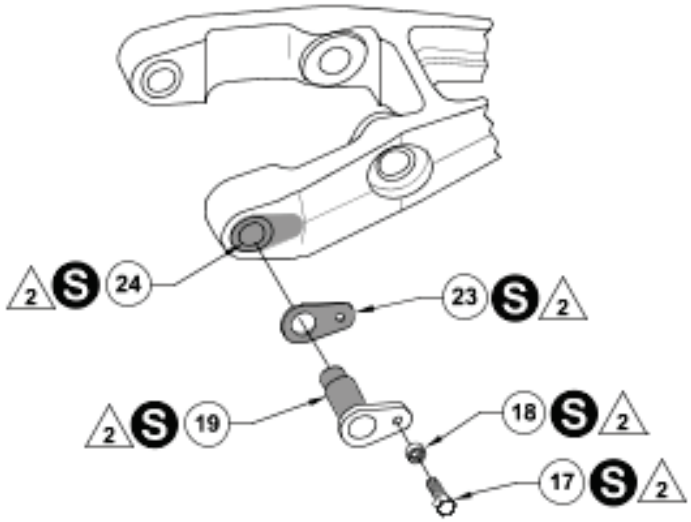
1. Cotter pin
2. Nut
3. Washer (AN960-816)
4. Spacer
5. Collective lever
6. Trunnion
7. Bolt
8. Washer (AN960-C516)
9. Clevis
10. Washer (AN960-C516)
11. Washer (AN960-C516L)
12. Cotter pin
13. Nut
14. Bolt
15. Washer (140-007-335-28C4)
16. Spacer
17. Bolt
18. Steel washer
19. Collective lever pin
20. Shim (helicopters not modified by TB 412-88-68)
21. Bearing and liner
22. Support
23. Shim (helicopters modified by TB 412-88-68)
24. Collective lever bore

ICN-412-A-623000-B-97499-00040-A-001-01

Figure 1. Collective Lever (Sheet 1 of 2)



PRE TB 412-88-68
S/N 33001 THROUGH 33167
S/N 34001 THROUGH 34024



POST TB 412-88-68
S/N 33168 AND SUBSEQUENT
S/N 34025 AND SUBSEQUENT

DETAIL C

S SEALANT (C-352)

NOTES

1. Ensure that no sealant is allowed to enter the bearing.
2. Apply sealant MIL-R-46082, type I (C-352) to faying surfaces as indicated by shading.

ICN-412-A-623000-B-97499-00041-A-001-01

Figure 1. Collective Lever (Sheet 2 of 2)

Collective Lever Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Denatured alcohol	C-326	AR	
Sealant	C-352	AR	
Lockwire	C-405	AR	

Procedure

NOTE

It is possible to install collective lever (5, Figure 1) in inverted position. Ensure collective lever (5) is installed with slotted area for installation of trunnion (6) facing down as illustrated.

1. Install collective lever (5, Figure 1) on support (22) as follows:

2. Install bolt (14) with washer (15) under bolt head, and then install bushing (16).

3. Install bushing (4), washer (3), and nut (2) on bolt (14).

4. Torque nut (2) 300 to 400 inch-pounds (33.9 to 45.1 Nm). Install cotter pin (1).

5. Position collective lever pins (19), without shims (23), through either side of collective lever (5) into bearing (21).

6. Install bolts (17) and washers (18) finger tight.

7. Push collective lever (5) to one side and hold pressure.

8. Measure distance between collective lever (5) and hold down flange of collective lever pins (19). Record dimension and remove collective lever pins.

9. Prepare two shims (23, Detail B) of equal thickness to within 0.005 inch (0.13 mm). Total thickness of two shims shall be equal to value recorded in Step 8. minus up to 0.004 inch (0.10 mm). Maintaining these dimensions will result in a 0.000 to 0.002 inch (0.00 to 0.05 mm) clamp up on each bearing when pins are installed.

10. Clean lever bores (24), inside diameter of bearing (21) and outside diameter of pins (19) with denatured alcohol (C-326).



MAKE SURE THAT NO SEALANT IS ALLOWED TO ENTER BEARINGS.

11. Refer to Detail C. Coat all faying surfaces of shims (23), pins (19), lever bores (24), washers (18) and bolts (17) with sealant (C-352).

NOTE

Accomplish the following steps while sealant (C-352) is still wet.

12. Install shims (23) prepared in Step 9. equally on pins (19) and install pins in each side of collective lever (5).

13. Ensure end of each pin is engaged in collective lever sleeve bearings (21).



WHEN INSTALLING BOLT (17), MAKE SURE THAT THE BOLT HAS NOT BOTTOMED OUT IN THE COLLECTIVE LEVER OR THE GRIP HAS BECOME THREAD BOUND. EITHER CONDITION WOULD RESULT IN LOWER THAN SPECIFIED TORQUE ON THE PIN EVEN THOUGH TORQUE HAS BEEN VERIFIED.

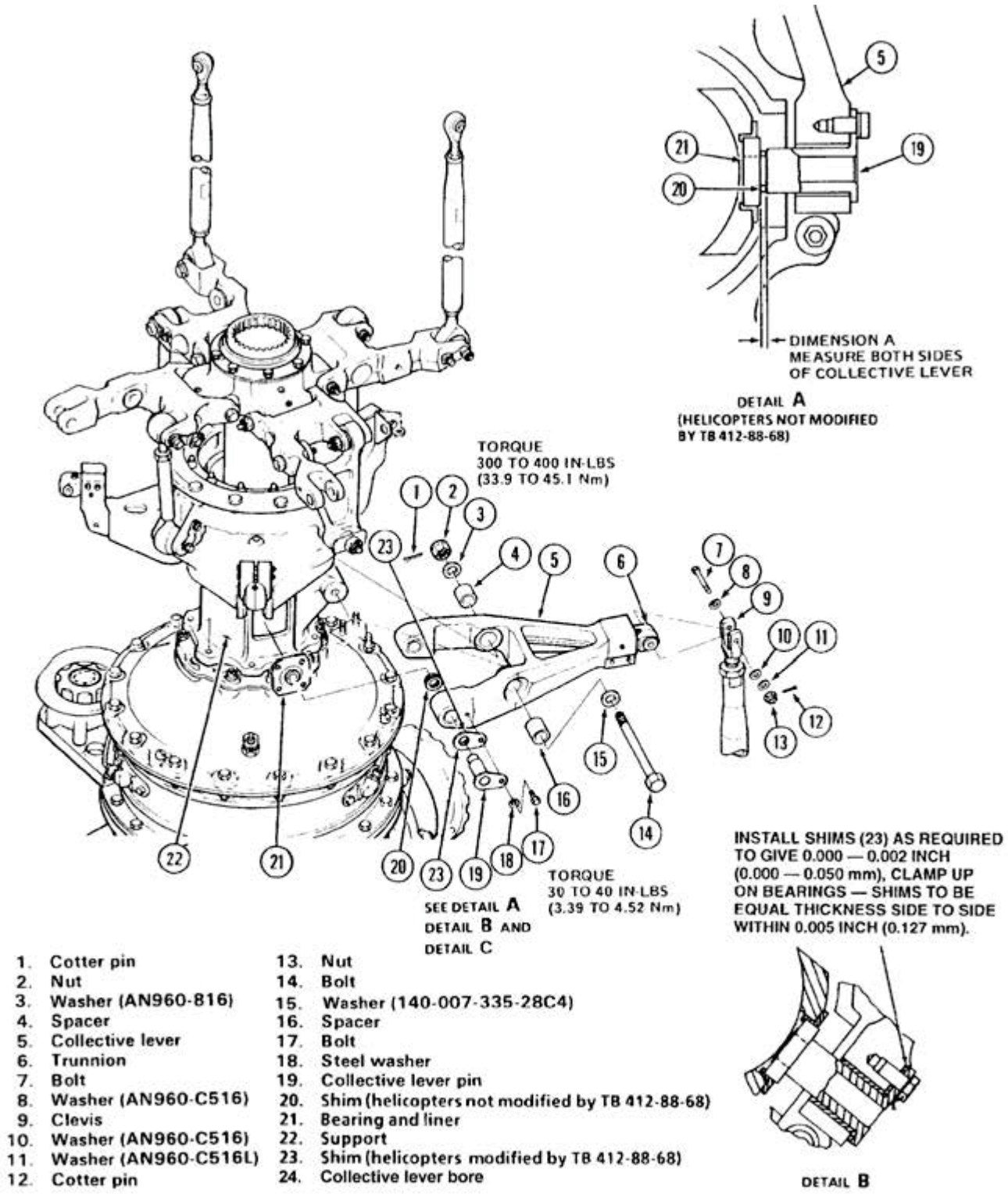
IF REQUIRED, INSTALL ONE WASHER NAS1149F0432P UNDER BOLT HEAD.

14. Install bolts (17) and washers (18).

15. Torque bolts (17) 30 to 40 inch-pounds (3.4 to 4.5 Nm) and secure bolts with lockwire (C-405).

16. Position clevis (9) on trunnion (6). Install bolt (7) with washer (8) under bolt head.

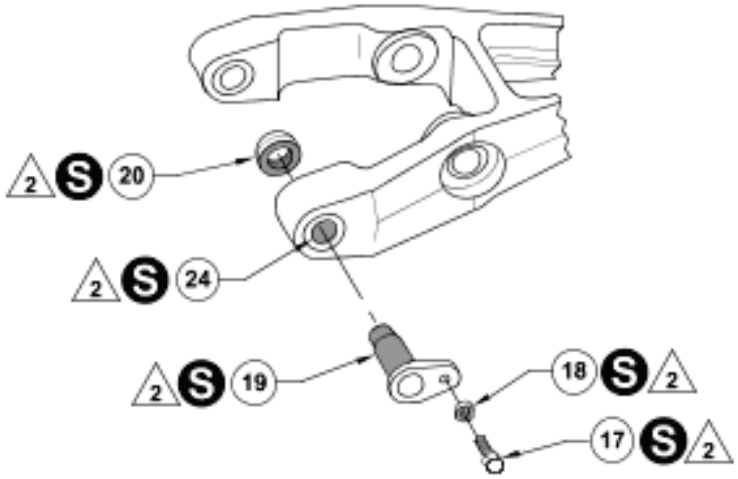
17. Install nut (13) with washers (10 and 11) as required to attain proper thread engagement for cotter pin (12). Install cotter pin.



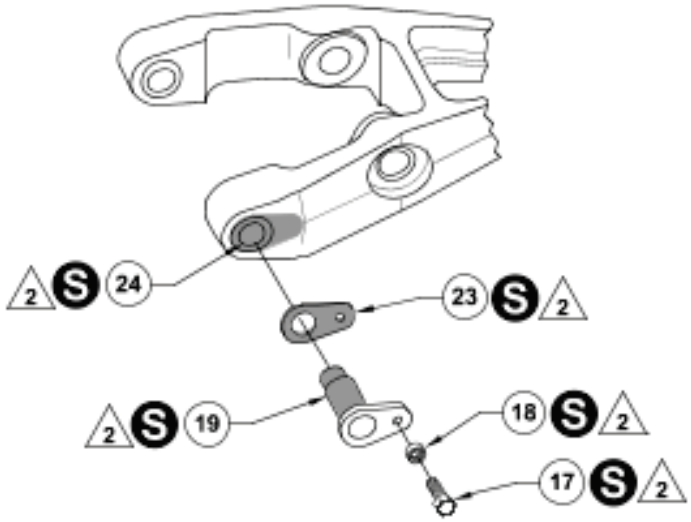
1. Cotter pin
2. Nut
3. Washer (AN960-816)
4. Spacer
5. Collective lever
6. Trunnion
7. Bolt
8. Washer (AN960-C516)
9. Clevis
10. Washer (AN960-C516)
11. Washer (AN960-C516L)
12. Cotter pin
13. Nut
14. Bolt
15. Washer (140-007-335-28C4)
16. Spacer
17. Bolt
18. Steel washer
19. Collective lever pin
20. Shim (helicopters not modified by TB 412-88-68)
21. Bearing and liner
22. Support
23. Shim (helicopters modified by TB 412-88-68)
24. Collective lever bore

ICN-412-A-623000-B-97499-00040-A-001-01

Figure 1. Collective Lever (Sheet 1 of 2)



PRE TB 412-88-68
S/N 33001 THROUGH 33167
S/N 34001 THROUGH 34024



POST TB 412-88-68
S/N 33168 AND SUBSEQUENT
S/N 34025 AND SUBSEQUENT

DETAIL C

S SEALANT (C-352)

NOTES

1. Ensure that no sealant is allowed to enter the bearing.
2. Apply sealant MIL-R-46082, type I (C-352) to faying surfaces as indicated by shading.

ICN-412-A-623000-B-97499-00041-A-001-01

Figure 1. Collective Lever (Sheet 2 of 2)

CHAPTER 63
MAIN ROTOR DRIVE SYSTEM

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number		Issue date	Applicable to
Main Rotor Drive - Description	DMC-412-A-63-00-00-00A-028A-A / 00988		2023-05-31	ALL
Foreign Particle Contamination - Inspection	DMC-412-A-63-00-00-01A-280A-A / 00989		2023-05-31	ALL
Foreign Particles - Identification	DMC-412-A-63-00-00-01A-310A-A / 00990		2023-05-31	ALL
Engine-to-Transmission (Main) Driveshaft - Description	DMC-412-A-63-10-01-00A-028A-A / 00991		2023-05-31	ALL
Engine-to-Transmission (Main) Driveshaft - Troubleshooting	DMC-412-A-63-10-01-00A-422A-A / 00992		2023-05-31	ALL
Engine-to-Transmission (Main) Driveshaft - Maintenance	DMC-412-A-63-10-01-01A-913A-A / 00993		2023-05-31	ALL
Engine-to-Transmission (Main) Driveshaft - Removal	DMC-412-A-63-10-01-01A-520A-A / 00994	C	2025-11-18	ALL
Engine-to-Transmission (Main) Driveshaft - Cleaning	DMC-412-A-63-10-01-01A-251A-A / 00995		2023-05-31	ALL
Engine-to-Transmission (Main) Driveshaft - Inspection	DMC-412-A-63-10-01-01A-280A-A / 00996		2023-05-31	ALL
Engine-to-Transmission (Main) Driveshaft - Repair	DMC-412-A-63-10-01-01A-685A-A / 00997		2023-05-31	ALL
Engine-to-Transmission (Main) Driveshaft - Lubrication	DMC-412-A-63-10-01-01A-240A-A / 00998		2023-05-31	ALL
Engine-to-Transmission (Main) Driveshaft - Installation	DMC-412-A-63-10-01-01A-720A-A / 00999	C	2025-11-18	ALL
Engine-to-Transmission (Main) Driveshaft - Balancing	DMC-412-A-63-10-01-01A-042A-A / 01000	C	2025-11-18	ALL
Engine-to-Driveshaft Coupling Adapter - Removal	DMC-412-A-63-10-01-02A-520A-A / 01001	C	2025-11-18	ALL
Engine-to-Driveshaft Coupling Adapter - Inspection	DMC-412-A-63-10-01-02A-280A-A / 01002	C	2025-11-18	ALL
Engine-to-Driveshaft Coupling Adapter - Repair	DMC-412-A-63-10-01-02A-685A-A / 01003		2023-05-31	ALL
Engine-to-Driveshaft Coupling Adapter - Installation	DMC-412-A-63-10-01-02A-720A-A / 01004	C	2025-11-18	ALL
Main Rotor Mast - Description	DMC-412-A-63-21-01-00A-028A-A / 01005	C	2025-11-18	ALL
Main Rotor Mast - Removal	DMC-412-A-63-21-01-00A-520A-A / 01006	C	2025-11-18	ALL
Main Rotor Mast - Cleaning	DMC-412-A-63-21-01-00A-251A-A / 01007	C	2025-11-18	ALL
Main Rotor Mast - Inspection and Repair	DMC-412-A-63-21-01-00A-280A-A / 01008		2023-05-31	ALL
Main Rotor Mast - Installation	DMC-412-A-63-21-01-00A-720A-A / 01009	C	2025-11-18	ALL
Torque Tube Assembly - Alignment	DMC-412-A-63-21-01-00A-272A-A / 01010	C	2025-11-18	ALL

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Transmission - Description	DMC-412-A-63-21-02-00A-028A-A / 01011		2023-05-31	ALL
Transmission - Troubleshooting	DMC-412-A-63-21-02-00A-422A-A / 01012		2023-05-31	ALL
Transmission - Removal	DMC-412-A-63-21-02-00A-520A-A / 01013	C	2025-11-18	ALL
Transmission - Inspection	DMC-412-A-63-21-02-00A-280A-A / 01014		2023-05-31	ALL
Transmission - Build-up	DMC-412-A-63-21-02-00A-720A-A / 01015	C	2025-11-18	ALL
Transmission - Installation	DMC-412-A-63-21-02-00A-720B-A / 01016	C	2025-11-18	ALL
Transmission - Alternate Run-in Procedures	DMC-412-A-63-21-02-00A-322A-A / 01017		2024-12-19	ALL
Mast Bearing Electric Chip Detector - Description	DMC-412-A-63-22-00-01A-042A-A / 01018		2023-05-31	ALL
Mast Bearing Electric Chip Detector - Removal	DMC-412-A-63-22-00-01A-520A-A / 01019		2023-05-31	ALL
Mast Bearing Electric Chip Detector - Inspection	DMC-412-A-63-22-00-01A-280A-A / 01020		2023-05-31	ALL
Mast Bearing Electric Chip Detector - Cleaning	DMC-412-A-63-22-00-01A-251A-A / 01021		2023-05-31	ALL
Mast Bearing Electric Chip Detector - Installation	DMC-412-A-63-22-00-01A-720A-A / 01022		2023-05-31	ALL
Main Input Gear Quill Electric Chip Detector - Description	DMC-412-A-63-22-00-02A-042A-A / 01023		2023-05-31	ALL
Main Input Gear Quill Electric Chip Detector - Removal	DMC-412-A-63-22-00-02A-520A-A / 01024		2023-05-31	ALL
Main Input Gear Quill Electric Chip Detector - Inspection	DMC-412-A-63-22-00-02A-280A-A / 01025		2023-05-31	ALL
Main Input Gear Quill Electric Chip Detector - Cleaning	DMC-412-A-63-22-00-02A-251A-A / 01026		2023-05-31	ALL
Main Input Gear Quill Electric Chip Detector - Installation	DMC-412-A-63-22-00-02A-720A-A / 01027		2023-05-31	ALL
Sump Case Electric Chip Detector - Description	DMC-412-A-63-22-00-03A-042A-A / 01028		2023-05-31	ALL
Sump Case Electric Chip Detector - Removal	DMC-412-A-63-22-00-03A-520A-A / 01029		2023-05-31	ALL
Sump Case Electric Chip Detector - Inspection	DMC-412-A-63-22-00-03A-280A-A / 01030		2023-05-31	ALL
Sump Case Electric Chip Detector - Cleaning	DMC-412-A-63-22-00-03A-251A-A / 01031		2023-05-31	ALL
Sump Case Electric Chip Detector - Installation	DMC-412-A-63-22-00-03A-720A-A / 01032		2023-05-31	ALL
Planetary Oil Jets - Description	DMC-412-A-63-22-01-01A-042A-A / 01033		2023-05-31	ALL
Planetary Oil Jets - Removal	DMC-412-A-63-22-01-01A-520A-A / 01034		2023-05-31	ALL
Planetary Oil Jets - Inspection	DMC-412-A-63-22-01-01A-280A-A / 01035		2023-05-31	ALL

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Planetary Oil Jets - Cleaning	DMC-412-A-63-22-01-01A-251A-A / 01036	2023-05-31	ALL
Planetary Oil Jets - Repair	DMC-412-A-63-22-01-01A-685A-A / 01037	2023-05-31	ALL
Planetary Oil Jets - Installation	DMC-412-A-63-22-01-01A-720A-A / 01038	2023-05-31	ALL
Oil Jet No. 1 - Removal	DMC-412-A-63-22-01-02A-520A-A / 01039	2023-05-31	ALL
Oil Jet No. 1 - Inspection	DMC-412-A-63-22-01-02A-280A-A / 01040	2023-05-31	ALL
Oil Jet No. 1 - Cleaning	DMC-412-A-63-22-01-02A-251A-A / 01041	2023-05-31	ALL
Oil Jet No. 1 - Repair	DMC-412-A-63-22-01-02A-685A-A / 01042	2023-05-31	ALL
Oil Jet No. 1 - Installation	DMC-412-A-63-22-01-02A-720A-A / 01043	2023-05-31	ALL
Oil Jet No. 2 - Removal	DMC-412-A-63-22-01-03A-520A-A / 01044	2023-05-31	ALL
Oil Jet No. 2 - Inspection	DMC-412-A-63-22-01-03A-280A-A / 01045	2023-05-31	ALL
Oil Jet No. 2 - Cleaning	DMC-412-A-63-22-01-03A-251A-A / 01046	2023-05-31	ALL
Oil Jet No. 2 - Repair	DMC-412-A-63-22-01-03A-685A-A / 01047	2023-05-31	ALL
Oil Jet No. 2 - Installation	DMC-412-A-63-22-01-03A-720A-A / 01048	2023-05-31	ALL
Oil Jet No. 3 - Removal	DMC-412-A-63-22-01-04A-520A-A / 01049	2023-05-31	ALL
Oil Jet No. 3 - Inspection	DMC-412-A-63-22-01-04A-280A-A / 01050	2023-05-31	ALL
Oil Jet No. 3 - Cleaning	DMC-412-A-63-22-01-04A-251A-A / 01051	2023-05-31	ALL
Oil Jet No. 3 - Repair	DMC-412-A-63-22-01-04A-685A-A / 01052	2023-05-31	ALL
Oil Jet No. 3 - Installation	DMC-412-A-63-22-01-04A-720A-A / 01053	2023-05-31	ALL
Oil Jet No. 4 - Removal	DMC-412-A-63-22-01-05A-520A-A / 01054	2023-05-31	ALL
Oil Jet No. 4 - Inspection	DMC-412-A-63-22-01-05A-280A-A / 01055	2023-05-31	ALL
Oil Jet No. 4 - Cleaning	DMC-412-A-63-22-01-05A-251A-A / 01056	2023-05-31	ALL
Oil Jet No. 4 - Repair	DMC-412-A-63-22-01-05A-685A-A / 01057	2023-05-31	ALL
Oil Jet No. 4 - Installation	DMC-412-A-63-22-01-05A-720A-A / 01058	2023-05-31	ALL
Oil Jet No. 5 - Removal	DMC-412-A-63-22-01-06A-520A-A / 01059	2023-05-31	ALL
Oil Jet No. 5 - Inspection	DMC-412-A-63-22-01-06A-280A-A / 01060	2023-05-31	ALL

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Oil Jet No. 5 - Cleaning	DMC-412-A-63-22-01-06A-251A-A / 01061	2023-05-31	ALL
Oil Jet No. 5 - Repair	DMC-412-A-63-22-01-06A-685A-A / 01062	2023-05-31	ALL
Oil Jet No. 5 - Installation	DMC-412-A-63-22-01-06A-720A-A / 01063	2023-05-31	ALL
Oil Jet No. 6 - Removal	DMC-412-A-63-22-01-07A-520A-A / 01064	2023-05-31	ALL
Oil Jet No. 6 - Inspection	DMC-412-A-63-22-01-07A-280A-A / 01065	2023-05-31	ALL
Oil Jet No. 6 - Cleaning	DMC-412-A-63-22-01-07A-251A-A / 01066	2023-05-31	ALL
Oil Jet No. 6 - Repair	DMC-412-A-63-22-01-07A-685A-A / 01067	2023-05-31	ALL
Oil Jet No. 6 - Installation	DMC-412-A-63-22-01-07A-720A-A / 01068	2023-05-31	ALL
Oil Jet No. 7 - Removal	DMC-412-A-63-22-01-08A-520A-A / 01069	2023-05-31	ALL
Oil Jet No. 7 - Inspection	DMC-412-A-63-22-01-08A-280A-A / 01070	2023-05-31	ALL
Oil Jet No. 7 - Cleaning	DMC-412-A-63-22-01-08A-251A-A / 01071	2023-05-31	ALL
Oil Jet No. 7 - Repair	DMC-412-A-63-22-01-08A-685A-A / 01072	2023-05-31	ALL
Oil Jet No. 7 - Installation	DMC-412-A-63-22-01-08A-720A-A / 01073	2023-05-31	ALL
Oil Pressure Regulating Valve - Removal	DMC-412-A-63-22-01-09A-520A-A / 01074	2023-05-31	ALL
Oil Pressure Regulating Valve - Cleaning	DMC-412-A-63-22-01-09A-251A-A / 01075	2023-05-31	ALL
Oil Pressure Regulating Valve - Inspection	DMC-412-A-63-22-01-09A-280A-A / 01076	2023-05-31	ALL
Oil Pressure Regulating Valve - Repair	DMC-412-A-63-22-01-09A-685A-A / 01077	2023-05-31	ALL
Oil Pressure Regulating Valve - Installation	DMC-412-A-63-22-01-09A-720A-A / 01078	2023-05-31	ALL
Oil Pressure Regulating Valve - Adjustment	DMC-412-A-63-22-01-09A-271A-A / 01078.1	2024-12-19	ALL
Oil Temperature Manifold - Removal	DMC-412-A-63-22-01-10A-520A-A / 01079	2023-05-31	ALL
Oil Temperature Manifold - Installation	DMC-412-A-63-22-01-10A-720A-A / 01080	2023-05-31	ALL
Oil Pressure Transmitter - Removal	DMC-412-A-63-22-01-11A-520A-A / 01081	2023-05-31	ALL
Oil Pressure Transmitter - Inspection	DMC-412-A-63-22-01-11A-280A-A / 01082	2023-05-31	ALL
Oil Pressure Transmitter - Installation	DMC-412-A-63-22-01-11A-720A-A / 01083	2023-05-31	ALL
Oil Pressure Switch - Removal	DMC-412-A-63-22-01-12A-520A-A / 01084	2023-05-31	ALL

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Oil Pressure Switch - Inspection	DMC-412-A-63-22-01-12A-280A-A / 01085	2023-05-31	ALL
Oil Pressure Switch - Installation	DMC-412-A-63-22-01-12A-720A-A / 01086	2023-05-31	ALL
Oil Pressure Manifold - Removal	DMC-412-A-63-22-01-13A-520A-A / 01087	2023-05-31	ALL
Oil Pressure Manifold - Cleaning	DMC-412-A-63-22-01-13A-251A-A / 01088	2023-05-31	ALL
Oil Pressure Manifold - Inspection	DMC-412-A-63-22-01-13A-280A-A / 01089	2023-05-31	ALL
Oil Pressure Manifold - Repair	DMC-412-A-63-22-01-13A-685A-A / 01090	2023-05-31	ALL
Oil Pressure Manifold - Installation	DMC-412-A-63-22-01-13A-720A-A / 01091	2023-05-31	ALL
Filler Neck, Scupper, and Mounting Bracket - Removal	DMC-412-A-63-22-01-14A-520A-A / 01092	2023-05-31	ALL
Filler Neck, Scupper, and Mounting Bracket - Cleaning	DMC-412-A-63-22-01-14A-251A-A / 01093	2023-05-31	ALL
Filler Neck, Scupper, and Mounting Bracket - Inspection	DMC-412-A-63-22-01-14A-280A-A / 01094	2023-05-31	ALL
Filler Neck, Scupper, and Mounting Bracket - Installation	DMC-412-A-63-22-01-14A-720A-A / 01095	2023-05-31	ALL
Oil Pump - Removal	DMC-412-A-63-22-01-15A-520A-A / 01096	2023-05-31	ALL
Oil Pump - Installation	DMC-412-A-63-22-01-15A-720A-A / 01097	2023-05-31	ALL
Oil Pump Inlet Screen - Removal	DMC-412-A-63-22-01-16A-520A-A / 01098	2023-05-31	ALL
Oil Pump Inlet Screen - Inspection	DMC-412-A-63-22-01-16A-280A-A / 01099	2023-05-31	ALL
Oil Pump Inlet Screen - Cleaning	DMC-412-A-63-22-01-16A-251A-A / 01100	2023-05-31	ALL
Oil Pump Inlet Screen - Installation	DMC-412-A-63-22-01-16A-720A-A / 01101	2024-12-19	ALL
Sump Case Internal Filter - Description	DMC-412-A-63-22-01-17A-042A-A / 01102	2023-05-31	ALL
Sump Case Internal Filter - Removal	DMC-412-A-63-22-01-17A-520A-A / 01103	2023-05-31	SN: 33001-33214 TB 412-91-94: Pre
Sump Case Internal Filter - Inspection	DMC-412-A-63-22-01-17A-280A-A / 01104	2023-05-31	SN: 33001-33214 TB 412-91-94: Pre
Sump Case Internal Filter - Cleaning	DMC-412-A-63-22-01-17A-251A-A / 01105	2023-05-31	SN: 33001-33214 TB 412-91-94: Pre
Sump Case Internal Filter - Installation	DMC-412-A-63-22-01-17A-720A-A / 01106	2023-05-31	SN: 33001-33214 TB 412-91-94: Pre

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Full Flow Debris Monitor - Description	DMC-412-A-63-22-01-18A-042A-A / 01107	C	2025-11-18	SN: 36020-Subsequent (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed or TB 412-91-94: Post)
Full Flow Debris Monitor - Removal	DMC-412-A-63-22-01-18A-520A-A / 01108	C	2025-11-18	SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed or TB 412-91-94: Post
Full Flow Debris Monitor - Inspection	DMC-412-A-63-22-01-18A-280A-A / 01109	C	2025-11-18	SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed or TB 412-91-94: Post
Full Flow Debris Monitor - Cleaning	DMC-412-A-63-22-01-18A-251A-A / 01110	C	2025-11-18	SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed or TB 412-91-94: Post

Document title	Data module code / Seq number		Issue date	Applicable to
Full Flow Debris Monitor - Installation	DMC-412-A-63-22-01-18A-720A-A / 01111	C	2025-11-18	SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed or TB 412-91-94: Post
Sight Glasses - Description	DMC-412-A-63-22-01-19A-042A-A / 01112		2023-05-31	ALL
Sight Glasses - Removal	DMC-412-A-63-22-01-19A-520A-A / 01113		2023-05-31	ALL
Sight Glasses - Cleaning	DMC-412-A-63-22-01-19A-251A-A / 01114		2023-05-31	ALL
Sight Glasses - Inspection	DMC-412-A-63-22-01-19A-280A-A / 01115		2023-05-31	ALL
Sight Glasses - Installation	DMC-412-A-63-22-01-19A-720A-A / 01116		2023-05-31	ALL
Transmission Oil Cooling System - Description	DMC-412-A-63-22-02-00A-028A-A / 01117		2023-05-31	ALL
External Oil Filter - Description	DMC-412-A-63-22-02-01A-042A-A / 01118		2023-05-31	ALL
External Oil Filter - Removal	DMC-412-A-63-22-02-01A-520A-A / 01119		2024-12-19	ALL
External Oil Filter - Inspection	DMC-412-A-63-22-02-01A-280A-A / 01120		2023-05-31	ALL
External Oil Filter - Cleaning	DMC-412-A-63-22-02-01A-251A-A / 01121		2023-05-31	ALL
External Oil Filter - Installation	DMC-412-A-63-22-02-01A-720A-A / 01122		2024-12-19	ALL
Thermal Bypass Valve - Description	DMC-412-A-63-22-02-02A-042A-A / 01123		2023-05-31	ALL
Thermal Bypass Valve - Removal	DMC-412-A-63-22-02-02A-520A-A / 01124		2023-05-31	ALL
Thermal Bypass Valve - Cleaning	DMC-412-A-63-22-02-02A-251A-A / 01125		2023-05-31	ALL
Thermal Bypass Valve - Installation	DMC-412-A-63-22-02-02A-720A-A / 01126		2023-05-31	ALL
Oil Drain Valves - Description	DMC-412-A-63-22-02-03A-042A-A / 01127		2023-05-31	ALL
Oil Drain Valves - Removal	DMC-412-A-63-22-02-03A-520A-A / 01128		2023-05-31	ALL
Oil Drain Valves - Cleaning	DMC-412-A-63-22-02-03A-251A-A / 01129		2023-05-31	ALL
Oil Drain Valves - Inspection	DMC-412-A-63-22-02-03A-280A-A / 01130		2023-05-31	ALL

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Oil Drain Valves - Installation	DMC-412-A-63-22-02-03A-720A-A / 01131		2023-05-31	ALL
Transmission Oil Cooler and Blower - Maintenance	DMC-412-A-63-22-02-04A-913A-A / 01132		2023-05-31	ALL
Transmission Quills - Description	DMC-412-A-63-21-02-00A-028B-A / 01133		2023-05-31	ALL
Main Input Quill - Description	DMC-412-A-63-21-02-01A-042A-A / 01134		2023-05-31	ALL
Main Input Quill - Removal	DMC-412-A-63-21-02-01A-520A-A / 01135		2023-05-31	ALL
Main Input Quill - Cleaning	DMC-412-A-63-21-02-01A-251A-A / 01136		2023-05-31	ALL
Main Input Quill - Inspection	DMC-412-A-63-21-02-01A-280A-A / 01137		2023-05-31	ALL
Main Input Quill - Repair	DMC-412-A-63-21-02-01A-685A-A / 01138		2023-05-31	ALL
Main Input Quill - Installation	DMC-412-A-63-21-02-01A-720A-A / 01139		2023-05-31	ALL
Main Input Quill Magnetic Seal - Removal	DMC-412-A-63-21-02-02A-520A-A / 01140	C	2025-11-18	SN: 33001-33213 or 36001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed))
Main Input Quill Magnetic Seal - Removal	DMC-412-A-63-21-02-02A-520B-A / 01141	C	2025-11-18	SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Main Input Quill Magnetic Seal - Cleaning	DMC-412-A-63-21-02-02A-251A-A / 01142		2023-05-31	ALL
Main Input Quill Magnetic Seal - Inspection	DMC-412-A-63-21-02-02A-280A-A / 01143		2023-05-31	ALL
Main Input Quill Magnetic Seal - Repair	DMC-412-A-63-21-02-02A-685A-A / 01144		2023-05-31	ALL

Document title	Data module code / Seq number		Issue date	Applicable to
Main Input Quill Magnetic Seal - Installation	DMC-412-A-63-21-02-02A-720A-A / 01145	C	2025-11-18	SN: 33001-33213 or 36001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed))
Main Input Quill Magnetic Seal - Installation	DMC-412-A-63-21-02-02A-720B-A / 01146	C	2025-11-18	SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Main Input Quill Lip Seal - Removal	DMC-412-A-63-21-02-03A-520A-A / 01147		2023-05-31	ALL
Main Input Quill Lip Seal - Cleaning	DMC-412-A-63-21-02-03A-251A-A / 01148		2023-05-31	ALL
Main Input Quill Lip Seal - Inspection	DMC-412-A-63-21-02-03A-280A-A / 01149		2023-05-31	ALL
Main Input Quill Lip Seal - Repair	DMC-412-A-63-21-02-03A-685A-A / 01150		2023-05-31	ALL
Main Input Quill Lip Seal - Installation	DMC-412-A-63-21-02-03A-720A-A / 01151		2023-05-31	ALL
No. 1 Hydraulic Pump Drive Quill - Description	DMC-412-A-63-21-02-04A-042A-A / 01152		2023-05-31	ALL
No. 1 Hydraulic Pump Drive Quill - Removal	DMC-412-A-63-21-02-04A-520A-A / 01153		2023-05-31	ALL
No. 1 Hydraulic Pump Drive Quill - Cleaning	DMC-412-A-63-21-02-04A-251A-A / 01154		2023-05-31	ALL
No. 1 Hydraulic Pump Drive Quill - Inspection	DMC-412-A-63-21-02-04A-280A-A / 01155		2023-05-31	ALL
No. 1 Hydraulic Pump Drive Quill - Repair	DMC-412-A-63-21-02-04A-685A-A / 01156		2023-05-31	ALL
No. 1 Hydraulic Pump Drive Quill - Installation	DMC-412-A-63-21-02-04A-720A-A / 01157		2023-05-31	ALL
No. 2 Hydraulic Pump Drive Quill - Description	DMC-412-A-63-21-02-05A-042A-A / 01158		2023-05-31	ALL
No. 2 Hydraulic Pump Drive Quill - Removal	DMC-412-A-63-21-02-05A-520A-A / 01159		2023-05-31	ALL
No. 2 Hydraulic Pump Drive Quill - Cleaning	DMC-412-A-63-21-02-05A-251A-A / 01160		2023-05-31	ALL
No. 2 Hydraulic Pump Drive Quill - Inspection	DMC-412-A-63-21-02-05A-280A-A / 01161		2023-05-31	ALL

Document title	Data module code / Seq number		Issue date	Applicable to
No. 2 Hydraulic Pump Drive Quill - Repair	DMC-412-A-63-21-02-05A-685A-A / 01162		2023-05-31	ALL
No. 2 Hydraulic Pump Drive Quill - Installation	DMC-412-A-63-21-02-05A-720A-A / 01163		2023-05-31	ALL
Rotor Tachometer Generator - Description	DMC-412-A-63-21-02-06A-042A-A / 01164		2023-05-31	ALL
Rotor Tachometer Generator - Removal	DMC-412-A-63-21-02-06A-520A-A / 01165		2023-05-31	ALL
Rotor Tachometer Generator - Cleaning	DMC-412-A-63-21-02-06A-251A-A / 01166		2023-05-31	ALL
Rotor Tachometer Generator - Inspection	DMC-412-A-63-21-02-06A-280A-A / 01167		2023-05-31	ALL
Rotor Tachometer Generator - Repair	DMC-412-A-63-21-02-06A-685A-A / 01168		2023-05-31	ALL
Rotor Tachometer Generator - Installation	DMC-412-A-63-21-02-06A-720A-A / 01169		2023-05-31	ALL
Tail Rotor Drive Quill - Description	DMC-412-A-63-21-02-07A-042A-A / 01170	C	2025-11-18	ALL
Tail Rotor Drive Quill - Removal	DMC-412-A-63-21-02-07A-520A-A / 01171		2023-05-31	ALL
Tail Rotor Drive Quill - Inspection	DMC-412-A-63-21-02-07A-280A-A / 01172	C	2025-11-18	ALL
Tail Rotor Drive Quill - Cleaning	DMC-412-A-63-21-02-07A-251A-A / 01173		2023-05-31	ALL
Tail Rotor Drive Quill - Repair	DMC-412-A-63-21-02-07A-685A-A / 01174		2023-05-31	ALL
Tail Rotor Drive Quill - Installation	DMC-412-A-63-21-02-07A-720A-A / 01175		2023-05-31	ALL
Rotor Brake Quill - Description	DMC-412-A-63-21-02-08A-042A-A / 01176		2023-05-31	ALL
Rotor Brake Quill - Removal	DMC-412-A-63-21-02-08A-520A-A / 01177		2023-05-31	ALL
Rotor Brake Quill - Cleaning	DMC-412-A-63-21-02-08A-251A-A / 01178		2023-05-31	ALL
Rotor Brake Quill - Inspection	DMC-412-A-63-21-02-08A-280A-A / 01179		2023-05-31	ALL
Rotor Brake Quill - Repair	DMC-412-A-63-21-02-08A-685A-A / 01180		2023-05-31	ALL
Rotor Brake Quill - Installation	DMC-412-A-63-21-02-08A-720A-A / 01181	C	2025-11-18	ALL
Transmission Quill Port Covers - Description	DMC-412-A-63-21-03-00A-028A-A / 01182		2023-05-31	ALL
Transmission Quill Port Covers - Removal	DMC-412-A-63-21-03-00A-520A-A / 01183		2023-05-31	ALL
Transmission Quill Port Covers - Installation	DMC-412-A-63-21-03-00A-720A-A / 01184		2023-05-31	ALL
Transmission Mounts - Description	DMC-412-A-63-21-04-00A-028A-A / 01185		2023-05-31	ALL
Aft Isolation Mount and Friction Dampers - Description	DMC-412-A-63-21-04-01A-042A-A / 01186		2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Aft Isolation Mount and Friction Dampers - Removal	DMC-412-A-63-21-04-01A-520A-A / 01187	2024-05-09	ALL
Transmission Mounts - Inspection	DMC-412-A-63-21-04-00A-280A-A / 01188	2023-08-28	ALL
Aft Mount Friction Dampers - Inspection	DMC-412-A-63-21-04-01A-280A-A / 01189	2025-05-13	ALL
Aft Isolation Mount and Friction Dampers - Installation	DMC-412-A-63-21-04-01A-720A-A / 01190	2024-05-09	ALL
Forward Isolation Mounts - Removal	DMC-412-A-63-21-04-02A-520A-A / 01191	2024-05-09	ALL
Forward Isolation Mounts - Inspection	DMC-412-A-63-21-04-02A-280A-A / 01192	2023-05-31	ALL
Forward Isolation Mounts - Installation	DMC-412-A-63-21-04-02A-720A-A / 01193	2024-05-09	ALL
Lift Link - Removal	DMC-412-A-63-21-04-03A-520A-A / 01194	2023-05-31	ALL
Lift Link - Cleaning	DMC-412-A-63-21-04-03A-251A-A / 01195	2023-05-31	ALL
Lift Link - Inspection and Repair	DMC-412-A-63-21-04-03A-280A-A / 01196	2023-05-31	ALL
Lift Link - Installation	DMC-412-A-63-21-04-03A-720A-A / 01197	2023-05-31	ALL
Lift Link Fitting - Removal	DMC-412-A-63-21-04-04A-520A-A / 01198	2023-05-31	ALL
Lift Link Fitting - Cleaning	DMC-412-A-63-21-04-04A-251A-A / 01199	2023-05-31	ALL
Lift Link Fitting - Inspection and Repair	DMC-412-A-63-21-04-04A-280A-A / 01200	2023-05-31	ALL
Lift Link Fitting - Installation	DMC-412-A-63-21-04-04A-720A-A / 01201	2023-05-31	ALL
Serviceability Checks - Transmission, Tail Rotor Gearbox, and Intermediate Gearbox - Description	DMC-412-A-63-00-00-00A-028B-A / 01202	2023-05-31	ALL
Intermediate or Tail Rotor Gearbox - Serviceability Check	DMC-412-A-63-00-00-00A-280B-A / 01203	2023-05-31	ALL
Transmission - Serviceability Check	DMC-412-A-63-00-00-00A-280A-A / 01204	2023-05-31	ALL
Rotor Brake System - Description	DMC-412-A-63-23-00-00A-028A-A / 01205	2023-05-31	ALL
Rotor Brake System - Troubleshooting	DMC-412-A-63-23-00-00A-422A-A / 01206	2023-05-31	ALL
Brake Assembly (P/N 9450103) - Bleeding	DMC-412-A-63-23-00-01A-231A-A / 01207	2023-05-31	ALL
Brake Assembly (P/N 9450103-1) - Bleeding	DMC-412-A-63-23-00-01A-231B-A / 01208	2023-05-31	ALL
Master Cylinder - Removal	DMC-412-A-63-23-00-02A-520A-A / 01209	2023-05-31	ALL
Master Cylinder - Inspection	DMC-412-A-63-23-00-02A-280A-A / 01210	2023-05-31	ALL
Master Cylinder - Repair	DMC-412-A-63-23-00-02A-685A-A / 01211	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Master Cylinder - Installation	DMC-412-A-63-23-00-02A-720A-A / 01212	2023-05-31	ALL
Brake Assembly - Description	DMC-412-A-63-23-00-03A-042A-A / 01213	2023-05-31	ALL
Brake Assembly (P/N 9450103) - Removal	DMC-412-A-63-23-00-03A-520A-A / 01214	2023-05-31	ALL
Brake Assembly (P/N 9450103-1) - Removal	DMC-412-A-63-23-00-03A-520B-A / 01215	2023-05-31	ALL
Brake Assembly - Inspection	DMC-412-A-63-23-00-03A-280A-A / 01216	2023-05-31	ALL
Brake Assembly - Repair	DMC-412-A-63-23-00-03A-685A-A / 01217	2023-05-31	ALL
Brake Assembly (P/N 9450103) - Installation	DMC-412-A-63-23-00-03A-720A-A / 01218	2023-05-31	ALL
Brake Assembly (P/N 9450103-1) - Installation	DMC-412-A-63-23-00-03A-720B-A / 01219	2023-05-31	ALL
Rotor Brake Disc - Removal	DMC-412-A-63-23-00-04A-520A-A / 01220	2023-05-31	ALL
Rotor Brake Disc - Inspection	DMC-412-A-63-23-00-04A-280A-A / 01221	2024-12-19	ALL
Rotor Brake Disc - Installation	DMC-412-A-63-23-00-04A-720A-A / 01222	2023-05-31	ALL
Flange Adapter - Removal	DMC-412-A-63-23-00-05A-520A-A / 01223	2023-05-31	ALL
Flange Adapter - Installation	DMC-412-A-63-23-00-05A-720A-A / 01224	2023-05-31	ALL

Main Rotor Drive - Description

This chapter contains maintenance instructions for the drive system components, which transmit power from the engine to the main rotor assembly and to transmission mounted accessories. This includes the engine-to-transmission (main) driveshaft, transmission and related components, and main rotor mast assembly.

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Foreign Particle Contamination - Inspection

Procedure

1. Foreign particles found in transmission oil, filters, or accumulated on chip detectors may indicate impending failure of an internal part of the transmission. However, the presence of metal particles is not necessarily an indication the transmission or a component is no longer serviceable. If in doubt, do a serviceability check (412-A-63-00-00-00A-028B-A / 01202). To determine the serviceability of a component, following must be considered
   - The quantity, source, form, and type of metal, together with the service history
   - The amount of time in service since the component was new or overhauled
   - Previous failures
   - Type of operation

2. The particles found may be steel, aluminum, magnesium, silver, copper, bronze, or phenolic in various shapes and quantities. If particles found are ferrous and magnetic, refer to the Figure 1 for identifications and recommendations. Non-ferrous particles found in transmission oil, screens or filters is not a cause for the transmission removal from service, however the particles should be identified to determine the proper maintenance action.

3. If silver and copper particles are found in the transmission oil, screens, or filters the origin may be a bearing or other component such as the planetary retainers. The presence of silver or copper in the transmission oil, screens, or filters is not a removal from service criteria for the transmission or the mast bearing assembly.

4. If non-significant quantity of silver or copper particles are found in the full found debris monitor/sump case internal filter, remove and clean the filter element. Non-significant quantity of debris is defined as less than approximately 1/3 of the filtering element surface area.

5. If significant quantity of silver or copper particles are found in the full found debris monitor/sump case internal filter, remove and clean the filter element of the full flow debris monitor/sump case internal filter and replace the external transmission oil filter.

6. If significant quantity of silver or copper particles are found in the full found debris monitor/sump case internal filter and the external transmission oil filter was found with an impending bypass indication, do the following
   - Change transmission oil
   - Flush all transmission oil system lines
   - Remove and clean all oil jets
   - Remove and clean the full flow debris monitor/sump case internal filter
   - Remove and clean the oil pump inlet screen
   - Replace the transmission external oil filter

7. A change in the color of the transmission oil could also indicate an oil contamination. If the oil is found dark or has an unusual color, examine the chip detectors for unwanted debris. Drain the oil and examine fro debris. If no debris is found, do a serviceability check (412-A-63-00-00-00A-028B-A / 01202).

NOTE:

The following information contained in this figure is provided for the evaluation of contamination found on the drive system chip detectors.

It is impossible to depict the shapes of all possible particulates contamination, especially those particles not generated within a gearbox, as a result of component wear or failure. However, the following basic assumptions will permit a rather accurate assessment of any debris found, even though the particles may be severely distorted as a result of passing through gear meshes.

DESCRIPTION:

Material, from the surfaces of gear teeth or rolling element bearings, is quite hard. Particles broken from these surfaces may have razor-sharp edges but rarely have sharp pointed ends. With few exceptions, surfaces of such particles will not appear smeared (as if cut by a shear) under magnification; a grainy appearance on the fracture surfaces is more common.

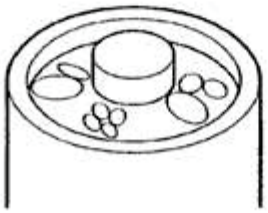
When damaged by passing through gear meshes, these hard particles tend to break up with relatively little deformation.

Foreign ferrous material introduced into gearboxes is often much softer than the surfaces of gear teeth and bearings. The same is true of most nongear and nonbearing components within gearboxes. The softer materials are usually quite ductile and malleable, i.e., particles from such materials can be readily bent or rolled into a wide range of shapes without fracturing.

Fragments of the softer material are capable of being torn or sheared from the parent component without suffering brittle fracture (easily broken or snapped). Instead, during separation the fragments are distorted and stretched such that they often exhibit surfaces that appear stretched. Ends of soft material fragments are often sharp-pointed; the ductile nature of the material permits such a condition.

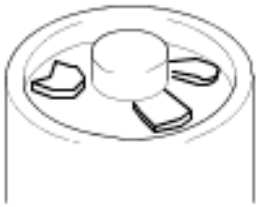
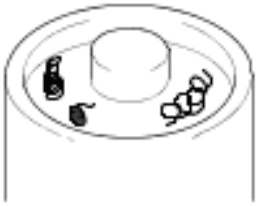
Each type of debris has been identified as significant debris or insignificant debris at the beginning of the explanation of significance to aid in troubleshooting.

CAUTION:
RECURRING ACCUMULATION OF INSIGNIFICANT DEBRIS WILL REQUIRE OVERHAUL/REPAIR OF GEARBOX.

TYPICAL APPEARANCE	DESCRIPTION	SIGNIFICANCE
	A few moderately-sized and/or numerous small nearly-flat flakes: Under magnification one side of flakes appears very smooth. Flakes are silvery in color with an occasional black side. Common term for this condition is flakes.	(SIGNIFICANT DEBRIS) This is a classic indication of rolling element bearing failure. Although less common, this can also indicate spalling of gear teeth. OVERHAUL/REPAIR GEARBOX

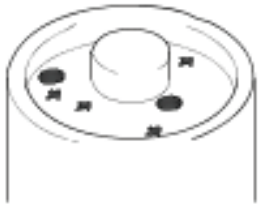
ICN-412-A-630000-B-97499-00001-A-001-01

Figure 1. Drive System Chip Detector Debris - Analysis (Sheet 1 of 3)

TYPICAL APPEARANCE	DESCRIPTION	SIGNIFICANCE
	Irregularly-shaped ferrous chunks of various sizes and shapes: Under magnification, one or more sides of particles appears rough and grainy. Color of particles is silvery-gray often with one or more black sides. This condition has no common term.	SIGNIFICANT DEBRIS Existence of this type of debris most likely indicates gear and/or bearing damage within the drive system component. OVERHAUL/REPAIR GEARBOX
	Spiral curls or comma-shaped particles: When examined under magnification, particles are often smooth and shiny on their convex surfaces and quite rough on the other surfaces and edges. The color of the particles is usually dull gray on the rough surfaces. The common term for this condition is manufacturing debris.	INSIGNIFICANT DEBRIS Particles are fragments of chips or shavings produced during the machining of ferrous components. Such contamination is often introduced into the drive system components on tools, during dusting operations within the component assembly area using compressed air. No corrective action is required following discovery of such material.
	Hair-like ferrous debris: May have a rectangular or triangular cross sections. Generally 0.030 inch (0.76 mm) or less in thickness. Length may range from 0.10 inch (2.54 mm) to over 1.0 inch (25.4 mm). Color of debris is usually light gray, although one or more sides may have a black appearance. Common term for condition is hairs.	INSIGNIFICANT DEBRIS Debris commonly are scrapings, produced as components, and are interference-fitted together during assembly of the drive system components. Debris of this general shape is also produced, after long term operation, at the corners of wear surfaces of such parts as transmission planetary supports. This type debris can be introduced into the component during periodic servicing. Hairs can appear on chip detectors at any point in the life of the drive system component. However, discovery of such material does not necessitate any corrective action.

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Figure 1. Drive System Chip Detector Debris - Analysis (Sheet 2 of 3)

TYPICAL APPEARANCE	DESCRIPTION	SIGNIFICANCE
	Irregularly-shaped ferrous particles: Usually triangular in cross sections. Often spike-like in appearance. Under magnification, one side of a triangular section will usually appear sheared. Color may be silvery-gray or black with one or more silver sides. Existence of two or more particles of this type on the chip detector at one time is rare. Common term for condition is manufacturing debris.	INSIGNIFICANT DEBRIS Particles of this type are commonly the result of tool slippage during assembly of the drive system component. An example would be the silver of metal torn from the edge of a screw slot as a screwdriver blade slips out of place. This type of debris would most likely appear during the first 50 hours of operation following component assembly or extensive maintenance. No corrective actions are required following discovery of such material.
	Tiny whisker-like particles or groups of dark microscopic particles: When removed with tape or paper towel, apparently-large chips disappear into a black smudge. Common term for this condition is smudge or fuzz.	INSIGNIFICANT DEBRIS Generally microscopic wear particles produced by normal wear within drive system component. Particles are often grouped by the field of the chip detector magnet to assume the shape of apparently large chips. Such material is common in gearboxes having several hundred operating hours. In such cases, this condition does not necessitate corrective action. Appearance of this type of debris in relatively new or recently-overhauled gearboxes may indicated bearing or gear micropitting. In such cases, detector should be inspected more frequently for signs of progressing damage.

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Figure 1. Drive System Chip Detector Debris - Analysis (Sheet 3 of 3)

Foreign Particles - Identification

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydrochloric acid	C-431	AR	
Nitric acid	C-432	AR	
Sodium chloride	C-501	AR	

Procedure



WHEN PARTICLES FOUND ARE READILY IDENTIFIABLE AS FRAGMENTS OF TRANSMISSION PARTS SUCH AS GEARS, HARDWARE, BEARINGS, LOCKWIRE, OR OTHER INTERNAL COMPONENTS, REPLACE TRANSMISSION ASSEMBLY,, MAST ASSEMBLY, OIL COOLER, AND EXTERNAL OIL FILTER ELEMENTS. CLEAN AND THOROUGHLY FLUSH ALL OIL LINES AND OIL JETS WITH OPERATING OIL (412-A-12-00-00-00A-040A-A / 00126).

1. Visually examine the particles. If you cannot identify the particles by color and hardness. Use the equipment that follows to identify them

- A permanent magnet
- 50% by volume of nitric acid (C-432) and water solution
- 50% by weight of sodium chloride (C-501) and water solution
- hydrochloric acid (C-431)
- A test tube

2. Put a small amount of 50% nitric acid (C-432) solution into a test tube and add a particle of the material you wish to test. If the particle does not dissolve, warm the solution slightly. If the particle dissolves after you warm the solution, add a few drops of the 5% of the sodium chloride (C-501) solution. If white precipitate forms are in the test tube, the particle is silver.

NOTE

Phenolic and aluminium do not react noticeably with nitric acid.

3. Put a small amount of hydrochloric acid (C-431) in a test tube and add a particle of the material you wish to test. If the solution bubbles quickly, the particle disintegrated and a black residue forms, the particle is probably aluminium or magnesium. Put a small amount of nitric acid (C-432) solution into a test tube and add a second particle of the same material. If there is no reaction with the nitric acid, the particle is probablyaluminium.

4. To find the difference between copper, bronze, and magnesium. Put a small amount of the nitric acid (C-432) solution into a test tube and add a particle of the material you wish to test. If bright green color forms in the acid solution, the particle is copper or bronze. If the solution bubbles quickly when you drop the particle into the acid, the particle is probably magnesium.

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Engine-to-Transmission (Main) Driveshaft - Description

The main driveshaft is installed between an adapter on the engine combining (reduction) gearbox and input drive quill on the transmission. The driveshaft is a steel tube with grease lubricated, spherical, splined coupling assemblies on either end to provide flexibility.

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Engine-to-Transmission (Main) Driveshaft - Troubleshooting

Procedure

1. Refer to Table 1.

Table 1. Engine-to-Transmission (Main) Driveshaft - Troubleshooting

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
High frequency vibration	Improperly assembled driveshaft.	Remove, disassemble, inspect, and assemble driveshaft (BHT-412-CR&O).
	Loss of grease from coupling.	Remove, disassemble, inspect, assemble, and install coupling (BHT-412-CR&O).
	Loose engine adapter or adapter bolt.	Remove, inspect, and install engine adapter.
	Dirt or burr on mounting diameters of driveshaft.	Remove, inspect, clean, and install driveshaft.
	Loose coupling bolt(s).	Loosen, then retighten coupling bolts in sequence.
	Driveshaft out of balance.	Balance driveshaft (412-A-18-11-00-07A-271A-A / 00171).
	Improperly assembled driveshaft coupling.	Remove, disassemble, inspect, assemble, and install coupling (BHT-412-CR&O).
Grease leakage	Faulty packings or seals.	Replace packings or seals (BHT-412-CR&O).
Over-temperature indicator (TEMP-PLATE) shows discoloration (from white to black)	Possible overheat condition and/or component degradation.	Refer to 412-A-63-10-01-01A-280A-A / 00996.

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Engine-to-Transmission (Main) Driveshaft - Maintenance

Procedure

1. This paragraphs provide maintenance instructions for the main driveshaft. Vibration analysis and balancing is contained in 412-A-18-11-00-00A-042B-A / 00169.

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Engine-to-Transmission (Main) Driveshaft - Removal

Procedure

1. Remove left and right air inlet fairings.

2. Open left engine top cowling. Disconnect fire detector leads at firewall. Disconnect left starter generator air hose from vent in engine cowl.

3. Remove left engine top cowling.

Applicable: (SN: 33001-33213 or SN: 36001-36019) (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)):

4. *Do the steps that follow:*

4.1. *If applicable, put a mark to identify the locations of the balancing weights (2 and 8, Figure 1).*

4.2. *Hold the driveshaft (6) and remove nuts (4 and 11), washers (3 and 10), bolts (1 and 7), and balancing weights (2 and 8), if installed.*

4.3. *Put a mark on the flexible couplings for installation in the same position.*

4.4. *Push the driveshaft (6) forward until the aft end is free.*

4.5. *Remove the driveshaft (6).*

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

5. *(412Classic/SP to 412HP Upgrade), do the steps that follow:*

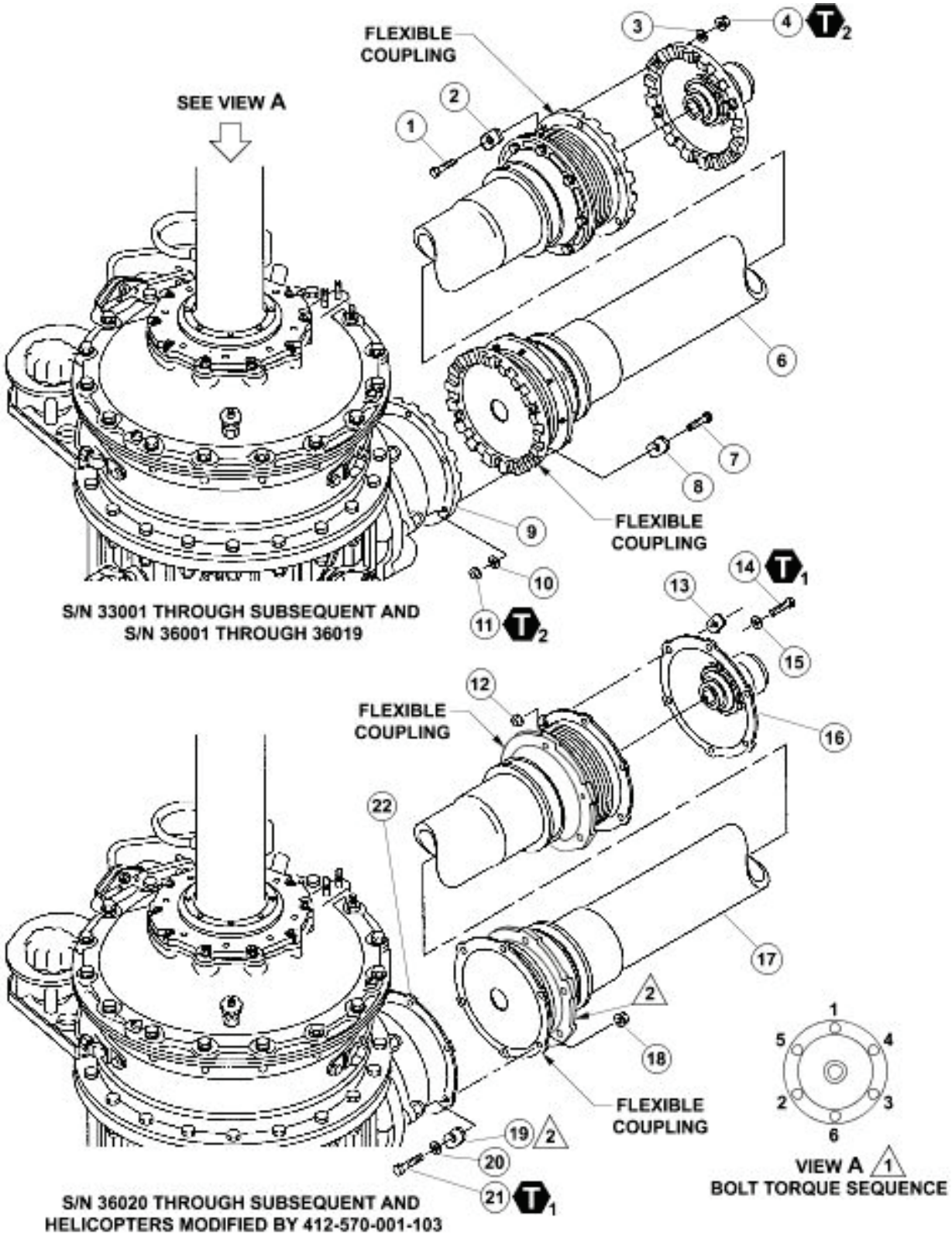
5.1. *Put marks to identify the location of the balancing weights (13 and 19).*

5.2. *Hold the driveshaft (17) and remove the nuts (12 and 18), balancing weights (13 and 19) (where applicable), washers (15 and 20), and bolts (14 and 21).*

5.3. *Put a mark on the flexible couplings for installation in the same position.*

5.4. *Push the driveshaft (17) forward until the aft end is free.*

5.5. *Remove the driveshaft (17).*



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Figure 1. Engine-to-Transmission (Main) Driveshaft (Sheet 1 of 2)

1. Bolt
2. Balancing weight
3. Washer
4. Nut
5. Engine-to-driveshaft curvic coupling adapter
6. Driveshaft
7. Bolt
8. Balancing weight
9. Main input quill curvic coupling adapter
10. Washer
11. Nut
12. Nut
13. Balancing weight
14. Bolt
15. Washer
16. Engine-to-driveshaft curvic coupling adapter
17. Driveshaft
18. Nut
19. Balancing weight
20. Washer
21. Bolt
22. Main input quill curvic coupling adapter

T1 70 TO 90 IN-LBS (7.91 TO 10.16 Nm)

T2 100 TO 140 IN-LBS (11.30 TO 15.81 Nm)

NOTES

1. Torque bolts in three steps; 1/3 full torque, 2/3 full torque, and full torque respectively. Tighten bolts in sequence as shown in Detail A. Repeat sequence at each torque step.

2. Use aft flange of coupling for balance adjustment when air conditioning is installed.

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Figure 1. Engine-to-Transmission (Main) Driveshaft (Sheet 2 of 2)

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Engine-to-Transmission (Main) Driveshaft - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Prior to cleaning, inspect driveshaft (4, Figure 1) for lubricant leakage.

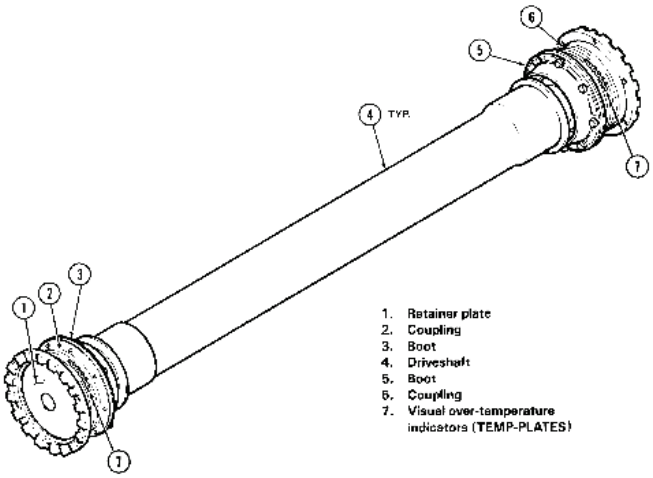


DO NOT IMMERSE DRIVESHAFT IN drycleaning solvent (C-304) OR ANY CLEANING AGENTS.

2. Clean driveshaft parts, except boots (3 and 5), with a clean, lint-free cloth dampened with drycleaning solvent (C-304).

3. Dry driveshaft (4) with filtered, compressed air.

4. Clean boots (3 and 5) with a clean, dry, lint-free cloth.



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Figure 1. Main Driveshaft - Damage Limits (Sheet 1 of 2)

NOTES

1. Evidence of lubricant leakage at couplings (2) or (6) is cause for overhaul evaluation of driveshaft. Refer to BHT-412-CR&O manual.

2. Mechanical and corrosion damage limits on driveshaft (4) are defined as follows:

MECHANICAL/CORROSION DAMAGE	LIMITS
Nicks and scratches running parallel to, or within 15° of parallel to shaft axis.	Not exceeding 0.004 inch (0.101 mm) in depth are acceptable without repair. Not exceeding 0.008 inch (0.203 mm) in depth or 1.5 inches (38 mm) in length are acceptable if polished out, provided the total polished area does not exceed 10% of circumference at any point.
Nicks and scratches not running within 15° of shaft axis.	Not exceeding 0.004 inch (0.101 mm) in depth are acceptable on up to 100% of the circumference of shaft, if polished out. Not exceeding 0.008 inch (0.203 mm) in depth or 1.5 inches (38 mm) in length are acceptable provided the total polished area does not exceed 10% of circumference at any point.
Corrosion pitting.	Not exceeding 0.008 inch (0.203 mm) in depth is acceptable provided it is polished out. Maximum length of rework at any point must not exceed 1.5 inches (38 mm) nor cover more than 10% of the circumference at any point. The maximum acceptable total reworked area is 4.0 square inches (2580 mm^2).

3. Mechanical and/or corrosion damage on retainer plate (1) or similar plate in couplings (6) that could result in lubricant leakage is not acceptable.

4. Mechanical and/or corrosion damage on boots (3 and 5) that could result in lubricant leakage is not acceptable.

5. Superficial mechanical damage on couplings (2 and 6) is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion damage is cause for overhaul evaluation of driveshaft (4). Refer to the BHT-412-CR&O.

6. Visual over-temperature indicator (TEMP-PLATES) for discoloration and overheat condition. A change in color of TEMP-PLATES dots from white to black will indicate a possible overheat condition and/or component degradation, and cause of indicator (TEMP-PLATES) dots' discoloration should be determined prior to condition operation. Refer to the BHT-412-CR&O for Inspection and Lubrication procedures.

ICN-412-A-631001-B-97499-00004-A-001-01

Figure 1. Main Driveshaft - Damage Limits (Sheet 2 of 2)

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Engine-to-Transmission (Main) Driveshaft - Inspection

Procedure

1. Inspect driveshaft in accordance with Figure 1.

2. Refer to the BHT-412-CR&O for detailed inspection information.

3. Perform serviceability check of over-temperature indicator (TEMP-PLATE) (7) as follows:

3.1. Over-temperature indicators (TEMP-PLATE) (7) must not show evidence of over-temperature, deterioration, debonding, or discoloration of the epoxy coating that prevents interpretation of the indicating dots. If any of these conditions exist, proceed to Step 3.2..

3.2. If only one over-temperature indicator (TEMP-PLATE) (7) of a given color is missing, and no dot on any other over-temperature indicator (TEMP-PLATE) (on the same coupling) is discolored or shows mechanical damage or degradation of the epoxy overcoating, the helicopter may be returned to service. The discrepant over-temperature indicator (TEMP-PLATE) should be replaced as soon as practical.

3.3. If any indicator dot on either the red or yellow bordered over-temperature indicator (TEMP-PLATE) (7) has changed color to black, see Table 1 for probable cause and required corrective action.

Table 1. Main Driveshaft TEMP-PLATE Condition and Corrective Action

ONE YELLOW TEMP-PLATE	OTHER YELLOW TEMP-PLATE	ONE RED TEMP-PLATE	OTHER RED TEMP-PLATE	PROBABLE CAUSE	SEE NOTE
Good	Black	Good	Good	Defect/Instl.	1
Black	Black	Good	Good	Elevated Temp.	2
Good	Good	Black	Good	Defect/Instl.	1
Good	Good	Black	Black	Defect/Instl.	4
Black	Black	Black	Black	Overtemp	3
Part Black	Good	Good	Good	Chem. Contamination	1
Part Black	Part Black	Good	Good	Chem. Contamination	1
Good	Good	Part Black	Good	Chem. Contamination	1
Good	Good	Part Black	Part Black	Chem. Contamination	4
Missing	Good	Good	Good	Defect/Instl.	1
Missing	Missing	Good	Good	Possible Elevated Temperature	2
Good	Good	Missing	Good	Defect/Instl.	1
Good	Good	Missing	Missing	Defect/Instl.	4
Missing	Missing	Missing	Missing	Possible Overtemp	3

NOTES:

1. Resume operation. In the event of defective TEMP-PLATE or improper installation, replace defective TEMP-PLATE as soon as practical in accordance with the BHT-412-CR&O.

2. Elevated coupling temperature is indicated. Determine probable cause of elevated temperature indication and take corrective action prior to continued operation. Accomplish checks in step a and step b. If probable cause is not revealed from step a or step b, perform step c and step d. It is recommended that the affected TEMP-PLATE be replaced in accordance with the BHT-412-CR&O prior to resuming operation.

a. Check the driveshaft coupling for any sign of grease leakage. If leakage is detected, the coupling must be serviced prior to returning the helicopter to service.

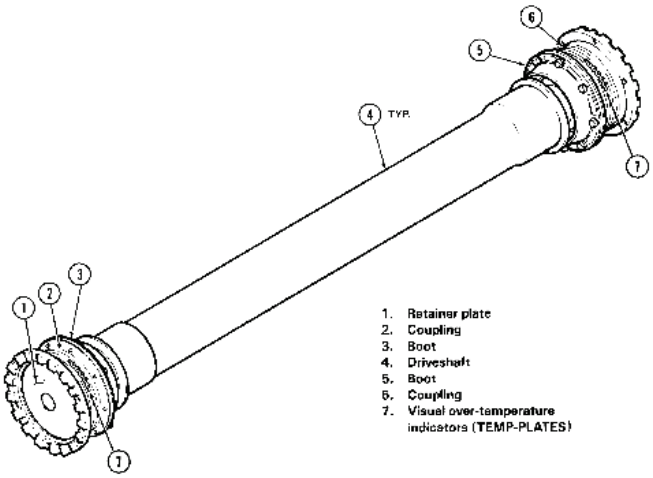
b. Review current helicopter operating conditions to determine the probable cause of elevated coupling temperature, such as extended operation at CG limits, extended operation at high power, repeated heavy lift, or high ambient temperature. Make appropriate adjustment to correct condition.

c. Inspect engine mounts and transmission pylon mounts for condition.

d. Verify engine-to-transmission alignment.

Table 1. Main Driveshaft TEMP-PLATE Condition and Corrective Action (continued)

ONE YELLOW TEMP-PLATE	OTHER YELLOW TEMP-PLATE	ONE RED TEMP-PLATE	OTHER RED TEMP-PLATE	PROBABLE CAUSE	SEE NOTE
3. Coupling overtemperature condition is very likely. Remove driveshaft and perform overtemperature inspection in accordance with the BHT-412-CR&O.					
4. Same as Note 1 except during operations before TEMP-PLATE replacement, a yellow-bordered TEMP-PLATE indication shall be considered as an overtemperature indication and Notes 2 and 3 must be accomplished.					



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Figure 1. Main Driveshaft - Damage Limits (Sheet 1 of 2)

NOTES

1. Evidence of lubricant leakage at couplings (2) or (6) is cause for overhaul evaluation of driveshaft. Refer to BHT-412-CR&O manual.

2. Mechanical and corrosion damage limits on driveshaft (4) are defined as follows:

MECHANICAL/CORROSION DAMAGE	LIMITS
Nicks and scratches running parallel to, or within 15° of parallel to shaft axis.	Not exceeding 0.004 inch (0.101 mm) in depth are acceptable without repair. Not exceeding 0.008 inch (0.203 mm) in depth or 1.5 inches (38 mm) in length are acceptable if polished out, provided the total polished area does not exceed 10% of circumference at any point.
Nicks and scratches not running within 15° of shaft axis.	Not exceeding 0.004 inch (0.101 mm) in depth are acceptable on up to 100% of the circumference of shaft, if polished out. Not exceeding 0.008 inch (0.203 mm) in depth or 1.5 inches (38 mm) in length are acceptable provided the total polished area does not exceed 10% of circumference at any point.
Corrosion pitting.	Not exceeding 0.008 inch (0.203 mm) in depth is acceptable provided it is polished out. Maximum length of rework at any point must not exceed 1.5 inches (38 mm) nor cover more than 10% of the circumference at any point. The maximum acceptable total reworked area is 4.0 square inches (2580 mm^2).

3. Mechanical and/or corrosion damage on retainer plate (1) or similar plate in couplings (6) that could result in lubricant leakage is not acceptable.

4. Mechanical and/or corrosion damage on boots (3 and 5) that could result in lubricant leakage is not acceptable.

5. Superficial mechanical damage on couplings (2 and 6) is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion damage is cause for overhaul evaluation of driveshaft (4). Refer to the BHT-412-CR&O.

6. Visual over-temperature indicator (TEMP-PLATES) for discoloration and overheat condition. A change in color of TEMP-PLATES dots from white to black will indicate a possible overheat condition and/or component degradation, and cause of indicator (TEMP-PLATES) dots' discoloration should be determined prior to condition operation. Refer to the BHT-412-CR&O for Inspection and Lubrication procedures.

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Figure 1. Main Driveshaft - Damage Limits (Sheet 2 of 2)

Engine-to-Transmission (Main) Driveshaft - Repair

Preliminary requirements

Consumables, materials and expendables

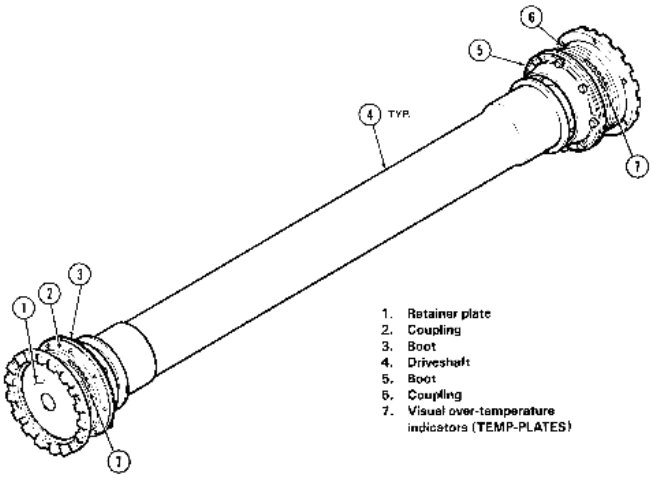
Nomenclature	Identification No.	Qty	Remarks
Epoxy polyamide primer	C-204	AR	

Procedure

1. Polish out acceptable defects on driveshaft (4, Figure 1). Use minimum radius of 0.500 inch (1.270 mm) in polishing out surface damage. Ensure damage limits are not exceeded.

2. Refinish polished area with two coats of epoxy polyamide primer (C-204).

3. Replace over-temperature indicator (TEMP-PLATE) (7) if mutilated or discolored (BHT-412-CR&O).



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Figure 1. Main Driveshaft - Damage Limits (Sheet 1 of 2)

NOTES

1. Evidence of lubricant leakage at couplings (2) or (6) is cause for overhaul evaluation of driveshaft. Refer to BHT-412-CR&O manual.

2. Mechanical and corrosion damage limits on driveshaft (4) are defined as follows:

MECHANICAL/CORROSION DAMAGE	LIMITS
Nicks and scratches running parallel to, or within 15° of parallel to shaft axis.	Not exceeding 0.004 inch (0.101 mm) in depth are acceptable without repair. Not exceeding 0.008 inch (0.203 mm) in depth or 1.5 inches (38 mm) in length are acceptable if polished out, provided the total polished area does not exceed 10% of circumference at any point.
Nicks and scratches not running within 15° of shaft axis.	Not exceeding 0.004 inch (0.101 mm) in depth are acceptable on up to 100% of the circumference of shaft, if polished out. Not exceeding 0.008 inch (0.203 mm) in depth or 1.5 inches (38 mm) in length are acceptable provided the total polished area does not exceed 10% of circumference at any point.
Corrosion pitting.	Not exceeding 0.008 inch (0.203 mm) in depth is acceptable provided it is polished out. Maximum length of rework at any point must not exceed 1.5 inches (38 mm) nor cover more than 10% of the circumference at any point. The maximum acceptable total reworked area is 4.0 square inches (2580 mm^2).

3. Mechanical and/or corrosion damage on retainer plate (1) or similar plate in couplings (6) that could result in lubricant leakage is not acceptable.

4. Mechanical and/or corrosion damage on boots (3 and 5) that could result in lubricant leakage is not acceptable.

5. Superficial mechanical damage on couplings (2 and 6) is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion damage is cause for overhaul evaluation of driveshaft (4). Refer to the BHT-412-CR&O.

6. Visual over-temperature indicator (TEMP-PLATES) for discoloration and overheat condition. A change in color of TEMP-PLATES dots from white to black will indicate a possible overheat condition and/or component degradation, and cause of indicator (TEMP-PLATES) dots' discoloration should be determined prior to condition operation. Refer to the BHT-412-CR&O for Inspection and Lubrication procedures.

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Figure 1. Main Driveshaft - Damage Limits (Sheet 2 of 2)

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Engine-to-Transmission (Main) Driveshaft - Lubrication

Procedure

1. Refer to the BHT-412-CR&O for coupling lubrication instructions.

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Engine-to-Transmission (Main) Driveshaft - Installation

Procedure

NOTE

The two ends of the driveshaft can be installed forward. It is recommended that the flexible coupling with the identification plate be installed in the forward position.

Applicable: SN: 33001-Subsequent or 36001-36019:

1. *Do the steps that follow:*

1.1. *Put the driveshaft (6, Figure 1) in position in the center firewall.*

NOTE

It is necessary to push the flexible couplings inboard to accomplish the step that follows. If possible, compress the flexible couplings equally to avoid moving the lubricant inside.

Use the marks from the removal to help put the flexible couplings in the initial position.

1.2. *Put the flexible couplings in the correct position on the main input quill curvic coupling adapter (9) and the engine-to-driveshaft curvic coupling adapter (5).*

NOTE

Use the marks from the removal to put the balancing weights in the initial position.

1.3. *If necessary, install the balancing weights (2 and 8) where applicable.*

1.4. *Install bolts (1 and 7), washers (3 and 10), and nuts (4 and 11).*

NOTE

Refer to Figure 1 for torque sequence.

1.5. *Torque nut (4 and 11)* .

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

2. *Do the steps that follow:*

2.1. *Put the driveshaft (17) in position in the center firewall.*

NOTE

It is necessary to push the flexible couplings inboard to accomplish the step that follows. If possible, compress the flexible couplings equally to avoid moving the lubricant inside.

Use the marks from the removal to help put the flexible couplings in the initial position.

2.2. *Put the flexible couplings in the correct couplings in the correct position on the main input quill curvic coupling adapter (22) and the engine-to-driveshaft curvic coupling adapter (16).*

NOTE

Use the marks from the removal to put the balancing weights in the initial position.

2.3. *Install the bolts (14 and 21), washers (15 and 20), balancing weights (13 and 19) (where applicable), and nuts (12 and 18).*

NOTE

Refer to Figure 1 for torque sequence.

2.4. *Torque the bolts (14 and 21)* .

3. Connect the starter-generator air hose and the fire detector leads.

4. Install the air inlet fairings and the engine top cowling.

NOTE

The ground run and leak check must be performed following the installation of new, repaired, or relubricated driveshaft flexible couplings.

5. *Perform a ground run of the helicopter for approximately 5 minutes at 100% rotor RPM. Refer to applicable Flight Manual.*

6. After the rotor is stopped, inspect the flexible coupling and surrounding structure for signs of grease leakage.

7. If there is leakage, remove the driveshaft (6 or 17) and inspect as per BHT-412-CR&O.

NOTE

SN: 33001-33213 or 36001-36019 (412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)):

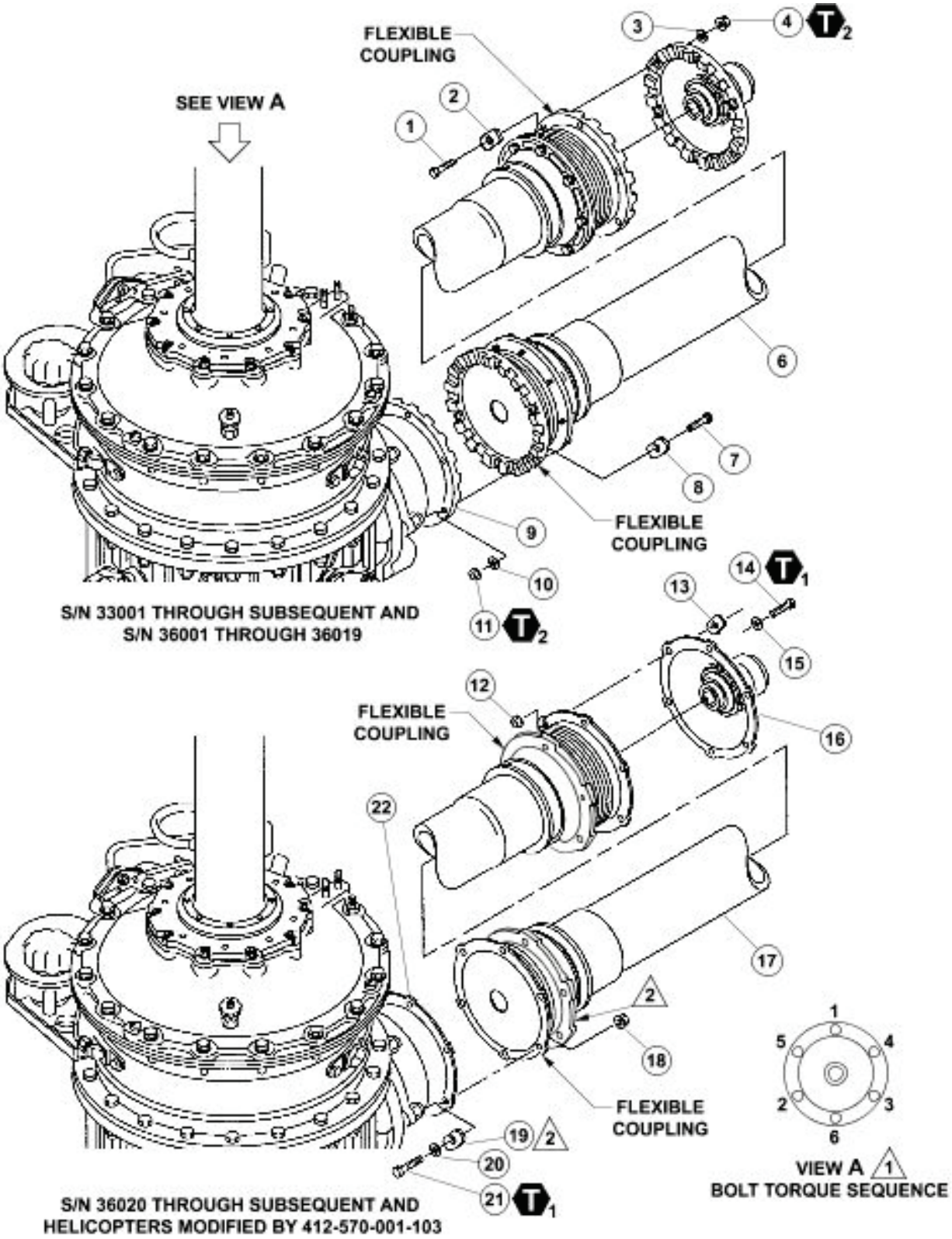
The step that follows is optional and not necessary for component life or serviceability requirements.

NOTE

SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

*The driveshaft must be balanced anytime it has been removed (*412-A-18-11-00-07A-271A-A / 00171*).*

8. If necessary, balance the driveshaft (6 or 17) (412-A-63-10-01-01A-042A-A / 01000).



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Figure 1. Engine-to-Transmission (Main) Driveshaft (Sheet 1 of 2)

1. Bolt
2. Balancing weight
3. Washer
4. Nut
5. Engine-to-driveshaft curvic coupling adapter
6. Driveshaft
7. Bolt
8. Balancing weight
9. Main input quill curvic coupling adapter
10. Washer
11. Nut
12. Nut
13. Balancing weight
14. Bolt
15. Washer
16. Engine-to-driveshaft curvic coupling adapter
17. Driveshaft
18. Nut
19. Balancing weight
20. Washer
21. Bolt
22. Main input quill curvic coupling adapter

T1 70 TO 90 IN-LBS (7.91 TO 10.16 Nm)

T2 100 TO 140 IN-LBS (11.30 TO 15.81 Nm)

NOTES

1. Torque bolts in three steps; 1/3 full torque, 2/3 full torque, and full torque respectively. Tighten bolts in sequence as shown in Detail A. Repeat sequence at each torque step.

2. Use aft flange of coupling for balance adjustment when air conditioning is installed.

ICN-412-A-631001-B-97499-00002-A-001-01

Figure 1. Engine-to-Transmission (Main) Driveshaft (Sheet 2 of 2)

Engine-to-Transmission (Main) Driveshaft - Balancing

NOTE

SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

Perform main driveshaft balancing whenever the driveshaft is removed and reinstalled.

Refer to 412-A-18-11-00-07A-271A-A / 00171 for balancing procedures.

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Engine-to-Driveshaft Coupling Adapter - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Wrench Set	T101588	1	
SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed: Wrench Set	*T103063*	*1*	*(412Classic/SP to 412HP Upgrade)*

Procedure

1. Remove the main driveshaft (412-A-63-10-01-01A-520A-A / 00994).

2. Straighten the tang of the washer (3, Figure 1) to permit removal of the retaining nut (4).

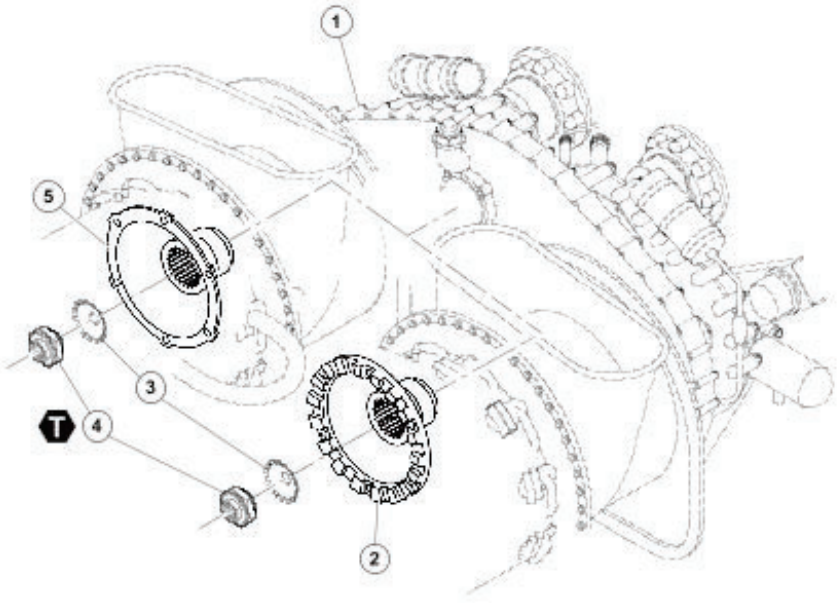
3. Position the plate assembly (7 or 8, Figure 2), as applicable, on the adapter (1) and install the nut (4).

4. Install the extension bar (6). Hold the plate assembly (7 or 8), as applicable, and turn the extension bar to loosen the retaining nut (5).

5. Remove the extension bar (6), plate assembly (7 or 8), as applicable, and retaining nut (5).

6. Remove the engine-to-driveshaft coupling adapter (2 or 5, Figure 1).

7. Discard the washer (3).



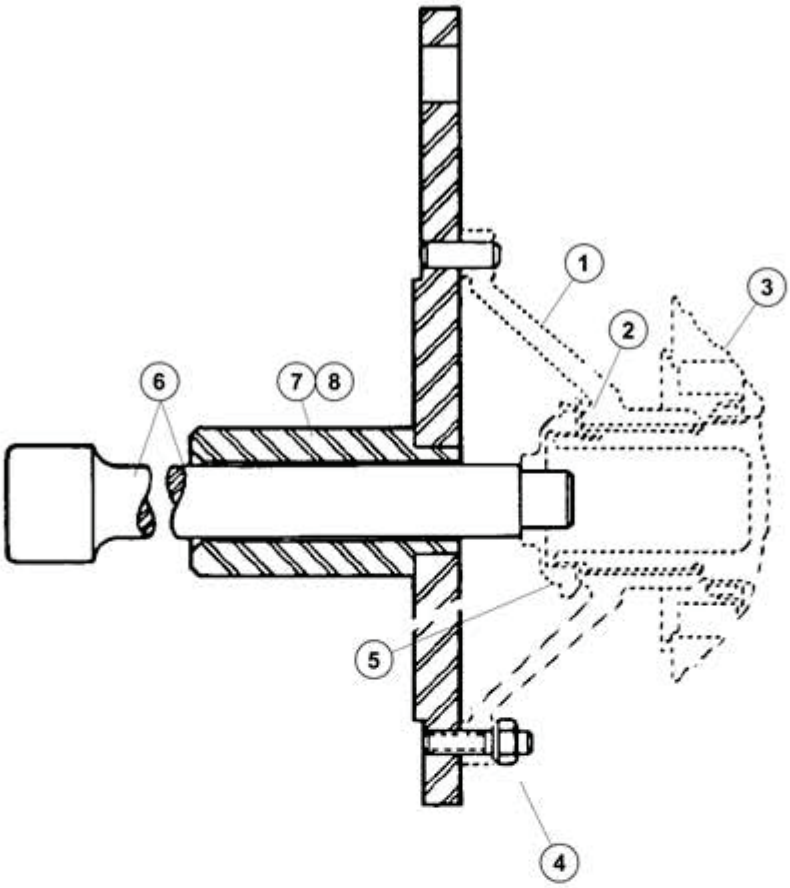
1. Engine
2. Engine-to-driveshaft curvic coupling adapter
3. Washer
4. Retaining nut
5. Engine-to-driveshaft coupling adapter

T 250 TO 275 FT-LBS
(339 TO 373 Nm)
INITIAL TORQUE

125 TO 200 FT-LBS
(170 TO 271 Nm)
FINAL TORQUE

ICN-412-A-631001-B-97499-00005-A-001-01

Figure 1. Engine-to-Driveshaft Coupling Adapter (Sheet 1 of 1)



1. Adapter
2. Washer
3. Engine installation reference
4. Nut
5. Retaining nut
6. Extension bar (L-122)
7. Plate assembly (T101588)
8. Plate assembly (T103063)

NOTES

1. Install T101588 plate assembly (7) on adapter (1). Secure with nut (4).
2. Insert extension bar (6) into plate assembly (7 or 8). Ensure that extension bar is seated all the way into retainer nut (5) before attempting to loosen nut.
3. Install T103063 plate assembly (8) on adapter (1) for model 412 S/N 36020 and subsequent, and helicopters modified by 412-570-001-103 (412SP to 412HP) (BHT-412-SI-74) (412Classic to 412HP) (BHT-412-SI-97).

ICN-412-A-631001-B-97499-00006-A-002-01

Figure 2. Engine-to-Driveshaft Coupling Wrench Set - Tool Application (Sheet 1 of 1)

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Engine-to-Driveshaft Coupling Adapter - Inspection

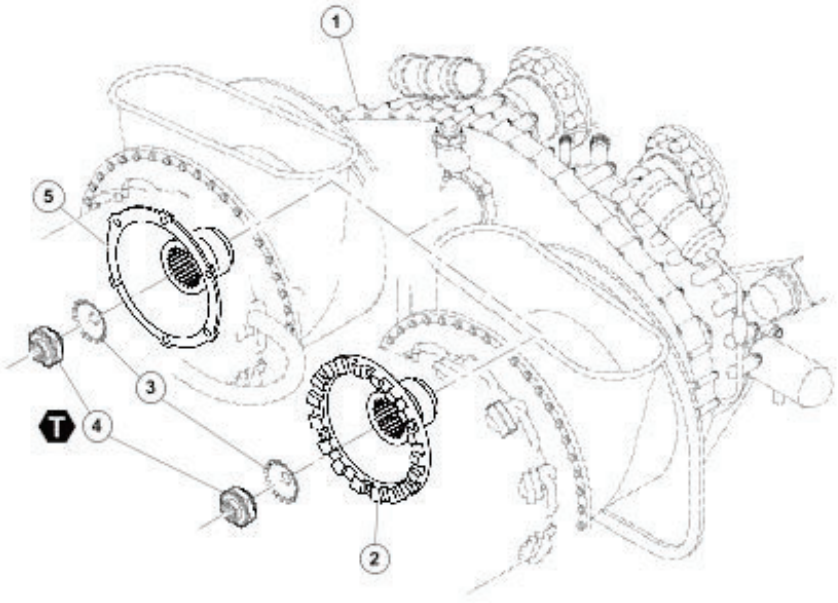
Procedure

Applicable: SN: 33001-Subsequent or 36001-36019 or 412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed):

1. *Inspect the engine-to-driveshaft curvic coupling adapter (2, Figure 1) as per information in Figure 2.*

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

2. *Inspect the engine-to-driveshaft coupling (5, Figure 1) adapter as per information in Figure 3.*



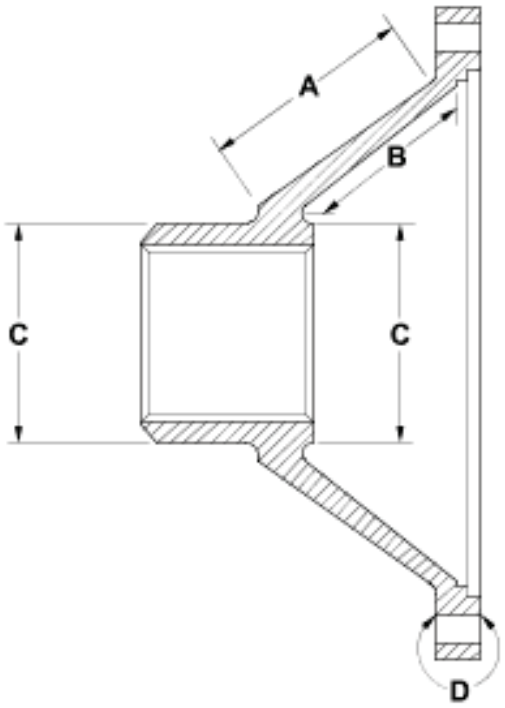
1. Engine
2. Engine-to-driveshaft curvic coupling adapter
3. Washer
4. Retaining nut
5. Engine-to-driveshaft coupling adapter

T 250 TO 275 FT-LBS
(339 TO 373 Nm)
INITIAL TORQUE

125 TO 200 FT-LBS
(170 TO 271 Nm)
FINAL TORQUE

ICN-412-A-631001-B-97499-00005-A-001-01

Figure 1. Engine-to-Driveshaft Coupling Adapter (Sheet 1 of 1)



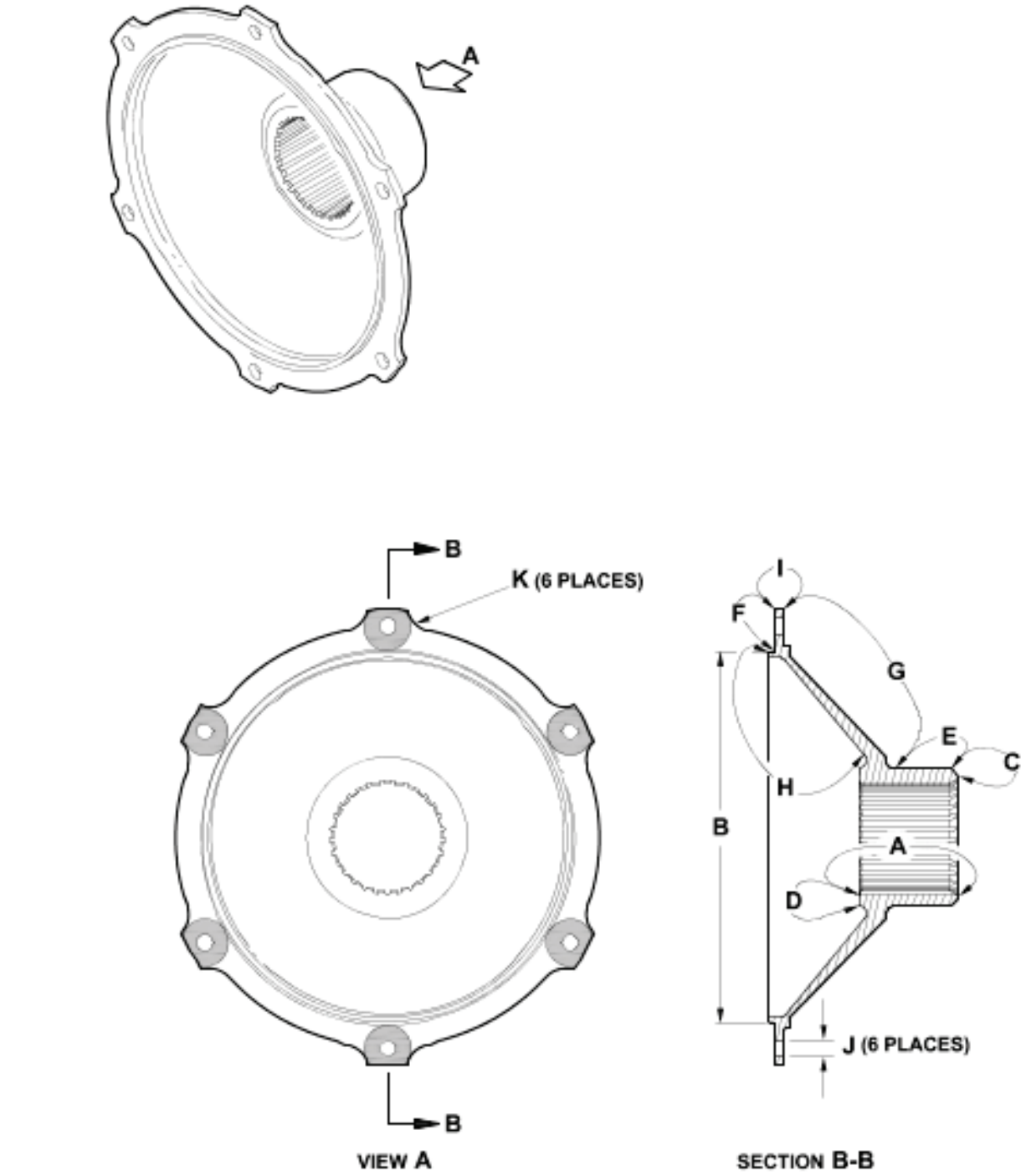
NO.	REF LTR	CHARACTERISTIC	INSPECTION METHOD	LIMIT
1.	Spline	Wear	Measure	1.6687 inches (42.385 mm) maximum dimension between flatted 0.1080 inch (2.7432 mm) diameter pins.
2.	All radiuses	Mechanical/ corrosion damage	Visual	None permitted.
3.	A	Mechanical/ corrosion damage	Visual/measure	0.010 inch (0.254 mm) maximum depth and no more than 20% of surface area. 1
4.	B	Mechanical/ corrosion damage	Visual/measure	0.010 inch (0.254 mm) maximum depth and no more than 20% of surface area. 1
5.	C	Mechanical/ corrosion damage	Visual/measure	0.010 inch (0.254 mm) maximum depth and no more than 10% of surface area. 1
6.	D and spline	Mechanical/ corrosion damage	Visual	Only superficial corrosion that can be removed by hand polishing with abrasive pad (C-407) is permitted. Mechanical damage is restricted to polishing out minor damage with a fine India Stone (C-464).

NOTES

1. No overlapping repair to outside diameter and inside diameter is permitted.
2. Brush cadmium plate on reworked areas.

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Figure 2. Engine-to-Driveshaft Curvic Coupling Adapter (S/N 33001 and Subsequent and 36001 through 36019) - Damage Limits (Sheet 1 of 1)



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Figure 3. Engine-to-Driveshaft Coupling Adapter (S/N 36020 and Subsequent) - Damage Limits (Sheet 1 of 3)

NO.	REF LTR	CHARACTERISTIC	INSPECTION METHOD	LIMIT	
1.	A	Spline wear	Measure between 0.1080 inch (2.743 mm) diameter pins	1.6687 inches (42.385 mm) maximum diameter.	
2.	A	Mechanical damage	Visual	None permitted.	
3.	B	Mechanical damage	Visual/measure	0.005 inch (0.13 mm) maximum depth, 0.500 inch (12.70 mm) maximum length, and 1.50 inches (38.1 mm) minimum spacing between damage.	1
4.	B	Wear	Measure	6.0795 inches (154.419 mm) minimum diameter.	
5.	C	Mechanical damage	Visual/measure	0.005 inch (0.13 mm) maximum depth, 0.500 inch (12.70 mm) maximum length, and 0.750 inch (19.05 mm) minimum spacing between damage.	1
6.	D	Nicks, dents, scratches	Visual/measure	0.005 inch (0.13 mm) maximum depth.	1
7.	D	Circular wear grooves	Visual/measure	0.002 inch (0.05 mm) maximum depth.	1
8.	E	Mechanical damage	Visual/measure	Fine axial scratches of 0.001 inch (0.03 mm) maximum depth.	
9.	A, B, C, D, E	Corrosion damage	Visual	Only superficial corrosion permitted.	2
10.	F	Corrosion damage	Visual/measure	0.002 inch (0.05 mm) maximum depth and no more than 50% of total surface area between two holes.	3
11.	G, H	Mechanical damage	Visual	0.010 inch (0.25 mm) maximum depth and no more than 20% of surface area.	1
12.	G, H	Corrosion damage	Visual	0.010 inch (0.25 mm) maximum depth and no more than 20% of surface area.	2
13.	I	Corrosion damage	Visual/measure	0.003 inch (0.08 mm) maximum depth.	

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Figure 3. Engine-to-Driveshaft Coupling Adapter (S/N 36020 and Subsequent) - Damage Limits (Sheet 2 of 3)

NO.	REF LTR	CHARACTERISTIC	INSPECTION METHOD	LIMIT	
14.	I	Mechanical damage	Visual/measure	0.010 inch (0.25 mm) maximum depth and 0.500 inch (12.70 mm) maximum length.	2 3
15.	J	Corrosion damage	Visual/measure	0.010 inch (0.25 mm) maximum depth and no more than 50% of total wall area.	3
16.	J	Bore Wear	Measure	0.275 inch (6.98 mm) maximum diameter.	
17.	J	Elongation	Measure	0.279 inch (7.08 mm) maximum diameter.	
18.	K	Corrosion damage	Visual/measure	0.002 inch (0.05 mm) maximum depth and 0.50 inch (12.7 mm) maximum distance around each hole.	
19.		Cracks	MPI	None permitted.	
20.	All radii	Mechanical/corrosion damage	Visual	None permitted.	

NOTES

1. Polish with a fine india stone (C-464) to remove any raised material.
2. Polish with a fine steel wool (C-411) or abrasive pad (C-407).
3. Remove corrosion with abrasive glass beads (C-470). Refer to the CSSD-PSE-87-001, Chapter 20 for procedure.
4. Brush cadmium plate reworked areas per BHT-ALL-SPM.
5. All dimensions, wear, and damage limits provided are limits required after repair except where indicated.
6. Driveshaft applicable to helicopters S/N 36020 and subsequent and helicopters modified by 412-570-001-103 or post BHT-412-SI-74 (412SP to 412HP upgrade) or BHT-412-SI-97 (412Classic to 412HP upgrade).

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Figure 3. Engine-to-Driveshaft Coupling Adapter (S/N 36020 and Subsequent) - Damage Limits (Sheet 3 of 3)

Engine-to-Driveshaft Coupling Adapter - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
India stone	C-464	AR	

Procedure

1. Repair of the adapter is restricted to polishing out minor damage on the face that contacts the driveshaft. Polish out repairable damage with a fine india stone (C-464).

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